Lectures on Optics

"To Understand the universe we must understand the light"

Saturday 5th September, 2026

LECTURES ON OPTICS

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Abstract

The field of optics is a captivating and continually evolving branch of science that plays an indispensable role in our modern world. "Advances in Optics: Exploring the Boundaries of Light and Vision" presents a comprehensive overview of the latest developments, theories, and applications in optics, spanning from fundamental principles to cutting-edge technologies.

This book delves into the fundamental properties of light, offering a clear and concise explanation of wave-particle duality, the nature of electromagnetic waves, and the behavior of light in various media. It explores the principles of geometric optics, including reflection, refraction, and the formation of images, while also addressing wave optics phenomena such as interference, diffraction, and polarization.

In addition to the foundational aspects, "Advances in Optics" goes beyond traditional optics to cover emerging topics and technologies. Readers will find detailed discussions on advanced optical materials, including metamaterials and photonic crystals, and their revolutionary applications in the manipulation of light for imaging, communication, and sensing. Quantum optics, an exciting frontier in optical science, is also explored, with a focus on quantum entanglement, quantum cryptography, and quantum computing.

Moreover, this book highlights the practical applications of optics in various fields, including medicine, telecommunications, astronomy, and environmental monitoring. It provides insights into state-of-the-art optical instruments and techniques, such as lasers, optical fibers, and adaptive optics systems, and their pivotal roles in advancing technology and scientific discovery.

"Advances in Optics" is designed to cater to a broad readership, from students and educators seeking a comprehensive textbook to researchers and professionals looking for the latest developments in the field. Its accessible style, accompanied by illustrative examples and diagrams, ensures that both newcomers and experts can appreciate the wonders of optics and its ever-expanding frontiers.

As we venture further into the 21st century, the exploration of light and vision promises to yield groundbreaking discoveries and transformative technologies. "Advances in Optics" serves as an invaluable resource for anyone curious about the principles, innovations, and limitless potential of optics in shaping our future.

Chapter 1

The loving Story of Optics

"The light is the God's shadow"
– Saint Agustin

When we study physics beyond doing long calculations, formulate complicated theories or doing extremely elaborated experiments in an advanced laboratory there is a really important fact many people ignore, the *story of physics* if we want to study physics we must have understand where was born the physics and more important, how was it done. Here we are going to check the story of physics from the geometrical optics, to corpuscular optics to wave optics, to electromagnetic optics and the most recent the quantum optics.

1.1 Optics from Euclid to Huygens

Here I'm going to copy and paste one of the best articles I've read about the story of optics (Herzberger, 1966).

If I were endowed with dictatorial powers, I would require everyone receiving a degree in a scientific subject to know its history and to have read the classical papers relating to it. Historical knowledge is important because it stimulates creative thinking. The man who first struggled with an idea, trying to find a law, looked at the situation with different eyes than do we who accept the law as a matter of course. He considered alternatives to the law, and different interpretations, and some of these alternatives and interpretations may still be stimulating and worth thinking about.

I was fortunate during the eight years of my affiliation with the Zeiss works at the beginning of my career to have access to their extensive library; it contained the originals, or at least translations, of nearly all the writings in optics from Euclid to contemporary times. Over thirty years ago, I wrote a historical survey describing the content of important papers on geometrical optics. This task was immensely facilitated by the extensive bibliography in the third edition of Czapski-Eppenstein,I to which I had contributed my small part as a work student at the firm of Zeiss. Lately, I have looked through most of these papers again, as far as they were available to me, and have discovered many important details that I had not understood at that time. I should like to share these with you. I emphasize that my interest is not basically historical. I am not interested so much in questions of priority and originality of documents. My object is to entice you to read and to enjoy the old scriptures, and maybe take along an idea or two that might be of interest.

1.1.1 Ancient times

I am sure that the Egyptians and, even more, the Babylonians, with their great interest in astronomy,must have had some knowledge of optical laws. But the first important mention of one of the phenomena of refraction I found was in Plato's Republic.' Plato(427-347 B.C.) quotes Socrates as saying (Jowett's translation): " . . . the same object appears straight when looked at out of the water, and crooked when in the water; and the concave becomes convex, owing to the illusion about colours to which the sight is liable. Thus every sort of confusion is revealed within us; and this is that weakness of the human mind on which the art of conjuring and of deceiving by light and shadow and other ingenious devices imposes, having an effect upon us like magic.¨ Our senses give an erroneous picture of reality; they are untrustworthy;alone can lead us to truth pure thinking. This idea persisted for two millennia under the influence of the misinterpreted teachings of Aristotle.

How we see is a question that long occupied the Greek philosophers, and, through them, the Neo-Platonists and the Scholastics. (Indeed, it cannot be said that the question is settled, and in recent times Ronchi has set forth some cogent ideas.) Even before Plato there was speculation in the writings of the Greek philosophers about the process of seeing. Most of the Greek thinkers believed that the eye sends out rays to the object, and Aristotle (384-322 B.C.) seems to have been the first to question this belief. If, says Aristotle, the eye sends out fiery rays that stream from it as from a lantern, why can we not see in the dark? His explanation, that the eye contains a transparent substance and that seeing is associated with the motion of something between the object and the eye, seems quite modern.

After the conquest of Greece by Alexander, Alexandria (in Egypt) and Sicily became the strongholds of Greek culture. The first great textbook of optics was written by Euclid of Alexandria, who lived about 300 B.C. And is better known for his Elements of Geometry. This famous work is still must reading for every scientist and is available in several editions. When King Archelaos once complained about the difficulty of learning mathematics and asked whether there was not an easier way, Euclid's answer is supposed to have been: "There is no royal route to geometry¨. When one of his pupils asked him what of material value could be gained by learning geometry, he told a slave, "Give the man threepence and tell him to go away¨.

Under Euclid's name we have two books of interest in the present connection: Optics and Catoptrics. The first deals with the laws of vision and perspective and the second with reflection. An English translation of the first was published in the Journal of the Optical Society of America twenty years ago. 7 It is a striking fact that one of the definitions in this book contains an idea that could be used to develop many problems of diffraction theory.

I quote some of Euclid's definitions, slightly paraphrased from the Optical Society of America translation: "Let it be assumed that lines drawn directly from the eye pass through a space of great extent [not infinite]; that the form of this space is the mantle of a cone with its apex in the eye and its base at the limits of our vision; that those things upon which the cone falls are seen and those things upon which the cone does not fall are not seen; and that things seen through several cones simultaneously appear to be more distinct.¨ From this, Euclid concludes, among other things, that nothing can be seen in its entirety. The same objects located nearby are seen more distinctly than if they were at a distance. Every object that is visible has a certain limit of distance. If that is reached-that is, if the object lies entirely inside the mantle of the cone of vision-it is no longer seen. These are the axioms from which he derives the laws of perspective.

In Catoptrics the law of reflection is stated, namely, that incoming and outgoing rays form the same angle with the surface normal. From this he derives the exact position of the image behind a plane mirror and the principle that left- and right-hand sides are interchanged. He generalizes these laws for spherical mirrors, both concave and convex, describing virtual images as well as stating that concave mirrors can give real images. As the position of the image is formed by reflection at curved surfaces, he suggests the point of intersection of the ray with the axis (shades of my definition of diapoint!). He shows the existence of a focal point for spherical mirrors, although he errs with respect to its position. One of his theorems says that an eye at the center of a spherical mirror sees only itself. Another states that you can light a fire at the focus of a concave mirror. He states that straight lines appear convex in a convex mirror. One can see the same object through many plane mirrors if they are positioned along the sides of a regular polygon with two more sides than there are mirrors.

According to Roman historians, the greatest mathematician of ancient times, Archimedes (287-212 B.C.), is supposed to have used mirrors to destroy the Roman fleet besieging Syracuse. Like most scientists, I have doubted this story, but a personal experience made me doubt my doubts.

Two years ago I had an optical accident. My wife had me get a large bottle of distilled water, and I left it in my auto in front of our laboratory. Driving home that evening I smelled smoke and stopped at a garage. After thoroughly investigating all the mechanical details and finding them in order, the mechanics discovered that smoke was coming out of the upholstery. The cylindrical bottle had acted like a lens and the sun had made a hole in the upholstery and ignited the stuffing inside. After this personal experience, I am somewhat more reluctant to doubt the story of Archimedes' mirrors.

The next person I should like to discuss, Hero of Alexandria, lived sometime between about 150 B.C. and A.D. 250. We have two outside dates for this philosopher: he mentions Hipparchos, who flourished between 146 B.C. and 126 B.C., but the mathematician Pappus (ca. A.D. 300) follows Hero's treatment in his own writings on mechanics. Be this as it may, Hero's Catoptrics is one of the most interesting books on optics. He is not content with Euclid's statement that light travels in straight lines, nor is he content with the form of the reflection law. He looks for a reason for both laws and derives it from the assumption that light chooses the shortest possible path between two points. We know that this minimum principle is not quite correct: the path is, in general, only an extremum. It may or may not be significant that Hero's book illustrates the concept with a picture of a convex mirror, for which the minimum principle is correct-not a concave mirror, for which it can be wrong. We shall find the same trick worked later by Fermat, who extended Hero's law to derive the refraction law. However, Fermat was called on the carpet for this by Abbé Mersenne, one of his contemporaries.

Hero of Alexandria-who, by the way, in his Mechanics described the so-called seven simple machines known to us-not only discussed spherical mirrors in his Catoptrics but, as an early practical experimentalist, also showed what can be done with cylindrical mirrors. If you go into an amusement parlor and see yourself in mirrors of all kinds (I especially like those that elongate the height and diminish the girth), you know the proprietor has read Hero of Alexandria. If you see a play in a tanagra theater, where the actors appear on the stage, by the skillful use of mirrors, as beautifully dressed dwarfs, you know that Hero of Alexandria wrote the prescription. If you want to see around the corner or look at a person without letting him see you, Hero of Alexandria tells you how to do it with mirrors.

Ptolemy of Alexandria, who lived about A.D. 150, is best known for his Almagest, but he was also the first to make a serious attempt to study the law of refraction. He wrote a five-part volume entitled Optics, several copies of which have come down to us. This work, of which the first part has been lost (although we have a description of its contents), can be summarized as follows.

The first part deals with the relations between the eye and light, and the second with the conditions of visibility: size, form, color, and motion of bodies. Ptolemy explains why we usually see only one image of an object, although we have two eyes, and describes the conditions under which we see more than one image. The third part deals with the laws of reflection by plane and convex mirrors. Ptolemy gives probably the first attempt at an explanation of why the moon and the sun appear larger at the horizon than at the zenith a question that is still being discussed.

In the fifth part, Ptolemy tries to find a law for refraction. First he observes that for small incident angles the angles of incidence and of refraction are proportional to each other. He also remarks that, in transition from a thinner to a denser medium, the refracted ray comes earer to the perpendicular, the opposite being true for the reverse direction of travel. He uses an ingenious contraption to measure the incident and the refracted angles for the transition from air into water, air into glass, and water into glass. This instrument consists of a circular disk divided into 3600 with a colored marker in the center and two movable markers on the periphery. You put the instrument into water, line up the three markers, and read the two positions, one giving the incident angle and the other the refracted angle. Ptolemy gives refraction tables in 100 intervals. My friend, the late Hans Boegehold, studied hese tables carefully and found that the values for 500 and 600 incident angles are surprisingly accurate, but that the other angles were obviously calculated by a second-degree interpolation formula. (The first differences are exactly 1 30'.14) Ptolemy applies the same method for the transition of light from air into glass and from water into glass. He also discusses atmospheric refraction, the cause of which he assumes to be that the densities of air and of ether vary with height. He points out that only for a star in the zenith do the true and the apparent positions coincide.

1.1.2 The arabain period

After the fall of Greece, scientific progress ceased on the continent of Europe for nearly a century and a half, but Alexandria remained the center of scientific discovery. When the Arabs took over, they brought their scientific inheritance and knowledge of mathematics and the mathematical sciences into northern Africa and Spain. It is remarkable that, at a time when magic and mysticism and misinterpretation of Aristotelian doctrines prevented scientific progress over most of Europe, the

Arabs held high the torch of science in their countries.

The next big contribution to our science was by Ibn al-Haitham, or Alhazen of Cairo, whose date is frequently set around 1100, although some sources tell us he lived from 965 to 1038. Alhazen's Optics" consists of seven parts. The first gives an anatomical description of the eye, which, he maintains, contains three liquids separated by four membranes. The names which he gives to them the watery medium, the crystalline medium, and the glassy medium are still used today. The phenomenon of binocular vision is explained by assuming that the pictures from the two eyes are transported to one visual nerve. Alhazen discusses the pupil of the eye and speaks of a pyramid of rays (rather than a cone) coming from the object to the eye.

The second part, like the corresponding book by Ptolemy, deals with the properties of bodies, and he distinguishes no fewer than twenty-two different properties.

The third part deals with optical illusions. He draws attention to the illusions arising from a false interpretation of a picture by the human mind and imagination. The fourth, fifth, and sixth parts deal with problems of reflection and mirrors. One of the problems studied in great detail has since been called after Alhazen: it consists in finding the point on a mirror where the incident and the reflected rays meet. Alhazen corrects errors of the Greeks about the position of the images in spherical, conical, and cylindrical mirrors. He also emphasizes the fact that the incident and reflected rays lie in the same plane.

Alhazen's seventh part deals with refraction. He proposes methods for measuring incident and refracted rays through different media, but he gives no tables. He knows that a glass sphere magnifies. He explains the apparently large size of the sun at the horizon by the assumption that we compare the sun with objects of known size lying along the line of sight. He investigates binocular vision and makes a good guess at the height of the atmosphere. He breaks with the idea of the Greeks that light rays originate at the eye and go to the objects, supporting his claim by the circumstance that sunlight can hurt the eye and that we can see afterimages.

Other manuscripts show that Alhazen used the pinhole to view solar eclipses and that he knew how the mage becomes blurred and finally becomes circular as the pinhole increases in size. He did not write as though this were a discovery of his own, and Aristotle had noted that the spot formed by the rays from the uneclipsed sun is circular when the rays pass through a rectangular hole in wickerwork if the hole is small, but becomes rectangular when the hole is large. However, Alhazen's is the first known unequivocal description of the camera obscura. Alhazen's works were of enormous influence in the development of optics.

1.1.3 The middle ages

The knowledge of the Arabs was transmitted to Europe via Spain and Sicily and set in motion the intellectual revival that culminated in the Renaissance. The first of the European scientists to play a role in optics was Vitello, a Polish mathematician who lived and studied in Italy about 1270. His book in ten parts is probably the most voluminous optical work ever written. It contains about five hundred closely printed folio pages. Unfortunately, he is more famous for his errors than for his contributions.

He reissued the tables of Ptolemy and added tables for the passage of light from water into air, from glass into air, and from glass into water. It is unfortunate that he never tells us how he arrived at his values, for they include impossible data, being in contradiction with the then unknown phenomenon of total reflection.

It was this error that later prevented Kepler from finding the correct form of the refraction law. Nevertheless, Vitello made two notable contributions. He explained the colors of the rainbow as originating from refraction and reflection by the water drops, and he made the proposal to collect the light of the sun into a point by using parabolic mirrors.

The Englishman Roger Bacon (1215-1294) discovered the spherical aberration of a mirror; he further stated that rays forming a cone around the axis in object space also form a cone around the axis in image space. From the magnifying property of a sphere, he speculated that it might be possible to invent instruments in which a man would look like a mountain and a small group like a big army. However, since he did not say how this could be achieved, it is doubtful whether he should be called (as he was by Sir John Herschel) the inventor of the telescope. His writings are so sketchy and in definite that it is impossible to attribute specific inventions to him or to reach any conclusions about what he actually knew.

1.1.4 The renaissance

Coming to the Renaissance, we must mention Leonardo da Vinci (1452-1519). His notebooks show that he studied the eye anatomically and made a model of it, although he supposed the rays to cross twice in it to form an upright image. 18 , 9 Immediately afterward he described the camera obscura but correctly showed the image inverted!20 He described, with sketches, a Rumford-type photometer and a machine for grinding concave mirrors. Unfortunately for his optical reputation and the development of optics, he never organized his notes into publishable form, so they were practically unknown until Venturi published them in 1797.

About half a century later came another Italian we should mention, Francesco Maurolico (1494-1577), who solved one of the problems that had bothered previous thinkers: that the image of the sun is round even if formed by a square pinhole. He improved our knowledge of vision by suggesting that the crystalline lens of the eye works like a double-convex lens. This enabled him to attribute myopia and hyperopia to an eye whose lens has a larger or smaller curvature than is proper for the length of the eyeball. He emphasized for the first time that the rays from an object point envelop a surface that is now called the caustic surface and that this surface is the place of the highest light concentration. And he explained the colors of the rainbow by multiple reflection in the water drops. All this information was concisely organized by 1567 into a work entitled Photismi de lumine , although it was not published for some thirty years (1611) after the author's death, and then only as a consequence of the interest in optics engendered by Galileo's telescopic discoveries.

In Italy, under the influence of the philosophy of the time, scientific progress and magic were mixed in a way that led to interesting reading. The development of mathematical methods, however, served to gradually separate speculation and imagination from the solid basis of observation.

One of the most colorful books of the period was the Magia naturalis of Giovanni Battista della Porta (ca. 1538 or 1543 -1615). In his youth, della Porta had written a four-part volume entitled De miraculis rerum naturalium (1558) and, subsequently, many books and plays, including De refractione (Fig. 6) in nine parts in 1593. The Magia naturalis, which was an expansion of the earlier De iraculis into twenty volumes, appeared in 1589. It became so famous that it was soon translated from Latin into the vernacular languages Italian, French, Spanish, and English. It is a strange medley. Only the seventeenth part concerns optics, but some of the others are also worth noting for our amusement if nothing else. I mention the titles of a few of the twenty parts, to give you an idea of their contents. The first is entitled, "The Causes of the Miraculous. It deals with astrology, sympathy and antipathy, and the influence of place and time on events. The second deals with the origin of animals. He quotes many authorities who agree that snakes are made from the marrow of men and the hair of women. The third deals with garden plants; the fourth, with how to become wealthy by good housekeeping; the fifth, with how to make precious stones artificially; and the ninth, with make-up and other ways of improving the beauty of women (as if this were necessary!).

Finally, the seventeenth part deals with focal mirrors and other mirrors and explains what one can do with them. This book exhibits progress in the understanding of optical phenomena. In it are discussed the same tricks that Hero had dealt with 1400 years earlier, namely, how to use mirrors to obtain desired effects. However, we find here the first exact theory of multiple mirrors, the first complete description of the camera obscura, with pinhole or lens, and a discussion of combinations of positive and negative lenses designed to improve vision. Porta compares the eye and its pupil to the camera obscura and its stop. (Leonardo had anticipated him here, but less definitely and accurately.) At the end of this century also falls the invention of the telescope and the microscope.

Eyeglasses had apparently been invented toward the end of the thirteenth century, but they seem not to have attracted the attention of writers on optics until the practical Leonardo discussed methods of making them. Some lens grinder-more likely, several independentlymust have discovered the principle of the Galilean telescope by accident, and certain passages in Leonardo's notebooks a hundred years before Galileo can be interpreted as instructions for making such a telescope. Be that as it may, a Dutchman, Zacharias Jansen (1588-1632), was said by his son to have made a telescope in 1604 according to the model of an Italian dated 1590, and soon telescopes became well known in Holland. Galileo Galilei (1564-1642) heard of them and, apparently independently, made an instrument that he exhibited in 1609, and in his lifetime made over a hundred. The Dutch instruments were so imperfect that a magnifying power of more than 3X was impractical. Galileo made instruments with a useful power of approximately 30 X, an order of magnitude greater. He was by no means the inventor of the telescope or even the first scientist to describe it; Vavilov reminds us that Leonardo may have anticipated him and that Porta described, three months after Galileo's demonstration, an instrument he had made independently. But Galileo is outstanding on two counts: he revolutionized man's ideas of the uni- verse by his applications of the telescope and, by improving it by an order of magnitude, Ronchi points out that Galileo is entitled to be considered the father of the art of fine optics.

Moreover, from telescope to microscope is a short optical step, and by 1624 Galileo was using lenses of short focal length and thus making precursors of the modern compound microscope. 9 A few decades thereafter, the Dutchman Antonj van Leeuwenhoek (1632-1723) improved the quality of simple lenses used for magnifiers or simple microscopes, and in a letter of 1674 he began the descriptions of the discoveries for which he became justly famous.24 Meanwhile, Robert Hooke (1635-1703) in England also improved the quality of lenses in order to make the compound microscope practical, and in 1665 the Royal Society published his Micrographia.

Aside from publications, which were in an embryonic state, the seventeenth century brought a lively exchange of ideas and thoughts between scientists, mostly friendly but sometimes polemical and frequently expressed in code to conceal them from the uninitiated. It is therefore necessary to study not only the books of these scientists but also their correspondence (cf. Leeuwenhoek's letters quoted by Dobell).

The same period witnessed the development of scientific knowledge to the system that is still important.Indeed, scientific development in the seventeenth century was greater than in any century before or since. Scientific giants like Kepler, Descartes, Pascal, Fermat, Newton, Leibniz, and Huygens appeared and left their marks in all fields of science, including optics. Let us sketch some of their contributions to geometrical optics.

Johannes Kepler (1571-1630) wrote two books on optics: Supplements to Vitelo in 1604 and Dioptrice in 1611. The object of the first is to find the law of refraction. Kepler nearly succeeds, though he does not look for a relation between the incident angle and the refracting angle but between the incident angle and the deviation. He asks himself what form a surface should have to refract a parallel beam of light exactly into a point image. He studies the hyperboloid, which, as we know, fulfills the condition; but he rejects it because it disagrees with the values that Vitello gave for the relation between the angles of the incident ray and the refracted ray. If only the Kepler of 1611 and the Dioptrice, in which he discovered the law of total reflection, had reread his book of 1604, he would have seen that the data of Vitello had been erroneous and thus he might have found the correct form of the law of refraction.

Kepler investigated the function of the eye more profoundly than his predecessors. He discovered that the eye is not at rest but moves around a point inside it, and he investigated the importance of this motion on our vision. He drew attention to the fact that the image on the retina is inverted.

All this is in his Supplements to Vitello.

His Dioptrice , although a relatively small book, was of importance to the development of optics. In it, Kepler assumes that for small angles the angle of refraction is proportional to the angle of incidence. From this assumption, he develops the laws of firstorder optics for thin lenses and systems of thin lenses with finite separations. He describes, among other things, the Keplerian telescope, in which two positive lenses are placed so that the back focus of the first is at the front focus of the second, and compares it with the form named after Galileo, consisting of a positive and a negative lens placed so that the focus of the former s at the virtual focus of the latter. He gives the focal length correctly for all forms of lenses. His book contains many practical proposals for lens design and is highly recommended to those who are interested in these problems.

Rene Descartes (1569-1650) must be considered as one of the discoverers of the correct law of refraction, for he published the sine law in his Dioptrique in 1637. After his death, he was violently attacked by Isaac Voss, who accused him of stealing the law from Willebrord Snel* (1591-1626). Descartes had indeed visited Leyden in 1630, where Snell, a professor of physics, had written a book on optics containing the refraction law. Although the book had never appeared, Voss, in his posthumous attack, claimed that Descartes had seen it on his visit after Snell's death, and Voss quoted Snell's proof.

The question of priority has been much discussed in the literature. A recent study has been published by Boegehold, who refers to a paper by P. Kramer. Boegehold found that Descartes, in some letters written in 1627, had suggested making plano-convex lenses with hyperbolic surfaces. Since such a lens would give a sharp point image for an infinite object if and only if one assumes the sine law, it is fair to conclude that Descartes knew of the law three years before his visit to Leyden in 1630. It is strange that Voss should have years after waited for twenty-five years-twelve Descartes's death-to make his accusation. Credit for discovering the law of refraction should therefore be divided between Descartes and Snel, with a fair share going to Johannes Kepler, who failed to find the sine law only because he trusted the tables of Vitello.

Descartes's Dioptrique is beautifully written. It tries to explain the law of refraction by means of the theory of elasticity. A light ray is compared to the path of a ball, first being bounced off the ground, then being slowed down as it goes into a denser medium. Descartes was the first to suggest determining the velocity of light from the position of the moon at an eclipse.

He also made experiments with human and animal eyes. If the eye is removed and put in the opening of a window in a darkened room, one can see the images of distant objects on the retina, and by pressing the eye a little, one can see the image of near-by objects, showing that the eye can accommodate.

In the eighth chapter of his book, Descartes investigates surfaces that image a point sharply-the socalled Cartesian surfaces, of which the conic sections constitute a special case. He proposes machines for grinding lenses and demonstrates the colors of the rainbow with small glass spheres filled with water.

But much of Descartes's reasoning in connection with the refraction law was faulty and was rejected by the great mathematician Pierre de Fermat (1601-1665), who pointed out that water would have to be less dense than air to fit Descartes's explanation. Fermat returned to the formulation of the refraction law given by Hero, deriving it from the assumption that light has different velocities in different media and takes the quickest path between any two points. Like Hero, he illustrates this idea with a drawing of a convex surface, for which the minimum principle is correct, rather than a concave surface, for which it can be wrong. But he was called on the carpet by Abbé Mersenne, as was mentioned earlier, one of the great scientists and letter writers of his day. Fermat's answer is a classic example of double talk: he says that the path would be a minimum if the surface were a plane, and the incoming ray sees so little of the surface that it cannot distinguish whether the surface is plane or curved!

1.1.5 Newton and Huygens

Sir Isaac Newton (1642-1727), in his Lectiones opticae of 1669-1671 (published posthumously), showed how to calculate the spherical aberration of a plano-convex lens for an infinite object point. This book contains the formula for locating the image, still known by his name, in terms of the distances of the object and the image from the focal points. It also contains a thorough investigation of reflection in prisms, including the position of minimum deviation. It even goes into some aberrations, in particular, image errors as functions of color, and it compares the size of the color aberration with the spherical aberration of a given lens and prism.

Although Newton's best known work in optics is his Opticcs , I should like to draw your attention especially to his beautiful paper on color, in which he describes the separation of white light into the colors of the spectrum by means of prisms. This charming paper should be read by every student of optics, especially since the writer not only gives his final conclusions but describes many false attempts at interpreting his experiments and the refutation of these attempts.

Newton was misled into believing that all materials have the same relative dispersion, since he found this to be true for such different media as glass and water. The conclusion that he drew-that telescopes should not contain refracting surfaces-was erroneous. But it led him to study mirror telescopes and, with his teacher Gregory, he contributed much to the development of these instruments.

Christian Huygens (1629-1695) wrote two books on optics: Dioptrical in 1653, published posthumously , and Traite de la lumiere , written in 1678 but not published for twelve years. The first book shows enormous advances with respect to the understanding of optical image formation. The second proposes a new and revolutionary theory of light, deriving refraction and reflection laws from a completely new principle.

This principle, set forth earlier in the Dioptrica, is a starting point from which all the laws of Gaussian optics can be derived for lens systems with finite thicknesses. It is expressed as follows: if we interchange object and eye in an optical system, the object will have the same apparent size and orientation as before. The law is known to us in the form that the product of lateral magnification and angular magnification is constant in an optical system, and equal to the ratio of the refractive indices.

Huygens found, before Newton, the law that, for a single surface or for a single lens, the image distance is proportional to the object distance when the distances are measured from the respective principal foci. He knew of the aplanatic surface. He was acquainted with the important property of the optical center of a lens (the point which divides the distance between the vertices in the ratio of the radii), namely, that a ray through it emerges parallel to its original direction. He had an approximate formula for the spherical aberration of a thick lens that enables the form for minimum spherical aberration to be determined.

In his Traite de la lumiere, which should be required reading for every physicist and is now readily available, Huygens derives optical laws from the fact that the light emerging from an object point P forms a wave surface S at any given time t. The light rays are then the normals to this surface and are not straight unless the velocity is constant throughout the medium.

The surface can also be constructed in the following way. At an arbitrary time , light from the point source P forms a wave surface S. (assumed to be spherical in the figure). Considering each point of S, as a source sending out a new wave, we see that the envelope of all the waves at the time gives the original wave surface S.

With the help of this concept, Huygens derives the straight paths of light in a medium of constant velocity. He derives not only the reflection and refraction laws but also the law of refraction in the atmosphere and in doubly refracting crystals. He clearly defines the caustic as the evolute of the rays and investigates it after reflection and refraction, in addition to investigating the degenerate form of the wave surfaces in the points of the caustic. He proves Malus' law: that rays form a normal system for a single reflection and refraction.

It should be emphasized, however, that this formulation is mathematically identical with ray optics, as Hamilton showed in his paper of 1831. The assumption that light has a finite wavelength and that it can be described as a sine wave going along a ray is foreign to Huygens. However, once this theory was established, the methods of Huygens enabled Fresnel to derive all the laws of diffraction in a most elegant manner.

Chapter 2

Geometrical Optics

The Euclidean Geometry exists thanks to light. Euclid wanted to undestand the light rays, so he discovered the plane geometry.

"Music is the arithmetic of sounds as optics is the geometry of light."
– Claude Debussy

2.1 The postulates of geometrical optics and the dark Room

We already know the history of the geometrical optics, well before starting the study of this theory we must take back the contributions of Al Azhen and talk about the **postulates of geometrical physics** which are

1. The speed of light is finite.
2. The light propagates in a straight line.
3. The light rays obey the Euclid's postulates of plane geometry.
   (a) A straight line segment can be drawn joining any two points.
   (b) A finite straight line (line segment) can be extended continuously in a straight line indefinitely.
   (c) A circle can be drawn with any point as its center and any distance as its radius.
   (d) All right angles are equal to one another, regardless of their orientation.
   (e) If a straight line falling on two straight lines makes the interior angles on the same side less than two right angles, the two lines, if extended indefinitely, will eventually meet on that same side.
4. In the matter the light's speed is less than in vacuum.

Now, before starting to use these postulates we must understand the physical meaning of them. The first postulate, **The speed of light is finite** is fundamental, it is incredibly important.

How could Hero of Alexandria understand that the speed of light must be finite? To be honest, I don't really know his motivation but, we could get an idea about why this should be like that from simple phenomenological reasoning without the use a mathematics or advanced physics. *Lets imagine a world where the speed of light is infinite*

How are going to be the nights in such world? It should be a complete chaos. Every star in the sky is a source of light, a source of infinite rays which each one travels at an infinite speed. There wont be any night, all the day is going to be full of light and even every event in the universe is going to be seen simultaneously, the big bang, the birth and death of a star. *Each past and present event is going to be seen at the same time.*

Other effect that only can be explained in a universe with light with finite speed. Imagine you're standing in the rain, which is falling vertically. If you start running, the rain hits you in the face. This happens because your movement changes the apparent direction of the rain. An astronomer, James Bradley, observed that the stars didn't appear to be exactly in their calculated positions. To see a star directly overhead, he had to tilt his telescope slightly in the direction of Earth's movement around the Sun, the light from the star "falls" toward Earth at a finite speed. Earth is moving. For the light beam to travel from the telescope lens to the eyepiece (and for us to see it), Earth has moved a little. Therefore, you have to point the telescope at a slight angle toward where Earth will be when the light reaches the eyepiece, not where it is at the moment it enters the lens.

If the speed of light were infinite, this effect wouldn't exist. The rain (light) would fall so fast that your (Earth's) movement would be irrelevant. The fact that we have to "aim ahead" of the target is irrefutable proof that light travels at a finite speed.

There are many other consequences but lets continue with all the other postulates. Now lets talk about **The light propagates in a straight line**. This does have sense because it is what is seen in any normal day, all the source of light can be seen as sources of infinite many rays which propagates at a finite speed. With tools purely geometrical there is no way to give a proof of this, however, from a wave theory for light, Huygens was able to prove that the rays are, in fact, the normal vectors of the wavefront are the rays, and, from a quantum perspective, Richard Feynman proved that the rays are the superposition of the "watches" that represent the probability amplitudes.

Often we see that light is not an straight line, for example, an easy experiment consists on a translucid bowl full of salted water, we're going to see that the ray presents a deflection, we're going to see that this curves can be generated as a composition of infinitely many straight refractions.

Because we've postulated that the light travels in straight lines, so we have to give a theory for the propagation so **The light rays obey the Euclid's postulates of plane geometry**. A point is going to be a source, a place where two rays meet, the place where two media separates, etc.

The extension of a line segment to a continuously straight line is a way to make or represent ray tracing. The circle postulate talks about the sphere that propagates alongside all the rays (which behave as the radius). The right angles are just a geometrical formalism which does not have a dramatic optical interpretation.

The last postulate is really important, from a physical point of view it tell us that all the rays are going to be propagate on a completely plane space, so we could ask ourselves if we want to study the light's behavior under the gravitational effects of compact objects we must change the fifth Euclid's postulate, and the short answer is yup, we have to. Under gravitational effects the differential geometry is a better tool to study the light-type observers on the curved space-time.

So, we have many arguments that supports the idea the light's speed must be finite, and if it is finite there should be a way to measure it, right? The answer is yes, and here we're going to use the postulates to follow the Fizeau's ideas on how to measure the light's speed.

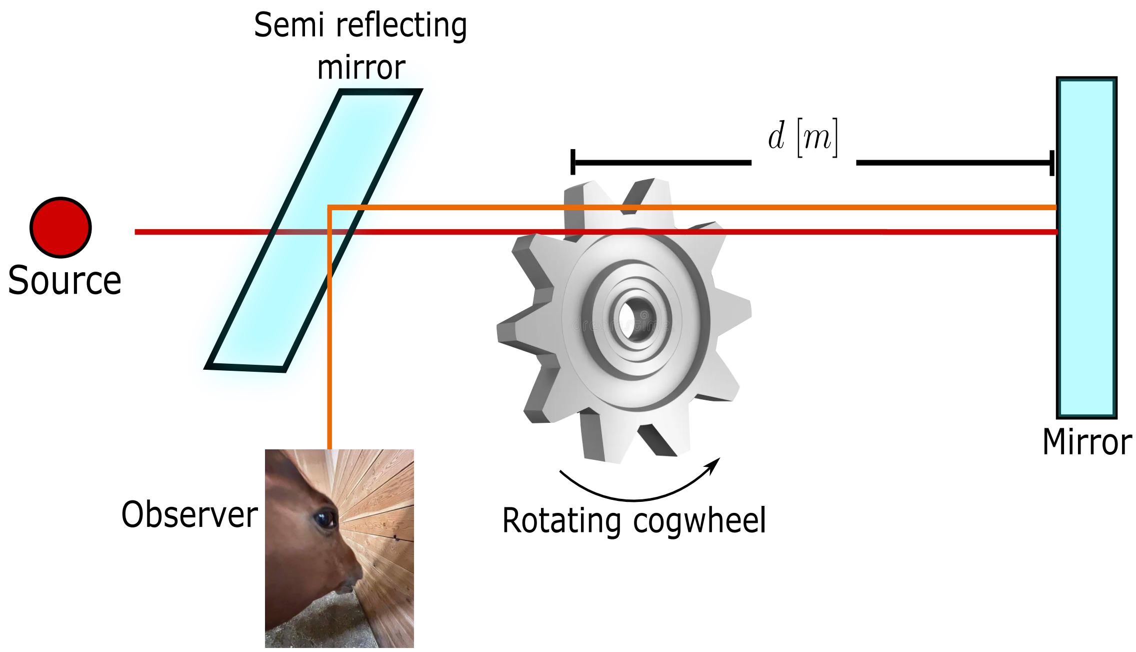


Figure 2.1: Schematic representation of the Fizeau's experiment pretended to measure the light's speed.

In 1849, Hippolyte Fizeau performed a groundbreaking experiment that provided the first non-astronomical measurement of the speed of light. His ingenious setup transformed the problem of measuring an extremely short time interval into a problem of measuring a rotation speed, which could be done with much greater precision. Imagine a very fast runner who starts from a point A, runs to a point B and immediately returns to A. To measure the runner's speed, you could place a door at A that opens and closes at regular intervals. If you open the door just when the runner leaves, then close it and open it again periodically, you might catch a glimpse of him when he returns if the timing is right. But if the runner is extremely fast, the door might be closed when he gets back. This is exactly the idea Fizeau used with light, replacing the door with a rapidly rotating toothed wheel and the runner with a light pulse.

Fizeau set up his apparatus on two hills in the outskirts of Paris, with the light source and observer on one hill (Montmartre) and a mirror on the other (Mont Valérien), about 8.633 kilometers away. A beam of light from a lamp was directed toward a partially reflecting mirror (a beam splitter) that sent part of the light toward a toothed wheel. The wheel could be spun at a controlled speed and had a large number of teeth and gaps – typically 720 teeth, meaning 720 gaps as well. After passing through a gap, the light traveled to the distant mirror, was reflected, and returned along the same path. On its way back, it would again have to pass through a gap in the wheel to reach the observer's eye via the beam splitter.

If the wheel were stationary, the returning light would pass through the same gap and be seen. But as the wheel began to rotate, the situation changed. When the wheel turned slowly, the gap through which the light had originally passed would have moved only slightly during the light's round trip, so most of the returning light still made it through. As the speed increased, the gap shifted more, and the light began to be partially blocked by the edge of a tooth. At a certain critical speed, the gap moved exactly so that by the time the light returned, the adjacent tooth had taken its place, completely eclipsing the light. The observer would see the light disappear for the first time. By knowing the rotation speed at which this first extinction occurred, the number of teeth, and the distance to the mirror, Fizeau could deduce the time it took for light to travel to the mirror and back.

Let us analyze the situation mathematically. In the figure 2.1 let d be the distance from the wheel to the distant mirror. The total path for the light is $2d$. The wheel has N teeth and consequently N gaps, so the angular width of one tooth (or one gap) is π/N radians (since a full circle has 2π radians and there are $2N$ segments – teeth and gaps – each of equal angular size). For the first extinction, the time required for the light to make the round trip must equal the time needed for the wheel to rotate by the angle corresponding to one tooth (i.e., to replace a gap with a tooth). If the wheel rotates with angular velocity ω (radians per second), the time to turn through an angle π/N is

$$t_{\text{rot}} = \frac{\pi/N}{\omega}. \tag{2.1}$$

The light travel time is

$$t_{\text{light}} = \frac{2d}{c}, \tag{2.2}$$

and equation these gives us

$$\frac{2d}{c} = \frac{\pi}{N\omega}. \tag{2.3}$$

But $\omega = 2\pi f$ where f is the rotation frequency in revolutions per second. Substituting,

$$\frac{2d}{c} = \frac{\pi}{N \cdot 2\pi f} = \frac{1}{2Nf}. \tag{2.4}$$

Solving for c yields

$$c = 4dNf. \tag{2.5}$$

Fizeau used $d = 8.633 \times 10^3$ m, $N = 720$, and observed the first extinction at $f \approx 12.6$ revolutions per second. Plugging in these numbers,

$$c = 4 \times 8.633 \times 10^3 \times 720 \times 12.6 \approx 3.13 \times 10^8 \text{ m/s}, \tag{2.6}$$

a value about 5% higher than the modern accepted value of 2.99792458×10^8 m/s, but remarkably accurate for the time. The discrepancy arose mainly from uncertainties in the distance measurement and the precise determination of the extinction condition. Fizeau's experiment was a triumph of experimental physics, demonstrating that light travels at a finite speed and providing a method that would be refined by later researchers, most notably Léon Foucault, who replaced the toothed wheel with a rotating mirror to achieve even greater precision. The conceptual beauty of Fizeau's approach lies in its translation of an infinitesimal time interval (on the order of 1/20000 of a second) into a measurable mechanical rotation, a perfect illustration of the deep connection between optics and mechanics.

Now we know the speed of light is finit, so lets use these postulates to understand one of the biggest contribution of Al Azhen, the dark room, see figure 2.2.

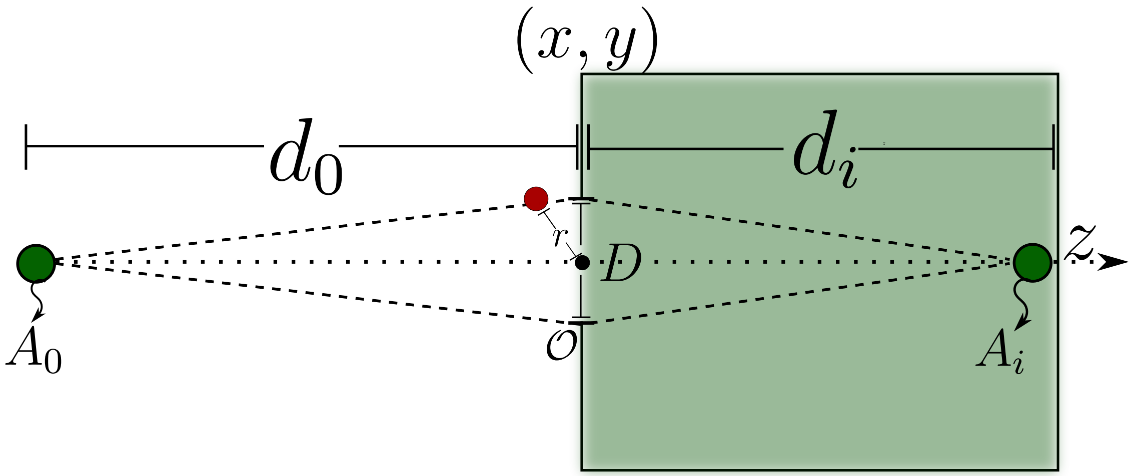


Figure 2.2: Graphical representation of a generic dark room which is a box of an arbitrary material which has a side opened and while it satisfies the Rayleigh's inequality i will form an inverted image.

Before develop mathematically the dark room we're going to set an important convention

- The light propagates from left to right along the z direction.
- A distance measured in the direction of light is positive and a distance measured against the direction of light is negative.

Once defined the convention lets do it. A source in the point A_o is at a distance d_o respect to the opening O of the dark room, (d_o is in fact negative because it is measure against the light direction from the point O), an image if formed at a distance d_i from O thus we must find out a relation between these parameters and the diameter D of the DR. First we're going to calculate the distances $\overline{A_oOA_i}$, $\overline{A_oRA_i}$ where R is the radius of the opening, therefore

$$\overline{A_oOA_i} = d_i - d_o \;\; d_i = \overline{OA_i} \;\; d_0 = \overline{OA_o}, \tag{2.7}$$

$$\overline{A_oRA_i} = \left[d_i^2 + x^2 + y^2\right]^{1/2} - \left[d_o^2 + x^2 + y^2\right]^{1/2}, \tag{2.8}$$

having on count our convention implies that $\sqrt{d_o^2} = -d_o$ so

$$\overline{A_oRA_i} = d_i\left[1 + \frac{r^2}{d_i^2}\right]^{1/2} - d_o\left[1 + \frac{r^2}{d_o^2}\right]^{1/2}. \tag{2.9}$$

Historically it was observed in the dark room that the image that was formed depended on the diameter of the opening and the distance of the object to view, it was observed that the image was formed for far objects, but how far is far? It is relative to the radius of the opening, then the ratios must satisfy that

$$\frac{r^2}{d_o^2} \ll 1\,,\ \ \frac{r^2}{d_i^2} \ll 1, \tag{2.10}$$

using this and using the Taylor's theorem for a polynomial expansion of the function $\sqrt{1+x}$ then

$$\overline{A_oRA_i} \approx d_i\left[1 + \frac{1}{2}\frac{r^2}{d_i^2} + \mathcal{O}(\delta^2)\right] - d_o\left[1 + \frac{1}{2}\frac{r^2}{d_o^2} + \mathcal{O}(\delta^2)\right], \tag{2.11}$$

$$\overline{A_oRA_i} \approx (d_i - d_o) + \frac{r^2}{2}\left(\frac{1}{d_i} - \frac{1}{d_o}\right) = \overline{A_oOA_i} + \frac{r^2}{2}\left(\frac{1}{d_i} - \frac{1}{d_o}\right) \tag{2.12}$$

with this we can express this in terms the difference of the light's path and the diameter of the dark room

$$\triangle L = \frac{r^2}{2}\left(\frac{1}{d_i} - \frac{1}{d_o}\right) = \frac{D^2}{8}\left(\frac{1}{d_i} - \frac{1}{d_o}\right). \tag{2.13}$$

In the equation 2.13 lies the hearth of the dark room, it was found by Rayleigh a certain relation between the light's wavelength and the light's path difference between the upper and lower part of the opening of the dark room, this relation (which will be discussed physically in a moment) is:

$$\triangle L < \frac{\lambda}{4}. \tag{2.14}$$

Using this relation we will find the maximum diamater for the dark room

$$\frac{D^2}{8}\left(\frac{1}{d_i} - \frac{1}{d_o}\right) < \frac{\lambda}{4}, \tag{2.15}$$

$$D_{max} = \left(\frac{2\lambda d_i d_o}{d_o - d_i}\right)^{1/2}. \quad (2.16)$$

In the real life there is no monochromatic light, it is forbidden by the laws of quantum mechanics, we can have quasi-monochromatic light but it is only in advanced laboratories so for a polychromatic source in the equation 2.16 we should do the calculations with the maximum wavelength that conforms the beam the origin of this expression have many origins which converges to the same phenomena. If we want to form an image the electromagnetic field coming from the object must arrive at the same time at the image point, if there is some kind of phase difference there will be aberrations as astigmatism or chromatic aberration (even if it is not a lens we can see it from the equation 2.16) from a quantum point of view this is related to an interpretation of the *size* of a photon (Santos et al., 2018).

Now we're going to analyze how does the light interact (from a simple and geometrical point of view) with transparent media. Suppose a source radiates from a point A and the ray light arrives at B, see figure 2.3.

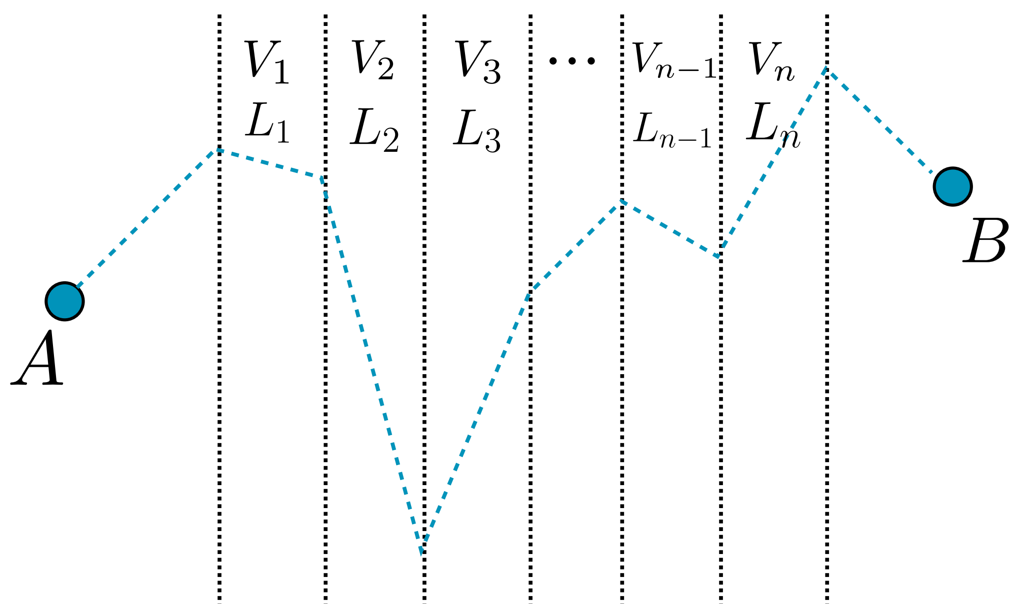


Figure 2.3: A schematic representation of a ray beam from A to B which goes through n different transparent media and in each one it had a different speed and traveled different length.

There are n different transparent media each one hast a different thickness and the light in each one is different, therefore the time that takes the light to go from A to B is

$$t_{AB} = \sum_{i=1}^{n} \frac{L_i}{v_i} = \sum_{i=1}^{n} \frac{L_i C}{v_i}\frac{1}{C}\ [s]. \quad (2.17)$$

Then we define the refraction index in the ith medium as the relation between the speed of light in vacuum and the speed of light in the ith medium. *It is a scalar in linear and homogeneous and isotropic media but mostly it is a tensor.*

$$t_{AB} = \frac{1}{C}\sum_{i=1}^{n} n_i L_i = \frac{L_{AB}}{C}\ [s], \tag{2.18}$$

Definition 1 *(Optical path): The optical path L_{AB} is the distance that must travel in vacuum the light to have the same time to go from a point A to a point B across n different media each one characterized by a different (or equal) refraction index.*

The optical path has a length greater than the geometrical path of the light, that is why it is called optical path. This can be easily generalized to continuous media using a Lebesgue's integral (or Riemann integral if it is well behaved)

$$L_{AB} = \int_A^B n(S)ds\ [m]. \tag{2.19}$$

Before going further we must have to set some definitions.

Definition 2 *(Phase): The phase is the relative angular displacement the ray has.*

Definition 3 *(Wavefront): The geometrical space where the phase of the light is constant is called the wavefront of the light.*

2.2 The Snell-Descartes law and the Fermat's principle

As we wad discussed in the historical chapter it was Heron of Alexandria who realizes that in light reflection it travels the least distance and Al Azhen proposes a maximum time, it was Fermat who formalizes it as a principle the **Fermat's principle**

Principle 1 *(Fermat's principle): The light travels a trajectory such that the optical path is stationary*

$$\delta L = 0. \tag{2.20}$$

The Fermat's principle can be "deduced¨ from the Hamilton principle or even more from a variational principle, lets give a look to it, lets look for an stationary curve for the time the light travels

$$t = \int_{t_0}^{t_f} dt = \int_A^B \frac{1}{C} dL = \int_{\tau_0}^{\tau_f} L d\tau, \tag{2.21}$$

from here we see that, in a mathematical way, if we want to find the relation between the trajectory of the light and the time it takes to go from A to B we will have to solve the Euler-Lagrange equation. Now lets use the Fermat's principle to see how the light interacts with a diopter (a refracting surface) in a vector way, known as the *vectorial Snell-Descartes law.*

Definition 4 *(Diopter): A Diopter is a surface $\Sigma \subset \mathbb{R}^2$ such that it separates two media with different refraction index.*

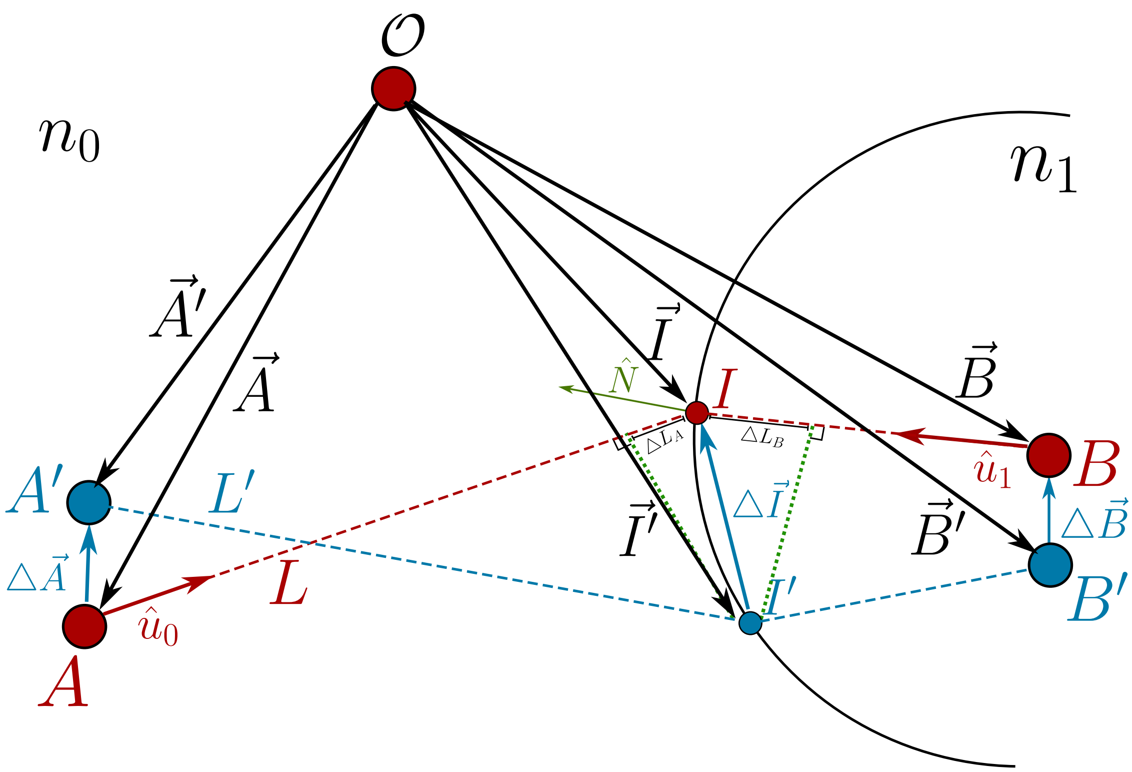


Figure 2.4: two media separated by a diopter which an observer O measures the path the light will follow from A to B and proposes a displacement of the source A to A' to see how must be the light path.

In the figure 2.4 we see two media separated by a transparent surface (diopter) with refractive index n_o,n_1, respectively, a source A emits a beam which reaches the diopter in I and it is refracted to B, from A there is an unitary vector $\hat{u}_o$ in the direction $\overline{AI}$ and the observer O analyzes a variation of A as A' and the incident point on the diopter is I', then lets analyze it vectorially

$$L = \left|\left|\vec{I} - \vec{A}\right|\right|, \tag{2.22}$$

$$\triangle\vec{I} = \vec{I'} - \vec{I}, \tag{2.23}$$

$$\triangle\vec{A} = \vec{A}' - \vec{A}, \tag{2.24}$$

with this we can calculate the path difference $L - L'$ vectorially as a proyection of the unitary vector in the direction of L given by $\hat{u}_o$ along $\triangle\vec{I} - \triangle\vec{A}$

$$\triangle L_A = \frac{\left(\vec{I} - \vec{A}\right) \cdot \left(\triangle\vec{I} - \triangle\vec{A}\right)}{\left|\left|\vec{I} - \vec{A}\right|\right|} = \hat{u}_o \cdot \left(\triangle\vec{I} - \triangle\vec{A}\right), \tag{2.25}$$

now for the optical path

$$\mathcal{L}_A = n_o \triangle L = n_o \hat{u}_o \left(\triangle\vec{I} - \triangle\vec{A}\right), \tag{2.26}$$

doing the same for the right side of the diopter but for the point B we then have got the total optical path difference as

$$\triangle\mathcal{L} = n_o \hat{u}_o \cdot \left(\triangle\vec{I} - \triangle\vec{A}\right) + n_1 \hat{u}_1 \cdot \left(\triangle\vec{B} - \triangle\vec{I}\right) \tag{2.27}$$

because the points A and B are fixed then $\triangle\vec{A} = 0$ and $\triangle\vec{B} = 0$ thus

$$\triangle\mathcal{L} = (n_o \hat{u_o} - n_1 \hat{u}_1) \cdot \triangle\vec{I}. \tag{2.28}$$

If the Fermat's principle is satisfied the equation 2.28 implies straightfully that $\triangle\vec{I}$ is a tangent vector to the diopter this implies then

$$n_o \hat{u_o} - n_1 \hat{u_1} = k\hat{N}. \tag{2.29}$$

The equation 2.29 is known as the vectorial Snell-Descartes law, from this equation we can arrive to the law of reflection and refraction to do it we only have to do the cross product of the equation 2.29

$$\hat{N} \times (n_0 \hat{u_0} - n_1 \hat{u_1}) = k\hat{N} \times \hat{N} = 0, \tag{2.30}$$

$$n_0 sin(\theta_0) = n_1 sin(\theta_1). \tag{2.31}$$

2.2.1 The critical angle, and the refraction by a prism

We had deduced the vectorial Snell-Descartes law, which is a strong law and we are going to apply it to some physical situations, but before doing that we are going to talk briefly about where is the physical regime where this law can work on.

We are working under the postulates of the geometrical optics, but lets analyze a bit the second one, *the light propagates in a straight line* we haven't say anything about the light's shape, if we say it propagates in a straight line we can image light rays as circular cylinders but what about the radius of this ray? Well, in fact there are more shapes we can imagine of the wavefront and the beam's profile, but we are going to talk about it in the chapter of electromagnetic optics, for now keep with the cylinder image.

If we make this ray to inside over a perfectly polished mirror as shown in 2.5 then we can see an prove easily experimentally the Snell-Descartes law will be fulfilled but what happen if we have not got a polished surface but an extremely irregular surface? we will have reflections in all the points over the surface as shown in the figure 2.6.

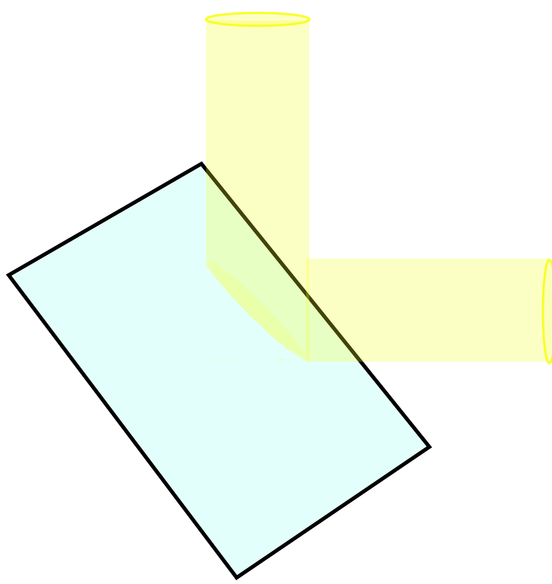


Figure 2.5: Representation of a cylindrical ray light being reflected by a perfectly polished mirror.

The pehonomena shown in the figure 2.6 is, often, called *diffuse refraction* but it is a kind of macroscopic effect of the Snell-Descartes law but there are other behaviors of the refraction index which give a particular behavior to the SD law, which are related to the electromagnetic properties of the matter, in fact, the refraction indez n is a complex quantity, which its real part is related to the refractive properties of the material and the imaginary part is related to absorption an losses in

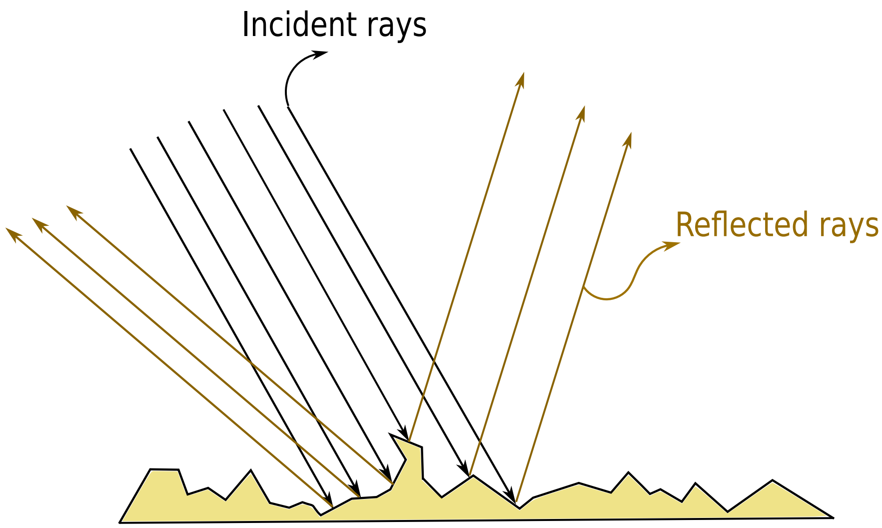


Figure 2.6: Representation of a cylindrical bunch of rays being reflected by an inhomogeneus surface.

the energy but we will work on this later.

The are some other special materials where the equations 2.29 has an amazing behavior. The meta materials which has a negative refraction index, but how can we interpret this? It is easy to see in the equation 2.29 that a variable changes of $n \to -n$ implies a physical effect of $\theta \to -\theta$ this means, the rays will be reflected or refracted to the other side!

Well then, lets come back to our well defined Behavior, where the SD law 2.29 will be fulfilled. Can we find a condition between two material such that there is no refraction? It is worth to ask about it because in nature the reflection and refraction happen at the same time both simultaneously because a translucent surface is not perfectly transparent and a refractive surface is not perfectly reflective this is related to the Fresnel's coefficients but this will be hard work of the chapter of electromagnetic optics, for now have on mind reflection and refraction happen at the same time. To try to kill the refraction we have to study two cases between two media.

1. $n_1 > n_2$
2. $n_1 < n_1$

For the first case then we have from 2.29

$$n_1 sin\theta_1 = n_2 sin\theta_2, \tag{2.32}$$

$$\frac{n_1}{n_2} sin\theta_1 = sin\theta_2. \tag{2.33}$$

If $\theta_1 = 0$ then $\theta_2 = 0$ thus if the incident ray is perfectly parallel to the normal vector to surface there wont be any refracted ray, only a completely reflected in the same plane as the incident but with opposite direction, this is said to get a phase shift of π. If the ray's incidence angle is perpendicular to the normal vector to surface $\theta_1 = \pi/2$ then we will have

$$\frac{n_1}{n_2} = sin\theta_2, \tag{2.34}$$

but $n_1 > n_2$ then $n_1/n_2 > 1$ which has no sense with the right hand side of the equation 2.34 because $0 \leq sin\theta_2 \leq 1$ therefore for $\theta_1 = \pi/2$ there is no refracted ray this is illustrated in the figure 2.7.

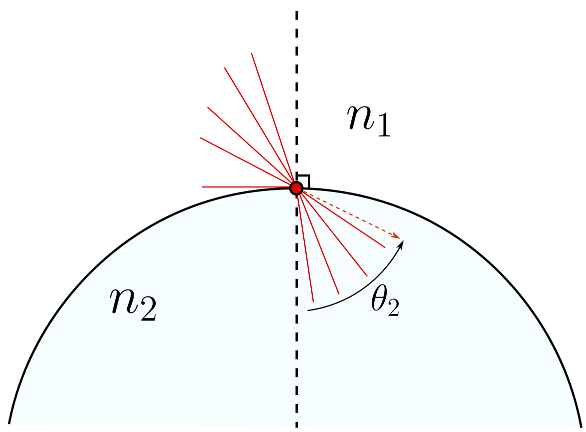


Figure 2.7: Representation of all the incident and refracted rays from a media with refraction index n_1 to another with refraction index n_2 such that $n_1 > n_2$. The angle θ_2 here shown wont never reach a value equal to $asin\left(\frac{n_1}{n_2}\right)$.

Now, we have to consider and study the case where $n_1 < n_2$ the obvious case is when $\theta_1 = 0$ therefore $\theta_2 = 0$ and there wont be any refracted ray, but the interesting thing comes when $\theta_1 = \pi/2$ then again we have got the same equation 2.34 but here is the main difference, now we've got $n_1 < n_2$ or $n_1/n_2 < 1$ therefore there exists this angle and it is called the *total reflection angle* or *critical angle* θ_c this is illustrated in the figure 2.8.

2.3 The Descartes's ovoid and diopters

From here on we are going to start to study the theory of image formation where the concepts of **object** and **image** are really important because there are real and virtual objects and image, this depends on the relation between these points ant the diopter, in fact, something is called virtual if we put a photo detector in that point and we wont measure any field if an object is to the left of a diopter is called real and if it is to the right is called virtual, in the same order of ideas if the image

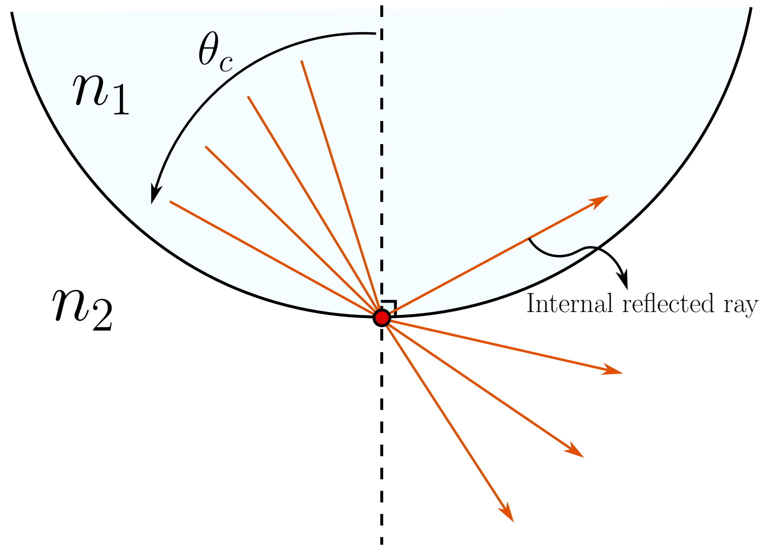


Figure 2.8: Representation of all the incident and refracted rays from a media with refraction index n_1 to another with refraction index n_2 such that $n_1 < n_2$. The angle θ_c here separates the rays that will be refracted , (angles less than θ_c), and the rays that will be totally reflected internally.

is to the left of the diopter it is virtual and if it is to the right is real.

Before starting we have to give some fundamental definitions

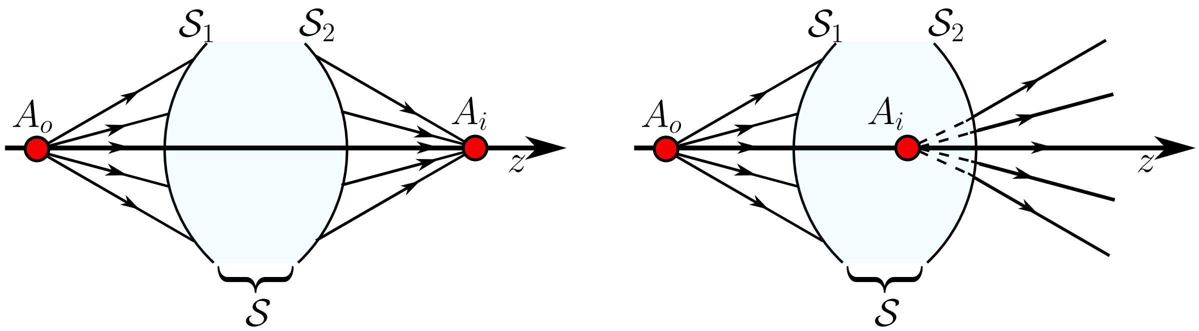
Definition 5 *(Optical system) an optical system is a set of translucid media (diopters) and reflective media (catopters) and we say that the* ***system is centered*** *if all the diopters are revolution surfaces and shared the same symmetry axis.*

The systems conformed only by refractive surfaces all called ***dioptric systems*** *and the system conformed only by reflective surfaces are called* ***catoptric systems*** *and the systems formed by these two kind of surfaces are called* ***catadioptrics systems****.*

Definition 6 *(Real and virtual points) Given a optical system $\mathcal{S}$ we say that an object point A_o is real if it is to the left and outside the optical system and we way that the object point is virtual if it is to the right and inside the optical system.*

In the same way, we say that an image point A_i is real if it is to the right and outside the optical system, and we way that the image point is virtual if it is inside the optical system.

To understand better these two definitions lets follow the figures 2.9 there we can see two examples of optical systems. In the first one we can see an object point located to the left and outside of the system $\mathcal{S}$ so it is a real object point, and the point A_i is located to the right and outside the optical system, therefore it is a real image point. In the second optical system we can see that the same



(a) Optical system $\mathcal{S}$ formed by two diopters $\mathcal{S}_1, \mathcal{S}_2$ where the point A_o is a real object and A_i is a real image point.

(b) Optical system $\mathcal{S}$ formed by two diopters $\mathcal{S}_1, \mathcal{S}_2$ where the point A_o is a real object and A_i is a virtual image point.

Figure 2.9: Representation of two optical systems where appear real object points, real and virtual image points.

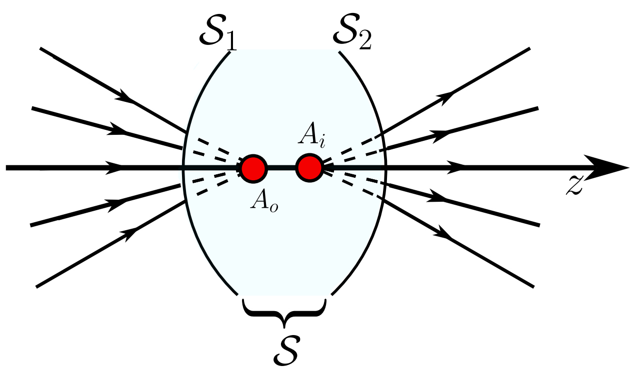


Figure 2.10: Optical system $\mathcal{S}$ formed by two diopters $\mathcal{S}_1, \mathcal{S}_2$ where the point A_o is a virtual object and A_i is a virtual image point.

real object point and that the point A_i is located inside the optical system, therefore it is a virtual image point for A_o.

In the figure 2.10 we can see two diopter $\mathcal{S}_1, \mathcal{S}_2$ which forms the optical system $\mathcal{S}$ where the point A_o is located inside the optical system, then it behaves as a virtual object point and A_i is to the right of A_o and also inside the optical systen, therefore it is a virtual image point.

In the optical system 2.11 we can see three optical systems $\mathcal{S}_1, \mathcal{S}_2, \mathcal{S}_3$ and a configuration of many points. The first one, A_o is completely to the left and outside any optical system, so it is a real object point, and the point A' is inside the optical system $\mathcal{S}_1$ so it is the **the virtual image of the real object** A_o but, as we can see, A' is to the left and outside the optical system $\mathcal{S}_2$ so it is a real object point for the rest of the total optical system. The point A'' is inside the optical system $\mathcal{S}_3$ so it is

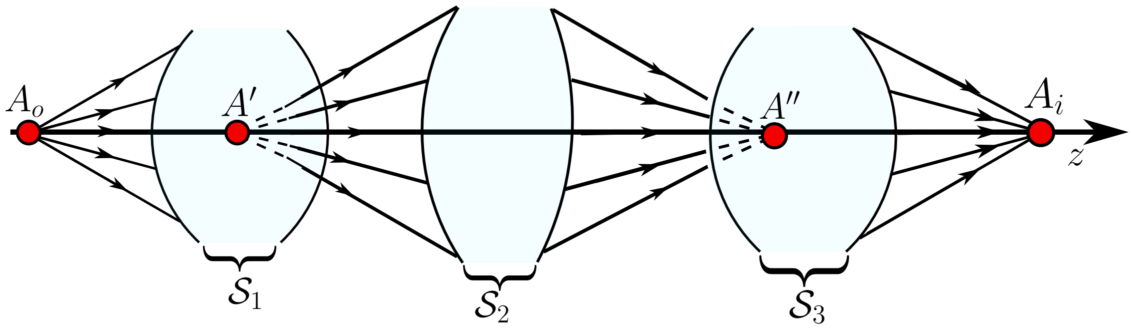


Figure 2.11: Optical system $\mathcal{S}$ formed by three optical systems $\mathcal{S}_1, \mathcal{S}_2, \mathcal{S}_3$ where the point A_o is a real object point, A' is a virtual image point for $\mathcal{S}_1$, but a real object point for $\mathcal{S}_2, \mathcal{S}_3$, A'' is a virtual image for the real object A' and A_i is the real image of the virtual object A''.

the virtual image of the real object A', and in the very last there is the point A_i which is completely to the right and outside any optical system, so it is the real image point of the virtual object point A''.

Definition 7 *(Stigmatism) From a **geometrical point of view** we say that an optical system $\mathcal{S}$ is **stigmatic** if for any given point source, all the rays that pass through $\mathcal{S}$ converge to a point source. In other words, a optical system $\mathcal{S}$ is stigmatic if for any object point there is one unique image point.*

We can also, define the stigmatism from a wavefront perspective. An rigorously stigmatic system is one that transforms an spherical wavefront $\mathcal{A}$ to a new spherical wavefront $\mathcal{A}'$ whose curvature centers, A_o and A_i are the rigorously stigmatic points (see figure 2.12). Thus, the spherical wavefront $\mathcal{A}'$ is the image of the spherical wavefront $\mathcal{A}$.

Understanding that an image-forming optical system is one that maps each point source of an object's electromagnetic field to the point sources of the image's electromagnetic field, corresponding to a homothety between these fields, the interest arises in studying what these optical systems are and what their characteristics are. In general, these systems can be composed of multiple elements, but initially we will study the simplest configuration for two reasons:

1. because it most easily allows us to understand the principles of these systems.

2. because later, more complex systems can be understood as compositions of these.

These elementary systems are made up of a single optical surface, either refractive or reflective, which is capable of perfectly focusing all the rays emitted by a point source into a perfect point image. In geometric optics, this means that all the rays travel the same optical path length. Initially, this property associated with a single pair of points (stigmatism) will be studied, and then the conditions under which this property becomes invariant to the translation of these points, which is known as

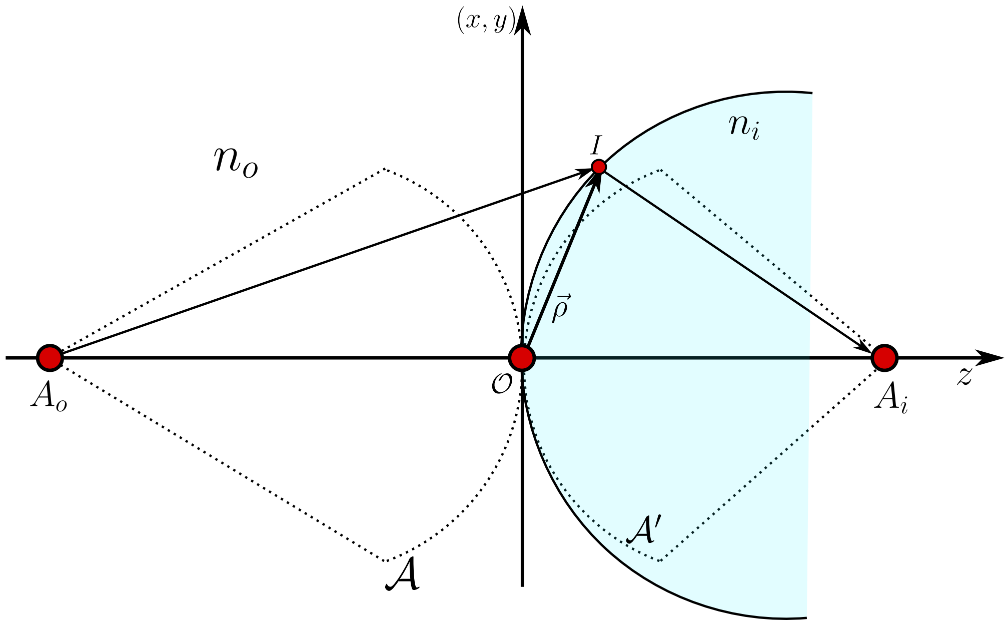


Figure 2.12: The stigmatic system transform the incident wavefront $\mathcal{A}$ to another new spherical wavefront $\mathcal{A}'$.

aplanetism, will be studied.

Now we are going to develope the *rigurous stigmatism* thes are system capable of making perfect images, this means from a point source there is a perfect point image. Let a diopter which separates two media with refraction index n_o and n_i to the left of the diopter there is an object A_o at a point $[0, 0, z_o]$ the field from A_o reaches a point I, measured from the diopter's vertex v, from the point I the beam is refracted and forms an image A_i at a point $[0, 0, z_i]$ then we are going to find the relations between all these parameters to make all the rays from A_o arrive at A_i making a perfect image. From the Fermat's principle we have got

$$n_0 l_0 + n_i l_i = cte \tag{2.35}$$

the, as we can see in the figure 2.13 we then have

$$-n_0[x^2 + y^2 + (z - z_o)^2]^{1/2} + n_i[x^2 + y^2 + (z - z_i)^2]^{1/2} = cte \tag{2.36}$$

and, remembering our sign convention $\sqrt{z_o^2} = -zo$, $z_i^2 = z_i$ and to find the constant we can see that if we want to make all the rays from A_o arrive to A_i then the rays that travel parallel to z satisfy also the Fermat's principle, thus

$$n_i z_i - n_o z_o = cte, \tag{2.37}$$

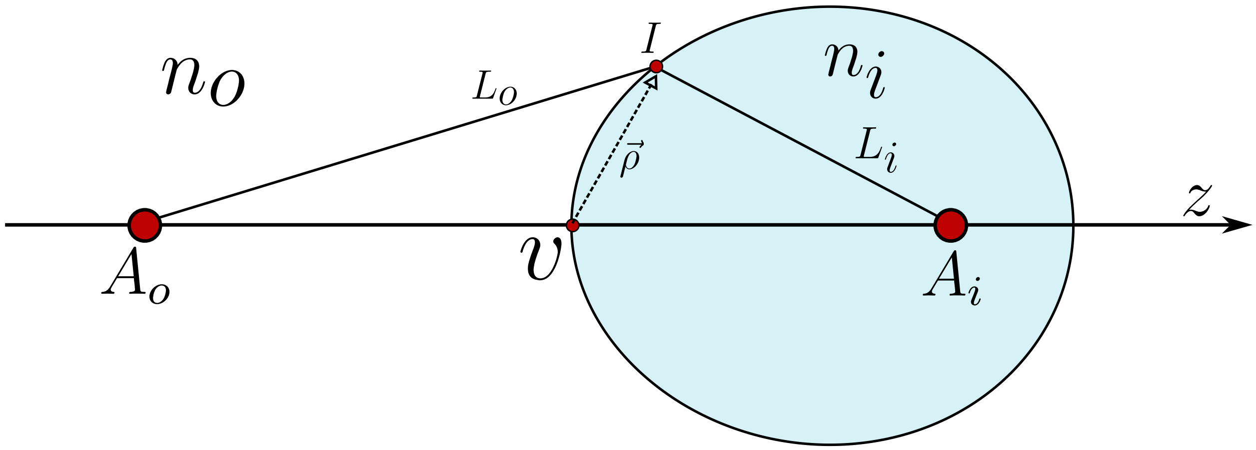


Figure 2.13: Scheme representation of the development of a rigorously stigmatic system, from a point source A_o at a position $[0, 0, z_o]$ the diopter with a vertice v forms a perfect image A_i at $[0, 0, z_i]$.

$$n_i[x^2 + y^2 + (z - z_i)^2]^{1/2} - n_0[x^2 + y^2 + (z - z_o)^2]^{1/2} = n_i z_i - n_o z_o, \tag{2.38}$$

now, we are going to do the following change of variables given by

$$\rho^2 = x^2 + y^2 + z^2, \tag{2.39}$$

which represents the square of the vector from the vertex to the interface shown in 2.13, so, having this on count, the equation becomes

$$n_i \left[\rho^2 + z_i^2 - 2zz_i\right]^{1/2} - n_0 \left[\rho^2 + z_0^2 - 2zz_0\right]^{1/2} = n_i z_i - n_0 z_0. \tag{2.40}$$

Now, the main goal is to transform the equation 2.40 into a more explicit and useful algebraic equation, so, the first thing we're going to do is to square both side of the equation 2.40 and then we'll have

$$n_o^2 \left[\rho^2 + z_o^2 - 2zz_o\right] + n_i^2 \left[\rho^2 + z_i - 2zz_i\right] - 2n_o n_1 \left[\rho^2 + z_o^2 - 2zz_o\right]^{1/2} \left[\rho^2 + z_i^2 - 2zz_i\right]^{1/2} = (n_i z_i - n_o z_o)^2$$

Now, we're going to re-order all the equation for convenience as follows

$$n_o^2 \left[\rho^2 + z_o^2 - 2zz_o\right] + n_i^2 \left[\rho^2 + z_i - 2zz_i\right] - (n_i z_i - n_o z_o)^2 = 2n_o n_1 \left[\rho^2 + z_o^2 - 2zz_o\right]^{1/2} \left[\rho^2 + z_i^2 - 2zz_i\right]^{1/2},$$

on the left hand side we have got the following

$$\begin{aligned}
&n_o^2\left[\rho^2+z_o^2-2zz_o\right]+n_i^2\left[\rho^2+z_i-2zz_i\right]-\left(n_iz_i-n_oz_o\right)^2=\\
&n_o^2\left[\rho^2+z_o^2-2zz_o\right]+n_i^2\left[\rho^2+z_i-2zz_i\right]-\left(n_i^2z_i^2+n_o^2z_o^2-2n_on_iz_oz_i\right)=\\
&n_o^2\left[\rho^2-2zz_o\right]+n_i^2\left[\rho^2-2zz_i\right]+2n_on_iz_oz_i,
\end{aligned}$$

where we just only simplified the common terms. Having this on count, the equation reduces to

$$n_o^2\left[\rho^2-2zz_o\right]+n_i^2\left[\rho^2-2zz_i\right]+2n_on_iz_oz_i=2n_on_1\left[\rho^2+z_o^2-2zz_o\right]^{1/2}\left[\rho^2+z_i^2-2zz_i\right]^{1/2}. \tag{2.41}$$

Now, we must square again, the whole equation as follows

$$\begin{aligned}
&\left[n_o^2\left(\rho^2-2zz_o\right)+n_i^2\left(\rho^2-2zz_i\right)\right]^2+\left(2n_on_iz_oz_i\right)^2+4n_on_iz_oz_i\left[n_o^2\left(\rho^2-2zz_o\right)+n_i^2\left(\rho^2-2zz_i\right)\right]=\\
&4n_o^2n_i^2\left(\rho^2+z_o^2-2zz_o\right)\left(\rho^2+z_i^2-2zz_i\right),
\end{aligned}$$

now we are going to expand the product on the right hand side in the following way

$$\begin{aligned}
&\left[n_o^2\left(\rho^2-2zz_o\right)+n_i^2\left(\rho^2-2zz_i\right)\right]^2+\left(2n_on_iz_oz_i\right)^2+4n_on_iz_oz_i\left[n_o^2\left(\rho^2-2zz_o\right)+n_i^2\left(\rho^2-2zz_i\right)\right]=\\
&4n_o^2n_i^2\left(\rho^2-2zz_o\right)\left(\rho^2-2zz_i\right)+4n_o^2n_i^2\left[z_o^2\left(\rho^2-2zz_i\right)+z_i^2\left(\rho^2-2zz_o\right)+z_o^2z_i^2\right].
\end{aligned}$$

Now, we must do a simplification, to do this, we must expand the first term of the left hand side like this

$$\left[n_o^2\left(\rho^2-2zz_o\right)+n_i^2\left(\rho^2-2zz_i\right)\right]^2=\left[n_o^2\left(\rho^2-2zz_o\right)\right]^2+\left[n_i^2\left(\rho^2-2zz_i\right)\right]^2+2n_o^2n_i^2\left(\rho^2-2zz_o\right)\left(\rho^2-2zz_i\right),$$

so the first term of the left hand side have on common in its expansion the same term than the expansion of the term on the right hand side, so, we could simplify then and look at the binomial is the same but with a minus sign instead of a plus sign, thus

$$\begin{aligned}
&\left[n_o^2\left(\rho^2-2zz_o\right)-n_i^2\left(\rho^2-2zz_i\right)\right]^2+\left(2n_on_iz_oz_i\right)^2+4n_on_iz_oz_i\left[n_o^2\left(\rho^2-2zz_o\right)+n_i^2\left(\rho^2-2zz_i\right)\right]=\\
&4n_o^2n_i^2\left[z_o^2\left(\rho^2-2zz_i\right)+z_i^2\left(\rho^2-2zz_o\right)+z_o^2z_i^2\right].
\end{aligned}$$

Now, we are going to forget, momentarily the first term of the left hand side and focus on all the rest of the equations because there are going to be many important simplifications. Then, having this on count we'll have the following:

$$\left(2n_on_iz_oz_i\right)^2+4n_on_iz_oz_i\left[n_o^2\left(\rho^2-2zz_o\right)+n_i^2\left(\rho^2-2zz_i\right)\right]-4n_o^2n_i^2\left[z_o^2\left(\rho^2-2zz_i\right)+z_i^2\left(\rho^2-2zz_o\right)+z_o^2z_i^2\right]=0$$

$$
\begin{aligned}
&4n_on_i\left[z_oz_in_o^2\left(\rho^2-2zz_o\right)+z_oz_in_i^2\left(\rho^2-2zz_i\right)-n_on_iz_o^2\left(\rho^2-2zz_i\right)-n_on_iz_i^2\left(\rho^2-2zz_o\right)\right.\\
&\left.-n_on_izo^2z_i^2+n_on_iz_o^2z_i^2\right],
\end{aligned}
$$

now, we have to factorize the common factors

$$
\begin{aligned}
&4n_on_i\left[\left(\rho^2-2zz_o\right)\left(z_oz_in_o^2-n_on_iz_i^2\right)+\left(\rho^2-2zz_i\right)\left(z_oz_in_i^2-n_on_iz_o^2\right)\right]=\\
&4n_on_i\left[z_in_o\left(\rho 2-2zz_o\right)\left(z_on_o-n_iz_i\right)+z_on_i\left(\rho^2-2zz_i\right)\left(z_in_i-z_on_o\right)\right]=\\
&4n_on_i\left[n_iz_i-n_oz_o\right]\left[n_iz_o\left(\rho^2-2zz_i\right)-n_oz_i\left(\rho^2-2zz_o\right)\right]=\\
&4n_on_i\left[n_iz_i-n_oz_o\right]\left[\rho^2\left(n_iz_o-n_oz_i\right)+2zz_iz_o\left(n_o-n_i\right)\right].
\end{aligned}
$$

So, joining this result with the first term we will have the following

$$
\left[n_o^2\left(\rho^2-2zz_o\right)-n_i^2\left(\rho^2-2zz_i\right)\right]^2+4n_on_i\left[n_iz_i-n_oz_o\right]\left[\rho^2\left(n_iz_o-n_oz_i\right)+2zz_iz_o\left(n_o-n_i\right)\right]=0,
$$

so, the last things we have to do are is re organize in the following way

$$
\left[\left(n_i^2-n_o^2\right)\rho^2-2z\left(n_i^2z_i-n_o^2z_o\right)\right]^2-4n_in_o\left[n_iz_i-n_oz_o\right]\left[\left(n_oz_i-n_iz_o\right)\rho^2+2zz_oz_i\left(n_i-n_o\right)\right]=0. \tag{2.42}
$$

The equation 2.42 is the algebraic expression for the Descartes Ovoid the rigorously stigmatic surface we wanted.

2.4 The spherical diopter

Now, we are going to see that the Descartes Ovoid 2.42 contains the information of all the other diopters known. So, the first one we are going to deduce is the spherical diopter this is obtained in a vicinity of the vertex shown in the figure 2.13. Therefore, to deduce this particular case, we are going to take the equation 2.42 and rewrite in the following convenient way

$$
\left[\left(n_o^2-n_i^2\right)\rho^2+2zz_oz_i\left(\frac{n_i^2}{z_o}-\frac{n_o^2}{z_i}\right)\right]^2-4n_in_o\left[n_iz_i-n_oz_o\right]\left[\left(n_oz_i-n_iz_o\right)\rho^2+2zz_oz_i\left(n_i-n_o\right)\right]=0. \tag{2.43}
$$

So, the main idea is to construct the sphere that is located in the vertex v. To do this, we must see that the squared term in 2.43 looks very similar to the right factor term in the second term, so, we want to equate them and then simplify, which leads us to the following equations

$$
n_o^2-n_i^2=c\left(n_oz_i-n_iz_o\right), \tag{2.44}
$$

$$
\frac{n_i^2}{z_o}-\frac{n_o^2}{z_i}=c\left(n_i-n_o\right) \tag{2.45}
$$

where c is a constant we must calculate. The second thing we are going to do, is to re-order the equation 2.44 in the following way

$$n_o^2 - cz_i n_o = n_i^2 - cz_o n_i, \tag{2.46}$$

and if we re-order the equation 2.45 we'll have

$$z_i n_i^2 - z_o n_o^2 = cz_o z_i(n_i - n_o),$$
$$z_i n_i^2 - cz_o z_i n_i = z_o n_o^2 - cz_o z_i n_o,$$
$$z_o \left(n_o^2 - cz_i n_o\right) = z_i \left(n_i^2 - cz_o n_i\right). \tag{2.47}$$

So if we use 2.46 on 2.47 we have got the following two equations

$$(z_o - z_i)\left(n_i^2 - cz_o n_i\right) = 0, \tag{2.48}$$
$$(z_o - z_i)\left(n_o^2 - cz_i n_o\right) = 0, \tag{2.49}$$

using the equations 2.48 and 2.49 we will have

$$z_o = \frac{n_i}{c}, \qquad z_i = \frac{n_0}{c}, \tag{2.50}$$

and using the equation 2.44 we will have

$$z_0 = z_i \qquad z_{0,i} = \frac{n_o + n_i}{c}. \tag{2.51}$$

Now, before going further analyzing this results, we have to use the equations 2.44, 2.45 on 2.43

$$\left[c\left(n_o z_i - n_i z_o\right)\rho^2 + 2zz_o z_i c(n_i - n_o)\right]^2 - 4n_i n_o\left[n_i z_i - n_o z_o\right]\left[\left(n_o z_i - z_i z_o\right)\rho^2 + 2zz_o z_i\left(n_o - n_o\right)\right] = 0,$$
$$\left[\left(n_o z_i - n_i z_o\right)c\rho^2 + 2zz_o z_i c(n_i - n_o)\right]\left[\left(n_o z_i - n_i z_o\right)\rho^2 + 2zz_o z_i(n_i - n_o) - 4n_i n_o\left(n_i z_i - n_o z_o\right)\right] = 0,$$

and by the fundamental theorem of Algebra, we have got the equation for the spherical diopter

$$\left(n_o z_i - n_i z_o\right)\rho^2 + 2zz_o z_i(n_i - n_o) = 0. \tag{2.52}$$

2.4.1 The Young-Weirstrass points

From the case 2.50 we have that

$$n_i z_i - n_o z_o = 0,$$

and using the equation 2.52 we will have

$$\rho^2\left(n_o^2 - n_i^2\right) + 2czz_o n_i(n_i - n_o) = 0, \tag{2.53}$$

now, dividing by $n_o^2 - n_i^2$ and then we must complete squares as follows

$$x^2 + y^2 + z^2 + \frac{2czz_oz_i}{n_i + n_o} = x^2 + y^2 + z^2 + \frac{2czz_oz_i}{n_i + n_o} + \left(\frac{cz_oz_i}{n_i + n_o}\right)^2 = \left(\frac{cz_oz_i}{n_i + n_o}\right)^2,$$

$$x^2 + y^2 + \left(z - c\frac{z_oz_i}{n_o + n_i}\right)^2 = c^2\frac{z_o^2z_i^2}{(n_i + n_o)^2}, \tag{2.54}$$

which corresponds to the equation of an sphere whose radius is given by

$$R = c\frac{z_oz_i}{n_o + n_i} \longrightarrow x^2 + y^2 + (z - R)^2 = R^2, \tag{2.55}$$

and using 2.50 $z_o = n_i/c,\ z_i = n_o/c$ we will have

$$R = c\frac{z_in_i}{c(n_o + n_i)} \longrightarrow z_i = R\left(1 + \frac{n_o}{n_i}\right), \tag{2.56}$$

$$R = c\frac{z_on_o}{c(n_o + n_i)} \longrightarrow z_i = R\left(1 + \frac{n_i}{n_o}\right) \tag{2.57}$$

which are known as **Young-Weirstrass** points or the aplanetic points. One of these points is inside and the other outside the sphere as is shown in 2.14.

2.4.2 The second spherical diopter

So, for the second case we have 2.51 and the equation 2.52 then let z_c given by

$$z_c = z_o = z_i = \frac{n_i + n_o}{c},$$

then we will have

$$\rho^2\left(n_oz_c - n_iz_c\right) + 2zz_c^2(n_i - n_o) = 0, \tag{2.58}$$

$$\rho^2 - 2zz_c = 0, \tag{2.59}$$

again we must complete squares in the following way

$$x^2 + y^2 + z^2 - 2zz_c + z_c^2 = z_c^2, \tag{2.60}$$

$$x^2 + y^2 + (z - z_c)^2 = z_c^2. \tag{2.61}$$

The equation 2.61 is an spherical diopter where the object point is on the center and the image point is also on the center.

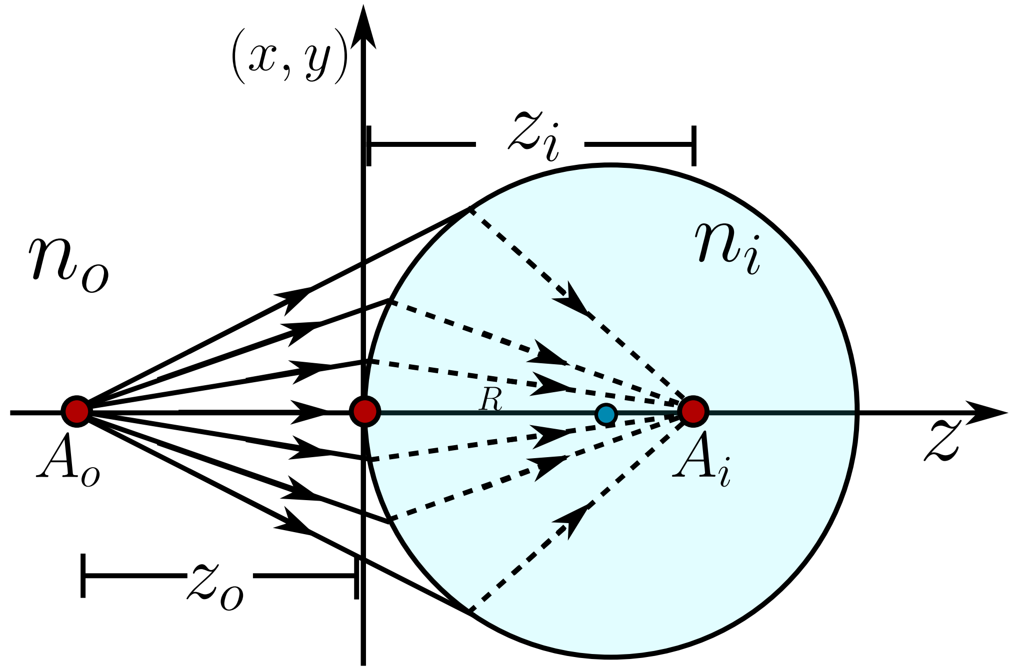


Figure 2.14: The spherical Diopter for the first case, where the object point is real and it is outside the sphere but the image point is virtual and inside the diopter.

2.5 Objects and images to the infinity

Some other cases arise when either the object or the image is at infinity, which is a convenient condition when the object is very far away, for example, in telescopes.

But even in microscopy, systems are designed to form the image at infinity, since under these conditions, an eyepiece facilitates visual accommodation for observation.

For an object in the infinity we divide by z_o^2 and take the limit $z_o \to -\infty$ in 2.42 in the following way

$$
\begin{aligned}
&\lim_{z_o\to-\infty}\left[\frac{\rho^2\left(n_o^2-n_i^2\right)}{z_o}+2z\frac{n_o^2z_o+n_i^2z_i}{z_o}\right]^2-\lim_{z_o\to-\infty}\frac{4n_on_i\left(n_oz_o+n_iz_i\right)}{z_o}\left[\frac{\rho^2\left(n_oz_i+n_iz_o\right)}{z_o}\right.\\
&\left.-\frac{2zz_oz_i(n_i-n_o)}{z_o}\right]=0,\\
&(2zn_o^2)^2-4n_o2n_i^2\left[\rho^2n_i-2zz_i(n_i-n_o)\right]=0,\\
&4z^2n_o^4-4n_o^2n_i^2\rho^2+8zz_in_o^2n_i^2-8zz_in_o^3n_i=0,\qquad \text{Dividing all the equation by } 4n_o^2,\\
&z^2\left(n_o^2-n_i^2\right)+2zn_iz_i(n_i-n_o)-n_i(x^2+y^2)=0,\\
&[n_o-n_i]\left[z^2\left(n_o+n_i\right)-2zn_iz_i\right]-n_i(x^2+y^2)=0.
\end{aligned}
$$

Now, to complete the algebraic expression we must multiply all the equation by (n_o+n_i) and complete the squares as follows

$$
\begin{aligned}
&(n_o-n_i)\left[(n_o+n_i)(n_o+n_i)z^2-2zn_iz_i(n_o+n_i)\right]-n_i^2(n_o+n_i)(x^2+y^2)=0,\\
&(n_o-n_i)\left[(n_o+n_i)^2z^2-2zn_iz_i(n_o+n_i)+n_i^2z_i^2-n_i^2(n_o+n_i)(x^2+y^2)\right]=(n_o-n_i)n_i^2z_i^2\ ,\\
&(n_o-n_i)\left[(n_o+n_i)z-n_iz_i\right]^2-n_i^2(n_o+n_i)(x^2+y^2)=(n_o-n_i)n_i^2z_i^2
\end{aligned}
$$

and dividing all by $(n_o-n_i)n_i^2z_i^2$ we will have

$$
\left(\frac{n_i+n_o}{n_iz_i}\right)^2\left[z-\frac{n_iz_i}{n_i+n_o}\right]^2+\frac{n_i+n_o}{z_i^2(n_i-n_o)}(x^2+y^2)=1. \tag{2.62}
$$

The result 2.62 represents a conical section of revolution around the z-axis from this equation we are going to obtain the Ellipsoid and the hyperboloid.

2.5.1 Revolution Ellipsoid

if $n_i>n_o$ we are going to define the following scalars

$$
a=\left(\frac{n_i-n_o}{n_i+n_o}\right)^{1/2}|z_i|,\qquad b=\left(\frac{n_i}{n_i+n_o}\right)|z_i|, \tag{2.63}
$$

and notice that $b>a$ so we will have **prolate ellipsoid** (see figure 2.15 given by

$$
\frac{1}{b^2}\left(z-b\right)^2+\frac{1}{a^2}(x^2+y^2)=1. \tag{2.64}
$$

We have got two cases, when $z_i>0$ and $z_i<0$. For both cases what is next is to study the eccentricity of the ellipse and the foci of the ellipsoids.

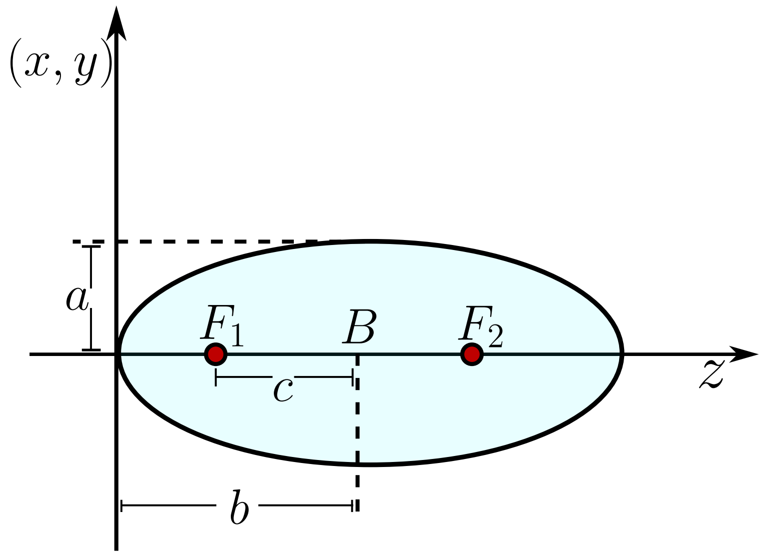


Figure 2.15: Prolate Ellipsoid centered and $z = B$.

When $z_i > 0$ lets define the segment $c = \overline{F_1B}$ such that the eccentricty is given by

$$\epsilon = \frac{c}{b} = \frac{\sqrt{b^2 - a^2}}{b} = \sqrt{1 - \frac{a^2}{b^2}} = \sqrt{\frac{n_o^2}{n_i^2}} = \frac{n_o}{n_i} < 1, \tag{2.65}$$

therefore the foci F_1 is centered to a distance

$$f_1 = b - c = b - \epsilon b = (1 - \epsilon)b = \frac{n_i - n_o}{n_i + n_o} z_i, \tag{2.66}$$

and the second foci is at

$$f_2 = b + c = b + \epsilon b = (1 + \epsilon)b = z_i. \tag{2.67}$$

This means, that for this first case that all the rays that comes parallel to the z-axis from infinity converges to the second Foci of the ellipsoid.

If $z_i < 0$ by the convention we have that $\sqrt{z_i^2} = -z_i$ and we define

$$b = -\frac{n_i}{n_i + n_o} z_i, \tag{2.68}$$

then the conical section equation 2.62 becomes

$$\frac{1}{b^2}(z + b)^2 + \frac{1}{a^2}(x^2 + y^2) = 1. \tag{2.69}$$

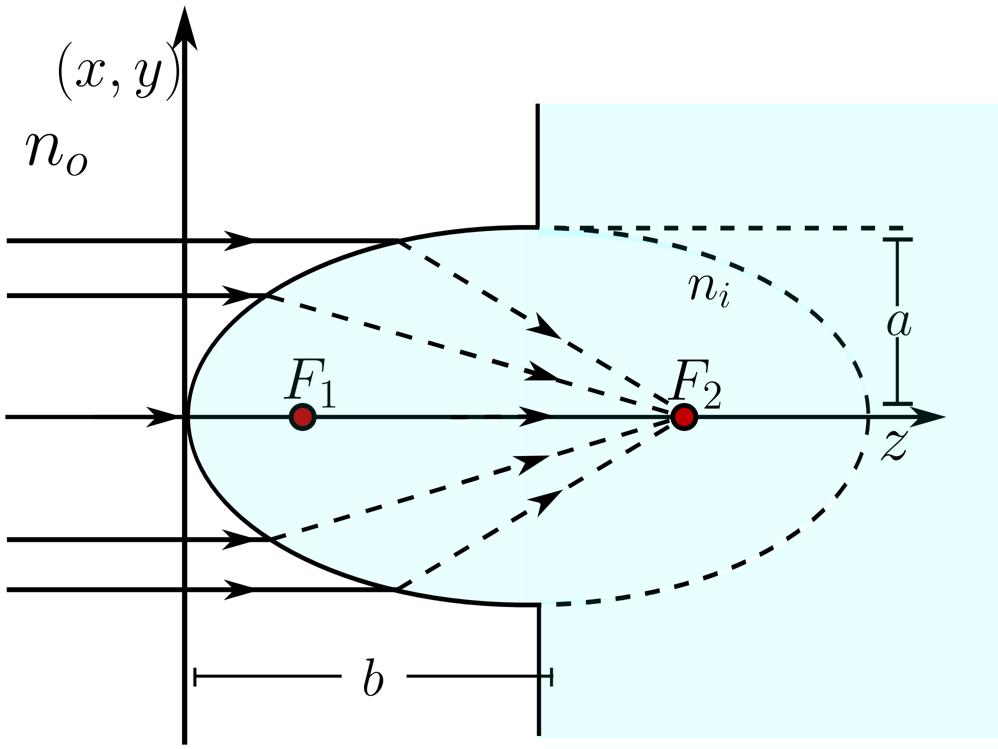


Figure 2.16: Ellipsoid diopter when $n_i > n_o$.

Again, we define $c = \overline{F_1 B}$ so the eccentricty is given by

$$\epsilon = \frac{c}{b}\frac{\sqrt{b^2 - a^2}}{b} = \sqrt{1 - \frac{a^2}{b^2}} = \frac{n_o}{n_i} < 1,$$

and the foci is now going to be to a distance

$$f_1 = -b - c = -b - \epsilon b = -b(1 + \epsilon) = z_i, \tag{2.70}$$

and the second one is at

$$f_2 = -b + c = -b + \epsilon b = (\epsilon - 1)b = \frac{n_i - n_o}{n_i + n_o} z_i. \tag{2.71}$$

Having all these results on count, we can see in figure 2.17 that all the rays that becomes parallel to the z-axis and from infinity diverges as they would come from a point source from F_1.

2.5.2 Revolution Hyperboloid

In the case $n_i < n_o$ we have got that the equation 2.62 becomes an hyperboloid and, again, there are two possibilities, $z_i < 0, z_i > 0$, so the conical equation becomes

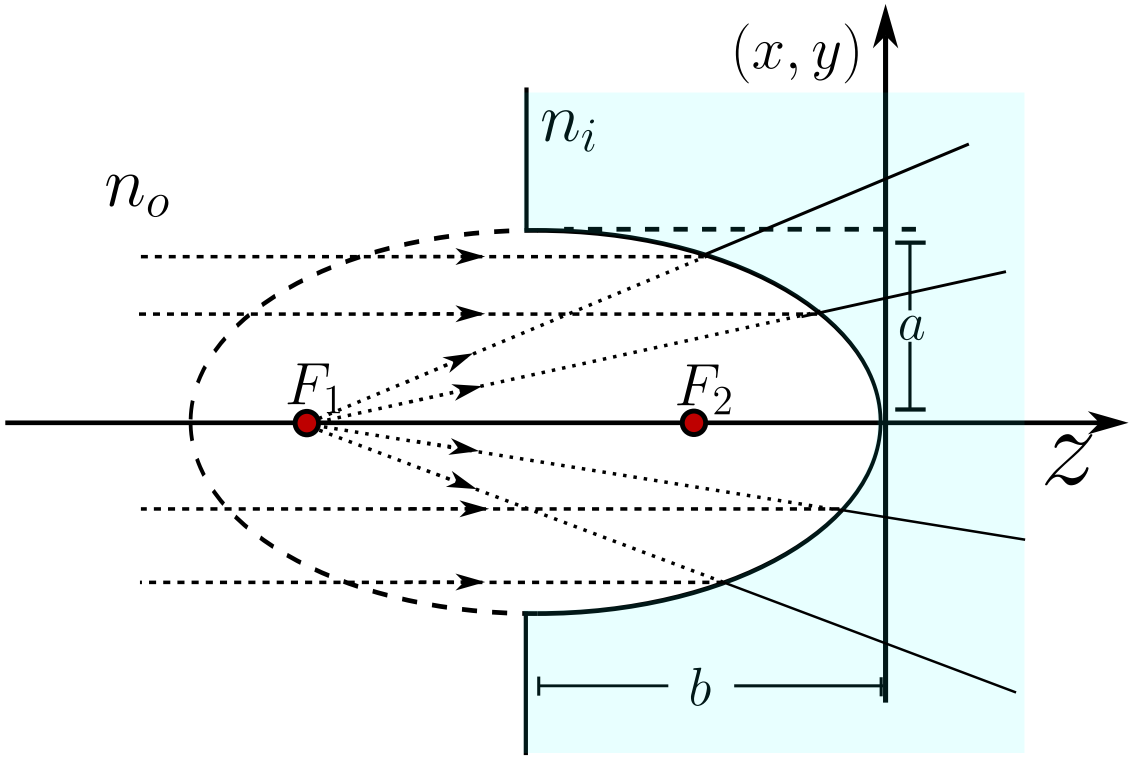


Figure 2.17: Ellipsoid diopter when $n_i > n_o$ and $z_i < 0$.

$$\left(\frac{n_i + no}{n_i z_i}\right)^2 \left[z - \frac{n_i z_i}{n_i + n_o}\right] - \frac{n_i + n_o}{z_i^2 (n_o - n_i)}(x^2 + y^2) = 1. \tag{2.72}$$

If $z_i > 0$ we define again the following scalars

$$b^2 = \frac{n_i^2}{(n_i + n_o)^2} z_i^2,$$
$$a^2 = \frac{n_o - n_i}{(n_i + n_o)^2} z_i^2,$$

so the equation of the hyperboloid (see figure 2.18) becomes

$$\frac{1}{b^2}(z - b)^2 - \frac{1}{a^2}(x^2 + y^2) = 1. \tag{2.73}$$

Again, to give a characterization of the conical surface we are going to define the segment $c = \overline{F_1 B}$ such that the eccentricity is given by

$$\epsilon = \frac{c}{b} = \frac{\sqrt{a^2 + b^2}}{b} = \sqrt{1 + \frac{a^2}{b^2}} = \frac{n_o}{n_i} > 1.$$

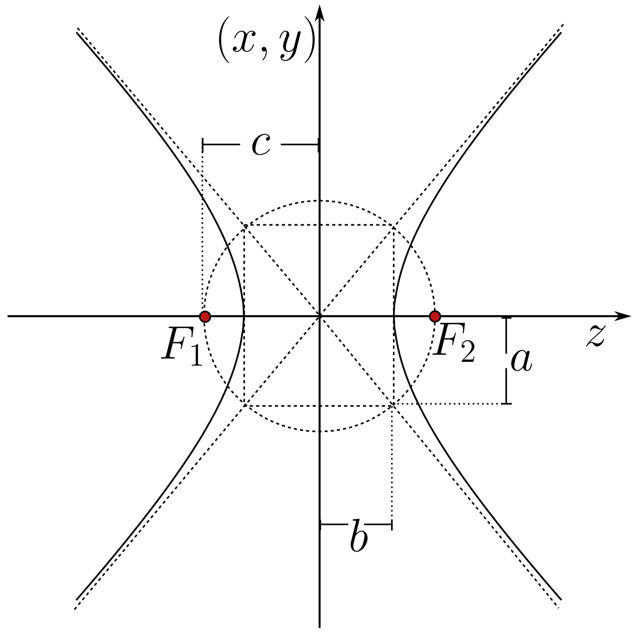


Figure 2.18: Geometrical representation of the two leaf hyperboloid with the parameters a, b and foci F_1 and F_2.

Thus the two Foci F_1 and F_2 are located at the distances

$$f_1 = b - c = b - \epsilon b = (1-\epsilon) b = \frac{n_i - n_o}{n_o + n_i} z_o, \tag{2.74}$$

$$f_2 = b + c = b + \epsilon b = (1+\epsilon) b = z_i. \tag{2.75}$$

In the figure 2.19 we can see the geometrical behavior of this first case, all the rays comes from the infinity and converges to a point which is located in the second foci F_2.

Now, for the second case, if $z_i < 0$ we define

$$b = -\frac{n_i}{n_i + n_o} z_o,$$

in such way that $b > 0$ with this we write

$$\frac{1}{b^2}(z+b)^2 - \frac{1}{a^2}(x^2 + y^2) = 1.$$

Again, we define the segment $c = \overline{F_1 B}$ such that the eccentricity is given by

$$\epsilon = \frac{c}{b} = \frac{\sqrt{a^2 + b^2}}{b} = \sqrt{1 + \frac{a^2}{b^2}} = \frac{n_o}{n_i} > 1.$$

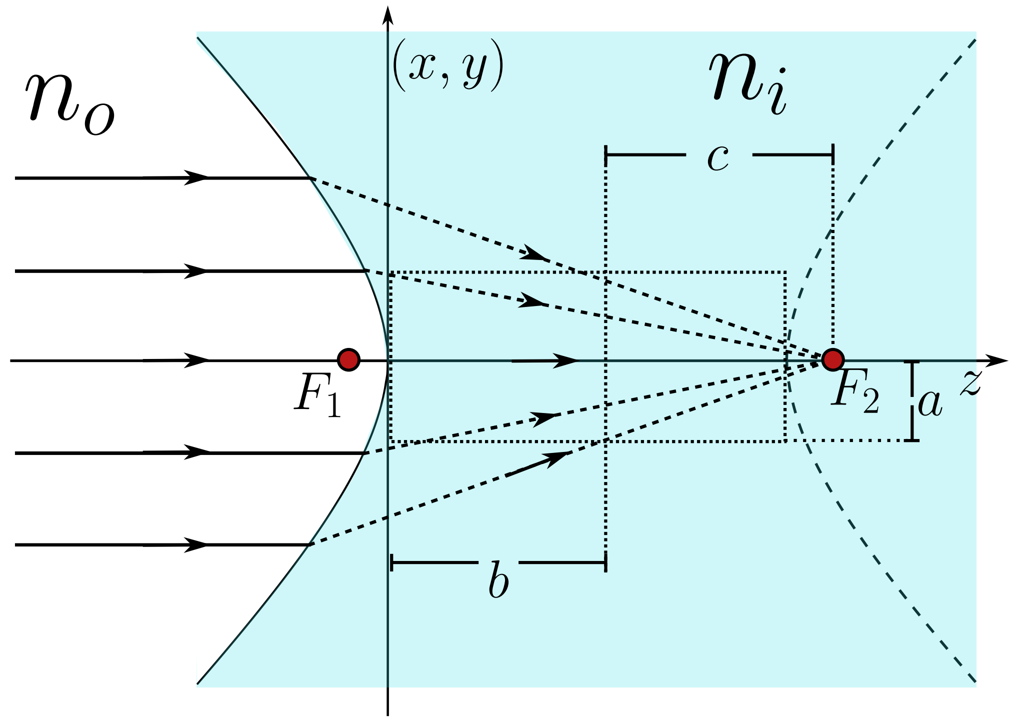


Figure 2.19: Geometrical representation of the first case of a two leaf hyperboloid.

As we can see in the figure 2.20 all the rays come from the infinity and interact with the hyperboloid diopter an the rays refracted in such geometrical way that they behave as they come from a object point source located in the foci of the first leaf at F_1.

2.6 Rigurously stigmatic surfaces by reflection

The reflective surfaces are the main components in the morden giant telescopes, for example, the James Webb telecopes is an array of many reflective surfaces which are electronically displaced such that the image formation gets the least aberration possible. Also, the reflective surfaces are better than the refractive ones in certain cases, because the optical systems can be smaller and don't have chromatic aberrations.

Tradionally, the reflective surface are developed indepentdenly from the refractive surface. Under this formulations, the rigurously stigmatic surfaces by reflection can be obtained doing $n_i = -n_o$, where the Descartes ovals become a conic surface. To this case, the equation 2.42 becomes

$$\left\{\left[(-n_o)^2 - n_o^2\right]\rho^2 - 2z\left[\left(n_o^2 z_o - (-n_o)^2 z_i\right]\right\}^2$$
$$- 4(-n_o)n_o\left[(-n_o)z_i - n_o z_o\right]\left\{\left[n_o z_i - (-n_o)z_o\right]\rho^2 + 2zz_o z_i\left[(-n_o) - n_o\right]\right\} = 0,$$

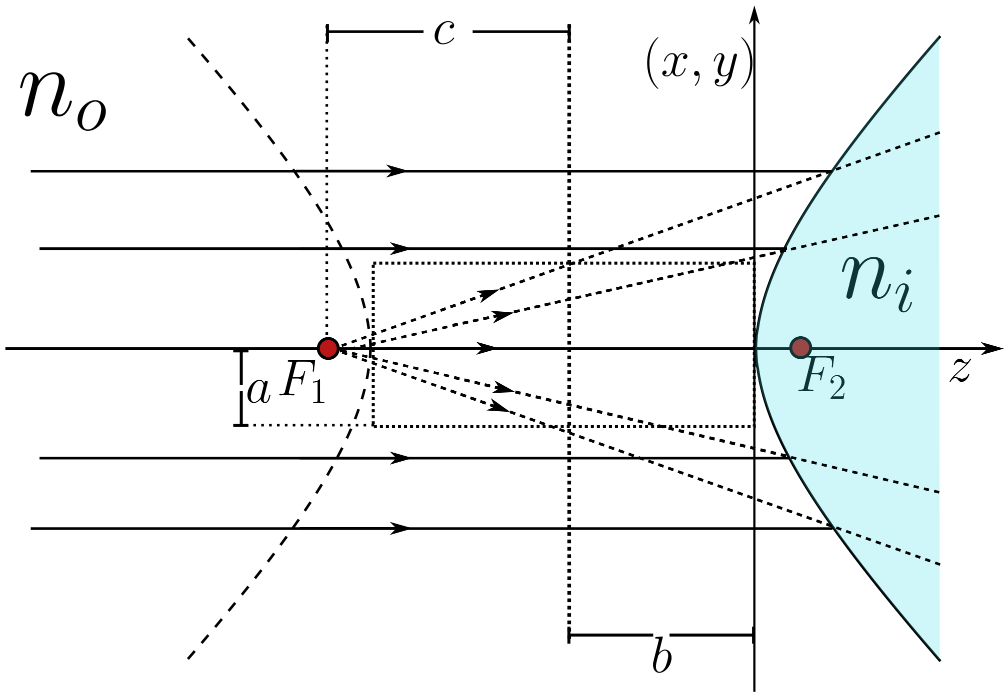


Figure 2.20: Geometrical representation of the first case of a two leaf hyperboloid.

simplyfing we've got

$$\left\{-2zn_o^2\left[z_o - z_i\right]\right\}^2 - 4n_o^3\left[z_i + z_o\right]\left\{n_o\left[z_i + z_o\right]\rho^2 - 4zz_oz_in_o\right\} = 0,$$
$$4z^2n_o^4\left(z_o - z_i\right)^2 - 4n_o^4\left(z_o + z_i\right)^2\rho^2 + 16n_o^4zz_oz_i\left(z_o + z_i\right) = 0,$$

dividing all by $-16n_o^4z_oz_i$ an having on count that $\rho^2 = x^2 + y^2 + z^2$ we've got

$$\frac{(z_o + z_i)^2}{4z_oz_i}\left(x^2 + y^2\right) + \frac{(z_o - z_i)^2}{4z_oz_i}z^2 - \frac{(z_o + z_i)^2}{4z_oz_i}z^2 - z(z_o + z_i) = 0,$$
$$\frac{(z_o + z_i)^2}{4z_oz_i}\left(x^2 + y^2\right) + z^2 - z(z_o + z_i) = 0,$$

and completing the squares we've got

$$\left(z - \frac{z_o + z_i}{2}\right)^2 + \frac{(z_o + z_i)^2}{4z_oz_i}\left(x^2 + y^2\right) = \left(\frac{z_o + z_i}{2}\right)^2, \tag{2.76}$$

which is precisely the equation of a conic of revolution around the z axis, exactly as it happened with the refractive surfaces, only that now the two indices have collapsed into a single reflective condition. The whole family of stigmatic mirrors is hidden inside 2.76, and the sign of the product z_oz_i is what decides which conic we are looking at. When $z_oz_i > 0$ the coefficient multiplying $x^2 + y^2$

is positive and the surface closes onto itself as an ellipsoid, whereas when $z_o z_i < 0$ that coefficient turns negative and the surface opens as a hyperboloid. There is also a beautiful limit waiting in 2.76, because if we push the object to infinity, $z_o \to -\infty$, the conic degenerates into a paraboloid, which is the case that rules every telescope. We are going to walk through each of these situations, since each one corresponds to a real optical instrument.

2.6.1 The ellipsoidal mirror

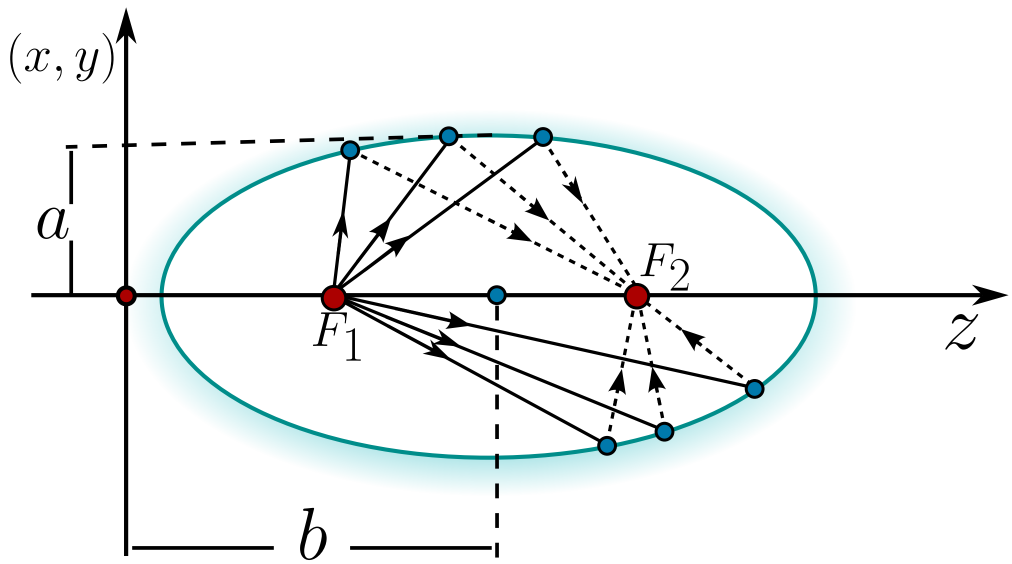


Figure 2.21: Ellipsoidal mirror. The object F_1 and the image F_2 coincide with the two foci of the ellipse, so every ray leaving one focus is reflected exactly onto the other.

Lets start with $z_o z_i > 0$, so that the object and the image sit on the same side of the vertex and the mirror is an ellipsoid. Dividing 2.76 by $\left(\frac{z_o+z_i}{2}\right)^2$ the equation takes the canonical shape

$$\frac{\left(z - \frac{z_o+z_i}{2}\right)^2}{\left(\frac{z_o+z_i}{2}\right)^2} + \frac{x^2+y^2}{z_o z_i} = 1, \tag{2.77}$$

and if we baptise the two semi-axes as

$$b^2 = \frac{(z_o+z_i)^2}{4}, \qquad a^2 = z_o z_i, \tag{2.78}$$

we recognise a prolate ellipsoid centered at $z = b = \frac{z_o+z_i}{2}$, that is

$$\frac{1}{b^2}(z-b)^2 + \frac{1}{a^2}(x^2+y^2) = 1. \tag{2.79}$$

That the ellipsoid comes out prolate, and not oblate, is not an accident, because

$$b^2 - a^2 = \frac{(z_o + z_i)^2}{4} - z_o z_i = \frac{(z_o - z_i)^2}{4} \geq 0,$$

so the semi-axis along the optical axis is always the largest one and the mirror is stretched along z. The distance from the center to each focus is then $c = \sqrt{b^2 - a^2} = \frac{|z_o - z_i|}{2}$, and the eccentricity becomes

$$\epsilon = \frac{c}{b} = \frac{\sqrt{b^2 - a^2}}{b} = \sqrt{1 - \frac{a^2}{b^2}} = \frac{|z_o - z_i|}{z_o + z_i} < 1, \tag{2.80}$$

which confirms once more that we are dealing with an ellipse. The really interesting thing is where the foci land, because measured from the vertex they sit at $z = b \pm c$, that is

$$b - c = \frac{z_o + z_i}{2} - \frac{|z_o - z_i|}{2}, \qquad b + c = \frac{z_o + z_i}{2} + \frac{|z_o - z_i|}{2}, \tag{2.81}$$

and no matter which of the two conjugate points is the larger one, these two numbers are nothing but z_o and z_i themselves. **The two foci of the ellipsoid are exactly the object point and the image point.**

This is the whole magic of the ellipsoidal mirror and the reason it is rigorously stigmatic. An ellipse has the geometric property that any ray leaving one focus is reflected straight into the other focus, because the sum of the two focal distances is constant along the whole curve, and a constant optical path is exactly the Fermat's principle at work. Placing a point source at one focus therefore produces a perfect point image at the other one, with no aberration at all. This is the principle behind the elliptical reflectors used to concentrate the light coming from one focus onto a detector sitting at the other, and it is the same property that makes the whispering galleries carry a voice from one focus of the room to the other.

When both z_o and z_i are negative the algebra is identical once we define $b = -\frac{z_o + z_i}{2} > 0$, so that the ellipsoid is now centered at $z = -b$ and its equation reads $\frac{1}{b^2}(z + b)^2 + \frac{1}{a^2}(x^2 + y^2) = 1$. The eccentricity is the same and the foci again fall on z_o and z_i, the only difference being that both conjugate points now live on the opposite side of the vertex.

2.6.2 The hyperboloidal mirror

Now we let the product change sign, $z_o z_i < 0$, which happens when the object and the image sit on opposite sides of the mirror and one of them is virtual. The coefficient of $x^2 + y^2$ in 2.76 becomes negative, so writing $a^2 = -z_o z_i > 0$ while keeping $b^2 = \frac{(z_o + z_i)^2}{4}$ the canonical form is now

$$\frac{1}{b^2}(z - b)^2 - \frac{1}{a^2}(x^2 + y^2) = 1, \tag{2.82}$$

a hyperboloid of two sheets around the optical axis. The focal distance is measured with a plus sign this time, $c = \sqrt{a^2 + b^2}$, and a short computation gives again $c = \frac{|z_o - z_i|}{2}$, so the eccentricity is

$$\epsilon = \frac{c}{|b|} = \frac{\sqrt{a^2 + b^2}}{|b|} = \sqrt{1 + \frac{a^2}{b^2}} = \frac{|z_o - z_i|}{|z_o + z_i|} > 1, \tag{2.83}$$

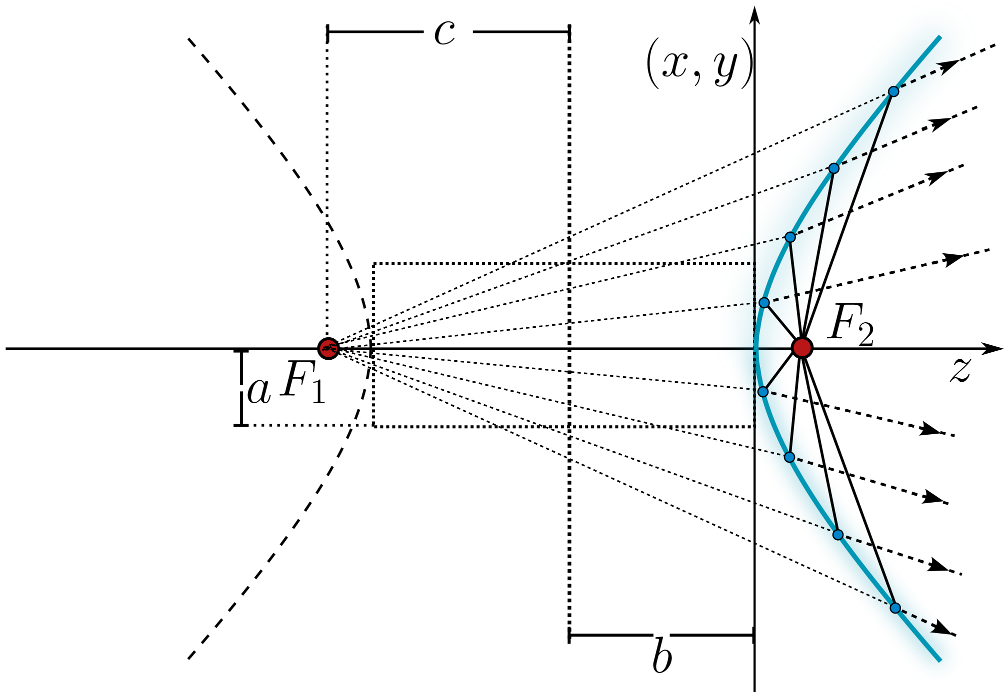


Figure 2.22: Hyperboloid mirror. The rays from the object F_1 diverge as they were emited by a virtual image locaed in F_2.

which is greater than one exactly because $z_o z_i < 0$ forces $|z_o - z_i| > |z_o + z_i|$. And just as before, the foci $z = b \pm c$ collapse onto z_o and z_i, so once more the two conjugate points are the two foci of the conic.

The hyperboloidal mirror is therefore stigmatic between its two foci as well, but with a twist, because since the foci lie on opposite sheets one of the conjugate points is always virtual. This is precisely the behavior exploited by the convex secondary mirror of a Cassegrain telescope, where the light that is already converging toward the focus of the primary mirror is intercepted by a hyperboloidal secondary and redirected, without adding any aberration, to a more comfortable focus behind the primary. The stigmatic pair of the hyperboloid is what lets these compact folded telescopes keep a perfect image on the axis.

2.6.3 Special cases: the spherical and the plane mirror

Two very familiar mirrors are hiding inside 2.76 as degenerate cases. The first one appears when the object and the image coincide, $z_o = z_i = R$. The coefficient of $x^2 + y^2$ becomes $\frac{(2R)^2}{4R^2} = 1$ and the conic reduces to $z^2 - 2Rz + x^2 + y^2 = 0$, and completing the square by adding R^2 to both sides we get

$$(z - R)^2 + x^2 + y^2 = R^2, \tag{2.84}$$

which is simply a sphere of radius R centered at $z = R$. This tells us that a spherical mirror is rigorously stigmatic for one very special pair of points, both of them sitting together at the center of the sphere, since a source placed at the center sends every ray along a radius, the ray strikes the surface perpendicularly and comes straight back on itself, forming a perfect image right on top of the source. For any other position the sphere is only approximately stigmatic, and this is the origin of the spherical aberration that spoils cheap mirrors and that the conics above were designed to cure.

The second degenerate case is $z_o = -z_i$, for which the sum $z_o + z_i$ vanishes. Both the linear term and the whole right hand side of 2.76 die at once, and the equation collapses to $z^2 = 0$, that is, the plane $z = 0$. We recover the ordinary plane mirror, which is stigmatic for every point and always produces an image symmetric to the object at $z_i = -z_o$, which is nothing more than the everyday statement that a plane mirror puts your reflection as far behind the glass as you stand in front of it.

2.6.4 The parabolic mirror and objects at infinity

The most important mirror of all comes from letting the object recede to infinity, $z_o \to -\infty$, which is the situation of a telescope pointed at a star. Going back to 2.76 and expanding the square on the left, the terms $\frac{(z_o+z_i)^2}{4}$ appear on both sides and cancel,

$$z^2 - z(z_o + z_i) + \frac{(z_o + z_i)^2}{4} + \frac{(z_o + z_i)^2}{4z_o z_i}(x^2 + y^2) = \frac{(z_o + z_i)^2}{4},$$

leaving the expanded conic

$$z^2 - z(z_o + z_i) + \frac{(z_o + z_i)^2}{4z_o z_i}(x^2 + y^2) = 0.$$

Dividing everything by z_o and taking the limit $z_o \to -\infty$, the term z^2/z_o vanishes, the factor $\frac{z_o+z_i}{z_o} \to 1$ and $\frac{(z_o+z_i)^2}{4z_o^2 z_i} \to \frac{1}{4z_i}$, so we are left with the remarkably clean surface

$$x^2 + y^2 = 4z_i z, \tag{2.85}$$

a paraboloid of revolution whose opening is fixed by the sign of z_i.

The paraboloid is the crown jewel of the reflective optics. A parabola has the defining property that every ray arriving parallel to its axis is reflected exactly through the focus, so a bundle of parallel rays coming from a star, which is an object at infinity, is gathered into a single perfect point at $z = z_i$, the focus of the mirror. There is no chromatic aberration, because the reflection does not depend on the wavelength, and there is no spherical aberration, because the surface is a true paraboloid instead of a sphere. This is why every serious reflecting telescope, from the first design of Newton to the segmented primary of the James Webb, uses a parabolic mirror as its light collecting surface, and it is also why the same shape reappears in the satellite dishes and the searchlights, where the ray tracing simply runs the other way and a source placed at the focus is turned into a parallel beam.

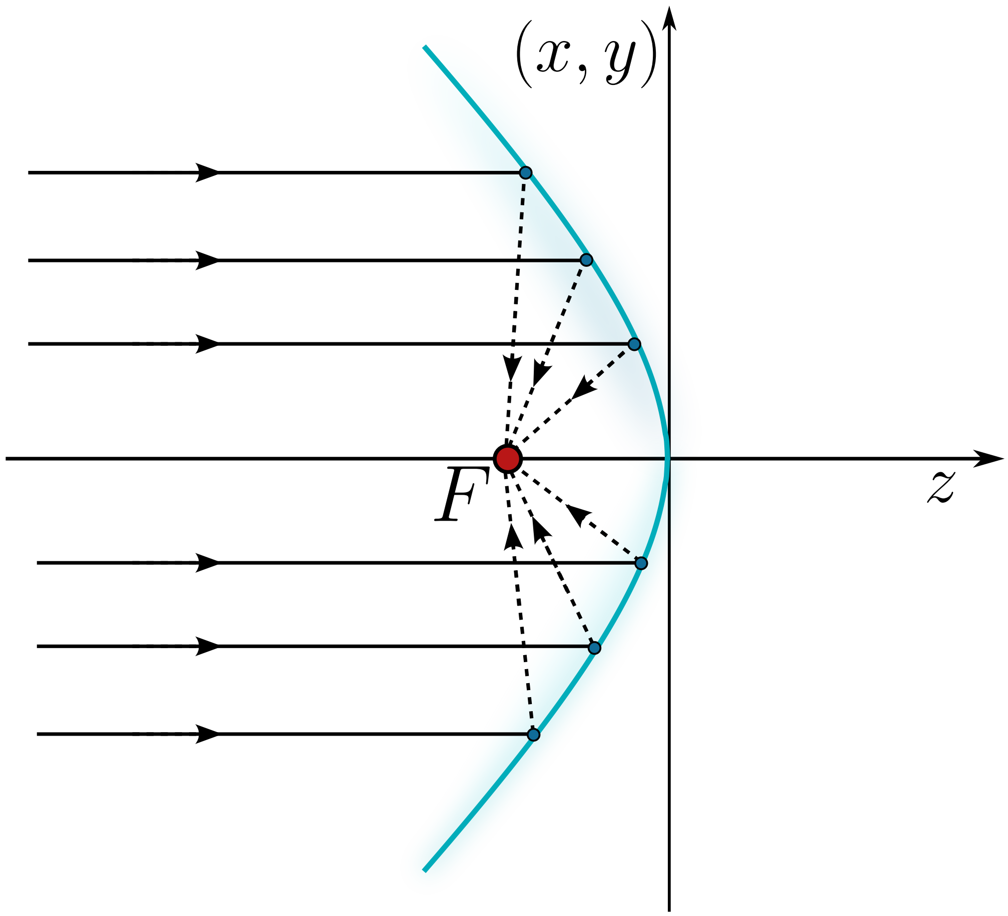


Figure 2.23: Parabolic mirror obtained as the limit $z_o \to -\infty$. The rays coming parallel to the axis from an object at infinity are all focused onto the single point $z = z_i$.

2.7 Explicit formulation of the Descartes ovoids

Everything we have built so far has left us with the Descartes ovoid written in the implicit form 2.42, that is, as the zero set of a quartic polynomial $F(x, y, z) = 0$. That form is beautiful for proving things, because every degenerate case we studied, the sphere, the ellipsoid, the hyperboloid, the paraboloid and the plane, was obtained simply by looking at what the polynomial does under a particular choice of n_o, n_i, z_o, z_i. But the implicit form is nearly useless for the person who has to actually build the surface. An optical designer, a lens grinder or a diamond turning machine does not want to know the set of points that annihilate a quartic, they want a height, they want to be told how much material to remove at a distance ρ from the axis. In other words, they want the *sag* of the surface, the explicit function $z(\rho)$.

So the goal of this section is purely practical, we want to solve 2.42 for z. The obstacle is that the equation is quartic in z, and a brute force attack on a quartic is a nightmare. The trick, which is the whole content of this section, is to notice that the ovoid is secretly a much simpler object, a *superconic*, and that when written in superconic form the equation becomes quadratic in z, which we can solve with the formula we learned as children.

2.7.1 The Descartes ovoid as a superconic

A superconic is a surface of revolution around the optical axis defined by

$$c_0 G z^2 - 2\left(1 + b_1\rho^2 + b_2\rho^4 + \cdots\right) z + \left(c_0 + c_1\rho^2 + c_2\rho^4 + \cdots\right)\rho^2 = 0, \tag{2.86}$$

where G is the conic constant, also called the Schwarzschild constant (Schwarzschild, 1905), and c_0 is the paraxial curvature, that is, the curvature the surface has right at the vertex. If we kill every deformation coefficient, $b_j = c_j = 0$ for $j \geq 1$, the surface collapses to

$$c_0 G z^2 - 2z + c_0\rho^2 = 0, \tag{2.87}$$

which is the standard conic of revolution. So a superconic is nothing but a conic that has been allowed to deform, term by term, in even powers of ρ.

Now comes the claim that makes everything work, *the Descartes ovoid is a superconic truncated at the very first deformation order*. To see it, take the ovoid 2.42

$$\left[\left(n_i^2 - n_o^2\right)\rho^2 - 2z\left(n_i^2 z_i - n_o^2 z_o\right)\right]^2 - 4n_i n_o\left(n_i z_i - n_o z_o\right)\left[\left(n_o z_i - n_i z_o\right)\rho^2 + 2zz_o z_i\left(n_i - n_o\right)\right] = 0,$$

and divide the whole equation by the constant

$$\mathcal{N} = 4n_i n_o z_i z_o\left(n_i - n_o\right)\left(n_i z_i - n_o z_o\right). \tag{2.88}$$

Let us do it piece by piece, because this is where the four shape parameters are born. The second term of the ovoid divided by $\mathcal{N}$ gives

$$\begin{aligned} &-\frac{4n_i n_o\left(n_i z_i - n_o z_o\right)\left[\left(n_o z_i - n_i z_o\right)\rho^2 + 2zz_o z_i\left(n_i - n_o\right)\right]}{4n_i n_o\left(n_i z_i - n_o z_o\right) z_i z_o\left(n_i - n_o\right)} = \\ &-\frac{\left(n_o z_i - n_i z_o\right)\rho^2}{z_i z_o\left(n_i - n_o\right)} - \frac{2z z_o z_i\left(n_i - n_o\right)}{z_i z_o\left(n_i - n_o\right)} = \frac{n_i z_o - n_o z_i}{z_i z_o\left(n_i - n_o\right)}\rho^2 - 2z, \end{aligned}$$

which already has the shape $O\rho^2 - 2z$ of the superconic 2.87 if we baptise

$$O = \frac{n_i z_o - n_o z_i}{z_i z_o\left(n_i - n_o\right)}. \tag{2.89}$$

The first term of the ovoid is a perfect square, so we expand it before dividing,

$$\begin{aligned} &\left[\left(n_i^2 - n_o^2\right)\rho^2 - 2z\left(n_i^2 z_i - n_o^2 z_o\right)\right]^2 = \\ &\left(n_i^2 - n_o^2\right)^2\rho^4 - 4z\left(n_i^2 - n_o^2\right)\left(n_i^2 z_i - n_o^2 z_o\right)\rho^2 + 4z^2\left(n_i^2 z_i - n_o^2 z_o\right)^2, \end{aligned}$$

and now we divide each of these three pieces by $\mathcal{N}$. The coefficient of z^2 becomes

$$\frac{4\left(n_i^2 z_i - n_o^2 z_o\right)^2}{4 n_i n_o z_i z_o \left(n_i - n_o\right)\left(n_i z_i - n_o z_o\right)} = \underbrace{\frac{\left(n_i^2 z_i - n_o^2 z_o\right)^2}{n_i n_o \left(n_i z_i - n_o z_o\right)\left(n_i z_o - n_o z_i\right)}}_{G} \underbrace{\frac{n_i z_o - n_o z_i}{z_i z_o \left(n_i - n_o\right)}}_{O} = OG,$$

where we have simply multiplied and divided by $(n_i z_o - n_o z_i)$ in order to make the factor O appear. The coefficient of $z\rho^2$ becomes

$$-\frac{4\left(n_i^2 - n_o^2\right)\left(n_i^2 z_i - n_o^2 z_o\right)}{4 n_i n_o z_i z_o \left(n_i - n_o\right)\left(n_i z_i - n_o z_o\right)} = -\frac{\left(n_i + n_o\right)\left(n_i - n_o\right)\left(n_i^2 z_i - n_o^2 z_o\right)}{n_i n_o z_i z_o \left(n_i - n_o\right)\left(n_i z_i - n_o z_o\right)} = -2\underbrace{\frac{\left(n_i + n_o\right)\left(n_i^2 z_i - n_o^2 z_o\right)}{2 n_i n_o z_i z_o \left(n_i z_i - n_o z_o\right)}}_{S},$$

and finally the coefficient of ρ^4 becomes

$$\frac{\left(n_i^2 - n_o^2\right)^2}{4 n_i n_o z_i z_o \left(n_i - n_o\right)\left(n_i z_i - n_o z_o\right)} = \frac{\left(n_i + n_o\right)^2 \left(n_i - n_o\right)^2}{4 n_i n_o z_i z_o \left(n_i - n_o\right)\left(n_i z_i - n_o z_o\right)} = \underbrace{\frac{\left(n_i - n_o\right)\left(n_i + n_o\right)^2}{4 n_i n_o z_i z_o \left(n_i z_i - n_o z_o\right)}}_{T}.$$

Collecting the four pieces, the Descartes ovoid has been transformed into

$$f(z,\rho) = OGz^2 - 2\left(1 + S\rho^2\right) z + \left(O + T\rho^2\right)\rho^2 = 0, \tag{2.90}$$

which is exactly the superconic 2.86 truncated at b_1 and c_1, with $b_1 = S$ and $c_1 = T$. The four quantities we found along the way,

$$G = \frac{\left(n_i^2 z_i - n_o^2 z_o\right)^2}{n_i n_o \left(n_i z_i - n_o z_o\right)\left(n_i z_o - n_o z_i\right)}, \tag{2.91}$$

$$O = \frac{n_i z_o - n_o z_i}{z_i z_o \left(n_i - n_o\right)}, \tag{2.92}$$

$$T = \frac{\left(n_i - n_o\right)\left(n_i + n_o\right)^2}{4 n_i n_o z_i z_o \left(n_i z_i - n_o z_o\right)}, \tag{2.93}$$

$$S = \frac{\left(n_i + n_o\right)\left(n_i^2 z_i - n_o^2 z_o\right)}{2 n_i n_o z_i z_o \left(n_i z_i - n_o z_o\right)}, \tag{2.94}$$

are called the **shape parameters** $GOTS$ of the ovoid, and they carry the whole physics of the surface, because the two refraction indices and the two conjugate positions enter the geometry only through these four numbers.

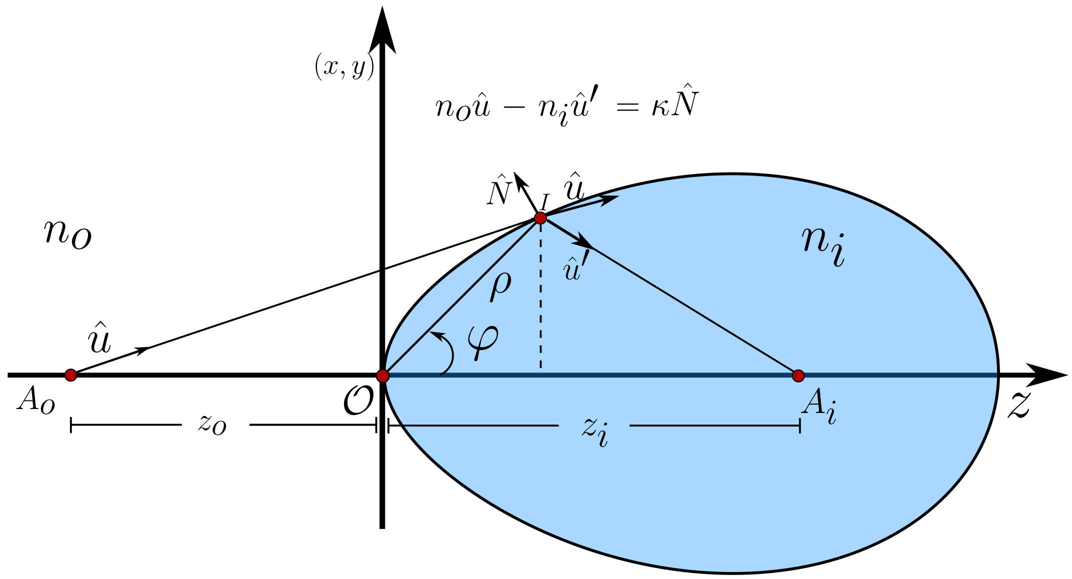


Figure 2.24: Rigorously stigmatic quartic surface described in the meridional plane by the polar parameters ρ and φ measured from the vertex $\mathcal{O}$. The object point A_o at z_o and the image point A_i at z_i are the strictly stigmatic pair, and at the incidence point I the vectorial Snell-Descartes law $n_o\hat{u} - n_i\hat{u}' = \kappa\hat{N}$ is satisfied.

These four parameters are not independent. If we compute the product GOT we find

$$GOT = \left[\frac{\left(n_i^2 z_i - n_o^2 z_o\right)^2}{n_i n_o z_i z_o \left(n_i - n_o\right)\left(n_i z_i - n_o z_o\right)}\right]\left[\frac{\left(n_i - n_o\right)\left(n_i + n_o\right)^2}{4 n_i n_o z_i z_o \left(n_i z_i - n_o z_o\right)}\right] =$$
$$\frac{\left(n_i + n_o\right)^2\left(n_i^2 z_i - n_o^2 z_o\right)^2}{4 n_i^2 n_o^2 z_i^2 z_o^2 \left(n_i z_i - n_o z_o\right)^2} = \left[\frac{\left(n_i + n_o\right)\left(n_i^2 z_i - n_o^2 z_o\right)}{2 n_i n_o z_i z_o \left(n_i z_i - n_o z_o\right)}\right]^2 = S^2,$$

where in the first line we already wrote OG as the single fraction we obtained above. Therefore the shape parameters obey the constraint

$$S^2 = GOT, \tag{2.95}$$

and this ligature is precisely what distinguishes a Descartes ovoid from a generic superconic. A superconic with four freely chosen parameters is just a deformed conic with no particular optical virtue, while a superconic whose parameters satisfy 2.95 is rigorously stigmatic. The constraint is therefore the algebraic fingerprint of perfect imaging.

It is worth pausing on what the constraint costs us. We have four parameters and one equation between them, so only three are free, and this matches the physical counting, because the surface is determined by the ratio of the indices and by the two conjugate distances, that is, by three independent numbers. Notice also that if we send either conjugate point to infinity, $z_o \to \infty$ or $z_i \to \infty$, both S and T carry a $z_i z_o$ in the denominator and therefore vanish, and 2.90 degenerates into the standard conic 2.87. This is the algebraic version of everything we found in the previous sections, the conics are the ovoids with one conjugate point at infinity.

2.7.2 The explicit formulation

Now that the ovoid is quadratic in z, the rest is elementary. Reading 2.90 as $az^2 + bz + c = 0$ with $a = OG$, $b = -2(1 + S\rho^2)$ and $c = (O + T\rho^2)\rho^2$, the quadratic formula gives

$$z = \frac{2\left(1 + S\rho^2\right) \pm \sqrt{4\left(1 + S\rho^2\right)^2 - 4OG\left(O + T\rho^2\right)\rho^2}}{2OG} = \\ \frac{\left(1 + S\rho^2\right) \pm \sqrt{\left(1 + S\rho^2\right)^2 - OG\left(O + T\rho^2\right)\rho^2}}{OG},$$

and expanding the discriminant

$$\left(1 + S\rho^2\right)^2 - OG\left(O + T\rho^2\right)\rho^2 = 1 + 2S\rho^2 + S^2\rho^4 - O^2G\rho^2 - OGT\rho^4 = \\ 1 + \left(2S - O^2G\right)\rho^2 + \left(S^2 - GOT\right)\rho^4,$$

so that the sag of the surface is

$$z = \frac{1 + S\rho^2 \pm \sqrt{1 + \left(2S - O^2G\right)\rho^2 + \left(S^2 - GOT\right)\rho^4}}{OG}. \tag{2.96}$$

Two things must be settled now. The first one is which of the two signs is the physical branch. The surface has to pass through the vertex $\mathcal{O}$, which is the origin of our coordinates, therefore we must demand $z \to 0$ when $\rho \to 0$. Putting $\rho = 0$ in 2.96 gives $z = (1 \pm 1)/OG$, which is $2/OG$ for the plus sign and 0 for the minus sign. Hence *the Descartes ovoid is the branch with the minus sign*, and the branch with the plus sign is the outer sheet of the quartic, the second surface hidden inside the same polynomial.

The second thing is the ligature 2.95. Because $S^2 - GOT = 0$, the quartic term inside the radical dies and the expression collapses to

$$z = \frac{1}{OG}\left(1 + S\rho^2 - \sqrt{1 + \left(2S - O^2G\right)\rho^2}\right). \tag{2.97}$$

This is already explicit, but it is numerically ugly, because for small ρ it is the difference of two numbers that are both close to one, divided by OG, which can itself be small. The cure is to rationalise, multiplying and dividing by the conjugate,

$$z = \frac{1}{OG}\frac{\left(1 + S\rho^2 - \sqrt{1 + \left(2S - O^2G\right)\rho^2}\right)\left(1 + S\rho^2 + \sqrt{1 + \left(2S - O^2G\right)\rho^2}\right)}{1 + S\rho^2 + \sqrt{1 + \left(2S - O^2G\right)\rho^2}} = \\ \frac{1}{OG}\frac{\left(1 + S\rho^2\right)^2 - \left[1 + \left(2S - O^2G\right)\rho^2\right]}{1 + S\rho^2 + \sqrt{1 + \left(2S - O^2G\right)\rho^2}},$$

and the numerator simplifies beautifully,

$$1 + 2S\rho^2 + S^2\rho^4 - 1 - 2S\rho^2 + O^2G\rho^2 = S^2\rho^4 + O^2G\rho^2 =$$
$$\rho^2\left(O^2G + S^2\rho^2\right) = \rho^2\left(O^2G + GOT\rho^2\right) = OG\rho^2\left(O + T\rho^2\right),$$

where in the second line we used the ligature $S^2 = GOT$ once more. The factor OG cancels against the one sitting outside, and we arrive at the result we were after,

$$z(\rho) = \frac{\left(O + T\rho^2\right)\rho^2}{1 + S\rho^2 + \sqrt{1 + (2S - O^2G)\,\rho^2}}. \tag{2.98}$$

Equation 2.98 is the explicit formulation of the Descartes ovoids. It is worth appreciating how much has happened here. We started from a quartic implicit surface and we ended with a closed formula that gives the height of the surface directly in terms of the radial coordinate and four numbers built out of n_o, n_i, z_o and z_i. There is no series, no iteration and no numerical root finding, the expression is exact.

Since our radial variable was defined as $\rho^2 = x^2 + y^2 + z^2$, that is, the distance from the vertex to the point on the surface and not the transverse height, we recover the transverse coordinate from

$$r(\rho) = \pm\sqrt{\rho^2 - z^2(\rho)}, \qquad r^2 = x^2 + y^2, \tag{2.99}$$

so the pair $(z(\rho), r(\rho))$ traces the meridional profile of the ovoid, and the full surface is obtained by revolving that profile around the optical axis.

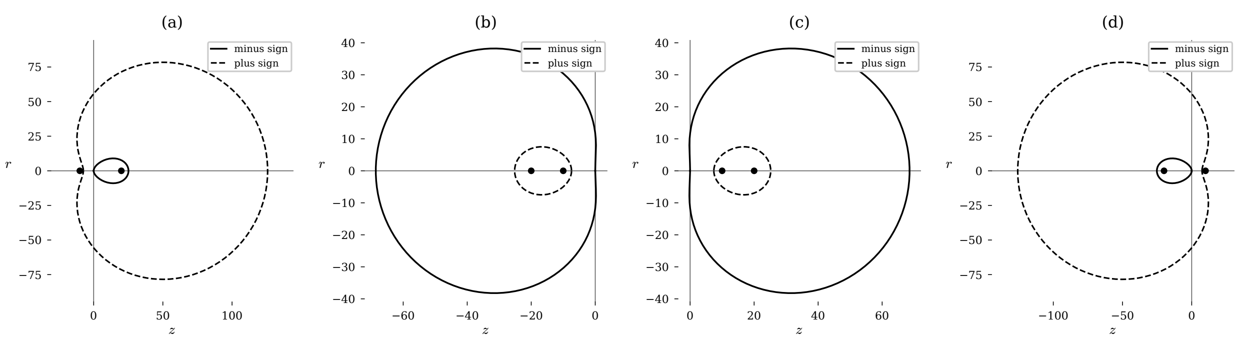


Figure 2.25: Meridional sections of Cartesian surfaces with the refraction indices fixed at $n_o = 1$ and $n_i = 1.7$ while the object and image positions are varied. The solid curves are the solutions with the minus sign in 2.96, which pass through the origin and are the Descartes ovoids, and the dashed curves are the solutions with the plus sign, the outer sheet of the same quartic. (a) $z_o = -10$ and $z_i = 20$. (b) $z_o = -10$ and $z_i = -20$. (c) $z_o = 10$ and $z_i = 20$. (d) $z_o = 10$ and $z_i = -20$.

In the figure 2.25 we can see how the two branches behave. The solid curves, which are the ones we selected, always start at the origin, while the dashed ones close the quartic from outside, and together they show how the surface makes a transition from the inner to the outer sheet. The four panels also show that the formulation handles without any modification the cases where the object or the image is virtual, that is, the cases where $z_o > 0$ or $z_i < 0$, which is exactly the situation that the older explicit formulations in the literature were unable to treat in a unified way. Depending on the signs, the surface opens to the right or to the left, and both are legitimate optical surfaces.

Surface	n_i	z_o	z_i	G	O	T	S
Ellipsoidal refractive	$n_i > n_o$	∞	z_i	$-\left(\frac{n_o}{n_i}\right)^2$	$\frac{n_i}{z_i(n_i-n_o)}$	0	0
Hyperboloidal refractive	$n_i < n_o$	∞	z_i	$-\left(\frac{n_o}{n_i}\right)^2$	$\frac{n_i}{z_i(n_i-n_o)}$	0	0
Hyperboloidal refractive	$n_i > n_o$	z_o	∞	$-\left(\frac{n_i}{n_o}\right)^2$	$-\frac{n_o}{z_o(n_i-n_o)}$	0	0
Ellipsoidal refractive	$n_i < n_o$	z_o	∞	$-\left(\frac{n_i}{n_o}\right)^2$	$-\frac{n_o}{z_o(n_i-n_o)}$	0	0
Spherical refractive	n_i	z_o	z_o	$\frac{(n_i+n_o)^2}{n_i n_o}$	$\frac{1}{z_o}$	$\frac{(n_i+n_o)^2}{4n_i n_o z_o^3}$	$\frac{(n_o+n_i)^2}{2n_i n_o z_o^2}$
Ellipsoidal reflective	$-n_o$	$z_o > 0$	$z_i > 0$	$-\left(\frac{z_i-z_o}{z_i+z_o}\right)^2$	$\frac{z_i+z_o}{2z_i z_o}$	0	0
Ellipsoidal reflective	$-n_o$	$z_o < 0$	$z_i < 0$	$-\left(\frac{z_i-z_o}{z_i+z_o}\right)^2$	$\frac{z_i+z_o}{2z_i z_o}$	0	0
Hyperboloidal reflective	$-n_o$	$z_o > 0$	$z_i < 0$	$-\left(\frac{z_i-z_o}{z_i+z_o}\right)^2$	$\frac{z_i+z_o}{2z_i z_o}$	0	0
Hyperboloidal reflective	$-n_o$	$z_o < 0$	$z_i > 0$	$-\left(\frac{z_i-z_o}{z_i+z_o}\right)^2$	$\frac{z_i+z_o}{2z_i z_o}$	0	0
Paraboloidal reflective	$-n_o$	∞	z_i	-1	$\frac{1}{2z_i}$	0	0
Paraboloidal reflective	$-n_o$	z_o	∞	-1	$\frac{1}{2z_o}$	0	0

Figure 2.26: Degenerate cases of refractive and reflective rigorously stigmatic surfaces according to the refraction indices and the object and image positions. For each combination of these parameters the values of the shape parameters $GOTS$ of the corresponding conic of revolution are obtained. Note that $T = S = 0$ in every conic case, and that the reflective surfaces are recovered by the substitution $n_i \to -n_o$.

The table 2.26 closes the circle with everything we did in the previous sections. Every single surface we deduced by hand, the ellipsoid, the hyperboloid, the sphere, the paraboloid and their reflective versions, appears there as a particular choice of the shape parameters. Notice that in all the conic rows we have $T = S = 0$, which is the algebraic statement that one of the conjugate points has been sent to infinity, and notice that the only row where T and S survive is the spherical refractive one, where $z_o = z_i$, that is, the second spherical diopter 2.61 where object and image sit together at the centre of the sphere.

2.7.2.1 The Schwarzschild equation

There is one more limit worth writing down explicitly, because it connects our formula with the equation that every optical design program in the world uses. When $T = S = 0$, that is, for the conic surfaces, the expression 2.98 reduces to

$$z(\rho) = \frac{O\rho^2}{1 + \sqrt{1 - O^2 G \rho^2}}, \tag{2.100}$$

which is still written in terms of the distance ρ from the vertex, and not in terms of the transverse height r. To change variables we use $\rho^2 = z^2 + r^2$ in the conic 2.87,

$$\begin{aligned} &OGz^2 - 2z + O\left(z^2 + r^2\right) = 0, \\ &O\left(G + 1\right) z^2 - 2z + Or^2 = 0, \end{aligned}$$

which is again quadratic in z, and solving it and choosing once more the branch through the vertex,

$$z = \frac{1 - \sqrt{1 - \left(G + 1\right) O^2 r^2}}{O\left(G + 1\right)},$$

and rationalising exactly as before, the numerator becomes $1 - 1 + (G + 1)O^2 r^2$, so that

$$z(r) = \frac{Or^2}{1 + \sqrt{1 - \left(K + 1\right) O^2 r^2}}, \qquad K = G, \tag{2.101}$$

which is the Schwarzschild formula for the standard conic surfaces (Schwarzschild, 1905), where $K = -e^2$ is the conic constant written in terms of the eccentricity and O is the curvature at the vertex.

This tells us how to read our own result. The equation 2.98 is a *generalisation of the Schwarzschild equation* for the Cartesian surfaces, where G generalises the Schwarzschild constant and we may call it the Descartes ovoid constant, O is the paraxial curvature of the ovoid at the vertex, and T is the parameter that labels the different families of ovoids. The parameter S adds nothing new, since the ligature 2.95 fixes it once the other three are given.

2.8 Special Descartes ovoids

With the shape parameters at hand we can now classify the ovoids in the same spirit in which one classifies the conics by their eccentricity. The classification is governed by G, and the reason is immediate from 2.91, since the numerator is a perfect square and $n_i n_o > 0$, the sign of G is nothing but the sign of the product

$$\left(n_i z_i - n_o z_o\right)\left(n_i z_o - n_o z_i\right), \tag{2.102}$$

so the whole taxonomy is decided by the relative position of the two conjugate points weighted by the indices.

2.8.1 Spherovoids, $G = 0$

From 2.91 the constant G vanishes if and only if the numerator vanishes,

$$\left(n_i^2 z_i - n_o^2 z_o\right)^2 = 0 \quad \Longleftrightarrow \quad n_i^2 z_i = n_o^2 z_o, \tag{2.103}$$

and the very same factor sits in the numerator of S, so this condition forces $G = S = 0$ simultaneously. The surface is then characterised by O and T alone, and 2.98 becomes

$$z(\rho) = \frac{\left(O + T\rho^2\right)\rho^2}{1 + \sqrt{1}} = \frac{1}{2}\left(O + T\rho^2\right)\rho^2. \tag{2.104}$$

These are the **spherovoids**, and the name is justified by what happens if we additionally switch off T. With $T = 0$ we get $z = O\rho^2/2$, and substituting $\rho^2 = z^2 + r^2$,

$$\begin{aligned} &2z = O\left(z^2 + r^2\right) \Longrightarrow z^2 - \frac{2z}{O} + r^2 = 0, \\ &z^2 - \frac{2z}{O} + \frac{1}{O^2} + r^2 = \frac{1}{O^2} \Longrightarrow \left(z - \frac{1}{O}\right)^2 + x^2 + y^2 = \frac{1}{O^2}, \end{aligned}$$

which is a sphere of radius $1/O$ centred at $z = 1/O$, in perfect agreement with the interpretation of O as the curvature at the vertex. So the sphere is the spherovoid with no deformation, and T is precisely the parameter that deforms the sphere into the rest of the family.

2.8.2 Parovoids, $G = -1$

Setting $G = -1$ in 2.91 gives the condition

$$\left(n_i^2 z_i - n_o^2 z_o\right)^2 = -n_i n_o \left(n_i z_i - n_o z_o\right)\left(n_i z_o - n_o z_i\right), \tag{2.105}$$

and since $G = -1$ makes $-O^2 G = +O^2$, the sag reads

$$z(\rho) = \frac{\left(O + T\rho^2\right)\rho^2}{1 + S\rho^2 + \sqrt{1 + \left(2S + O^2\right)\rho^2}}. \tag{2.106}$$

These are the **parovoids**, the ovoidal relatives of the paraboloid, which is recovered when $T = S = 0$ and $K = G = -1$ in 2.101, since then the radical becomes $\sqrt{1}$ and $z = Or^2/2$, the parabola of curvature O.

2.8.3 Hyperovoids, $G < -1$

When

$$\left(n_i^2 z_i - n_o^2 z_o\right)^2 < -n_i n_o \left(n_i z_i - n_o z_o\right)\left(n_i z_o - n_o z_i\right), \tag{2.107}$$

the constant satisfies $G < -1$ and the sag keeps the general form

$$z(\rho) = \frac{\left(O + T\rho^2\right)\rho^2}{1 + S\rho^2 + \sqrt{1 + (2S - O^2G)\,\rho^2}}, \tag{2.108}$$

with $-O^2G > O^2$. These are the **hyperovoids**, the deformed versions of the two sheet hyperboloid we obtained in the section of objects at infinity, and they are the surfaces one uses when the conjugate pair demands an eccentricity larger than one.

2.8.4 Prolate and oblate ellipsovoids

Finally, if the product 2.102 is positive,

$$(n_i z_i - n_o z_o)\,(n_i z_o - n_o z_i) > 0, \tag{2.109}$$

then $G > 0$ and the surface is a **prolate ellipsovoid**, stretched along the optical axis, while in the intermediate window

$$0 > G > -1, \tag{2.110}$$

the surface is an **oblate ellipsovoid**, flattened along the axis. In both cases the sag is given by 2.108, only the sign and the magnitude of G change.

It is illuminating to line up the whole family with the conic constant of the standard surfaces, because the parallel is exact. A conic with $K > 0$ is an oblate ellipsoid, $K = 0$ is a sphere, $-1 < K < 0$ is a prolate ellipsoid, $K = -1$ is a paraboloid and $K < -1$ is a hyperboloid. The ovoids repeat the same ladder with G in the role of K, with the difference that the ovoid carries in addition the deformation T, which is what allows it to be rigorously stigmatic for a *pair of finite points* instead of only for a point and infinity. That extra parameter is exactly the price of bringing the second conjugate point from infinity to a finite distance.

2.9 Descartes ovoids and aspheric surfaces

We have now two different ways of writing surfaces that are better than the sphere, and it is important to understand how they relate to each other, because they answer two different questions.

The rigorously stigmatic surfaces we have built are perfect, but their perfection is extremely local, it holds for one pair of points and for nothing else, with the single exception of the sphere at its aplanatic points, which we will meet again in a moment. If the object is moved away from that privileged position, the image immediately stops being a point, because each ray leaving the object is refracted towards a slightly different place. In practice the position of the object is arbitrary, so a theory that only knows how to be perfect at one point is not enough, and the study of aberrations becomes mandatory.

The standard engineering answer is to give up rigorous stigmatism and instead deform a conic until the residual distortion is small over a whole range of positions. This is done by adding higher order terms in r to the Schwarzschild equation 2.101, called deformation terms,

$$z(r) = \frac{c_0' r^2}{1 + \sqrt{1 - (1 + K')\, c_0'^2 r^2}} + \sum_{j=2}^{\infty} A_{2j} r^{2j}, \tag{2.111}$$

where the numbers A_{2j} are the **asphericity coefficients** and $K' = -e^2$ is the Schwarzschild constant of the base conic. The conic surfaces are then the degenerate case of the aspheric surfaces in which every A_{2j} vanishes.

Here lies the essential difference. The aspheric surfaces 2.111 are *open* surfaces, they are a conic plus a polynomial correction that grows without bound, whereas the Descartes ovoids are *closed* quartic surfaces. They are therefore not equivalent, and an aspheric surface with $A_{2j} \neq 0$ is in general not rigorously stigmatic for any pair of points at all. What it is good at is something else, it is good at flattening the residual wavefront error over an extended field, which is why aspheres are used to correct aberrated wavefronts and to design isoplanatic systems. The Descartes ovoids, on the contrary, are particular cases of the superconics, and they buy exactness at one pair of points at the cost of losing it everywhere else.

Both languages are needed. The ovoid tells us what perfect imaging looks like and gives us the reference against which everything else is measured, and the asphere tells us how to distribute the imperfection when perfection is impossible. The rest of this chapter is devoted to the second question, and we shall attack it in the place where it matters most, the spherical diopter, which is by far the most used surface in optics for the simple historical reason that it is the easiest one to grind by hand.

2.10 Approximate stigmatism in the spherical diopter

Rigorous stigmatism and the aplanatism conditions give us the principles that govern distortion free image formation, but their usefulness is limited, because they can only be applied when the object sits at one very specific position with respect to the diopter. In practice the position of the object is arbitrary, and what we want is to describe what an instrument does under those conditions, where distortions appear that are, in principle, the consequence of the stigmatism being only approximate.

Among all the diopters the spherical one is the one with the largest number of applications, partly for historical reasons, since it is the surface that was easiest to obtain by hand or by artisanal methods, and that is precisely what pushed the study of its distortions.

Let us take a pair of conjugate points O and O' such that the optical path $L(O, O')$ does not depend on the point I where the ray crosses the spherical diopter, see figure 2.27, where we write $d = z_o$ and $d' = z_i$ for the object and image distances measured from the vertex. Putting the vertex at the origin, the centre of curvature sits at $z = R$ and the incidence point is

$$I = (R \sin\varphi,\ 0,\ R\,(1 - \cos\varphi))\,, \tag{2.112}$$

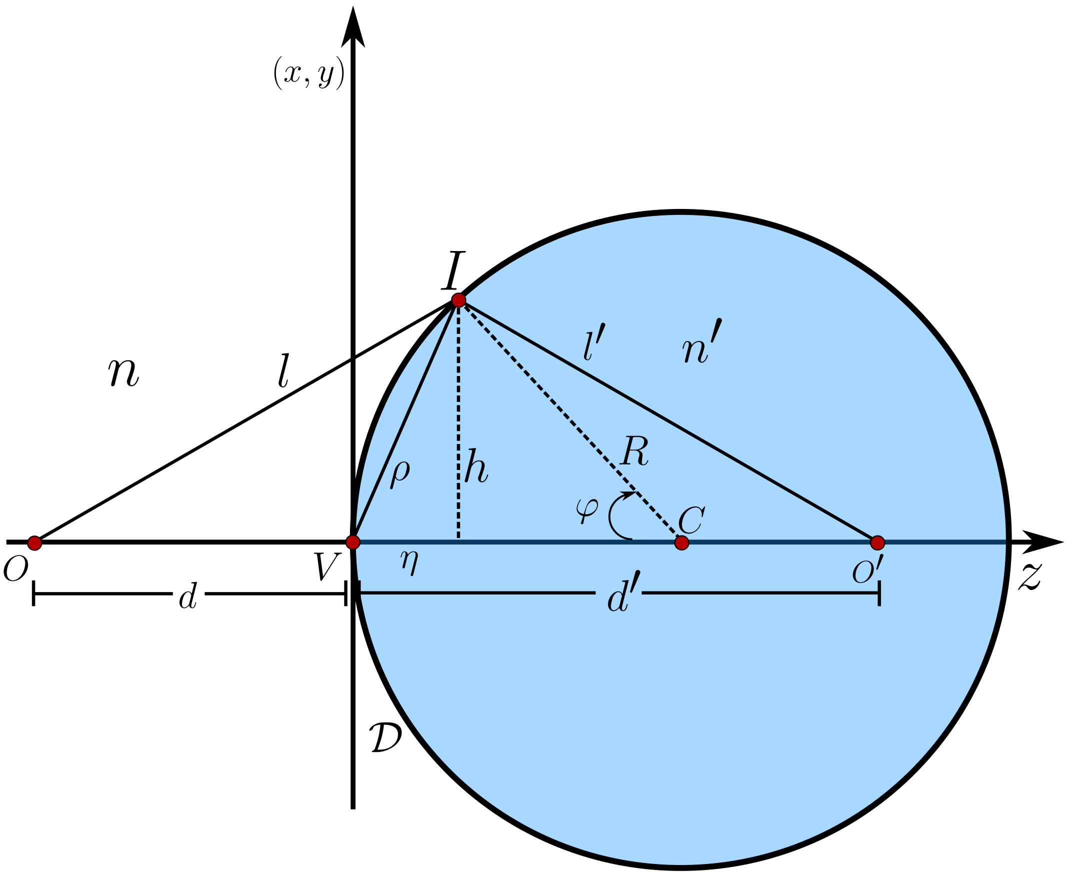


Figure 2.27: Image formation by a spherical diopter $\mathcal{D}$ of radius R, vertex V and centre of curvature C. The object O is at a distance $d = z_o$ from the vertex and the image O' at $d' = z_i$, and the ray reaches the surface at the point I, which is located by the angle φ measured at the centre of curvature. The chord $\rho = \overline{VI}$, the transverse height h and the sagitta η are the three geometrical quantities that will organise the whole calculation.

so that the two segments are

$$l_o^2 = [R\,(1 - \cos\varphi) - z_o]^2 + R^2 \sin^2\varphi =$$
$$R^2\left(1 - 2\cos\varphi + \cos^2\varphi + \sin^2\varphi\right) - 2Rz_o\,(1 - \cos\varphi) + z_o^2,$$

where the two terms in blue add up to one, and therefore

$$l_o = \left[R^2 + (z_o - R)^2 + 2R\,(z_o - R)\cos\varphi\right]^{1/2}, \tag{2.113}$$

$$l_i = \left[R^2 + (z_i - R)^2 + 2R\,(z_i - R)\cos\varphi\right]^{1/2}, \tag{2.114}$$

the second one being obtained in exactly the same way on the image side. Both expressions can be written more compactly as $l_o^2 = z_o^2 + 2R(R - z_o)(1 - \cos\varphi)$ and $l_i^2 = z_i^2 + 2R(R - z_i)(1 - \cos\varphi)$, and in this form it is evident that the whole dependence on the incidence point enters only through $1 - \cos\varphi$.

The optical path is $L(OO') = n_o l_o + n_i l_i$, and if we demand the Fermat's principle for the ray that passes through I, that is, if we demand that small variations of φ produce no first order change in L, we must impose $dL/d\varphi = 0$. Differentiating 2.113 and 2.114,

$$2l_o\frac{dl_o}{d\varphi} = -2R\,(z_o - R)\sin\varphi \Longrightarrow \frac{dl_o}{d\varphi} = \frac{R\,(R - z_o)}{l_o}\sin\varphi,$$

and identically for l_i, so the stationarity condition reads

$$n_o\frac{R\,(z_o - R)}{l_o}\sin\varphi + n_i\frac{R\,(z_i - R)}{l_i}\sin\varphi = 0, \tag{2.115}$$

where we have changed the global sign, which is harmless since the right hand side is zero. Dividing by $R\sin\varphi$, which is legitimate for any ray that is not the axial one, and separating the terms,

$$\begin{aligned}\frac{n_o\,(z_o - R)}{l_o} + \frac{n_i\,(z_i - R)}{l_i} &= 0,\\ \frac{n_o z_o}{l_o} - \frac{n_o R}{l_o} + \frac{n_i z_i}{l_i} - \frac{n_i R}{l_i} &= 0,\end{aligned}$$

so that we finally obtain the exact conjugation relation of the spherical diopter,

$$\frac{n_i}{l_i} + \frac{n_o}{l_o} = \frac{1}{R}\left[n_i\frac{z_i}{l_i} + n_o\frac{z_o}{l_o}\right]. \tag{2.116}$$

It is very important to be clear about what 2.116 does and does not say. We have not demanded $L(O,O') = cte$ for every ray, which is what rigorous stigmatism would require, we have only demanded that the Fermat's principle hold for the ray that goes through the particular point I, that is, that for a given angle φ small variations $\Delta\varphi$ produce negligible variations of $L(O,O')$. Therefore O and O' are not necessarily the Young-Weierstrass points. From here on we shall no longer speak of rigorous stigmatism but of approximate stigmatism, aplanatism will be replaced by the weaker notion of isoplanatism, and in general no perfect image is formed at all.

2.10.1 Gauss optics, the paraxial approximation

An enormous simplification is obtained if we restrict the rays that are allowed through the system so that $\sin\varphi \approx \varphi$ and $\cos\varphi \approx 1$. This is the paraxial or Gaussian approximation, and geometrically it means that we keep only the rays that hug the optical axis, so that the diopter is replaced by its tangent plane as far as the distances are concerned. In this regime $l_o \approx -z_o$, the minus sign coming from our convention since $z_o < 0$ while l_o is a positive length, and $l_i \approx z_i$. Substituting into 2.116,

$$\frac{n_i}{z_i} + \frac{n_o}{-z_o} = \frac{1}{R}\left[n_i\frac{z_i}{z_i} + n_o\frac{z_o}{-z_o}\right] = \frac{n_i - n_o}{R},$$

that is,

$$\frac{n_i}{z_i} - \frac{n_o}{z_o} = \frac{n_i - n_o}{R}, \tag{2.117}$$

which is the conjugation formula of the spherical diopter in the paraxial approximation, and we say that O and O' are conjugate points, O' being the paraxial image of O. The right hand side depends only on the surface and the media, never on the position of the object, so it deserves a name of its own, the **vergence**

$$V = \frac{n_i - n_o}{R}. \tag{2.118}$$

The focal distances are defined by sending either the object or the image to infinity. Taking $z_i \to \infty$ in 2.117 the left hand side keeps only the object term and we obtain the object focus, and taking $z_o \to \infty$ we obtain the image focus,

$$f_o = \frac{n_o R}{n_o - n_i}, \tag{2.119}$$

$$f_i = \frac{n_i R}{n_i - n_o}. \tag{2.120}$$

Notice that f_o and f_i are not equal unless $n_o = n_i$, and that their ratio is $f_i/f_o = -n_i/n_o$. This asymmetry, which does not exist for a mirror, is the geometrical signature of the fact that the light travels at different speeds on the two sides of the surface.

Under these conditions the optical path is approximately independent of the point I, and this is why we speak of approximate stigmatism. The image is formed with distortions which on one hand are minimal, and on the other hand are invariant under a translation in a transverse plane, and this last property deserves its own name.

Definition 8 *(Isoplanatism) An approximately stigmatic system whose distortions remain invariant under a transverse translation of the object is called* ***isoplanatic****.*

The physical content of isoplanatism is worth spelling out, because it is the bridge between geometrical optics and the Fourier optics we will need much later. If the defects of the instrument are the same at every point of the field, then the instrument acts on the object as a linear and space invariant filter, and the image is the convolution of the object with a single impulse response. If the defects change from point to point the system stops being a convolution and the whole machinery of transfer functions collapses. So isoplanatism is exactly the condition under which an optical instrument can be treated as a linear filter.

By restricting the rays that are allowed through the system, then, we have managed to impose conditions under which the spherical diopter is approximately stigmatic even away from its Young-Weierstrass points. The same approximations can be taken for the hyperboloidal and ellipsoidal diopters, and for the reflective surfaces, with no change in the reasoning.

2.10.2 Spherical aberration

Paraxial optics is a beautiful lie. It is exact only in the limit of a vanishingly thin pencil of rays around the axis, and any real instrument has a finite aperture. So let us now allow the rays that depart moderately from the paraxial condition to also pass through the system, which means we must increase the precision of the approximations and keep the first correction to the paraxial result.

The first step is purely geometrical. From the figure 2.27 we read $\cos\varphi = (R-\eta)/R = 1-\eta/R$, where η is the sagitta, the axial distance from the vertex to the foot of the incidence point. Since $\rho^2 = \eta^2 + h^2$ and $R^2 = (R-\eta)^2 + h^2$, subtracting the second from the first,

$$\rho^2 - R^2 = \eta^2 - (R-\eta)^2 = \eta^2 - R^2 + 2R\eta - \eta^2 = 2R\eta - R^2,$$
$$\rho^2 = 2R\eta \Longrightarrow \frac{\eta}{R} = \frac{\rho^2}{2R^2},$$

and therefore

$$\frac{\rho^2}{2R^2} = 1 - \cos\varphi. \tag{2.121}$$

This little relation is the key of the entire section, because it converts the angular variable φ, which is how the geometry is naturally written, into the aperture variable ρ, which is how the aberrations are naturally measured. Substituting 2.121 into the compact form of 2.113,

$$l_o^2 = z_o^2 + 2R^2(1-\cos\varphi) - 2Rz_o(1-\cos\varphi) = z_o^2 + 2R^2\frac{\rho^2}{2R^2} - 2Rz_o\frac{\rho^2}{2RR},$$
$$l_o^2 = z_o^2 + \rho^2 - \frac{z_o\rho^2}{R} = z_o^2\left[1+\frac{\rho^2}{z_o}\left(\frac{1}{z_o}-\frac{1}{R}\right)\right],$$

so that, remembering $\sqrt{z_o^2} = -z_o$,

$$l_o = -z_o\left[1+\frac{\rho^2}{z_o}\left(\frac{1}{z_o}-\frac{1}{R}\right)\right]^{1/2}, \tag{2.122}$$
$$l_i = z_i\left[1+\frac{\rho^2}{z_i}\left(\frac{1}{z_i}-\frac{1}{R}\right)\right]^{1/2}. \tag{2.123}$$

These two expressions are still exact, no approximation has been made yet, we have merely traded φ for ρ. Substituting them into the exact conjugation 2.116 and using $n_o/l_o = -(n_o/z_o)[\cdots]^{-1/2}$,

$$\frac{n_i}{z_i}\left[1+\frac{\rho^2}{z_i}\left(\frac{1}{z_i}-\frac{1}{R}\right)\right]^{-1/2} - \frac{n_o}{z_o}\left[1+\frac{\rho^2}{z_o}\left(\frac{1}{z_o}-\frac{1}{R}\right)\right]^{-1/2} =$$
$$\frac{1}{R}\left\{n_i\left[1+\frac{\rho^2}{z_i}\left(\frac{1}{z_i}-\frac{1}{R}\right)\right]^{-1/2} - n_o\left[1+\frac{\rho^2}{z_o}\left(\frac{1}{z_o}-\frac{1}{R}\right)\right]^{-1/2}\right\}. \tag{2.124}$$

Now, and only now, we approximate. For $\rho \ll |z_o|$, $\rho \ll z_i$ and $\rho \ll |R|$ the quantity inside the brackets is close to one, and the binomial expansion $(1+x)^{-1/2} \approx 1 - x/2$ gives

$$\frac{1}{l_o} = -\frac{1}{z_o}\left[1 + \frac{\rho^2}{z_o}\left(\frac{1}{z_o} - \frac{1}{R}\right)\right]^{-1/2} \approx -\frac{1}{z_o}\left[1 - \frac{\rho^2}{2z_o}\left(\frac{1}{z_o} - \frac{1}{R}\right)\right], \tag{2.125}$$

$$\frac{1}{l_i} = \frac{1}{z_i}\left[1 + \frac{\rho^2}{z_i}\left(\frac{1}{z_i} - \frac{1}{R}\right)\right]^{-1/2} \approx \frac{1}{z_i}\left[1 - \frac{\rho^2}{2z_i}\left(\frac{1}{z_i} - \frac{1}{R}\right)\right], \tag{2.126}$$

where the improvement over the Gauss optics is transparent, in the paraxial case we simply put $1/l_o = -1/z_o$ and here we keep the first correction in ρ^2. Substituting into 2.124,

$$\begin{aligned} &\frac{n_i}{z_i}\left[1 - \frac{\rho^2}{2z_i}\left(\frac{1}{z_i} - \frac{1}{R}\right)\right] - \frac{n_o}{z_o}\left[1 - \frac{\rho^2}{2z_o}\left(\frac{1}{z_o} - \frac{1}{R}\right)\right] = \\ &\frac{1}{R}\left\{n_i\left[1 - \frac{\rho^2}{2z_i}\left(\frac{1}{z_i} - \frac{1}{R}\right)\right] - n_o\left[1 - \frac{\rho^2}{2z_o}\left(\frac{1}{z_o} - \frac{1}{R}\right)\right]\right\}, \end{aligned} \tag{2.127}$$

and expanding both sides and moving everything that does not contain ρ^2 to the left,

$$\begin{aligned} \frac{n_i}{z_i} - \frac{n_o}{z_o} = \frac{n_i - n_o}{R} - \frac{1}{R}\frac{n_i\rho^2}{2z_i}\left(\frac{1}{z_i} - \frac{1}{R}\right) + \frac{1}{R}\frac{n_o\rho^2}{2z_o}\left(\frac{1}{z_o} - \frac{1}{R}\right) \\ + \frac{n_i}{z_i}\frac{\rho^2}{2z_i}\left(\frac{1}{z_i} - \frac{1}{R}\right) - \frac{n_o}{z_o}\frac{\rho^2}{2z_o}\left(\frac{1}{z_o} - \frac{1}{R}\right). \end{aligned} \tag{2.128}$$

The four terms in ρ^2 pair up two by two, the first with the third and the second with the fourth, and each pair produces the same binomial squared,

$$\begin{aligned} \frac{n_i\rho^2}{2z_i}\left(\frac{1}{z_i} - \frac{1}{R}\right)\left[\frac{1}{z_i} - \frac{1}{R}\right] &= \frac{n_i\rho^2}{2z_i}\left(\frac{1}{z_i} - \frac{1}{R}\right)^2, \\ -\frac{n_o\rho^2}{2z_o}\left(\frac{1}{z_o} - \frac{1}{R}\right)\left[\frac{1}{z_o} - \frac{1}{R}\right] &= -\frac{n_o\rho^2}{2z_o}\left(\frac{1}{z_o} - \frac{1}{R}\right)^2, \end{aligned}$$

so that the whole calculation collapses into

$$\frac{n_i}{z_i} - \frac{n_o}{z_o} = \frac{n_i - n_o}{R} + \rho^2\left[\frac{n_i}{2z_i}\left(\frac{1}{z_i} - \frac{1}{R}\right)^2 - \frac{n_o}{2z_o}\left(\frac{1}{z_o} - \frac{1}{R}\right)^2\right]. \tag{2.129}$$

Equation 2.129 is the conjugation formula of the spherical diopter corrected to first order in the aperture. It is the paraxial formula 2.117 plus one extra term proportional to ρ^2, and that extra term is the **spherical aberration**. Its meaning is direct, since the right hand side now depends on ρ, the position z_i where the ray crosses the axis depends on how far from the vertex the ray struck the surface. Rays through different zones of the aperture cross the axis at different points, and there is no single image point at all.

The same relation can be rearranged by putting each medium on its own side,

$$n_i\left[\frac{1}{z_i}-\frac{1}{R}-\frac{\rho^2}{2z_i}\left(\frac{1}{z_i}-\frac{1}{R}\right)^2\right]=n_o\left[\frac{1}{z_o}-\frac{1}{R}-\frac{\rho^2}{2z_o}\left(\frac{1}{z_o}-\frac{1}{R}\right)^2\right], \tag{2.130}$$

which says that a certain quantity has the same value on both sides of the diopter, and this is what we call a metaxial invariant in the presence of aberrations.

Definition 9 *(Marginal invariant) For a spherical refracting surface we define*

$$\mathcal{M}=n\left[\frac{1}{z}-\frac{1}{R}-\frac{\rho^2}{2z}\left(\frac{1}{z}-\frac{1}{R}\right)^2\right], \tag{2.131}$$

which takes the primary aberrations into account, and which for $\rho \ll |z|$ and $\rho \ll |R|$ reduces to the Abbe invariant of the paraxial optics

$$\mathcal{A}=n\left(\frac{1}{z}-\frac{1}{R}\right). \tag{2.132}$$

2.10.2.1 The wavefront aberration and the quartic wavefront

So far we have described the aberration as a defect in the ray positions. There is a second and much more powerful description, in terms of the shape of the wavefront that leaves the surface, and it is the one that will connect this chapter with the physical optics of the following ones. Following Hopkins (Hopkins, 1950), we define the **wave front aberration** as the residual phase

$$W(\vec{\rho})=\frac{1}{2}\left[\frac{n_i}{2z_i}\left(\frac{1}{z_i}-\frac{1}{R}\right)^2-\frac{n_o}{2z_o}\left(\frac{1}{z_o}-\frac{1}{R}\right)^2\right]\rho^4=C_{40}\rho^4, \tag{2.133}$$

where C_{40} is the quartic aberration coefficient for a spherical wavefront passing through a spherical refracting surface. The bracket is exactly one half of the aberration bracket of 2.129, which is the usual statement that the transverse ray aberration is the gradient of the wavefront aberration.

The geometry behind 2.133 is shown in the figure 2.28, and it is worth reading carefully. *A spherical refracting surface converts a spherical wavefront into a quartic wavefront*, and that is the whole origin of the geometric aberrations. The marginal rays converge at a point A' instead of the point A_p where the paraxial rays arrive, because the quartic surface Q that emerges from the diopter moves away from the Gaussian spherical surface of reference $\mathcal{A}'$ by the amount $W(\vec{\rho})$. The rays, being normal to the wavefront, no longer meet at a single point but are tangent to an envelope, the **caustic**, which is the bright cusped curve one sees at the bottom of a cup of coffee.

This gives us a very clean way of restating everything we did in the first half of the chapter. When a point source is placed at a strictly stigmatic point of a refracting quartic surface, that is, of a Descartes ovoid, the emerging wavefront is a perfect spherical wave converging to the conjugate

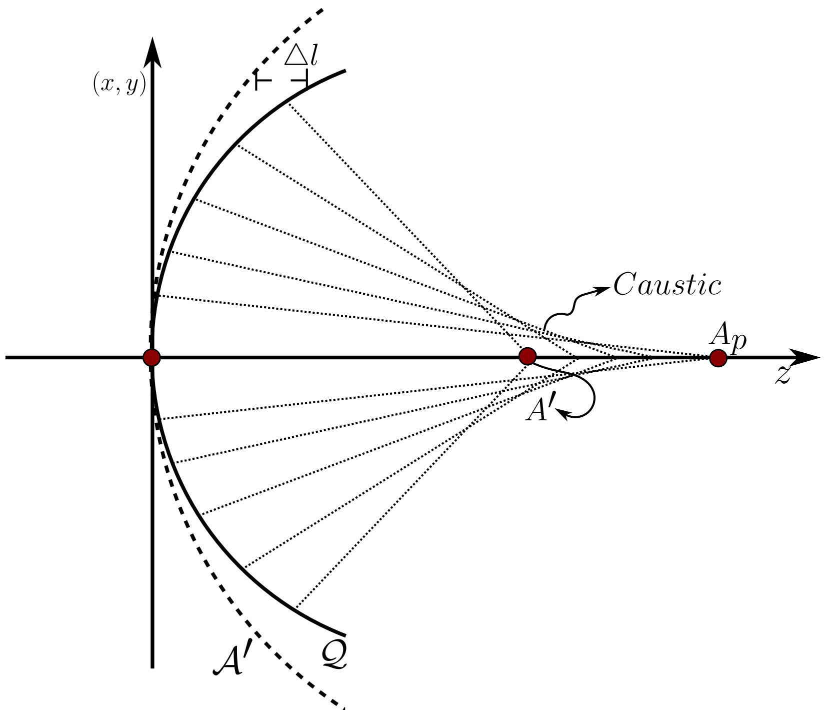


Figure 2.28: Quartic surface Q of the wavefront emerging from a spherical refracting surface. The marginal rays converge at A' instead of the paraxial point A_p, because the quartic wavefront Q departs from the Gaussian spherical reference $\mathcal{A}'$ by the amount $\Delta l = W(\vec{\rho})$, and the envelope of the refracted rays forms the caustic.

point. Those points are in general unique for a given quartic surface, and if the object is removed from them the emerging wavefront ceases to be spherical and becomes a quartic phase wavefront. The sphere is a particular case of the Descartes ovoids where the strictly stigmatic pair are the Young-Weierstrass points, and therefore any object placed anywhere else gives rise to a wavefront with quartic geometry and, by the above, to the spherical geometric aberrations.

2.10.2.2 The Seidel aberrations

Up to now the object has been on the axis. If we let it move off axis the aberration function acquires a much richer structure, and this is where the classical five aberrations appear. Let $\vec{A}$ be the vector from the vertex V to the reference point V_0 on the axis, let $\vec{r}$ be the position on the surface measured from V_0, and let $\vec{\rho} = \vec{r} + \vec{A}$ as before. Calling $a = |\vec{A}|$ and θ the angle between $\vec{A}$ and $\vec{r}$, the cosine rule gives $\rho^2 = a^2 + r^2 + 2ar\cos\theta$, and the aberration function measured relative to the axial point is

$$\begin{aligned} W &= -C_{40}\left(\rho^4 - a^4\right) \\ &= -C_{40}\left(r^4 + 4ar^3\cos\theta + 4a^2r^2\cos^2\theta + 2a^2r^2 + 4a^3r\cos\theta\right), \end{aligned} \tag{2.134}$$

where the a^4 generated by squaring the cosine rule has cancelled against the a^4 we subtracted. Since the triangles V_0VC and $A'B'C$ are similar, a is proportional to the transverse image size $\sigma' = \overline{A'B'}$, that is $a = g\sigma'$ with $g = R/(z_i - R)$, and substituting we may write

$$W\left(r,\theta,\sigma'\right) = {}_0C_{40}r^4 + {}_1C_{31}\sigma' r^3\cos\theta + {}_2C_{22}\sigma'^2r^2\cos^2\theta + {}_2C_{20}\sigma'^2r^2 + {}_3C_{11}\sigma'^3r\cos\theta, \tag{2.135}$$

where the coefficient ${}_aC_{bc}$ carries the powers of σ'^a, r^b and $\cos^c\theta$ respectively. These five terms are the monochromatic or **Seidel aberrations** (Welford, 1986), and each one deserves a word on what it actually looks like and where an instrument builder runs into it,

1. **Spherical aberration**, r^4. This is the only term with no power of σ' at all, so it survives even for a point on the axis, exactly the effect we derived earlier in this section, rays through different zones of the aperture come to a focus at different points along the axis. It blurs a point into a small disc surrounded by a soft halo, and it is why fast lenses, telescopes and camera objectives with a large aperture need aspheric surfaces or multi-element corrections to stay sharp.

2. **Coma**, $\sigma' r^3\cos\theta$. One power of the field and three of the aperture means it grows quickly as we move off axis, and it is the aberration that turns a point image into the comet-shaped smear that gives it its name, because rays passing through different annular zones of the aperture land at different transverse positions and with different magnifications. It is the dominant defect of fast, wide field instruments such as Schmidt cameras, and it is precisely what the Abbe sine condition of section 2.11.1 is designed to eliminate.

3. **Astigmatism**, $\sigma'^2r^2\cos^2\theta$. Here an off-axis point does not focus to a point at all, but to two separate, mutually perpendicular line images at two different distances along the axis, the tangential and the sagittal foci, because a tilted pencil of rays sees an effectively different curvature of the surface in the two perpendicular planes. It is the same astigmatism that a badly shaped cornea produces in the human eye.

4. **Petzval field curvature**, σ'^2r^2. This term carries no dependence on θ, so it does not blur the image at all, it simply relocates where the sharpest focus falls. The consequence is that even a system free of every other aberration forms its image on a curved surface, the Petzval surface, rather than on a plane, so a flat detector such as film or a sensor can only be in focus over a small region of the field while the edges turn soft. This is why photographic objectives use dedicated field-flattening elements.

5. **Distortion**, $\sigma'^3r\cos\theta$. Being linear in r, this term displaces the image point transversally without spreading it out at all, the image stays perfectly sharp but the magnification changes

with field position. This is the origin of the familiar barrel and pincushion warping of straight lines seen near the edges of wide angle photographs, an effect with no blur, only geometric displacement.

The structure of 2.135 is very instructive. The first term does not contain σ' at all, which is why spherical aberration survives even for an object on the axis, and it is the only one of the five that does. The last term is linear in r, which means that it displaces the image without blurring it, and this is why distortion changes the shape of a picture but keeps it sharp. The three terms in between mix aperture and field, and they are the ones that make an instrument that is perfect at the centre of the field become useless at its edge.

2.10.2.3 Aberrations close to the paraxial condition

In practice it is convenient to have the aberration term of 2.129 written in terms of only one of the two conjugate distances, and this will be essential when we compose two surfaces into a lens. Since our solution is close to the paraxial condition, we may use the paraxial formula 2.117 inside the aberration term, which is already a small correction, so that the error we commit is of higher order.

The key identity is the following. From 2.117 we have $1/z_i = (1/n_i)\left[n_o/z_o + (n_i - n_o)/R\right]$, and therefore

$$\frac{1}{z_i} - \frac{1}{R} = \frac{1}{n_i}\left[\frac{n_o}{z_o} + \frac{n_i - n_o}{R}\right] - \frac{1}{R} = \frac{1}{n_i}\left[\frac{n_o}{z_o} + \frac{n_i - n_o}{R} - \frac{n_i}{R}\right] = \frac{n_o}{n_i}\left(\frac{1}{z_o} - \frac{1}{R}\right),$$

a remarkably clean relation which says that the Abbe invariants on the two sides are proportional. Writing $u = 1/z_o - 1/R$ for brevity, the aberration bracket becomes

$$\frac{n_i}{2z_i}\left(\frac{n_o}{n_i}\right)^2 u^2 - \frac{n_o}{2z_o}u^2 = \frac{u^2}{2}\left[\frac{n_o^2}{n_i z_i} - \frac{n_o}{z_o}\right] = \frac{u^2}{2}\frac{n_o^2}{n_i^2}\left[\frac{n_o}{z_o} + \frac{n_i - n_o}{R}\right] - \frac{u^2}{2}\frac{n_o}{z_o},$$

where in the last step we used the paraxial formula once more to eliminate $1/z_i$. Factoring n_o/n_i^2,

$$\frac{u^2}{2}\frac{n_o}{n_i^2}\left[\frac{n_o^2}{z_o} + \frac{n_o\left(n_i - n_o\right)}{R} - \frac{n_i^2}{z_o}\right] = \frac{u^2}{2}\frac{n_o}{n_i^2}\left[\frac{n_o^2 - n_i^2}{z_o} + \frac{n_o\left(n_i - n_o\right)}{R}\right] =$$
$$\frac{u^2}{2}\frac{n_o}{n_i^2}\left(n_i - n_o\right)\left[\frac{n_o}{R} - \frac{n_i + n_o}{z_o}\right],$$

and rewriting the prefactor as $n_o/n_i^2 = \left(n_o/n_i\right)^2/n_o$ we obtain the identity we wanted,

$$\frac{n_i}{2z_i}\left(\frac{1}{z_i} - \frac{1}{R}\right)^2 - \frac{n_o}{2z_o}\left(\frac{1}{z_o} - \frac{1}{R}\right)^2 \approx \frac{1}{2}\left(\frac{n_o}{n_i}\right)^2\left(\frac{1}{z_o} - \frac{1}{R}\right)^2\left(\frac{n_i - n_o}{n_o}\right)\left(\frac{n_o}{R} - \frac{n_i + n_o}{z_o}\right), \quad (2.136)$$

which is the Airy formula for the aberration at refraction by a single surface. Therefore the conjugation formula written in terms of the object distance alone is

$$\frac{n_i}{z_i} - \frac{n_o}{z_o} = \frac{n_i - n_o}{R} + \rho^2\frac{1}{2}\left(\frac{n_o}{n_i}\right)^2\left(\frac{1}{z_o} - \frac{1}{R}\right)^2\left(\frac{n_i - n_o}{n_o}\right)\left(\frac{n_o}{R} - \frac{n_i + n_o}{z_o}\right) = V + \Delta V_+, \quad (2.137)$$

and repeating the whole argument with the roles of the two media exchanged, that is, using $1/z_o - 1/R = (n_i/n_o)\,(1/z_i - 1/R)$, the same expression written in terms of the image distance alone is

$$\frac{n_i}{z_i} - \frac{n_o}{z_o} = \frac{n_i - n_o}{R} - \rho^2 \frac{1}{2}\left(\frac{n_i}{n_o}\right)^2 \left(\frac{1}{z_i} - \frac{1}{R}\right)^2 \left(\frac{n_o - n_i}{n_i}\right)\left(\frac{n_i}{R} - \frac{n_i + n_o}{z_i}\right) = V - \Delta V_-, \quad (2.138)$$

where V is the vergence 2.118 and $\Delta V_\pm$ is the change in the vergence produced by the marginal rays.

It is very convenient to absorb these corrections into an effective radius. Defining

$$\frac{1}{R_+} = \frac{\rho^2}{2n_o}\left(\frac{n_o}{n_i}\right)^2 \left(\frac{1}{z_o} - \frac{1}{R}\right)^2 \left(\frac{n_o}{R} - \frac{n_i + n_o}{z_o}\right), \quad (2.139)$$

$$\frac{1}{R_-} = \frac{\rho^2}{2n_i}\left(\frac{n_i}{n_o}\right)^2 \left(\frac{1}{z_i} - \frac{1}{R}\right)^2 \left(\frac{n_i}{R} - \frac{n_i + n_o}{z_i}\right), \quad (2.140)$$

both 2.137 and 2.138 collapse into the single compact statement

$$\frac{n_i}{z_i} - \frac{n_o}{z_o} = (n_i - n_o)\left(\frac{1}{R} + \frac{1}{R_\pm}\right), \quad (2.141)$$

and we define the **marginal rays vergence** as $V_\pm = (n_i - n_o)\,(1/R + 1/R_\pm)$.

This is the cleanest way of understanding what the marginal rays do. *To include the marginal rays in the study of the spherical refracting surface, all we have to do is to add to the inverse of the radius the term* $1/R_\pm$, according to whether we are working in object or in image coordinates. The surface behaves, for those rays, as if it had a different radius of curvature, and since $R_\pm$ depends on ρ^2, each zone of the aperture sees a different sphere. That is spherical aberration in one sentence.

2.10.3 Focal length for marginal rays

Since the aberration term depends on the distance ρ at which the rays cross the surface, different rays are deviated differently and consequently the system does not have a single focal length, but a value that depends on ρ for each ray. Let us compute it.

We take 2.137 in the form $1/z_i = n_o/(n_i z_o) + (n_i - n_o)/(n_i R) + \rho^2 \Delta V_+/n_i$ and send the object to infinity, $z_o \to \infty$, which is the situation of a telescope pointed at a star. In that limit the first term vanishes, $1/z_o \to 0$ inside the squared bracket so that it becomes $1/R^2$, and the last parenthesis becomes n_o/R, hence

$$\frac{1}{f_{M_+}} = \frac{n_i - n_o}{n_i R} + \frac{\rho^2}{n_i}\frac{1}{2}\left(\frac{n_o}{n_i}\right)^2 \frac{1}{R^2}\left(\frac{n_i - n_o}{n_o}\right)\frac{n_o}{R} = \frac{n_i - n_o}{n_i R} + \frac{\rho^2 n_o^2\,(n_i - n_o)}{2 n_i^3 R^3},$$

and factoring the paraxial piece,

$$\frac{1}{f_{M_+}} = \frac{n_i - n_o}{n_i R}\left[1 + \frac{\rho^2}{2R^2}\left(\frac{n_o}{n_i}\right)^2\right], \quad (2.142)$$

or, using 2.121 to go back to the angular variable,

$$\frac{1}{f_+} = \frac{n_i - n_o}{n_i R}\left[1 + \left(\frac{n_o}{n_i}\right)^2 - \left(\frac{n_o}{n_i}\right)^2 \cos\varphi\right]. \tag{2.143}$$

We can now read off the two limits. The focus depends on φ, so we write $f_+(\varphi)$, and the paraxial focus is recovered as $f_{P_+} = \lim_{\varphi\to 0} f_+(\varphi)$,

$$f_{P_+} = \frac{n_i R}{n_i - n_o}, \tag{2.144}$$

in agreement with 2.120, as it must be. For the marginal rays we adopt the reasonable working condition $\varphi \leq \pi/3$, which corresponds to $\rho \leq |R|$, and taking $f_{M_+} = \lim_{\varphi\to\pi/3} f_+(\varphi)$, where $\cos\varphi = 1/2$ and therefore $1 - \cos\varphi = 1/2$,

$$\frac{1}{f_{M_+}} = \frac{1}{f_{P_+}}\left[1 + \frac{1}{2}\left(\frac{n_o}{n_i}\right)^2\right] = \frac{1}{f_{P_+}}\frac{2n_i^2 + n_o^2}{2n_i^2},$$

that is,

$$f_{M_+} = \frac{2n_i^2}{2n_i^2 + n_o^2} f_{P_+}. \tag{2.145}$$

Since the fraction is smaller than one, the marginal focus always falls *short* of the paraxial focus, which is the well known fact that the outer zones of a positive spherical surface focus too soon. The gap between the two is called the **longitudinal spherical aberration**, $\Delta f_+ = f_{P_+} - f_{M_+}$, and it is given by

$$\Delta f_+ = f_{P_+}\left[1 - \frac{2n_i^2}{2n_i^2 + n_o^2}\right] = \frac{n_o^2}{2n_i^2 + n_o^2} f_{P_+}, \tag{2.146}$$

which is an explicit closed expression for the longitudinal spherical aberration of a spherical refracting surface, see figure 2.29.

It is worth noticing how much physics is packed into 2.146. The aberration is proportional to the paraxial focal length, so a fast surface with a short focal length is not automatically better, and it depends only on the ratio of the two indices, so the only way to reduce it while keeping the surface spherical is to increase n_i with respect to n_o. For a glass of $n_i = 1.5$ in air the ratio $n_o^2/(2n_i^2 + n_o^2)$ is about 0.18, which means that the marginal focus sits almost a fifth of the focal length in front of the paraxial one, a completely unacceptable amount for any serious instrument. This single number explains why the whole history of optical design is the history of fighting spherical aberration.

2.10.3.1 Longitudinal object defocus

The same calculation can be run on the object side. Taking 2.138 and sending the image to infinity, $z_i \to \infty$, the squared bracket becomes $1/R^2$ and the last parenthesis becomes n_i/R, so that

$$\frac{1}{f_-} = \frac{n_o - n_i}{n_o R}\left[1 + \frac{\rho^2}{2R^2}\left(\frac{n_i}{n_o}\right)^2\right], \tag{2.147}$$

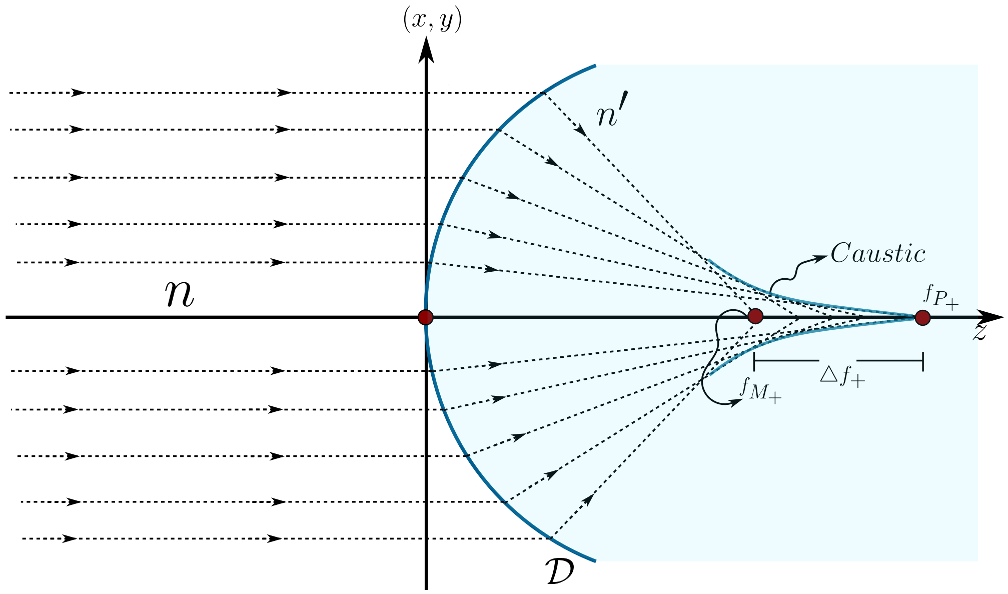


Figure 2.29: Illustration of the longitudinal spherical aberration. The rays arriving parallel to the axis from an object at infinity are focused at f_{M_+} if they are marginal and at f_{P_+} if they are paraxial, and the separation between the two is Δf_+. The envelope of the refracted rays is the caustic.

or in terms of the angle,

$$\frac{1}{f_-} = \frac{n_o - n_i}{n_o R}\left[1 + \left(\frac{n_i}{n_o}\right)^2 - \left(\frac{n_i}{n_o}\right)^2 \cos\varphi\right]. \tag{2.148}$$

Exactly as before, the paraxial object focus is $f_{P_-} = \lim_{\varphi\to 0} f_-(\varphi)$ and the marginal one is $f_{M_-} = \lim_{\varphi\to\pi/3} f_-(\varphi)$,

$$f_{P_-} = \frac{n_o R}{n_o - n_i}, \tag{2.149}$$

$$f_{M_-} = \frac{2n_o^2}{2n_o^2 + n_i^2} f_{P_-}, \tag{2.150}$$

and we define the **longitudinal object defocus** by $\Delta f_- = f_{M_-} - f_{P_-}$, so that

$$\Delta f_- = -\frac{n_i^2}{2n_o^2 + n_i^2} f_{P_-}, \tag{2.151}$$

see figure 2.30.

The interest of this second definition is justified when we consider compound optical systems, as in the case of a lens, because there the object space of one subsystem is the image space of the previous one, so that speaking of the aberrations in the object takes on a very concrete meaning.

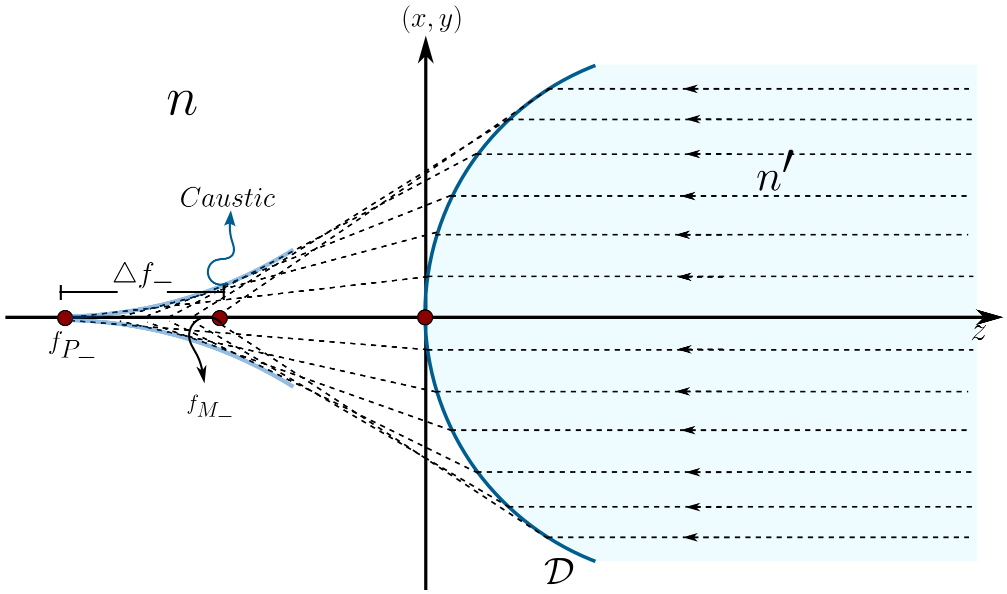


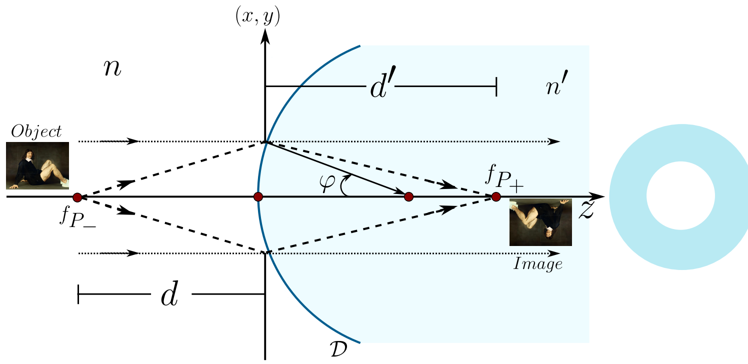
Figure 2.30: Illustration of the defocus in the object space. The rays that emerge parallel to the axis in the image space come, for the marginal zone, from f_{M-} and, for the paraxial zone, from f_{P-}, and the separation between the two is Δf_-.

2.10.3.2 Image formation in the marginal ray regime

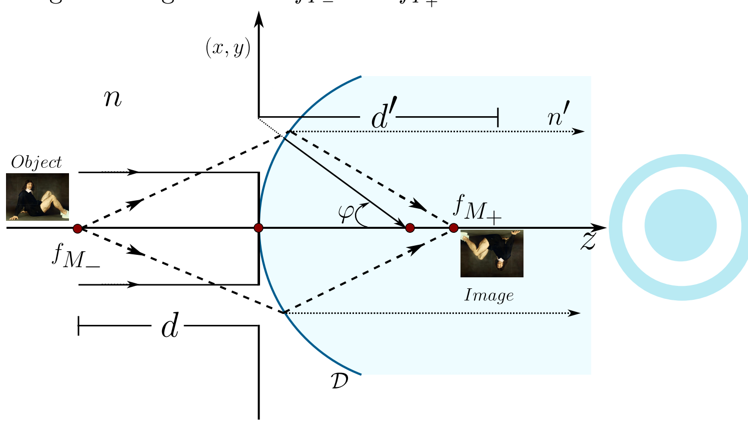
Consequently the marginal and the paraxial rays have different focal lengths, so their images are formed at different distances and with different magnifications, and the superposition of the two is precisely what deteriorates the final composite image.

This suggests a way out which is as simple as it is elegant, shown in the figure 2.31. The optical system could form an image of the same quality with the marginal rays as with the paraxial ones if the two sets of rays were *separated* instead of superposed, which is done by putting a pupil that selects only one annular zone of the aperture. For this we take $\Delta\varphi$ as small as necessary so that the Fermat's principle 2.116 is fulfilled separately in each pupil, around some angle φ, and in this way each zone has a well defined focal length, both in the object and in the image space.

The physical lesson is that spherical aberration is not a defect of any individual ray, every ray obeys the Snell-Descartes law perfectly, it is a defect of the *superposition* of rays coming from zones with different focal lengths. Break the superposition and the blur disappears, at the price of throwing away most of the light. This is exactly the trade off that a photographer makes when stopping down a lens, and it is also the reason why the resolution of an instrument cannot be improved indefinitely by closing the aperture, because at some point the diffraction we shall study in a later chapter takes over.



(a) With paraxial rays, selected by a central pupil, the object is imaged through the foci f_{P_-} and f_{P_+}.



(b) With marginal rays, selected by an annular pupil, the same object is imaged through f_{M_-} and f_{M_+}.

Figure 2.31: Image formation systems. Each family of rays alone forms a sharp image, and it is only their superposition that produces the aberrated one.

2.10.4 The thin lens with aberrations

A lens is two spherical diopters back to back, so everything we have done can be composed. Let the first surface have radius R_1 and form, from the object at z_o in the medium n_o, an intermediate image at d_1 inside the glass of index n_i, and let the second surface of radius R_2 take that intermediate image, now sitting at d_2, and form the final image at z_i back in the medium n_o. Using 2.137 for the first surface and 2.138 for the second,

$$\frac{n_i}{d_1} - \frac{n_o}{z_o} = \frac{n_i - n_o}{R_1} + \rho_1^2 \frac{1}{2}\left(\frac{n_o}{n_i}\right)^2 \left(\frac{1}{z_o} - \frac{1}{R_1}\right)^2 \left(\frac{n_i - n_o}{n_o}\right)\left(\frac{n_o}{R_1} - \frac{n_i + n_o}{z_o}\right), \tag{2.152}$$

$$\frac{n_o}{z_i} - \frac{n_i}{d_2} = \frac{n_o - n_i}{R_2} - \rho_2^2 \frac{1}{2}\left(\frac{n_o}{n_i}\right)^2 \left(\frac{1}{z_i} - \frac{1}{R_2}\right)^2 \left(\frac{n_i - n_o}{n_o}\right)\left(\frac{n_o}{R_2} - \frac{n_i + n_o}{z_i}\right). \tag{2.153}$$

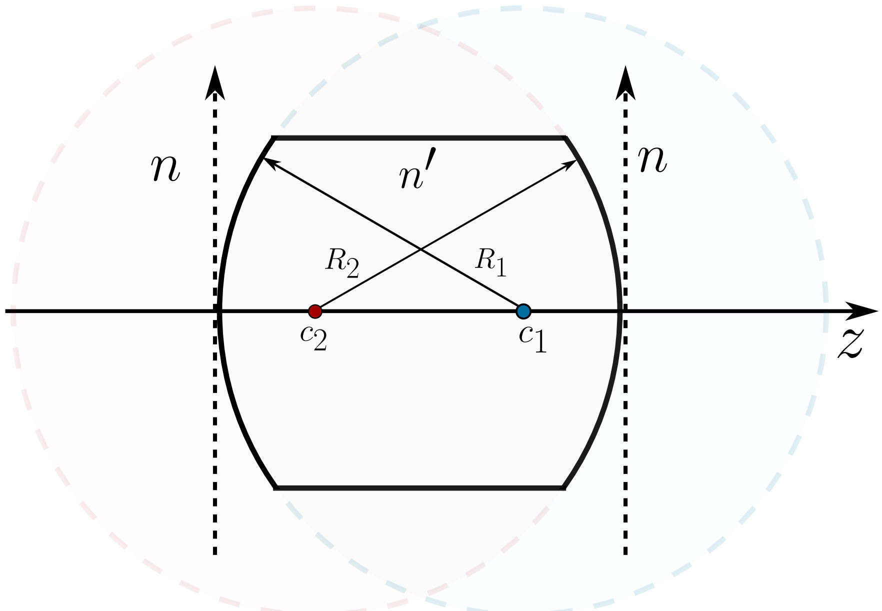


Figure 2.32: Thin lens with aberrations, formed by two spherical diopters of radii R_1 and R_2 separating the medium n from the glass n'. The thickness Δ between the two vertices is taken to zero at the end of the calculation.

Calling $\Delta = d_1 - d_2$ the thickness of the lens, see figure 2.32, and adding 2.152 and 2.153, the intermediate distances combine as

$$n_i\left(\frac{1}{d_1} - \frac{1}{d_2}\right) = n_i\frac{d_2 - d_1}{d_1 d_2} = -\frac{n_i\Delta}{d_2\left(\Delta + d_2\right)},$$

so that

$$\begin{aligned}\frac{n_o}{z_i} - \frac{n_o}{z_o} &= (n_i - n_o)\left(\frac{1}{R_1} - \frac{1}{R_2}\right) + \frac{n_i\Delta}{d_2\left(\Delta + d_2\right)}\\ &- \rho_2^2\frac{1}{2}\left(\frac{n_o}{n_i}\right)^2\left(\frac{1}{z_i} - \frac{1}{R_2}\right)^2\left(\frac{n_i - n_o}{n_o}\right)\left(\frac{n_o}{R_2} - \frac{n_i + n_o}{z_i}\right)\\ &+ \rho_1^2\frac{1}{2}\left(\frac{n_o}{n_i}\right)^2\left(\frac{1}{z_o} - \frac{1}{R_1}\right)^2\left(\frac{n_i - n_o}{n_o}\right)\left(\frac{n_o}{R_1} - \frac{n_i + n_o}{z_o}\right).\end{aligned} \tag{2.154}$$

For a thin lens we take the limit $\Delta \to 0$, which kills the second term and makes $\rho_1 = \rho_2 = \rho$, and we define the paraxial focal distance by

$$\frac{1}{f_p} = \frac{n_i - n_o}{n_o}\left(\frac{1}{R_1} - \frac{1}{R_2}\right), \tag{2.155}$$

which is the lens maker's formula. With this,

$$\frac{n_o}{z_i} - \frac{n_o}{z_o} = \frac{n_o}{f_p} + \rho^2 \frac{1}{2} \left(\frac{n_o}{n_i}\right)^2 \left(\frac{n_i - n_o}{n_o}\right) \left[\left(\frac{1}{z_o} - \frac{1}{R_1}\right)^2 \left(\frac{n_o}{R_1} - \frac{n_i + n_o}{z_o}\right) - \left(\frac{1}{z_i} - \frac{1}{R_2}\right)^2 \left(\frac{n_o}{R_2} - \frac{n_i + n_o}{z_i}\right) \right]. \tag{2.156}$$

For moderate diameters we may replace, on the right hand side only, the paraxial relation $1/z_i = 1/z_o - 1/f_p$, and then send the object to infinity, $z_o \to \infty$. In that limit $1/z_o \to 0$, so the first bracket becomes $(1/R_1)^2 (n_o/R_1) = n_o/R_1^3$ and the second becomes $(1/f_p + 1/R_2)^2 (n_o/R_2 + (n_i + n_o)/f_p)$, therefore the focal distance for marginal rays in a thin lens is

$$\frac{1}{f'_m} = \frac{1}{f_p} + \rho^2 \frac{1}{2} \left(\frac{n_o}{n_i}\right)^2 \left(\frac{n_i - n_o}{n_o}\right) \left[\frac{1}{R_1^3} - \left(\frac{1}{f_p} + \frac{1}{R_2}\right)^2 \left(\frac{1}{R_2} + \frac{n_i + n_o}{n_o} \frac{1}{f_p}\right) \right]. \tag{2.157}$$

Defining the aberration of the thin lens as the whole ρ^2 term,

$$\frac{1}{f_a} = \rho^2 \frac{1}{2} \left(\frac{n_o}{n_i}\right)^2 \left(\frac{n_i - n_o}{n_o}\right) \left[\frac{1}{R_1^3} - \left(\frac{1}{f_p} + \frac{1}{R_2}\right)^2 \left(\frac{1}{R_2} + \frac{n_i + n_o}{n_o} \frac{1}{f_p}\right) \right] = \frac{n_i - n_o}{n_o R_{1+}} - \frac{n_i - n_o}{n_o R_{2-}}, \tag{2.158}$$

where the two effective radii are

$$\frac{1}{R_{1+}} = \frac{\rho^2}{2} \left(\frac{n_o}{n_i}\right)^2 \frac{1}{R_1^3}, \tag{2.159}$$

$$\frac{1}{R_{2-}} = \frac{\rho^2}{2} \left(\frac{n_o}{n_i}\right)^2 \left(\frac{1}{f_p} + \frac{1}{R_2}\right)^2 \left(\frac{1}{R_2} + \frac{n_i + n_o}{n_o f_p}\right), \tag{2.160}$$

the whole result takes the memorable form

$$\frac{1}{f'_m} = \frac{1}{f_p} + \frac{1}{f_a}. \tag{2.161}$$

The aberrated lens behaves, for the marginal rays, as a thin lens of focal length f_p in parallel with a second, purely aberrational lens of focal length f_a. And here appears something the single surface could not offer, the bracket in 2.158 contains R_1 and R_2 separately, not only through the combination $1/R_1 - 1/R_2$ that fixes f_p. Two lenses with the same focal length but different *bending*, that is, with the same difference of curvatures but different individual curvatures, therefore have different spherical aberration. This is the origin of the classical technique of lens bending, and it is why a simple magnifier is always ground as a meniscus with its convex face towards the object rather than as a symmetric biconvex lens (Kingslake and Johnson, 2010).

2.11 Aplanatism

Everything up to this point has been about a single pair of points on the axis. Rigorous stigmatism gives us a perfect image of one point, and approximate stigmatism gives us a slightly imperfect image of one point. But no one photographs a point. What we actually want to image is an *extended* object, and so the question that closes this chapter is the following, under what conditions does the good behaviour of a stigmatic pair survive when we move the object off the axis?

Consider a centred optical system $\mathcal{S}$ and an object AB normal to the axis, with image $A'B'$. The system $\mathcal{S}$ is said to be **aplanatic** if for every point M of AB a ray emerging from M and crossing $\mathcal{S}$ passes through M', the image of M, so that M and M' are a stigmatic pair. The word comes from the Greek *stigma*, point, and the requirement is that the whole segment, not just its centre, be imaged point by point.

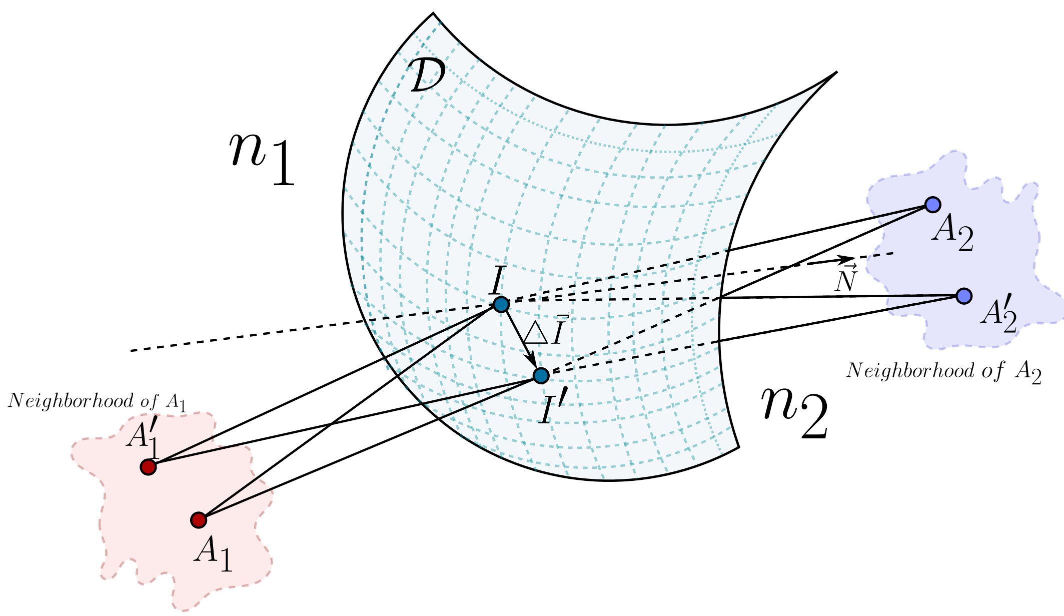


Figure 2.33: The aplanatism condition. Two pairs of rigorously stigmatic points (A_1, A_1') and (A_2, A_2') are considered, with A_1' in a neighbourhood of A_1 and A_2' in a neighbourhood of A_2, the diopter D separating the media n_1 and n_2, and $\Delta\vec{I}$ being the displacement of the incidence point on the surface. The Spanish label *vecindad* in the figure means neighbourhood.

Let us suppose two pairs of points (A_1, A_1') and (A_2, A_2') that are rigorously stigmatic, see figure 2.33, in such a way that $L(A_1, A_2) = cte$ and $L(A_1', A_2') = cte$, where A_1' is close to A_1 and A_2' is close to A_2. In the language of topology, A_1' lies in a neighbourhood of A_1 and A_2' lies in a neighbourhood of A_2.

Using exactly the same variational argument that gave us the vectorial Snell-Descartes law 2.29, we write the difference of the two optical paths as

$$\Delta L\left(A_1,A_2\right)=L\left(A_1',A_2'\right)-L\left(A_1,A_2\right)=\left(n_1\vec{u}_1-n_2\vec{u}_2\right)\cdot d\vec{I}+n_2\vec{u}_2\cdot\Delta\vec{A}_2-n_1\vec{u}_1\cdot\Delta\vec{A}_1=cte, \quad (2.162)$$

where $\Delta\vec{A}_1=\vec{A}_1'-\vec{A}_1$ and $\Delta\vec{A}_2=\vec{A}_2'-\vec{A}_2$. The first term is the one that gave us the law of refraction, and it vanishes identically,

$$\left(n_1\vec{u}_1-n_2\vec{u}_2\right)\cdot d\vec{I}=0,$$

because by the Snell-Descartes law 2.29 the vector $n_1\vec{u}_1-n_2\vec{u}_2$ is parallel to the normal $\hat{N}$, while $d\vec{I}$ is tangent to the diopter. We are therefore left with the **aplanatism condition**

$$\Delta L\left(A_1,A_2\right)=n_2\vec{u}_2\cdot\Delta\vec{A}_2-n_1\vec{u}_1\cdot\Delta\vec{A}_1=cte. \quad (2.163)$$

This deserves a comment before we specialise it. Equation 2.163 is a statement about *derivatives* of the optical path with respect to the position of the object, not about the optical path itself. Rigorous stigmatism says that L does not change when we move the incidence point I over the surface. Aplanatism says something much stronger, that the first variation of L with respect to a displacement of the object is also independent of I. The first condition makes one point sharp, the second makes the whole neighbourhood of that point sharp.

2.11.1 The Abbe sine condition

Let us now specialise 2.163 to the case that matters for an object normal to the axis, that is, the case where the displacement is *transverse*. Then $\Delta\vec{A}_1$ and $\Delta\vec{A}_2$ are perpendicular to the axis, and the scalar products pick up the sines of the angles that the rays make with the axis,

$$n_2\overline{A_2A_2'}\sin\theta_2-n_1\overline{A_1A_1'}\sin\theta_1=cte. \quad (2.164)$$

The constant is fixed by the cheapest possible ray, the one that travels along the axis. For that ray $\theta_1=\theta_2=0$, both sines vanish, and therefore the constant is zero. Since the constant must be the same for every ray, we obtain

$$\frac{\sin\theta_1}{\sin\theta_2}=\frac{n_2}{n_1}\frac{\overline{A_2A_2'}}{\overline{A_1A_1'}}, \quad (2.165)$$

or, writing the **transverse magnification** as

$$g_t=\frac{\overline{A_2A_2'}}{\overline{A_1A_1'}}, \quad (2.166)$$

the condition takes the form in which Abbe stated it in 1873 (Abbe, 1873),

$$n_1\overline{A_1A_1'}\sin\theta_1=n_2\overline{A_2A_2'}\sin\theta_2 \qquad \Longleftrightarrow \qquad g_t=\frac{n_1\sin\theta_1}{n_2\sin\theta_2}. \quad (2.167)$$

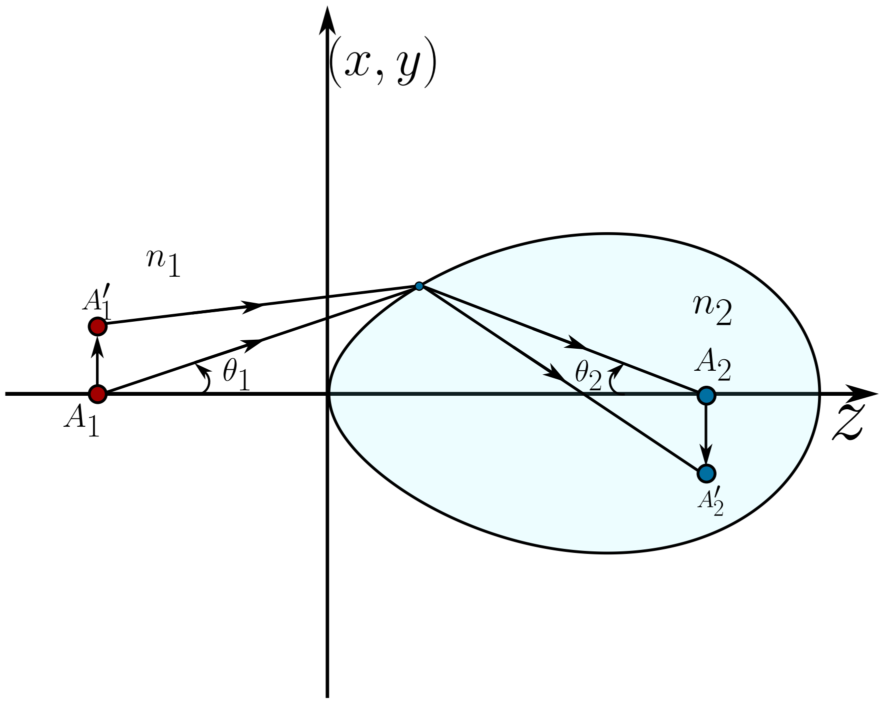


Figure 2.34: Geometry of the Abbe sine condition. The transverse object $\overline{A_1A_1'}$ in the medium n_1 is imaged as $\overline{A_2A_2'}$ in the medium n_2, and θ_1 and θ_2 are the angles that a given ray makes with the axis in the object and image spaces.

Definition 10 *(Aplanatism) A rigorously stigmatic system whose stigmatism is maintained for extended objects, with a magnification that is invariant under a transverse translation, is called **aplanatic**.*

The Abbe sine condition is one of those results whose statement is far shorter than its consequences, so it is worth unpacking what it actually demands. The quantity g_t on the left of 2.167 is a single number, fixed once the conjugate planes are chosen. The quantity on the right involves θ_1 and θ_2, which change from ray to ray as we move over the aperture. So the condition says that *the ratio* $\sin\theta_1/\sin\theta_2$ *must be the same for every ray in the pencil*, from the one that grazes the axis to the one that grazes the rim. If it is not, then rays coming from different zones of the aperture image the off axis point at different transverse positions, and the point spreads into the comet shaped blur that gives coma its name. The sine condition is therefore, in practice, the condition for the absence of coma.

It is very illuminating to compare it with the paraxial result. In the paraxial regime $\sin\theta \approx \theta$ and 2.167 degenerates into $n_1y_1\theta_1 = n_2y_2\theta_2$, which is the Lagrange-Helmholtz invariant of Gauss optics. So the sine condition is not a new law, it is the exact version of a relation we already knew in its linearised form, and the whole difficulty of optical design lies in the gap between θ and $\sin\theta$.

There is one more consequence which is worth stating because it closes a circle we opened many pages ago. We proved, when we studied the spherical diopter, that a sphere is rigorously stigmatic for the pair of Young-Weierstrass points

$$z_o = R\left(1 + \frac{n_i}{n_o}\right), \qquad z_i = R\left(1 + \frac{n_o}{n_i}\right),$$

and one can check that at those two points the sphere also satisfies the sine condition exactly, which is precisely why they are called the *aplanatic points* of the sphere. This is the only situation in which a spherical surface is simultaneously free of spherical aberration and of coma, and it is not a curiosity of the textbooks, it is the working principle of every high numerical aperture microscope objective. The front element of an oil immersion objective is a hemisphere used exactly at its aplanatic points, and the immersion oil is there to make n_o match the glass so that the geometry is preserved. In this sense the whole of modern microscopy rests on a pair of points that Young and Weierstrass found in a sphere.

Finally, the sine condition also explains why resolution and aperture are the same problem. If we write $\mathrm{NA} = n \sin\theta$ for the numerical aperture, then 2.167 says that the product of the object size by the object side numerical aperture is invariant, $y_1 \mathrm{NA}_1 = y_2 \mathrm{NA}_2$. To resolve smaller details we must shrink y_1, and the invariant then forces NA_1 to grow. Abbe understood this in the same years, and it led him to the diffraction limit that will occupy us in the chapter on diffraction. Geometrical optics, pushed to its own boundary, ends up pointing at the wave theory that must replace it.

Chapter 3

Electromagnetic Optics and Polarisation

Light is the slender thread that ties electricity and magnetism into a single weave.

"We can scarcely avoid the inference that light consists in the transverse undulations of the same medium which is the cause of electric and magnetic phenomena."

– James Clerk Maxwell

3.1 A historical context of electromagnetic optics and polarisation

The idea that light is an electromagnetic phenomenon - and that its transverse vibration carries a hidden orientation we call polarisation - is one of the great unifications of nineteenth-century physics. Before light could be understood as a travelling disturbance of electric and magnetic fields, it had to shed two older skins: Newton's corpuscles and the purely mechanical aether of the early wave theorists. The story of polarisation, in particular, is the story of how a seemingly minor curiosity - light reflected from a window behaving differently from ordinary light - forced physicists to accept that the optical vibration is transverse, and ultimately vectorial.

The first quantitative encounter with polarisation came in 1808, when Étienne-Louis Malus, looking through a calcite crystal at sunlight reflected from the windows of the Luxembourg Palace in Paris, noticed that the intensity of the transmitted beam changed as he rotated the crystal. From these observations he extracted the law that bears his name, $I = I_0 \cos^2 \theta$, relating the transmitted intensity to the angle between the analyser and the plane of polarisation Malus (1809). Malus had discovered that reflection could endow light with the same directional property that double refraction revealed, without invoking any crystal at all. A few years later, in 1815, David Brewster established the law that the polarisation upon reflection is complete at a particular angle, the Brewster angle, for which the reflected and refracted rays are perpendicular Brewster (1815).

These facts were deeply puzzling within the longitudinal wave picture inherited from sound. The resolution came from Augustin-Jean Fresnel and François Arago, who between 1816 and 1819 showed experimentally that two beams polarised at right angles do not interfere. Fresnel drew the inevitable, audacious conclusion: the optical vibration must be *transverse* to the direction of propagation. From this single hypothesis he derived the laws governing the amplitudes of reflected and refracted light - the Fresnel equations - which remain in use today. Transversality made polarisation a natural attribute of light: a wave oscillating in a plane perpendicular to its motion can do so along infinitely many directions, and the choice of that direction *is* its state of polarisation.

A decisive step toward a quantitative description of partially polarised light was taken by George Gabriel Stokes in 1852. Recognising that realistic beams are never perfectly polarised, Stokes introduced four measurable quantities - today called the Stokes parameters - that completely characterise the state of polarisation, including the degree of polarisation, of any beam, coherent or not Stokes (1852). Decades later, Henri Poincaré gave these parameters a beautiful geometric home: every state of polarisation corresponds to a point on the surface of a sphere, the Poincaré sphere, on which the poles represent circular polarisation and the equator linear polarisation. This geometric picture is the one that reappears, in the scalar reduction of diffraction theory, when one decomposes the field onto two opposite states of the sphere.

The grand synthesis arrived with James Clerk Maxwell. In his 1865 memoir *A Dynamical Theory of the Electromagnetic Field* Maxwell (1865), Maxwell showed that the equations governing electricity and magnetism admit wave solutions propagating at a speed numerically equal to the measured speed of light, and concluded that "light is an electromagnetic disturbance." Polarisation was thereby identified with the direction of the electric field vector $\vec{E}$, and the whole edifice of physical optics became a chapter of electromagnetism. The experimental confirmation came in 1887, when Heinrich

Hertz generated and detected electromagnetic waves in the laboratory, verifying that they reflect, refract and polarise exactly as light does Hertz (1893).

In the twentieth century the description of polarisation was recast in the language of linear algebra. R. Clark Jones, in a celebrated series of papers beginning in 1941, introduced the calculus of two-component complex vectors and 2×2 matrices that now bears his name, ideally suited to fully polarised, coherent light Jones (1941). For partially polarised or depolarising systems, the complementary 4×4 Mueller calculus acting on the Stokes vector provides the appropriate tool. The modern treatment of all these matters - transversality, the Fresnel equations, the StokesPoincaré formalism and the Jones and Mueller calculi - is presented with classic completeness in Born and Wolf's *Principles of Optics* Born and Wolf (1999), the reference against which later treatments are still measured.

From Malus's rotating crystal to the polarisation-encoded qubits of quantum information, the orientation of the optical field has proven to be far more than an ornament: it is a genuine degree of freedom of light, vectorial in nature, and the bridge between the scalar optics of rays and the full electromagnetic description developed in the chapters that follow.

3.2 The vector behavior of the electromagnetic radiation

Until this part of the lectures, we've studied the light as a ray, however, a better model for the light becomes with the 4 Maxwell's equations, which are:

$$\boldsymbol{\nabla} \cdot \vec{E} = \frac{\rho}{\epsilon_0}, \qquad \textit{Gauss Law for the electric field} \tag{3.1}$$

$$\boldsymbol{\nabla} \cdot \vec{B} = 0, \qquad \textit{Gauss Law for the magnetic field} \tag{3.2}$$

$$\boldsymbol{\nabla} \times \vec{E} = -\frac{\partial}{\partial t}\vec{B}, \qquad \textit{Faraday-Lenz law} \tag{3.3}$$

$$\boldsymbol{\nabla} \times \vec{B} = \mu_0\epsilon_0\frac{\partial}{\partial t}\vec{E} + \mu_0\vec{J}, \qquad \text{Ampére-Maxwell Law..} \tag{3.4}$$

The equations from 3.1 to 3.4 have got such elegant form thanks to the vector notation given by the physicist Oliver Heaviside and thanks to some of the most useful theorems of the mathematical analysis, the divergence (Gauss) and curl (Stokes) theorems, which allow a smoth transition from the integral form of the Maxwell equations to the differential ones.

The Gauss Law for the electric field 3.1 is really easy to read when we know the geometric interpretatio4n of the divergence operator $\nabla \cdot$ which quantifies how much the electric field $\vec{E}$ goes into (also gets out) from a point, but the right hand side of the equation 3.1 tells us that the presence of any *charge density* ρ produces sources and sinks of electric field (positive charge densities produce sources and negative charge densities produce sinks of electric field).

The Gauss Law for the magnetic field 3.2 tells us that **alwyas the sources of magnetic fields are compensated by sinks of magnetic fields**, also it is interpreted as **there are no magnetic fields monopoles**.

The equation 3.3 is known as the Faraday-Lenz law, the Faraday law tells us that every circulation of electric field produces a variation of magnetic field and every time variation of magnetic field produces circulations of electric field, however there is something importart, because if we take the equation 3.3 as $\nabla \times \vec{E} = \partial_t \vec{B}$ without the minus sign there wont be energy conservation, so the minus sign is introduced by Lenz and this equation can describe the beahavior of eddy currents.

The fourth equation 3.4 describes the circulations of magnetic fields, due to the Gauss Law 3.2 there wont be any polar magnetic field, it is only possible to have axial-like magnetic fields so, this circulations can only be created by two ways, from the presence of density currents $\vec{J}$ or by time variation of the electric fields, Maxwell added this term such that the continuity equation

$$\frac{\partial}{\partial t}\rho + \boldsymbol{\nabla} \cdot \vec{J} = 0, \tag{3.5}$$

is satisfied.

Now,because we're interested in the study of the electromagnetic radiation *light* we must look for a wave equation, to do this, we're going to use the following mathematical identity:

$$\boldsymbol{\nabla} \times \left(\boldsymbol{\nabla} \times \vec{B}\right) = \boldsymbol{\nabla}\left(\boldsymbol{\nabla} \cdot \vec{A}\right) - \nabla^2 \vec{A}. \tag{3.6}$$

From the Faraday-Lenz law 3.3 we've got applying the curl to both sides

$$\boldsymbol{\nabla}\left(\boldsymbol{\nabla} \times \vec{E}\right) = -\boldsymbol{\nabla} \times \left(\frac{\partial}{\partial t}\vec{B}\right),$$

now, befores we're going further we're going to suppose that the electromagnetic fields are smooth vector functions, or at least of class $\mathcal{C}^2 := \left\{f : D^2 f \in \mathcal{C}\right\}$ (this means all the second derivaties are continuos), therefore because these are continuous we can use the identity 3.6 and interchange the curl with the time derivative on $\vec{B}$, thus

$$\boldsymbol{\nabla}\left(\boldsymbol{\nabla} \cdot \vec{E}\right) - \nabla^2 \vec{E} = -\frac{\partial}{\partial t}\left(\boldsymbol{\nabla} \times \vec{B}\right),$$

to solve the left hand side we are going to use the Gauss law of the electric field 3.1 and for the right hand side we're going to use the Ampére-Maxwell law 3.4 such that we've got

$$\boldsymbol{\nabla}\left(\frac{\rho}{\epsilon_0}\right) - \nabla^2 \vec{E} = -\frac{\partial}{\partial t}\left(\mu_0 \epsilon_0 \frac{\partial}{\partial t}\vec{E} + \mu_0 \vec{J}\right),$$

$$\nabla^2 \vec{E} - \frac{1}{C^2}\frac{\partial^2}{\partial t^2}\vec{E} = \frac{1}{\epsilon_0}\boldsymbol{\nabla}\rho + \mu_0 \frac{\partial}{\partial t}\vec{J}, \tag{3.7}$$

where $C^2 = \frac{1}{\epsilon_0 \mu_0}$.

The 3.7 is the formal wave equation for the electric field. In the left hand side we find the traditional well known wave equation but on the right hand side we can see some physical information about how can those waves be generated. We can see that any gradient of electric charge densities produces electric waves and that time variations of the current density source them too, with a plus sign, because the derivation used the minus sign of the Lenz law twice: once when $\partial_t \vec{B}$ was replaced by $-\boldsymbol{\nabla} \times \vec{E}$ in going from 3.4 to the line above, and a second time when the whole equation was rearranged to bring $\nabla^2 \vec{E} - \frac{1}{C^2}\partial_t^2 \vec{E}$ to the left, and the two minus signs cancel each other. Now, we've shown that there are electric waves so we're going to proof that there are also magnetic waves.

From the Ampére-Maxwell law 3.4 we've got applying the curl on both sides

$$\boldsymbol{\nabla}\left(\boldsymbol{\nabla} \times \vec{B}\right) = \boldsymbol{\nabla} \times \left(\mu_0 \epsilon_0 \frac{\partial}{\partial t}\vec{E} + \mu_0 \vec{J}\right),$$

now, under our assumptions and also $\rho, \vec{J}$ are at leat $\mathcal{C}^2$, then we can apply the identity 3.6 and interchange the time derivatide with the curl on $\vec{E}$, thus

$$\boldsymbol{\nabla}\left(\boldsymbol{\nabla} \cdot \vec{B}\right) - \nabla^2 \vec{B} = \mu_0 \epsilon_0 \frac{\partial}{\partial t}\left(\boldsymbol{\nabla} \times E\right) + \mu_0 \boldsymbol{\nabla} \times \vec{J}.$$

On the left side we can use the Gauss law for the magnetic field 3.2, and on the right hand side we can use the Faraday-Lenz law thus

$$-\nabla^2 \vec{B} = \mu_0 \epsilon_0 \frac{\partial}{\partial t}\left(-\frac{\partial}{\partial t}\vec{B}\right) + \mu_0 \boldsymbol{\nabla} \times \vec{J},$$

$$\nabla^2 \vec{B} - \frac{1}{C^2}\frac{\partial^2}{\partial t^2}\vec{B} = -\mu_0 \boldsymbol{\nabla} \times \vec{J}. \tag{3.8}$$

The equation 3.8 is the wave equation for the magnetic field, on the left hand side we can recognize the traditional wave terms and on the right hand side we can see the sources of magnetic waves, because there are no magnetic monopoles there is not term for gradients of such charges, only the circulations of density currents produces this magnetic waves.

Befores going any further we must stop and think about something important, if the light is both electric and magnetic radiation, why the equations 3.7 and 3.8 are only coupled by the density current, or even more, in the vacuum, both equation are independent, so if we think on photons we can question ourselves, ***Where are the photons? Are the photons on the electric or magnetic field*** We are so used to learn they're on both, but how?

3.3 Maxwell equations are the Schrödinger equation for photons

The sentence *photons are in the electromagnetic radiation* is easy to beliave but hard to proof from a classical point of view. We could be tempted by the devil and add both equation 3.7 and 3.8 and get something like

$$\nabla^2\left(\vec{E} + \vec{B}\right) - \frac{1}{C^2}\frac{\partial^2}{\partial t^2}\left(\vec{E} + \vec{B}\right) = \frac{1}{\epsilon_0}\boldsymbol{\nabla}\rho + \mu_0\left(\frac{\partial}{\partial t} - \boldsymbol{\nabla}\times\right)\vec{J}, \tag{3.9}$$

which is correct, completely correct **in a world where does not exist physics and analysis**.

From an analysis point of view, the electric field $\vec{E}$ and the magnetic field $\vec{B}$ have got completely different natures, the electric field is a polar vector function, and the magnetic field is an axial vector function, so both fields live on differente vector spaces and the addition $\vec{E} + \vec{B}$ does not have any sense.

To solve this problem, there is an interesing vector field called the ***Riemann-Silberstein field*** which is a correct mathematical (and physical as we we'll see) superposition of the electric and magnetic fields given by

$$\vec{F} = \vec{E} + iC\vec{B}. \tag{3.10}$$

That imaginary unit i may be seen innocent, however adding this complex topology to $\vec{E}, \vec{B}$ solves the superposition problem. It is called for Riemann because he was the first one who studied (alongside Gauss) the electromagnetism using complex numbers, and for Riemann because he used the vector field 3.10 to rewrite the Einstein's field equations in a quaternions formulation. This may be seen as an innocent superpostion but we're going to see that this will allow us, from a classical point of view, to arrive to the correct *photon wave function* and showing that the Maxwell equations acting on 3.10 (which behaves as the photon field) are also quantum equations.

Lets rewrite the Maxwell equations 3.1-3.4 in terms of the photon field $\vec{F}$. First, just multiply the Gauss law for the magnetic field 3.2 by iC and add to the electric Gauss law 3.1 and we'll have

$$\boldsymbol{\nabla} \cdot \left(\vec{E} + iC\vec{B}\right) = \frac{\rho}{\epsilon_0}. \tag{3.11}$$

And, also multiply the Ampére-Maxwell law 3.4 by iC and add it to the Faraday-Lenz law 3.3 we will have

$$\boldsymbol{\nabla} \times \left(\vec{E} + iC\vec{B}\right) = \frac{i}{C}\frac{\partial}{\partial t}\left(\vec{E} + iC\vec{B}\right) + iC\mu_0\vec{J}. \tag{3.12}$$

Joning both equations and using the definitions of $\vec{F}$ we'll wave that the four Maxwell's equations becomes two

$$\boldsymbol{\nabla} \cdot \vec{F} = \frac{\rho}{\epsilon_0}, \qquad \text{Gauss law for the photon field} \tag{3.13}$$

$$\boldsymbol{\nabla} \times \vec{F} = \frac{i}{C}\frac{\partial}{\partial t}\vec{F} + iC\mu_0\vec{J}, \qquad \text{Faraday-Lenz-Ampére-Maxwell laws for the photon field.} \tag{3.14}$$

The physical interpretaion of the equations 3.13 and 3.14 becomes clear. The Gauss law 3.13 tells us that the presence of charge densities produces sources and sinks of photon fields (electromagnetic fields), and the second equation 3.14 says that time variatons of the photon field produces circulations of photon fields and also, the mere presence of density currents produces also circulations of this photon fields. Now it is becoming clearer the relation between where are the photons and how can it be seen in the elctromagnetic field at classical levels.

Now, we're going to proof that the set of equations 3.13 and 3.14 contains two wave equation. One traditional wave equation for $\vec{F}$ and an Schrödigner wave equation for $\vec{F}$, showing this beauty behavior both classical and quantum of the Maxwells equation. Lets apply the curl of the second law 3.14 to both sides as follows

$$\boldsymbol{\nabla} \times \left(\boldsymbol{\nabla} \times \vec{F}\right) = \frac{i}{C}\boldsymbol{\nabla} \times \left(\frac{\partial}{\partial t}\vec{F}\right) + iC\mu_0\boldsymbol{\nabla} \times \vec{J},$$

applying the identity 3.6 on the left han side and permutting th time derivative and the curl in the right hand side as

$$\boldsymbol{\nabla}\left(\boldsymbol{\nabla} \cdot \vec{F}\right) - \nabla^2\vec{F} = \frac{i}{C}\frac{\partial}{\partial t}\left(\boldsymbol{\nabla} \times \vec{F}\right) + iC\mu_0\boldsymbol{\nabla} \times \vec{J},$$

using the Gauss law 3.13 and the FLAM law 3.14 we'll have

$$\frac{1}{\epsilon_0}\boldsymbol{\nabla}\rho - \nabla^2\vec{F} = \frac{i}{C}\frac{\partial}{\partial t}\left(\frac{i}{C}\frac{\partial}{\partial t}\vec{F} + iC\mu_0\vec{J}\right) + iC\mu_0\boldsymbol{\nabla} \times \vec{J},$$

carrying out the time derivative on the right hand side, and using $\frac{i}{C} \cdot \frac{i}{C} = -\frac{1}{C^2}$ and $\frac{i}{C} \cdot iC\mu_0 = -\mu_0$, we've got

$$\frac{1}{\epsilon_0}\boldsymbol{\nabla}\rho - \nabla^2\vec{F} = -\frac{1}{C^2}\frac{\partial^2}{\partial t^2}\vec{F} - \mu_0\frac{\partial}{\partial t}\vec{J} + iC\mu_0\boldsymbol{\nabla} \times \vec{J},$$

and moving $\nabla^2\vec{F}$ and $\frac{1}{C^2}\partial_t^2\vec{F}$ to the same side, that is multiplying the whole equation by -1 and swapping it around, we arrive at the *classical wave equation for the photon field*

$$\nabla^2\vec{F} - \frac{1}{C^2}\frac{\partial^2}{\partial t^2}\vec{F} = \frac{1}{\epsilon_0}\boldsymbol{\nabla}\rho + \mu_0\left(\frac{\partial}{\partial t} - iC\boldsymbol{\nabla}\times\right)\vec{J}. \tag{3.15}$$

As a check, notice that $\nabla^2 - \frac{1}{C^2}\partial_t^2$ is a linear operator and $\vec{F} = \vec{E} + iC\vec{B}$ is a linear combination, so 3.15 must equal 3.7 plus iC times 3.8, and indeed $\left[\frac{1}{\epsilon_0}\boldsymbol{\nabla}\rho + \mu_0\partial_t\vec{J}\right] + iC\left[-\mu_0\boldsymbol{\nabla} \times \vec{J}\right] = \frac{1}{\epsilon_0}\boldsymbol{\nabla}\rho + \mu_0\left(\partial_t\vec{J} - iC\boldsymbol{\nabla} \times \vec{J}\right)$, exactly the right hand side above.

We can see that looks very similar to the sinister equation 3.9 but that was a really forced one insted of our beauty 3.15 which was calculated from the Maxwell's equations, and we see the origin of this electromagnetic or photon fields, these waves are originated as from charge density gradientes, and time variations and circulations of the density current, now the behavior of the photons, from a classical perspective can be seen in the presence of charges and currents.

Now, to find out the quantum wave equation from the classical equation 3.14 we are going to use, the following algebraic identity:

$$\vec{A} \times \vec{B} = -i\left(\vec{A} \cdot \vec{S}\right)\vec{B}, \qquad \vec{S} = S_x\hat{i} + S_y\hat{j} + S_z\hat{k} \tag{3.16}$$

$$S_x = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & -i \\ 0 & i & 0 \end{bmatrix}, \ S_y = \begin{bmatrix} 0 & 0 & i \\ 0 & 0 & 0 \\ -i & 0 & 0 \end{bmatrix}, \ S_z = \begin{bmatrix} 0 & -i & 0 \\ i & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

The matrices S_x, S_y, S_z are called teh photon spin 1 matrices, and $\vec{A}, \vec{B}$ are arbitrary vectors. Uusing the identity 3.16 on 3.14 we'll have

$$\begin{aligned}
\boldsymbol{\nabla} \times \vec{F} &= -i\left(\boldsymbol{\nabla} \cdot \vec{S}\right)\vec{F}, \quad \text{using the product rule}\\
\boldsymbol{\nabla} \times \vec{F} &= -i\left(\vec{S} \cdot \boldsymbol{\nabla}\right)\vec{F},\\
\frac{i}{C}\frac{\partial}{\partial t}\vec{F} &= -i\left(\vec{S} \cdot \boldsymbol{\nabla}\right)\vec{F} - iC\mu_0\vec{J},
\end{aligned}$$

multipliying both sides by $C\hbar$ we'll have

$$i\hbar\frac{\partial}{\partial t}\vec{F} = C\left(\vec{S} \cdot \frac{\hbar}{i}\boldsymbol{\nabla}\right)\vec{F} - i\hbar C^2\mu_0\vec{J}, \tag{3.17}$$

which is the Schrödinger-like equation for the photon field $\vec{F}$ with the perturbation of the density currents $\vec{J}$ if we're on the vacuum, we'll have the well known as the ***Birula's photon wave function*** Białynicki-Birula (1994)

$$i\hbar\frac{\partial}{\partial t}\vec{F} = C\left(\vec{S} \cdot \frac{\hbar}{i}\boldsymbol{\nabla}\right)\vec{F}. \tag{3.18}$$

Therefore the Maxwells equation by themselves are classical and quantum, contains the wave functions that describes the photon behavior and, as we we'll see further in this lectures, we can carry on this result to general relativity using a tensor that I named **Riemann-Silberstein tenstor** which even condensates the four equations to just one.

3.4 Properties of the electromagnetic waves

From now on we're going to focus on the electric wave equation 3.7 and not the complete electromagnetic wave equation 3.15 for three reasons:

1. For now, we're going to focus on the radiation propagation on the vacuum, so the electric field intensity $||\vec{E}||$ is a lot bigger than the magnetic field intensity $||\vec{B}||$ with a relation approximadelety like $||\vec{E}|| \approx C||\vec{B}||$.

2. The interaction between light and matter is really hard to study when we consider the electric polarization and even more when we consider all the magnetic properties and effects. This topics are going to be study in the lectures on quantum optics.

3. This is still a lecture on fundamentals, so we're going to start from the basic notions and getting richer models while we advance.

3.4.1 Principal solutions of the electric wave equation

Due to the last reasons, we're going to to focus on the study of the electric wave equation

$$\nabla^2 \vec{E}(\vec{r},t) = \frac{1}{C^2}\frac{\partial^2 \vec{E}(\vec{r},t)}{\partial t^2}. \tag{3.19}$$

We're going to solver this equation in a physical way (latter on this lectures we're going to solve it in a more sophisticaed way). We know from a physical point of view that $\vec{E}$ should be a propagation function with a constant velcity $\vec{v}$ (because there is no matter so the momentum is conserved), therefore we could suppose the electric wave have got the following expression

$$\vec{E}(\vec{r},t) = \vec{u}(\vec{k}\cdot\vec{r} \pm \omega t, t), \tag{3.20}$$

where we have got that $\omega^2/k^2 = C^2$. The two most popular solutions in the basics literature are the plane wave equation

$$\vec{E}(\vec{r},t) = E(\vec{r},t)e^{i\left(\vec{k}\cdot r - \omega t + \phi\right)}\hat{e}(\vec{r},t), \tag{3.21}$$

where $E(\vec{r},t)$ is the amplitude, $\vec{k}$ is the waver number, a kind of spatial frequency, ω is the angular frequency, ϕ is the proper fase, an $\hat{e}$ is the unit vector that describes the vector behavior of the electric wave, also we will identify $\hat{e}$ as the *polarisation.*

The other one, very popular is the spherical wave

$$\vec{E}(\vec{r},t) = \frac{E(t)}{||\vec{r}||}e^{i(kr-\omega t+\phi)}\hat{e}. \tag{3.22}$$

Both expressions are usually written down as if they had fallen from the sky, but each of them is a genuine solution of the wave equation 3.19 and can be obtained directly from it. Let us do it carefully, first for the plane wave and then for the spherical one.

In the vacuum the wave equation 3.19 acts on every Cartesian component of $\vec{E}$ in exactly the same way and without mixing them, so for the moment we can forget the vector character, write $\vec{E}(\vec{r},t) = \psi(\vec{r},t)\hat{e}$ with $\hat{e}$ a fixed unit vector, and solve the scalar equation

$$\nabla^2\psi = \frac{1}{C^2}\frac{\partial^2\psi}{\partial t^2}. \tag{3.23}$$

A natural way to attack it is to separate the space and time dependence, proposing $\psi(\vec{r},t) = R(\vec{r})T(t)$. Substituting into 3.23 and dividing by RT we get

$$\frac{\nabla^2 R}{R} = \frac{1}{C^2}\frac{\ddot{T}}{T}.$$

The left hand side depends only on the position and the right hand side only on the time, so the only way for them to be equal for every $\vec{r}$ and every t is that both are the same constant. We call that constant $-k^2$, where the minus sign is chosen so that the time part stays bounded and oscillatory, as a wave must be. The time equation then becomes

$$\ddot{T} + C^2 k^2 T = 0,$$

and defining $\omega = Ck$, which is precisely the relation $\omega^2/k^2 = C^2$ we anticipated, its solution is $T(t) = e^{\mp i\omega t}$. The spatial part becomes the Helmholtz equation

$$\nabla^2 R + k^2 R = 0. \tag{3.24}$$

Now we ask for the simplest possible geometry of the wavefronts. If we want the surfaces of constant phase to be planes, the field must depend on the position only through a single linear combination of the coordinates, that is, through $\vec{k} \cdot \vec{r}$ for some fixed vector $\vec{k}$. This suggests

$$R(\vec{r}) = e^{i\vec{k}\cdot\vec{r}},$$

and since $\nabla^2 e^{i\vec{k}\cdot\vec{r}} = -\left|\left|\vec{k}\right|\right|^2 e^{i\vec{k}\cdot\vec{r}}$, the Helmholtz equation 3.24 is satisfied as long as $\left|\left|\vec{k}\right|\right| = k$. The vector $\vec{k}$ that was born as a mere separation constant turns out to point along the direction of propagation and to have a length fixed by the frequency. Putting the space and time parts back together, restoring the amplitude E_0 and a constant phase ϕ, and recovering the polarisation vector, we arrive at

$$\vec{E}(\vec{r}, t) = E_0\, e^{i\left(\vec{k}\cdot\vec{r} - \omega t + \phi\right)}\hat{e},$$

which is exactly the plane wave 3.21. If instead of the constant E_0 and the fixed $\hat{e}$ we allow a slowly varying amplitude $E(\vec{r}, t)$ and polarisation $\hat{e}(\vec{r}, t)$, we recover the modulated waves that we will use later, whose wavefronts are still approximately planes over regions small compared with the modulation.

There is still one condition we have not used, and it is the one that makes the wave truly electromagnetic. In the vacuum the Gauss law 3.1 reads $\boldsymbol{\nabla} \cdot \vec{E} = 0$, and evaluating it on our solution

$$\boldsymbol{\nabla} \cdot \left(E_0\, e^{i\left(\vec{k}\cdot\vec{r} - \omega t + \phi\right)}\hat{e}\right) = i\, E_0 \left(\vec{k} \cdot \hat{e}\right) e^{i\left(\vec{k}\cdot\vec{r} - \omega t + \phi\right)} = 0,$$

which can only hold if $\vec{k} \cdot \hat{e} = 0$. The polarisation is forced to be perpendicular to the direction of propagation, and we recover, now as a theorem instead of a hypothesis, the transversality that Fresnel had to postulate a century and a half earlier.

The plane wave describes a field whose wavefronts are infinite parallel planes, an idealisation that is excellent far from the source but says nothing about how the radiation spreads out from a small emitter. For that situation the natural symmetry is not translational but spherical, so we now look for a field that depends on the position only through the distance $r = ||\vec{r}||$ to the source. Working again with the scalar amplitude $\psi(r, t)$, we need the Laplacian of a function that depends on r alone, which in spherical coordinates is

$$\nabla^2 \psi = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2 \frac{\partial \psi}{\partial r}\right).$$

This expression hides a very useful identity. If we expand it and compare it with the second derivative of the product $r\psi$, we find

$$\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial\psi}{\partial r}\right) = \frac{1}{r}\frac{\partial^2}{\partial r^2}\left(r\psi\right), \tag{3.25}$$

which turns the awkward radial Laplacian into an ordinary second derivative acting on $r\psi$. Using 3.25, the wave equation 3.23 for a spherically symmetric field becomes

$$\frac{1}{r}\frac{\partial^2}{\partial r^2}\left(r\psi\right) = \frac{1}{C^2}\frac{\partial^2\psi}{\partial t^2}.$$

Multiplying both sides by r and noticing that r does not depend on the time, so it can go inside the time derivative on the right, we get

$$\frac{\partial^2}{\partial r^2}\left(r\psi\right) = \frac{1}{C^2}\frac{\partial^2}{\partial t^2}\left(r\psi\right).$$

If we now baptise the combination $u(r,t) = r\,\psi(r,t)$, this is nothing but the one dimensional wave equation

$$\frac{\partial^2 u}{\partial r^2} = \frac{1}{C^2}\frac{\partial^2 u}{\partial t^2}, \tag{3.26}$$

whose general solution, by d'Alembert, is the sum of a disturbance travelling outwards and one travelling inwards,

$$u(r,t) = f(r - Ct) + g(r + Ct).$$

The term $g(r + Ct)$ represents a wave collapsing towards the source, so for radiation escaping from an emitter we keep only the outgoing part $u(r,t) = f(r - Ct)$. Undoing the change $u = r\psi$ we finally obtain

$$\psi(r,t) = \frac{f(r - Ct)}{r},$$

and the factor $1/r$, which we had to guess before, now appears on its own as a direct consequence of the geometry. Choosing a monochromatic profile $f(r - Ct) = E_0\, e^{ik(r-Ct)} = E_0\, e^{i(kr-\omega t)}$, again with $\omega = Ck$, and restoring the phase and the polarisation we reach the spherical wave 3.22,

$$\vec{E}(\vec{r},t) = \frac{E_0}{r}\, e^{i(kr-\omega t+\phi)}\hat{e}.$$

It is worth pausing on the $1/r$. The amplitude of the field falls off as the inverse of the distance, so its square, which measures the intensity, falls off as $1/r^2$. Since the area of a spherical wavefront grows exactly as r^2, the total power crossing every sphere centred on the source is the same, and energy is conserved as the wave expands. This is why the field goes as $1/r$ and not as $1/r^2$: the inverse square law belongs to the intensity, not to the field itself.

The plane wave and the spherical wave look like very different objects, and yet they are intimately related, because far enough from the source the spherical wave is indistinguishable from a plane wave. To see it, place ourselves at a large distance r_0 from the emitter along a fixed direction $\hat{n} = \vec{r}_0/r_0$, and look at the field only inside a small region around that point, writing $\vec{r} = \vec{r}_0 + \vec{\delta}$ with $\left|\left|\vec{\delta}\right|\right| \ll r_0$. The distance to the source is then

$$r = \left|\left|\vec{r}_0 + \vec{\delta}\right|\right| = r_0\sqrt{1 + \frac{2\,\hat{n}\cdot\vec{\delta}}{r_0} + \frac{\left|\left|\vec{\delta}\right|\right|^2}{r_0^2}} \approx r_0 + \hat{n}\cdot\vec{\delta},$$

where we kept only the first order in $\vec{\delta}/r_0$. The phase of the spherical wave becomes, up to the constant kr_0,

$$kr \approx kr_0 + k\,\hat{n}\cdot\vec{\delta} = kr_0 + \vec{k}\cdot\vec{\delta}, \qquad \vec{k} = k\,\hat{n},$$

so it depends on the position through the single linear combination $\vec{k}\cdot\vec{\delta}$, which is exactly the signature of a plane wave travelling along $\hat{n}$. The amplitude, on the other hand, is $E_0/r \approx E_0/r_0$, which over the small region hardly changes and can be treated as the constant E_0/r_0. Putting both together, in the neighbourhood of the point the field reads

$$\vec{E}(\vec{r},t) \approx \frac{E_0}{r_0}\,e^{i\left(kr_0 + \vec{k}\cdot\vec{\delta} - \omega t + \phi\right)}\hat{e},$$

which is precisely a plane wave 3.21 of amplitude E_0/r_0 and wavevector $\vec{k} = k\hat{n}$. In physical terms, a small patch of a sphere of enormous radius is almost flat, and the observer far from the source sees the curved wavefronts as if they were planes. This is the regime in which the plane wave, despite carrying infinite energy across an infinite front, becomes a legitimate and extremely convenient approximation, and it is the reason why so much of optics can be done with plane waves even though every real source radiates spherically.

3.4.2 Beams with a particular phase and structured light

The plane wave and the spherical wave are the two solutions that every textbook writes down first, and for a good reason, since they are the simplest geometries the wave equation admits. It would be a serious mistake, however, to believe that they are the only solutions, or even the most important ones. Both are idealisations that no laboratory ever produces exactly, because the plane wave has an infinite transverse extent and carries an infinite power, and the pure spherical wave demands a perfectly point like source radiating in every direction at once. The light we actually generate and use, above all the light of a laser, comes in the form of *beams*, that is, fields well confined around a propagation axis that carry a finite power. What distinguishes one beam from another, and what grants each one its remarkable properties, is not so much its amplitude as the way its *phase* is organised across the wavefront. This subsection is a short tour through the most relevant of these structured beams.

Before writing any of them it helps to have a common equation from which they all descend. Almost every beam of interest travels mostly along one axis, which we take to be z, with only a gentle spreading in the transverse plane, and this is what we call the *paraxial regime*. We therefore look for solutions of the Helmholtz equation 3.24 of the form

$$\vec{E}(\vec{r},t) = u(x,y,z)\,e^{i(kz-\omega t)}\hat{e},$$

where the exponential carries the fast oscillation along the axis and u is a slowly varying envelope that sculpts the beam. Substituting this into the Helmholtz equation and cancelling the common factor we obtain

$$\nabla_\perp^2 u + \frac{\partial^2 u}{\partial z^2} + 2ik\frac{\partial u}{\partial z} = 0, \qquad \nabla_\perp^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2},$$

and because the envelope varies slowly over a wavelength we may neglect $\partial_z^2 u$ against $2k\,\partial_z u$, arriving at the *paraxial wave equation*

$$i\frac{\partial u}{\partial z} = -\frac{1}{2k}\nabla_\perp^2 u. \tag{3.27}$$

It is impossible not to notice that 3.27 is formally identical to the Schrödinger equation of a free particle in two dimensions, with the propagation coordinate z playing the role of the time and $1/k$ playing the role of $\hbar/m$. This is not a coincidence but another face of the deep link between Maxwell and Schrödinger we met earlier in this chapter, and it tells us that the whole catalogue of structured beams is, in a precise mathematical sense, the catalogue of the wave packets of a free particle.

The simplest and most important solution of 3.27 is the one whose transverse profile is a Gaussian. Writing $\rho^2 = x^2 + y^2$ for the transverse radius, its electric field is

$$\vec{E} = E_0\,\frac{w_0}{w(z)}\exp\left(-\frac{\rho^2}{w(z)^2}\right)\exp\left(i\left[kz - \omega t + \frac{k\rho^2}{2R(z)} - \zeta(z)\right]\right)\hat{e}, \tag{3.28}$$

where the three functions that govern the beam are the width, the radius of curvature and the Gouy phase,

$$w(z) = w_0\sqrt{1+\left(\frac{z}{z_R}\right)^2}, \qquad R(z) = z\left[1+\left(\frac{z_R}{z}\right)^2\right], \qquad \zeta(z) = \arctan\frac{z}{z_R}, \tag{3.29}$$

and $z_R = \frac{kw_0^2}{2} = \frac{\pi w_0^2}{\lambda}$ is the *Rayleigh range*, the distance over which the beam stays approximately collimated, while w_0 is the *waist*, the minimal radius the beam reaches at $z = 0$. Reading the phase of 3.28 term by term makes the physics appear. The piece kz is the ordinary plane wave advance along the axis. The quadratic piece $k\rho^2/2R(z)$ is exactly the phase that a spherical wave of radius $R(z)$ carries near the axis, so it tells us that the wavefronts are neither planes nor full spheres but gently curved caps, flat at the waist where $R \to \infty$, most strongly curved around the Rayleigh range, and flattening again far away where $R(z) \approx z$ and the beam finally looks spherical. The last piece, the *Gouy phase* $\zeta(z)$, is an extra amount of phase, close to π in total, that the beam accumulates as it crosses the focus, an anomaly with no counterpart in the plane wave and measurable by interferometry. The amplitude factor $w_0/w(z)$ in front of the Gaussian keeps the power constant while the beam widens, since as $w(z)$ grows the peak drops so that the integrated intensity is conserved. The beam spreads with a divergence half angle $\theta = \lambda/\pi w_0$, and this reciprocal relation between waist and divergence is nothing but the diffraction limit and, once more, an uncertainty relation between the transverse position and the transverse momentum of the field. The Gaussian beam is the fundamental transverse mode of a laser resonator, the famous TEM_{00}, and it is the most confined beam that diffraction permits, which is why it is the default description of any well behaved laser and the starting point for the design of almost every optical system that works with coherent light.

The Gaussian is only the lowest member of a whole family. Since 3.27 is linear, it admits complete sets of higher order modes that share the same Gaussian envelope but decorate it with nodal structure. In Cartesian coordinates these are the *Hermite-Gauss* modes, indexed by two integers and built from Hermite polynomials, which display a rectangular grid of bright lobes. In cylindrical coordinates the natural family is that of the *Laguerre-Gauss* modes, indexed by a radial number p and an azimuthal number ℓ, whose field is

$$\begin{aligned}\vec{E}_{p\ell} = E_0\, \frac{w_0}{w(z)} \left(\frac{\rho\sqrt{2}}{w(z)}\right)^{|\ell|} L_p^{|\ell|}\left(\frac{2\rho^2}{w(z)^2}\right) \exp\left(-\frac{\rho^2}{w(z)^2}\right) e^{i\ell\varphi} \\ \times \exp\left(i\left[kz - \omega t + \frac{k\rho^2}{2R(z)} - (2p + |\ell| + 1)\,\zeta(z)\right]\right)\hat{e},\end{aligned} \tag{3.30}$$

where $L_p^{|\ell|}$ are the associated Laguerre polynomials and φ is the azimuthal angle. Two features of 3.30 deserve attention. The Gouy phase now grows in proportion to the mode order $2p + |\ell| + 1$, so different modes advance their phase at different rates, a fact that governs how a superposition of them rotates and reshapes as it propagates. Far more striking is the azimuthal factor $e^{i\ell\varphi}$, which winds the phase through ℓ complete turns around the axis and forces the field to vanish exactly on the axis, giving these beams their characteristic doughnut profile with a dark core. The wavefront is then no longer a stack of caps but a set of intertwined helices, an *optical vortex* of topological charge ℓ. This helical phase is not a mere decoration, because a field carrying $e^{i\ell\varphi}$ transports orbital angular momentum, and each of its photons carries exactly $\ell\hbar$ of it, a quantity entirely separate from the spin angular momentum $\pm\hbar$ associated with circular polarisation. For this reason the Laguerre-Gauss beams have become one of the central tools of modern optics. They are used in the *optical tweezers* to trap microscopic particles and set them spinning, in super resolution microscopy where the dark core of a doughnut beam switches off the fluorescence everywhere except at the very centre, and in classical and quantum communications alike, where the unbounded integer ℓ offers a new and in principle limitless alphabet that lets a single photon encode much more than a single bit.

All the beams written so far spread as they travel, because they are paraxial and diffraction is unavoidable for any field of finite width. It is natural to ask whether the exact Helmholtz equation admits a beam that does not spread at all, and it does. If we demand a field whose transverse intensity profile is literally independent of z, separation of variables in cylindrical coordinates leads to the *Bessel beam*

$$\vec{E} = E_0\, J_\ell(k_\perp\rho)\, e^{i\ell\varphi}\, e^{i(k_z z - \omega t)}\hat{e}, \qquad k_\perp^2 + k_z^2 = k^2, \tag{3.31}$$

where J_ℓ is the Bessel function of order ℓ. Its transverse pattern, a bright central spot when $\ell = 0$ or a vortex ring when $\ell \neq 0$, surrounded by concentric rings, is exactly the same at every plane z, so the beam is said to be *non-diffracting*. The picture behind 3.31 is illuminating, since it is the coherent superposition of plane waves whose wavevectors all lie on a cone of fixed half angle $\arctan(k_\perp/k_z)$, and the interference of that cone regenerates the same profile plane after plane. This conical construction also explains the beam's most surprising property, its *self-healing*, because if an obstacle blocks the central spot the outer rays that build the cone continue past it and reconstruct the pattern a short distance downstream. The ideal Bessel beam pays for these gifts with an infinite power, since its rings extend forever and each one carries a comparable share of energy, so in practice it is produced with an axicon or a ring aperture and remains non-diffracting only over a finite range. Its long, thin and self-repairing focus makes it invaluable for light sheet microscopy, for precise optical alignment and for material processing whenever a large depth of focus is required.

There is a last beam whose behaviour borders on the paradoxical. Looking again at the paraxial equation 3.27, now in a single transverse dimension, one finds a solution written in terms of the Airy function,

$$u(x, z = 0) = \mathrm{Ai}\left(\frac{x}{x_0}\right) e^{\,ax/x_0}, \qquad a > 0, \tag{3.32}$$

where x_0 sets the transverse scale and the exponential aperture e^{ax/x_0} is introduced to make the total energy finite. As it propagates, the *Airy beam* shows two properties that seem to contradict everything the plane wave taught us. It is non-diffracting, keeping its intensity profile over long distances, and it is *self-accelerating*, since its main lobe does not travel in a straight line but follows a parabolic trajectory $x \propto z^2$, so the beam bends sideways even though no force and no index gradient act on it. Nothing is being violated here, because the centre of mass of the whole field does travel straight, and it is only the intensity pattern that curves, in the same way that the crest of a water wave may move differently from the water itself. Like the Bessel beam it is also self-healing, and these curved, resilient trajectories have been used to route light around obstacles, to guide electric discharges along curved plasma channels, to sweep microscopic particles along a chosen path and, once again, in advanced microscopy.

Seen together, this family teaches a lesson that the plane and the spherical waves, taken alone, keep hidden, namely that what organises a beam and endows it with a useful behaviour is the shape of its phase across the wavefront. A phase linear in the transverse coordinate, $e^{i\vec{k}_\perp\cdot\vec{\rho}}$, simply tilts the beam. A phase quadratic in ρ, $e^{ik\rho^2/2R}$, focuses or defocuses it and is exactly what a thin lens imprints. A phase linear in the azimuth, $e^{i\ell\varphi}$, gives it a vortex and orbital angular momentum. A conical phase produces the non-diffracting Bessel beam, and a cubic phase produces the self bending Airy beam. Modern devices such as the spatial light modulators and the metasurfaces let us paint almost any of these phase profiles onto a wavefront at will, and this ability to engineer the phase is the whole discipline known today as *structured light*. The plane wave and the spherical wave remain useful as building blocks and as limiting cases, but from here on the reader should keep in mind that they are only two convenient points in a vastly richer landscape of solutions.

3.4.3 From 3D wavefronts to 2D wavefronts

As we've seen the electromagnetic radiation has got many presentations, however, there is a very strong formulation where the light is studied in $2D$, in fact, the following chapter treats the electromagnetic radiation as a $2D$ field.

From the mathematical point of view there is a whole theory about arbitrary surfaces or complex geometries, here, we're going to use the properties of *smooth manifolds*[1] This is thanks to the nature

[1]A smooth manifold if a set which can be completely covered by subsets and there are smooth functions from the manifold to $\mathbb{R}^n$.

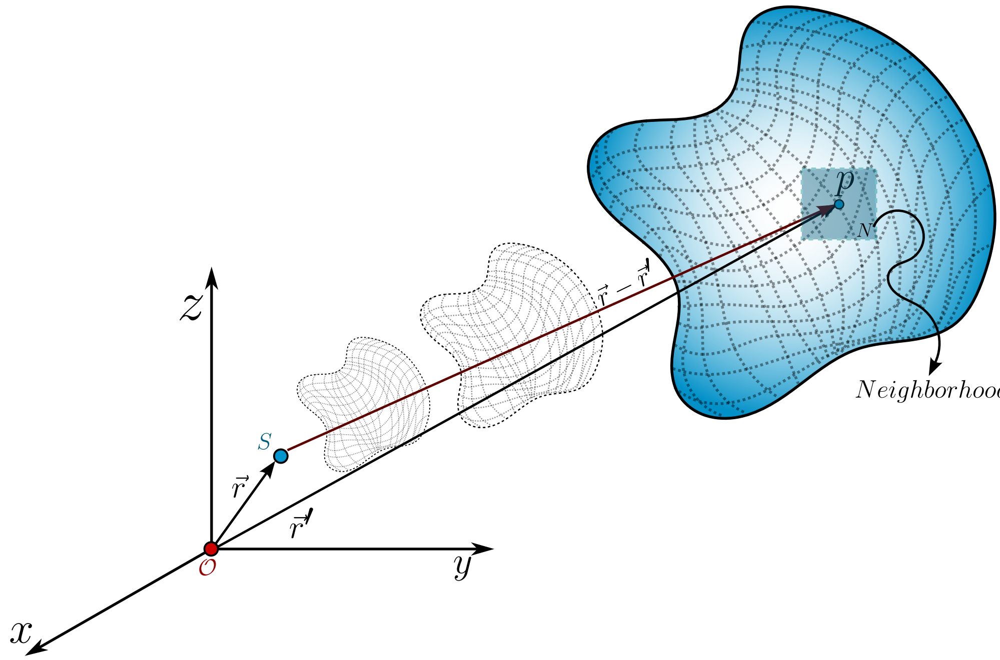


Figure 3.1: A source $\mathcal{S}$ produces an arbitrary kind of wave which propagates in the space-time, then, for any given time and position we can take an arbitrary point p and, due to the smooth manifold nature of this wave, there exists a neighborhood N of p where the wavefront is plane.

of the physical solution of the wave equations, so, given any arbitrary wavefront as shown 3.1 we can choose any point a set and observer on a neighborhood N, and because the wavefront behaves as an smooth manifold then this neighborhood is $\mathbb{R}^2$.

What we are doing is thinkg on this surface like the earth, because we're son tiny we see the earth as plane but it is not. The same phenomena is here, we step on and observer across the point p on the wavefront and for us there is a neighborhood N where it behaves exactly as $\mathbb{R}^2$ ad that is going to allow us to start the study of the polarisation.

3.5 Light's polarisation

In the electromagnetic theory the word *polarisation* is used as a tendency of the electric dipoles to orientate in a given microscopic volume. However, this word also is used to describe the **electric field orientation** in its propagation. Those two definitions then seem completely indepent, however, as we are going to show in the microscopic theory of radiation these two are completely related.

The polarisation of light is one of the most usefull properties of light, there are many applications on real life which uses this property, for example, the cancer cells stole the glucose from the healthy ones, so there is a higher sugar concentration in cancer cells than the healthy ones, so if we irradiate a sample with linearly polarized light and study the transmited radiation, we're going to see that the light which went across the cancer cells changed their polarization, rotating a given agle (which depends on the concentration of glucose) so we can detected the presence of cancer cells using the polarisation.

3.5.1 Monochromatic and deterministic formulation: Jones spinors and the polarization ellipse

As we've said, we're going to focus on the electric field behavior in a $2D$ formulation, so the electric field is going to be a vector field $\vec{E} \in \mathcal{C}^2\left(\mathbb{C}^2\right)$ which means it is *a complex vector function which their second derivatives are continuous.* Its general form is given by

$$\vec{E}(\vec{r}, t; \omega) = \begin{bmatrix} \mathcal{E}_1 e^{i\left(\vec{k}\cdot\vec{r}-\omega t+\phi_1\right)} \\ \mathcal{E}_2 e^{i\left(\vec{k}\cdot\vec{r}-\omega t+\phi_2\right)} \end{bmatrix}, \tag{3.33}$$

where we're considering only one frequency ω so this describes the electric field of a monocrhomatic field and as we can see, this vector field is a plane wave.

We can factor the propagation phase $\exp\left[i\left(\vec{k}\cdot\vec{r}-\omega t\right)\right]$ in both components ad we will have the following vector field

$$\vec{e} = \begin{bmatrix} \mathcal{E}_1 e^{i\phi_1} \\ \mathcal{E}_2 e^{i\phi_2} \end{bmatrix}. \tag{3.34}$$

The vector field 3.34 is known as the *Jones vector* but, in fact, their true nature ist vectorial, the field 3.34 is a **spinor**[2] a we're going to see this spinor behavior in this chapter.

The Jones spinor 3.34 is a vector that describes how the electric field oscillates, in general, under our assumptions[3] is an ellipse, well known as the polarization ellipse. To get the ellipse from 3.34 we're going to define the following electric field componentes, given by

$$\frac{E_1}{\mathcal{E}_1} e^{-i\phi_2} = \frac{1}{2} e^{i\triangle\phi}, \tag{3.35}$$

$$\frac{E_2}{\mathcal{E}_2} e^{-i\phi_1} = \frac{1}{2} e^{-i\triangle\phi}, \tag{3.36}$$

$$\triangle\phi = \phi_1 - \phi_2.$$

Now, we're going to calculate the magnitude of the difference between 3.35 and 3.36, and because these are complex number, it is given by

$$\left(\frac{E_1}{\mathcal{E}_1} e^{-i\phi_2} - \frac{E_2}{\mathcal{E}_2} e^{-i\phi_1}\right)\left(\frac{E_1}{\mathcal{E}_1} e^{-i\phi_2} - \frac{E_2}{\mathcal{E}_2} e^{-i\phi_1}\right)^* = \frac{1}{4}\left(e^{i\triangle\phi} - e^{-i\triangle\phi}\right)\left(e^{i\triangle\phi} - e^{-i\triangle\phi}\right)^*,$$

[2] We can see a spinor a $1/2$ tensor. From a geometric point of view, we say that a vector $\vec{v}$ is, in fact, and spinor if it needs two complete rotation of 360° to arrive its original position

[3] deterministic, $2D$, monocrhomatic, space-time independent, propagation on vacuum

where $*$ represents the complex conjugate. In the right hand side we can see tha the difference between those two exponentials are the sine of the phase difference $\triangle$ thus

$$\frac{1}{4}\left(e^{i\triangle\phi}-e^{-i\triangle\phi}\right)\left(e^{i\triangle\phi}-e^{-i\triangle\phi}\right)^{*}=\frac{1}{4}\left(2i\sin\triangle\phi\right)\left(-2i\sin\triangle\phi\right)=\sin^{2}\triangle\phi. \tag{3.37}$$

In the left hand side we've got the following

$$\left(\frac{E_1}{\mathcal{E}_1}e^{-i\phi_2}-\frac{E_2}{\mathcal{E}_2}e^{-i\phi_1}\right)\left(\frac{E_1}{\mathcal{E}_1}e^{-i\phi_2}-\frac{E_2}{\mathcal{E}_2}e^{-i\phi_1}\right)^{*}=$$
$$\frac{|E_1|^2}{\mathcal{E}_1^2}-\frac{E_1E_2^*}{\mathcal{E}_1\mathcal{E}_2}e^{i\triangle\phi}-\frac{E_2E_1^*}{\mathcal{E}_2\mathcal{E}_1}e^{-i\triangle\phi}+\frac{|E_2|^2}{\mathcal{E}_2^2},$$

the crossed terms in blue corresponds to the same quantity but one is the complex conjugate of the other, so if we factor the minus sign we'll have the real part of that complex number, so, the left hand side is given by

$$\frac{|E_1|^2}{\mathcal{E}_1^2}+\frac{|E_2|^2}{\mathcal{E}_2^2}-2\frac{Re\left\{E_1E_2^*\right\}}{\mathcal{E}_1\mathcal{E}_2}\cos\triangle\phi. \tag{3.38}$$

Combining the LHS 3.38 and the RHS 3.37 we will have the *polarisation ellipse*

$$\frac{|E_1|^2}{\mathcal{E}_1^2}+\frac{|E_2|^2}{\mathcal{E}_2^2}-2\frac{Re\left\{E_1E_2^*\right\}}{\mathcal{E}_1\mathcal{E}_2}\cos\triangle\phi=\sin^2\triangle\phi. \tag{3.39}$$

Then we've shown the electric field lives on an geometric space on $\mathcal{C}^2$ which is a real rotated elllipse, before going further we should notice something.

The Jones spinor 3.34 is not an unique thing of the electric field, also describes the magnetic field under these assumptions but also **describes any complex field with 2 degrees of freedom**, for example the gravitational waves have got a polarisation ellipse, $2D$ phonons have polarisation ellipse, and so on.

From a mathematical point of view the origin of the ellipse is obvious, the Jones spinor 3.34 corresponds to two unit complex circles which rotate asincronically, so the general figure is an ellipse not a circle, but, there is a further a more fundamental question, and it is **why does the nature choose an elllipse as the fundamental description of the orientation of the electric field** The short answer is: $No, the nature does not choose that$, the long answer is going to be shown in the theory of coherence chapter.

3.5.1.1 How to build the polarisation ellipse: Phasors

Given any Jones spinor 3.34 how can we construct the polarisation ellipse? To answer this we're going to use phasors which is a formal word for a polar form of a complex number.

Because we've got a $2D$ vector we've got two components, two polar complex numbers $\mathcal{E}_m e^{i\phi_m}$ with $m = 1, 2$ so we can describe these two phasors as two circunferences as is shown in 3.2. The

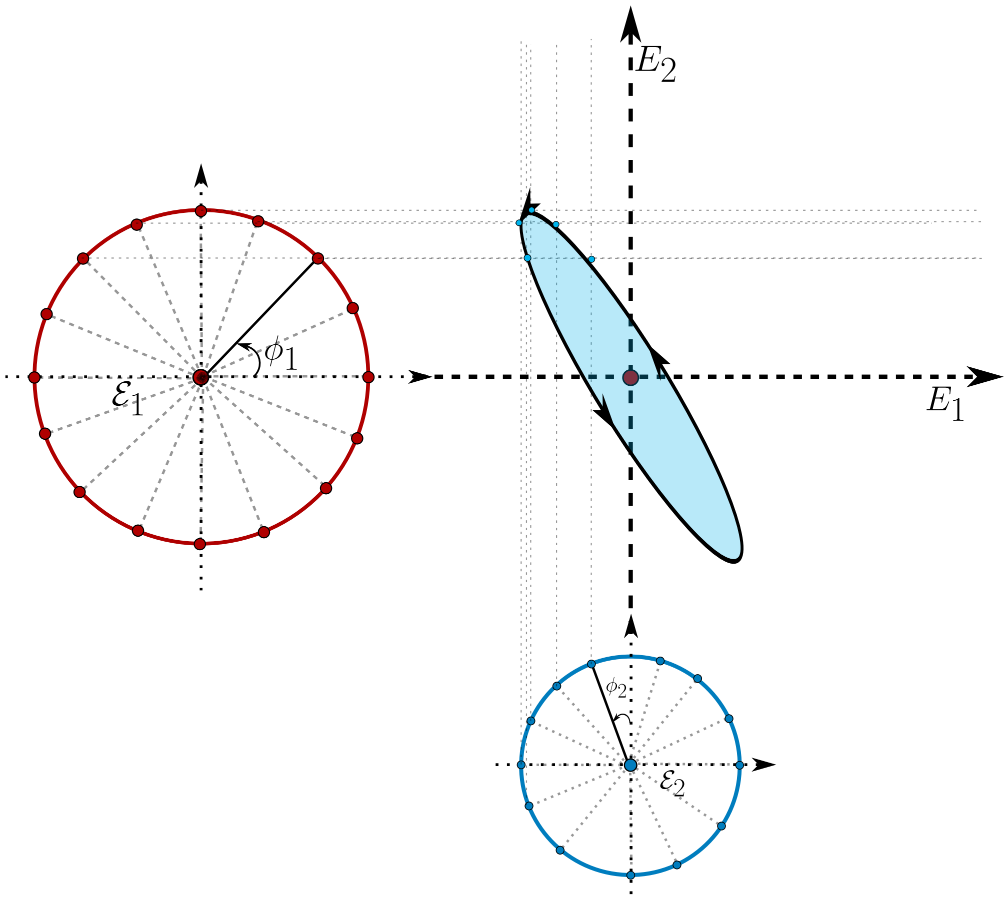


Figure 3.2: The polarisation ellipse can be reconstructred taking all the possible rotations of the two phasors (which corresponds to the compoenentes of the Jones spinor).

x component or $1-st$ component is a circunference with radius $\mathcal{E}_1$ and the initial point on this circunference is located at an angle ϕ_1 respect to the $x-axis$ (shown as the red circunference in the figure 3.2). The second phasor, corresponding to the y component of the jones spinor is another circunference but the radius is $\mathcal{E}_2$ and the initial point is located at an angle of ϕ_2 but respect to the $y-axis$.

To construct the polarisation ellipse, we start with the two initial points an draw two straight lines, for the horizontal phasor (the red one) the lines are horizontal, and for the vertical phasor (the blue one) the lines are vertical. In the intersection of the two rects we draw a point, and we desplace both initial points by the same angle, lets call it $\delta\phi$ until we arrive at the initial point. The set of all intersection points draw the polarisation ellipse, which their semi-axes are going to be $\mathcal{E}_1, \mathcal{E}_2$ and the orientation is going to depends on $\triangle\phi$ if $\triangle\phi = \frac{\pi}{2} + n\pi$ with $n \in \mathbb{Z}$ the ellipse isn't going to be orientated also $\triangle\phi$ conditionates if the electric field is going to rotate clockwise or counterclockwise.

3.5.1.2 How to build the polarisation ellipse: Ellipse angles

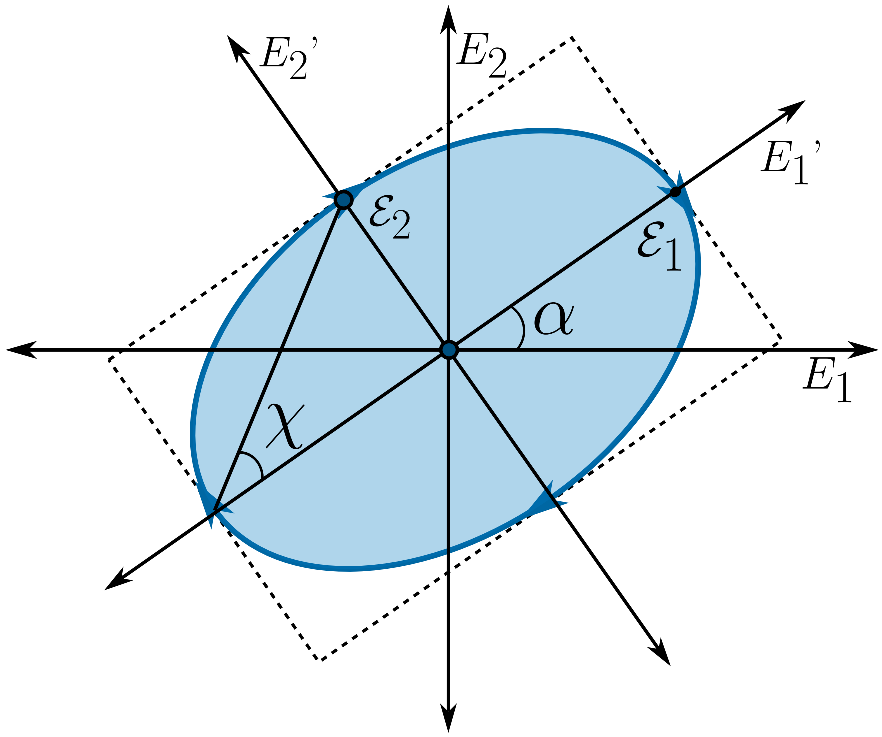


Figure 3.3: The general form of the polarisation ellipse can be described with only two angles; the orientation angle α and the elipticity angle χ.

It is not very comfortable to work with phasor and even we can't see the ellipse inmediately with the Jones spinor, so we would like a more visual or efficient way to interpretate the polarisation. First we must notice that the electric field amplitudes only change the size of the ellipse, but the relative phases ϕ_1, ϕ_2 are the ones who determine the form of the ellipse, so we can work out with a Jones spinor as an unit vector, this means $\vec{e} \to \hat{e}$ such that $\hat{e} \cdot \hat{e} = 1$.

The general polarisation ellipse can be seen in 3.3 which can be fully described using just two angles α and χ.

Definition 11 *(Characteristic polarisation angles): We define the two characteristic polarisation angles as the orientation angle $-\pi \leq \alpha \leq \pi$ which describes how orientated is the ellipse respect to the horizontal axis, and the elipticity angle $-\pi/2 \leq \chi \leq \pi/2$ which describes completily the form of the ellipse.*

How can we translate the Jones spinor with characteristic angles? To do this, lets remind that without any loss of generality we can say the Jones spinor is unitary ($\hat{e} \cdot \hat{e} = 1$) and lets

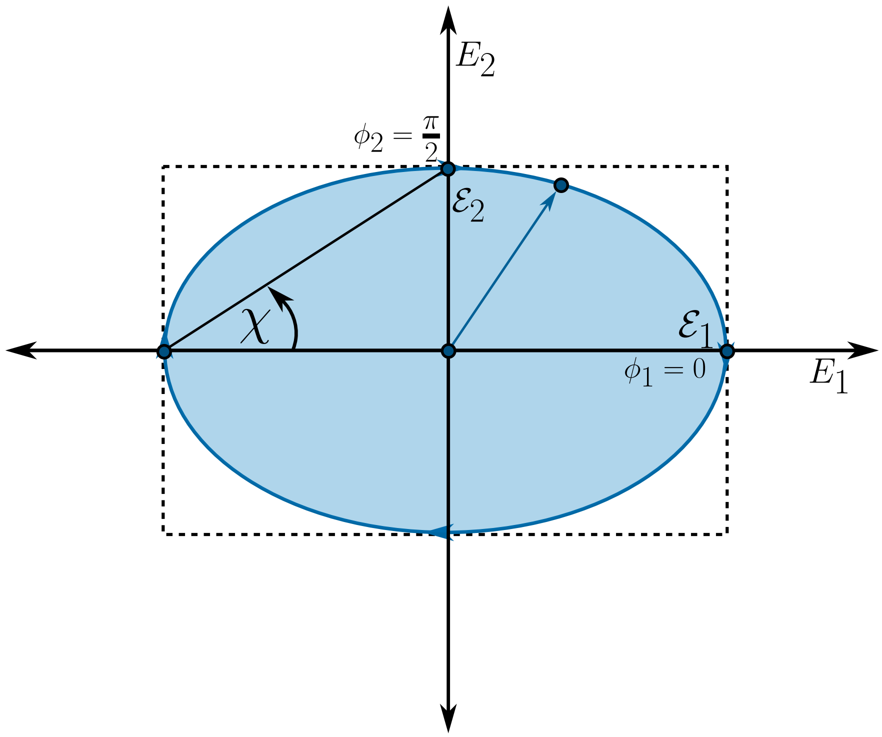


Figure 3.4: The polarisation ellipse is completely unrotated due to the relative phases between the componentes of the electric field.

consider the following phases $\phi_1 = 0$ and $\phi_2 = \pi/2$ and we'll have a polarization elllipse as in shown in 3.4.

As we can see, the polarization vector can be describe using a polar parametrization with the angle χ, so the Jones spinor becomes

$$\hat{e}(\chi) = \begin{bmatrix} \cos\chi \\ i\sin\chi \end{bmatrix}, \tag{3.40}$$

the i term comes from the phase $\phi_2 = \pi/2$ where $\exp(\pi/2) = i$. Now, to construct the full polarization states shown in 3.3 we only have to rotate the Jones spinor 3.40 an angle α, so, the polarisation states is given by

$$\hat{e}(\alpha,\chi) = \begin{bmatrix} \cos\alpha & -\sin\alpha \\ \sin\alpha & \cos\alpha \end{bmatrix} \begin{bmatrix} \cos\chi \\ i\sin\chi \end{bmatrix},$$
$$\hat{e}(\alpha,\chi) = \begin{bmatrix} \cos\alpha\cos\chi - i\sin\alpha\sin\chi \\ \sin\alpha\cos\chi + i\cos\alpha\sin\chi \end{bmatrix}$$

Definition 12 *(Polarisation convention): From now on in this lectures whenever we're working with polarisation states in terms of the angles* $(\alpha.\chi)$ *instead of using* $\hat{e}(\alpha,\chi)$ *we're going to use the ket* *notation, therefore* $\hat{e}(\alpha,\chi) \longmapsto |\alpha;\chi\rangle$ *where we use the semicolon* ; *instead of the comma* , *because the angles are two parameters and does not represente a two level ket* ($|\alpha,\chi\rangle$) *(however polarisation is a two level system).*

Having on count this convention, we've got that the general description of a polarisation state in terms of the characteristic angles is given by

$$|\alpha;\chi\rangle = \begin{bmatrix} \cos\alpha\cos\chi - i\sin\alpha\sin\chi \\ \sin\alpha\cos\chi + i\cos\alpha\sin\chi \end{bmatrix}. \tag{3.41}$$

Because the polarisation is a $2D$ vector in this formulation it can be written as the superposition of, at leat, two linearly independent polarisation states. There is a convention, whenever we talk about polarisation we always write it as the superposition of two ortognal polarisation states, so, how can we build this orthogonal states? We have got two answers, using the orientation angle or the elipticity angle. Using the orientation angle is the most obvious and inmediate way, because we only have to put the two ellipses orthogonal, this means taking the state $|\alpha;\chi\rangle$ and rotate it like $|\alpha + \frac{\pi}{2};\chi\rangle$.

To think about orthogonal states using elipticity we only have got to think about that given any two orthogonal polarisation states $|0;\chi\rangle$, $|0;\chi'\rangle$ the following two states $|\alpha;\chi\rangle$ and $|0;\chi'\rangle$ are also orthogonal because the rotation transformation preservers the relative angle between vectors. Having this on count, lets suppose that $\chi' = \chi + \theta$ and by orthogonality lets calculate the value of the angle θ that guarantees the orthogonality between polarisation states. Lets calculate the dot product

$$\begin{aligned}
&\langle 0;\chi|0;\chi+\theta\rangle = [\cos\chi, -i\sin\chi]\begin{bmatrix} cos(\chi+\theta) \\ i\sin(\chi+\theta)\end{bmatrix} = 0,\\
&\cos\chi\cos(\chi+\theta) + \sin\chi\sin(\chi+\theta) = 0,\\
&\cos\left(\chi+\theta-\chi\right) = 0,\\
&\cos\theta = 0 \longrightarrow \theta = (2n-1)\frac{\pi}{2} \quad n\in\mathbb{Z},
\end{aligned}$$

where we have used the cosine of the sum of two angles, so for the cosine of θ to vanish it must ve an odd multiple of $pi/2$, without any loss of generality lets take $n = 1$, so $\theta = \pi/2$ and the orthogonal state can be obtained just taking a displacemente of $\pi/2$ on the elipticity, then any polarization state can be writeen as

$$|\alpha;\chi\rangle = c_1\,|\alpha';\chi'\rangle + c_2\left|\alpha';\chi' + \frac{\pi}{2}\right\rangle. \tag{3.42}$$

3.5.1.3 The canonical polarisation states: horizontal, vertical, diagonal and circular

Equation 3.41 already contains the whole catalogue of polarisation states, so it is worth stopping here and finding explicitly the handful of states that we are going to meet again and again in the laboratory: the two linear states aligned with the coordinate axes, the two diagonal linear states, and the two circular states. For every one of them we want three things: the amplitudes and phases

$\mathcal{E}_1, \mathcal{E}_2, \phi_1, \phi_2$ of the original spinor 3.34, normalised so that $\mathcal{E}_1^2 + \mathcal{E}_2^2 = 1$, and the pair of characteristic angles (α, χ) that reproduce that spinor in 3.41.

Since $|\alpha;\chi\rangle$ has got a real first term in its combination and a purely imaginary second term whenever we compare it componentwise with a target spinor $\vec{e} = [e_1, e_2]^\mathsf{T}$, it is convenient to split 3.41 into its real and imaginary parts once and for all,

$$\cos\alpha\cos\chi = \mathrm{Re}\{e_1\}, \quad -\sin\alpha\sin\chi = \mathrm{Im}\{e_1\}, \quad \sin\alpha\cos\chi = \mathrm{Re}\{e_2\}, \quad \cos\alpha\sin\chi = \mathrm{Im}\{e_2\}. \tag{3.43}$$

Every state below is obtained by feeding a particular $\vec{e}$ into the system 3.43 and solving for α, χ inside their allowed ranges $-\pi \leq \alpha \leq \pi$ and $-\pi/2 \leq \chi \leq \pi/2$.

Horizontal linear polarisation. The field only oscillates along the x axis, so the second component of the spinor must vanish, $\mathcal{E}_2 = 0, \mathcal{E}_1 = 1$. A common phase multiplying the whole spinor has got no physical meaning, since it is only a redefinition of the origin of time, so we are free to set $\phi_1 = 0$; the phase ϕ_2 plays no role at all because it multiplies a null amplitude. The normalised spinor is

$$\vec{e}_H = \begin{bmatrix}1\\0\end{bmatrix}, \qquad \phi_1 = 0, \quad \mathcal{E}_2 = 0. \tag{3.44}$$

Plugging $e_1 = 1, e_2 = 0$ into 3.43 gives $\cos\alpha\cos\chi = 1$ together with $\sin\alpha\sin\chi = \sin\alpha\cos\chi = \cos\alpha\sin\chi = 0$, and because $\cos\alpha\cos\chi = 1$ is the maximum value that a product of two cosines can reach, it forces $\cos\alpha = \cos\chi = 1$, that is

$$\boxed{\alpha = 0, \qquad \chi = 0} \qquad \Longrightarrow \qquad |0;0\rangle = \begin{bmatrix}1\\0\end{bmatrix}. \tag{3.45}$$

Vertical linear polarisation. By the same argument, now the field only oscillates along y, so $\mathcal{E}_1 = 0, \mathcal{E}_2 = 1$ and we set $\phi_2 = 0$ (again, ϕ_1 is irrelevant),

$$\vec{e}_V = \begin{bmatrix}0\\1\end{bmatrix}, \qquad \phi_2 = 0, \quad \mathcal{E}_1 = 0. \tag{3.46}$$

Feeding $e_1 = 0, e_2 = 1$ into 3.43 gives $\sin\alpha\cos\chi = 1$, which forces $\sin\alpha = \cos\chi = 1$, so

$$\boxed{\alpha = \frac{\pi}{2}, \qquad \chi = 0} \qquad \Longrightarrow \qquad |\pi/2;0\rangle = \begin{bmatrix}0\\1\end{bmatrix}. \tag{3.47}$$

Diagonal linear polarisation, at $\pi/4$ from the horizontal. Now both components carry the same amplitude, $\mathcal{E}_1 = \mathcal{E}_2 = 1/\sqrt{2}$, and, this being the new ingredient, they also share the same phase, $\phi_1 = \phi_2 = 0$, so that $\triangle\phi = \phi_1 - \phi_2 = 0$ and

$$\vec{e}_D = \frac{1}{\sqrt{2}}\begin{bmatrix}1\\1\end{bmatrix}, \qquad \phi_1 = \phi_2 = 0. \tag{3.48}$$

Here 3.43 reads $\cos\alpha\cos\chi = \sin\alpha\cos\chi = 1/\sqrt{2}$ and $\sin\alpha\sin\chi = \cos\alpha\sin\chi = 0$. If $\sin\chi$ were different from zero the last two equations would force $\sin\alpha = \cos\alpha = 0$ at once, which is impossible, so we must have $\chi = 0$; the first two equations then reduce to $\cos\alpha = \sin\alpha = 1/\sqrt{2}$, giving

$$\boxed{\alpha = \frac{\pi}{4}, \qquad \chi = 0}. \tag{3.49}$$

Antidiagonal linear polarisation, at $-\pi/4$ from the horizontal. The amplitudes are the same as in the diagonal case, but now the two components oscillate in opposition, $\phi_1 = 0, \phi_2 = \pi$, so that $e^{i\phi_2} = -1$, $\triangle\phi = -\pi$, and

$$\vec{e}_A = \frac{1}{\sqrt{2}}\begin{bmatrix}1\\-1\end{bmatrix}, \qquad \phi_1 = 0, \quad \phi_2 = \pi. \tag{3.50}$$

The same reasoning used above forces again $\chi = 0$, and now $\cos\alpha = 1/\sqrt{2}, \sin\alpha = -1/\sqrt{2}$, so

$$\boxed{\alpha = -\frac{\pi}{4}, \qquad \chi = 0}. \tag{3.51}$$

It is worth pausing on the four linear states 3.45, 3.47, 3.49 and 3.51: all of them share $\chi = 0$, and this is not a coincidence, since the ellipticity angle measures how far the ellipse has departed from a straight segment, and every linear state is nothing but a degenerate ellipse of zero ellipticity. The orientation angle α is the only one that changes from state to state, and it does exactly what its name promises, it tells us the tilt of that segment with respect to the x axis.

Circular polarisations. We now ask for the opposite extreme, the states in which the two orthogonal oscillations have got the same amplitude, $\mathcal{E}_1 = \mathcal{E}_2 = 1/\sqrt{2}$, but are shifted by a quarter of a cycle rather than being in phase or in opposition, $\phi_1 = 0, \phi_2 = \mp\pi/2$, so that $e^{i\phi_2} = \pm i$, $\triangle\phi = \pm\pi/2$, and

$$\vec{e}_\pm = \frac{1}{\sqrt{2}}\begin{bmatrix}1\\\pm i\end{bmatrix}, \qquad \phi_1 = 0, \quad \phi_2 = \mp\frac{\pi}{2}. \tag{3.52}$$

Plugging $e_1 = 1/\sqrt{2}, e_2 = \pm i/\sqrt{2}$ into 3.43, the real part of the second line, $\sin\alpha\cos\chi = 0$, together with the imaginary part of the first line, $\sin\alpha\sin\chi = 0$, can only be satisfied inside the allowed ranges by $\alpha = 0$ (the case $\cos\chi = 0$ is discarded because it would force the left hand side of $\cos\alpha\cos\chi = 1/\sqrt{2}$ to vanish). With $\alpha = 0$ the remaining equations collapse to $\cos\chi = 1/\sqrt{2}$ and $\sin\chi = \pm 1/\sqrt{2}$, so

$$\boxed{\alpha = 0, \qquad \chi = \pm\frac{\pi}{4}} \qquad \Longrightarrow \qquad |0; \pm\pi/4\rangle = \frac{1}{\sqrt{2}}\begin{bmatrix}1\\\pm i\end{bmatrix}. \tag{3.53}$$

Notice that the orientation angle α is completely undetermined by the shape of the state, and indeed it must be so, a circle has got no preferred direction to be tilted towards, so setting $\alpha = 0$ is only a matter of convenience. What really distinguishes the two circular states is the sign of χ, and this sign is directly inherited from the sign of $\triangle\phi = \phi_1 - \phi_2$, exactly as anticipated when we built the ellipse from the two phasors: a positive $\triangle\phi$ produces $\chi < 0$ and a negative $\triangle\phi$ produces $\chi > 0$. To see the rotation itself, we restore the time dependence at a fixed point in space and take the real part of the spinor 3.52,

$$E_1(t) = \frac{1}{\sqrt{2}}\cos(\omega t), \qquad E_2(t) = \frac{1}{\sqrt{2}}\cos\left(\omega t \mp \frac{\pi}{2}\right) = \pm\frac{1}{\sqrt{2}}\sin(\omega t),$$

so that at $\omega t = 0$ the vector (E_1, E_2) points along $+x$, and an instant later it has acquired a small positive component along $+y$ for the upper sign and along $-y$ for the lower sign. In other words, the state with $\chi = +\pi/4$ traces a counterclockwise circle and the state with $\chi = -\pi/4$ traces a clockwise circle in the x-y plane as time goes on. Whether counterclockwise is called left or right circular polarisation depends on whether the observer is defined to look along the propagation direction or against it, a convention we postpone fixing until the Poincaré sphere is introduced; for the moment we simply keep the two states labelled by the unambiguous quantity $\chi = \pm\pi/4$, the largest ellipticity angle allowed, exactly as $\chi = 0$ was the smallest.

Together, the six states 3.45, 3.47, 3.49, 3.51 and 3.53 already reveal the whole geometric role of the pair (α, χ): α only rotates a state that has got some elongation, while χ interpolates continuously between a straight segment at $\chi = 0$ and a circle at $|\chi| = \pi/4$, with its sign fixing the sense in which the tip of the electric field turns; every other value of χ in between will describe, as we should expect by now, a genuine ellipse of intermediate eccentricity.

3.5.2 Time evolution of the polarisations: Stokes parameters and the Poincaré sphere

The mathematics developed so far describes a beam of light whose polarisation is perfectly defined at every instant, the situation formalised by the Jones spinor 3.34. Real light is seldom so cooperative: sunlight, the light of an incandescent lamp, or even laser light after it has bounced once off a rough surface, is generally a superposition of many independent wavetrains whose relative phase drifts randomly and far too fast to be tracked. Nothing in the deterministic polarisation ellipse built in the previous subsection can describe such a beam, and it was precisely this gap that George Gabriel Stokes set out to close in 1852, in a paper read before the Cambridge Philosophical Society and modestly titled *On the composition and resolution of streams of polarized light from different sources* (Stokes, 1852). Stokes, who a few years later became Lucasian Professor of Mathematics at Cambridge (the same chair later held by Dirac and, more recently, by Hawking), and whose name today is more readily attached to the Navier-Stokes equations of fluid motion, to Stokes' theorem of vector calculus and to the Stokes shift of fluorescence, proposed four real quantities, built from simple combinations of intensities measured behind a polariser and a quarter wave plate held at a handful of fixed orientations, that could characterise the polarisation of any beam whatsoever, fully polarised, fully unpolarised, or anything in between. Unlike the Jones spinor 3.34, which needs a single coherent phase to be written down at all, Stokes' four numbers are honest time averages, and they remain perfectly well defined even when no such phase exists.

It is a curious fact that this construction, so evidently useful and so close in spirit to the density matrix that quantum mechanics would introduce some seventy years later for exactly the same reason (to describe mixtures that a single wavefunction cannot capture), attracted almost no attention for the rest of the nineteenth century. Stokes himself never returned to it: he went on to serve as secretary and later as president of the Royal Society, absorbed by the administrative life such offices demanded, and his scientific attention drifted to fluorescence, to the wave theory of light in the aether, and above all to hydrodynamics. Nor did the wider community pick it up. Optics in the second half of the nineteenth century was still occupied with problems that were, by construction, coherent and deterministic, interference, diffraction, the propagation of well defined wavefronts, and the statistical treatment of an ensemble of independent wavetrains had no natural home in that programme. Even the form in which Stokes left his result, four numbers without an accompanying algebra of matrices to act on them, made it look more like a table of measurements than like a theory. His four coefficients, not yet known by the name *Stokes parameters* that later authors would give them, waited in the pages of the *Transactions of the Cambridge Philosophical Society* for the better part of a century.

The first person to give this problem a genuinely new geometric life, although without building explicitly on Stokes' algebra, was Henri Poincaré. Poincaré (1854 to 1912) is often remembered as the last of the great universalist mathematicians, equally at home in celestial mechanics, where his study of the three body problem contained, without his fully realising it, the first seeds of chaos theory, in topology, where the conjecture that bears his name would only be settled a century later, and in the foundations of mathematical physics, where his work on the Lorentz group anticipated several of the kinematic ingredients of special relativity. In 1892, in the second volume of his lecture course *Théorie Mathématique de la Lumière*, Poincaré proposed representing every state of polarisation as a single point on the surface of a sphere (Poincaré, 1892). He built this sphere directly from the orientation and ellipticity angles of the polarisation ellipse, the very angles α and χ of the ket 3.41, so that the poles of the sphere fall on the two circular states, its equator is entirely covered by the linear states, exactly as anticipated in the historical overview at the beginning of this chapter, and every other point corresponds to one particular ellipse. It was only later, once the sphere and Stokes' four numbers were laid side by side, that it became clear the coordinates Poincaré had been plotting are, up to normalisation by the total intensity, precisely the three last Stokes parameters: two men, working four decades apart with entirely different tools, one through operational measurements of intensity and the other through the geometry of spherical trigonometry, had arrived at the same object from opposite directions. The sphere has carried Poincaré's name ever since, even though, as we shall see, its natural coordinates are Stokes'.

The parameters finally found the audience they deserved when a new generation of physicists needed to follow partially polarised light through processes that destroy coherence outright, above all multiple scattering in planetary atmospheres, in colloidal suspensions, and in the interstellar medium, a regime where the Jones calculus (Jones, 1941), powerful as it is, simply cannot be used, since it presupposes a single coherent amplitude that scattering does not leave behind. Soleillet, in 1929, and Francis Perrin, in a 1942 study of light scattered by opalescent media (Perrin, 1942), independently rebuilt essentially the same four component, matrix based description that the problem demanded, at first without fully recognising it as Stokes' own. The decisive act of recognition, and the paper usually credited with rescuing Stokes' parameters from a century of obscurity, restoring his original notation and giving the whole subject the name by which it is still taught, was published by Walter McMaster in the *American Journal of Physics* in 1954 (McMaster, 1954). Subrahmanyan Chandrasekhar's monumental treatise on radiative transfer, appearing in the same decade, adopted the Stokes vector as the natural object to propagate through a scattering atmosphere and cemented its place in astrophysics (Chandrasekhar, 1950). From ellipsometry and remote sensing to the polarisation maps of the cosmic microwave background and to modern quantum optics, where the Stokes parameters reappear as the expectation values of the photon's Pauli operators, the formalism that Stokes wrote down in 1852, that the nineteenth century forgot, and that Poincaré unknowingly rediscovered forty years later in the language of spheres, is today one of the most used tools in the whole of optics.

Let us now build these four parameters explicitly, starting from nothing more than the polarisation ellipse 3.39 itself, and see why Stokes was forced to define them as time averages rather than simply reading them off the field.

The ellipse 3.39 is a fixed curve, but the physical field is not a curve, it is a point running around that curve once every optical period. To say this with equations, the real components E_1, E_2 that satisfy 3.39 at every instant are themselves functions of time,

$$E_1(t) = \mathcal{E}_1 \cos(\omega t - \phi_1), \qquad E_2(t) = \mathcal{E}_2 \cos(\omega t - \phi_2), \tag{3.54}$$

the real part of the electric field 3.34 at a fixed point in space (its constant spatial phase $\vec{k} \cdot \vec{r}$ has simply been absorbed into ϕ_1, ϕ_2). Therefore the expression for the polarisation ellipse is given by

$$\frac{|E_1(t)|^2}{\mathcal{E}_1^2} + \frac{|E_2(t)|^2}{\mathcal{E}_2^2} - 2\frac{Re\left\{E_1(t)E_2^*(t)\right\}}{\mathcal{E}_1\mathcal{E}_2} \cos \triangle\phi(t) = \sin^2 \triangle\phi(t) \tag{3.55}$$

One can check directly that eliminating t between the two equations 3.54, exactly as was done with the phasors 3.35 and 3.36, returns 3.39; the ellipse is nothing but the trace, frozen in space, left behind by the point $(E_1(t), E_2(t))$ as it oscillates.

A single period of visible light lasts a few femtoseconds, $T = \lambda/C \sim 2 \times 10^{-15}$ s for green light, while even the fastest photodetector available in a laboratory needs picoseconds to respond, and the human eye needs tens of milliseconds. In either case an enormous number of optical cycles, somewhere between 10^6 and 10^{13} of them depending on the instrument, go by before any detector can register a single reading; the point $(E_1(t), E_2(t))$ has swept around the ellipse an enormous number of times before any instrument manages to read anything at all. No detector, in other words, can ever measure the instantaneous field $E(t)$ itself, only some average of it taken over a great many periods. This is not a limitation that better technology could one day remove, it is what intensity means: the energy flux carried by a wave is proportional to the square of the field, so what a detector physically integrates is always $\left\langle E^2 \right\rangle$, never E. This is exactly the operation Stokes performed, building his four parameters out of quadratic, time averaged combinations of the oscillating field rather than out of the ellipse itself, and it is why they, unlike the Jones spinor, remain meaningful even when no single coherent phase exists to write down. Over one period $T = 2\pi/\omega$, define the time average of any function $f(t)$ as $\langle f \rangle \equiv \frac{1}{T}\int_0^T f(t)\,dt$. The first two averages we need are the mean square amplitudes,

$$\langle E_1^2 \rangle = \frac{\mathcal{E}_1^2}{T}\int_0^T \cos^2(\omega t - \phi_1)\,dt = \frac{\mathcal{E}_1^2}{2T}\int_0^T \left[1 + \cos(2\omega t - 2\phi_1)\right] dt = \frac{\mathcal{E}_1^2}{2T}\left[T + 0\right] = \frac{\mathcal{E}_1^2}{2},$$

where we used $\cos^2\theta = \frac{1}{2}(1+\cos 2\theta)$ and the fact that $\cos(2\omega t - 2\phi_1)$ completes exactly two full oscillations as t runs from 0 to T, so its integral over that stretch vanishes; the identical computation gives $\langle E_2^2\rangle = \mathcal{E}_2^2/2$.

The remaining piece of information lives in how E_1 and E_2 oscillate together, and for that we need the cross average

$$\begin{aligned}\langle E_1E_2\rangle &= \frac{\mathcal{E}_1\mathcal{E}_2}{T}\int_0^T \cos(\omega t-\phi_1)\cos(\omega t-\phi_2)\,dt\\ &= \frac{\mathcal{E}_1\mathcal{E}_2}{2T}\int_0^T [\cos(\phi_1-\phi_2)+\cos(2\omega t-\phi_1-\phi_2)]\,dt = \frac{\mathcal{E}_1\mathcal{E}_2}{2}\cos\triangle\phi,\end{aligned}$$

where we used $\cos A\cos B = \frac{1}{2}[\cos(A-B)+\cos(A+B)]$, the phase difference $\triangle\phi = \phi_1-\phi_2$ already defined near 3.35, and, once again, that the piece oscillating at 2ω averages to zero over a full period.

This single average, however, only ever gives us $\cos\triangle\phi$, and a beam with $\triangle\phi$ and one with $-\triangle\phi$ (opposite handedness, exactly as when we built the circular states 3.53) would be indistinguishable from $\langle E_1E_2\rangle$ alone. To tell them apart we need a second, independent combination, and the natural one to try is E_2 sampled a quarter of a period earlier,

$$\tilde{E}_2(t) \equiv E_2\left(t-\frac{T}{4}\right) = \mathcal{E}_2\cos\left(\omega t-\frac{\pi}{2}-\phi_2\right) = \mathcal{E}_2\sin(\omega t-\phi_2),$$

using $\cos(x-\pi/2)=\sin x$ and $\omega T/4=\pi/2$; physically, this quarter period delay is exactly what a birefringent plate cut to a quarter of a wavelength does to whichever component crosses it, so $\tilde{E}_2$ is not a mere mathematical trick but a genuinely measurable field. Its cross average with E_1 is

$$\begin{aligned}\langle E_1\tilde{E}_2\rangle &= \frac{\mathcal{E}_1\mathcal{E}_2}{T}\int_0^T \cos(\omega t-\phi_1)\sin(\omega t-\phi_2)\,dt\\ &= \frac{\mathcal{E}_1\mathcal{E}_2}{2T}\int_0^T [\sin(2\omega t-\phi_1-\phi_2)+\sin\triangle\phi]\,dt = \frac{\mathcal{E}_1\mathcal{E}_2}{2}\sin\triangle\phi,\end{aligned}$$

where now we used $\cos A\sin B = \frac{1}{2}[\sin(A+B)-\sin(A-B)]$ with $A-B=\phi_2-\phi_1=-\triangle\phi$, together with the vanishing, as always, of the term oscillating at 2ω.

Having these time averages the time-dependent polarisation ellipse 3.55 becomes

$$\frac{\mathcal{E}_1^2}{2\mathcal{E}_1^2}+\frac{\mathcal{E}_2^2}{2\mathcal{E}_2^2}-\frac{2\mathcal{E}_1\mathcal{E}_2}{2\mathcal{E}_1\mathcal{E}_2}\cos^2\triangle\phi = \sin^2\triangle\phi,$$

multiplying both side by $4\mathcal{E}_1^2\mathcal{E}_2^2$ we'll have

$$2\mathcal{E}_1^2\mathcal{E}_2^2+2\mathcal{E}_1^2\mathcal{E}_2^2-(2\mathcal{E}_1\mathcal{E}_2\cos\triangle\phi)^2 = (2\mathcal{E}_1\mathcal{E}_2\sin\triangle\phi)^2,$$

by mathematical convenience we're going to add zero as $0=\mathcal{E}_1^4+\mathcal{E}_2^4-\mathcal{E}_1^4-\mathcal{E}_2^4$ as follows

$$\mathcal{E}_1^4 + \mathcal{E}_2^4 + 2\mathcal{E}_1^2\mathcal{E}_2^2 - \mathcal{E}_1^4 - \mathcal{E}_2^4 + 2\mathcal{E}_1^2\mathcal{E}_2^2 - (2\mathcal{E}_1\mathcal{E}_2\cos\triangle\phi)^2 = (2\mathcal{E}_1\mathcal{E}_2\sin\triangle\phi)^2\,,$$
$$\left(\mathcal{E}_1^2 + \mathcal{E}_2^2\right)^2 - \left(\mathcal{E}_1^2 - \mathcal{E}_2^2\right)^2 - (2\mathcal{E}_1\mathcal{E}_2\cos\triangle\phi)^2 = (2\mathcal{E}_1\mathcal{E}_2\sin\triangle\phi)^2$$

We finally have everything we need. Stokes defined his four parameters, up to the overall factor of two needed to cancel the one half that every average above carries, as

$$S_0 \equiv 2\langle E_1^2\rangle + 2\langle E_2^2\rangle = \mathcal{E}_1^2 + \mathcal{E}_2^2, \tag{3.56}$$
$$S_1 \equiv 2\langle E_1^2\rangle - 2\langle E_2^2\rangle = \mathcal{E}_1^2 - \mathcal{E}_2^2, \tag{3.57}$$
$$S_2 \equiv 4\langle E_1 E_2\rangle = 2\mathcal{E}_1\mathcal{E}_2\cos\triangle\phi, \tag{3.58}$$
$$S_3 \equiv 4\langle E_1 \tilde{E}_2\rangle = 2\mathcal{E}_1\mathcal{E}_2\sin\triangle\phi. \tag{3.59}$$

Four numbers built out of only two amplitudes and one phase difference cannot be independent, and indeed squaring and adding 3.57 to 3.59 gives

$$\begin{aligned} S_1^2 + S_2^2 + S_3^2 &= \left(\mathcal{E}_1^2 - \mathcal{E}_2^2\right)^2 + 4\mathcal{E}_1^2\mathcal{E}_2^2\cos^2\triangle\phi + 4\mathcal{E}_1^2\mathcal{E}_2^2\sin^2\triangle\phi \\ &= \mathcal{E}_1^4 - 2\mathcal{E}_1^2\mathcal{E}_2^2 + \mathcal{E}_2^4 + 4\mathcal{E}_1^2\mathcal{E}_2^2\left(\cos^2\triangle\phi + \sin^2\triangle\phi\right) \\ &= \mathcal{E}_1^4 + 2\mathcal{E}_1^2\mathcal{E}_2^2 + \mathcal{E}_2^4 = \left(\mathcal{E}_1^2 + \mathcal{E}_2^2\right)^2, \end{aligned}$$

where the whole phase difference $\triangle\phi$ has disappeared through the single identity $\cos^2\triangle\phi + \sin^2\triangle\phi = 1$, leaving, by 3.56,

$$S_1^2 + S_2^2 + S_3^2 = S_0^2. \tag{3.60}$$

This is the equality Stokes was really after: of the four numbers describing a fully polarised beam, only three are independent, exactly as only three are needed to fix a point on a sphere of given radius, a hint of the Poincaré sphere we will make precise shortly.

Every one of these four combinations is quadratic in the field, and this is precisely what lets us rewrite them, exactly, in terms of the complex components $e_1 \equiv \mathcal{E}_1 e^{i\phi_1}$ and $e_2 \equiv \mathcal{E}_2 e^{i\phi_2}$ of the spinor 3.34, without a single cosine or sine left in sight. Since $e_1e_1^* = \mathcal{E}_1^2$, $e_2e_2^* = \mathcal{E}_2^2$, and $e_1e_2^* = \mathcal{E}_1\mathcal{E}_2 e^{i(\phi_1-\phi_2)} = \mathcal{E}_1\mathcal{E}_2\left(\cos\triangle\phi + i\sin\triangle\phi\right)$, comparing real and imaginary parts term by term with 3.58 and 3.59 shows that

$$\mathcal{E}_1^2 \pm \mathcal{E}_2^2 = e_1e_1^* \pm e_2e_2^*, \qquad 2\mathcal{E}_1\mathcal{E}_2\cos\triangle\phi = e_1e_2^* + e_2e_1^*, \qquad 2\mathcal{E}_1\mathcal{E}_2\sin\triangle\phi = i\left(e_1^*e_2 - e_1e_2^*\right),$$

the last equality following from $e_1^*e_2 - e_1e_2^* = \mathcal{E}_1\mathcal{E}_2\left(e^{-i\triangle\phi} - e^{i\triangle\phi}\right) = -2i\,\mathcal{E}_1\mathcal{E}_2\sin\triangle\phi$, so multiplying through by i removes the leftover $i^2 = -1$ and leaves a real number, exactly as a physical intensity must be.

Definition 13 *(Stokes parameters): Given the Jones spinor $\vec{e} = [e_1, e_2]^{\mathsf{T}}$ of a fully polarised,* *monochromatic beam 3.34, its Stokes parameters are the four real numbers*

$$S_0 = e_1 e_1^* + e_2 e_2^*, \tag{3.61}$$
$$S_1 = e_1 e_1^* - e_2 e_2^*, \tag{3.62}$$
$$S_2 = e_1 e_2^* + e_2 e_1^*, \tag{3.63}$$
$$S_3 = i\left(e_1^* e_2 - e_1 e_2^*\right). \tag{3.64}$$

Each of the four carries its own, immediate physical meaning. $S_0 = \mathcal{E}_1^2 + \mathcal{E}_2^2$ is simply the total intensity of the beam, the same quantity we normalised to one when we wrote the Jones ket 3.41 as a unit vector. $S_1 = \mathcal{E}_1^2 - \mathcal{E}_2^2$ measures how much the beam prefers the horizontal axis over the vertical one, positive for a beam closer to $|0;0\rangle$ and negative for one closer to $|\pi/2;0\rangle$. $S_2 = 2\mathcal{E}_1\mathcal{E}_2 \cos\triangle\phi$ measures the same kind of preference, only rotated by forty five degrees, between the two diagonal states 3.49 and 3.51. And $S_3 = 2\mathcal{E}_1\mathcal{E}_2 \sin\triangle\phi$ measures the preference between the two senses of circulation 3.53.

Together, (S_1, S_2, S_3) are nothing but three Cartesian coordinates. Writing the components of the general ket 3.41 as $e_1 = \cos\alpha\cos\chi - i\sin\alpha\sin\chi$ and $e_2 = \sin\alpha\cos\chi + i\cos\alpha\sin\chi$, for which $S_0 = 1$ since 3.41 is already a unit vector, and substituting into 3.62 to 3.64, a short calculation using $\cos^2\theta - \sin^2\theta = \cos 2\theta$ and $2\sin\theta\cos\theta = \sin 2\theta$ on each term gives

$$S_1 = S_0\cos 2\alpha\cos 2\chi, \qquad S_2 = S_0\sin 2\alpha\cos 2\chi, \qquad S_3 = -S_0\sin 2\chi, \tag{3.65}$$

so that $(S_1, S_2, S_3)/S_0$ is, quite literally, a point on a sphere of unit radius, parametrised by twice the orientation angle and twice the ellipticity angle: this is exactly the Poincaré (see the figure 3.5) sphere of the historical discussion above, and 3.65 is the dictionary between its Cartesian coordinates and the angles α, χ used throughout this chapter.

As a consistency check, squaring and adding the three equalities 3.65 reproduces the identity 3.60 once more, $S_1^2 + S_2^2 + S_3^2 = S_0^2\cos^2 2\chi\left(\cos^2 2\alpha + \sin^2 2\alpha\right) + S_0^2\sin^2 2\chi = S_0^2\left(\cos^2 2\chi + \sin^2 2\chi\right) = S_0^2$, this time because every fully polarised beam corresponds to a single, definite pair of angles α, χ, that is, to a single point sitting exactly on the surface of the Poincaré sphere.

Real light, as we already remarked in the historical discussion, is rarely so cooperative: sunlight, or light that has scattered even once, is better described as a rapidly and randomly fluctuating superposition of many such states, with $\alpha(t)$ and $\chi(t)$ themselves wandering on a timescale far shorter than any detector can follow, so that what actually gets measured is not the Stokes vector of one ellipse but a further time average over many different ones. Averaging a family of unit vectors that point in different directions on the sphere never gives a vector as long as the individual ones were, so for such partially polarised light the equality 3.60 is generally replaced by the inequality $S_1^2 + S_2^2 + S_3^2 < S_0^2$, with the extreme case $S_1 = S_2 = S_3 = 0$ describing completely unpolarised light, a beam with a perfectly well defined intensity but no preferred point on the Poincaré sphere at all. We will not develop this idea further here, since it belongs properly to the theory of coherence taken up in 6, but it is worth keeping in mind already that the four numbers built in this subsection say everything there is to say about polarisation, deterministic or not, while the single ket 3.41 only ever describes the former.

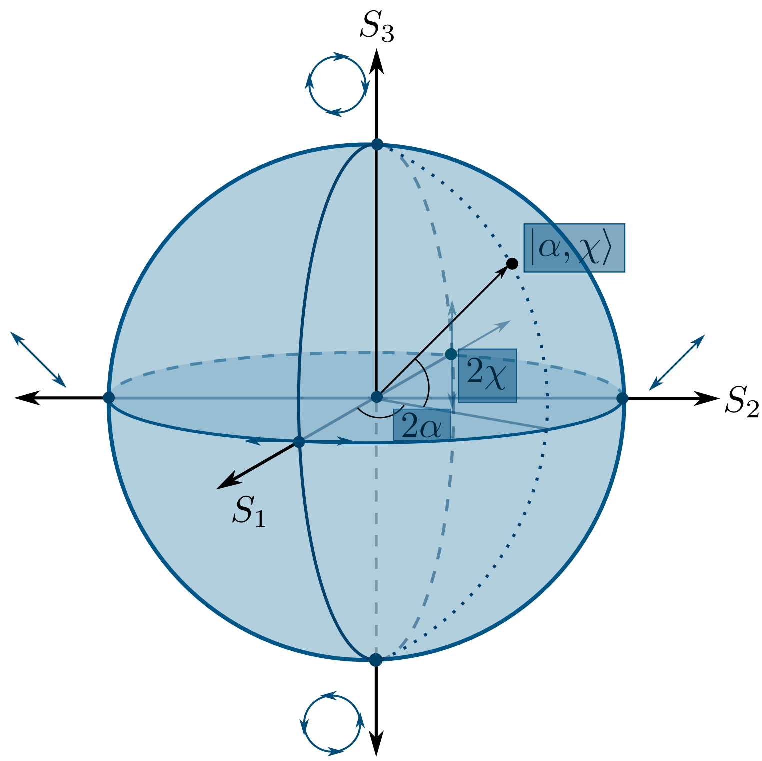


Figure 3.5: In the Poincaré any polarisation state is a point on the surface on the sphere, howerver if the state is partially polarized it would be a pointe inside the poincaré sphere whose distance from the center is $\sqrt{S_1^2 + S_2^2 + S_3^2}$.

In the poincaré sphere 3.5 and the Stokes parameters 3.65 we see an interesting relation, given any state $|\alpha; \chi\rangle$ on the Poincaré sphere the characteristic angles aren't the same, these two are the double, so a full roation on the Jones spinor becomes two full rotations of the Stokes vector $\vec{S} = [S_1, S_2, S_3]^T$. In fact, we've proven in 3.42 that two orthogonal polarisation states are separated an angle of $\pi/2$ but, on the Poincaré sphere this angle is the double, so two orthogonal states are separated π from the other, this means **any two orthogonal polarization states are diametrally opposite on the Poincaré sphere**.

3.5.2.1 Why the Stokes vector is not a vector

It is customary, and we shall keep the custom, to arrange the four numbers 3.61 to 3.64 into a column and to call the result the *Stokes vector*. The name is convenient and completely standard, but taken literally it is wrong, and the reason is not a pedantic one, since it is the very same doubling of angles that makes the Poincaré sphere work in the first place.

In physics an object does not earn the name of vector by being a list of numbers. It earns it by transforming in a definite way under the symmetry group of the space it is supposed to live in, so that a rotation of the laboratory axes by an angle θ takes its components v_i into $R_{ij}(\theta)v_j$. The triple (S_1, S_2, S_3) fails this test at the first attempt. Turn the laboratory axes by θ about the propagation direction; the orientation of every ellipse is now measured from the new x axis, so $\alpha \to \alpha - \theta$, and 3.65 gives at once

$$S_1' = S_1 \cos 2\theta + S_2 \sin 2\theta, \qquad S_2' = -S_1 \sin 2\theta + S_2 \cos 2\theta, \qquad S_3' = S_3. \tag{3.66}$$

The pair (S_1, S_2) does rotate, but by twice the angle through which the laboratory was turned. The cleanest way to see how fatal this is, is to turn the laboratory by half a turn, $\theta = \pi$. Every polarisation ellipse is centrally symmetric, so it is carried exactly onto itself, no state of light has changed, and every measurable quantity must come back unchanged; and indeed 3.66 returns $S_i' = S_i$, because $2\theta = 2\pi$. A genuine vector of the transverse plane would have come back as $-\vec{v}$ instead. So (S_1, S_2, S_3) is not a vector of the physical space in which the experiment sits, and there is no direction one could point at in the laboratory and call the direction of S_3.

The second failure is algebraic rather than geometric. A vector lives in a vector space, and a vector space is closed under multiplication by any real number, in particular by -1. But $-\vec{S}$ has got $S_0 < 0$, a beam of negative intensity, which is not a state of anything, so no physical Stokes vector possesses an additive inverse. What the physical Stokes vectors form is not a vector space but a convex cone, the region carved out by $S_0 \geq 0$ together with the inequality $S_0^2 \geq S_1^2 + S_2^2 + S_3^2$ discussed above. Even the addition that this cone does admit is not the one a physicist would naively expect: adding two Stokes vectors describes the superposition of two mutually *incoherent* beams, which was precisely Stokes' purpose, and it is emphatically not the coherent superposition of two fields, for which one must add the Jones spinors first and only afterwards form the four averages. The Stokes parameters are quadratic in the field, and quadratic quantities never add the way the field itself does.

If the four numbers are not the components of a vector, then components of what are they? Of an operator. It is enough to collect the very same products that appear in 3.61 to 3.64 into a single array.

Definition 14 *(Polarisation matrix): The polarisation matrix, also called the coherence matrix, of a beam with Jones spinor $\vec{e} = [e_1, e_2]^{\mathsf{T}}$ is the Hermitian, positive semidefinite matrix*

$$\Phi \equiv \left\langle \vec{e}\,\vec{e}^{\dagger} \right\rangle = \begin{bmatrix} \langle e_1 e_1^* \rangle & \langle e_1 e_2^* \rangle \\ \langle e_2 e_1^* \rangle & \langle e_2 e_2^* \rangle \end{bmatrix}, \tag{3.67}$$

the averages being taken over the response time of the detector, exactly as in the construction of the Stokes parameters themselves.

Take now the four Hermitian matrices

$$\sigma_0 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \qquad \sigma_1 = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}, \qquad \sigma_2 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \qquad \sigma_3 = \begin{bmatrix} 0 & i \\ -i & 0 \end{bmatrix}, \tag{3.68}$$

the identity together with the three Pauli matrices, ordered and signed to match the conventions we have been using. Any Hermitian 2×2 matrix can be expanded in this set, and carrying out the four traces term by term against 3.67 reproduces exactly the definitions 3.61 to 3.64,

$$S_\mu = \operatorname{tr}\left(\Phi\sigma_\mu\right), \qquad \Phi = \frac{1}{2}\sum_{\mu=0}^{3} S_\mu\sigma_\mu. \tag{3.69}$$

The S_μ are therefore the coordinates of a Hermitian *operator* in the basis $\{\sigma_\mu\}$, which is a tensorial object living in an internal, abstract space, and not a vector of the laboratory. This reading also explains, in one line, the inequality that partially polarised light forced upon us, because writing 3.69 out gives

$$\Phi = \frac{1}{2}\begin{bmatrix} S_0 + S_1 & S_2 + iS_3 \\ S_2 - iS_3 & S_0 - S_1 \end{bmatrix}, \qquad \det\Phi = \frac{1}{4}\left(S_0^2 - S_1^2 - S_2^2 - S_3^2\right) \geq 0,$$

the determinant being non negative precisely because Φ is positive semidefinite, that is, because no measurement of intensity may ever return a negative number. That single physical demand is the entire content of $S_1^2 + S_2^2 + S_3^2 \leq S_0^2$, with equality, a vanishing determinant and a rank one matrix, describing fully polarised light.

There is, finally, one honest sense in which the word vector can be rescued. The combination $S_0^2 - S_1^2 - S_2^2 - S_3^2$ that has just appeared is exactly a Minkowski interval, with S_0 in the role of the time component and (S_1, S_2, S_3) in that of the spatial one, and the physical states fill the forward light cone of an abstract four dimensional space: fully polarised light sits on the cone itself, where the interval vanishes, partially polarised light strictly inside it, and completely unpolarised light on the time axis, where the three spatial components vanish together. The real matrices that describe the action of optical elements on this four tuple, the Mueller matrices to which we will come later, then turn out to belong to a group closely related to the Lorentz group, an observation already made by McMaster when he rescued the formalism (McMaster, 1954). The Stokes four tuple is a four vector of that abstract space, never a three vector of the laboratory, and keeping the traditional name is harmless as long as we remember which space it is a vector of.

3.5.3 The bridge between Jones and Stokes: the stereographic projection

We now own two different languages for one and the same physical object. The Jones formalism describes a polarisation state with two complex numbers 3.34, and the Stokes formalism describes it with four real ones, 3.61 to 3.64, which for fully polarised light are constrained by 3.60 to live on the surface of the Poincaré sphere 3.5. Two languages that speak about the same thing must be translatable into each other, and the translation here is not a mere change of symbols: it is a piece of geometry, and a very old and very beautiful one, the stereographic projection. What follows is a deliberately phenomenological tour of that bridge, close in spirit to the way Azzam and Bashara present it in their treatise on polarised light (Azzam and Bashara, 1977), where the reader will also find all the fine print we happily skip here.

3.5.3.1 What the stereographic projection is

Stereographic projection is the oldest recipe known for drawing a sphere on a flat sheet of paper. Take a sphere resting on a plane, single out the point N of the sphere diametrically opposite to the point of contact, and call it the centre of projection. To find where a point P of the sphere lands on the paper, draw the straight line joining N to P and follow it until it pierces the plane; the piercing point Q is the image of P. That is the whole construction, and figure 3.6 shows it at a glance.

Definition 15 *(Stereographic projection): Let S^2 be a sphere resting on a plane $\mathbb{R}^2$, and let $N \in S^2$ be the point diametrically opposite to the point of contact. The stereographic projection from N assigns to every point $P \in S^2$, with $P \neq N$, the unique point $Q \in \mathbb{R}^2$ at which the straight line through N and P meets the plane. The map is one to one, it sends circles drawn on the sphere to circles or straight lines drawn on the plane, and it preserves the angle at which any two curves cross. The centre N alone has no image, so the plane is completed with a single extra point, the point at infinity, declared to be the image of N.*

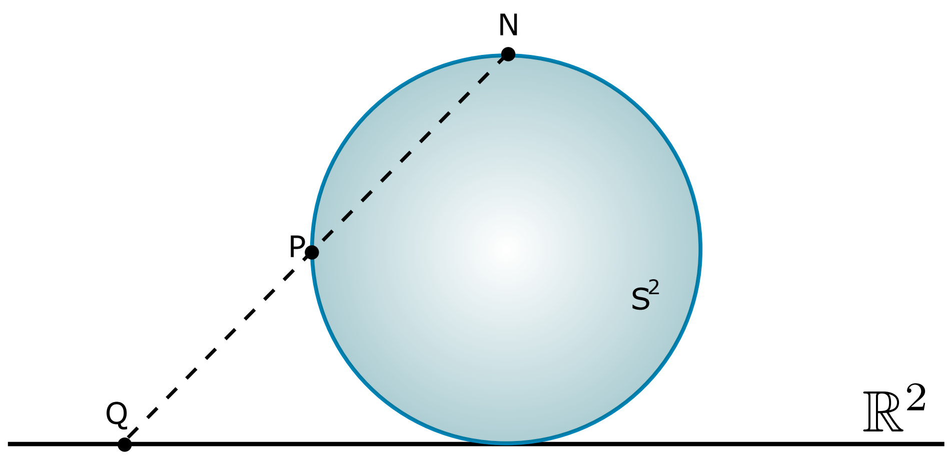


Figure 3.6: The stereographic projection. A ray leaving the centre of projection N and passing through a point P of the sphere S^2 meets the plane $\mathbb{R}^2$ at a single point Q. Points near the contact point fall near the origin, and points near N fly off towards infinity.

The last clause of the definition is the one that carries all the physics. The plane by itself is not enough to hold the whole sphere, since one point of the sphere always escapes; but as soon as we agree that all the ways of running off to infinity on the plane are one and the same destination, the accounting becomes exact and the sphere and the completed plane become the same object seen twice. This completed plane, $\mathbb{C} \cup \{\infty\}$, is what mathematicians call the extended complex plane, and the sphere carrying it is what they call the Riemann sphere. The construction is far older than that name: Hipparchus and Ptolemy already used it to draw star charts, and the medieval astrolabe is nothing but a stereographic map of the celestial sphere engraved on a brass disc, which works precisely because the projection turns circles into circles and keeps angles honest.

3.5.3.2 Why a polarisation state is a single complex number

Nothing in the previous paragraph mentions light, so we must now ask what a sphere and a plane have to do with polarisation. The answer starts with a simple count. The Jones spinor 3.34 carries two complex numbers, that is, four real ones, the two amplitudes $\mathcal{E}_1, \mathcal{E}_2$ and the two phases ϕ_1, ϕ_2; but two of those four say nothing whatsoever about polarisation. The overall size of the spinor is the intensity of the beam, and dimming the light does not change the shape of its ellipse; the overall phase is only a choice of the origin of time, and starting the clock a little earlier does not change that shape either. Multiplying the whole spinor by any complex number therefore leaves the polarisation state untouched, and what survives such a multiplication is only the *ratio* of the two components.

Definition 16 *(Complex polarisation ratio): Given the Jones spinor $\vec{e} = [e_1, e_2]^{\mathsf{T}}$ of a fully polarised beam 3.34, its complex polarisation ratio is the number*

$$w \equiv \frac{e_2}{e_1} = \frac{\mathcal{E}_2}{\mathcal{E}_1}\, e^{i(\phi_2 - \phi_1)} = \frac{\mathcal{E}_2}{\mathcal{E}_1}\, e^{-i\triangle\phi} \;\in\; \mathbb{C} \cup \{\infty\}, \tag{3.70}$$

with the value $w = \infty$ reserved for the case $e_1 = 0$. Two Jones spinors describe the same polarisation state if and only if they share the same w.

Both pieces of w are quantities we have already been using throughout this chapter, only now packed into one symbol. Its modulus $|w| = \mathcal{E}_2/\mathcal{E}_1$ is the ratio of the two amplitudes, which decides how elongated the ellipse is and towards which axis it leans, and its argument $\arg w = -\triangle\phi$ is the phase lag between the two oscillations, which decides how open the ellipse is and in which sense the field turns. Feeding the six canonical states of the previous subsection into 3.70 gives, immediately,

$$w_H = 0, \qquad w_V = \infty, \qquad w_D = 1, \qquad w_A = -1, \qquad w_{\pm} = \pm i, \tag{3.71}$$

read off from the spinors 3.44, 3.46, 3.48, 3.50 and 3.52. Notice that the list already forces the point at infinity upon us: vertical polarisation has got no finite ratio at all, since its first component vanishes, and it is a perfectly ordinary state of light that no honest map is allowed to leave out. So the natural home of polarisation is not the plane $\mathbb{C}$ but the completed plane $\mathbb{C} \cup \{\infty\}$, and by the previous subsubsection that home is a sphere.

3.5.3.3 The Poincaré sphere is that sphere

The sphere we have just been handed is not a new one. Rest the Poincaré sphere on the complex plane of w, so that it touches the plane at the origin, and project from the point N on top, as in figure 3.7. Every reading of the resulting picture is a statement about light.

The point of contact carries $w = 0$, and by 3.71 that is horizontal polarisation; the centre of projection is sent to infinity, and that is vertical polarisation. The two states that our coordinate axes single out therefore sit at the two poles of the picture, diametrically opposite each other, exactly as they should, since they are orthogonal states and we already knew that orthogonal states are

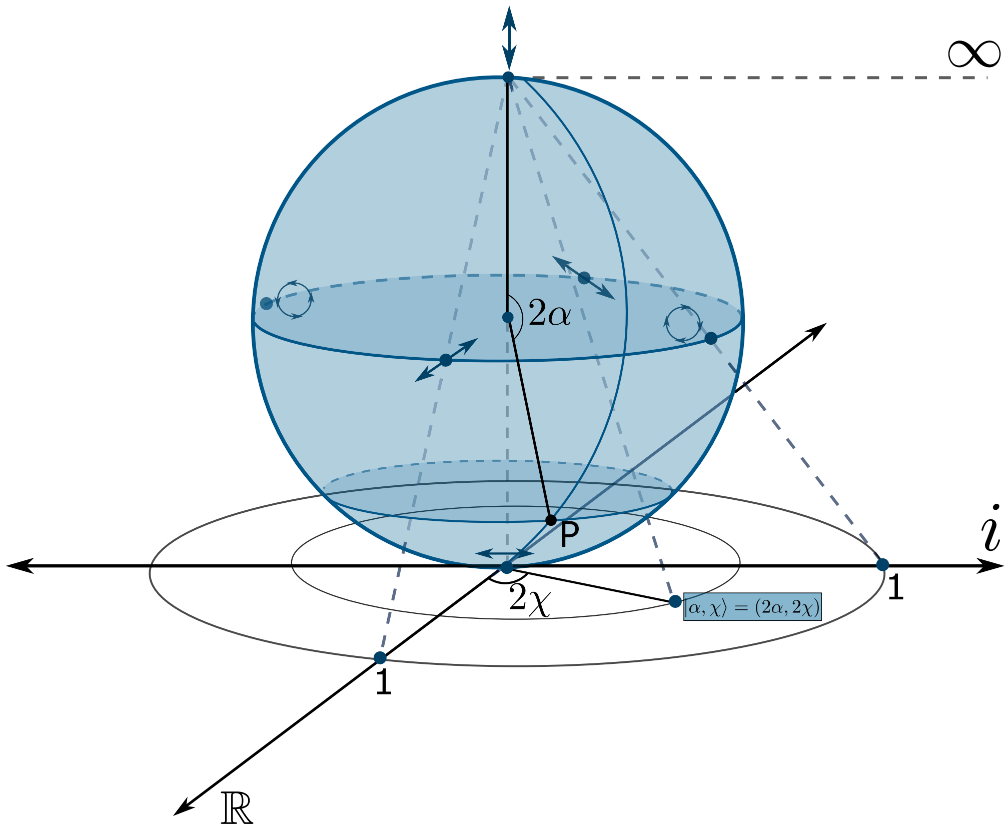


Figure 3.7: The Poincaré sphere resting on the complex plane of the ratio w. Horizontal polarisation sits at the point of contact, $w = 0$, vertical polarisation sits at the centre of projection and is sent to the point at infinity, and the equator, where the two amplitudes are equal, is mapped onto the unit circle $|w| = 1$. The angles 2α and 2χ that locate a state P on the sphere are twice the orientation and ellipticity angles of its ellipse.

antipodal on the Poincaré sphere. Halfway between them lies the equator, the set of states for which the two amplitudes are equal, and it is mapped onto the unit circle $|w| = 1$. On that circle sit the diagonal state at $w = 1$, the antidiagonal at $w = -1$, and the two circular states at $w = \pm i$, each pair again antipodal, again orthogonal. Moving radially outwards from the origin means pouring amplitude into the second component and turning a horizontal state gradually into a vertical one; going around at fixed radius means sliding the phase difference $\triangle\phi$ and opening or closing the ellipse without touching its amplitudes.

The antipodal rule survives the trip down to the plane in a form worth recording. The state orthogonal to $\vec{e} = [e_1, e_2]^\mathsf{T}$ is $[-e_2^*, e_1^*]^\mathsf{T}$, whose ratio is $e_1^*/(-e_2^*)$, so on the plane the antipodal map reads

$$w \;\longmapsto\; -\frac{1}{w^*}, \tag{3.72}$$

which indeed exchanges 0 with ∞, 1 with -1, and $+i$ with $-i$, that is, horizontal with vertical, diagonal with antidiagonal, and each circular state with the other one. A statement that on the sphere was a plain geometric fact, going to the other side, becomes on the plane a short algebraic rule.

The dictionary between the two formalisms is now explicit. Dividing the Stokes parameters 3.61 to 3.64 by S_0 and writing them in terms of the ratio 3.70 gives

$$\frac{S_1}{S_0} = \frac{1-|w|^2}{1+|w|^2}, \qquad \frac{S_2}{S_0} = \frac{2\,\mathrm{Re}\{w\}}{1+|w|^2}, \qquad \frac{S_3}{S_0} = \frac{-2\,\mathrm{Im}\{w\}}{1+|w|^2}, \tag{3.73}$$

and these three expressions are, letter for letter, the classical formulae of the inverse stereographic projection, the ones that lift a point of the plane back up to the sphere. The sign carried by S_3 is only the mirror convention of the drawing, and says which of the two circular states we agree to place on the upper half of the imaginary axis. Nothing else in 3.73 is a matter of convention: the Jones ratio and the Stokes vector are the same information, once written on the plane and once written on the sphere.

3.5.3.4 A flat atlas of every polarisation state

Because the projection loses nothing, we may throw the sphere away and keep only the map. Figure 3.8 does exactly that: it is the complex plane of w with the polarisation ellipse of each state drawn at the point that represents it, a complete atlas of polarised light on a single sheet.

Three features of the atlas deserve to be read out loud. First, the whole real axis is filled with linear states: a real w means $\triangle\phi = 0$ or $\triangle\phi = \pi$, the two components oscillate in step or in opposition, and the ellipse collapses to a segment, whose tilt sweeps through every possible orientation as w runs from one end of the axis to the other. Second, the imaginary axis carries the states whose ellipse is aligned with the coordinate axes, growing from the flat horizontal segment at the origin, through wider and wider ellipses, to a perfect circle at $w = \pm i$, and then closing again into the vertical segment far away. Third, and most striking, the edges of the atlas in every direction, $+\infty$, $-\infty$, $+i\infty$ and $-i\infty$ alike, all carry the same vertical polarisation, because on the sphere all those escape routes converge on the single point N. The plane looks infinite, but it is not: it closes up on itself, and the object it really depicts is compact, bounded and without edges, a sphere.

This is the sense in which the Jones and the Stokes formalisms are one theory. A polarisation state is a point; call it w and you are doing Jones calculus on the plane, call it $(S_1, S_2, S_3)/S_0$ and you are doing Stokes calculus on the sphere, and 3.73 carries you back and forth at will. The choice between them is a choice of convenience, and each has got its own: the sphere makes orthogonality and the doubling of angles visible at a glance, while the plane reduces a state to a single number that can simply be divided or multiplied. The next subsection exploits precisely this, because once a state is a single complex number, what an optical element does to light becomes a transformation of the plane onto itself, and the geometry of the sphere tells us in advance which transformations are allowed.

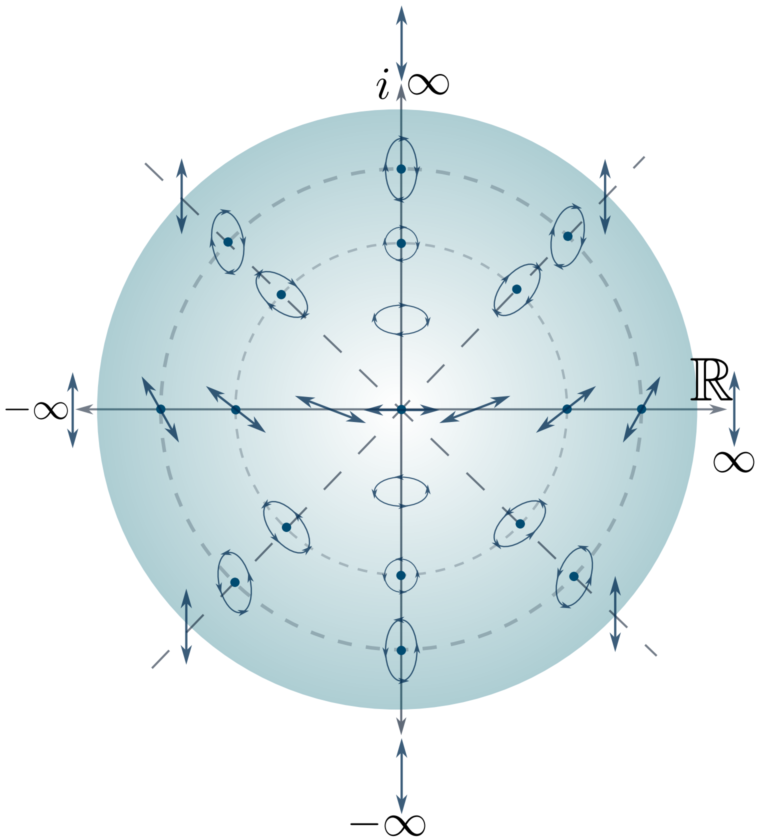


Figure 3.8: The flattened Poincaré sphere. Each point of the complex plane of w carries the ellipse of the state it represents. The origin is horizontal polarisation, the real axis carries every linear state, the imaginary axis carries the ellipses aligned with the coordinate axes together with the two circular states, and every direction of escape to infinity leads to one and the same state, the vertical one.

3.5.4 Transformations of Polarisation: Jones matrices

Every polarisation state built so far is a fixed vector in $\mathbb{C}^2$, but light does not stay the same forever: it crosses lenses, reflects off mirrors, is dimmed by tinted glass, or travels through crystals whose very atomic arrangement singles out a preferred direction in space. Every one of these processes, as long as it acts linearly and does not depend on where or when the light arrives (that is, as long as it is a genuine, homogeneous optical element and not, say, a nonlinear medium or a time varying modulator), can only do one thing to the Jones spinor 3.34: multiply it by a fixed matrix.

Definition 17 *(Jones matrix): A linear, homogeneous optical element acting on a fully polarised, monochromatic beam is represented by a matrix $J \in \mathbb{C}^{2\times 2}$, its Jones matrix, such that the outgoing spinor 3.34 is related to the incoming one by*

$$\vec{e}\,' = J\,\vec{e}.$$

Almost every optical element used in a real laboratory falls into one of three families, according to what kind of matrix J it is: retarders, rotators and dichroics. We build all three from the same simple idea, that of writing J in the basis of its own eigenstates, and we let the physics of each device tell us what those eigenvalues should be.

3.5.4.1 Retarders

Physically, a retarder, also called a wave plate, is cut from a birefringent crystal, quartz, calcite or mica, whose internal atomic arrangement is not the same along every direction, so that the refractive index it presents to light depends on how that light is polarised. Every birefringent plate has, built into it, two mutually orthogonal privileged directions, its fast axis and its slow axis, and light polarised exactly along one of them propagates through unchanged in form, picking up only an ordinary propagation phase, but at a different rate along the fast axis, index n_f, than along the slow one, index n_s. After a thickness d, the component travelling along the slow axis has fallen behind the one along the fast axis by a phase

$$\delta = \frac{2\pi d}{\lambda}\left(n_s - n_f\right), \tag{3.74}$$

called the retardance, while any other input, being a superposition of the two axes, comes out with its ellipse reshaped, because the two halves of the superposition have rotated relative to each other in the complex plane.

Definition 18 *(Retarder): Let $|\psi_1\rangle = |\alpha;\chi\rangle$ and $|\psi_2\rangle = |\alpha+\pi/2;-\chi\rangle$ be a pair of orthogonal Jones kets 3.41*[4]*. A retarder of eigenstates $|\psi_1\rangle, |\psi_2\rangle$ and retardance δ is the Jones matrix*

$$W(\alpha,\chi;\delta) = e^{i\delta/2}\,|\psi_1\rangle\langle\psi_1| + e^{-i\delta/2}\,|\psi_2\rangle\langle\psi_2|\,. \tag{3.75}$$

Equation 3.75 is exactly the spectral decomposition of W: $|\psi_1\rangle, |\psi_2\rangle$ are, by construction, its eigenvectors, with eigenvalues $e^{\pm i\delta/2}$ of unit modulus, so W is unitary and preserves the total intensity $\vec{e}^{\dagger}\vec{e}$ of any beam it acts on, exactly as an element that only dephases, and never absorbs, should.

The case $\chi = 0$ is by far the most common, since it is what an ordinary birefringent plate, cut with its fast axis in the plane of the crystal, produces; we call it a linear retarder, because both eigenstates $|\alpha;0\rangle = [\cos\alpha, \sin\alpha]^{\mathsf{T}}$ and $|\alpha+\pi/2;0\rangle = [-\sin\alpha, \cos\alpha]^{\mathsf{T}}$ are linearly polarised. Substituting them into 3.75 and writing $e^{\pm i\delta/2} = \cos(\delta/2) \pm i\sin(\delta/2)$, the four entries work out to

$$\begin{aligned}
W_{11} &= \cos^2\alpha\, e^{i\delta/2} + \sin^2\alpha\, e^{-i\delta/2} = \cos\frac{\delta}{2} + i\sin\frac{\delta}{2}\cos 2\alpha,\\
W_{22} &= \sin^2\alpha\, e^{i\delta/2} + \cos^2\alpha\, e^{-i\delta/2} = \cos\frac{\delta}{2} - i\sin\frac{\delta}{2}\cos 2\alpha,\\
W_{12} = W_{21} &= \sin\alpha\cos\alpha\left(e^{i\delta/2} - e^{-i\delta/2}\right) = i\sin\frac{\delta}{2}\sin 2\alpha,
\end{aligned}$$

[4] Writing out the components of $|\alpha;\chi\rangle$ and $|\alpha+\pi/2;-\chi\rangle$ and multiplying term by term shows that $\langle\psi_1|\psi_2\rangle = 0$ identically, for every α and every χ, so this pair is always orthogonal.

so that the linear retarder of fast axis at α and retardance δ is

$$W(\alpha, 0; \delta) = \begin{bmatrix} \cos\frac{\delta}{2} + i\sin\frac{\delta}{2}\cos 2\alpha & i\sin\frac{\delta}{2}\sin 2\alpha \\ i\sin\frac{\delta}{2}\sin 2\alpha & \cos\frac{\delta}{2} - i\sin\frac{\delta}{2}\cos 2\alpha \end{bmatrix}. \tag{3.76}$$

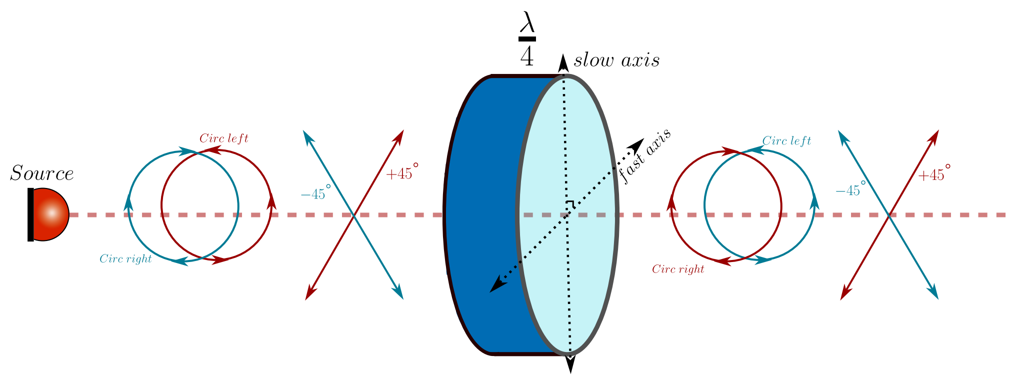


Figure 3.9: A quarter wave plate retarder is a retarder with $\delta = \pi/2$ and when it has got its fast axis oriented with the horizontal one it transforms the polarization in a very specific way. The $\pm 45°$ diagonal states transform into circular polarization states, and the circular polarization states transform into $\pm 45°$ diagonal states.

Two particular values of δ are so useful that they have earned their own names. When $\delta = \pi/2$, a quarter of a full cycle, we speak of a quarter wave plate (see figure 3.9); taking the fast axis horizontal, $\alpha = 0$, equation 3.76 becomes, up to the irrelevant global phase $e^{i\pi/4}$,

$$W_{\lambda/4} = \begin{bmatrix} 1 & 0 \\ 0 & -i \end{bmatrix}, \tag{3.77}$$

whose action on the diagonal state 3.49 is immediate to check: $|\pi/4; 0\rangle = \frac{1}{\sqrt{2}}[1, 1]^\mathsf{T}$ is sent to $\frac{1}{\sqrt{2}}[1, -i]^\mathsf{T} = |0; -\pi/4\rangle$, one of the circular states 3.53. This is exactly why a quarter wave plate is the standard tool for turning linear polarisation into circular, and back.

When instead $\delta = \pi$, half a full cycle, we get the half wave plate,

$$W(\alpha, 0; \pi) = i\begin{bmatrix} \cos 2\alpha & \sin 2\alpha \\ \sin 2\alpha & -\cos 2\alpha \end{bmatrix}, \tag{3.78}$$

and, up to the global phase i, this matrix is a reflection: acting on a linear state it sends $|\theta; 0\rangle$ to $|2\alpha - \theta; 0\rangle$, reflecting its orientation about the fast axis, so a half wave plate rotates any linear input by twice the angle between the input and the fast axis, and it is the tool of choice whenever an arbitrary rotation of a linear polarisation is needed. Acting instead on a circular state, a short calculation with 3.78 shows that $|0; \pi/4\rangle$ is sent to $|0; -\pi/4\rangle$ for every value of α, so a half wave plate always reverses the handedness of a circular input, whatever its own orientation happens to be.

Nothing prevents us from choosing genuinely elliptical eigenstates, $0 < |\chi| < \pi/4$: the resulting elliptical retarder is a real, if less common, device, built from a birefringent material whose axes have additionally been twisted by a small optical activity, so that its privileged states are ellipses rather than straight lines, and it is worth constructing its Jones matrix properly, both because it is the fully general case of 3.75 and because doing so will let us obtain the rotator, further on, in a single line.

Write the first eigenstate $|\psi_1\rangle = |\alpha;\chi\rangle$ of 3.41 as $|\psi_1\rangle = [a, b]^\mathsf{T}$ with

$$a = \cos\alpha\cos\chi - i\sin\alpha\sin\chi, \qquad b = \sin\alpha\cos\chi + i\cos\alpha\sin\chi,$$

so that $|a|^2 + |b|^2 = 1$, since $|\psi_1\rangle$ is a unit vector. Comparing component by component with $|\psi_2\rangle = |\alpha + \pi/2; -\chi\rangle$ shows the compact relation $|\psi_2\rangle = [-b^*, a^*]^\mathsf{T}$, so that the matrix built from the two orthonormal eigenstates as columns,

$$U(\alpha,\chi) = \begin{bmatrix} |\psi_1\rangle & |\psi_2\rangle \end{bmatrix} = \begin{bmatrix} a & -b^* \\ b & a^* \end{bmatrix}, \tag{3.79}$$

is unitary, $U^\dagger U = UU^\dagger = I$, precisely because $|\psi_1\rangle, |\psi_2\rangle$ are orthonormal. Equation 3.79 is nothing but the change of basis matrix from the eigenbasis $\{|\psi_1\rangle, |\psi_2\rangle\}$, in which the retarder is diagonal by definition, to the standard basis in which we write the Jones spinor 3.34. Writing $\Lambda(\delta) = \operatorname{diag}\left(e^{i\delta/2}, e^{-i\delta/2}\right)$ for the retarder expressed in its own eigenbasis, this change of basis gives a second, completely explicit way of writing 3.75,

$$W(\alpha,\chi;\delta) = U(\alpha,\chi)\,\Lambda(\delta)\,U(\alpha,\chi)^\dagger, \tag{3.80}$$

and carrying out the product explicitly,

$$U\Lambda U^\dagger = \begin{bmatrix} a & -b^* \\ b & a^* \end{bmatrix} \begin{bmatrix} e^{i\delta/2} & 0 \\ 0 & e^{-i\delta/2} \end{bmatrix} \begin{bmatrix} a^* & b^* \\ -b & a \end{bmatrix} = \begin{bmatrix} |a|^2 e^{i\delta/2} + |b|^2 e^{-i\delta/2} & ab^*\left(e^{i\delta/2} - e^{-i\delta/2}\right) \\ a^* b\left(e^{i\delta/2} - e^{-i\delta/2}\right) & |b|^2 e^{i\delta/2} + |a|^2 e^{-i\delta/2} \end{bmatrix}.$$

Squaring and multiplying the two expressions for a, b above gives $|a|^2 - |b|^2 = \cos 2\alpha\cos 2\chi$, $|a|^2 + |b|^2 = 1$ and $ab^* = \frac{1}{2}\sin 2\alpha\cos 2\chi - \frac{i}{2}\sin 2\chi$, exactly the same three combinations already met while building the linear retarder; using them, together with $e^{\pm i\delta/2} = \cos(\delta/2) \pm i\sin(\delta/2)$, the elliptical retarder takes the closed form

$$W(\alpha,\chi;\delta) = \begin{bmatrix} \cos\dfrac{\delta}{2} + i\sin\dfrac{\delta}{2}\cos 2\alpha\cos 2\chi & \sin\dfrac{\delta}{2}\sin 2\chi + i\sin\dfrac{\delta}{2}\sin 2\alpha\cos 2\chi \\ -\sin\dfrac{\delta}{2}\sin 2\chi + i\sin\dfrac{\delta}{2}\sin 2\alpha\cos 2\chi & \cos\dfrac{\delta}{2} - i\sin\dfrac{\delta}{2}\cos 2\alpha\cos 2\chi \end{bmatrix}. \tag{3.81}$$

Setting $\chi = 0$ in 3.81 gives $\cos 2\chi = 1, \sin 2\chi = 0$ and collapses it exactly back to the linear retarder 3.76, as it must: the fully general elliptical case was, after all, only ever hiding the extra angle χ inside the change of basis matrix 3.79.

3.5.4.2 Rotators

Let us push the general elliptical retarder 3.81 to the opposite extreme from the linear case and choose circular eigenstates, $\alpha = 0, \chi = \pi/4$, that is $|\psi_1\rangle = |0; \pi/4\rangle$ and $|\psi_2\rangle = |0; -\pi/4\rangle$ of 3.53. Since $\cos 2\chi = \cos(\pi/2) = 0$ and $\sin 2\chi = \sin(\pi/2) = 1$, every term in 3.81 carrying a factor of $\cos 2\chi$ vanishes identically and we are left with

$$W(0, \pi/4; \delta) = \begin{bmatrix} \cos\frac{\delta}{2} & \sin\frac{\delta}{2} \\ -\sin\frac{\delta}{2} & \cos\frac{\delta}{2} \end{bmatrix}, \tag{3.82}$$

and the right hand side is exactly the rotation matrix already used in 3.41 to build every ellipse out of the unrotated form 3.40. A retarder of circular eigenstates, in other words, is nothing but a rotation of the whole Jones plane: it sends $|\alpha; \chi\rangle$ to $|\alpha - \delta/2; \chi\rangle$ for every α and every χ, leaving the ellipticity untouched and shifting only the orientation.

Definition 19 *(Rotator): A rotator by an angle θ is the Jones matrix*

$$R(\theta) = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}, \tag{3.83}$$

equivalently the circular retarder 3.82 of retardance $\delta = -2\theta$, so that $R(\theta)\,|\alpha; \chi\rangle = |\alpha + \theta; \chi\rangle$ for every α, χ.

Physically, this is exactly what happens inside an optically active medium: a solution of chiral sugar molecules, or a slab of quartz cut along its optic axis, where it is not the two linear directions that see different refractive indices, but the two circular ones. The phenomenon is called circular birefringence, and a beam of any polarisation crossing a thickness d of such a medium has its ellipse rotated, rigidly and without distortion, by an angle proportional to d; the same net effect, a pure rotation with no change of ellipticity, is produced by a Faraday rotator, a piece of glass or crystal placed in a strong magnetic field along the beam, in which it is the magnetic field rather than molecular chirality that splits the two circular eigenstates. Notice that $R(\theta)$, unlike a generic retarder, is a real matrix: rotators are the only non-trivial polarisation transformations whose Jones matrix has got no imaginary part at all, a reflection of the fact that neither optical activity nor the Faraday effect privileges any particular linear direction in space, only a sense of circulation.

3.5.4.3 Dichroics and polarisers

Every element considered so far is unitary: it reshapes the polarisation ellipse but never changes the total intensity. A dichroic element breaks this on purpose. Physically, it is again built from an anisotropic material, but now one whose two privileged directions are absorbed differently rather than merely retarded differently, because the microscopic charges responsible for the absorption, free electrons in a metal wire, iodine chains in a stretched polymer, the delocalised electrons of an elongated organic molecule, can oscillate much more easily along one direction than along the perpendicular one. Mathematically we only have to repeat the construction 3.75 once more, replacing the two phase factors of unit modulus by two real, non-negative amplitude transmittances.

Definition 20 *(Dichroic): A dichroic element of eigenstates $|\psi_1\rangle, |\psi_2\rangle$ (orthogonal, as in the retarder) and amplitude transmittances $t_1, t_2 \in \mathbb{R}$, $t_1, t_2 \geq 0$, is the Jones matrix*

$$D = t_1 |\psi_1\rangle \langle\psi_1| + t_2 |\psi_2\rangle \langle\psi_2| . \tag{3.84}$$

Because t_1, t_2 are real, D in 3.84 is Hermitian rather than unitary, unless $t_1 = t_2$, the trivial case in which nothing happens to the polarisation, only to the overall brightness: light along $|\psi_1\rangle$ comes out multiplied by t_1 and light along $|\psi_2\rangle$ by t_2, and any state in between comes out reshaped towards whichever eigenstate survives best.

Taking $\chi = 0$ once again gives the linear dichroic, and repeating the same short calculation that led to 3.76, now with t_1, t_2 in place of $e^{\pm i\delta/2}$,

$$D(\alpha, 0; t_1, t_2) = \begin{bmatrix} t_1 \cos^2\alpha + t_2 \sin^2\alpha & (t_1 - t_2)\sin\alpha\cos\alpha \\ (t_1 - t_2)\sin\alpha\cos\alpha & t_1 \sin^2\alpha + t_2\cos^2\alpha \end{bmatrix}. \tag{3.85}$$

The most important member of this family is obtained by pushing the contrast between the two eigenstates to its limit, $t_1 = 1$ and $t_2 = 0$: the surviving axis is transmitted perfectly and the other is extinguished completely. Equation 3.85 then collapses to

$$P(\alpha) = \begin{bmatrix} \cos^2\alpha & \sin\alpha\cos\alpha \\ \sin\alpha\cos\alpha & \sin^2\alpha \end{bmatrix} = |\alpha; 0\rangle \langle\alpha; 0| , \tag{3.86}$$

which we recognise, on the right, as the orthogonal projector onto the linear state $|\alpha; 0\rangle$.

Definition 21 *(Ideal linear polariser): An ideal linear polariser of transmission axis at angle α is the limiting linear dichroic 3.85 with $t_1 = 1, t_2 = 0$, that is, the orthogonal projector $P(\alpha) = |\alpha; 0\rangle \langle\alpha; 0|$ of 3.86.*

This is the deepest sense in which a polariser is a simple device: it does not really transform a polarisation state, it projects it, discarding every component along $|\alpha + \pi/2; 0\rangle$ and keeping only the shadow that the incoming ellipse casts onto its own transmission axis. Every partial linear dichroic 3.85 with $t_2 \neq 0$ is only an imperfect version of this same projection, letting some of the unwanted component leak through.

Circular dichroism is built the same way, with $|\psi_1\rangle = |0; \pi/4\rangle$ and $|\psi_2\rangle = |0; -\pi/4\rangle$ in 3.84. Specialising a, b from the change of basis matrix 3.79 to $\alpha = 0, \chi = \pi/4$ gives $a = 1/\sqrt{2}$, $b = i/\sqrt{2}$, so

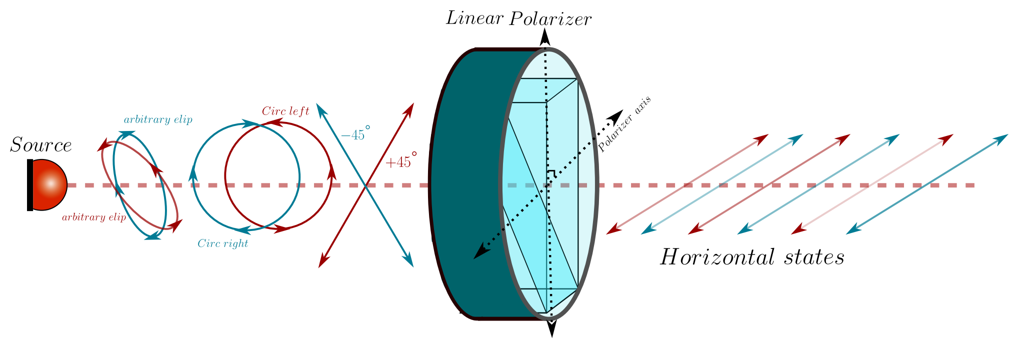


Figure 3.10: A linear polariser with its polariser axis oriented parallel to the horizontal axis proyects all the incident states to horizontal ones, however the transmited intensity is reduced by the relative angle between the incident state $|\alpha;\chi\rangle$ and the polariser axis α_{pol} in this case $\alpha_{pol} = 0°$.

that $|a|^2 = |b|^2 = 1/2$ and $ab^* = -i/2$; repeating, with these values, the same matrix multiplication that led to 3.81, but with the real transmittances t_1, t_2 in place of the phases $e^{\pm i\delta/2}$,

$$D_{\circlearrowleft}(t_1, t_2) = \begin{bmatrix} \dfrac{t_1 + t_2}{2} & -i\dfrac{t_1 - t_2}{2} \\ i\dfrac{t_1 - t_2}{2} & \dfrac{t_1 + t_2}{2} \end{bmatrix}, \tag{3.87}$$

describing a material that absorbs its two circular eigenstates differently. Circular dichroism is a much smaller effect than linear dichroism in most materials, but it is an extremely sensitive probe of molecular chirality, proteins, sugars and DNA are all optically active and mildly circularly dichroic, which is why circular dichroism spectroscopy is a standard tool in structural biology and chemistry.

Stepping back from the algebra, what actually makes a material dichroic is always some built in anisotropy at the microscopic level: an elongated molecule, a stretched polymer chain, or a crystal lattice, along which charges are freer to respond to the oscillating electric field of light in one direction than in the perpendicular one, so the material absorbs, rather than merely slows down, one polarisation component more strongly than the other. A polariser, more generally, is simply any device, dichroic or not, designed to output light of a well controlled polarisation state from an input that may not have one at all; dichroism is only one of the physical mechanisms that nature and engineering have found to build one.

Three real polarisers illustrate how different those mechanisms can be. Tourmaline, the mineral in which dichroism was first noticed, is naturally dichroic, but the polariser found today in almost every pair of sunglasses, camera filter and liquid crystal display is Edwin Land's Polaroid sheet, a film of polyvinyl alcohol stretched in one direction to align its long polymer chains and then doped with iodine, whose mobile electrons run freely along the chains and absorb the field component parallel to them, the linear dichroic 3.85 realised in a plastic film. A wire grid polariser applies the same physical idea to a metal instead of a dye: an array of parallel conducting wires, spaced much closer together than the wavelength of the light, lets the free electrons in the wires oscillate and reflect the field component parallel to them, while the perpendicular component, unable to drive a current across the gaps between wires, passes through almost unaffected; because it works by reflection rather than absorption, it survives far higher optical powers than a dyed polymer sheet, and it is the polariser of choice in the infrared and in microwave optics. Finally, the Glan-Thompson prism takes a completely different route, using no absorption at all: two prisms of calcite, a strongly birefringent crystal, are cut and cemented together so that, at the internal interface, the ordinary ray strikes beyond its critical angle and is totally internally reflected out of the beam, while the extraordinary ray, seeing a different refractive index, is transmitted straight through. Being lossless for the transmitted beam and free of any absorbing dye, this kind of polariser can handle the intense, focused light of a laser without being damaged, which the simple dichroic sheet cannot.

3.5.5 Transformations of Polarisation: Mueller matrices

The Jones calculus of the previous subsection is exact, elegant and, for a large class of problems, entirely sufficient. It is also, in a very precise sense, too narrow. A Jones matrix acts on the spinor 3.34, and that spinor exists only when the beam has got a single, well defined phase relation between its two components, that is, only when the light is fully polarised. The moment we point our instrument at sunlight, at a cloud, at a sheet of paper, at a biological tissue or at any surface rough on the scale of a wavelength, the object we must transform is no longer a spinor but the four Stokes parameters 3.61 to 3.64, and no 2×2 complex matrix will do the job.

There is a second and deeper failure, and it is worth stating before we build anything. A Jones matrix can never *depolarise*. Whatever it does to a fully polarised beam, the outgoing beam is again fully polarised, so a formalism built on Jones matrices alone cannot describe the single most common thing that real matter does to light. We shall prove this in a moment, and the proof will also tell us exactly what has to be added.

The formalism that repairs both defects was assembled by Hans Mueller at the Massachusetts Institute of Technology in the early nineteen forties, building on the earlier scattering work of Soleillet and of Perrin (Perrin, 1942), and it was the same wave of activity that rescued the Stokes parameters from the obscurity described at the beginning of this section. Its later systematisation owes a great deal to McMaster (McMaster, 1954) and to Chandrasekhar's treatment of radiative transfer (Chandrasekhar, 1950), where polarised light must be propagated through a scattering atmosphere and where Jones spinors are simply not available.

3.5.5.1 From the Jones matrix to the coherence matrix

The bridge between the two calculi is the polarisation matrix 3.67, and building it is a one line calculation whose consequences occupy the rest of this subsection. Let a linear, homogeneous element with Jones matrix J act on the beam, so that $\vec{e}\,' = J\vec{e}$. The outgoing polarisation matrix is, by definition,

$$\Phi' = \left\langle \vec{e}\,'\vec{e}\,'^{\dagger} \right\rangle = \left\langle J\vec{e}\,(J\vec{e})^{\dagger} \right\rangle = \left\langle J\,\vec{e}\,\vec{e}^{\,\dagger} J^{\dagger} \right\rangle = J \left\langle \vec{e}\,\vec{e}^{\,\dagger} \right\rangle J^{\dagger} = J\,\Phi\,J^{\dagger}. \tag{3.88}$$

Every step is elementary, but the third one is the whole point and deserves to be said out loud: the matrix J could be taken outside the time average because the *element* is deterministic even when the *light* is not. The waveplate does not flicker; only the field does. This is precisely why 3.88 remains valid for partially polarised and even for completely unpolarised light, where the spinor $\vec{e}$ has got no meaning at all and only the averages survive.

Notice also the shape of the transformation. The field transforms linearly, $\vec{e} \to J\vec{e}$, but the polarisation matrix transforms by a **congruence**, $\Phi \to J\Phi J^\dagger$, with J appearing twice. Since the Stokes parameters are themselves quadratic in the field, this is exactly what we should have expected, and it is the algebraic reason why the Mueller matrix we are about to build is quadratic in the Jones matrix.

3.5.5.2 The Mueller matrix

Because the Stokes parameters are linear coordinates of Φ through 3.69, and because 3.88 is linear in Φ, the map induced on the four Stokes parameters must be linear as well.

Definition 22 *(Mueller matrix): A linear, homogeneous optical element acting on a beam of arbitrary degree of polarisation is represented by a real 4×4 matrix M, its Mueller matrix, such that the outgoing Stokes vector is related to the incoming one by*

$$S'_\mu = \sum_{\nu=0}^{3} M_{\mu\nu} S_\nu, \qquad \textit{that is} \qquad \vec{S}\,' = M\vec{S}. \tag{3.89}$$

When the element also possesses a Jones matrix, the two descriptions must agree, and forcing them to agree hands us the Mueller matrix explicitly. Insert 3.88 into the first of 3.69 and then expand the incoming Φ with the second,

$$S'_\mu = \mathrm{tr}\left(\Phi'\sigma_\mu\right) = \mathrm{tr}\left(J\Phi J^\dagger\sigma_\mu\right) = \mathrm{tr}\left(J\left[\frac{1}{2}\sum_{\nu=0}^{3} S_\nu\sigma_\nu\right]J^\dagger\sigma_\mu\right) = \sum_{\nu=0}^{3}\left[\frac{1}{2}\mathrm{tr}\left(\sigma_\mu J\sigma_\nu J^\dagger\right)\right]S_\nu,$$

where the trace has been taken through the finite sum and the cyclic property $\mathrm{tr}(AB) = \mathrm{tr}(BA)$ has been used to bring σ_μ to the front. Comparing with 3.89 we can read off the coefficients at once.

Definition 23 *(Mueller matrix of a Jones element): An element whose action on fully polarised light is given by the Jones matrix J has got the Mueller matrix*

$$\boxed{M_{\mu\nu} = \frac{1}{2}\, tr\left(\sigma_\mu\, J\, \sigma_\nu\, J^\dagger\right)} \tag{3.90}$$

*with σ_μ the four matrices 3.68. A Mueller matrix that can be written in this form is called **non depolarising**, or a **Mueller-Jones** matrix.*

Formula 3.90 is the dictionary we were after, and three of its properties are worth extracting immediately.

Taking the complex conjugate of 3.90 and using $\overline{\operatorname{tr} A} = \operatorname{tr} A^\dagger$ together with the hermiticity $\sigma_\mu^\dagger = \sigma_\mu$,

$$\overline{M_{\mu\nu}} = \frac{1}{2}\operatorname{tr}\left(\left[\sigma_\mu J \sigma_\nu J^\dagger\right]^\dagger\right) = \frac{1}{2}\operatorname{tr}\left(J\sigma_\nu J^\dagger \sigma_\mu\right) = M_{\mu\nu},$$

the last step being once more the cyclic property. So M is a real matrix, as it must be, since it maps four measurable intensities into four measurable intensities.

Replacing J by $e^{i\theta}J$ leaves 3.90 untouched, because the phase from J cancels against the conjugate phase from $J^\dagger$. The map $J \mapsto M$ is therefore not injective: all Jones matrices differing by an overall phase collapse onto the same Mueller matrix. This is not a defect but an improvement, since that phase was never measurable in the first place, and we shall see in a moment that it is the same two to one correspondence that produces the doubling of angles on the Poincaré sphere.

If light crosses first the element J_1 and then the element J_2, the combined Jones matrix is $J_2 J_1$, and substituting it into 3.90 while inserting the resolution $\frac{1}{2}\sum_\lambda \sigma_\lambda \operatorname{tr}(\sigma_\lambda \cdot\,)$ between the two factors shows that

$$M\left(J_2 J_1\right) = M\left(J_2\right) M\left(J_1\right), \tag{3.91}$$

so that 3.90 is a group homomorphism and an optical bench is described, exactly as in the Jones calculus, by the ordered product of the matrices of its elements.

3.5.5.3 Retarders are rotations of the Poincaré sphere

Let us now compute a Mueller matrix explicitly, and let us choose the simplest retarder, the linear one with its fast axis along x, whose Jones matrix is 3.76 with $\alpha = 0$, that is $W = \operatorname{diag}\left(e^{i\delta/2}, e^{-i\delta/2}\right)$. We need the four objects $W\sigma_\nu W^\dagger$. The first two are immediate, since W is unitary and diagonal and therefore commutes with σ_0 and σ_1,

$$W\sigma_0 W^\dagger = WW^\dagger = \sigma_0, \qquad W\sigma_1 W^\dagger = \sigma_1.$$

The interesting ones are the other two. Multiplying out,

$$W\sigma_2 W^\dagger = \begin{bmatrix} 0 & e^{i\delta} \\ e^{-i\delta} & 0 \end{bmatrix} = \cos\delta \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} + \sin\delta \begin{bmatrix} 0 & i \\ -i & 0 \end{bmatrix} = \cos\delta\,\sigma_2 + \sin\delta\,\sigma_3,$$

where we simply split $e^{\pm i\delta} = \cos\delta \pm i\sin\delta$ and recognised the two Pauli matrices, and in exactly the same way

$$W\sigma_3 W^\dagger = \begin{bmatrix} 0 & ie^{i\delta} \\ -ie^{-i\delta} & 0 \end{bmatrix} = -\sin\delta\,\sigma_2 + \cos\delta\,\sigma_3.$$

Since $\frac{1}{2}\operatorname{tr}(\sigma_\mu\sigma_\nu) = \delta_{\mu\nu}$, the coefficients just found are literally the columns of M, and we obtain

$$M_W(\delta) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & \cos\delta & -\sin\delta \\ 0 & 0 & \sin\delta & \cos\delta \end{bmatrix}. \tag{3.92}$$

Read this matrix slowly, because it is the central geometrical fact of the whole subsection. The first row and the first column are those of the identity, so S_0 is untouched: a retarder absorbs nothing, exactly as its unitarity promised. The remaining 3×3 block leaves S_1 alone and rotates the pair (S_2, S_3) by the angle δ. In other words, on the Poincaré sphere the element performs a **rigid rotation by the retardance δ about the S_1 axis**, and the S_1 axis is precisely the diameter joining the two eigenstates of this retarder, the horizontal and vertical states 3.45 and 3.47. The eigenstates are the fixed points of the motion, as fixed points of a rotation must be, and every other state is carried around them on a circle of constant latitude.

Nothing in this conclusion depends on our having chosen $\alpha = 0$. Repeating the computation with the general elliptical retarder 3.81 is a longer but entirely mechanical exercise, and it yields the following statement, which we record as the geometrical heart of the Mueller calculus.

Theorem 1 *(Retarders as rotations): The Mueller matrix of a retarder of retardance δ whose eigenstates are $|\alpha;\chi\rangle$ and its orthogonal partner is*

$$M_W = \begin{bmatrix} 1 & \vec{0}^{\mathsf{T}} \\ \vec{0} & R\,(\hat{n}, \delta) \end{bmatrix}, \tag{3.93}$$

where $R(\hat{n}, \delta)$ is the ordinary 3×3 rotation by the angle δ about the unit vector

$$\hat{n} = (\cos 2\alpha \cos 2\chi,\ \sin 2\alpha \cos 2\chi,\ -\sin 2\chi)\,, \tag{3.94}$$

which by 3.65 is exactly the point of the Poincaré sphere representing the eigenstate $|\alpha;\chi\rangle$.

Every waveplate on an optical bench is therefore a rotation of the Poincaré sphere (see figure 3.11), and the two numbers that specify it have got transparent meanings: the orientation of its axis fixes *about which diameter* the sphere turns, and its retardance fixes *by how much*. Three familiar devices become obvious.

A **quarter wave plate**, $\delta = \pi/2$, turns the sphere by a quarter of a full turn. Starting from a linear state on the equator and choosing the axis at $\pi/4$ from it, a quarter turn lands exactly on a pole, which is the statement that a quarter wave plate converts linear light into circular light, as we found by direct matrix multiplication in the previous subsection.

A **half wave plate**, $\delta = \pi$, turns the sphere by half a full turn about an equatorial axis. A rotation by π about the axis $\hat{n}$ sends any vector $\vec{v}$ to $2(\hat{n} \cdot \vec{v})\hat{n} - \vec{v}$, so a linear state at azimuth 2θ on the equator, acted on by a plate whose axis sits at azimuth 2α, is carried to azimuth $4\alpha - 2\theta$. Dividing by two, the physical polarisation direction goes from θ to $2\alpha - \theta$: **a half wave plate reflects the plane of polarisation about its own fast axis**. Applying the same rule to the poles, $\vec{v} = (0, 0, \pm 1)$ with $\hat{n}$ equatorial gives $\vec{v} \to -\vec{v}$, so the two circular states are exchanged and the handedness of any beam is reversed.

A **rotator** 3.83 is, as we proved, a circular retarder, so its axis 3.94 is the polar axis $\hat{n} = (0, 0, -1)$ and it spins the sphere about the line joining the two circular states. A rotator that turns the

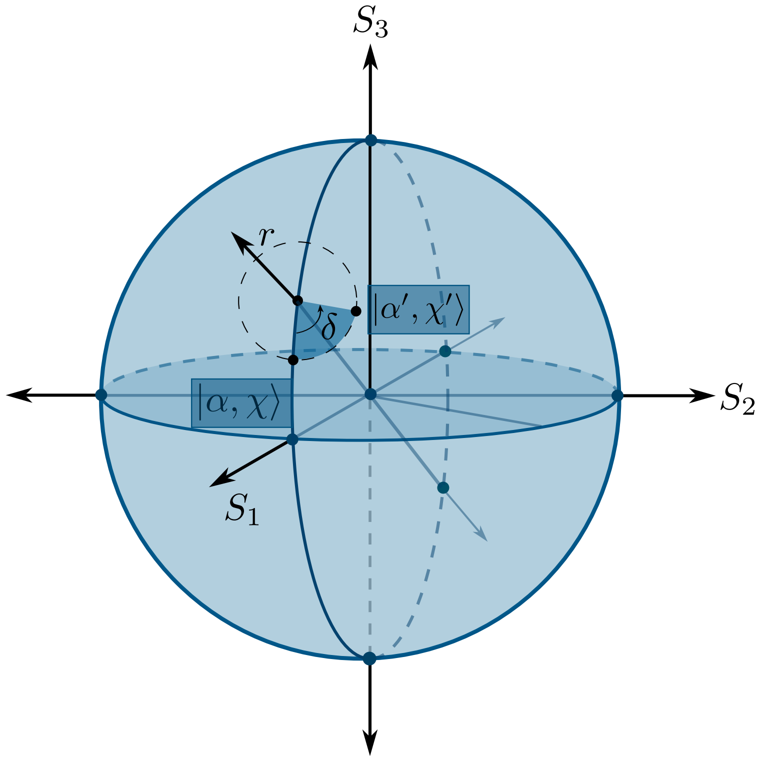


Figure 3.11: An initial state $|\alpha;\chi\rangle$ under the effect of a rotated retarder with retar δ and eigenstate r is rotated an angle of δ arounde the r-axis to a final state $|\alpha'.\chi'\rangle$.

physical polarisation by an angle θ has got retardance $\delta = 2\theta$, so it rotates the Poincaré sphere by 2θ. This is the doubling of angles once more, and we can now see its true origin: the map 3.90 restricted to unitary Jones matrices is the classical two to one homomorphism from $SU(2)$ onto the rotation group $SO(3)$. Turning a waveplate through a full physical revolution, $\theta : 0 \to \pi$, returns the Jones matrix not to $+I$ but to $-I$, while the Mueller matrix has already completed a full 2π rotation and returned to the identity. The polarisation spinor is a spinor in exactly the sense that an electron wavefunction is, and the Poincaré sphere is its Bloch sphere.

3.5.5.4 Polarisers, dichroics and the Malus law

Not every element is unitary, and the non unitary ones move the sphere in a completely different way. Take first the ideal linear polariser 3.86, whose Jones matrix is the projector $J = |\psi\rangle\langle\psi|$ onto the transmitted state. Feeding it into 3.90 and writing $\hat{n}$ for the Stokes vector of that state, the result is the strikingly simple

$$M_P = \frac{1}{2}\begin{bmatrix} 1 & \hat{n}^{\mathsf{T}} \\ \hat{n} & \hat{n}\,\hat{n}^{\mathsf{T}} \end{bmatrix}. \tag{3.95}$$

This matrix has got rank one in a strong sense: whatever comes in, the outgoing Stokes vector is proportional to $(1, \hat{n})$, that is, the entire Poincaré sphere is crushed onto the single point $\hat{n}$. The map is catastrophically non invertible, which is the geometric expression of the obvious physical fact that a polariser destroys information and cannot be undone.

The Malus law now drops out with no effort at all. Send in linear light at an angle θ, whose Stokes vector is $\vec{S} = (1, \cos 2\theta, \sin 2\theta, 0)$, through a polariser whose axis is at α, so that $\hat{n} = (\cos 2\alpha, \sin 2\alpha, 0)$. The transmitted intensity is the zeroth component of 3.95 acting on $\vec{S}$,

$$S_0' = \frac{1}{2}\left(1 + \hat{n} \cdot \vec{S}_{123}\right) = \frac{1}{2}\left[1 + \cos 2\alpha \cos 2\theta + \sin 2\alpha \sin 2\theta\right] = \frac{1}{2}\left[1 + \cos\left(2\theta - 2\alpha\right)\right] = \cos^2\left(\theta - \alpha\right),$$

which is the Malus law, obtained here as a statement about the angle between two vectors on the Poincaré sphere rather than as a statement about a projection in the laboratory. Feeding unpolarised light $\vec{S} = (1, 0, 0, 0)$ into the same formula gives $S_0' = 1/2$ together with a fully polarised output, which is the equally familiar fact that a polariser transmits exactly half of natural light.

The general linear dichroic 3.85, with real transmittances t_1 and t_2 along its two eigenstates, interpolates between the retarder and the polariser and is more interesting still. Taking $J = \mathrm{diag}(t_1, t_2)$ in 3.90 and computing the four congruences as before,

$$M_D = \begin{bmatrix} a & b & 0 & 0 \\ b & a & 0 & 0 \\ 0 & 0 & c & 0 \\ 0 & 0 & 0 & c \end{bmatrix}, \qquad a = \frac{t_1^2 + t_2^2}{2}, \quad b = \frac{t_1^2 - t_2^2}{2}, \quad c = t_1 t_2, \tag{3.96}$$

and a one line computation shows that these three numbers are not independent,

$$a^2 - b^2 = \frac{\left(t_1^2 + t_2^2\right)^2 - \left(t_1^2 - t_2^2\right)^2}{4} = t_1^2 t_2^2 = c^2.$$

The relation $a^2 - b^2 = c^2$ is the signature of a hyperbolic rotation. Writing $a = c \cosh\zeta$ and $b = c \sinh\zeta$, which the constraint permits, 3.96 becomes

$$M_D = t_1 t_2 \begin{bmatrix} \cosh\zeta & \sinh\zeta & 0 & 0 \\ \sinh\zeta & \cosh\zeta & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \qquad \tanh\zeta = \frac{t_1^2 - t_2^2}{t_1^2 + t_2^2} \equiv D, \tag{3.97}$$

an overall attenuation multiplying a **Lorentz boost** along the S_1 axis, with rapidity ζ and with the quantity D, called the diattenuation, playing the part of the velocity β.

This is the promise made at the end of the discussion of why the Stokes vector is not a vector, now delivered. There we found that the physical states fill the forward light cone of an abstract Minkowski space, with the interval $S_0^2 - S_1^2 - S_2^2 - S_3^2$ measuring the departure from full polarisation. We can now name the transformations: **retarders are the rotations of that Minkowski space and dichroics are its boosts**, and together they generate the proper orthochronous Lorentz group.

The geometry is faithful in every detail. A boost along $\hat{n}$ drags points of the sphere toward $\hat{n}$, crowding them ever closer as the rapidity grows, and this is exactly what a partial polariser does: it does not turn a state, it pulls it toward its own transmission axis. The ideal polariser 3.95 is the limit $t_2 \to 0$, that is $D \to 1$ and $\zeta \to \infty$, the infinite boost that carries every point to the same destination, which is why the sphere collapses.

3.5.5.5 What no Jones matrix can do: depolarisation

We can now prove the claim made at the very beginning. Suppose the incoming light is fully polarised, so that by the discussion of the Stokes identity its polarisation matrix has $\det \Phi = 0$, equivalently $\Phi = \vec{e}\,\vec{e}^{\dagger}$ has rank one. Then by 3.88 and the multiplicative property of the determinant,

$$\det \Phi' = \det\left(J\Phi J^{\dagger}\right) = \left|\det J\right|^2 \det \Phi = 0, \tag{3.98}$$

so the outgoing light is fully polarised as well. **A Jones element maps pure states to pure states and can never reduce the degree of polarisation.** The Poincaré sphere may be rotated, boosted or crushed, but a point on its surface never moves into its interior.

Yet the interior is where most of the light in the universe lives. The resolution is that a depolarising sample is not one Jones matrix but a *statistical ensemble* of them, fluctuating faster than the detector can follow or varying from point to point across the illuminated spot: the rough surface presents a different microfacet to each part of the beam, the tissue offers a different scattering history to each photon, the turbulent atmosphere a different birefringence at each instant. Averaging 3.88 over that ensemble and using the linearity of 3.90 in the pair $(J, J^{\dagger})$,

$$M = \langle M\left(J\right)\rangle = \sum_k p_k\, M\left(J_k\right), \qquad p_k \geq 0, \quad \sum_k p_k = 1, \tag{3.99}$$

a convex combination of non depolarising matrices. Convex sums of rotations are not rotations, and this is where the degree of polarisation is lost, in exactly the way that averaging unit vectors pointing in different directions was found to shorten the Stokes vector. The archetype is the ideal depolariser

$$M_{\Delta} = \operatorname{diag}\left(1, d_1, d_2, d_3\right), \qquad \left|d_j\right| \leq 1, \tag{3.100}$$

which shrinks the sphere into an ellipsoid with semiaxes d_1, d_2, d_3 centred on the origin, and which for $d_1 = d_2 = d_3 = 0$ collapses everything onto the centre, that is, onto completely unpolarised light of unchanged intensity.

A parameter count makes the size of the gap plain. A general Mueller matrix carries 16 real numbers. A Jones matrix carries 8, of which one is the unmeasurable global phase, so the non depolarising Mueller matrices form a family of only 7 parameters sitting inside a 16 dimensional space. The overwhelming majority of physically admissible Mueller matrices therefore have got no Jones counterpart whatsoever, and this is the precise sense in which the Mueller calculus is not a reformulation of the Jones calculus but a genuine enlargement of it.

However, we'll see in the chapter of they of coherence that our work in the GOTS can use the Jones matrices to explain with great precission the depolarisation phenomena.

3.5.5.6 Why this matters

It is worth closing with the reason this formalism is used far outside the optics laboratory. A Mueller matrix is 16 real numbers, and every one of them is measurable: illuminate the sample with four known input states, analyse each output with four known projections, and the resulting 16 intensities invert to give M. The technique is called Mueller polarimetry, and what it delivers is a complete linear optical fingerprint of the sample, containing its birefringence, its dichroism and, crucially, its depolarising power, all of them separable from one another by decomposing M into a product of a retarder, a dichroic and a depolariser.

The depolarising part is often the most informative, precisely because it is the part Jones could not describe. Healthy and cancerous tissue differ markedly in how strongly they depolarise, since the difference in the arrangement of collagen fibres changes the statistics of the ensemble in 3.99, and polarimetric imaging is being developed as a label free diagnostic on exactly this basis. In remote sensing the depolarisation of scattered sunlight distinguishes spherical droplets from irregular ice crystals in a cloud, which is why the Stokes formalism entered atmospheric physics through the radiative transfer of Chandrasekhar (Chandrasekhar, 1950). In semiconductor metrology, ellipsometry measures the Mueller matrix of a wafer to determine film thicknesses to sub nanometre precision (Azzam and Bashara, 1977). And in astronomy the polarisation of starlight, scrambled by interstellar dust, maps the galactic magnetic field.

There is finally a structural lesson that reaches well beyond optics. The polarisation matrix Φ is a 2×2 Hermitian, positive semidefinite matrix of unit trace once normalised, which is precisely the definition of the density matrix of a two level quantum system; the Poincaré sphere is its Bloch sphere; a Jones matrix is a unitary evolution; and a general Mueller matrix is the classical counterpart of a quantum channel, a completely positive map, with 3.99 playing the role of the ensemble decomposition of that channel. Depolarisation is decoherence. Everything we have built in this subsection, the rotations, the boosts, the convex mixing that destroys purity, reappears unchanged in the theory of open quantum systems, and a bench with a laser and two waveplates is one of the cheapest and most transparent laboratories in which those ideas can be watched happening.

3.5.6 The Jones theorems

Jones did not merely propose a convenient notation. In the series of papers that runs through the *Journal of the Optical Society of America* between 1941 and 1956, and above all in the second of them, written jointly with Henry Hurwitz (Hurwitz and Jones, 1941), he proved a handful of structural theorems that answer a question no amount of matrix multiplication answers by itself: given that an optical bench may contain an arbitrary number of waveplates, polarisers and rotators arranged in any order, **what is the most general thing such a bench can possibly do to a beam of light, and how few elements are needed to do it?**

The answers are remarkably restrictive, and that is exactly what makes them useful. They tell the instrument designer that certain constructions are impossible, that certain others are redundant, and that any device whatsoever, however baroque, is equivalent to a short standard chain. Before stating them we need to fix what equivalence means and to classify the primitive elements by the kind of matrix they possess.

Definition 24 *(Optical equivalence): Two optical systems are* ***optically equivalent*** *if their Jones matrices differ at most by a global phase factor, that is, if $J_2 = e^{i\gamma} J_1$ for some real γ. Equivalently, by 3.90, they possess the same Mueller matrix, and therefore no measurement of any kind performed on the outgoing light can distinguish them.*

The three primitive devices built in the previous subsections fall into sharply distinct algebraic classes, and the whole of what follows is a consequence of that classification. A retarder 3.75 is **unitary** and, once the irrelevant global phase is removed, **unimodular**, $\det W = 1$; the reader may check this directly on the linear retarder 3.76, whose determinant is $\cos^2 \frac{\delta}{2} + \sin^2 \frac{\delta}{2} \left(\cos^2 2\alpha + \sin^2 2\alpha\right) = 1$. A rotator 3.83 is **real orthogonal** with unit determinant, hence a particular case of a retarder, as we already proved. A linear dichroic 3.85, and its limiting case the ideal polariser 3.86, is **real symmetric and positive semidefinite**, while a general elliptical dichroic 3.84 is **Hermitian and positive semidefinite**. Finally, a cascade is the ordered product of the matrices of its elements. Every theorem below is a statement about what such products can look like.

3.5.6.1 Theorem I: every system has an eigenpolarisation

Theorem 2 *(Existence of eigenpolarisations): Every optical system with Jones matrix J possesses at least one polarisation state that it leaves unchanged in form, altering it at most in amplitude and phase. Generically there are exactly two such states, and they are mutually orthogonal if and only if J is a normal matrix, $JJ^\dagger = J^\dagger J$.*

Proof. The eigenvalue equation $J\vec{e} = \lambda\vec{e}$ has a non trivial solution precisely when $\det(J - \lambda I) = 0$. Writing $J = \begin{bmatrix} j_{11} & j_{12} \\ j_{21} & j_{22} \end{bmatrix}$, this determinant is the quadratic

$$\lambda^2 - (j_{11} + j_{22})\,\lambda + (j_{11}j_{22} - j_{12}j_{21}) = \lambda^2 - (\operatorname{tr} J)\,\lambda + \det J = 0, \tag{3.101}$$

a polynomial of degree two with complex coefficients. By the fundamental theorem of algebra it possesses at least one root $\lambda_1 \in \mathbb{C}$, and for that root the matrix $J - \lambda_1 I$ is singular, so its kernel contains a non zero vector $\vec{e}_1$. That vector is a Jones spinor 3.34 and therefore a genuine polarisation state, and by construction $J\vec{e}_1 = \lambda_1\vec{e}_1$: the outgoing spinor is the incoming one multiplied by a single complex number, whose modulus is an amplitude change and whose argument is a phase change, neither of which alters the shape or the orientation of the ellipse. When the discriminant of 3.101 does not vanish the two roots are distinct and the two eigenvectors are linearly independent, giving two eigenpolarisations.

For the second part, recall the spectral theorem: a complex square matrix admits an *orthonormal* basis of eigenvectors if and only if it is normal. Since orthogonality of the two eigenspinors in the

Hermitian product is precisely orthogonality of the two polarisation states, the equivalence follows at once. ■

The theorem has got more content than it first appears. All three primitive elements are normal: a retarder is unitary and a dichroic is Hermitian, and both classes satisfy $JJ^\dagger = J^\dagger J$ trivially. Their eigenstates are therefore always orthogonal, which is why we were able to build every one of them in the previous subsections from a pair of orthogonal states $|\psi_1\rangle, |\psi_2\rangle$ using a spectral decomposition. **A cascade, however, need not be normal.** Take a linear retarder of retardance $\delta = 1.2$ with its axis at $\alpha = 0.4$, followed by a linear dichroic with transmittances $t_1 = 0.9$ and $t_2 = 0.3$ along the coordinate axes. The commutator $[J, J^\dagger]$ of the product does not vanish, and the two eigenpolarisations turn out to satisfy $|\langle e_1|e_2\rangle| \approx 0.46$, which is very far from zero.

Elements whose eigenpolarisations are orthogonal are called **homogeneous**, and the others **inhomogeneous**. The lesson is worth stating plainly, because it is a common source of confusion: a waveplate followed by a polariser is not, in general, describable as some other waveplate followed by some other polariser with the same axes, and the reason is that the composite object has stopped being normal. The eigenstates it does possess are no longer a pair of orthogonal ellipses but a skew pair, and this is one of the ways in which real optical systems misbehave.

3.5.6.2 Theorem II: retarders and rotators

Theorem 3 *(First equivalence theorem): An optical system containing any number of retarders and rotators, in any order, is optically equivalent to a system containing only two elements: one linear retarder followed by one rotator.*

Proof. Every retarder and every rotator has got a unitary, unimodular Jones matrix, that is, an element of the group $SU(2)$. Since $SU(2)$ is a group it is closed under multiplication, so the whole cascade, whatever its length and order, is again a single element $U \in SU(2)$. The theorem therefore reduces to a purely algebraic claim: every $U \in SU(2)$ can be written as $U = R(\theta)\, W(\alpha, 0; \delta)$ for suitable θ, α, δ.

The cleanest way to see this is to move to the Poincaré sphere, where we already know what these objects do. By the theorem on retarders as rotations, the linear retarder $W(\alpha, 0; \delta)$ acts on the sphere as a rotation by δ about the equatorial axis 3.94 sitting at azimuth 2α, since $\chi = 0$ places $\hat{n}$ on the equator; and the rotator $R(\theta)$, being a circular retarder, acts as a rotation about the polar axis. So the claim becomes: *every rotation of the sphere is a rotation about an equatorial axis followed by a rotation about the polar axis.*

This is the Euler angle decomposition in disguise. Write the general rotation in the ZYZ convention as $\mathcal{R} = Z(\phi)\, Y(\vartheta)\, Z(\psi)$, which is possible for any element of $SO(3)$. Now insert the identity $Z(\psi)Z(-\psi)$ and regroup,

$$\mathcal{R} = Z(\phi)Y(\vartheta)Z(\psi) = \underbrace{Z\,(\phi + \psi)}_{\text{polar}}\ \underbrace{[Z(-\psi)\, Y(\vartheta)\, Z(\psi)]}_{\text{equatorial}},$$

and observe that the bracket is a rotation by the angle ϑ about the axis $Z(-\psi)\hat{y}$, which lies in the equatorial plane because $\hat{y}$ does and Z preserves that plane. The first factor is a rotation about the polar axis. Since ϕ, ϑ, ψ range over all of $SO(3)$, so do the three parameters θ, α, δ of our two element chain. Finally, the Mueller map 3.90 restricted to $SU(2)$ is onto $SO(3)$, and its only ambiguity is the sign of U, which is a global phase and hence optically invisible. The two systems are therefore optically equivalent. ∎

The physical reading is the one that matters. **No matter how many waveplates you stack, at whatever angles, the result is a single elliptical retarder.** It can absorb nothing, because unitarity is preserved under multiplication, and it can depolarise nothing, by 3.98. Everything a pile of birefringent crystal can do is to rotate the Poincaré sphere once, and three numbers suffice to say how. This is why the elliptical retarder we constructed by a change of basis is not an academic curiosity but a piece of hardware: any (α, χ, δ) you like can be realised with one ordinary waveplate and one rotator, and in the laboratory the combination of two quarter wave plates and a half wave plate is the standard universal polarisation controller built on exactly this theorem.

3.5.6.3 Theorem III: partial polarisers and rotators

Theorem 4 *(Second equivalence theorem): An optical system containing any number of linear partial polarisers and rotators, in any order, is optically equivalent to a system containing only two elements: one linear partial polariser followed by one rotator.*

Proof. A linear partial polariser 3.85 has a *real symmetric* matrix, and a rotator 3.83 has a *real orthogonal* one. Products of real matrices are real, so the entire cascade is described by some real matrix A, non singular as long as no element is an ideal polariser. Furthermore $\det A > 0$, because every factor has positive determinant, namely $t_1 t_2 > 0$ for a partial polariser and $+1$ for a rotator.

We now invoke the real polar decomposition, and since it is the engine of both this theorem and the next, let us prove it rather than quote it. Consider the matrix AA^{T}. It is symmetric, since $\left(AA^{\mathsf{T}}\right)^{\mathsf{T}} = AA^{\mathsf{T}}$, and it is positive definite, because for any non zero real vector $\vec{v}$ we have $\vec{v}^{\mathsf{T}} AA^{\mathsf{T}} \vec{v} = \left\|A^{\mathsf{T}}\vec{v}\right\|^2 > 0$, the strict inequality holding because A^{T} is invertible. A real symmetric positive definite matrix can be diagonalised by an orthogonal matrix, $AA^{\mathsf{T}} = V \operatorname{diag}(\mu_1, \mu_2)\, V^{\mathsf{T}}$ with $\mu_1, \mu_2 > 0$, so we may define

$$S \equiv V \operatorname{diag}\left(\sqrt{\mu_1}, \sqrt{\mu_2}\right) V^{\mathsf{T}}, \tag{3.102}$$

which is again real, symmetric, positive definite, invertible, and satisfies $S^2 = AA^{\mathsf{T}}$. Setting $R \equiv S^{-1}A$ we verify directly that

$$RR^{\mathsf{T}} = S^{-1}A\, A^{\mathsf{T}} S^{-1} = S^{-1}S^2 S^{-1} = I,$$

where we used that S^{-1} is symmetric. So R is real orthogonal, and $\det R = \det A / \det S > 0$ forces $\det R = +1$, making R a proper rotation, that is, a rotator $R(\theta)$. We have obtained $A = SR$ with S

a real symmetric positive definite matrix, which is exactly a linear partial polariser with some axis and some pair of transmittances, followed by a rotator. ■

Note where each hypothesis was used, because the theorem is sharp. Reality of the matrices is what forced the unitary factor to be a mere rotation instead of a general retarder; had we allowed elliptical partial polarisers, the argument would have produced a retarder in its place and the statement would fail. Physically this says something clean: **a stack of absorbing but non birefringent elements can never manufacture birefringence**, it can only attenuate anisotropically and turn the plane of polarisation. Dichroism and birefringence do not generate each other.

3.5.6.4 Theorem IV: the general equivalence theorem

Theorem 5 *(Third equivalence theorem): Every non singular Jones matrix J factorises uniquely as*

$$J = H\,U, \tag{3.103}$$

with H Hermitian positive definite and U unitary. Consequently any optical system whatsoever, containing any number of retarders, partial polarisers and rotators in any order, is optically equivalent to one elliptical partial polariser followed by one elliptical retarder, and therefore, expanding each of these through Theorem II, to a chain of four primitive elements: two retarders, one partial polariser and one rotator.

Proof. The construction copies the real case with daggers in place of transposes. Consider $JJ^\dagger$. It is Hermitian, since $\left(JJ^\dagger\right)^\dagger = JJ^\dagger$, and positive definite, because for any non zero $\vec{v} \in \mathbb{C}^2$,

$$\vec{v}^\dagger JJ^\dagger \vec{v} = \left(J^\dagger \vec{v}\right)^\dagger \left(J^\dagger \vec{v}\right) = \left\|J^\dagger \vec{v}\right\|^2 > 0,$$

strictly, because $J^\dagger$ is invertible and so cannot annihilate $\vec{v}$. By the spectral theorem a Hermitian matrix is diagonalisable by a unitary one, $JJ^\dagger = V\,\mathrm{diag}\,(\mu_1, \mu_2)\,V^\dagger$ with $\mu_1, \mu_2 > 0$ real, and we define its positive square root

$$H \equiv V\,\mathrm{diag}\,(\sqrt{\mu_1}, \sqrt{\mu_2})\,V^\dagger, \tag{3.104}$$

which is Hermitian, positive definite, invertible and satisfies $H^2 = JJ^\dagger$. Put $U \equiv H^{-1}J$. Then, using $H^\dagger = H$ and hence $\left(H^{-1}\right)^\dagger = H^{-1}$,

$$UU^\dagger = H^{-1}J\,J^\dagger H^{-1} = H^{-1}H^2H^{-1} = I,$$

so U is unitary and $J = HU$ as claimed. For uniqueness, suppose $J = H'U'$ with H' Hermitian positive definite and U' unitary. Then $JJ^\dagger = H'U'U'^\dagger H' = H'^2$, and since a positive definite Hermitian matrix has got exactly one positive definite Hermitian square root, $H' = H$, whence $U' = H^{-1}J = U$. ■

The dictionary back to hardware is immediate. The factor H, being Hermitian and positive definite, is by 3.84 an elliptical partial polariser, with its two eigenstates and its two real transmittances

$\sqrt{\mu_1}, \sqrt{\mu_2}$. The factor U, being unitary, is by 3.75 an elliptical retarder, which Theorem II lets us build from one linear retarder and one rotator. And H itself, an elliptical partial polariser, is a linear partial polariser conjugated by a retarder that carries the linear eigenstates onto the elliptical ones. Counting the free parameters confirms that nothing has been lost or invented: a general $J \in GL(2,\mathbb{C})$ has got eight real parameters, matched exactly by four for H plus four for U in 3.103, and equally by the hardware chain of two retarders at two parameters each, one partial polariser at three, and one rotator at one, totalling eight.

This is the deepest of the three theorems and it deserves a sentence of its own. **Every linear, non depolarising optical system in existence, however many elements it contains, does exactly two things and in this order: it attenuates the two components of the field unequally, and then it retards one relative to the other.** Diattenuation followed by birefringence, and nothing else is possible. In the language of the previous subsection the statement is even more striking, because a Hermitian positive H maps to a Lorentz boost and a unitary U to a rotation: 3.103 is nothing but the familiar decomposition of a Lorentz transformation into a boost times a rotation, transplanted to the Poincaré sphere. The result is not a historical curiosity either. Its descendant at the level of Mueller matrices, in which a measured M is factorised into a depolariser, a retarder and a diattenuator, is the standard tool by which a modern polarimeter turns sixteen raw numbers into three physically interpretable objects.

3.5.6.5 Theorem V: reversibility

The last theorem concerns what happens when the light is sent back through the system, and it is the one with the most conspicuous practical consequence.

Theorem 6 *(Reversibility): If a reciprocal optical system has Jones matrix J for propagation in one direction, then for propagation in the opposite direction its Jones matrix is the transpose J^{T}, the transverse axes being referred to the direction of travel.*

Proof. Reversing the beam does two things: it traverses the elements in the opposite order, and it traverses each individual element backwards. The first is automatic in the calculus, since if the forward system is $J = J_n \cdots J_2 J_1$ then the reversed one is $\tilde{J}_1 \tilde{J}_2 \cdots \tilde{J}_n$, where $\tilde{J}_k$ denotes the kth element traversed backwards. The identity $(J_n \cdots J_1)^{\mathsf{T}} = J_1^{\mathsf{T}} \cdots J_n^{\mathsf{T}}$ reverses the order automatically, so the theorem holds for the whole system as soon as it holds for each element, $\tilde{J}_k = J_k^{\mathsf{T}}$. It therefore suffices to check the primitives.

A linear retarder 3.76 is a **symmetric** matrix, its two off diagonal entries being equal to $i \sin\frac{\delta}{2} \sin 2\alpha$, so $W^{\mathsf{T}} = W$: a waveplate looks the same from either side, which is exactly what a slab of birefringent crystal must do, since its fast and slow axes are properties of the crystal and not of the direction of travel. A linear dichroic 3.85 is likewise symmetric, and for the same reason. A rotator 3.83 is orthogonal, so $R(\theta)^{\mathsf{T}} = R(\theta)^{-1} = R(-\theta)$: run backwards, an optically active medium rotates the plane of polarisation by the opposite angle. This is the correct physics, because in a sugar solution the sense of rotation is tied to the handedness of the molecules and to the direction of propagation together, so reversing the beam reverses the apparent sense. Every primitive obeys the rule, and therefore so does every system built from them. ■

The consequences are best seen in a double pass, the situation of a beam that goes through a system, reflects off a mirror and returns. Ignoring the mirror itself, the round trip matrix is $J^{\mathsf{T}}J$. For a quarter wave plate with its axis along x this gives, since W is symmetric,

$$W\left(0,0;\tfrac{\pi}{2}\right)^{\mathsf{T}} W\left(0,0;\tfrac{\pi}{2}\right) = W\left(0,0;\tfrac{\pi}{2}\right)^2 = \begin{bmatrix} i & 0 \\ 0 & -i \end{bmatrix} = W\left(0,0;\pi\right), \tag{3.105}$$

precisely a half wave plate: **double passing a retarder doubles its retardance**, which is the everyday trick by which a quarter wave plate placed in front of a mirror rotates linear polarisation by ninety degrees and lets a polarising beam splitter separate the outgoing beam from the returning one. For a rotator, by contrast,

$$R(\theta)^{\mathsf{T}} R(\theta) = R(-\theta)R(\theta) = I, \tag{3.106}$$

so **double passing an optically active medium undoes the rotation exactly**. Send polarised light through a tube of sugar solution, reflect it and send it back, and it emerges with its original azimuth.

And here is the payoff. A Faraday rotator, mentioned when we introduced the rotators, looks identical to an optically active medium on a single pass, yet its sense of rotation is fixed by the external magnetic field alone and not by the direction of propagation. On the return pass it therefore contributes $R(\theta)$ again rather than $R(-\theta)$, so the round trip gives $R(\theta)R(\theta) = R(2\theta)$ instead of the identity. The Faraday rotator **violates the reversibility theorem**, and it is allowed to do so precisely because a magnetic field breaks the time reversal symmetry on which reciprocity rests. This single fact is the entire working principle of the optical isolator, the device that lets light out of a laser but not back into it, without which no high power laser or fibre optic communication link would be stable. A theorem about transposes, proved in three lines, is what tells us that such a device cannot be built from waveplates and sugar and must be built from magnetised glass.

3.5.6.6 What the theorems say together

Standing back, the five results form a single picture. Theorem I says every system has a state it preserves, but warns that cascades of well behaved elements need not be well behaved themselves. Theorems II and III say that the two families of devices are closed and do not generate each other: birefringence stacks into birefringence and dichroism into dichroism. Theorem IV says that when the two families are mixed, the outcome is still astonishingly restricted, one diattenuator and one retarder and nothing more, eight real numbers in total. Theorem V says that this whole structure is blind to the direction of travel unless a magnetic field is present to break the symmetry.

The unifying statement is group theoretic, and it is worth making explicit because it will reappear whenever linear systems are studied. The non depolarising optical elements form the group $GL(2,\mathbb{C})$, of which the retarders form the unitary subgroup and the dichroics the positive Hermitian cone, and

3.103 is the statement that the group is the product of the two. Everything a bench can do to fully polarised light is an element of that group, and everything it can do to partially polarised light is, by the ensemble argument 3.99, a convex mixture of such elements. The Jones theorems are, in the end, the complete answer to the question of what polarisation optics is capable of.

3.5.7 Generation of elliptic states: Composition of Waveplates

Every theorem proved in the previous subsection took an optical bench as already built and asked what such a bench does to an arbitrary beam. It is time to turn the question the other way around. Suppose the source on the table delivers a fixed, fully polarised beam, most often plain linear light straight out of a laser or through a first polariser, and the task set is to deliver at the far end an elliptical state $|\alpha; \chi\rangle$ chosen in advance, whatever α and χ happen to be. Theorem II already contains the answer in embryo: a cascade of retarders and rotators, however long and in whatever order, is optically equivalent to a single elliptical retarder, so the sphere can always be carried from the starting point to the chosen one by a fixed, short chain of birefringent plates. What that theorem left unsaid is how, given the two points, the axis and the retardance of such a chain are actually to be dialled in, and that is the question this subsection answers.

The decision to build the whole apparatus out of waveplates alone, and out of nothing else, is not a matter of taste but the entire point of the exercise. A retarder is unitary, so by 3.98 it cannot depolarise the beam, and by 3.93 its Mueller matrix leaves S_0 fixed, so it cannot absorb from it either: whatever power is measured at the entrance of the first plate is exactly the power measured at the exit of the last one, merely redistributed between the two field components rather than thrown away. Composing waveplates is therefore **the one method of preparing an arbitrary ellipse that costs nothing in intensity**, a pure rotation of the Poincaré sphere rather than the boost toward a fixed point that a polariser, as the next subsection will show, cannot avoid performing. This is precisely why the pair of quarter wave plates flanking a half wave plate, already singled out at the end of Theorem II as the standard universal polarisation controller, is the instrument reached for whenever the prepared state must carry as much of the source's power as possible, from the routine calibration of a polarimeter to the encoding of a qubit in the polarisation of a single photon, a setting in which every photon lost on the way to the desired ellipse is a photon that no later stage of the experiment can recover.

3.5.7.1 The linear birrefringents law

Every element built in the previous subsections was described by numbers fixed once and for all, the pair of eigenstates and the retardance, and the algebra never asked what happens when the experimenter walks up to the bench and *turns* one of them. Turning a waveplate is, however, the commonest gesture in a polarisation laboratory and by far the cheapest degree of freedom available, since a plate in a rotation stage costs nothing beyond the plate itself, so it is the natural place to begin. The question we settle here is therefore the following: **if one single plate is turned bodily through a complete revolution, which polarisation states come out?** The answer is a theorem with a very clean geometrical content, formulated in this form by Salazar Ariza in her doctoral thesis (Salazar Ariza, 2021), whose statement and proof we follow here, adapted to the conventions of these notes.

Before stating it we must write down properly a naming convention that has been used implicitly ever since the retarder was built, because it is a permanent source of confusion for the beginner.

Definition 25 *(Linear, elliptical and circular birefringents): A birefringent element, or retarder 3.75, is named after the polarisation of its own eigenstates* $|\psi_1\rangle = |\alpha; \chi\rangle$ *and* $|\psi_2\rangle = |\alpha + \pi/2; -\chi\rangle$*, and never after the light it happens to be fed with. It is a* ***linear birefringent*** *when* $\chi = 0$*, so that both eigenstates are linear and the element is 3.76; a* ***circular birefringent*** *when* $\chi = \pm\pi/4$*, so that both eigenstates are the circular states 3.53 and the element is the rotator 3.83; and an* ***elliptical birefringent*** *for every intermediate value* $0 < |\chi| < \pi/4$*, the general case 3.81.*

On the Poincaré sphere the three names have got an immediate reading through 3.94, since the rotation axis of a retarder is the point representing its own eigenstate: a linear birefringent turns the sphere about an *equatorial* axis, a circular one about the *polar* axis, and an elliptical one about an axis of intermediate latitude. An ordinary waveplate cut from quartz or mica is a linear birefringent, and it is the only one of the three that can be bought off the shelf and screwed into a rotation mount, which is why it is the one whose rotation we study.

Let the plate be turned in its own plane so that its fast axis makes an angle θ with the horizontal, and let it act on a fixed, fully polarised input $|\alpha; \chi\rangle$ whose Stokes vector 3.65, normalised to $S_0 = 1$, we write as

$$\vec{S} = (S_1, S_2, S_3) = (\cos 2\alpha \cos 2\chi,\ \sin 2\alpha \cos 2\chi,\ -\sin 2\chi)\,. \tag{3.107}$$

Two remarks about the parameter θ are in order before anything is computed. First, the angle enters the Jones matrix 3.76 only through 2θ, so the plate at θ and the plate at $\theta + \pi$ are the same object, as they must be: the fast axis is an unoriented line drawn on a piece of crystal and not an arrow, and half a turn of the mount puts the very same crystal back in the beam. Consequently **a physical rotation from $0°$ to $360°$ traverses the whole family of accessible states exactly twice**, and half a revolution already exhausts it. Second, the retardance δ does not change while the plate turns, since 3.74 depends only on the thickness and on the two refractive indices, both of them properties of the crystal and not of its orientation.

Definition 26 *(Enantiogyre states): Two fully polarised states are* ***enantiogyres*** *if their ellipses have got the same orientation and the same shape but are traced in opposite senses, that is, if they are the pair* $|\alpha; \chi\rangle$ *and* $|\alpha; -\chi\rangle$*. By 3.65 the operation relating them is the reflection of the Poincaré sphere in its equatorial plane,*

$$\vec{S} = (S_1, S_2, S_3) \longmapsto \vec{S}^{\,e} = (S_1, S_2, -S_3)\,, \tag{3.108}$$

the name being coined in (Salazar Ariza, 2021) from the ancient Greek enantios, opposite, and gyros, turn.

The enantiogyre must not be confused with the orthogonal state, which we proved to be the *antipode* $(-S_1, -S_2, -S_3)$ of the sphere: the enantiogyre is only a mirror image in the equator, it keeps the azimuth α untouched and flips nothing but the handedness. The two coincide only for the circular states, and a linear state is its own enantiogyre, since $\chi = 0$ gives $S_3 = 0$ and the reflection 3.108 leaves it where it was.

Theorem 7 *(Law of the rotating linear birefringent): Let a fully polarised beam of Stokes vector 3.107 cross a linear birefringent of fixed retardance δ whose fast axis lies at an angle θ with the horizontal, and let $\vec{S}\,'(\theta)$ be the Stokes vector of the emerging beam. As the plate is rotated bodily through a complete revolution, $0 \leq \theta \leq 2\pi$, the locus of $\vec{S}\,'(\theta)$ is the closed curve in which the Poincaré sphere is cut by the circular cone*

$$\boxed{\left(S_1' - S_1\right)^2 + \left(S_2' - S_2\right)^2 = \tan^2\frac{\delta}{2}\left(S_3' + S_3\right)^2} \tag{3.109}$$

whose vertex is the enantiogyre 3.108 of the input state, whose axis is parallel to the S_3 axis, and whose half opening angle is half the retardance, $\delta/2$. The curve is traversed twice in a full turn of the plate.

Proof. Nothing has to be multiplied out to see that the plate moves the beam by rotating the sphere, because that work was already done when we proved that every retarder is a rigid rotation 3.93 by its retardance about the axis 3.94 representing its own eigenstate. Our plate is linear, $\chi = 0$, and its fast axis sits at azimuth θ, so putting $\alpha = \theta$ and $\chi = 0$ in 3.94, where $\cos 2\chi = 1$ and $\sin 2\chi = 0$ kills the third component, its axis is the equatorial unit vector

$$\hat{n}(\theta) = (\cos 2\theta,\ \sin 2\theta,\ 0)\,, \tag{3.110}$$

which sweeps the entire equator once while θ runs over half a turn. The emerging state is therefore $\vec{S}\,'(\theta) = R\left(\hat{n}(\theta), \delta\right)\vec{S}$, and since a rotation preserves lengths, $\vec{S}\,'$ never leaves the unit sphere.

Expanding that rotation in laboratory coordinates is what would make the computation long, so let us instead carry a frame along with the plate. Take

$$\hat{u} = \hat{n}(\theta) = (\cos 2\theta, \sin 2\theta, 0)\,, \qquad \hat{v} = \hat{z} \times \hat{u} = (-\sin 2\theta, \cos 2\theta, 0)\,, \qquad \hat{z} = (0,0,1)\,, \tag{3.111}$$

three mutually orthogonal unit vectors with $\hat{u} \times \hat{v} = \hat{z}$, so the frame is right handed. It is the natural frame here because $\hat{u}$ is the rotation axis, which stays put, and the pair $(\hat{v}, \hat{z})$ spans the plane in which the turning actually happens. Notice at once the asymmetry that will produce the whole theorem: $\hat{u}$ **and** $\hat{v}$ **turn with the plate, but** $\hat{z}$ **is nailed to the laboratory**.

Decomposing $\vec{S} = p\,\hat{u} + q\,\hat{v} + S_3\hat{z}$ in that frame, the three coefficients are the projections $p = \vec{S} \cdot \hat{u}$, $q = \vec{S} \cdot \hat{v}$ and $\vec{S} \cdot \hat{z} = S_3$, that is

$$\begin{aligned} p &= S_1 \cos 2\theta + S_2 \sin 2\theta = \cos 2\chi \left[\cos 2\alpha \cos 2\theta + \sin 2\alpha \sin 2\theta\right] = \cos 2\chi \cos\left(2\alpha - 2\theta\right)\,, \\ q &= -S_1 \sin 2\theta + S_2 \cos 2\theta = \cos 2\chi \left[\sin 2\alpha \cos 2\theta - \cos 2\alpha \sin 2\theta\right] = \cos 2\chi \sin\left(2\alpha - 2\theta\right)\,, \end{aligned}$$

where we substituted 3.107, took the common factor $\cos 2\chi$ out of both brackets and recognised in them the cosine and the sine of the difference $2\alpha - 2\theta$. Only q will survive to the end.

A right handed rotation by δ about $\hat{u}$ leaves $\hat{u}$ untouched and turns the other two basis vectors into $\hat{v} \to \cos\delta\,\hat{v} + \sin\delta\,\hat{z}$ and $\hat{z} \to -\sin\delta\,\hat{v} + \cos\delta\,\hat{z}$, so acting on $\vec{S}$ term by term,

$$\begin{aligned} \vec{S}\,' &= p\,\hat{u} + q\,(\cos\delta\,\hat{v} + \sin\delta\,\hat{z}) + S_3\,(-\sin\delta\,\hat{v} + \cos\delta\,\hat{z}) \\ &= p\,\hat{u} + (q\cos\delta - S_3\sin\delta)\,\hat{v} + (q\sin\delta + S_3\cos\delta)\,\hat{z}. \end{aligned} \tag{3.112}$$

This is the whole of the physics of the problem; everything that follows is trigonometry, and it starts with the one idea the proof needs: measure the output from the enantiogyre, not from the origin. The enantiogyre 3.108 of the input, $\vec{S}^{\,\mathrm{e}} = (S_1, S_2, -S_3)$, has got exactly the same equatorial part as $\vec{S}$, and therefore in the moving frame 3.111 the same two coefficients p and q just found, because the reflection touches nothing but the third component,

$$\vec{S}^{\,\mathrm{e}} = p\,\hat{u} + q\,\hat{v} - S_3\hat{z}.$$

Subtracting this from 3.112 component by component,

$$\begin{aligned} \vec{S}\,' - \vec{S}^{\,\mathrm{e}} &= (\cancel{p-p})\,\hat{u} + (q\cos\delta - S_3\sin\delta - q)\,\hat{v} + (q\sin\delta + S_3\cos\delta + S_3)\,\hat{z} \\ &= [q\,(\cos\delta - 1) - S_3\sin\delta]\,\hat{v} + [q\sin\delta + S_3\,(\cos\delta + 1)]\,\hat{z}, \end{aligned}$$

and the very first thing to notice is that the component along the rotation axis $\hat{u}$ has cancelled identically, for every θ and every δ. The segment joining the fixed point $\vec{S}^{\,\mathrm{e}}$ to the moving point $\vec{S}\,'$ is perpendicular to the axis of the plate, and half of the theorem is already proved.

Insert now the three identities $\cos\delta - 1 = -2\sin^2\frac{\delta}{2}$, $\cos\delta + 1 = 2\cos^2\frac{\delta}{2}$ and $\sin\delta = 2\sin\frac{\delta}{2}\cos\frac{\delta}{2}$, which turn each bracket into a product,

$$\begin{aligned} [q\,(\cos\delta - 1) - S_3\sin\delta] &= -2q\sin^2\frac{\delta}{2} - 2S_3\sin\frac{\delta}{2}\cos\frac{\delta}{2} = -2\sin\frac{\delta}{2}\left(q\sin\frac{\delta}{2} + S_3\cos\frac{\delta}{2}\right), \\ [q\sin\delta + S_3\,(\cos\delta + 1)] &= 2q\sin\frac{\delta}{2}\cos\frac{\delta}{2} + 2S_3\cos^2\frac{\delta}{2} = \quad 2\cos\frac{\delta}{2}\left(q\sin\frac{\delta}{2} + S_3\cos\frac{\delta}{2}\right), \end{aligned}$$

where in each line we simply took out the common factor, $-2\sin\frac{\delta}{2}$ in the first and $2\cos\frac{\delta}{2}$ in the second. **The very same bracket has appeared twice**, so it deserves a name,

$$K(\theta) \equiv q\sin\frac{\delta}{2} + S_3\cos\frac{\delta}{2} = \cos 2\chi\sin(2\alpha - 2\theta)\sin\frac{\delta}{2} - \sin 2\chi\cos\frac{\delta}{2}, \tag{3.113}$$

the last expression following from the value of q found above and from $S_3 = -\sin 2\chi$. With it, the whole displacement collapses into a single vector,

$$\vec{S}\,' - \vec{S}^{\,\mathrm{e}} = 2K(\theta)\underbrace{\left[-\sin\frac{\delta}{2}\,\hat{v} + \cos\frac{\delta}{2}\,\hat{z}\right]}_{\hat{w}(\theta)}, \tag{3.114}$$

and $\hat{w}$ is a unit vector, since $\hat{v}$ and $\hat{z}$ are orthogonal and $\sin^2\frac{\delta}{2}+\cos^2\frac{\delta}{2}=1$.

It only remains to read off the geometry that 3.114 is describing. Project $\hat{w}$ on the fixed vertical direction,

$$\hat{w}\cdot\hat{z} = -\sin\frac{\delta}{2}\,\cancel{(\hat{v}\cdot\hat{z})} + \cos\frac{\delta}{2}\,\underbrace{(\hat{z}\cdot\hat{z})}_{=1} = \cos\frac{\delta}{2},$$

where the first term vanished because $\hat{v}$ is equatorial and $\hat{z}$ is polar. This number does not depend on θ, even though $\hat{w}$ itself does, and that is the crux of the matter: the direction in which the output sits, *as seen from the enantiogyre*, may turn about the vertical as the plate turns, but it never changes its inclination. The angle between $\vec{S}\,' - \vec{S}^{\,\mathrm{e}}$ and the S_3 axis is $\delta/2$ whenever $K>0$ and $\pi-\delta/2$ whenever $K<0$, which is precisely the description of the two nappes of a right circular cone of vertex $\vec{S}^{\,\mathrm{e}}$, axis $\hat{z}$ and half opening $\delta/2$. Writing it in coordinates, the equatorial part of 3.114 is $-2K\sin\frac{\delta}{2}\hat{v}$, of length $2\,|K|\sin\frac{\delta}{2}$, and its vertical part is $2K\cos\frac{\delta}{2}$, so dividing one by the other,

$$\frac{\sqrt{\left(S_1'-S_1\right)^2+\left(S_2'-S_2\right)^2}}{|S_3'+S_3|} = \frac{\cancel{2\,|K|}\sin\frac{\delta}{2}}{\cancel{2\,|K|}\cos\frac{\delta}{2}} = \tan\frac{\delta}{2},$$

since the common factor $2\,|K|$, the only θ dependent quantity left, cancels between numerator and denominator. Squaring both sides gives 3.109, and as the output also lies on the unit sphere, $\vec{S}\,'(\theta)$ belongs to the intersection of the two surfaces.

One thing is still owed: that the plate reaches every point of that intersection, and nothing outside it. Cut the picture at a fixed height $S_3'=z$. The cone meets that horizontal plane in a circle of radius $\tan\frac{\delta}{2}\,|z+S_3|$ centred at (S_1,S_2) and the sphere meets it in a circle of radius $\sqrt{1-z^2}$ centred at the origin, and two distinct circles of a plane cross at most twice, so the intersection offers at most two points at each height. On the other hand the height actually reached by the beam is, by the third component of 3.112,

$$S_3'(\theta) = q\sin\delta + S_3\cos\delta = \cos 2\chi\sin\left(2\alpha-2\theta\right)\sin\delta + S_3\cos\delta, \tag{3.115}$$

a sinusoid of 2θ, so every height strictly between its two extremes is attained at exactly two orientations of the plate within half a revolution. The two points the plate supplies are the two points the geometry allows, the locus closes on itself, and by the earlier remark on the periodicity of θ a full turn runs over it twice. ■

The theorem is worth testing against the two plates we already know. For a **half wave plate**, $\delta=\pi$, the half opening becomes $\pi/2$ and $\tan\frac{\delta}{2}$ diverges, so 3.109 must be read after dividing by $\tan^2\frac{\delta}{2}$, which leaves $\left(S_3'+S_3\right)^2=0$: the cone has opened out into the horizontal plane through its own vertex, and the trajectory is the full circle of constant latitude $S_3'=-S_3$. Whatever the orientation of a half wave plate, the emerging ellipse has got the same shape as the incoming one and the opposite handedness, $\chi'=-\chi$, while the plate angle controls its azimuth and nothing else. This contains, as the particular case $\chi=\pm\pi/4$, the fact already extracted from 3.78, that a half wave plate always exchanges the two circular states.

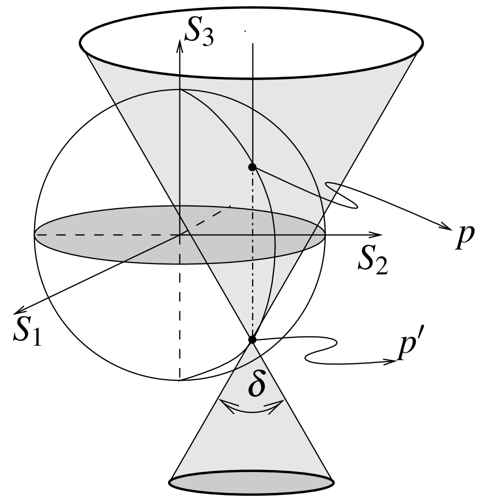


Figure 3.12: The law of the rotating linear birefringent on the Poincaré sphere. The input state p and its enantiogyre p' share the azimuth and the shape of the ellipse and differ only in the handedness, so the segment joining them is parallel to the S_3 axis and fixes the axis of the cone, whose vertex sits at p' and whose opening angle is the retardance δ. Every state that emerges while the plate is turned lies on the curve where this double cone cuts the sphere. In the particular case $p = p'$ the input state must be linear. Figure taken from (Salazar Ariza, 2021).

For a **quarter wave plate**, $\delta = \pi/2$, the cone has a half opening of 45°. If the input is linear, $\chi = 0$ makes $S_3 = 0$ and the vertex $\vec{S}^{\mathrm{e}} = \vec{S}$ is the input state itself, sitting on the equator; the trajectory passes through it at $\theta = \alpha$ and $\theta = \alpha + \pi/2$, which is no surprise, since a plate aligned with, or crossed to, the input polarisation has that polarisation as an eigenstate and returns it untouched. Its shadow on the equatorial plane is then a circle of radius 1/2 through the origin and through $\vec{S}$, so the trajectory lies on a circular cylinder as well as on the cone, and this is exactly the characterisation Azzam gave of the rotating quarter wave plate (Azzam, 2000). The cylinder, however, is an accident of linear input: as soon as the incoming light is elliptical the vertex leaves the equator, the shadow acquires an inner loop and becomes a genuine limaçon of Pascal, and the cylinder description collapses while 3.109 survives untouched. Closing that gap was precisely the motivation of (Salazar Ariza, 2021).

Two further readings of 3.113 are worth recording. The trajectory reaches the vertex itself, that is, the plate delivers the enantiogyre of the input, exactly when $K(\theta) = 0$, which by 3.113 means

$$\sin(2\alpha - 2\theta) = \tan 2\chi \cot \frac{\delta}{2}, \tag{3.116}$$

an equation that has got a solution only when $|\tan 2\chi| \leq \left|\tan \frac{\delta}{2}\right|$, and then it has got two of them in half a revolution, so that the closed curve crosses itself at the vertex. A linear input satisfies this for any retardance, since $\tan 2\chi = 0$, and this is why the cone of figure 3.12 is drawn with its vertex on the trajectory; a strongly elliptical input crossing a weak retarder does not, and its trajectory is then a small closed loop that never reaches the vertex at all.

And here is the answer to the question that opened this subsection, the one that matters for generating elliptical states. **One plate, however patiently it is turned, does not give access to the whole Poincaré sphere.** From a given input only a single closed curve is reachable, and the two numbers that fix that curve are both out of the experimenter's hands once the plate has been bought and the source switched on: the input state fixes the vertex of the cone and the retardance fixes its opening. Turning the plate moves the state along the curve and does absolutely nothing else. To reach a target that does not happen to lie on it there are exactly two ways out, and they are the subjects of what follows: make δ itself adjustable, which is the tunable birefringent of the next subsubsection, or add a second plate and use the freedom that Theorem II guarantees to any pair of retarders.

3.5.7.2 The full tunable birrefringent

The law just proved closed one door and opened two, and this subsubsection walks through both of them with the same pair of plates. The first question is about *states*: given that one plate is confined to a curve, can two plates reach every point of the Poincaré sphere, and if so with what settings? The second is about *elements*: the retardance 3.74 of a plate is welded into it by the thickness of the crystal, so can two plates be assembled into a single birefringent whose retardance is not welded into anything and can be dialled by hand? The answers are yes and yes, and they are the two halves of what an optical bench calls a polarisation state generator and a tunable birefringent.

Take the first question. A polariser at the entrance delivers the horizontal state 3.45, and the target is an arbitrary $|\alpha;\chi\rangle$. The two plates that will do it are **one half wave plate and one quarter wave plate**, in that order, and to see why we first need to know exactly what each of them does, which is a matter of two matrix multiplications.

Start with the quarter wave plate, and let its fast axis sit at an angle β. Setting $\delta = \pi/2$ in 3.76, where $\cos\frac{\pi}{4} = \sin\frac{\pi}{4} = \frac{1}{\sqrt{2}}$ makes both entries share the same factor,

$$W\left(\beta, 0; \frac{\pi}{2}\right) = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 + i\cos 2\beta & i\sin 2\beta \\ i\sin 2\beta & 1 - i\cos 2\beta \end{bmatrix}, \tag{3.117}$$

and acting with it on the horizontal ket only the first column survives, since the second column is multiplied by the vanishing second component of $[1,0]^{\mathsf{T}}$,

$$\vec{e}\,' = W\left(\beta, 0; \frac{\pi}{2}\right) \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 + i\cos 2\beta \\ i\sin 2\beta \end{bmatrix}. \tag{3.118}$$

Which ellipse is this? The quickest way to recognise a spinor is to compute its Stokes parameters and read the angles off 3.65. The two moduli are $|e_1'|^2 = \frac{1}{2}\left(1 + \cos^2 2\beta\right)$ and $|e_2'|^2 = \frac{1}{2}\sin^2 2\beta$, so by 3.62

$$S_1' = \frac{1}{2}\left(1 + \cos^2 2\beta - \sin^2 2\beta\right) = \frac{1}{2}\left(1 + \cos 4\beta\right) = \cos^2 2\beta,$$

where we recognised $\cos^2 2\beta - \sin^2 2\beta = \cos 4\beta$ and then used $1 + \cos 4\beta = 2\cos^2 2\beta$. For the other two we need the crossed product

$$e_1' e_2'^* = \frac{1}{2}\left(1 + i\cos 2\beta\right)\left(-i\sin 2\beta\right) = \frac{1}{2}\left(\sin 2\beta\cos 2\beta - i\sin 2\beta\right),$$

whose real part doubled is, by 3.63, and whose imaginary part gives, by 3.64,

$$S_2' = 2\,\mathrm{Re}\left(e_1' e_2'^*\right) = \sin 2\beta\cos 2\beta, \qquad S_3' = -2\,\mathrm{Im}\left(e_1'^* e_2'\right) = -\sin 2\beta.$$

Comparing term by term with 3.65, the third one gives $\sin 2\chi' = \sin 2\beta$ at once, so $\chi' = \beta$, and with $\cos 2\chi' = \cos 2\beta$ the first two become $\cos 2\alpha' \cos 2\beta = \cos^2 2\beta$ and $\sin 2\alpha' \cos 2\beta = \sin 2\beta \cos 2\beta$, where the common factor $\cos 2\beta$ cancels on both sides and leaves $\cos 2\alpha' = \cos 2\beta$ together with $\sin 2\alpha' = \sin 2\beta$, that is $\alpha' = \beta$. The conclusion is worth boxing, because it is the whole reason the quarter wave plate is the ellipticity knob of every optical bench,

$$\boxed{W\left(\beta, 0; \frac{\pi}{2}\right)|0;0\rangle = |\beta;\beta\rangle} \tag{3.119}$$

up to the usual irrelevant global phase. The emerging ellipse has got its major axis along the plate's own fast axis and an ellipticity angle numerically equal to that same angle. At $\beta = 0$ nothing happens, as it must, since the input is then an eigenstate of the plate; at $\beta = \pi/4$ we land on $|\pi/4;\pi/4\rangle$, a circular state, which is the fact drawn in figure 3.9; and for everything in between we get the one parameter family $|\beta;\beta\rangle$, which is nothing but the cone curve of the previous subsubsection for a linear input, drawn here in the language of kets instead of Stokes vectors.

Now the half wave plate, whose action we have so far only checked on linear and on circular inputs. Take 3.78 with its fast axis at η, drop the global phase i, and let it act on a completely general ket 3.41. The first component of the product is

$$\begin{aligned}
&\cos 2\eta\left(\cos\alpha\cos\chi - i\sin\alpha\sin\chi\right) + \sin 2\eta\left(\sin\alpha\cos\chi + i\cos\alpha\sin\chi\right)\\
&\quad = \cos\chi\left[\cos 2\eta\cos\alpha + \sin 2\eta\sin\alpha\right] + i\sin\chi\left[\sin 2\eta\cos\alpha - \cos 2\eta\sin\alpha\right]\\
&\quad = \cos\chi\cos\left(2\eta - \alpha\right) + i\sin\chi\sin\left(2\eta - \alpha\right),
\end{aligned}$$

where we grouped the real and the imaginary parts and recognised in each bracket the cosine and the sine of the difference $2\eta - \alpha$, and the second component is, by the same two steps,

$$\begin{aligned}
&\sin 2\eta\left(\cos\alpha\cos\chi - i\sin\alpha\sin\chi\right) - \cos 2\eta\left(\sin\alpha\cos\chi + i\cos\alpha\sin\chi\right)\\
&\quad = \cos\chi\sin\left(2\eta - \alpha\right) - i\sin\chi\cos\left(2\eta - \alpha\right).
\end{aligned}$$

Put the two together and compare with the general ket 3.41: writing $\cos\chi = \cos(-\chi)$ and $\sin\chi = -\sin(-\chi)$, the pair is exactly $|2\eta - \alpha; -\chi\rangle$, so that

$$\boxed{W\left(\eta, 0; \pi\right)|\alpha;\chi\rangle = |2\eta - \alpha;\, -\chi\rangle} \tag{3.120}$$

for every input whatsoever. A half wave plate reflects the azimuth about its own fast axis and reverses the handedness, which is the reflection of the Poincaré sphere in the plane containing its axis, and it is the general statement of which the two facts noted under 3.78 were particular cases.

The two rules 3.119 and 3.120 are all we need. Send the horizontal state first through the half wave plate at η, which by 3.120 with $\alpha = \chi = 0$ delivers the linear state $|2\eta; 0\rangle$, and then through the quarter wave plate at β. That second step is not 3.119 as it stands, because the plate now meets a linear state of azimuth 2η and not the horizontal one, but nothing has to be recomputed: a rotator carries the whole construction with it. Since $R(\psi)\,|\alpha;\chi\rangle = |\alpha+\psi;\chi\rangle$ by 3.83, and since conjugating a retarder by a rotator merely turns its eigenstates, hence its fast axis,

$$R(-\psi)\,W(\beta,0;\delta)\,R(\psi) = W(\beta-\psi,0;\delta)\,, \qquad \text{that is} \qquad W(\beta,0;\delta)\,R(\psi) = R(\psi)\,W(\beta-\psi,0;\delta)\,, \tag{3.121}$$

we may slide the rotator through the plate and use 3.119 untouched,

$$\begin{aligned} W\left(\beta,0;\tfrac{\pi}{2}\right)|2\eta;0\rangle &= W\left(\beta,0;\tfrac{\pi}{2}\right)R(2\eta)\,|0;0\rangle = R(2\eta)\,W\left(\beta-2\eta,0;\tfrac{\pi}{2}\right)|0;0\rangle \\ &= R(2\eta)\,|\beta-2\eta;\;\beta-2\eta\rangle\,, \end{aligned}$$

and the last rotator only adds 2η to the azimuth and leaves the ellipticity alone. The whole chain therefore reads

$$W\left(\beta,0;\frac{\pi}{2}\right)W(\eta,0;\pi)\,|0;0\rangle = |\beta;\;\beta-2\eta\rangle\,, \tag{3.122}$$

and since the two angles β and $\beta-2\eta$ can be given any values we please, the construction is complete.

Theorem 8 *(Universal polarisation state generator): A half wave plate with its fast axis at η followed by a quarter wave plate with its fast axis at β transforms horizontally polarised light into the state $|\beta;\beta-2\eta\rangle$. Consequently every fully polarised state $|\alpha;\chi\rangle$ whatsoever is produced, without any loss of intensity, by the settings*

$$\boxed{\beta = \alpha, \qquad \eta = \frac{\alpha-\chi}{2}} \tag{3.123}$$

The recipe is easier to remember than to derive. **The quarter wave plate is simply set at the azimuth wanted for the target ellipse, and the half wave plate is then turned until the difference between the two plate angles produces the required ellipticity**, since 3.122 says $\chi = \beta - 2\eta$. Geometrically, the half wave plate chooses which meridian of the Poincaré sphere the state will climb and the quarter wave plate decides how far up it climbs, and because both elements are retarders the beam arrives at its destination carrying every photon it started with, which was the whole point of building the generator out of waveplates in the first place. Notice also that the solution 3.123 is not unique, since η and $\eta + \pi/2$ describe the same plate up to a sign that the global phase absorbs, and that is a convenience rather than a defect: on a real mount one takes whichever of the equivalent settings the rotation stage can reach.

We turn now to the second question, which is a different kind of request altogether. The generator we have just built changes the *state* of the light while the elements themselves stay what they are; what we want now is to change the *element*. Looking at 3.74, the retardance of a plate is fixed by the thickness of the crystal and by two refractive indices, none of which we can touch once the plate is cut, so a variable retarder normally means either a Babinet or Soleil compensator, in which a pair of wedges is slid to change the effective thickness, or a liquid crystal cell, in which a voltage changes the indices. There is however a purely mechanical alternative, and it is remarkably economical: **two quarter wave plates whose fast axes are simply turned against each other behave as one single birefringent whose retardance depends only on the angle between them**. This biplate was characterised in detail by Pabón, Hernandez and Torres (Pabón et al., 2023), who worked it out with the geometric algebra of Pauli vectors; we shall obtain the same results with nothing but the Jones matrices of this chapter.

The algebra becomes almost effortless once the retarder is written in vector form, so let us do that first. Take the general elliptical retarder 3.81, separate the part proportional to the identity from the rest, and compare what is left with the Pauli matrices 3.68. Since

$$\hat{n}\cdot\vec{\sigma} \equiv n_1\sigma_1 + n_2\sigma_2 + n_3\sigma_3 = \begin{bmatrix} n_1 & n_2 + in_3 \\ n_2 - in_3 & -n_1 \end{bmatrix},$$

the combination $\cos\frac{\delta}{2}\sigma_0 + i\sin\frac{\delta}{2}\,(\hat{n}\cdot\vec{\sigma})$ has got $\cos\frac{\delta}{2} + i\sin\frac{\delta}{2}n_1$ in the upper left corner and $\sin\frac{\delta}{2}\,(in_2 - n_3)$ in the upper right one, which reproduce 3.81 entry by entry as soon as we identify $n_1 = \cos 2\alpha\cos 2\chi$, $n_2 = \sin 2\alpha\cos 2\chi$ and $n_3 = -\sin 2\chi$. That triple is not a new object: by 3.65 it is the Stokes vector of the eigenstate $|\alpha;\chi\rangle$, that is, the very axis 3.94 about which the element turns the Poincaré sphere. So

$$\boxed{W(\alpha,\chi;\delta) = \cos\frac{\delta}{2}\,\sigma_0 + i\sin\frac{\delta}{2}\,(\hat{n}\cdot\vec{\sigma}) = \exp\left(i\frac{\delta}{2}\,\hat{n}\cdot\vec{\sigma}\right),} \tag{3.124}$$

the exponential form following because $(\hat{n}\cdot\vec{\sigma})^2 = \sigma_0$ for a unit vector, so that the exponential series splits into the cosine and sine of $\delta/2$ exactly as it does for a number. Equation 3.124 is the Jones translation of the exponentials of Pauli vectors that (Pabón et al., 2023) handles with geometric algebra, and it says something we already knew geometrically, that a retarder is a rotation by δ about $\hat{n}$, only now written as a 2×2 matrix we can multiply.

For a linear plate, $\chi = 0$ leaves $\hat{n} = (\cos 2\theta, \sin 2\theta, 0)$, so it is convenient to abbreviate

$$\Sigma(\theta) \equiv \cos 2\theta\,\sigma_1 + \sin 2\theta\,\sigma_2 = \begin{bmatrix} \cos 2\theta & \sin 2\theta \\ \sin 2\theta & -\cos 2\theta \end{bmatrix}, \qquad \Sigma(\theta)^2 = \sigma_0, \tag{3.125}$$

and a quarter wave plate, for which $\cos\frac{\pi}{4} = \sin\frac{\pi}{4} = \frac{1}{\sqrt{2}}$, becomes the particularly simple

$$Q(\theta) \equiv W\left(\theta, 0; \frac{\pi}{2}\right) = \frac{1}{\sqrt{2}}\left[\sigma_0 + i\,\Sigma(\theta)\right]. \tag{3.126}$$

Let the beam meet the plate at θ_m first and the plate at θ_n second, and name the two angles that will turn out to matter,

$$\theta' \equiv \theta_n - \theta_m \quad \text{(the angle between the fast axes)}, \qquad \bar{\theta} \equiv \frac{\theta_m + \theta_n}{2} \quad \text{(their bisector)}. \tag{3.127}$$

The biplate is the ordered product of the two, which by 3.126 expands into four terms,

$$Q(\theta_n)Q(\theta_m) = \frac{1}{2}\left[\sigma_0 + i\Sigma_n\right]\left[\sigma_0 + i\Sigma_m\right] = \frac{1}{2}\left[\sigma_0 + i\left(\Sigma_n + \Sigma_m\right) - \Sigma_n\Sigma_m\right], \tag{3.128}$$

writing Σ_m for $\Sigma(\theta_m)$ and Σ_n for $\Sigma(\theta_n)$, and everything now depends on two elementary identities.

The product first. Multiplying out and using $\sigma_1^2 = \sigma_2^2 = \sigma_0$ on the two square terms together with the anticommutation $\sigma_2\sigma_1 = -\sigma_1\sigma_2$ on the two crossed ones,

$$\begin{aligned}\Sigma_n\Sigma_m &= \left(\cos 2\theta_n\sigma_1 + \sin 2\theta_n\sigma_2\right)\left(\cos 2\theta_m\sigma_1 + \sin 2\theta_m\sigma_2\right)\\ &= \underbrace{\left(\cos 2\theta_n \cos 2\theta_m + \sin 2\theta_n \sin 2\theta_m\right)}_{\cos(2\theta_n - 2\theta_m)}\sigma_0 - \underbrace{\left(\sin 2\theta_n \cos 2\theta_m - \cos 2\theta_n \sin 2\theta_m\right)}_{\sin(2\theta_n - 2\theta_m)}\sigma_1\sigma_2,\end{aligned}$$

and since $\sigma_1\sigma_2 = \left[\begin{smallmatrix} 0 & 1 \\ -1 & 0\end{smallmatrix}\right] = -i\sigma_3$ by direct multiplication of 3.68, this is

$$\Sigma_n\Sigma_m = \cos 2\theta'\,\sigma_0 + i\sin 2\theta'\,\sigma_3. \tag{3.129}$$

The third Pauli matrix has appeared out of two purely linear elements, and that single fact is the origin of everything peculiar about the biplate. Now the sum, where the textbook formulas $\cos A + \cos B = 2\cos\frac{A+B}{2}\cos\frac{A-B}{2}$ and $\sin A + \sin B = 2\sin\frac{A+B}{2}\cos\frac{A-B}{2}$ with $A = 2\theta_n$ and $B = 2\theta_m$ produce the same factor $\cos\theta'$ in both components, because $\cos\frac{A-B}{2} = \cos\theta'$ is even and therefore common to the two,

$$\Sigma_n + \Sigma_m = 2\cos\theta'\left[\cos 2\bar{\theta}\,\sigma_1 + \sin 2\bar{\theta}\,\sigma_2\right] = 2\cos\theta'\,\Sigma\left(\bar{\theta}\right). \tag{3.130}$$

Substituting 3.129 and 3.130 into 3.128, and inserting the two half angle identities $1 - \cos 2\theta' = 2\sin^2\theta'$ and $\sin 2\theta' = 2\sin\theta'\cos\theta'$, the factors of 2 cancel against the $\frac{1}{2}$ in front and the biplate collapses to

$$Q(\theta_n)Q(\theta_m) = \sin^2\theta'\,\sigma_0 + i\left[\cos\theta'\cos 2\bar{\theta}\,\sigma_1 + \cos\theta'\sin 2\bar{\theta}\,\sigma_2 - \sin\theta'\cos\theta'\,\sigma_3\right]. \tag{3.131}$$

This has exactly the shape of 3.124, which is no accident, since a product of unitary matrices is unitary and Theorem II already promised that any stack of retarders is a single elliptical retarder. Calling γ its retardance and (α, χ) its eigenstate, the coefficient of the identity gives the retardance immediately,

$$\boxed{\cos\frac{\gamma}{2} = \sin^2\theta'} \tag{3.132}$$

while the remaining three coefficients must equal $\sin\frac{\gamma}{2}\,\hat{n}$. Their squares add up to

$$\sin^2\frac{\gamma}{2} = \cos^2\theta'\underbrace{\left(\cos^2 2\bar{\theta} + \sin^2 2\bar{\theta}\right)}_{=1} + \sin^2\theta'\cos^2\theta' = \cos^2\theta'\left(1 + \sin^2\theta'\right),$$

which is the same statement as 3.132, since $1 - \sin^4\theta' = \left(1 - \sin^2\theta'\right)\left(1 + \sin^2\theta'\right)$, and therefore no new information; it is however exactly what we need to normalise. Restricting the angle between the axes to $0 \leq \theta' \leq \pi/2$, where $\cos\theta' \geq 0$, we have $\sin\frac{\gamma}{2} = \cos\theta'\sqrt{1+\sin^2\theta'}$, and dividing the three coefficients by it cancels the factor $\cos\theta'$ from every one of them,

$$\hat{n} = \frac{1}{\sqrt{1+\sin^2\theta'}}\left(\cos 2\bar{\theta},\ \sin 2\bar{\theta},\ -\sin\theta'\right). \tag{3.133}$$

Comparing 3.133 with the Stokes form $\hat{n} = (\cos 2\alpha\cos 2\chi, \sin 2\alpha\cos 2\chi, -\sin 2\chi)$ finishes the job. The first two components carry the same positive factor, so their ratio fixes the azimuth outright, $2\alpha = 2\bar{\theta}$, and their common factor gives $\cos 2\chi = 1/\sqrt{1+\sin^2\theta'}$, while the third gives $\sin 2\chi = \sin\theta'/\sqrt{1+\sin^2\theta'}$. Dividing the last two, the square root cancels, and we are left with a result of quite unreasonable simplicity,

$$\boxed{\alpha = \frac{\theta_m+\theta_n}{2}, \qquad \tan 2\chi = \sin\theta'} \tag{3.134}$$

Read together, 3.132 and 3.134 say everything there is to say about the device, and each of the three parameters is controlled by something different. **The retardance and the ellipticity depend only on the difference of the two plate angles, and the azimuth only on their sum**, which is precisely what makes the biplate practical: mounting the two plates so that they can be turned both together and against each other gives one knob that tunes the element and another that orients it, and the two do not interfere.

The retardance runs over the whole useful range. At $\theta' = 0$ the two plates are aligned, 3.132 gives $\cos\frac{\gamma}{2} = 0$ and hence $\gamma = \pi$, which is no surprise whatsoever, since two quarter waves stacked on the same axis are a half wave. At $\theta' = \pi/2$ the plates are crossed, $\cos\frac{\gamma}{2} = 1$ gives $\gamma = 0$, and the biplate is optically invisible, the second plate undoing exactly what the first one did. In between γ falls monotonically from a half wave to nothing: at $\theta' = 30°$ it is $151°$, at $\theta' = 45°$ it is exactly $120°$, at $\theta' = 54°$ it is $98°$. **Any retardance between zero and a half wave is available with a screwdriver**, and this is the tunable birefringent promised at the end of the previous subsubsection.

The price is written in the second half of 3.134, and it is the whole point of (Pabón et al., 2023). Except when the plates are aligned, the eigenstates of the biplate are *not* linear: the device is an elliptical birefringent in the sense of the definition that opened this subsection, and its ellipticity is dragged along with the retardance, since both are governed by θ' alone. How elliptical can it get? Since $|\sin\theta'| \leq 1$, equation 3.134 forces $|\tan 2\chi| \leq 1$, that is

$$|\chi| \leq \frac{\pi}{8} = 22.5°, \tag{3.135}$$

and the bound is reached only in the limit $\theta' \to \pi/2$, where by 3.132 the retardance has already gone to zero and the eigenstates of a transparent element mean nothing at all. **Two quarter wave plates can therefore never be assembled into a circular birefringent**: a rotator needs $\chi = \pi/4$ and the biplate cannot pass $\pi/8$, so optical activity is not something this construction can manufacture on its own. For a value in the middle of the range, $\theta' = 54°$, equation 3.134 predicts $\tan 2\chi = \sin 54° = 0.809$ and hence $\chi = 19.48°$, which is precisely the ellipticity measured on the bench in (Pabón et al., 2023) with a helium neon laser, two commercial quarter wave plates on a synchronised mount and a commercial polarimeter, the plates being stepped in two degree intervals over a half turn. Their arrangement is the one drawn in figure 3.13, and it is worth looking at because it is exactly the apparatus of the previous subsubsection with the single plate replaced by the pair: a laser, a collimating lens, a polariser that prepares the input state, the biplate itself, and a polarimeter that reads out the four Stokes parameters of whatever comes through.

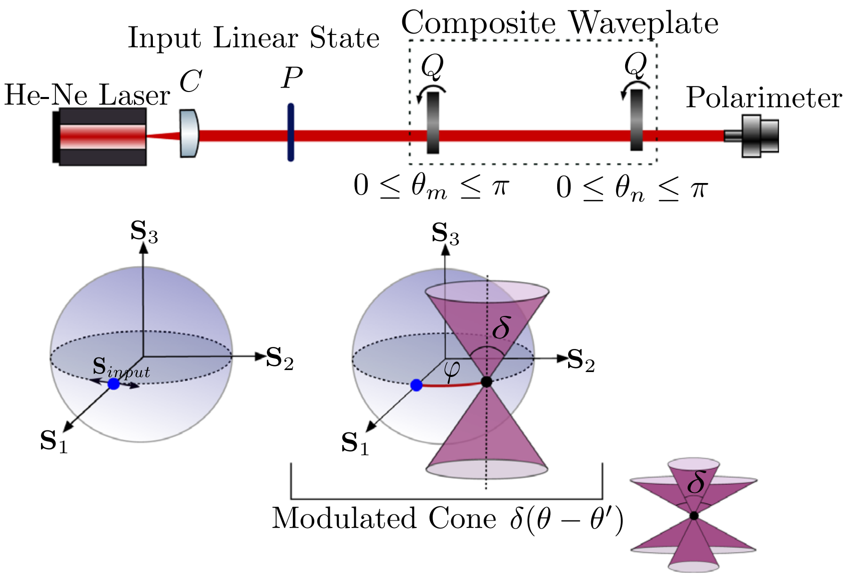


Figure 3.13: The bench used to measure the states generated by a biplate of two quarter wave plates, with the collimating lens C, the polariser P that prepares the input state, the two plates Q at angles θ_m and θ_n, and the polarimeter that measures the emerging Stokes vector. Turning the pair as a rigid unit sweeps a curve on the Poincaré sphere, again the intersection with a cone, but the cone is now rotated by the angle φ carried by the optical activity of 3.136, which is the subject of the next subsubsection. Figure taken from (Pabón et al., 2023).

One last reading closes the circle with the Jones theorems. Theorem II guarantees that our biplate, being a cascade of retarders, must also be writable as a single *linear* retarder followed by a rotator, and carrying that decomposition out on 3.131 gives

$$Q(\theta_n)Q(\theta_m) = R\left(\theta' - \frac{\pi}{2}\right) W\left(\theta_m + \frac{\pi}{4}, 0; \pi - 2\theta'\right), \tag{3.136}$$

a linear plate of retardance $\pi - 2\theta'$ with its fast axis at forty five degrees from the first of the two quarter wave plates, followed by a rotation of the plane of polarisation by $\theta' - \pi/2$. This is the mechanism laid bare: **the biplate is a linear birefringent with a genuine optical activity hidden inside it**, and it is that rotator, absent from either plate separately and manufactured by the pair, that tilts the eigenstates off the equator of the Poincaré sphere and makes them elliptical. When the plates are crossed both the retardance and the rotation vanish together, as they must, and when they are aligned the rotator becomes a quarter of a turn and 3.136 collapses back to a half wave plate at θ_m, as the reader may check with 3.120.

We have now got, in one small mount, a birefringent whose retardance, ellipticity and orientation are all under mechanical control. What we have not got is the law that governs it when it is turned: the theorem of the previous subsubsection was proved for a *linear* birefringent, and 3.134 has just told us that the biplate is not one. Its trajectories on the Poincaré sphere therefore fall outside that law, and finding the law that does cover them, for an arbitrary elliptical birefringent, is the subject of what follows.

3.5.7.3 The elliptic birrefringents law

The law of the rotating linear birefringent was proved under one hypothesis that we never questioned while we were using it: that the eigenstates of the turning element are linear, $\chi_q = 0$, so that its axis on the Poincaré sphere lies on the equator. The biplate of the previous subsubsection has just shown us that this hypothesis is not academic bookkeeping, since 3.134 makes the eigenstates of two crossed quarter wave plates genuinely elliptical for every θ' except zero. The device we have just built therefore falls outside the only law we possess, and the purpose of this subsubsection is to remove the hypothesis altogether and find what a *general* elliptical birefringent does when it is turned. The answer, obtained in the language of geometric algebra by Pabón, Hernandez and Torres (Pabón et al., 2023) and reproduced here entirely with Jones matrices, is that almost nothing changes: the trajectory is still the intersection of the sphere with a cone, but the cone has been *rotated*, and the angle it has been rotated through is exactly the optical activity hidden inside the element.

Before anything else we must be careful about what turning an elliptical element even means. A linear plate has got a fast axis painted on it and rotating it in its mount is unambiguous; a general birefringent has instead a pair of elliptical eigenstates, and rotating the crystal rotates those ellipses rigidly in the laboratory, tilting their azimuth while leaving their shape untouched. In the notation of 3.75 the element therefore goes from $W(\alpha_q, \chi_q; \gamma)$ to $W(\alpha_q + \theta, \chi_q; \gamma)$, with χ_q and γ frozen because they belong to the crystal and not to the mount. Since the rotator 3.83 does precisely $R(\theta)\,|\alpha; \chi\rangle = |\alpha + \theta; \chi\rangle$ to every state, an element built on rotated eigenstates is the original one seen from a rotated frame,

$$W\left(\alpha_q + \theta, \chi_q; \gamma\right) = R(\theta)\, W\left(\alpha_q, \chi_q; \gamma\right) R(-\theta), \tag{3.137}$$

which is the general form of the covariance 3.121 already used for linear plates, and which follows from the spectral definition in one line: conjugating $|\psi_1\rangle\langle\psi_1|$ by $R(\theta)$ turns it into $R(\theta)\,|\psi_1\rangle\langle\psi_1|\,R(\theta)^\dagger = |\alpha_q + \theta; \chi_q\rangle\langle\alpha_q + \theta; \chi_q|$, and the same for $|\psi_2\rangle$, while the two eigenvalues $e^{\pm i\gamma/2}$ are numbers and are not touched by the conjugation at all.

The whole proof now rests on splitting the elliptical element into a linear one and a rotator, which Theorem II guarantees is possible and which we shall carry out explicitly, because we need the three parameters and not merely the promise that they exist. The vector form 3.124 makes this a short computation. Start by writing the rotator itself in that language: it is the circular retarder 3.82 of retardance -2ψ, whose eigenstate $|0;\pi/4\rangle$ has by 3.65 the Stokes vector $\hat{n}=(0,0,-1)$, so 3.124 gives $\cos\psi\,\sigma_0 + i\sin(-\psi)\,(-\sigma_3)$, that is

$$R(\psi) = \cos\psi\,\sigma_0 + i\sin\psi\,\sigma_3 = \begin{bmatrix} \cos\psi & i\,(i\sin\psi) \\ i\,(-i\sin\psi) & \cos\psi \end{bmatrix} = \begin{bmatrix} \cos\psi & -\sin\psi \\ \sin\psi & \cos\psi \end{bmatrix}, \tag{3.138}$$

which is 3.83 itself, as it had to be. Keeping the abbreviation 3.125 for the equatorial combination, a linear plate is $W(\alpha_l,0;\delta) = \cos\frac{\delta}{2}\sigma_0 + i\sin\frac{\delta}{2}\Sigma(\alpha_l)$, and the product of the two is

$$\begin{aligned} R(\psi)\,W\,(\alpha_l,0;\delta) &= [\cos\psi\,\sigma_0 + i\sin\psi\,\sigma_3]\left[\cos\frac{\delta}{2}\sigma_0 + i\sin\frac{\delta}{2}\Sigma(\alpha_l)\right] \\ &= \cos\psi\cos\frac{\delta}{2}\,\sigma_0 + i\cos\psi\sin\frac{\delta}{2}\,\Sigma(\alpha_l) + i\sin\psi\cos\frac{\delta}{2}\,\sigma_3 - \sin\psi\sin\frac{\delta}{2}\,\sigma_3\Sigma(\alpha_l). \end{aligned} \tag{3.139}$$

Only the last term is unfamiliar, and it is disposed of by two products read straight off 3.68,

$$\sigma_3\sigma_1 = \begin{bmatrix} 0 & i \\ -i & 0 \end{bmatrix}\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} = \begin{bmatrix} 0 & -i \\ -i & 0 \end{bmatrix} = -i\sigma_2, \qquad \sigma_3\sigma_2 = \begin{bmatrix} 0 & i \\ -i & 0 \end{bmatrix}\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} i & 0 \\ 0 & -i \end{bmatrix} = i\sigma_1,$$

so that, substituting them into the definition 3.125 of Σ,

$$\sigma_3\Sigma(\alpha_l) = -i\cos 2\alpha_l\,\sigma_2 + i\sin 2\alpha_l\,\sigma_1 = i\left[\cos\left(2\alpha_l - \tfrac{\pi}{2}\right)\sigma_1 + \sin\left(2\alpha_l - \tfrac{\pi}{2}\right)\sigma_2\right] = i\,\Sigma\left(\alpha_l - \frac{\pi}{4}\right), \tag{3.140}$$

where we wrote $\sin 2\alpha_l = \cos\left(2\alpha_l - \frac{\pi}{2}\right)$ and $-\cos 2\alpha_l = \sin\left(2\alpha_l - \frac{\pi}{2}\right)$ in order to recognise another Σ. **Multiplying by σ_3 therefore rotates an equatorial direction by a quarter of a turn**, which is only to be expected, since σ_3 is the polar axis and the equator turns about it. Putting 3.140 back into 3.139, the two equatorial terms combine into

$$\sin\frac{\delta}{2}\left[\cos\psi\,\Sigma(\alpha_l) - \sin\psi\,\Sigma\left(\alpha_l - \frac{\pi}{4}\right)\right],$$

and writing both Σ's out with 3.125, the coefficient of σ_1 is $\cos\psi\cos 2\alpha_l - \sin\psi\sin 2\alpha_l = \cos\left(2\alpha_l + \psi\right)$ and the coefficient of σ_2 is $\cos\psi\sin 2\alpha_l + \sin\psi\cos 2\alpha_l = \sin\left(2\alpha_l + \psi\right)$, both of them by the addition formulas, so the bracket is one single Σ again,

$$R(\psi)\,W\,(\alpha_l,0;\delta) = \cos\psi\cos\frac{\delta}{2}\,\sigma_0 + i\left[\sin\frac{\delta}{2}\,\Sigma\left(\alpha_l + \frac{\psi}{2}\right) + \sin\psi\cos\frac{\delta}{2}\,\sigma_3\right]. \tag{3.141}$$

This has exactly the shape of 3.124, so the chain is a single elliptical retarder $W(\alpha_q,\chi_q;\gamma)$, and comparing the four coefficients one by one identifies it. The identity gives $\cos\frac{\gamma}{2} = \cos\psi\cos\frac{\delta}{2}$; the two equatorial ones give $\sin\frac{\gamma}{2}\cos 2\chi_q = \sin\frac{\delta}{2}$ together with $2\alpha_q = 2\alpha_l + \psi$; and the polar one gives $-\sin\frac{\gamma}{2}\sin 2\chi_q = \sin\psi\cos\frac{\delta}{2}$. Since it is the elliptical element that is handed to us and the pair that we must find, it is the inverted form we shall use: dividing the polar relation by the identity relation cancels $\cos\frac{\delta}{2}$, and the three of them become

$$\boxed{\sin\frac{\delta}{2} = \sin\frac{\gamma}{2}\cos 2\chi_q, \qquad \tan\psi = -\tan\frac{\gamma}{2}\sin 2\chi_q, \qquad \alpha_l = \alpha_q - \frac{\psi}{2}} \tag{3.142}$$

taking $0 \leq \gamma \leq \pi$, which costs nothing, since exchanging the two eigenstates of any retarder flips the sign of its retardance and brings it into that range. Equation 3.142 is the whole content of figure 3.14: **every elliptical birefringent is a linear birefringent of retardance δ followed by a rotator of angle ψ**, and the ellipticity of its eigenstates is precisely the fingerprint of that hidden rotator, since $\chi_q = 0$ forces $\psi = 0$ and $\delta = \gamma$, returning us to a plain waveplate.

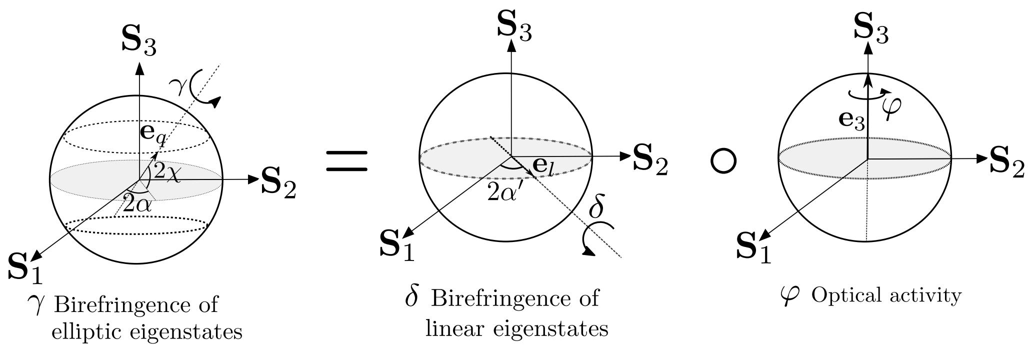


Figure 3.14: The decomposition 3.142 of a birefringent with elliptical eigenstates, of retardance γ and eigenstate angles (α, χ), into a birefringent with linear eigenstates of retardance δ and orientation α' and an optical activity of rotation $\varphi = 2\psi$. Figure taken from (Pabón et al., 2023).

It is worth checking 3.142 against the one elliptical birefringent we have built with our own hands. For the biplate, 3.132 and 3.134 give $\cos\frac{\gamma}{2} = \sin^2\theta'$ and $\tan 2\chi_q = \sin\theta'$, from which $\sin\frac{\gamma}{2} = \cos\theta'\sqrt{1+\sin^2\theta'}$ and $\cos 2\chi_q = 1/\sqrt{1+\sin^2\theta'}$, so the first of 3.142 gives $\sin\frac{\delta}{2} = \cos\theta'$ after the square roots cancel, that is $\delta = \pi - 2\theta'$, and the second gives $\tan\psi = -\tan\frac{\gamma}{2}\sin 2\chi_q = -\cot\theta'$, that is $\psi = \theta' - \frac{\pi}{2}$. Both agree exactly with the decomposition 3.136 obtained there by a different route.

Now comes the step that makes the theorem, and it is a single line. Rotate the element bodily by θ and use 3.137 with the decomposition just found,

$$\begin{aligned} W(\alpha_q+\theta, \chi_q; \gamma) &= R(\theta)\,R(\psi)\,W(\alpha_l, 0; \delta)\,R(-\theta) = R(\psi)\,\underbrace{R(\theta)\,W(\alpha_l, 0; \delta)\,R(-\theta)}_{W(\alpha_l+\theta,0;\delta)} \\ &= R(\psi)\,W(\alpha_l+\theta, 0; \delta)\,, \end{aligned} \tag{3.143}$$

where we first commuted the two rotators, which is legitimate because rotations of the plane about the same axis commute, $R(\theta)R(\psi) = R(\theta+\psi) = R(\psi)R(\theta)$, and then recognised the conjugation as the covariance 3.121. Read what 3.143 says: as the element turns, **its retardance δ and its**

rotator ψ stay exactly where they were and only the fast axis of the linear part travels. The reason is worth stating out loud, because it is the physical heart of this subsubsection: a rotator has got no orientation to change. Optical activity treats every direction of the laboratory alike, so turning the crystal that carries it changes nothing whatsoever, while the linear birefringence, which does single out a direction, turns with the mount.

Theorem 9 *(Law of the rotating elliptical birefringent): Let a fully polarised beam $|\alpha;\chi\rangle$ cross a birefringent of retardance γ whose eigenstates are the elliptical pair $|\alpha_q;\chi_q\rangle$, and let the element be turned bodily through a complete revolution, so that its eigenstate azimuth runs over $\alpha_q+\theta$ with $0\leq\theta\leq 2\pi$. Then the locus of the emergent Stokes vector is the closed curve in which the Poincaré sphere is cut by the circular cone*

$$\boxed{\left(S_1'-V_1\right)^2+\left(S_2'-V_2\right)^2=\tan^2\frac{\delta}{2}\left(S_3'-V_3\right)^2} \tag{3.144}$$

whose axis is parallel to the S_3 axis, whose half opening angle is $\delta/2$ with δ the retardance of the linear part 3.142, and whose vertex is the Stokes vector

$$\vec{V}=\vec{S}\left(|\alpha+\psi;-\chi\rangle\right)=\left(\cos\left(2\alpha+2\psi\right)\cos 2\chi,\ \sin\left(2\alpha+2\psi\right)\cos 2\chi,\ \sin 2\chi\right), \tag{3.145}$$

that is, the enantiogyre of the incident state after its azimuth has been advanced by the optical activity ψ of 3.142. The curve is traversed twice in a full turn.

Proof. By 3.143 the emerging spinor is $R(\psi)W\left(\alpha_l+\theta,0;\delta\right)\vec{e}$, so the beam meets first a linear birefringent of fixed retardance δ whose axis is being rotated exactly as in the previous subsubsection, and then one fixed rotator. Call $\vec{S}'_{\text{lin}}(\theta)$ the Stokes vector after that first stage. The law of the rotating linear birefringent applies to it word for word, so $\vec{S}'_{\text{lin}}(\theta)$ runs over the whole curve where the sphere meets the cone $\mathcal{C}$ of half opening $\delta/2$, axis parallel to $\hat{z}$, and vertex at the enantiogyre $\vec{S}^{\text{e}}$ of the incident state.

It remains to push that curve through the rotator, and for this we need the rotator not as a Jones matrix but as a motion of the sphere. A rotator is the circular retarder of retardance -2ψ, and by the theorem on retarders as rotations 3.93 its Mueller matrix is the rotation of the sphere by that retardance about the axis 3.94 of its own eigenstate, which is the south pole $\hat{n}=(0,0,-1)$. A rotation by -2ψ about $-\hat{z}$ is a rotation by $+2\psi$ about $+\hat{z}$, so $R(\psi)$ turns the whole Poincaré sphere rigidly about its polar axis by twice the angle it turns the ellipse through, which is the doubling of angles once more and which we could have read directly off 3.65, since sending α to $\alpha+\psi$ sends the Stokes azimuth 2α to $2\alpha+2\psi$ and leaves $S_3=-\sin 2\chi$ untouched.

A rigid rotation carries cones into cones and preserves every angle and every distance, so the image of $\mathcal{C}$ is again a circular cone of half opening $\delta/2$; its axis is unchanged, because the axis of $\mathcal{C}$ is the very axis about which we are turning; and its vertex is the image of the old vertex. That image is computed by the rule just established: the enantiogyre $\vec{S}^{\text{e}}$ is the Stokes vector of $|\alpha;-\chi\rangle$, and advancing its azimuth by ψ gives the Stokes vector of $|\alpha+\psi;-\chi\rangle$, which written out with 3.65

is precisely 3.145. Since the intersection of the sphere with $\mathcal{C}$ is swept entirely, and since $R(\psi)$ is a bijection of the sphere that maps that intersection onto the intersection of the sphere with the rotated cone, the emergent states sweep the latter entirely as well, and the periodicity argument is unchanged because $|\alpha_q + \pi; \chi_q\rangle$ and $|\alpha_q; \chi_q\rangle$ are the same state. ■

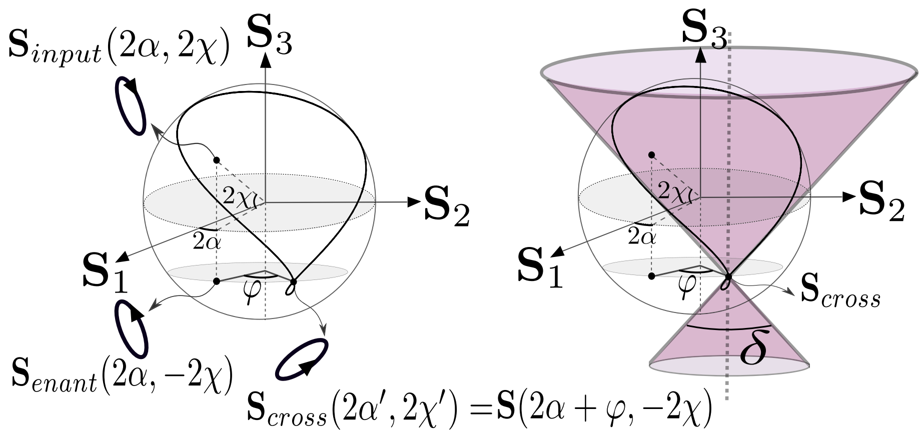


Figure 3.15: The law of the rotating elliptical birefringent. The incident state $\mathbf{S}_{\text{input}}$ and its enantiogyre $\mathbf{S}_{\text{enant}}$ are related by the reflection 3.108, and turning that enantiogyre in azimuth by the optical activity $\varphi = 2\psi$ produces the state $\mathbf{S}_{\text{cross}}$ of 3.145, which is the vertex of the cone. The emergent states then fill the curve where that cone cuts the sphere, exactly as in the linear case but with the cone rotated about the polar axis. Figure taken from (Pabón et al., 2023).

The state 3.145 is called the **crossed state** of the incident beam in (Pabón et al., 2023), and figure 3.15 shows how it is built out of the incident one in two moves, a reflection in the equator and a turn about the pole. The two moves commute, so it makes no difference whether the beam is first reflected and then turned or the other way round, and this is only the statement that the reflection acts on S_3 while the rotation acts on the azimuth.

Everything now falls into place, and the division of labour between the two ingredients of the element could not be cleaner. **The shape of the cone is fixed by the linear birefringence and its position by the optical activity**: δ opens or closes it, ψ swings it around the polar axis, and the retardance γ of the elliptical element never appears by itself, only through the pair 3.142 into which it is dissolved. Setting $\chi_q = 0$ collapses 3.144 back onto 3.109, since then $\psi = 0$ and $\delta = \gamma$ and the vertex returns to the plain enantiogyre, so the law we proved first was the equatorial special case of the law we have proved now.

For the biplate the two parameters are not independent, and it is instructive to see why. There we found $\delta = \pi - 2\theta'$ and $\psi = \theta' - \frac{\pi}{2}$, so that

$$2\psi = 2\theta' - \pi = -\left(\pi - 2\theta'\right) = -\delta, \tag{3.146}$$

and **the cone of a biplate of two quarter wave plates is always swung by exactly minus**

its own opening. Turning the two plates against each other by θ' therefore opens the cone and rotates it in the same single motion, which is why the drawing in figure 3.13 labels it a modulated cone: with only two quarter wave plates one cannot dial the opening and the swing separately, and reaching an arbitrary pair (δ, ψ) needs a third element.

The practical value of the law is that it can be run backwards. Send a known state into an unknown birefringent, turn the element through half a revolution and record the emerging states with a polarimeter, exactly as in figure 3.13: the curve that comes out determines a cone, the opening of that cone hands over the linear retardance δ, the azimuth of its vertex compared with the enantiogyre of the input hands over the optical activity ψ, and 3.142 then inverts to give the retardance γ and the eigenstate ellipticity χ_q of the sample. A single mechanical sweep and a single detector are enough to characterise the element completely, without ever having to find its eigenstates by trial and error, and this is how the biplate of the previous subsubsection was validated in (Pabón et al., 2023). Together with the linear law, of which it is the generalisation, 3.144 closes the question that opened this subsection: we know now exactly which states a rotating birefringent of any kind can deliver, and exactly which it cannot.

3.5.8 Generation of elliptic states: Polarizers as projection operators

The previous subsection reached every elliptical state through a pure rotation of the Poincaré sphere, at no cost in intensity, and it is fair to ask why one would ever trade so economical a method for an element that the whole of the Mueller matrix subsection went out of its way to characterise as destructive. The answer is that a projector sells something no retarder can offer at any price: **indifference to the input**. Send a beam whose polarisation has drifted, even slightly, from the state a fixed chain of waveplates was calibrated for, and by 3.93 the output ellipse drifts by exactly the same rotation, since a rotation carries every point of the sphere together and cannot single one out for special treatment. Send the same drifting beam through an elliptical polariser instead, and by 3.95 the outgoing state is still crushed onto the single point $\hat{n}$ fixed by the polariser's own transmission axis, for the simple reason that a projection onto $|\psi\rangle$ retains no memory of the input beyond how much of $|\psi\rangle$ it happened to contain. A polariser is therefore not merely a different route to an ellipse already reachable with waveplates, it is the only one of the two that erases the uncertainty of an imperfectly known or imperfectly polarised source, at the unavoidable price, quantified below by the elliptical Malus law, of transmitting less light than went in.

There is a second, more mundane reason to keep a stock of polarisers next to the waveplates, and it has got to do with colour rather than with information. The retardance of a waveplate is a phase accumulated over a fixed thickness, $\delta = 2\pi\Delta n\, d/\lambda$, so a given plate is exactly a quarter wave or a half wave at one wavelength only, and out of broadband light, an ultrashort laser pulse being the extreme case, it manufactures not one ellipse but a blur of slightly wrong ones, one for every wavelength under the pulse's spectrum. The action of an ideal polariser is instead a projection fixed purely by the geometry of its transmission axis, and to an excellent approximation over a very wide band it does not care what λ is. Whenever the state to be generated must be pure and achromatic rather than merely efficient, for instance to serve as the calibration standard of a polarimeter of the kind described by Azzam and Bashara (Azzam and Bashara, 1977), the polariser is the one device built in this chapter that can honestly promise it.

3.5.8.1 The Malus law

We begin with the oldest quantitative statement in the whole of polarisation optics. When Étienne-Louis Malus, in 1809, wrote down that the intensity transmitted by a crystal analyser falls as the square of the cosine of the angle it makes with the plane of polarisation (Malus, 1809), he was describing an experiment he could only read with his own eye, on a phenomenon nobody could yet explain, seven years before Fresnel would establish that light is a transverse wave and more than a century before anyone would suspect that the same formula also gives the probability that a single photon survives the crossing. The law has already appeared once in these notes, in the Mueller subsection, where it dropped out of 3.95 as a statement about the angle between two vectors of the Poincaré sphere. Here we prove it again, in the Jones language and from the projector itself, and then we do the two things that the Mueller derivation left undone: we extend it to light that has got no Jones spinor at all, and we ask what it means for one photon rather than for a beam.

Let the analyser be the ideal linear polariser 3.86 with its transmission axis at α_p, and let the incident beam be fully polarised and linear, $|\alpha;0\rangle$. Since the polariser is the projector $P(\alpha_p) = |\alpha_p;0\rangle\,\langle\alpha_p;0|$, its action needs no matrix multiplication whatsoever, only the associativity of the outer product,

$$\vec{e}\,' = P(\alpha_p)\,|\alpha;0\rangle = |\alpha_p;0\rangle\,\underbrace{\langle\alpha_p;0|\alpha;0\rangle}_{\text{a number}}, \tag{3.147}$$

so that **the outgoing state is the transmission axis itself**, always and whatever came in, multiplied by one complex number. That number is an ordinary scalar product of two real unit vectors, since a linear ket 3.41 with $\chi = 0$ reduces to $[\cos\alpha, \sin\alpha]^{\mathsf{T}}$,

$$\langle\alpha_p;0|\alpha;0\rangle = \begin{bmatrix}\cos\alpha_p & \sin\alpha_p\end{bmatrix}\begin{bmatrix}\cos\alpha \\ \sin\alpha\end{bmatrix} = \cos\alpha_p\cos\alpha + \sin\alpha_p\sin\alpha = \cos(\alpha-\alpha_p), \tag{3.148}$$

the last step being nothing but the cosine of a difference. The transmitted intensity is now read off 3.61, that is, from the squared norm of the outgoing spinor,

$$I = \vec{e}\,'^{\dagger}\vec{e}\,' = |\langle\alpha_p;0|\alpha;0\rangle|^2\,\underbrace{\langle\alpha_p;0|\alpha_p;0\rangle}_{=1} = \cos^2(\alpha-\alpha_p)\,,$$

where the norm of the transmission ket is one because every Jones ket is a unit vector, and calling $\theta = \alpha - \alpha_p$ the angle between the incident plane of polarisation and the analyser, and restoring the incident intensity I_0 that we had normalised away, this is

$$\boxed{I = I_0\cos^2\theta} \tag{3.149}$$

Three readings of this one line are worth pausing on. The first is that **Malus's law is the squared modulus of a scalar product**, $I = I_0\,|\langle\psi|e\rangle|^2$, and not really a statement about crystals at all; the cosine is the geometry of two unit vectors in $\mathbb{C}^2$ and nothing else, which is why the same formula will survive every generalisation we throw at it in the rest of this subsection. The second is that the projector is idempotent, $P^2 = P$, as 3.86 shows at once, so a second identical polariser placed behind the first transmits everything the first one passed and Malus's law does not apply twice; the light has already been made to forget where it came from. The third is the case $\theta = \pi/2$, where 3.149 vanishes identically: two crossed polarisers extinguish the beam completely, and it is exactly this null that makes the arrangement such a sensitive instrument, since anything at all placed between them that rotates or reshapes the ellipse, a stressed piece of glass, a sugar solution, a birefringent crystal, a biological tissue, lights up against a black background.

Nothing forces the incident light to be linear, and the same two lines handle the general fully polarised state. Feeding $|\alpha;\chi\rangle$ of 3.41 into 3.147, the scalar product now has got an imaginary part,

$$\begin{aligned}\langle\alpha_p;0|\alpha;\chi\rangle &= \cos\alpha_p\left(\cos\alpha\cos\chi - i\sin\alpha\sin\chi\right) + \sin\alpha_p\left(\sin\alpha\cos\chi + i\cos\alpha\sin\chi\right)\\ &= \cos\chi\,\underbrace{\left[\cos\alpha_p\cos\alpha + \sin\alpha_p\sin\alpha\right]}_{\cos(\alpha-\alpha_p)} + i\sin\chi\,\underbrace{\left[\sin\alpha_p\cos\alpha - \cos\alpha_p\sin\alpha\right]}_{-\sin(\alpha-\alpha_p)}\\ &= \cos\chi\cos\theta - i\sin\chi\sin\theta,\end{aligned}$$

having grouped the real and the imaginary parts and recognised in each bracket the cosine and the sine of the same difference $\theta = \alpha - \alpha_p$. Its squared modulus is the sum of the squares of the two parts,

$$I = \cos^2\chi\cos^2\theta + \sin^2\chi\sin^2\theta = \frac{(1+\cos 2\chi)(1+\cos 2\theta)}{4} + \frac{(1-\cos 2\chi)(1-\cos 2\theta)}{4},$$

where we inserted the two half angle identities $\cos^2 x = \frac{1+\cos 2x}{2}$ and $\sin^2 x = \frac{1-\cos 2x}{2}$ in all four factors, and expanding the two products the crossed terms $\cos 2\chi$ and $\cos 2\theta$ cancel in pairs and only $2 + 2\cos 2\chi\cos 2\theta$ survives in the numerator,

$$I = \frac{I_0}{2}\left[1 + \cos 2\chi\cos 2\theta\right]. \tag{3.150}$$

Setting $\chi = 0$ recovers 3.149 through $\frac{1}{2}(1+\cos 2\theta) = \cos^2\theta$, as it must. But look at what happens at the other extreme. For circular light, $\chi = \pm\pi/4$ makes $\cos 2\chi = 0$ and 3.150 collapses to $I = I_0/2$ *for every orientation of the analyser*: **a rotating linear polariser cannot distinguish circular light from unpolarised light**, because both give a flat, featureless half. This is not a defect of the calculation but a genuine limitation of the instrument, and it is precisely why Stokes' operational recipe, recalled when we introduced his four parameters, needs a quarter wave plate inserted before the analyser: the plate turns the invisible S_3 into a visible S_1 or S_2, and only then does rotating the polariser reveal it.

We come now to the case Malus himself was actually looking at, because the sunlight reflected from the windows of the Luxembourg Palace was never fully polarised. Here the Jones formalism appears to fail us at the first step, since 3.147 needs a spinor $\vec{e}$ and a partially polarised beam has not got one. The repair is the polarisation matrix 3.67, which exists for every beam whatsoever, and the calculation is if anything shorter. By 3.88 the outgoing polarisation matrix is $\Phi' = P\Phi P^\dagger$, and the transmitted intensity is its trace, by the first of 3.69 with σ_0,

$$I = \mathrm{tr}\,\Phi' = \mathrm{tr}\left(P\,\Phi\,P^\dagger\right) = \mathrm{tr}\left(P^\dagger P\,\Phi\right) = \mathrm{tr}\,(P\,\Phi)\,, \tag{3.151}$$

where we first used the cyclic property of the trace to bring $P^\dagger$ to the front and then collapsed $P^\dagger P = P^2 = P$, since a projector is Hermitian and idempotent. Writing the projector as $P = |\psi\rangle\langle\psi|$ for whatever state $|\psi\rangle$ it transmits, the trace of a rank one matrix is an expectation value,

$$I = \mathrm{tr}\,(|\psi\rangle\langle\psi|\,\Phi) = \langle\psi|\,\Phi\,|\psi\rangle\,,$$

and expanding Φ in the Pauli basis with the second of 3.69,

$$I = \frac{1}{2}\sum_{\mu=0}^{3} S_\mu\,\langle\psi|\,\sigma_\mu\,|\psi\rangle = \frac{1}{2}\left[S_0\,\underbrace{\langle\psi|\,\sigma_0\,|\psi\rangle}_{=1} + \sum_{i=1}^{3} S_i\,n_i\right],$$

because $\langle\psi|\,\sigma_\mu\,|\psi\rangle = \mathrm{tr}\,(|\psi\rangle\langle\psi|\,\sigma_\mu)$ is by 3.69 the μth Stokes parameter of the transmitted state itself, that is, $\hat{n}$ for the three spatial ones and unity for σ_0, since $|\psi\rangle$ is normalised. We have arrived at the general form of the law.

Theorem 10 *(Malus law for light of any degree of polarisation): An ideal polariser transmitting the state $|\psi\rangle$, whose Stokes vector is the unit vector $\hat{n}$, illuminated by a beam of arbitrary degree of polarisation and Stokes parameters $\left(S_0, \vec{S}\right)$, transmits the intensity*

$$\boxed{I = \frac{1}{2}\left(S_0 + \hat{n}\cdot\vec{S}\right)} \tag{3.152}$$

Equation 3.152 contains everything. It is manifestly the zeroth row of the Mueller matrix 3.95, which is the consistency we expected, but obtained here without ever leaving the Jones calculus. For unpolarised light, $\vec{S} = \vec{0}$, it gives $I = S_0/2$ whatever the analyser does, the familiar fact that a polariser transmits exactly half of natural light. For fully polarised light, $\left|\vec{S}\right| = S_0$, the scalar product is $S_0\cos\Theta$ with Θ the angle *on the Poincaré sphere* between the incident state and the transmitted one, so that $I = \frac{S_0}{2}(1+\cos\Theta) = S_0\cos^2\frac{\Theta}{2}$, which is Malus's law once again with the doubling of angles doing its usual work: the physical angle θ appears on the sphere as $\Theta = 2\theta$.

Take now the case that matters in the laboratory, a linear analyser rotated through all its orientations in front of a beam we know nothing about. Putting $\hat{n} = (\cos 2\alpha_p, \sin 2\alpha_p, 0)$ into 3.152,

$$I(\alpha_p) = \frac{1}{2}\left[S_0 + S_1\cos 2\alpha_p + S_2\sin 2\alpha_p\right] = \frac{S_0}{2}\left[1 + \mathcal{P}_L\cos 2\,(\alpha_M - \alpha_p)\right], \tag{3.153}$$

the second form following by writing the pair (S_1, S_2) in polar coordinates, $S_1 = S_0\mathcal{P}_L\cos 2\alpha_M$ and $S_2 = S_0\mathcal{P}_L\sin 2\alpha_M$, and collapsing the two terms with the cosine of a difference. **The Malus curve**

of any beam whatsoever is a raised cosine in twice the analyser angle: its period is π, as it must be since an analyser at α_p and at $\alpha_p + \pi$ are the same device, its phase α_M marks the azimuth the beam prefers, and its amplitude relative to its mean is the number $\mathcal{P}_L$ we have just been forced to introduce.

Definition 27 *(Degree of polarisation): The degree of polarisation of a beam with Stokes parameters* $\left(S_0, \vec{S}\right)$*, and its linear part, are the two dimensionless numbers*

$$\mathcal{P} = \frac{\sqrt{S_1^2 + S_2^2 + S_3^2}}{S_0}, \qquad \mathcal{P}_L = \frac{\sqrt{S_1^2 + S_2^2}}{S_0}, \tag{3.154}$$

both lying between zero and one by the inequality $S_1^2 + S_2^2 + S_3^2 \leq S_0^2$ *established with 3.67, with* $\mathcal{P} = 1$ *for fully polarised light, which sits on the surface of the Poincaré sphere, and* $\mathcal{P} = 0$ *for unpolarised light, which sits at its centre.*

The value of 3.153 is that it turns that definition into a measurement. The cosine runs between $+1$ and -1 as the analyser is turned, so the brightest and the dimmest readings of the whole sweep are $I_{\max} = \frac{S_0}{2}\left(1 + \mathcal{P}_L\right)$ and $I_{\min} = \frac{S_0}{2}\left(1 - \mathcal{P}_L\right)$, and forming their normalised difference cancels the factor $S_0/2$, which is exactly the quantity a detector of unknown gain cannot be trusted to give us,

$$\boxed{\frac{I_{\max} - I_{\min}}{I_{\max} + I_{\min}} = \frac{\cancel{\frac{S_0}{2}}\left[\left(1 + \mathcal{P}_L\right) - \left(1 - \mathcal{P}_L\right)\right]}{\cancel{\frac{S_0}{2}}\left[\left(1 + \mathcal{P}_L\right) + \left(1 - \mathcal{P}_L\right)\right]} = \mathcal{P}_L} \tag{3.155}$$

The contrast of the Malus curve is the degree of linear polarisation of the beam. This is the operational meaning of a quantity we had so far only defined algebraically, and it requires no calibrated detector, no absolute intensities and no knowledge of the source, only one polariser, one rotation stage and the ratio of two readings. It is worth being honest about what it does not give: the sweep is blind to S_3, exactly as 3.150 warned us, so 3.155 returns $\mathcal{P}_L$ and never $\mathcal{P}$, and a beam that is fully circularly polarised will report a contrast of zero, indistinguishable from natural light. Recovering the missing component is precisely the job of the quarter wave plate, and with it the same rotating analyser becomes the complete Stokes polarimeter of the previous subsections.

One more variant is worth recording, because no real polariser is the ideal projector 3.86. Sending fully polarised linear light at an angle θ through the honest dichroic 3.85, whose two axes transmit the amplitudes $t_1 \geq t_2$ rather than one and zero, the two components are attenuated separately and the intensities add,

$$I = t_1^2 \cos^2\theta + t_2^2 \sin^2\theta, \tag{3.156}$$

so the curve no longer reaches zero, and repeating the contrast computation 3.155 with $I_{\max} = t_1^2$ and $I_{\min} = t_2^2$ gives

$$\frac{I_{\max} - I_{\min}}{I_{\max} + I_{\min}} = \frac{t_1^2 - t_2^2}{t_1^2 + t_2^2} = D,$$

the diattenuation already met in 3.97. The same experimental curve therefore reads two different quantities depending on which end of it we trust: **with a known ideal analyser its contrast measures the polarisation of the beam, and with a known fully polarised beam it measures the quality of the analyser**. The ratio t_2^2/t_1^2, quoted by manufacturers as the extinction ratio, is what makes the crossed polariser null of the previous paragraph imperfect in practice, and a good sheet polariser reaches 10^{-3} while a Glan-Thompson prism reaches 10^{-6}.

It remains to say what all of this means for one photon, and this is where the innocent cosine turns out to have been hiding something. A photodetector does not measure a field, and it does not measure an intensity either: it counts photoelectrons, one at a time, at random instants. The semiclassical theory of photodetection (Mandel and Wolf, 1995), whose fully quantum version is due to Glauber (Glauber, 1963), states that the probability of registering one count in a detector element of area dA during a time dt is

$$dp = \frac{\eta}{\hbar\omega}\, I\left(\vec{r}, t\right) dA\, dt, \tag{3.157}$$

with η the quantum efficiency of the detector. **Intensity is therefore not a thing in itself but a detection probability density**, the mean number of photons arriving per unit area and per unit time, multiplied by the energy $\hbar\omega$ each of them carries. Every intensity written anywhere in this chapter, and in particular the I of 3.149, is a statement about how densely the clicks of a counter fall.

Read 3.149 through 3.157 and it says something far stronger than Malus could have meant. Since the transmitted rate is $\cos^2\theta$ times the incident rate, and since photons are neither created nor split by a sheet of polymer, the only way the beam can obey the law is for each individual photon to be transmitted with probability $\cos^2\theta$ and absorbed with probability $\sin^2\theta$. **A single photon never passes fractionally.** It passes or it does not, and the cosine squared is the weight of the coin. If N photons in the state $|\alpha; 0\rangle$ are sent one by one into an analyser at α_p, the number that emerge is a binomial variable of mean and variance

$$\langle n \rangle = N\cos^2\theta, \qquad \left\langle (\triangle n)^2 \right\rangle = N\cos^2\theta\,\sin^2\theta, \tag{3.158}$$

so the classical law 3.149 is the first of these two and nothing more, an average recovered in the limit of large N because the relative fluctuation $\triangle n / \langle n \rangle$ falls as $1/\sqrt{N}$. The second is invisible to Malus's eye and to any ordinary photodiode, but it is measurable with a counting detector, and it carries a signature the classical law cannot: the variance vanishes at $\theta = 0$ and at $\theta = \pi/2$, that is, a photon prepared in an eigenstate of the analyser is transmitted, or stopped, with no randomness whatsoever. Uncertainty appears only for the states in between, and it is maximal at $\theta = \pi/4$, where the photon is in an equal superposition and the analyser is being asked a question the photon has got no definite answer to.

The formula that governs this is $|\langle\psi|e\rangle|^2$, and a physicist trained after 1926 recognises it immediately: it is the Born rule, the postulate that the probability of finding a system in a state is the squared modulus of the amplitude connecting the two. Malus wrote the earliest instance of it down in 1809, one hundred and seventeen years early, for a phenomenon whose carrier had not been identified, in a formalism that did not exist. The rest of the dictionary is just as literal. The projector $P = |\psi\rangle \langle\psi|$ is the projection postulate, and 3.147 says that a photon which survives the crossing emerges in the state $|\psi\rangle$ having lost every memory of what it was before, which is exactly the indifference to the input that opened this subsection. The polarisation matrix Φ, normalised by its trace, is the density matrix of a two level system, so that 3.151 is the Born rule for mixed states and the degree of polarisation 3.154 is the purity of that state, unity for a pure qubit and zero for the maximally mixed one. And 3.152 is then the general expression for the probability of a projective measurement on a qubit, with the Poincaré sphere serving as the Bloch sphere, exactly as anticipated at the end of the Mueller subsection.

This is why the polarisation of light became the workhorse of experimental quantum information rather than any of the other degrees of freedom a photon possesses. Preparing a qubit is turning the two waveplates of 3.123, measuring it is turning a polariser and counting clicks, and the answer is always 3.152. The $\cos^2$ of the Malus curve is the shape that makes the correlations of an entangled pair violate the Bell inequalities, since a classical model in which each photon carries a definite instruction for every analyser angle cannot reproduce a cosine squared of the *relative* angle; and the contrast 3.155 of that curve, measured exactly as we have described it, is the visibility that every laboratory quotes as the figure of merit of its source. A law read off a calcite crystal by candlelight in Paris turns out to be, without a single symbol changed, the measurement postulate of quantum mechanics applied to the simplest system there is.

3.5.8.2 The elliptic polariser

Everything in the previous subsubsection was done with the linear polariser 3.86, and that is the only polariser one can buy. A sheet of Polaroid, a wire grid and a Glan-Thompson prism all single out a *direction* in the laboratory, because the microscopic anisotropy that makes them work, a stretched polymer chain, a row of wires, a crystal axis, is itself a direction; not one of them singles out an ellipse, and nothing in the catalogue of materials projects onto a state of definite handedness. Yet the whole argument that opened this subsection, that a projector is indifferent to its input and delivers one fixed state no matter what arrives, is far too valuable to be confined to the equator of the Poincaré sphere. If elliptical projectors are not sold, they must be assembled, and the purpose of this subsubsection is to assemble one and to find out exactly how far it can be pushed.

Two objects have to be told apart before we start, and since the distinction is the whole point of what follows we make it a definition.

Definition 28 *(Elliptic projector and elliptic polariser): An* ***elliptic projector*** *is an optical system, and equally the mathematical object representing it, whose Jones matrix is an orthogonal projector,* $\Pi = \Pi^\dagger = \Pi^2$*, of rank one, and whose transmitted state* $|\phi\rangle$ *is elliptical. Being a projector, it delivers* $|\phi\rangle$ *and nothing else, whatever the incident light may have been. An* ***elliptic polariser*** *is an elliptic projector equipped with free parameters such that, as those parameters are varied over their range, the transmitted state* $|\phi\rangle$ *sweeps the whole Poincaré sphere, so that any polarisation state whatsoever, linear, elliptical or circular, of either handedness, can be selected by adjusting the device.*

The difference is not pedantry. A given elliptic projector transmits one particular ellipse and is useless for any other; an elliptic polariser is a single instrument that can be set to project onto any state at all, which is what one actually wants on a bench. We shall see that a construction can be an elliptic projector for every setting of its retarders and still fail to be an elliptic polariser, and that the condition separating the two cases is remarkably sharp.

The construction we take as our starting point is due to Pellat-Finet, who has recently shown that three ordinary elements suffice, and whose quaternionic formulation of polarisation (Pellat-Finet, 1984) we have already met second hand, since it is the language in which (Pabón et al., 2023) is written. The arrangement is the following: a linear birefringent of retardance δ with its fast axis at θ_1; behind it a linear polariser with its transmission axis at θ_2; and behind that a second linear birefringent, *identical to the first*, but mounted with its fast axis at $\theta_1 + \pi/2$. Ordering the three matrices as the light meets them, the device is

$$\Pi = W\left(\theta_1 + \frac{\pi}{2}, 0; \delta\right) P\left(\theta_2\right) W\left(\theta_1, 0; \delta\right). \tag{3.159}$$

The first thing to establish is what the third element actually is, and the answer explains the whole construction. Replacing θ_1 by $\theta_1 + \pi/2$ in the linear retarder 3.76 changes only the two doubled angles, through $\cos\left(2\theta_1 + \pi\right) = -\cos 2\theta_1$ and $\sin\left(2\theta_1 + \pi\right) = -\sin 2\theta_1$, so every term carrying an $i\sin\frac{\delta}{2}$ reverses its sign,

$$W\left(\theta_1 + \frac{\pi}{2}, 0; \delta\right) = \begin{bmatrix} \cos\frac{\delta}{2} - i\sin\frac{\delta}{2}\cos 2\theta_1 & -i\sin\frac{\delta}{2}\sin 2\theta_1 \\ -i\sin\frac{\delta}{2}\sin 2\theta_1 & \cos\frac{\delta}{2} + i\sin\frac{\delta}{2}\cos 2\theta_1 \end{bmatrix} = \overline{W\left(\theta_1, 0; \delta\right)} = W\left(\theta_1, 0; \delta\right)^{\dagger}, \tag{3.160}$$

the last equality holding because a linear retarder is a **symmetric** matrix, as we established when proving the reversibility theorem, so that transposing it does nothing and its adjoint is its plain complex conjugate. Since W is unitary, $W^{\dagger} = W^{-1}$, and we have learnt something worth saying in words: **turning a waveplate through ninety degrees exchanges its fast and slow axes, which is the same as reversing the sign of its retardance**, $W(\theta + \frac{\pi}{2}, 0; \delta) = W(\theta, 0; -\delta)$, so the third element of 3.159 is not merely similar to the first, it is the first one *undone*. Writing W for $W(\theta_1, 0; \delta)$ from here on, the device is a conjugation,

$$\Pi = W^{\dagger}\, P\left(\theta_2\right)\, W. \tag{3.161}$$

That single observation makes the two defining properties immediate. For the Hermiticity, the adjoint of a product reverses the order and P is itself Hermitian, being a projector,

$$\Pi^{\dagger} = \left(W^{\dagger} P W\right)^{\dagger} = W^{\dagger} \underbrace{P^{\dagger}}_{=P} W = \Pi,$$

and for the idempotence the two inner factors annihilate each other,

$$\Pi^2 = W^{\dagger} P \underbrace{W W^{\dagger}}_{=\sigma_0} P W = W^{\dagger} \underbrace{P^2}_{=P} W = \Pi,$$

where $WW^\dagger$ collapses because the plate is unitary and P^2 collapses because the polariser is a projector. So 3.159 is an orthogonal projector, and it is worth pausing to see that **the third plate is what earns this**. Without it the chain would be $P(\theta_2)W$, which is neither Hermitian nor idempotent, and which by Theorem I is an inhomogeneous element whose two eigenpolarisations are not even orthogonal; worse, its output is $|\theta_2;0\rangle$, a *linear* state, for every input. The last plate converts that linear output into an elliptical one and, at the same time, repairs the algebra.

Since $P(\theta_2) = |\theta_2;0\rangle\langle\theta_2;0|$ has got rank one, so has the conjugated matrix, and the transmitted state can be read off directly by grouping the factors,

$$\Pi = W^\dagger |\theta_2;0\rangle\langle\theta_2;0| W = \left(W^\dagger |\theta_2;0\rangle\right)\left(W^\dagger |\theta_2;0\rangle\right)^\dagger = |\phi\rangle\langle\phi|, \qquad |\phi\rangle \equiv W^\dagger |\theta_2;0\rangle, \tag{3.162}$$

and $|\phi\rangle$ is automatically a unit vector, since $\langle\phi|\phi\rangle = \langle\theta_2;0|\underbrace{WW^\dagger}_{=\sigma_0}|\theta_2;0\rangle = 1$. By the definition of the ideal polariser, 3.159 is therefore **a perfect polariser whose transmission state is** $|\phi\rangle$, and everything now hinges on which state that is.

To compute it, move into the frame of the first plate, where the retarder is diagonal. The covariance 3.121 gives $W(\theta_1,0;\delta) = R(\theta_1)W(0,0;\delta)R(-\theta_1)$, so taking adjoints, and using that $R(\theta)^\dagger = R(-\theta)$ for the real orthogonal rotator,

$$|\phi\rangle = R(\theta_1)\, W(0,0;\delta)^\dagger \underbrace{R(-\theta_1)\,|\theta_2;0\rangle}_{|\theta_2-\theta_1;0\rangle} = R(\theta_1)\, W(0,0;-\delta)\,|\beta;0\rangle, \qquad \beta \equiv \theta_2 - \theta_1,$$

where the rotator only shifted the azimuth of the linear ket. **Only the angle β between the polariser and the first plate can matter**, which was to be expected, since θ_1 merely says where the whole apparatus has been bolted to the table. Setting $\alpha = 0$ in 3.76 makes the plate diagonal, $W(0,0;\delta) = \mathrm{diag}\left(e^{i\delta/2}, e^{-i\delta/2}\right)$, so that its inverse acts on $|\beta;0\rangle = [\cos\beta, \sin\beta]^\mathsf{T}$ as

$$W(0,0;-\delta)\,|\beta;0\rangle = \begin{bmatrix} e^{-i\delta/2}\cos\beta \\ e^{i\delta/2}\sin\beta \end{bmatrix} = e^{-i\delta/2}\begin{bmatrix} \cos\beta \\ e^{i\delta}\sin\beta \end{bmatrix}, \tag{3.163}$$

and the overall factor $e^{-i\delta/2}$ is a global phase and may be dropped. What is left is a spinor of amplitudes $\cos\beta$ and $\sin\beta$ carrying a relative phase δ between its two components, which is exactly the general form 3.34 we started the chapter with: **the polariser fixes the amplitudes and the plate fixes the phase**, and between them they can build any ellipse. To name it we compute its Stokes parameters with 3.62 to 3.64. The moduli are immediate,

$$S_1 = \cos^2\beta - \sin^2\beta = \cos 2\beta,$$

and for the other two we need the crossed product $e_1 e_2^* = \cos\beta\, e^{-i\delta}\sin\beta$, whose real and imaginary parts give

$$S_2 = 2\,\mathrm{Re}\,(e_1e_2^*) = 2\sin\beta\cos\beta\cos\delta = \sin 2\beta\cos\delta, \qquad S_3 = -2\,\mathrm{Im}\,(e_1^*e_2) = -\sin 2\beta\sin\delta,$$

using $2\sin\beta\cos\beta = \sin 2\beta$ in both. As a check, squaring and adding the three returns

$$S_1^2 + S_2^2 + S_3^2 = \cos^2 2\beta + \sin^2 2\beta \underbrace{\left(\cos^2\delta + \sin^2\delta\right)}_{=1} = 1,$$

so the state is fully polarised, as any output of a projector must be. The final rotator $R(\theta_1)$ then adds θ_1 to the azimuth without touching the ellipticity, and comparing with the dictionary 3.65 we can read the two characteristic angles off at once.

Theorem 11 *(Pellat-Finet elliptic projector): The chain 3.159, formed by a linear birefringent of retardance δ at θ_1, a linear polariser at θ_2 and the same birefringent at $\theta_1 + \pi/2$, is for every value of its three angles an orthogonal projector $\Pi = |\phi\rangle\langle\phi|$ onto the fully polarised state $|\phi\rangle = |\alpha;\chi\rangle$ whose characteristic angles are, writing $\beta = \theta_2 - \theta_1$,*

$$\boxed{\sin 2\chi = \sin 2\beta\,\sin\delta, \qquad \tan 2\left(\alpha - \theta_1\right) = \tan 2\beta\,\cos\delta} \tag{3.164}$$

It is therefore an elliptic projector whenever $\sin 2\beta \sin\delta \neq 0$.

Before pushing further it is worth checking the two degenerate cases against common sense. If the plates do nothing, $\delta = 0$, then 3.164 gives $\chi = 0$ and $\alpha = \theta_1 + \beta = \theta_2$, so the device collapses into the bare linear polariser it contains, as it must. If the polariser is aligned with the plate, $\beta = 0$, then again $\chi = 0$ and $\alpha = \theta_1$, because the incident light is being projected onto an eigenstate of the plates, which pass it untouched. Everything in between is a genuine ellipse.

Now the question that separates a projector from a polariser: as the three angles are turned, how much of the Poincaré sphere does $|\phi\rangle$ actually cover? The azimuth is no obstacle, since θ_1 appears in 3.164 as a free additive shift and can place α anywhere. The ellipticity is another matter. The first of 3.164 is a product of two sines, one of which we control through β and one of which is welded into the plates, and since $\sin 2\beta$ ranges over the whole of $[-1, 1]$ as β is turned, while $\sin\delta$ does not move at all,

$$|\sin 2\chi| = |\sin 2\beta|\,|\sin\delta| \leq |\sin\delta|\,, \qquad \text{that is} \qquad |\chi| \leq \frac{1}{2}\arcsin|\sin\delta|\,, \tag{3.165}$$

with the bound attained at $\beta = \pm\pi/4$, where the polariser bisects the axes of the plate. **The reachable states fill a band around the equator and nothing more**, a band whose half width is $\delta/2$ for $0 \leq \delta \leq \pi/2$ and $(\pi - \delta)/2$ beyond it, so that a device built with eighth wave plates, $\delta = \pi/4$, can never produce an ellipse more open than $\chi = \pi/8$, however patiently its three mounts are turned. The circular states sit at $|\chi| = \pi/4$, the extreme value the ellipticity angle can take, and 3.165 reaches them only when $\arcsin|\sin\delta| = \pi/2$. This is the condition we were after, and it is exact in both directions.

Theorem 12 *(Elliptic polariser): The Pellat-Finet chain 3.159 is an elliptic polariser, that is, its transmitted state covers the entire Poincaré sphere as θ_1 and θ_2 are varied,* ***if and only if its two birefringents are quarter wave plates****,*

$$|\sin\delta| = 1 \qquad \Longleftrightarrow \qquad \delta = \frac{\pi}{2} \pmod{\pi}. \tag{3.166}$$

When that condition holds the device reduces to the projector

$$\boxed{\Pi = |\theta_1;\, \theta_2 - \theta_1\rangle \langle\theta_1;\, \theta_2 - \theta_1|} \tag{3.167}$$

whose transmitted state has got azimuth $\alpha = \theta_1$ and ellipticity $\chi = \theta_2 - \theta_1$.

Proof. Necessity is 3.165: covering the sphere requires in particular reaching its two poles, $|\chi| = \pi/4$, which forces $\arcsin|\sin\delta| = \pi/2$ and hence $|\sin\delta| = 1$, that is $\delta = \pi/2$ modulo π, an odd number of quarter waves. No other retardance can reach the circular states, by however much the plates and the polariser are turned, since the bound 3.165 does not involve θ_1 or θ_2 at all.

For sufficiency, set $\delta = \pi/2$ in 3.163, where $e^{i\delta} = e^{i\pi/2} = i$, and the spinor before the final rotation becomes

$$\begin{bmatrix} \cos\beta \\ i\sin\beta \end{bmatrix},$$

which we have met before: it is precisely the unrotated ellipse 3.40 of ellipticity angle β, that is, the ket $|0;\beta\rangle$. Applying the last rotator, which by 3.83 only advances the azimuth,

$$|\phi\rangle = R(\theta_1)\,|0;\beta\rangle = |\theta_1;\beta\rangle = |\theta_1;\, \theta_2 - \theta_1\rangle\,,$$

and 3.167 follows from 3.162. The same conclusion drops out of 3.164 without any new work: $\cos\delta = 0$ kills the second relation, leaving $\alpha = \theta_1$, while $\sin\delta = 1$ reduces the first to $\sin 2\chi = \sin 2\beta$ and hence $\chi = \beta$. Since θ_1 and θ_2 may be given any values independently, the pair $(\alpha, \chi) = (\theta_1, \theta_2 - \theta_1)$ runs over every azimuth and every ellipticity, which is the whole sphere. ■

It is worth writing the elliptic polariser out as an explicit matrix, and since the object is a ket times a bra we may as well multiply the two out honestly, for a general pair of angles (α, χ), and only afterwards put in the values $\alpha = \theta_1$ and $\chi = \theta_2 - \theta_1$ delivered by the theorem. Abbreviate the two components of 3.41 as

$$|\alpha;\chi\rangle = \begin{bmatrix} a \\ b \end{bmatrix}, \qquad a = \cos\alpha\cos\chi - i\sin\alpha\sin\chi, \qquad b = \sin\alpha\cos\chi + i\cos\alpha\sin\chi,$$

so that the outer product is the array of the four products $|\alpha;\chi\rangle\langle\alpha;\chi| = \left[\begin{smallmatrix} a\bar{a} & a\bar{b} \\ b\bar{a} & b\bar{b} \end{smallmatrix}\right]$, and take them one at a time. The upper left entry is a modulus, so the imaginary parts contribute their squares,

$$\begin{aligned} |a|^2 = \cos^2\alpha\cos^2\chi + \sin^2\alpha\sin^2\chi &= \frac{(1+\cos 2\alpha)\,(1+\cos 2\chi)}{4} + \frac{(1-\cos 2\alpha)\,(1-\cos 2\chi)}{4} \\ &= \frac{1}{4}\left[1 + \cancel{\cos 2\alpha} + \cancel{\cos 2\chi} + \cos 2\alpha\cos 2\chi + 1 - \cancel{\cos 2\alpha} - \cancel{\cos 2\chi} + \cos 2\alpha\cos 2\chi\right] \\ &= \frac{1}{2}\left(1 + \cos 2\alpha\cos 2\chi\right), \end{aligned}$$

where we inserted the half angle identities $\cos^2 x = \frac{1+\cos 2x}{2}$ and $\sin^2 x = \frac{1-\cos 2x}{2}$ in all four factors and then the single cosines cancelled in pairs between the two products, leaving only the crossed term twice. The lower right entry costs nothing more, since $|a|^2 + |b|^2 = 1$ for a unit ket and therefore $|b|^2 = \frac{1}{2}(1 - \cos 2\alpha \cos 2\chi)$. For the off diagonal entry we multiply out in full,

$$\begin{aligned}
b\bar{a} &= (\sin\alpha\cos\chi + i\cos\alpha\sin\chi)(\cos\alpha\cos\chi + i\sin\alpha\sin\chi) \\
&= \sin\alpha\cos\alpha\cos^2\chi + i\sin^2\alpha\sin\chi\cos\chi + i\cos^2\alpha\sin\chi\cos\chi \underbrace{+\, i^2}_{=-1}\sin\alpha\cos\alpha\sin^2\chi \\
&= \sin\alpha\cos\alpha\left(\cos^2\chi - \sin^2\chi\right) + i\sin\chi\cos\chi\underbrace{\left(\sin^2\alpha + \cos^2\alpha\right)}_{=1} \\
&= \frac{1}{2}\sin 2\alpha\cos 2\chi + \frac{i}{2}\sin 2\chi,
\end{aligned}$$

having grouped the two real terms and the two imaginary ones and used $2\sin x\cos x = \sin 2x$ twice and $\cos^2\chi - \sin^2\chi = \cos 2\chi$ once, and the remaining entry is its conjugate, $a\bar{b} = \overline{b\bar{a}}$. Collecting the four,

$$|\alpha;\chi\rangle\langle\alpha;\chi| = \frac{1}{2}\begin{bmatrix} 1 + \cos 2\alpha\cos 2\chi & \sin 2\alpha\cos 2\chi - i\sin 2\chi \\ \sin 2\alpha\cos 2\chi + i\sin 2\chi & 1 - \cos 2\alpha\cos 2\chi \end{bmatrix} = \frac{1}{2}\left[\sigma_0 + \hat{n}\cdot\vec{\sigma}\right], \tag{3.168}$$

the last equality being read straight off 3.68 with $\hat{n} = (\cos 2\alpha\cos 2\chi, \sin 2\alpha\cos 2\chi, -\sin 2\chi)$, which is by 3.65 the Stokes vector of $|\alpha;\chi\rangle$ itself. So the brute force multiplication confirms what the Malus subsubsection told us in one line, that a rank one projector is half the identity plus half the Pauli expansion of the state it projects onto, and it confirms it entry by entry.

Substituting now the angles of the theorem, $\alpha = \theta_1$ and $\chi = \theta_2 - \theta_1$, the elliptic polariser stands written in terms of its two mount angles alone,

$$\boxed{\Pi = \frac{1}{2}\begin{bmatrix} 1 + \cos 2\theta_1\cos(2\theta_2 - 2\theta_1) & \sin 2\theta_1\cos(2\theta_2 - 2\theta_1) - i\sin(2\theta_2 - 2\theta_1) \\ \sin 2\theta_1\cos(2\theta_2 - 2\theta_1) + i\sin(2\theta_2 - 2\theta_1) & 1 - \cos 2\theta_1\cos(2\theta_2 - 2\theta_1) \end{bmatrix}} \tag{3.169}$$

or, in the compact Pauli form,

$$\Pi = \frac{1}{2}\left[\sigma_0 + \hat{n}\cdot\vec{\sigma}\right], \qquad \hat{n} = \begin{pmatrix} \cos 2\theta_1\cos(2\theta_2 - 2\theta_1) \\ \sin 2\theta_1\cos(2\theta_2 - 2\theta_1) \\ -\sin(2\theta_2 - 2\theta_1) \end{pmatrix}. \tag{3.170}$$

Setting $\theta_2 = \theta_1$ makes the two cosines equal to one and the sine vanish, and 3.169 collapses to $\frac{1}{2}\begin{bmatrix} 1+\cos 2\theta_1 & \sin 2\theta_1 \\ \sin 2\theta_1 & 1-\cos 2\theta_1 \end{bmatrix}$, which is the ordinary linear polariser 3.86 at θ_1 once the half angle identities are read backwards.

Three remarks close the construction. The first is about the knobs, and it repeats a pattern we have now seen three times: **the plates choose the azimuth and the polariser chooses the ellipticity**, since $\alpha = \theta_1$ and $\chi = \theta_2 - \theta_1$ depend on the common angle and on the difference separately. Turning the whole mount rigidly rotates the transmitted ellipse without deforming it; turning the polariser alone inside a fixed pair of plates opens the ellipse from a straight line at

$\theta_2 = \theta_1$, through every intermediate shape, to a circle at $\theta_2 = \theta_1 \pm \pi/4$, and back. It is the same separation of controls that made the biplate practical, and it is what makes this a usable instrument rather than a curiosity.

The second is the comparison that this whole subsection was built to make. The state generator of 3.123 and the elliptic polariser 3.167 deliver exactly the same catalogue of states, the entire Poincaré sphere, and they do it with a comparable pile of hardware, two plates against two plates and a polariser. They are not interchangeable. The generator is unitary and loses not one photon, but it must be told what state is coming in, and it transforms that state rather than fixing it; feed it something else and it delivers something else. The polariser 3.167 does not care in the slightest what arrives, and returns $|\theta_1; \theta_2 - \theta_1\rangle$ whether the input is a laser, a lamp or an unpolarised thermal source, at the cost, quantified by the Malus law of the next subsubsection, of throwing away everything that does not match.

The third is a caveat about colour that honesty requires, because the argument at the head of this subsection praised the polariser for being achromatic. Equation 3.166 has just made the whole construction depend on a retardance being exactly $\pi/2$, and by 3.74 that is true at one wavelength only. The elliptic polariser is therefore achromatic in the element at its heart and chromatic in the two that dress it, and its transmitted ellipticity drifts across a broadband spectrum exactly as the state generator's did. What survives untouched at every wavelength is the projector property itself, since 3.161 is Hermitian and idempotent for any δ whatsoever: broadband light emerges from the device fully polarised at every wavelength, just not in the same state at each.

3.5.8.3 The elliptic Malus law

We can now pay the debt left open twice, once when the subsection was introduced and once at the end of the last one, and write down the most general form the law of Malus can take. The arrangement is the oldest in polarisation optics, a polariser followed by an analyser, with the single change that both of them are now the elliptic polarisers 3.167 we have just learnt to build rather than the linear ones Malus had. Two identical devices are set to the same pair of parameters (α_p, χ_p); a beam in an arbitrary state $|\alpha; \chi\rangle$ falls on the first; and the second is then turned bodily, mount and all, through a complete revolution, $0 \le \theta \le 2\pi$. What does the detector behind it read?

The first thing to settle is what a rigid rotation does to an elliptic polariser, and the answer is already contained in 3.167. Turning the whole mount by θ turns every one of the three elements with it, so the two plate angles and the polariser angle all advance together, $\theta_1 \to \theta_1 + \theta$ and $\theta_2 \to \theta_2 + \theta$, and the transmitted state becomes

$$|\phi(\theta)\rangle = |\theta_1 + \theta;\, (\theta_2 + \theta) - (\theta_1 + \theta)\rangle = |\theta_1 + \theta;\, \theta_2 - \theta_1\rangle = |\alpha_p + \theta;\, \chi_p\rangle\,, \qquad (3.171)$$

where the rotation cancels identically inside the second slot, since the ellipticity is built out of the *difference* of two angles that have both been shifted by the same amount. Equivalently, and this is the general statement for any element, the device is conjugated by a rotator, $\Pi(\theta) = R(\theta)\Pi R(-\theta)$, exactly as in 3.137. **Turning an elliptic polariser advances the azimuth of the state it**

transmits and leaves its ellipticity frozen, so that on the Poincaré sphere the transmitted point does not wander freely but travels along a *circle of constant latitude*, the parallel at height $S_3 = -\sin 2\chi_p$. Everything peculiar about the law we are about to derive comes from that single geometrical fact.

Next we need the intensity delivered by two projectors in a row, and it is worth doing in general, for an input of any degree of polarisation, because the calculation is no harder. Let $\Pi_1 = |\phi_1\rangle\langle\phi_1|$ and $\Pi_2 = |\phi_2\rangle\langle\phi_2|$ be the two devices in the order the light meets them, so the cascade has got the Jones matrix $J = \Pi_2\Pi_1$, and let Φ be the polarisation matrix 3.67 of the incident beam. By 3.151 the transmitted intensity is the trace of the outgoing polarisation matrix,

$$\begin{aligned} I &= \operatorname{tr}\left(J\Phi J^\dagger\right) = \operatorname{tr}\left(\Pi_2\Pi_1\Phi\Pi_1\Pi_2\right) = \operatorname{tr}\left(\underbrace{\Pi_2^2}_{=\Pi_2}\Pi_1\Phi\Pi_1\right) \\ &= \operatorname{tr}\left(|\phi_2\rangle\langle\phi_2|\,|\phi_1\rangle\langle\phi_1|\,\Phi\,|\phi_1\rangle\langle\phi_1|\right) = \underbrace{\langle\phi_1|\phi_2\rangle\langle\phi_2|\phi_1\rangle}_{|\langle\phi_2|\phi_1\rangle|^2}\langle\phi_1|\,\Phi\,|\phi_1\rangle\,, \end{aligned} \tag{3.172}$$

where we used the cyclic property to bring the last Π_2 round to the front and collapsed $\Pi_2^2 = \Pi_2$, then wrote every projector as a ket times a bra and collected the two scalars that fell out of the middle. **The cascade factorises exactly**, into one number that depends on the incident light and on the first device and a second number that depends on the two devices alone. Nothing survives of the input in the second factor, which is the algebraic face of the indifference to the input we have been advertising since the beginning of this subsection.

The first factor is the general Malus law 3.152 already proved, $\langle\phi_1|\,\Phi\,|\phi_1\rangle = \frac{1}{2}\left(S_0 + \hat{n}_p\cdot\vec{S}\right)$, so for a fully polarised input everything reduces to scalar products of unit vectors on the sphere, and those are worth writing out. With $\hat{n} = (\cos 2\alpha\cos 2\chi, \sin 2\alpha\cos 2\chi, -\sin 2\chi)$ for the incident state and $\hat{n}_p$ the same expression in (α_p, χ_p), multiplying the two triples term by term,

$$\begin{aligned} \hat{n}_p\cdot\hat{n} &= \cos 2\chi_p\cos 2\chi\underbrace{\left[\cos 2\alpha_p\cos 2\alpha + \sin 2\alpha_p\sin 2\alpha\right]}_{\cos 2(\alpha-\alpha_p)} + \left(-\sin 2\chi_p\right)\left(-\sin 2\chi\right) \\ &= \cos 2\chi\cos 2\chi_p\cos 2\left(\alpha-\alpha_p\right) + \sin 2\chi\sin 2\chi_p, \end{aligned} \tag{3.173}$$

having taken the common factors out of the first two components and recognised the cosine of a difference. Equation 3.173 deserves to be recognised for what it is: reading 2χ as a latitude and 2α as a longitude, it is the **spherical law of cosines**, $\cos\Theta = \sin\varphi_1\sin\varphi_2 + \cos\varphi_1\cos\varphi_2\cos\triangle\lambda$, for the great circle arc Θ joining two points of the Poincaré sphere. The whole of Malus's law is spherical trigonometry.

The second factor is the same scalar product taken between the two polariser states, which by 3.171 share their latitude and differ only in longitude, by 2θ. Putting $\chi = \chi_p$ and $\alpha - \alpha_p = \theta$ into 3.173,

$$\hat{n}_p(\theta)\cdot\hat{n}_p = \cos^2 2\chi_p\cos 2\theta + \sin^2 2\chi_p,$$

and since for two fully polarised states the squared overlap is $|\langle\phi_2|\phi_1\rangle|^2 = \frac{1}{2}(1+\hat{n}_2\cdot\hat{n}_1)$, which is 3.152 with $S_0 = 1$,

$$|\langle\phi(\theta)|\phi\rangle|^2 = \frac{1}{2}\left[1+\sin^2 2\chi_p + \cos^2 2\chi_p \cos 2\theta\right] = \frac{1}{2}\left[2-\cos^2 2\chi_p + \cos^2 2\chi_p \cos 2\theta\right]$$
$$= 1 - \frac{\cos^2 2\chi_p}{2}\underbrace{(1-\cos 2\theta)}_{=2\sin^2\theta} = 1 - \cos^2 2\chi_p\,\sin^2\theta, \tag{3.174}$$

where we replaced $1+\sin^2 2\chi_p$ by $2-\cos^2 2\chi_p$, collected the two terms carrying $\cos^2 2\chi_p$, and used the half angle identity $1-\cos 2\theta = 2\sin^2\theta$. Multiplying the two factors together gives the law.

Theorem 13 *(Elliptic Malus law): Let a beam of intensity S_0 and Stokes vector $\vec{S}$ fall on an elliptic polariser of parameters (α_p, χ_p), and let a second, identical elliptic polariser behind it be turned bodily through the angle θ. The transmitted intensity is*

$$\boxed{I(\theta) = \frac{1}{2}\left(S_0 + \hat{n}_p\cdot\vec{S}\right)\left[1-\cos^2 2\chi_p\,\sin^2\theta\right]} \tag{3.175}$$

and in particular, for a fully polarised incident state $|\alpha;\chi\rangle$ of intensity I_0,

$$\boxed{I(\theta) = \frac{I_0}{2}\Big[1+\cos 2\chi\cos 2\chi_p\cos 2(\alpha-\alpha_p)+\sin 2\chi\sin 2\chi_p\Big]\Big[1-\cos^2 2\chi_p\sin^2\theta\Big]} \tag{3.176}$$

Every Malus law written so far is a corner of this one. Setting $\chi_p = 0$ makes both polarisers linear, $\cos^2 2\chi_p = 1$ collapses the second bracket to $1-\sin^2\theta = \cos^2\theta$, and the first bracket becomes $1+\cos 2\chi\cos 2(\alpha-\alpha_p)$, so that 3.176 reduces to 3.150 multiplied by $\cos^2\theta$; setting in addition $\chi = 0$, so that the incident light is linear too, leaves

$$I(\theta) = \frac{I_0}{2}\left[1+\cos 2(\alpha-\alpha_p)\right]\cos^2\theta = I_0\cos^2(\alpha-\alpha_p)\cos^2\theta,$$

which is Malus's own law applied twice in a row, once at the polariser and once at the analyser, exactly as the textbook cascade demands. And since $\sin^2\theta$ has got period π, **a full revolution of the mount traverses the curve twice**, the same periodicity we met in the rotating birefringent laws and for the same reason, that a polarisation state is unchanged by a half turn.

The interesting part of 3.176 is what happens when the polarisers are *not* linear, because then something the classical experiment takes for granted stops being true. Turning the analyser to $\theta = \pi/2$, the position we would call crossed, the second bracket does not vanish but leaves

$$\frac{I_{\min}}{I_{\max}} = 1-\cos^2 2\chi_p = \sin^2 2\chi_p, \tag{3.177}$$

so that **two identical elliptic polarisers can never be crossed**. A pair set to an ellipticity of 11.25° still passes fourteen per cent of the light at their darkest, a pair at 22.5° passes half of it, and a pair of circular polarisers, $\chi_p = \pi/4$, passes everything at every angle and shows no modulation whatsoever, the curve 3.174 flattening to a constant. The geometry says exactly why: extinction requires the two transmitted states to be orthogonal, that is antipodal on the sphere, and by 3.171 the rotated state is confined to its parallel of latitude, which contains the antipode of its starting point only when that parallel is the equator. The contrast of the curve, computed as in 3.155, decays accordingly,

$$\frac{I_{\max} - I_{\min}}{I_{\max} + I_{\min}} = \frac{1 - \sin^2 2\chi_p}{1 + \sin^2 2\chi_p} = \frac{\cos^2 2\chi_p}{1 + \sin^2 2\chi_p}, \tag{3.178}$$

falling from one for linear polarisers to zero for circular ones. None of this forbids extinction outright: the orthogonal state $|\alpha_p + \pi/2; -\chi_p\rangle$ is perfectly available to an elliptic polariser, by 3.167 one only has to set $\theta_1' = \theta_1 + \pi/2$ and $\theta_2' = 2\theta_1 - \theta_2 + \pi/2$. What the law forbids is reaching it *by turning the mount*, because a rigid rotation cannot change an ellipticity; the second device has to be taken apart and rebuilt with its internal angle reversed.

It remains to read the law at the level of single photons, and here the two factors of 3.172 say something worth dwelling on. Dividing by S_0, the law gives the probability that one photon crosses the whole apparatus, and by 3.157 that probability, multiplied by the incident flux, is literally the density of clicks the counter behind the analyser registers as the mount turns. That probability is a product,

$$p(\theta) = \underbrace{\frac{1}{2}\,(1 + \hat{n}_p \cdot \hat{n})}_{p_1} \cdot \underbrace{\left[1 - \cos^2 2\chi_p \sin^2\theta\right]}_{p_2}, \tag{3.179}$$

and a product of probabilities is the signature of two *independent* trials. The independence is not an assumption but a consequence of the projection postulate, and it is the whole content of the factorisation: a photon that survives the first device does not merely pass through it, **it is reprepared in the state $|\phi_1\rangle$ and loses every memory of the state it arrived in**, so the odds it faces at the second device cannot possibly depend on where it came from. This is why p_2 contains neither α nor χ. Classically one would have no right to expect it; a wave that merely had its unwanted component filtered away would still carry some trace of its history.

The same reading explains the oldest parlour trick in the subject. Two crossed linear polarisers pass nothing, $p = 0$; slide a third polariser at 45° between them and light reappears, with $p = \cos^2\frac{\pi}{4}\cos^2\frac{\pi}{4} = \frac{1}{4}$. Inserting an absorbing element has *increased* the transmission, which is impossible for a filter that only removes and possible only for a device that reprepares, and 3.172 is the formula that permits it, since each projector resets the state that the next one measures.

The residual transmission 3.177 carries the deepest of these readings. At the crossed position the surviving probability is $\sin^2 2\chi_p = |\langle\phi_2|\phi_1\rangle|^2$, the squared overlap of two states that are *not* orthogonal, and a photon prepared in $|\phi_1\rangle$ has that probability of being counted by a device asking whether it is in $|\phi_2\rangle$. The failure to extinguish is therefore not an imperfection of the hardware, which we have assumed ideal throughout, but the statement that **two non orthogonal quantum states cannot be told apart with certainty**, with the leaked intensity measuring exactly how far from orthogonal the two ellipses are. Perfect extinction, $\langle\phi_2|\phi_1\rangle = 0$, is available only for the antipodal pairs, and it is the same impossibility of discriminating non orthogonal states, exploited on purpose, that makes an eavesdropper detectable in quantum key distribution.

Finally, the counting statistics repeat the pattern of the previous subsubsection with the two probabilities compounded. Sending N photons one at a time, the number reaching the detector is binomial in $p(\theta)$,

$$\langle n\rangle = N\,p_1p_2, \qquad \left\langle (\triangle n)^2 \right\rangle = N\,p_1p_2\,(1-p_1p_2)\,,$$

so the elliptic Malus curve 3.176 is the mean of that distribution and nothing else, recovered in the classical limit of large N. Written in the language of states rather than intensities, 3.172 reads $p = |\langle\phi_2|\phi_1\rangle|^2\,\langle\phi_1|\,\rho\,|\phi_1\rangle$ with $\rho = \Phi/S_0$ the density matrix, which is the rule for two projective measurements performed one after the other on a qubit. The most general law of Malus is, once more and now in its final form, the measurement postulate of quantum mechanics applied twice.

Chapter 4

Interference and Interferometers

When two waves meet, light may add to light and yield darkness; in that paradox lies the whole of physical optics.

"When two undulations, from different origins, coincide either perfectly or very nearly in direction, their joint effect is a combination of the motions belonging to each."
– Thomas Young

4.1 A historical context of interference and interferometers

Interference is the most direct fingerprint of the wave nature of light: when two beams overlap, their amplitudes - not their intensities - add, and the result can be brighter than either beam alone or, astonishingly, complete darkness. From this simple superposition principle grew the interferometer, an instrument that turns the invisible phase of a light wave into a visible pattern of fringes, and which became one of the most precise measuring devices ever built.

Although Robert Hooke and Isaac Newton both studied the coloured fringes produced by thin films - the phenomenon still known as Newton's rings, carefully described in Newton's *Opticks* of 1704 Newton (1704) - neither interpreted them as evidence of waves. Newton's immense authority, and his corpuscular theory, held the field for a century. The decisive break came in 18011804 with Thomas Young, whose double-slit experiment showed that light passing through two narrow apertures produces a pattern of bright and dark bands that can only be explained if the two beams alternately reinforce and cancel one another Young (1804). Young introduced the very word "interference" and, with it, the principle of superposition that underlies all of wave optics.

Young's qualitative insight was forged into a quantitative theory by Augustin-Jean Fresnel, who combined the principle of interference with Huygens's secondary wavelets to compute diffraction and interference patterns in full detail, and whose recognition that light is a transverse wave (developed with Arago) tied interference firmly to polarisation: only components sharing a polarisation can interfere. By the middle of the nineteenth century the wave theory was secure, and interference had become a tool. Hippolyte Fizeau and Léon Foucault used interferometric methods to compare the speed of light in different media, and Fizeau exploited the visibility of fringes to infer stellar properties - an idea far ahead of its time.

The instrument that defines the modern era is the Michelson interferometer. Albert A. Michelson, seeking ever more precise measurements, built in the 1880s a device that splits a beam in two, sends the halves along perpendicular arms, and recombines them so that any difference in optical path appears as a shift of the fringes. With it, Michelson and Edward Morley performed in 1887 their famous experiment to detect the motion of the Earth through the luminiferous aether; the null result Michelson and Morley (1887) became one of the experimental pillars on which Einstein's special relativity would later rest. Michelson also turned his interferometer into a metrological standard, measuring the length of the standard metre in wavelengths of light, work for which he received the Nobel Prize in 1907.

Almost simultaneously, Charles Fabry and Alfred Perot introduced in 1899 an interferometer of a different kind, based on multiple reflections between two highly reflecting parallel surfaces Fabry and Perot (1899). The FabryPerot interferometer produces extremely sharp transmission fringes and offers very high spectral resolving power; it is the direct ancestor of the optical resonators and cavities that define the modes of every laser. Other configurations followed - the MachZehnder interferometer, with its separated paths ideal for probing transparent samples, and the Sagnac interferometer, sensitive to rotation and the heart of modern optical gyroscopes.

The visibility of the fringes in any of these instruments is not merely an experimental nuisance: it is a direct measure of the coherence of the light, and it was precisely the analysis of fringe visibility that motivated the statistical theory of coherence developed in a later chapter. In the twentieth century, interferometry reached almost unimaginable sensitivity: the Laser Interferometer Gravitational-Wave Observatory (LIGO), a pair of kilometre-scale Michelson interferometers, detected in 2015 the passage of a gravitational wave by measuring a change in arm length thousands of times smaller than a proton Abbott et al. (2016). From Young's two slits to the detection of colliding black holes, the interferometer has remained the same idea - read the phase of a wave by letting it interfere with itself - refined to extraordinary perfection.

4.2 How does interference occur?

Lets consider two sources of electromagnetic field, under the conditions we've mentioned on this lectueres, we're going only to focus on the electric component, therefore, lets suppose we've got the following two sources of electric field

$$\vec{E}_1\left(\vec{r},t\right) = \vec{E}_1(\vec{r})e^{-i\omega_1 t}, \tag{4.1}$$
$$\vec{E}_2\left(\vec{r},t\right) = \vec{E}_2(\vec{r})e^{-i\omega_2 t}, \tag{4.2}$$

where the complex field amplitudes are given by

$$\vec{E}_1(\vec{r}) = E_1(\vec{r})e^{i\varphi_1(\vec{r})}\hat{e}_1, \tag{4.3}$$
$$\vec{E}_2(\vec{r}) = E_2(\vec{r})e^{i\varphi_2(\vec{r})}\hat{e}_2. \tag{4.4}$$

Theses complex field amplitudes are almost general, here the polarization vectores $\hat{e}_1, \hat{e}_2$ are'nt space-time dependent, which, in general, is what it's observed. The phases $\varphi_1(\vec{r}), \varphi_1(\vec{r})$ represent an arbitrary wavefront behavior,also $E_1(\vec{r}), E_2(\vec{r})$ represent arbitrary amplitudes. Thus the expressions above are general experessions for arbitrary sources of radiation.

The interference appears in the superposition of two or more electric fields, therefore, to study the interference of these two field in a space-time point $(\vec{r},t)$ we're going to add these two as follows

$$\vec{E}(\vec{r},t) = \vec{E}_1(\vec{r},t) + \vec{E}_2(\vec{r},t). \tag{4.5}$$

As we've explained before, measuring the electric field is really hardo to do, so what we study and see are the irradiencies or radiations intensity, also the frequencies $\{\omega_1, \omega_2\}$ are too fast time oscillations for the human eye and most of the detectors, so we only can study their time averages. Having all this on count, lets calculate the average of the irradiances

$$\left\langle \left|\left|\vec{E}(\vec{r},t)\right|\right|^2 \right\rangle = \left\langle \left|\left|\vec{E}_1(\vec{r},t)\right|\right|^2 \right\rangle + \left\langle \left|\left|\vec{E}_2(\vec{r},t)\right|\right|^2 \right\rangle + 2\left\langle Re\left\{\vec{E}_1^*(\vec{r},t)\cdot\vec{E}_2(\vec{r},t)\right\}\right\rangle,$$
$$\left\langle \left|\left|\vec{E}(\vec{r},t)\right|\right|^2 \right\rangle = \left|\left|\vec{E}_1(\vec{r})\right|\right|^2 + \left|\left|\vec{E}_2(\vec{r})\right|\right|^2 + 2Re\left\{\vec{E}_1^*(\vec{r})\cdot\vec{E}_2(\vec{r})\left\langle e^{-i(\omega_1-\omega_2)t}\right\rangle\right\}.$$

The first two terms corresponds to **the total field intensity** and the third terms is called **the interference term**because it contains the interaction of both fields.

From a mathematical point of view, to make the interaction non zero, both frequencies must be the same $\omega_1 = \omega_2$, this, like most books say, should imply that the interference only ocurrs with two sources with the same frequency or on identical photons but, this is not completely correct. The frequency-difference exponential $\exp\left(-i\left[\omega_1 - \omega_2\right]t\right)$ is, in general, not zero, so **there are a lot ultrafast oscillations and interferences** what we see, and what we detected using measuring instruments is the time average, and in average, these are zero.

Before going further, lets define the field intensisties as

$$\left|\left|\vec{E}_1(\vec{r})\right|\right|^2 = I_1(\vec{r}), \tag{4.6}$$

$$\left|\left|\vec{E}_2(\vec{r})\right|\right|^2 = I_2(\vec{r}). \tag{4.7}$$

The interference terms is then given by

$$\vec{E}_1^*(\vec{r}) \cdot \vec{E}_2(\vec{r}) = E_1(\vec{r})E_2(\vec{r})e^{i[\phi_2(\vec{r})-\phi_1(\vec{r})]}\hat{e}_1^* \cdot \hat{e}_2, \tag{4.8}$$

then the field intensity is

$$I = I_1 + I_2 + 2Re\left\{\left(\hat{e}_1^* \cdot \hat{e}_2\right)e^{i\triangle\phi(\vec{r})}\right\}\underbrace{E_1(\vec{r})E_2(\vec{r})}_{(I_1 I_2)^{1/2}}. \tag{4.9}$$

The functional $\triangle\phi = \phi_2(\vec{r}) - \phi_1(\vec{r})$ is known as the phase-difference of the waves. The intereference term is zero if the **polarisations are orthogonal** this means $\hat{e}_1^* \cdot \hat{e}_2 = 0$. Without any loss of generality an only for mathematical convenience[1] we're going to consider both sources have got the same polarisation, this means $\hat{e}_1 = \hat{e}_2 \rightarrow \hat{e}_1^* \cdot \hat{e}_1 = 1$. Thus, the total intensity becomes

$$I = I_1 + I_2 + 2\left(I_1 I_2\right)^{1/2}\cos\triangle\phi = \left(I_1 + I_2\right)\left(1 + 2\frac{\left(I_1 I_2\right)^{1/2}}{I_1 + I_2}\cos\triangle\phi\right). \tag{4.10}$$

The equation 4.10 is the our most general expression for the interference. Lets notice that the intereference terms is composed in two parts: the first one is a relation between the two intensities I_1, I_2 and the second one only depends of the phase-difference of the two sources by a cosine, then, **the real interference becomes in the phases not in the intensities**, the first terms only **modulates the visibility** of the interference, and we then can call it $\mathcal{V}$ and write it as

$$\mathcal{V} = 2\frac{\left(I_1 I_2\right)^{1/2}}{I_1 + I_2} = \frac{I_{max} - I_{min}}{I_{max} + I_{min}}, \tag{4.11}$$

$$I_{min} = \left(I_1^{1/2} - I_2^{1/2}\right)^2, \quad I_{max} = \left(I_1^{1/2} + I_2^{1/2}\right)^2, \tag{4.12}$$

this term $\mathcal{V}$ is called the **visibility factor** which module the constrast on the interference fringes.

[1] In fact, the polarisation here is only an contrast modulator, however, when we study the spatial coherence, the polarisation is even more important and conditionates the behavior of the interference.

4.2.1 Interference of plane waves

The expression 4.10 is, as we said, the general interference expression, so lets apply it to two plane wave sources. For the plane wave we've got that the complex amplitudes are given by

$$E_1(\vec{r},t) = E_1 e^{i\left(\vec{r}\cdot\vec{k}+\varphi_1\right)}, \qquad E_2(\vec{r},t) = E_2 e^{i\left(\vec{r}\cdot\vec{k}+\varphi_2\right)}, \tag{4.13}$$

where, we are going to suppose, just for to make the maths a bit easier but the methodology is the same, that both amplitudes are the same $E_1 = E_2$.

Because, we are considering that the both waves have got the same frequency $\omega_1 = omega_2$ this implies that both have the same wave number amplitude, this means $||\vec{k}_1|| = ||\vec{k}_2|| = k$. Then, we will have that the phase-difference term is given by

$$\triangle\phi = \left(\vec{k}_2 - \vec{k}_1\right)\cdot\vec{r} + \varphi_2 - \varphi_1. \tag{4.14}$$

The interference 4.10 having on count that $I_1 = I_2 = I_0$ is then

$$I = (I_0 + I_0)\left(1 + 2\frac{(I_0 I_0)^{1/2}}{I_0 + I_0}\cos\triangle\phi\right) = 2I_0\left[1 + \cos\left(\vec{K}\cdot r + \triangle\varphi\right)\right], \tag{4.15}$$

where $\vec{K} = \vec{k}_2 - \vec{k}_1$ is the interferences vector. To understand bettwer the intereference, we're going to do a bit of geometry (see figure 4.1).

Because we're interested on the interference plane, we will take to $\vec{r} = X\hat{x}$ on the interference plane, where $\hat{x} = \vec{K}/||\vec{K}||$, then the spatial dependce gets the form

$$\vec{K}\cdot\vec{r} = KX = 2\kappa X\sin\left(\frac{\alpha}{2}\right), \tag{4.16}$$

and because K is a wave number vector on the interference plane, it can be written in terms of its wave-lenght,which orresponds to the interfringe distance Λ as

$$K = \frac{2\pi}{\Lambda}, \tag{4.17}$$

then

$$\vec{K}\cdot\vec{r} = \frac{2\pi}{\Lambda}X. \tag{4.18}$$

To calculate the relation between the interfringe distance Λ an the known parameters of the wave, we're going to equate the equations 4.16 and 4.18 therefore

$$\frac{2\pi}{\Lambda}X = \frac{2\pi}{\lambda}X\sin\left(\frac{\alpha}{2}\right), \tag{4.19}$$

$$\Lambda = \frac{\lambda}{2\sin\left(\frac{\alpha}{2}\right)}, \tag{4.20}$$

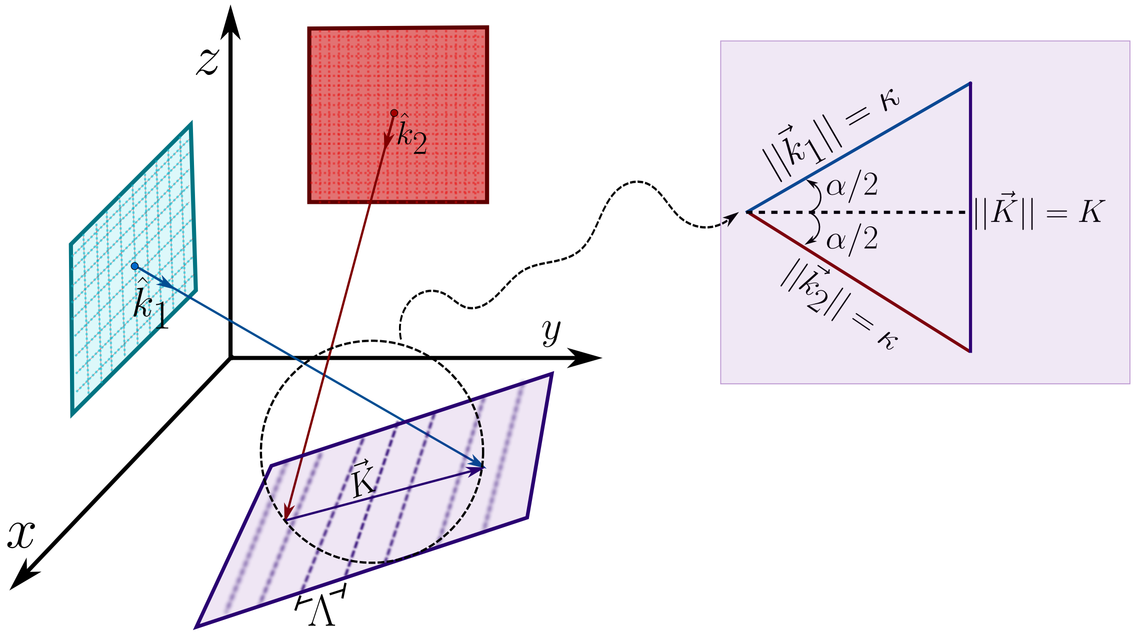


Figure 4.1: Two plane waves with unitary number vector $\hat{k}_1, \hat{k}_2$ propagate and interferece producing a pattern of interference characterized by a vector $\vec{K}$ and an interfringe distance Λ.

where we used that $\kappa = \frac{2\pi}{\lambda}$. With this term, we finally got that the total intensity in the interference of two plane wave is given by

$$I = 2I_0\left[1 + \cos\left(\frac{2\pi}{\Lambda}X + \phi\right)\right]. \tag{4.21}$$

We can see that the interfringe distance Λ behaves in a very logique way. In $\alpha \to 0$ then $\Lambda \to \infty$ this means, when the two wave number vector $\hat{k}_1, \hat{k}_2$ gets each time more parallel, the interfringe distance tends to infinity, this means, there won't be any interference pattern.

In the other way, if $\alpha \to \pi$ then $\Lambda \to min$ this means, when the two wave number vectors tend to be 180° separeted the interfrence distance is the minimum, being clrear for us to see the interference pattern.

4.2.2 Interference of spherical waves

In a similar way to the last section about plane waves, we're going to consider two radiation sources which propagte as spherical waves, so the electric field waves are given by

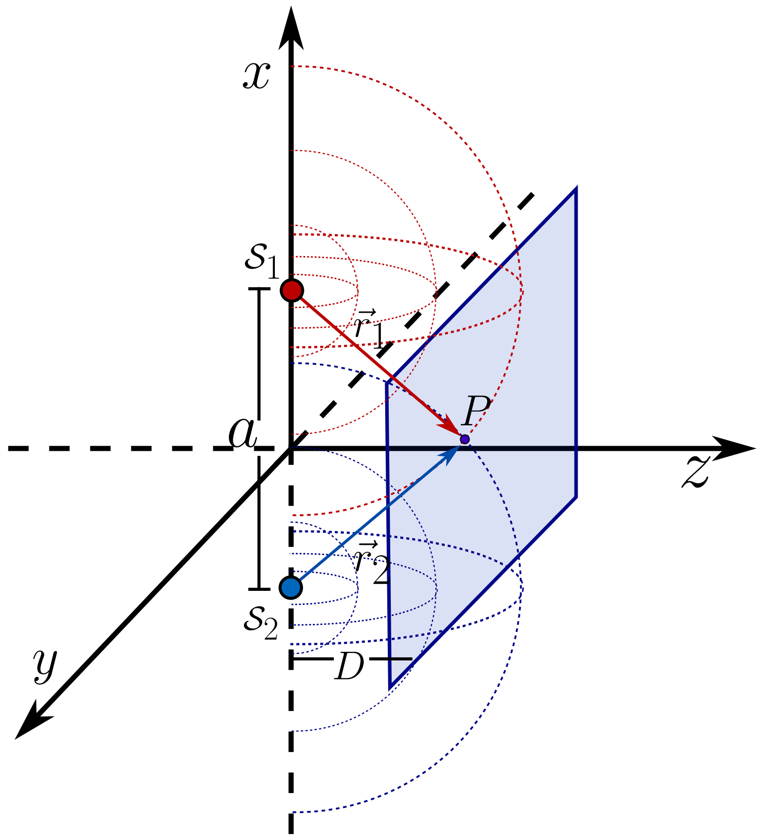


Figure 4.2: Two plane sources of spherical waves interfere in a point P located on the interference plane. The two sources are separated a distance a an the distance from both surface to the interference plane is D.

$$E_1(\vec{r}) = \frac{E_1}{r_1} e^{i(kr_1+\varphi_1)}, \tag{4.22}$$

$$E_2(\vec{r}) = \frac{E_2}{r_2} e^{i(kr_2+\varphi_2)}. \tag{4.23}$$

And, again, we're going to consider, without any loss of generality just for aesthetical purposes, that both have got the sample amplitude $E_1 = E_2 = E_0$. Also, lets suppose a point P on the interference plane and that the two point sources are separated a distance a $[m]$ and both sources are separated a dsitance D $[m]$ as is shown in the figure 4.2.

We have got that $r_1 = \overline{\mathcal{S}_1 P}$ and $r_2 = \overline{\mathcal{S}_2 P}$, then the fase difference is given by

$$\triangle\phi(r) = \kappa\,(r_2 - r_1) + \triangle\varphi. \tag{4.24}$$

And we are going to consider that the field amplitudes, in a neighborhood of P change very little with r_1 and r_2, and then

$$\frac{E_0}{r_1} \approx \frac{E_0}{r_2} = E, \tag{4.25}$$

the intensity is then given by

$$I = 2I_0 \left\{1 + \cos\left[\kappa\left(r_2 - r_1\right) + \triangle\varphi\right]\right\}, \tag{4.26}$$

before going further, see that the **surfaces of equal intensity $\mathbf{r_2} - \mathbf{r_1} = cte$ are hyperboloids with focii in $\mathcal{S}_1, \mathcal{S}_2$.**

In a plane parallel to $\overline{\mathcal{S}_1\mathcal{S}_2}$ at a distance $z = D$ two hyperbolids are produced. For $r_1 \approx r_2$ and $D \gg a$ this hiperboloids are practically planes. Therefore

$$r_1 = \left[D^2 + \left(x - \frac{a}{2}\right)^2 + y^2\right]^{1/2} = D\left[1 + \frac{x^2}{D^2} + \frac{y^2}{D^2} + \frac{a^2}{4D^2} - \frac{ax}{D^2}\right]^{1/2}, \tag{4.27}$$

applying taylor series to first order of $\sqrt{1+x} \approx 1 + x/2$ we've got

$$r_1 \approx D\left[1 + \frac{x^2 + y^2}{2D^2} + \frac{a^2}{2D^2} - \frac{ax}{2D^2}\right], \tag{4.28}$$

and in the same way for r_2

$$r_2 \approx D\left[1 + \frac{x^2 + y^2}{2D^2} + \frac{a^2}{2D^2} + \frac{ax}{2D^2}\right]. \tag{4.29}$$

Substracting 4.29 from 4.27 we then we will have that the phase difference is then

$$\triangle\phi = \kappa\left(r_2 - r_1\right) + \triangle\varphi \approx \frac{\kappa a X}{D} + \triangle\varphi, \tag{4.30}$$

then the total intensity in the interference is given by

$$I = 2I_0\left[1 + \cos\left(\kappa\frac{aX}{D} + \varphi_2 - \varphi_1\right)\right], \tag{4.31}$$

where again

$$KX = \frac{\kappa a X}{D} = \frac{2\pi}{\lambda}\frac{aX}{D},$$

but $K = 2\pi/\Lambda$, being again K the magnitude of the resultante wave number vector on the plane interference, and Λ the interfringe distance. Thus the relation between Λ an the known paramaters of both waves is

$$\frac{2\pi}{\Lambda} = \frac{2\pi a X}{D} \longrightarrow \Lambda = \frac{\lambda D}{a}.$$

With this result, we can see that when the two sources get closer $a \to 0$ the interfringe $\Lambda \to \infty$ this means, the interference is going to disappear when the two point sources get together.

For example, lets consider two monocrhomatic plane waves, which have got complex amplitudes

$$\vec{E}_1 = \mathcal{E} e^{i\varphi_1} \hat{e}_1, \qquad \vec{E}_2 = \mathcal{E} e^{i\varphi_2} \hat{e}_2, \tag{4.32}$$

where $\hat{e}_1, \hat{e}_2$ are thei polarisation vectors and their wave number vectors are given by

$$\vec{k}_1 = [k_{1x}, k_{1y}, k_{1z}]^T \qquad \vec{k}_2 = [k_{2x}, k_{2y}, k_{2z}]^T . \tag{4.33}$$

Now, lets answer the following questions:

1. What relation should be between $\vec{k}_1, \vec{k}_2$ to realize the interference?
2. What's the expression for the visibility?
3. What's the plane where the interference fringes have the least distance? What is the orientation of the plane and the interference fringes distances on that plane?

Lets answer each part following the methodology we have already built. The observable quantity is never the instantaneous field but the time average of the irradiance, which for our two waves reads

$$\left\langle \left|\left|\vec{E}\right|\right|^2 \right\rangle = I_1 + I_2 + 2\,\mathrm{Re}\left\{ \vec{E}_1^* \cdot \vec{E}_2 \left\langle e^{-i(\omega_1 - \omega_2)t} \right\rangle \right\} . \tag{4.34}$$

The average $\left\langle e^{-i(\omega_1-\omega_2)t} \right\rangle$ vanishes for any detector whose integration time T satisfies $|\omega_1 - \omega_2| T \gg 1$, so the interference term survives only when $\omega_1 = \omega_2 = \omega$. This first condition is **temporal, not geometric**, and the statement grants it by declaring both waves monochromatic and of the same nature. Its immediate translation to the space of wave number vectors, in a homogeneous and isotropic medium of index n, is

$$\left|\left|\vec{k}_1\right|\right| = \left|\left|\vec{k}_2\right|\right| = \kappa = \frac{n\omega}{c} = \frac{2\pi}{\lambda}. \tag{4.35}$$

This equality of the magnitudes is the only relation that the interference truly requires, and it does not constrain the directions at all. It is worth saying this out loud, because the wording of the question tempts us to answer that the vectors must be parallel, and it is exactly the opposite. If $\vec{k}_1 = \vec{k}_2$ the interference term does exist but no longer depends on the position, the illumination becomes uniform and there are no fringes. For an observable pattern we need the interference vector

$$\vec{K} \equiv \vec{k}_2 - \vec{k}_1 \neq \vec{0}, \tag{4.36}$$

whose magnitude follows at once from the equality of the moduli 4.35,

$$\left|\left|\vec{K}\right|\right|^2 = 2\kappa^2 (1 - \cos\alpha) = 4\kappa^2 \sin^2 \frac{\alpha}{2} \quad \Longrightarrow \quad K = 2\kappa \sin\frac{\alpha}{2}, \tag{4.37}$$

with α the angle between $\vec{k}_1$ and $\vec{k}_2$. A third condition, this time genuinely geometric, comes from the interference term carrying the factor $\hat{e}_1^* \cdot \hat{e}_2$, which vanishes for orthogonal polarisations. If we want the maximum contrast we need $\hat{e}_1 = \hat{e}_2 = \hat{e}$, and since both waves are transverse, so that $\hat{e} \perp \vec{k}_1$ and $\hat{e} \perp \vec{k}_2$, this common direction is fixed up to a sign,

$$\hat{e} = \frac{\vec{k}_1 \times \vec{k}_2}{\left\|\vec{k}_1 \times \vec{k}_2\right\|}. \tag{4.38}$$

In words, two non collinear plane waves reach unit visibility only when they are polarised perpendicular to the plane that contains $\vec{k}_1$ and $\vec{k}_2$, which is the quantitative form of the Arago and Fresnel statement we met before.

For the second part, with $\omega_1 = \omega_2$ the interference term survives the average and the total irradiance is

$$I(\vec{r}) = I_1 + I_2 + 2\sqrt{I_1 I_2}\, |\hat{e}_1^* \cdot \hat{e}_2| \cos\left[\triangle\phi(\vec{r}) + \delta\right], \qquad \triangle\phi(\vec{r}) = \vec{K} \cdot \vec{r} + \varphi_2 - \varphi_1, \tag{4.39}$$

where $I_j = |\mathcal{E}_j|^2$ and $\delta = \arg(\hat{e}_1^* \cdot \hat{e}_2)$. As the cosine sweeps the whole interval $[-1, 1]$ when $\vec{r}$ runs over the space, the extremes of the irradiance are read directly,

$$I_{\max} = I_1 + I_2 + 2\sqrt{I_1 I_2}\, |\hat{e}_1^* \cdot \hat{e}_2|, \qquad I_{\min} = I_1 + I_2 - 2\sqrt{I_1 I_2}\, |\hat{e}_1^* \cdot \hat{e}_2|, \tag{4.40}$$

and the Michelson visibility, the same one we use as the contrast modulator, follows without any approximation

$$\mathcal{V} = \frac{I_{\max} - I_{\min}}{I_{\max} + I_{\min}} = \frac{2\sqrt{I_1 I_2}}{I_1 + I_2}\, |\hat{e}_1^* \cdot \hat{e}_2|. \tag{4.41}$$

With it the irradiance takes the canonical form

$$I(\vec{r}) = (I_1 + I_2)\left[1 + \mathcal{V} \cos\triangle\phi(\vec{r})\right], \tag{4.42}$$

which cleanly separates the three roles, since $I_1 + I_2$ fixes the mean level, $\mathcal{V}$ fixes the contrast and $\triangle\phi$ carries all the physics of the interference. Notice that $\mathcal{V} = 1$ demands both equal intensities and parallel polarisations, but also that $\mathcal{V}$ is remarkably robust against an imbalance of the intensities, since a ratio as large as $I_1/I_2 = 2$ still leaves $\mathcal{V} = 0.943$. We will make use of this insensitivity in the next example.

For the third part, the surfaces of constant phase, and therefore the fringes seen as objects living in the three dimensional space, are the planes $\vec{K} \cdot \vec{r} = \text{cte}$, perpendicular to $\vec{K}$ and separated by

$$\Lambda_0 = \frac{2\pi}{K} = \frac{\lambda}{2\sin(\alpha/2)}. \tag{4.43}$$

What the question asks is on which observation plane P the cut of this family produces the closest fringes. Let $\hat{n}$ be the normal of P and decompose $\vec{K}$ into a part along $\hat{n}$ and a part $\vec{K}_\parallel$ lying inside the plane. Only $\vec{K}_\parallel$ modulates the field over P, so the observed fringes are straight lines perpendicular to $\vec{K}_\parallel$ with a separation

$$\Lambda(\hat{n}) = \frac{2\pi}{\left|\left|\vec{K}_\parallel\right|\right|} = \frac{\Lambda_0}{\sin\gamma}, \qquad \gamma = \angle(\vec{K}, \hat{n}). \tag{4.44}$$

Since $\Lambda(\hat{n}) \geq \Lambda_0$ for every orientation, with equality only when $\sin\gamma = 1$, the fringes are closest exactly when the observation plane contains the direction of $\vec{K}$,

$$P: \quad \hat{n}\cdot\vec{K} = 0, \qquad \Lambda_{\min} = \frac{2\pi}{\left|\left|\vec{k}_2 - \vec{k}_1\right|\right|} = \frac{\lambda}{2\sin(\alpha/2)}, \tag{4.45}$$

while any other tilt dilates the fringes by the factor $1/\sin\gamma$. This is a whole one parameter family of planes rather than a single one, and two of its members deserve a name. The first is the plane spanned by $\vec{k}_1$ and $\vec{k}_2$, whose normal is $\hat{n} = (\vec{k}_1 \times \vec{k}_2)/||\vec{k}_1 \times \vec{k}_2||$, and it contains $\vec{K}$ because $\vec{K}$ is a linear combination of $\vec{k}_1$ and $\vec{k}_2$, being moreover the same plane to which the polarisation had to be perpendicular in the first part. The second is the plane perpendicular to the bisector of the two beams, which is the one used as a screen in the laboratory, and it belongs to the optimal family thanks to the identity

$$\vec{K}\cdot(\vec{k}_1 + \vec{k}_2) = \left|\left|\vec{k}_2\right|\right|^2 - \left|\left|\vec{k}_1\right|\right|^2 = 0. \tag{4.46}$$

So the interference vector is always perpendicular to the bisector, the natural screen is already the optimal one and there is no need to tilt it. On it the fringes are straight lines parallel to $\vec{k}_1 + \vec{k}_2$ with the separation $\lambda/[2\sin(\alpha/2)]$.

4.3 Interferometers

4.3.1 Young interferometer

For most of the eighteenth century the nature of light seemed a settled matter. The authority of Newton, whose *Opticks* Newton (1704) accounted for reflection, refraction and the colours of thin films in terms of tiny corpuscles, had pushed into the background the wave picture that Huygens had proposed in his *Traité de la Lumière* Huygens (1690). A corpuscle travels in a straight line and carries momentum, so the rectilinear propagation of light and the sharpness of shadows looked like immediate consequences of the theory, and the enormous prestige of Newton did the rest. The wave idea survived only as a minority opinion, lacking the one thing that would have made it irresistible, a phenomenon that particles simply cannot imitate.

That phenomenon was interference, and it was Thomas Young who brought it to light. In his Bakerian lecture of 1801, published as *On the Theory of Light and Colours* Young (1802), Young stated what he called the principle of the superposition of undulations, the plain but revolutionary idea that where two portions of light meet their disturbances add, so that two bright beams can combine to give darkness. A few years later, in a second Bakerian lecture Young (1804), he turned the principle into an experiment, letting light pass through a small opening and then splitting the emerging beam so that its two halves overlapped on a screen, where they painted a regular pattern of bright and dark bands. In his *Course of Lectures on Natural Philosophy* Young (1807) the arrangement acquired the form we still teach, two closely spaced apertures illuminated by a common source, and from the spacing of the fringes Young was able to do something no one had done before, to measure the wavelength of light and to find it astonishingly small, a fraction of a thousandth of a millimetre.

The importance of this result is difficult to overstate. The dark bands were the direct proof that light added to light can produce less light, something no theory of independent particles can explain, and they gave the wave theory the empirical victory it had lacked for a century. The reception, however, was hostile, since to attack Newton in the England of 1804 was close to heresy, and Young's papers were ridiculed in the reviews of the day. Vindication came from the continent a decade later, when Fresnel and Arago placed the wave theory on a rigorous mathematical footing and extended it to diffraction and polarisation. The experiment itself never left the stage. It is the archetype of every wavefront division interferometer, the simplest device in which the phase difference between two coherent beams is turned into a visible intensity pattern, and its complete modern treatment belongs to any serious optics text Born and Wolf (1999). More than that, when the same experiment is repeated with electrons, atoms or single photons sent one at a time, the fringes still build up, and the two slit arrangement becomes the cleanest illustration of quantum superposition, the phenomenon that Feynman called the only mystery of quantum mechanics Feynman et al. (1965). The precision descendants of Young's screen reach today the kilometre sized interferometers that detect gravitational waves Abbott et al. (2016).

To see the whole machinery of the previous section at work on this instrument, lets analyse the Young interferometer quantitatively, with its two slits illuminated by a spherical wave coming from a point source, arranged as in the figure 4.3. We place the two slits $\mathcal{S}_1$ and $\mathcal{S}_2$ symmetrically at $x = \pm d/2$ on the plane $z = 0$, separated by a distance d, and the observation screen at $z = D$. The point source $\mathcal{S}$ sits at $(x, z) = (-a, -b)$, so that b is its distance to the slit plane and a its transverse displacement from the symmetry axis, with $a = 0$ meaning the source on the axis, and we measure the coordinate x on the screen from that axis.

The methodology of the chapter applies in two chained stages, and this is really the only structural idea of the problem. The first stage goes from the source to the slits and decides two things, namely with what amplitude and with what phase the field reaches each slit. The second stage goes from the slits to the screen and is always the interference of two secondary spherical waves, exactly the one we treated in the spherical wave section, with the slits acting as the sources $\mathcal{S}_1, \mathcal{S}_2$ separated by d and the screen at a distance D. Writing the field that leaves the slit j as $\mathcal{E}_j e^{i\psi_j}$ and the path from that slit to the point P on the screen as s_j, the total relative phase at P is

$$\triangle\phi(x) = \kappa(s_2 - s_1) + (\psi_2 - \psi_1), \tag{4.47}$$

and the irradiance obeys the canonical expression 4.42 with the visibility 4.41 evaluated on the amplitudes that reach each slit.

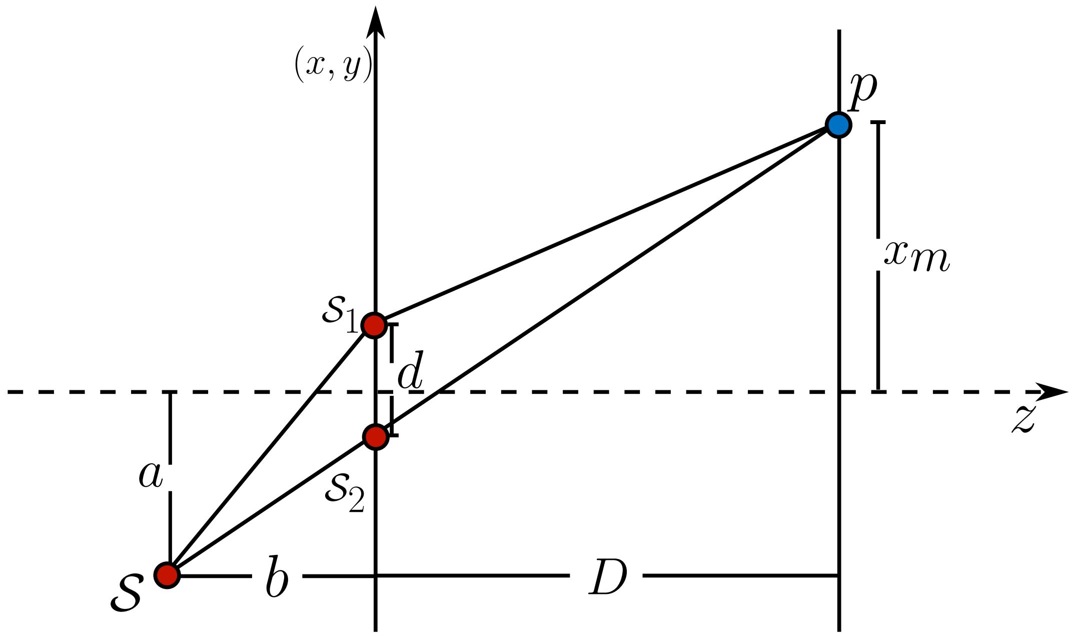


Figure 4.3: Young interferometer illuminated by a spherical wave from a point source $\mathcal{S}$ placed off axis. The source to slit stage sets the amplitude and the phase reaching each slit $\mathcal{S}_1, \mathcal{S}_2$, and the slit to screen stage is the interference of two secondary spherical waves.

The second stage is common to any illumination, so we dispatch it once. With $s_j = [D^2 + (x \mp d/2)^2]^{1/2}$ and the same Taylor expansion of the chapter,

$$s_2 - s_1 \approx \frac{(x + d/2)^2 - (x - d/2)^2}{2D} = \frac{xd}{D}, \tag{4.48}$$

valid in the paraxial regime $D \gg d, |x|$, from which

$$\kappa(s_2 - s_1) \approx \frac{2\pi}{\lambda}\frac{xd}{D} = \frac{2\pi}{\Lambda}x, \qquad \Lambda = \frac{\lambda D}{d}. \tag{4.49}$$

This is precisely the interfringe of the spherical wave section with $a \to d$. Everything that distinguishes one illumination from another sits, then, in the term $\psi_2 - \psi_1$.

For the spherical illumination of this example the source at $(-a, -b)$ emits the wave $E = (E_0/r)e^{i\kappa r}$ of the chapter, and now the two slits are no longer equivalent, because their distances to the source

$$r_1 = \left[b^2 + \left(a + \frac{d}{2}\right)^2\right]^{1/2}, \qquad r_2 = \left[b^2 + \left(a - \frac{d}{2}\right)^2\right]^{1/2} \tag{4.50}$$

differ as soon as $a \neq 0$, and this affects both the phase and the amplitude. In phase, expanding as in the chapter with $b \gg d, a$,

$$r_1 - r_2 \approx \frac{(a + d/2)^2 - (a - d/2)^2}{2b} = \frac{ad}{b} \quad \Longrightarrow \quad \psi_2 - \psi_1 = \kappa(r_2 - r_1) = -\frac{\kappa a d}{b}. \tag{4.51}$$

In amplitude, the slit j receives E_0/r_j, so $I_j \propto 1/r_j^2$ and the visibility 4.41 becomes

$$\mathcal{V} = \frac{2\sqrt{I_1 I_2}}{I_1 + I_2} = \frac{2/(r_1 r_2)}{1/r_1^2 + 1/r_2^2} = \frac{2r_1 r_2}{r_1^2 + r_2^2}. \tag{4.52}$$

Putting the two stages together in 4.47, the total phase is

$$\triangle\phi(x) = \frac{\kappa d}{D}x - \frac{\kappa a d}{b} = \frac{2\pi d}{\lambda}\left(\frac{x}{D} - \frac{a}{b}\right), \tag{4.53}$$

and the irradiance on the screen reads

$$I(x) = (I_1 + I_2)\left\{1 + \mathcal{V}\cos\left[\frac{2\pi d}{\lambda}\left(\frac{x}{D} - \frac{a}{b}\right)\right]\right\}, \qquad \mathcal{V} = \frac{2r_1 r_2}{r_1^2 + r_2^2}. \tag{4.54}$$

The qualitative difference with a plane illumination, for which $\mathcal{V} = 1$ always, is not in the position of the fringes but in the contrast, because an off axis spherical source gives $\mathcal{V} < 1$.

The first part asks for the value of a that reduces the contrast by 10% with respect to $a = 0$. At $a = 0$ we have $r_1 = r_2$ and $\mathcal{V} = 1$, and we ask for $\mathcal{V}(a) = 0.9$. Writing $t = r_1/r_2 \geq 1$, the condition $2t/(1+t^2) = 9/10$ is the quadratic $9t^2 - 20t + 9 = 0$, whose larger root is

$$t = \frac{10 + \sqrt{19}}{9} = 1.5954 \quad \Longrightarrow \quad \frac{r_1^2}{r_2^2} = 2.5454. \tag{4.55}$$

A contrast drop of only 10% already requires one slit to be some 60% farther from the source than the other, which shows how insensitive the visibility is to an amplitude imbalance, exactly as we anticipated in the second part of the first example. Substituting the distances 4.50,

$$\frac{b^2 + (a + d/2)^2}{b^2 + (a - d/2)^2} = t^2 \quad \Longleftrightarrow \quad a^2 - \frac{10}{\sqrt{19}}\, d\, a + b^2 + \frac{d^2}{4} = 0, \tag{4.56}$$

a quadratic in a whose formal solution is

$$a = \frac{5d}{\sqrt{19}} \pm \sqrt{\frac{81 d^2}{76} - b^2}. \tag{4.57}$$

Here the physics jumps out of the discriminant, since real solutions exist only if $b \leq \frac{9d}{2\sqrt{19}} = 1.0324\, d$. In a real Young interferometer one always has $b \gg d$, typically $d \sim 10^{-4}$ m and $b \sim 10^{-1}$ m, so the discriminant is negative and no displacement of a point source can degrade the contrast by 10%. The paraxial estimate confirms it, since with $t = 1 + \eta$ and $\eta \approx (r_1 - r_2)/b \approx ad/b^2$ the visibility is $\mathcal{V} \approx 1 - \eta^2/2 = 1 - \frac{1}{2}(ad/b^2)^2$, so $\mathcal{V} = 0.9$ would need $ad/b^2 = \sqrt{0.2} \approx 0.447$, that is $a \approx 0.45\, b^2/d$, a gigantic displacement that destroys the paraxial approximation long before it touches the contrast. The physical conclusion is that a strictly point source always produces fringes of essentially unit visibility, wherever it is placed, because moving it translates the pattern but does not erase it. The version of the question that does have a solution is the one where a is the width of an incoherent thermal source instead of the displacement of a point source, which belongs to the study of spatial coherence and gives a visibility $\mathcal{V}(a) = |\text{sinc}(\pi ad/\lambda b)|$ that falls to its first zero at $a = \lambda b/d$. The lesson to keep is the contrast between the two readings, since displacing a point source acts on the phase and slides the fringes, while broadening an incoherent source acts on the coherence and washes them out.

The second part asks for the position x_m of the m-th maximum. The maxima of the irradiance 4.54 are the zeros modulo 2π of the phase 4.53,

$$\frac{2\pi d}{\lambda}\left(\frac{x_m}{D} - \frac{a}{b}\right) = 2\pi m, \qquad m \in \mathbb{Z}, \tag{4.58}$$

from where

$$x_m = m\frac{\lambda D}{d} + \frac{aD}{b} = m\Lambda + \frac{aD}{b}. \tag{4.59}$$

The pattern keeps its full periodicity $\Lambda = \lambda D/d$ and is shifted as a rigid block by aD/b toward the side opposite to the displacement of the source, consistent with the figure, where the source lies below the axis while x_m is measured above. Geometrically, the zero order maximum sits where the total path difference vanishes, that is on the straight line that joins the source with the midpoint of the slits and continues to the screen, a line that leaves $(-a, -b)$, passes through the origin and reaches the screen at $x = aD/b$.

The third part asks for a condition between a and b so that the origin is a minimum. A minimum at $x = 0$ requires the total phase there to be an odd multiple of π. Evaluating 4.53 at $x = 0$,

$$\triangle\phi(0) = -\frac{2\pi d}{\lambda}\frac{a}{b} = -(2m+1)\pi, \qquad m = 0, 1, 2, \ldots \tag{4.60}$$

which is the same as demanding that the path difference introduced by the source to slits stage be an odd number of half wavelengths, $ad/b = (2m+1)\lambda/2$, that is

$$\frac{a}{b} = \frac{(2m+1)\lambda}{2d}, \qquad m = 0, 1, 2, \ldots \tag{4.61}$$

The condition depends on a and b only through their ratio, as it must, since the only thing a point source communicates to the system is the direction from which it illuminates it. The smallest case $m = 0$ gives $a/b = \lambda/2d$, which for typical numbers, $\lambda = 633\,\text{nm}$ and $d = 0.2\,\text{mm}$, is $a/b \approx 1.6 \times 10^{-3}$, so a displacement of about 1.6 mm for every metre of b is already enough to turn the central maximum into a minimum.

4.3.2 Fresnel biprism

The double mirror that Fresnel used to prove the wave nature of light Fresnel (1819) has one practical drawback, namely that two separate flat mirrors have to be tilted and held at an almost invisible angle between them, which makes the alignment tedious and unstable. Fresnel's own answer to this difficulty was to replace the two mirrors by a single piece of glass ground into two thin prisms joined base to base, with the shared edge running along the optical axis. This is the *Fresnel biprism*, and it produces exactly the same effect, two coherent virtual images of a single point source, but out of one rigid element instead of two separately adjustable ones.

Before setting up the biprism itself, it is worth pausing on the principle that makes this whole family of instruments work, since it is the same principle that will let us handle every wavefront division interferometer to come. When a wave leaves a real point source and passes through an optical element, be it a lens, a mirror or a prism, its wavefront is in general reshaped into something far too complicated to describe by hand. There is, however, a special situation in which nothing of interest is lost, namely when the element is *stigmatic* for the pair of points involved, in the sense already introduced in the previous chapter (see figure 2.12), a spherical wavefront leaving the object point is transformed into another spherical wavefront, indistinguishable in its geometry from the one that a genuine point source sitting at the image point would radiate. Whenever this holds, Fermat's principle guarantees that every ray reaching the image has travelled the same optical path, so the phase downstream of the element is exactly the phase of a fictitious spherical wave centred on that image, whether the image is real or virtual. This is the *method of images*: rather than propagating the true, complicated wavefront through the optical element, we replace the element by the point sources it images and let the machinery of two point source interference, already built in the previous section, take care of everything past that plane.

This single observation is what turns an ordinary mirror or prism into an interferometer. If one primary source is imaged twice, once through each of two branches of the same optical element, the observer downstream no longer sees one wave but two, radiated by two virtual sources $\mathcal{S}_1$ and $\mathcal{S}_2$. These two sources are coherent by construction rather than by assumption, since both wavefronts are literally two halves of the same original wave, carrying the same frequency and a phase relationship fixed once and for all by the geometry of the element; the condition $\omega_1 = \omega_2$ that opened this chapter is here automatically satisfied and never has to be imposed by hand, unlike the case of two genuinely independent sources. Once the positions of $\mathcal{S}_1$ and $\mathcal{S}_2$ are known, the problem stops being about mirrors or prisms altogether and becomes, exactly, the interference of two spherical waves solved earlier in this chapter, with the separation a and the source to screen distance D read directly off the geometry of the particular device. In this sense every wavefront division interferometer, the double mirror, the biprism, and the others we will meet, differs from the next only in how it manufactures its pair of virtual sources; the fringes that follow are always the same fringes.

The method is only as trustworthy as the stigmatism of the imaging step, and that has to be checked separately for every device, since it is never automatic. A plane mirror is rigorously and unconditionally stigmatic for every object point whatsoever, as we saw when it appeared as the degenerate conic $z_o = -z_i$ of the reflective surfaces 2.76, which is why Fresnel's original double mirror needs no further justification beyond the law of reflection. A prism is a different matter, because among refracting interfaces only the very special Cartesian ovoids are rigorously stigmatic, and an ordinary prism face is not one of them; it can only be stigmatic approximately, for a narrow paraxial bundle of rays clustered near a fixed angle of incidence. Showing that this approximate stigmatism holds well enough for a thin prism, and pinning down exactly how well, is therefore the first task in developing the biprism, and it is where we begin.

Let a point source $\mathcal{S}$ sit at a distance z_1 from the biprism, and let the screen of observation stand a further distance z_2 beyond it, so that $D = z_1 + z_2$ is the total distance from the source to the screen (see figure 4.4). Each half of the biprism is a thin prism of apex angle α and refractive index n, and the whole instrument is symmetric about the optical axis. The method of images consists of replacing the biprism entirely by the two virtual sources it creates, which turns the problem back into the two point source interference we already solved.

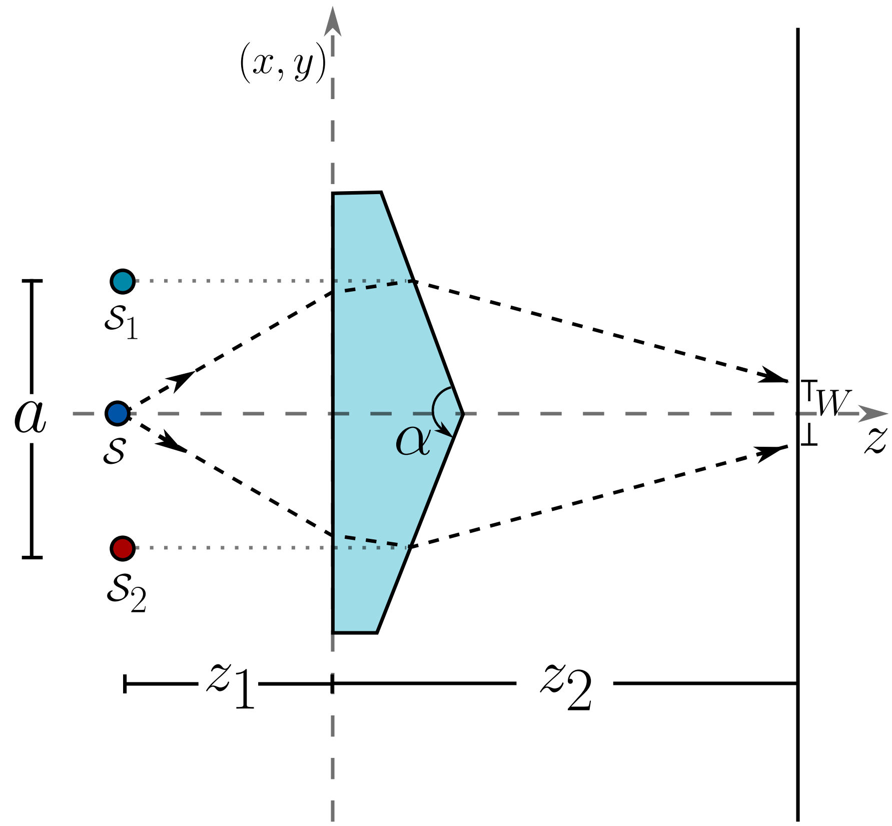


Figure 4.4: Method of images applied to the Fresnel biprism. Each half of the biprism produces a virtual point image of the source, $\mathcal{S}_1$ from the upper half and $\mathcal{S}_2$ from the lower half, separated by a and located in the same transverse plane as $\mathcal{S}$. Beyond the biprism the two beams overlap only inside the shaded strip of width W on the screen.

A thin prism deviates a ray by an angle that, to first order in the angle of incidence and near the condition of minimum deviation, does not depend on exactly where the ray crosses the prism nor, to that same order, on its precise angle of incidence,

$$\delta = (n - 1)\alpha. \tag{4.62}$$

This independence is the whole reason the method of images works here, because if the deviation changed from ray to ray the upper half of the biprism would not produce a point image but an aberrated smear. It is worth checking how good 4.62 really is, since later we will be pushed toward angles that are not so small. Comparing it with the exact deviation at symmetric incidence, $\delta_{\text{exact}} = 2\arcsin[n\sin(\alpha/2)] - \alpha$, for $n = 1.5$ the relative error is only 0.04% at $\alpha = 0.05\,\text{rad}$, 0.16% at $\alpha = 0.10\,\text{rad}$, 0.35% at $\alpha = 0.15\,\text{rad}$ and 0.63% at $\alpha = 0.20\,\text{rad}$, so the thin prism approximation survives well into the range we will need, though it stops being free.

Consider a ray leaving $\mathcal{S}$ with paraxial slope m and crossing the upper half of the biprism. It strikes the prism plane at a height $y_p = z_1 m$ and leaves with the new slope $m - \delta$, deviated toward the axis. Extending this outgoing ray backward by the same distance z_1 to the plane of the source, its apparent point of origin is at a height

$$y_v = y_p - z_1(m - \delta) = z_1 m - z_1 m + z_1\delta = z_1\delta, \tag{4.63}$$

which does not depend on m. Every ray that the upper half deviates, whatever its original slope, appears to come from the same point, so the upper half of the biprism produces a rigorously stigmatic virtual image $\mathcal{S}_1$ at a height $z_1\delta$ above the axis, in the very same transverse plane as the source. By the mirror symmetry of the instrument the lower half produces $\mathcal{S}_2$ at $-z_1\delta$. The biprism has disappeared from the problem, replaced by two mutually coherent point sources separated by

$$a = 2z_1\delta = 2z_1(n-1)\alpha, \tag{4.64}$$

sitting at a distance $D = z_1 + z_2$ from the screen. From here on the problem is precisely the interference of two spherical waves that we already solved, so the irradiance on the screen, with y the transverse coordinate, takes the canonical form

$$I(y) = (I_1 + I_2)\left[1 + \mathcal{V}\cos\left(\kappa\frac{a}{D}y\right)\right], \tag{4.65}$$

with a fringe spacing given directly by the two point source result

$$\Lambda = \frac{\lambda D}{a} = \frac{\lambda(z_1 + z_2)}{2z_1(n-1)\alpha}. \tag{4.66}$$

It helps to read 4.66 in the form that makes the underlying physics transparent, writing $\Lambda = \lambda/\beta$ with

$$\beta = \frac{a}{D} = 2(n-1)\alpha\,\frac{z_1}{z_1 + z_2}, \tag{4.67}$$

the local angle at which the two beams cross on the screen. This is exactly the same quantity that appeared in the interference of two plane waves, where $\Lambda = \lambda/[2\sin(\beta/2)]$, and the two expressions differ by only 0.12% at $\beta = 0.1\,\text{rad}$, so identifying $\Lambda \approx \lambda/\beta$ here is entirely legitimate. What 4.67 makes plain is that

$$\beta \leq 2(n-1)\alpha, \tag{4.68}$$

with equality only in the limits $z_2 \to 0$, the screen pressed against the biprism, or $z_1 \to \infty$, a collimated beam. The crossing angle can never exceed the full deviation the biprism is able to imprint on a beam, and the divergence of the spherical wave leaving a finite z_1 can only make it smaller. A biprism illuminated by a point source is, in this sense, always a weaker interferometer than the same biprism illuminated by a plane wave.

As a concrete example, take a biprism of index $n = 1.5$ illuminated by light of wavelength $\lambda = 632\,\text{nm}$, and ask for the geometry that produces a fringe spacing $\Lambda = 10\lambda$. Setting $\Lambda = 10\lambda$ in 4.66 means $\beta = 1/10$, and since $n = 1.5$ gives $2(n-1) = 1$ exactly, the condition collapses to something remarkably clean,

$$\alpha\,\frac{z_1}{z_1+z_2} = \frac{1}{10}\,\text{rad} \quad\Longleftrightarrow\quad \alpha = \frac{1}{10}\left(1+\frac{z_2}{z_1}\right)\text{rad}. \tag{4.69}$$

This is not a single value of α but a whole one parameter family, since fixing the ratio z_2/z_1 fixes α and vice versa, and every member of the family shares the same image separation $a = D/10$, a tenth of the total source to screen distance, which is a strikingly large separation for two virtual images meant to interfere. There is a bound hiding in 4.69 that is easy to miss, because $z_2/z_1 \geq 0$ forces

$$\alpha \geq \frac{1}{10}\,\text{rad} = 5.73^\circ, \tag{4.70}$$

with the minimum reached only in the collimated limit. A prism angle of nearly six degrees is not what one usually pictures under the instruction to keep α small, and this is worth stating openly rather than hiding it, because the two requirements pull in opposite directions. A fringe spacing of $10\lambda = 6.32\,\mu\text{m}$ is an extraordinarily fine spacing to demand, comparable to the period of a diffraction grating, and producing it requires the two beams to cross at $\beta = 0.1\,\text{rad}$, a full tenth of a radian, while a biprism can only ever supply a crossing angle of $2(n-1)\alpha$. The solution exists and is entirely consistent, but it sits at the edge of the thin prism regime rather than deep inside it, and the error check performed above shows that the approximation 4.62 still holds at $\alpha = 0.1\,\text{rad}$, but by a margin of only 0.16% rather than by a comfortable one. It is also worth noticing that λ never appears in the condition on α alone, which makes sense once we remember that Λ was specified in units of λ and the pure geometry of the biprism carries no memory of colour.

A condition on α is not yet the whole answer, because it says nothing about whether a useful pattern is actually observable. The two beams emerging from the biprism only overlap inside a finite strip of the screen, since the ray that grazes the shared edge from above is deviated downward by δ and reaches the screen at a height $-z_2\delta$, while the ray grazing the edge from below is deviated upward and reaches $+z_2\delta$. Outside this interval only one of the two beams arrives and there is no interference at all, so the field of overlap has a width

$$W = 2z_2\delta = 2z_2(n-1)\alpha, \tag{4.71}$$

and the number of fringes that fit inside it follows at once,

$$N = \frac{W}{\Lambda} = \frac{z_2(n-1)\alpha}{5\lambda}. \tag{4.72}$$

Evaluated on the design condition 4.69, with $2(n-1) = 1$ for $n = 1.5$, this becomes $N = z_2(z_1+z_2)/(100\,z_1\lambda)$, and it is instructive to see it on real numbers.

z_1	z_2	α	a	N
1 m	1 cm	$0.101\,\text{rad} = 5.79^\circ$	10.1 cm	160
0.5 m	5 mm	$0.101\,\text{rad} = 5.79^\circ$	5.05 cm	80
∞ (collimated)	1 cm	$0.100\,\text{rad} = 5.73^\circ$	–	158

With the screen a centimetre from the biprism and the source a metre away, the design delivers 160 fringes of $6.32\,\mu$m inside a field of overlap only about a millimetre wide, so resolving them in practice needs a microscope objective or a detector with sub micron pixels, a condition that deserves to be stated explicitly rather than left implicit in the numbers. The visibility over this whole field stays extremely close to unity, since on the axis of symmetry the two virtual sources are exactly equidistant from the observation point and $\mathcal{V} = 1$ exactly, and even away from the axis the two distances remain equal up to second order corrections, so the same robustness of the contrast against small imbalances that we established for the Young interferometer is at work here, only now it holds by the exact symmetry of the construction rather than by an accident of the numbers.

It remains to see what happens when the point source is displaced off the optical axis, since a real source is never perfectly centred. Repeat the same construction with the source at $\mathcal{S} = (-z_1, s)$, a small transverse offset s away from the axis. A ray leaving $\mathcal{S}$ with slope m strikes the upper half of the biprism at height $y_p = s + z_1 m$ and leaves with slope $m - \delta$, so its virtual origin is now at

$$y_v = s + z_1 m - z_1(m - \delta) = s + z_1\delta, \tag{4.73}$$

once again independent of m, so the images remain perfectly stigmatic under decentring. The two virtual sources simply follow the displacement,

$$\mathcal{S}_1 = s + z_1\delta, \qquad \mathcal{S}_2 = s - z_1\delta, \tag{4.74}$$

and three consequences follow from 4.74 at once.

The fringe spacing does not change. The separation is still $a = 2z_1\delta$ and the distance to the screen is still $D = z_1 + z_2$, both untouched by s, so $\Lambda = \lambda D/a$ is invariant to first order in the displacement. It is invariant to second order as well, and for a reason worth stating rather than taking for granted, since the deviation of a prism is stationary at the condition of minimum deviation, $d\delta/di = 0$ there, so tilting the incident beam slightly changes δ only at order $(\Delta i)^2$. The biprism is therefore intrinsically robust against a decentred source, more so than the Young interferometer with its fixed slits, where no such stationarity protects the spacing.

The whole fringe system translates rigidly by exactly s. Both virtual images move together with the source, so the plane of zero path difference, the perpendicular bisector of $\mathcal{S}_1\mathcal{S}_2$, is simply the plane $y = s$, and the zero order fringe sits there regardless of z_2. There is no geometric amplification of the kind we found for the Young interferometer with fixed slits, where the shift on the screen was sD/b rather than s itself, and the difference is that there the source and its two images move independently of the fixed slits, while here the source and both of its images move together as a rigid unit.

The field of overlap translates too, but in the opposite direction and by a different amount, $-z_2 s/z_1$. The ray grazing the shared edge from above now leaves $\mathcal{S}$ with slope $-s/z_1$ and acquires the extra deviation $\mp\delta$, reaching the screen at $y = -z_2 s/z_1 \mp z_2\delta$, so the window keeps its width $W = 2z_2\delta$ but is now centred at $-z_2 s/z_1$. Since the fringes and the window move in opposite directions, the central fringe eventually walks out of the observable field once

$$|s| > \frac{z_1 z_2 \delta}{z_1 + z_2} = \frac{z_2\beta}{2}. \tag{4.75}$$

With the numbers of the table above, $z_1 = 1\,\text{m}$, $z_2 = 1\,\text{cm}$ and $\beta = 0.1$, this bound is $|s| < 0.5\,\text{mm}$, a generous tolerance for the alignment of the source, yet a displacement of only $s = 6.3\,\mu\text{m}$ is already enough to slide the whole pattern by one complete fringe. The Fresnel biprism is therefore, at the same time, an interferometer remarkably tolerant of a coarse misalignment of the source, since the fringe spacing survives it exactly, and an extremely sensitive detector of a fine transverse displacement, since the fringe position tracks that displacement one to one, a duality that turns out to be useful whenever the instrument is repurposed as a position sensor rather than as a simple demonstration of interference.

4.3.3 Michelson and Morley interferometer

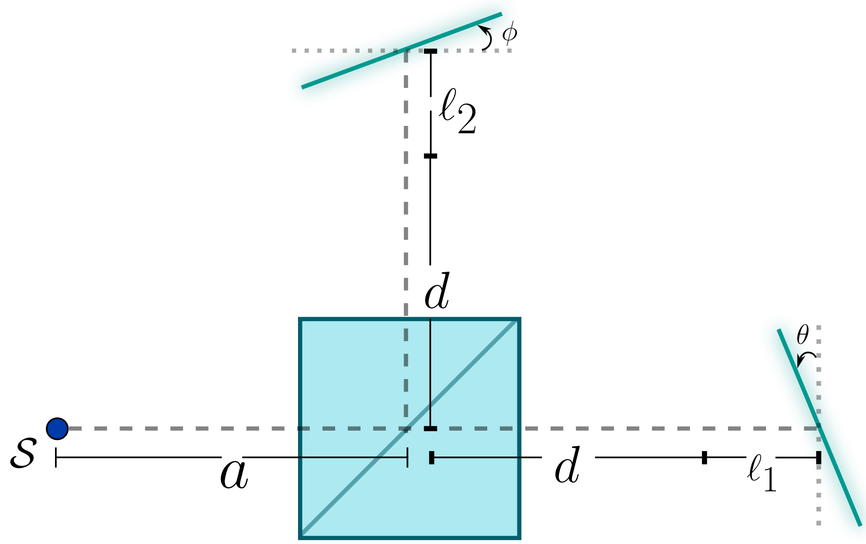


Figure 4.5: General configuration of the Michelson interferometer. A source placed a distance a from the beamsplitting cube illuminates it, and the beam is divided along two perpendicular arms. In arm 1 the light travels a total distance $d + \ell_1$ before striking mirror 1, tilted an angle θ away from the orientation that would send the beam straight back along itself; in arm 2 it travels a total distance $d + \ell_2$ before striking mirror 2, tilted an angle ϕ in the same sense. The two reflected beams return to the beamsplitter and recombine there, ready to interfere.

The configuration sketched above did not spring fully formed from the 1887 experiment already described at the opening of this chapter. Michelson built and tested a first, considerably simpler version of it in 1881, while on leave from the United States Naval Academy and working in Hermann von Helmholtz's laboratory in Berlin Michelson (1881). That prototype was mounted directly on a stone pier in the basement of the Astrophysical Observatory in Potsdam, in an attempt to damp out the ordinary tremors of the building, and it already embodied the essential idea sketched in figure 4.5, a beam split and sent along two perpendicular paths before being brought back together. Its single, short arm and its residual sensitivity to vibration, however, produced a fringe shift far too small and unsteady to draw any firm conclusion, and Alfred Potier and Lord Rayleigh soon pointed out besides that Michelson's original analysis had understated the expected shift by a factor of two, an error traceable to treating the motion of the Earth as though it affected both arms in the same way. Correcting the theory raised the predicted shift, but what the measurement truly needed was a much longer optical path and a mounting immune to vibration, and supplying both is exactly what occupied Michelson once he moved to the Case School of Applied Science and joined forces with Edward Morley.

The apparatus that resulted rested on a massive sandstone slab floated on a pool of mercury, so that it could be rotated smoothly about a vertical axis without flexing, and each arm was folded by a set of auxiliary mirrors to stretch its effective length to roughly eleven metres, some ten times longer than in the Potsdam attempt, which is what finally brought the sensitivity of the instrument within reach of the aether drift the theory predicted, only for none to appear Michelson and Morley (1887). The same principle, two beams recombined after independent and adjustable paths, proved useful well beyond that null result. Four decades later Michelson, together with Francis Pease, mounted a twenty foot version of essentially the same interferometer across the aperture of the 100 inch Hooker telescope at Mount Wilson, and used the visibility of the fringes it produced, precisely the visibility introduced earlier in this chapter, to measure for the first time the angular diameter of a star other than the Sun, Betelgeuse, from nothing more than the separation between the two outer mirrors at which the fringes first washed out Michelson and Pease (1921). The same architecture went on to become the working principle of Fourier transform spectroscopy, where a single continuously scanned mirror encodes an entire spectrum into one interferogram, and later still of optical coherence tomography, which uses exactly this idea to image tissue a few micrometres beneath the surface. Let us now solve the instrument. The difficulty of the Michelson interferometer is not conceptual but bookkeeping, because the light is folded twice in each arm and a direct ray tracing forces us to carry two reflection points, four propagation legs and the two tilts all at the same time. The method of images removes the folding entirely. The idea is that a plane mirror does nothing to a wave except to make it appear to come from somewhere else, so that instead of following the light around the corner we may replace the mirror by the image of the source in it and let the light travel in a straight line. Applying this once for every reflection the beam actually suffers, each arm collapses into a single point source radiating freely into the output space, and the whole interferometer is reduced to the problem of two point sources that we already solved in the section on the interference of spherical waves. Everything that follows is the systematic execution of that idea.

The tool we need is the reflection of a point in a plane. Let the plane pass through a point $\vec{P}$ and have unit normal $\hat{n}$, and let $\vec{X}$ be the point to be reflected. The quantity $\left(\vec{X}-\vec{P}\right)\cdot\hat{n}$ is the signed distance from $\vec{X}$ to the plane, positive on the side towards which $\hat{n}$ points, so that subtracting twice that distance along $\hat{n}$ carries the point across the plane and leaves it at the same distance on the other side. The image is therefore

$$\mathcal{R}\left(\vec{X}\right)=\vec{X}-2\left[\left(\vec{X}-\vec{P}\right)\cdot\hat{n}\right]\hat{n}, \tag{4.76}$$

and it is worth pausing on why this simple operation is legitimate optics rather than a drawing trick. The map 4.76 is an isometry, since it preserves the distance between any two points, and it is an involution, since applying it twice returns the original point. The first property is the important one, because the optical path from the source to any point of the output space, measured along the true folded trajectory, is exactly equal to the straight line distance from the image to that same point. The image is not an approximation of the source, it is the source as far as every optical path length is concerned, and since interference depends on nothing but optical path lengths, the two descriptions are rigorously equivalent.

We place the origin O at the centre of the beamsplitting cube, take the x axis along the first arm and the y axis along the second, and read off the data of figure 4.5. The source sits at

$$\mathcal{S} = (-a,\, 0)\,, \tag{4.77}$$

the first mirror has its vertex on the x axis at a distance $d + \ell_1$ from the origin and is tilted by θ away from the orientation perpendicular to its arm, so that its normal has turned by the same angle away from $\hat{x}$,

$$\vec{P}_1 = (d + \ell_1,\, 0)\,, \qquad \hat{n}_1 = (\cos\theta,\, \sin\theta)\,, \tag{4.78}$$

while the second mirror has its vertex on the y axis at a distance $d + \ell_2$ and is tilted by ϕ in the same sense, so that its normal has turned away from $\hat{y}$,

$$\vec{P}_2 = (0,\, d + \ell_2)\,, \qquad \hat{n}_2 = (-\sin\phi,\, \cos\phi)\,. \tag{4.79}$$

The splitting surface is the diagonal of the cube, a plane through the origin inclined at 45 degrees, whose unit normal we may take as

$$\vec{P}_B = (0,\, 0)\,, \qquad \hat{n}_B = \frac{1}{\sqrt{2}}\,(1,\, -1)\,. \tag{4.80}$$

To keep the expressions readable we abbreviate the two total arm lengths measured from the origin,

$$L_1 = d + \ell_1, \qquad L_2 = d + \ell_2, \tag{4.81}$$

and we note once and for all that transmission through the beamsplitter produces no image at all, since a beam that goes straight through a compensated plate emerges travelling along the same line it came in on, so only the genuine reflections have to be accounted for.

We begin with the first arm, in which the light crosses the splitter, travels to the first mirror, is reflected there, comes back and is reflected a second time at the splitter on its way out. Two reflections occur, first in M_1 and then in the beamsplitter, and the images must be taken in that same order, because the second mirror sees not the original source but the image already produced by the first.

The first reflection is in M_1. We need the signed distance from the source to the plane of that mirror, and using 4.77 and 4.78,

$$\mathcal{S} - \vec{P}_1 = (-a - L_1,\, 0)\,, \qquad \left(\mathcal{S} - \vec{P}_1\right) \cdot \hat{n}_1 = -\,(a + L_1)\cos\theta,$$

a quantity that is negative, as it must be, since the source lies on the side opposite to the one the normal points to. Feeding this into 4.76,

$$\begin{aligned}\mathcal{S}_1' &= (-a,\, 0) + 2\,(a + L_1)\cos\theta\;(\cos\theta,\, \sin\theta) = \\ &\left(-a + 2\,(a + L_1)\cos^2\theta,\;\; 2\,(a + L_1)\sin\theta\cos\theta\right),\end{aligned}$$

and the two double angle identities $2\cos^2\theta = 1 + \cos 2\theta$ and $2\sin\theta\cos\theta = \sin 2\theta$ turn this into

$$\begin{aligned}\mathcal{S}_1' &= (-a + (a + L_1) + (a + L_1)\cos 2\theta,\;\; (a + L_1)\sin 2\theta) = \\ &(L_1 + (a + L_1)\cos 2\theta,\;\; (a + L_1)\sin 2\theta)\,,\end{aligned}$$

where the two terms in red have cancelled, since $-a + (a + L_1) = L_1$. We therefore have the image of the source in the first mirror,

$$\mathcal{S}_1' = (L_1 + (a + L_1)\cos 2\theta,\;\; (a + L_1)\sin 2\theta)\,. \tag{4.82}$$

It is worth stopping to read 4.82 physically, because the tilt has entered doubled. This is the elementary and often forgotten fact that turning a mirror by an angle turns the reflected beam by twice that angle, and it appears here automatically, without our having invoked it, simply because the reflection operator acts twice on the component along the normal. Notice also that the distance from the origin to this image satisfies

$$\left|\left|\mathcal{S}_1'\right|\right|^2 = L_1^2 + (a + L_1)^2 + 2L_1\,(a + L_1)\cos 2\theta,$$

which is the law of cosines for a triangle of sides L_1 and $a + L_1$ enclosing the angle $\pi - 2\theta$. The law of cosines does govern this construction, but with the angle 2θ and not θ, and it emerges by itself rather than having to be guessed. As a check, setting $\theta = 0$ leaves $\mathcal{S}_1' = (a + 2L_1,\, 0)$, the image sitting on the axis of the arm at a distance $a + 2L_1$ from the origin, which is precisely the source retreated by twice the distance it stands from the mirror, as a mirror facing the beam squarely must do.

The second reflection is in the beamsplitter, and here the algebra becomes remarkably simple. For an arbitrary point $\vec{X} = (X_1,\, X_2)$ the signed distance to the splitting plane is $\vec{X}\cdot\hat{n}_B = (X_1 - X_2)\,/\sqrt{2}$, so that 4.76 gives

$$\mathcal{R}_B\left(\vec{X}\right) = (X_1,\, X_2) - \frac{2\,(X_1 - X_2)}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}}\,(1,\, -1) = (X_1,\, X_2) - (X_1 - X_2)\,(1,\, -1) =$$
$$(X_1 - X_1 + X_2,\;\; X_2 + X_1 - X_2) = (X_2,\, X_1)\,,$$

the terms in red cancelling in each component, so that

$$\mathcal{R}_B\,(X_1,\, X_2) = (X_2,\, X_1)\,, \tag{4.83}$$

that is, *reflection in the beamsplitter is nothing but the interchange of the two coordinates.* This is the analytic expression of the fact that a mirror at 45 degrees exchanges the two arms of the instrument, sending the axis of the first onto the axis of the second, and it is the single feature that makes the Michelson interferometer so symmetric. Applying 4.83 to 4.82 we obtain the virtual source of the first arm,

$$\mathcal{S}_1 = ((a + L_1)\sin 2\theta,\;\; L_1 + (a + L_1)\cos 2\theta)\,, \tag{4.84}$$

and the construction that leads to it is drawn in figure 4.6, where the perpendicular dropped from each point to the corresponding plane and the equality of the two distances on either side of it are marked explicitly.

We now turn to the second arm, and the first thing to notice is that the order of the operations is reversed. Here the light is reflected at the splitter on its way in, travels up to the second mirror, is reflected there, and finally crosses the splitter on its way out without being reflected again. The two reflections are therefore first in the beamsplitter and then in M_2, exactly the opposite of what happened in the first arm, and since reflections in different planes do not commute the order matters and must be respected.

The first reflection is in the beamsplitter, and 4.83 disposes of it immediately,

$$\mathcal{S}_2' = \mathcal{R}_B\,(-a,\, 0) = (0,\, -a)\,. \tag{4.85}$$

This deserves a comment, because the result is more eloquent than the calculation. The image of the source in the splitter sits on the axis of the second arm, at a distance a on the far side of the origin, which is to say that the light climbing towards the second mirror behaves in every respect as though it had been emitted by a source placed a distance a below the beamsplitter. That is the whole content of the folding, expressed in one line, and it is the reason the two arms of a Michelson interferometer can be compared at all.

The second reflection is in M_2. Using 4.79 and 4.85,

$$\mathcal{S}_2' - \vec{P}_2 = (0,\, -a - L_2)\,, \qquad \left(\mathcal{S}_2' - \vec{P}_2\right) \cdot \hat{n}_2 = -\,(a + L_2)\cos\phi,$$

and substituting into 4.76,

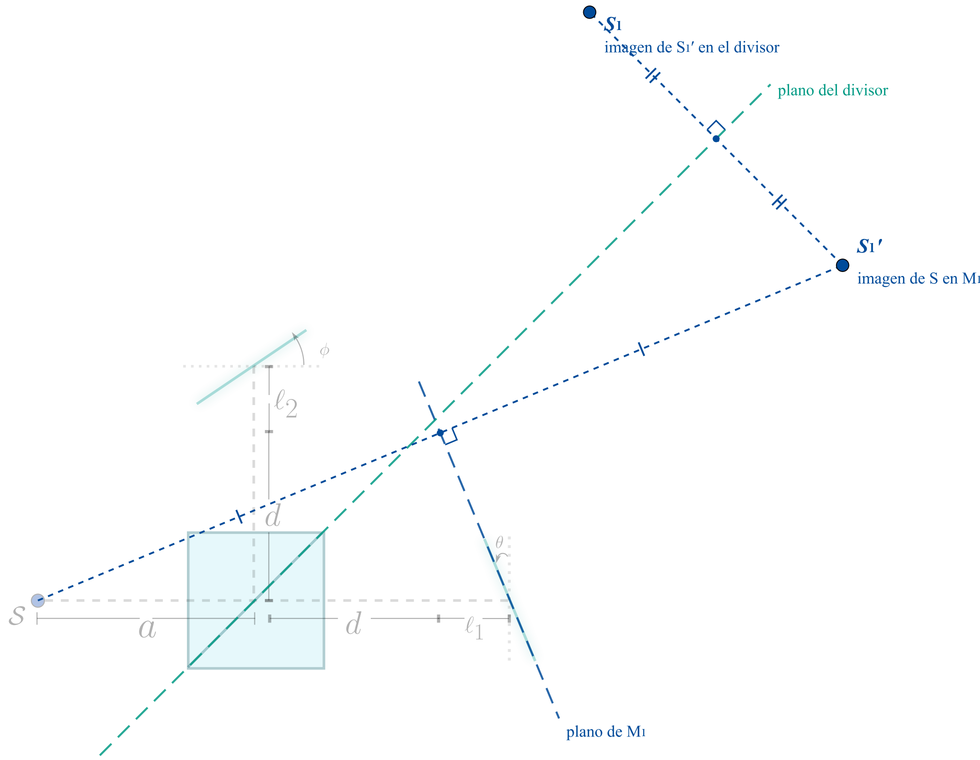


Figure 4.6: Construction of the virtual source of the first arm. The source $\mathcal{S}$ is first reflected in the plane of the mirror M_1, giving $\mathcal{S}_1'$, and this image is then reflected in the plane of the beamsplitter, giving $\mathcal{S}_1$. In each step the segment joining a point to its image is perpendicular to the corresponding plane and is bisected by it, which is what the right angles and the equal marks record. The interferometer of figure 4.5 is kept faded in the background so that the construction may be followed against the real geometry of the instrument.

$$\mathcal{S}_2 = (0,\, -a) + 2\,(a + L_2)\cos\phi\;(-\sin\phi,\, \cos\phi) =$$
$$\left(-2\,(a + L_2)\sin\phi\cos\phi,\;\; -a + 2\,(a + L_2)\cos^2\phi\right),$$

so that the same two double angle identities used before, together with the same cancellation $-a + (a + L_2) = L_2$, give the virtual source of the second arm,

$$\mathcal{S}_2 = \left(-\,(a + L_2)\sin 2\phi,\;\; L_2 + (a + L_2)\cos 2\phi\right), \tag{4.86}$$

whose construction is drawn in figure 4.7.

The two results 4.84 and 4.86 are the complete solution of the interferometer, and writing them together and restoring the lengths 4.81,

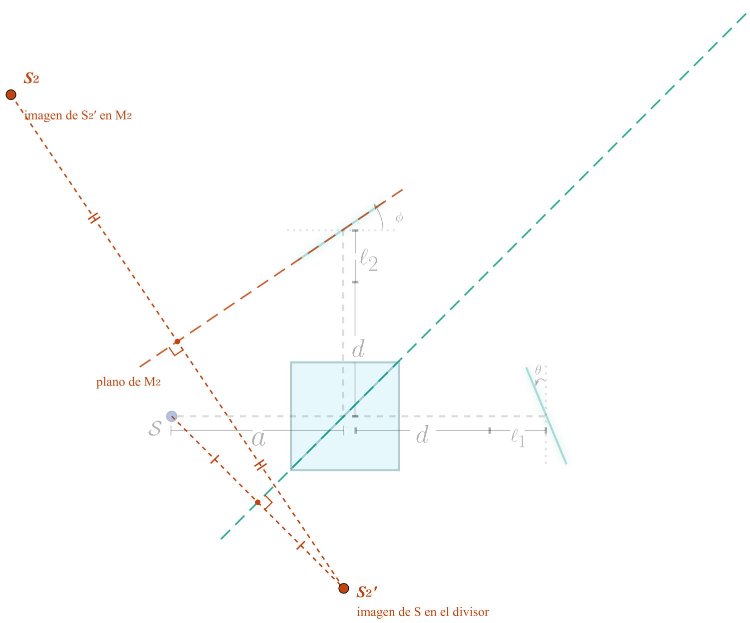


Figure 4.7: Construction of the virtual source of the second arm. Here the order is reversed with respect to the first arm, since the light meets the beamsplitter before the mirror. The source is first reflected in the plane of the splitter, giving $\mathcal{S}_2'$ on the axis of the second arm at a distance a below the origin, and this image is then reflected in the plane of the mirror M_2, giving $\mathcal{S}_2$.

$$\mathcal{S}_1 = \left((a+d+\ell_1)\sin 2\theta,\quad (d+\ell_1)+(a+d+\ell_1)\cos 2\theta\right), \tag{4.87}$$
$$\mathcal{S}_2 = \left(-(a+d+\ell_2)\sin 2\phi,\ \ (d+\ell_2)+(a+d+\ell_2)\cos 2\phi\right). \tag{4.88}$$

Two checks confirm that these expressions are right. The first is a symmetry. The interchange $x \leftrightarrow y$ together with $\theta \leftrightarrow \phi$ and $\ell_1 \leftrightarrow \ell_2$ carries 4.87 into 4.88 up to the sign of the first component, and that is exactly as it should be, because by 4.83 the interchange of the coordinates *is* the reflection in the beamsplitter, which is precisely the operation that exchanges the two arms of the instrument. A formula that failed this test would be describing an asymmetric interferometer. The second check is the aligned instrument. Setting $\theta = \phi = 0$ both cosines become unity and both sines vanish, leaving

$$\mathcal{S}_1 = (0,\ a+2\,(d+\ell_1)), \qquad \mathcal{S}_2 = (0,\ a+2\,(d+\ell_2)), \tag{4.89}$$

so that both virtual sources fall on the axis of the output arm, one behind the other, separated by

$$||\mathcal{S}_1 - \mathcal{S}_2|| = 2\,(\ell_1 - \ell_2),$$

which is the classical statement that a Michelson interferometer with aligned mirrors is equivalent to two point sources separated by twice the difference of the arm lengths. The factor of two, which every treatment of the instrument quotes, has come out of the algebra rather than having been asserted.

In the general case the two sources are neither coincident nor aligned with the output axis, and what governs the interference is their separation,

$$\vec{\Delta} = \mathcal{S}_1 - \mathcal{S}_2 = \left(\Delta_\perp,\ \Delta_\parallel\right), \tag{4.90}$$

whose two components, transverse and longitudinal with respect to the direction in which the light leaves the instrument, are

$$\Delta_\perp = (a + d + \ell_1)\sin 2\theta + (a + d + \ell_2)\sin 2\phi, \tag{4.91}$$

$$\Delta_\parallel = (\ell_1 - \ell_2) + (a + d + \ell_1)\cos 2\theta - (a + d + \ell_2)\cos 2\phi. \tag{4.92}$$

This separation of $\vec{\Delta}$ into two pieces is not a formality, because the two components produce interference patterns of entirely different appearance, as we are about to see. It is worth insisting that no symmetry relates $\mathcal{S}_1$ to $\mathcal{S}_2$. The two sources do not lie at the same height above the splitter, since that would require the right-hand side of 4.92 to vanish, nor at the same distance from the origin, since $||\mathcal{S}_1||$ and $||\mathcal{S}_2||$ are given by two laws of cosines with different sides and different angles. Each arm carries its own length and its own tilt into the answer, and the two only conspire in the particular configurations in which one of the components of $\vec{\Delta}$ happens to cancel. The complete construction, with both arms solved and the two equivalent virtual sources shown together with their separation, is collected in figure 4.8.

Everything that remains is to let the two equivalent sources interfere, and since the machinery for that was built in the first half of this chapter the work is short. We treat the two cases in turn.

Consider first the case in which the instrument is illuminated by **plane waves**, which is what happens when the source is placed at the focus of a collimating lens or, equivalently, when it is pushed to infinity. There are then no point sources to speak of, but the method of images applies just as well to the direction of propagation as to the position of the source, because the reflection of a wave vector in a plane obeys the very same operator 4.76 with the point $\vec{P}$ removed, that is $\vec{k} \to \vec{k} - 2\left(\vec{k}\cdot\hat{n}\right)\hat{n}$. Let the wave arrive along the first arm, $\vec{k} = \kappa\hat{x}$. In the first arm it is reflected in M_1, for which $\vec{k}\cdot\hat{n}_1 = \kappa\cos\theta$ and therefore

$$\vec{k} \longrightarrow \kappa\left(1 - 2\cos^2\theta,\ -2\sin\theta\cos\theta\right) = -\kappa\left(\cos 2\theta,\ \sin 2\theta\right),$$

and it is then reflected in the splitter, which by 4.83 merely interchanges the components, so that the wave leaving the first arm has

$$\vec{k}_1 = -\kappa\left(\sin 2\theta,\ \cos 2\theta\right). \tag{4.93}$$

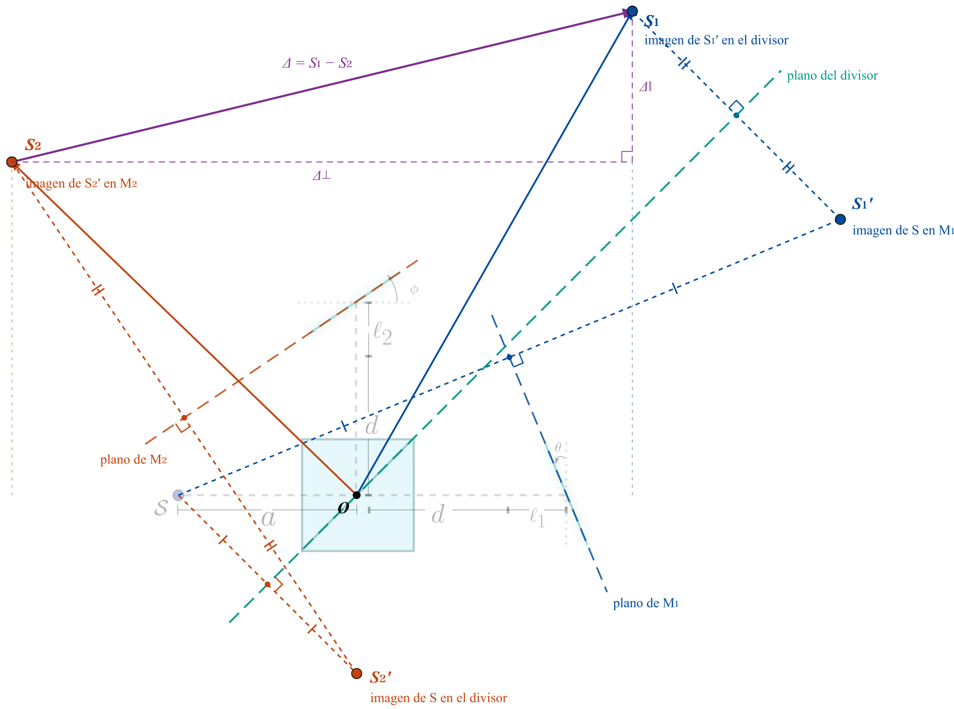


Figure 4.8: The interferometer completely solved by the method of images. The construction of both arms is superposed on the faded instrument, and the two resulting virtual sources $\mathcal{S}_1$ and $\mathcal{S}_2$ are shown with their position vectors measured from the origin O at the centre of the beamsplitter, together with the separation $\vec{\Delta}$ that governs the interference, resolved into the transverse and longitudinal components $\Delta_\perp$ and $\Delta_\parallel$ of 4.91 and 4.92. The configuration drawn is the generic one, with unequal arms and unequal tilts, so that the two sources are neither at the same height nor at the same distance from O and both components of $\vec{\Delta}$ are present at once. From the point of view of the output, the whole interferometer has been replaced by these two mutually coherent point sources.

In the second arm the order is again reversed. The splitter turns $\kappa\hat{x}$ into $\kappa\hat{y}$, the wave climbs to M_2, for which $\vec{k}\cdot\hat{n}_2 = \kappa\cos\phi$ and therefore

$$\kappa\,(0,\,1) \longrightarrow \kappa\left(2\sin\phi\cos\phi,\; 1-2\cos^2\phi\right) = \kappa\,(\sin 2\phi,\; -\cos 2\phi)\,,$$

and the final transmission leaves it untouched, so that

$$\vec{k}_2 = \kappa\,(\sin 2\phi,\; -\cos 2\phi)\,. \tag{4.94}$$

Both 4.93 and 4.94 reduce to $-\kappa\hat{y}$ when the mirrors are aligned, confirming that the two waves then leave the instrument along the same downward axis. The angle α between them follows from their scalar product,

$$\frac{\vec{k}_1 \cdot \vec{k}_2}{\kappa^2} = -\sin 2\theta \sin 2\phi + \cos 2\theta \cos 2\phi = \cos\left(2\theta + 2\phi\right),$$

so that, remarkably,

$$\alpha = 2\left(\theta + \phi\right), \tag{4.95}$$

the two tilts adding rather than subtracting, each one contributing twice its own value. Since the two waves have equal moduli, we are exactly in the situation studied at the beginning of this chapter, and the interference vector has the magnitude $K = 2\kappa \sin(\alpha/2)$ found there. The fringes are therefore straight, parallel to $\vec{k}_1 + \vec{k}_2$, and separated by

$$\Lambda = \frac{2\pi}{K} = \frac{\lambda}{2\sin\left(\theta + \phi\right)}, \tag{4.96}$$

with the irradiance

$$I = 2I_0 \left[1 + \cos\left(\frac{2\pi}{\Lambda} X + \frac{4\pi}{\lambda}\left(\ell_1 - \ell_2\right) + \delta_B\right)\right], \tag{4.97}$$

where X is measured on the observation plane along the direction of $\vec{K}$, the term $4\pi(\ell_1 - \ell_2)/\lambda$ is the phase accumulated by the difference $2(\ell_1 - \ell_2)$ of the two optical paths on the axis, and δ_B is the phase introduced by the splitter itself, equal to π for a symmetric beamsplitter, which is what makes the two output ports of the instrument complementary and conserves the energy between them. The behaviour of 4.96 is the expected one, since aligning the mirrors sends $\theta + \phi \to 0$ and therefore $\Lambda \to \infty$, the fringes widening without limit until the whole field is uniformly bright or uniformly dark according to the value of the arm difference, which is exactly what one observes when a Michelson interferometer is well aligned under collimated illumination.

Consider now the case of **spherical waves**, that is, of a genuine point source at a finite distance, which is the case the whole image construction was built for. The instrument has been replaced by the two mutually coherent point sources $\mathcal{S}_1$ and $\mathcal{S}_2$ of 4.87 and 4.88, and we observe the light on a screen placed perpendicular to the output axis at a distance D below the beamsplitter, with coordinates X in the plane of the figure and Z perpendicular to it. Writing $\mathcal{S}_j = (X_j, Y_j)$ the distances from each source to a point of the screen are

$$r_j = \left[\left(X - X_j\right)^2 + Z^2 + \left(D + Y_j\right)^2\right]^{1/2},$$

and since both sources lie close to the axis compared with D we expand exactly as we did for two spherical waves earlier in this chapter,

$$r_j \approx \left(D + Y_j\right) + \frac{\left(X - X_j\right)^2 + Z^2}{2\left(D + Y_j\right)}.$$

Subtracting the two, and replacing $D + Y_j$ by the mean distance $\bar{D} = D + (Y_1 + Y_2)/2$ inside the quadratic terms, which is legitimate because the difference of the two denominators only affects the result at the next order,

$$r_1 - r_2 \approx (Y_1 - Y_2) + \frac{(X - X_1)^2 + Z^2 - (X - X_2)^2 - Z^2}{2\bar{D}} =$$
$$\Delta_\parallel + \frac{X^2 - 2XX_1 + X_1^2 - X^2 + 2XX_2 - X_2^2}{2\bar{D}} = \Delta_\parallel - \frac{\Delta_\perp}{\bar{D}}X + \frac{X_1^2 - X_2^2}{2\bar{D}},$$

where the terms in red have cancelled, the transverse coordinate Z dropping out entirely because both sources lie in the plane $Z = 0$. The last term is a constant and merely displaces the pattern as a whole, and it disappears if the origin of X on the screen is taken at the midpoint of the projections of the two sources, which we do from here on. The phase difference is then

$$\triangle\phi = \kappa\,(r_2 - r_1) + \delta_B = \frac{2\pi}{\lambda}\frac{\Delta_\perp}{\bar{D}}X - \frac{2\pi}{\lambda}\Delta_\parallel + \delta_B, \tag{4.98}$$

and the irradiance on the screen is

$$I = 2I_0\left[1 + \cos\left(\frac{2\pi}{\lambda}\frac{\Delta_\perp}{\bar{D}}X - \frac{2\pi}{\lambda}\Delta_\parallel + \delta_B\right)\right]. \tag{4.99}$$

Equation 4.99 shows at once the different roles of the two components of the separation. The transverse component $\Delta_\perp$ is the only one that depends on the position on the screen, and it produces straight fringes perpendicular to X, that is, parallel to the direction Z, with the spacing

$$\Lambda = \frac{\lambda\bar{D}}{\Delta_\perp} = \frac{\lambda\bar{D}}{(a + d + \ell_1)\sin 2\theta + (a + d + \ell_2)\sin 2\phi}, \tag{4.100}$$

which is the result $\Lambda = \lambda D/a$ obtained for two point sources earlier in this chapter, with the separation of the sources now supplied by the geometry of the interferometer. These are the fringes of equal thickness, and they are what one sees when the arms are nearly equal but the mirrors are tilted. The longitudinal component $\Delta_\parallel$, on the contrary, does not depend on X at all and therefore does not create fringes in this approximation, it only adds the constant phase $-2\pi\Delta_\parallel/\lambda$ which decides whether the centre of the pattern is bright or dark.

That last statement is true only within the paraxial expansion, and abandoning it recovers the most familiar pattern of the instrument. When the mirrors are aligned we have $\Delta_\perp = 0$ and 4.100 gives an infinite spacing, meaning that no straight fringes exist, so the pattern must be sought at angles large enough for the expansion to fail. Two point sources separated by $\Delta_\parallel$ along the line of sight, observed in the direction making an angle ψ with that line, deliver a path difference equal to the projection $\Delta_\parallel \cos\psi$, and the condition for a bright fringe is therefore

$$\Delta_\parallel \cos\psi = m\lambda, \qquad m \in \mathbb{Z}, \tag{4.101}$$

which describes circles centred on the output axis, since all the directions sharing one value of ψ form a cone. These are the fringes of equal inclination, the celebrated circular fringes of the Michelson interferometer, and with the aligned value $\Delta_{\|} = 2(\ell_1 - \ell_2)$ the condition 4.101 becomes

$$2\,(\ell_1 - \ell_2)\cos\psi = m\lambda, \tag{4.102}$$

the textbook expression, here derived rather than quoted. As the arm difference is reduced the rings swell and are swallowed one by one at the centre, and at $\ell_1 = \ell_2$ the whole field becomes uniform, which is how the instrument is brought to zero path difference in practice.

The general case interpolates between the two, since both components of $\vec{\Delta}$ are usually present, and the pattern is then a family of curves which are neither straight lines nor circles but the intersections with the screen of the hyperboloids of revolution of foci $\mathcal{S}_1$ and $\mathcal{S}_2$ that we identified as the surfaces of equal intensity when we first studied two spherical sources. Straight fringes and circular fringes are simply the two limits in which one of the two components dominates, and this is why the practical alignment of a Michelson interferometer consists of watching the fringes straighten as the tilts are removed and then watching them widen as the arms are equalised.

It is worth closing with a consistency check between the two cases, which also shows that nothing has been lost along the way. Push the source to infinity, that is let a grow without bound, and take the tilts small. Then $\Delta_{\perp} \approx 2a(\theta + \phi)$ from 4.91, while the mean distance to the screen $\bar{D} \approx D + a$ is itself dominated by a, so that 4.100 gives

$$\Lambda \approx \frac{\lambda\,(D+a)}{2a\,(\theta+\phi)} \xrightarrow[a\to\infty]{} \frac{\lambda}{2\,(\theta+\phi)},$$

which is exactly the small angle limit of the result 4.96 obtained for plane waves. The two descriptions agree where they must, and the point source solution contains the collimated one as the particular case in which the source has been removed to infinity.

Chapter 5

Theory of Diffraction

Where geometry promised sharp shadows, light insisted on painting fringes.

“Diffraction is any deviation of light rays from rectilinear paths which cannot be interpreted as reflection or refraction.”
– Arnold Sommerfeld

5.1 A historical context of diffraction theory: from shadows to the vector world

For more than two thousand years, from Euclid to the threshold of the Renaissance, geometrical optics rested on a single, almost self-evident assumption: light travels in straight lines Herzberger (1966). A straight ray casts a straight-edged shadow, and every child who has ever played with their own silhouette on a wall has confirmed it a thousand times. This picture explained mirrors, lenses and eclipses so well that for centuries nobody thought to look closely at the edge of a shadow itself to ask what really happens in that thin borderland where light stops and darkness begins. Diffraction is the story of what was hiding in that borderland, and of how a single overlooked detail, pursued patiently over three centuries, forced physicists to rebuild their entire understanding of what light is. It is, in a real sense, the story of light refusing to be simplified.

The first person to look carefully enough was an Italian Jesuit named Francesco Maria Grimaldi. Sometime around 1660, working in a darkened room in Bologna, Grimaldi let a narrow beam of sunlight fall through a tiny aperture and cross the edge of a rod before striking a screen. Ordinary geometry made a firm prediction: a sharp-edged shadow, dark on one side and bright on the other, with nothing more to say. What Grimaldi saw instead was stranger and far more delicate—the boundary of the shadow was not a clean line but a soft fringe, laced with faint bands of colour, and the shadow itself was slightly wider than pure geometry allowed. Light was somehow reaching into the space it was not supposed to reach, and bending in ways no ray could bend. Grimaldi did not live to see his work in print; his observations appeared only after his death, in the 1665 book *Physico-Mathesis de Lumine* Grimaldi (1665). There he gave the effect its name, *diffractio*, from the Latin for "breaking into pieces," because it looked to him as though the light itself had been shattered at the edge of the obstacle. It was a remarkably prescient observation, made with nothing more than sunlight, a dark room and a sharp eye and for a long time, almost nobody noticed it had been made.

The reason nobody noticed is largely one man's name: Isaac Newton. Newton himself had seen delicate coloured rings form between a lens and a flat glass plate—the pattern we still call Newton's rings today and he described them meticulously in his own *Opticks* of 1704 Newton (1704). Yet Newton remained convinced that light was made of small particles, corpuscles, shot out in straight lines, and he folded these puzzling colours into that picture as best he could rather than let them overturn it. Newton's scientific authority in the eighteenth century was so enormous that to argue against him was close to professional suicide, and so Grimaldi's fringes, and the wave-like reasoning that might have explained them, were quietly set aside for more than a hundred years. It is one of the great cautionary tales of physics: the evidence for diffraction was sitting in plain sight the entire time, waiting for someone willing to take it seriously.

Someone did, although his idea arrived slightly ahead of its time and without quite enough machinery to finish the job. In 1690 the Dutch physicist Christiaan Huygens proposed, in his *Traité de la Lumière* Huygens (1690), an almost playful way of picturing how a wave moves forward. Imagine, Huygens said, that every single point on a wavefront is itself a tiny source, spraying out its own small circular ripple. A moment later, the new wavefront is simply the smooth envelope that wraps around all of those countless little ripples at once. This picture now known as Huygens's principle explained reflection and refraction beautifully, and it hinted that a wave could indeed sneak sideways into a geometrical shadow, since the ripples spreading from points near the edge of an obstacle have nowhere to go but into that forbidden space. What Huygens's construction could not yet explain was why the resulting light and dark fringes appeared exactly where they did, because he had no notion that the little ripples might reinforce each other in some places and cancel each other in others. That missing idea was interference, and supplying it would take more than a century.

It finally arrived in a remarkable few years at the start of the nineteenth century. In 1801, the English polymath Thomas Young performed the experiment that every optics student meets sooner or later: he let light pass through two narrow, closely spaced slits and watched the pattern it formed on a screen beyond Young (1804). Instead of two bright bands, he saw many—an entire sequence of bright and dark stripes, exactly as one would expect if two overlapping ripples on a pond could sometimes crest together and sometimes cancel to stillness. Light, Young concluded, must be a wave, and waves can add or subtract. It was a devastatingly simple experiment carrying an enormous conclusion, and for exactly that reason it was met in England with scorn rather than applause; attacking Newton was still not a popular sport. Vindication came from France. Between 1815 and 1820 the young engineer Augustin-Jean Fresnel took Huygens's wavelets and Young's interference and welded them into a single, quantitative principle—the Huygens-Fresnel principle—capable of predicting, for the first time, exactly how bright or dark any point in a diffraction pattern should be Fresnel (1819). Fresnel did not merely describe diffraction qualitatively, as Grimaldi and Huygens had; he calculated it, treating every point of an opening as a genuine source of secondary waves whose ripples had to be added together, crest against crest and trough against trough, to find the light at any point beyond.

Fresnel's theory then delivered one of the most celebrated moments in the history of physics. The French Academy of Sciences, still largely unconvinced, set diffraction as the subject of a prize competition, and the mathematician Siméon Denis Poisson, a confirmed sceptic of the wave picture, examined Fresnel's equations looking for an absurdity that would sink them for good. He found what he was looking for, or thought he had: Fresnel's own mathematics predicted that at the very centre of the shadow cast by a small circular disc, directly behind the obstacle where common sense guarantees total darkness, there should be a bright spot of light. Poisson presented this as a reductio ad absurdum, confident that no such thing could ever be observed. The academy's chairman, Dominique Arago, decided to actually perform the experiment rather than argue about it and there, glowing faintly at the heart of the disc's shadow, was the spot Poisson had predicted in mockery. The Poisson-Arago spot, as it is now known with a certain irony, became one of the most dramatic experimental confirmations any theory has ever received, and it settled the contest between particles and waves for the rest of the century.

Fresnel's success, however, was built on inspired intuition rather than rigorous derivation—his integral worked extremely well, but nobody had shown that it followed necessarily from the wave equation itself. That gap was closed in 1883 by the German physicist Gustav Kirchhoff, who used the mathematics of Green's theorem to derive Fresnel's diffraction integral directly from the wave equation, placing the whole construction on a firm mathematical foundation Kirchhoff (1883).

Kirchhoff's formulation became, and largely remains, the working horse of what we now call the scalar theory of diffraction: it treats light as a single scalar ripple rather than a full electromagnetic vector, an approximation that works remarkably well whenever the obstacles involved are much larger than the wavelength of light. There was, admittedly, a subtle logical wrinkle in Kirchhoff's argument—his boundary conditions demanded that both the field and its derivative vanish simultaneously behind an opaque screen, a requirement that is mathematically over-strict—and it was Henri Poincaré who first pointed this out. The wrinkle was ironed out in 1896 by Arnold Sommerfeld, who found an alternative, fully self-consistent solution for diffraction by a straight edge and a half-plane, removing the inconsistency without sacrificing any of Kirchhoff's practical success Sommerfeld (1896, 1954).

All of this, ingenious as it was, still treated light as a single wiggling number instead of the vector quantity that Maxwell's equations say it truly is—and by the early twentieth century that omission started to matter. In 1908 Gustav Mie solved Maxwell's equations exactly for light scattering off a small sphere, keeping the full vector and polarisation content of the field Mie (1908); his result explains, among many other things, why the sky is blue and why clouds and milk are white, and it remains indispensable wherever light interacts with particles comparable in size to its own wavelength. The following year Peter Debye extended the same vector thinking to the tight focus of a lens, showing that near a sharp focal point the scalar approximation simply breaks down. The synthesis of everything that had come before—Fresnel's integral, Kirchhoff and Sommerfeld's rigour, and the full vector corrections of Mie and Debye—was finally gathered into a single, monumental reference by Max Born and Emil Wolf, whose *Principles of Optics* (1959) has served as the field's bible ever since Born and Wolf (1999). Born and Wolf made explicit exactly when the simpler scalar picture is trustworthy and when it fails outright: in lenses of high numerical aperture, in anisotropic or chiral materials, and whenever the relevant structures shrink down to the scale of the wavelength itself, the vector nature of light can no longer be ignored.

The invention of the laser in 1960 gave diffraction theory an entirely new practical urgency, since it suddenly supplied a light source coherent and controllable enough to exploit diffraction on purpose rather than merely tolerate it. Joseph Goodman's *Introduction to Fourier Optics* (1968) reframed the whole subject in a beautifully economical way: an optical system, Goodman showed, can be treated as a kind of signal processor, and the diffraction pattern produced far from an aperture is, for all practical purposes, the same mathematical object as the Fourier transform of that aperture's shape Goodman (2005). This single reframing opened the door to holography, to diffractive lenses that focus light using etched patterns instead of curved glass, and to the computational correction of aberrations in modern cameras and telescopes.

The story is very much still being written. Today, engineers etch **metasurfaces**—arrays of nanoscale structures far smaller than the wavelength of light—that reshape a wavefront's phase and polarisation point by point, allowing a sheet barely thicker than a human hair to do the work of a stack of lenses Yu et al. (2011). Electron and X-ray diffraction, governed by exactly the same wave logic that Fresnel and Kirchhoff worked out for visible light, let scientists map the atomic architecture of crystals and molecules—it was X-ray diffraction, after all, that first revealed the double-helix structure of DNA. And diffraction sets an absolute, unbreakable limit on what any optical instrument can ever resolve: no telescope, however perfectly polished, and no microscope, however expensively built, can focus light down to a mathematical point, because the very act of passing light through a finite aperture spreads it out again. Every image a telescope or a microscope has ever produced, from the first blurred moons Galileo sketched to the sharpest pictures the largest observatories can deliver today, is diffraction's signature stamped on the light itself. Three and a half centuries after a Jesuit noticed that a shadow in a dark Bolognese room was not quite as sharp as geometry demanded, diffraction has grown from a curious imperfection into one of the deepest and most useful ideas in all of physics—and, as the chapters that follow will show in careful detail, it still has a great deal to teach us about the nature of light.

5.2 Scalar theory of diffraction

As we have proved in 3.7 and 3.8 the electromagnetic waves are very rich in their nature and the nature that can create waves. In this chapter we're going to study these wave behavior of the light, however, thiese are some introductory lectures, so solving these two equation are hard, and if we pretend to solve them analitically, well, that is completelety impossible.

To do physics it is very often to impose assumptions on the model to make it easier to understand and develope, or even, to calculate analytical solutions. In this chapter we are going to counter how many assumptions do we need to arrive to the scalar diffraction theory. Having this on count, lets start with the first one.

Assumption 1 *In the propagation of light the magnetic interaction is completely null.*

This must be done because, in nature, the magnetic interactions are richier than the electric ones, however, it may sound funnier but it makes the solutions of 3.8 a lot harder to be calculated. Therefore, the electromagnetic formulation can be understood only by means of the electric wave equation 3.7

$$\left(\nabla^2 - \frac{1}{C}^2 \frac{\partial^2}{\partial t^2}\right) \vec{E} = \frac{1}{\epsilon_0} \boldsymbol{\nabla} \rho + \mu_o \frac{\partial}{\partial t} \vec{J}.$$

However, this wave equation is really hard to solve exactly, most of the solution are going to be found in numerical ones. So to make the life a lot easier, we're going to use the next assumption

Assumption 2 *The electric wave propagates in the vacuum, there are no charge densities nor current densities.*

Under these assumptions, the wave quation adopts the well known form

$$\nabla^2 \vec{E} = \frac{1}{C^2} \frac{\partial^2}{\partial t^2} \vec{E} \tag{5.1}$$

In the chapter 3.2 we start the study of the vector behavior of the electromagnetic radiation and we arrived to the concept of polarisation, which describes the vector behavior of light, and we found that, in general, this vector behavior if confined on an ellipse, the well known as the polarisation ellipse (see figure 3.3). Thus, in general, the electric vector is given by

$$\vec{E}(\vec{r},t) = E(\vec{r},t)\,\hat{\vec{e}}(\vec{r},t), \tag{5.2}$$

where $E(\vec{r},t)$ denotes the amplitude of the field and $\hat{\vec{e}}(\vec{r},t)$ represents the polarisation vector, which besides varying in space may evolve in time in a stochastic manner.

The associated wave equation then takes the form

$$\nabla^2 \left[E(\vec{r},t)\hat{\vec{e}}(\vec{r},t)\right] - \mu\varepsilon \frac{\partial^2}{\partial t^2}\left[E(\vec{r},t)\hat{\vec{e}}(\vec{r},t)\right] = 0, \tag{5.3}$$

whose direct solution presents notable complexities. To tackle this problem one resorts to simplifying strategies, such as decomposing the field into scalar components. These scalar solutions may afterwards be combined to reconstruct the full vector behaviour.

The temporal variation of amplitude, phase and polarisation is analysed rigorously through coherence theory (explored in depth in Chapter 6). Nevertheless, under certain approximations the temporal dependence of the amplitude is attributed solely to the intrinsic oscillations of the field, while the polarisation is regarded as static. Also, from an electromagnetic point of view, the polarisation states is completely determintaed by the Fresnel Coefficients, therefore in each spatial position the electric wave can have a completelety different polarisation state, so, we are going to need to do some others assumptions:

Assumption 3 *The polarisation of the wave does not evolve in time.*

Assumption 4 *The polarisation of the wave does not change in any space point, it its completelty isotropic and homogeneous in the space.*

Under these two assumptions, the electric field then simplifies to

$$\vec{E}(\vec{r},t) = E(\vec{r},t)\,\hat{\vec{e}}(\vec{r}), \tag{5.4}$$

allowing its expansion in an orthogonal polarisation basis,

$$\vec{E}(\vec{r},t) = E_1(\vec{r},t)\hat{\vec{e}}_1 + E_2(\vec{r},t)\hat{\vec{e}}_2. \tag{5.5}$$

However, allowing this expression carries another assumption. As we've seen in the polarisation chapter it is an approximation that allows to study the electric field from a $3D$ perspective to a $2D$ this is due to the fact that electric and magnetic field, under our conditions, are orthogonal to the propagation direction, and the wavefront is smooth. Sumarizing, the assumption used to express the polarization as the superposition of two orthogonal states $\{\hat{e}_1, \hat{e}_2\}$ is

Assumption 5 *The electric and magnetic fields composing the electromagnetic wave are orthogonal to the propagation axis, so the vector behavior is given by* 2 *polarisation vectors.*

Each scalar component E_i individually satisfies the wave equation

$$\nabla^2 E_i - \mu\varepsilon \frac{\partial^2 E_i}{\partial t^2} = 0 \quad (i = 1, 2), \tag{5.6}$$

thereby decoupling the vector problem into manageable scalar equations.

In situations where one component is negligible (for instance $E_2 \to 0$), the field reduces to

$$\vec{E}(\vec{r}, t) = E(\vec{r}, t)\,\hat{e}, \tag{5.7}$$

defining a purely scalar case. This formalism underlines how the general vector description can emerge from the controlled superposition of scalar solutions.

5.3 The scalar wave equation (Helmholtz equation)

Let $\Psi(\mathbf{r}, t)$ be an electric polycrhomatic arbitrary field component. This quantity satisfies the classical wave equation

$$\nabla^2 \Psi(\mathbf{r}, t) - \mu\varepsilon \frac{\partial^2 \Psi(\mathbf{r}, t)}{\partial t^2} = 0. \tag{5.8}$$

The study of white light cambines many phenomna and from an experimental point of view it can be hard, also, from a mathematical and physical point of view we're not completely interested in the time evolution of the wave, we would like **to study the space propagation of a single mode wave**.

To ble able to isolate, mathematically, this single frequency we're going to impose the following assumption

Assumption 6 *The electric field componente $\Psi(\vec{r}, t)$ can be write as a time Fourier Ttransform.*

Thus the electric field amplitude is given by

$$\Psi(\mathbf{r}, t) = \int_{\mathbb{R}^+} U(\mathbf{r}, \nu)\, \exp(-2\pi \mathrm{i} \nu t)\, d\nu. \tag{5.9}$$

Substituting (5.9) into (5.8), the equation is re-expressed as

$$\int_{\mathbb{R}^+} \left[\nabla^2 U(\mathbf{r}, \nu) + \frac{\omega^2}{C^2} U(\mathbf{r}, \nu) \right] \exp(-2\pi \mathrm{i} \nu t)\, d\nu = 0, \tag{5.10}$$

where we have to integrate over all the positive frequencies, however, this is just a physical argument, from a mathematical point of view, it is more correcto to integrate over all $\mathbb{R}$ this because the negative frequencies are just mathematical extensions of the positivie ones. This integral leads directly to the Helmholtz equation for each spectral component

$$(\nabla^2 + k^2) U(\mathbf{r}, \nu) = 0. \tag{5.11}$$

Here $k = 2\pi/\lambda$ denotes the wavenumber, and $\lambda = c/(n\nu)$ is the wavelength in a medium of refractive index n. This formulation describes mathematically how each monochromatic component of the field (characterised by its complex amplitude U) obeys an equation independent of the temporal frequency.

Before going further, lets talk a bit about the way we're going to solve the Helmholtz equation. There are many ways to try to solve EDPs analitically, in fact, once a problem has been proved to have an unique solution given a boundary condition and an initial condition, no matter what method we use, because the solution is unique, all the methods get the same solution.

From the physical point of view, this diffraction problem is a mathematical model of how is the effect of the electric field constrained to the boundary of a surface on the electric field inside. Due to this physical meaning of the problem, we need a method to transform the Helmholtz equation to a integral equation integrated on a volume to later reduce it to a surface integral. This methodology is what conditionates the need of use the Kirchoff Scalar theory which need the Green-Kirchoff theorem.

5.4 The Kirchoff scalar boundary-value problem

The scalar theory of diffraction, based on solutions of the wave equation under boundary conditions, describes the propagation of fields in paraxial approximations or with smooth phase modulations. Despite its scalar formulation, the formalism developed by Kirchhoff by means of Green's functions is adaptable to vector fields.

The original inconsistency of Kirchhoff, pointed out by Poincaré, lies in the over-determination of the boundary conditions (prescribing both the field and its normal derivative on a plane), which violates the uniqueness theorems of differential equations. Sommerfeld resolved this by redefining the Green's function so as to eliminate the redundancy, guaranteeing mathematical consistency.

5.4.1 Green's theorem

Green's theorem, a direct corollary of Gauss's theorem, establishes fundamental relations between volume and surface integrals for scalar fields. Its derivation rests on the divergence theorem, and the divergence theorem is not valid for arbitrary functions: it demands that the vector field whose divergence is being integrated be differentiable at every interior point and continuous up to the boundary. Since the two functions we shall eventually insert are the optical field and a Green's function, that requirement is a physical statement about the light and about the screen, and it deserves to be recorded rather than assumed silently.

Assumption 7 *The field and the auxiliary function have continuous first and second derivatives throughout the volume V and on its boundary $\Sigma(V)$, so that no source, no discontinuity and no singularity lies on the surface of integration.*

Let $f(\vec{r})$ and $g(\vec{r})$ be two scalar functions meeting this requirement in a domain V. We start from the differential identity

$$\nabla \cdot [f(\vec{r})\nabla g(\vec{r})] = f(\vec{r})\nabla^2 g(\vec{r}) + \nabla f(\vec{r}) \cdot \nabla g(\vec{r}). \tag{5.12}$$

Subtracting the analogous expression with f and g interchanged, we obtain

$$\nabla \cdot [f\nabla g - g\nabla f] = f\nabla^2 g - g\nabla^2 f. \tag{5.13}$$

Integrating this relation over a volume V and applying the Gauss divergence theorem,

$$\int_V \nabla \cdot [f(\vec{r})\nabla g(\vec{r}) - g(\vec{r})\nabla f(\vec{r})]\, dV = \int_V \left(f\nabla^2 g - g\nabla^2 f\right) dV, \tag{5.14}$$

which allows the volume integral on the right-hand side to be converted into a surface integral over ∂V

$$\oint_{\Sigma(V)} [f(\vec{r})\nabla g(\vec{r}) - g(\vec{r})\nabla f(\vec{r})] \cdot d\vec{\sigma} = \int_V \left(f\nabla^2 g - g\nabla^2 f\right) dV, \tag{5.15}$$

where:

- $d\vec{\sigma} = \hat{n}\, d\sigma$ is the differential surface element oriented along the outward normal $\hat{n}$ to the surface $\Sigma(V)$.
- $\frac{\partial}{\partial n} = \hat{n} \cdot \nabla$ denotes the directional derivative along $\hat{n}$.
- The operator ∇ acts exclusively on the spatial coordinates $\vec{r}$.

This identity allows the solution of inhomogeneous differential equations (e.g. the Helmholtz equation) to be expressed in terms of integrals over the sources and the boundary conditions. The presence of $\hat{n}$ emphasises that the surface contribution depends critically on the geometry of the volume V.

5.4.2 The Helmholtz-Kirchoff integral theorem

The Helmholtz-Kirchoff integral theorem extends Green's formalism to electromagnetic fields. Starting from Green's identity and adding the term $k^2 fg$ to both sides, one obtains

$$\oint_{\Sigma(V)} [f(\vec{r})\nabla g(\vec{r}) - g(\vec{r})\nabla f(\vec{r})] \cdot d\vec{\sigma} = \int_V f(\vec{r}) \left(\nabla^2 + k^2\right) g(\vec{r})\, dV - \int_V g(\vec{r}) \left(\nabla^2 + k^2\right) f(\vec{r})\, dV, \tag{5.16}$$

where f and g are scalar functions. We now place the optical field in the slot of f, and in doing so we assert that the field obeys the homogeneous Helmholtz equation at every point of V, which is to say that the region in which we observe the light contains none of the matter that produced it. This is not a property of the equation but a description of the experiment, and it is the reason why the boundary data can determine the interior at all: if light were being created inside V no amount of information on $\Sigma(V)$ could account for it.

Assumption 8 *Every source of the field lies outside the volume V, so that inside it the amplitude satisfies the homogeneous equation $\left(\nabla^2+k^2\right)U=0$ and all the light observed within has entered through the boundary.*

Choosing then $f=U(\vec{r},\nu)$ and g as an arbitrary smooth function, the expression simplifies to

$$\oint_{\Sigma(V)}\left[U(\vec{r},\nu)\nabla g(\vec{r})-g(\vec{r})\nabla U(\vec{r},\nu)\right]\cdot d\vec{\sigma}=\int_V U(\vec{r},\nu)\left(\nabla^2+k^2\right)g(\vec{r})\,dV. \tag{5.17}$$

Everything now depends on what is put in the place of g, and the choice is not a matter of taste. It is worth stating plainly what we are trying to achieve, because the whole of the Kirchhoff theory is contained in that statement. We know, or we are willing to prescribe, the field on a closed surface, and we want the field at an arbitrary point of the region that the surface encloses. A partial differential equation does not answer such a question directly, since it only relates the field at a point to the field at the neighbouring points; to travel from the boundary to the interior one has to propagate that local relation across the whole region. What we want instead is a formula, an integral in which the boundary data enter as they stand and the field at the chosen interior point comes out.

The systematic way of manufacturing such a formula was invented by George Green in 1828, in a privately printed essay on electricity and magnetism that almost nobody read at the time (Green, 1828), and the idea is best appreciated through an algebraic analogy. A linear system $A\vec{x}=\vec{b}$ is solved once and for all as soon as the inverse matrix is known, and the inverse is characterised column by column by $A\vec{g}_j=\hat{e}_j$: the jth column of A^{-1} is the response of the system to a unit excitation concentrated in the jth slot, and the solution of the original problem is then the superposition $\vec{x}=\sum_j b_j\vec{g}_j$. A linear differential operator is the continuous limit of such a matrix. The discrete index becomes a point of space, the Kronecker delta $\hat{e}_j$ becomes the Dirac delta $\delta(\vec{r}-\vec{r}')$, the column $\vec{g}_j$ becomes a function $G(\vec{r},\vec{r}')$ of two points, and the sum over j becomes an integral over the source point.

Definition 29 *Let $\mathcal{L}$ be a linear differential operator acting on the variable $\vec{r}$. A Green's function of $\mathcal{L}$ is a solution, in the sense of distributions, of $\mathcal{L}\,G(\vec{r},\vec{r}')=\delta(\vec{r}-\vec{r}')$, up to a constant normalising the strength of the excitation; equivalently, it is the integral kernel of the inverse operator $\mathcal{L}^{-1}$, so that the solution of $\mathcal{L}u=f$ is $u(\vec{r}')=\int G(\vec{r},\vec{r}')f(\vec{r})\,dV$.*

Three features of that definition deserve emphasis. The first is that G is not a function in the ordinary sense but a distribution, and a singular one: it diverges where the excitation sits, and it acquires a meaning only under an integral sign, a notion that Dirac introduced in quantum mechanics (Dirac, 1930) and that Schwartz later placed on a rigorous footing (Schwartz, 1950). The second is that a Green's function is never unique, because any solution of the homogeneous equation $\mathcal{L}h=0$ may be added to it without spoiling the defining property; that freedom is not a defect but the resource which boundary conditions are there to consume, and we shall spend it deliberately in a moment. The third is that to know G is to have inverted the operator once and for all: the labour of solving the equation has been performed in advance for the most elementary excitation conceivable, and every other problem governed by the same operator is thereby reduced to a quadrature.

The reason why this is exactly what our problem asks for lies in the structure of Green's identity, which is a bilinear pairing: it couples the unknown U with a second function that we are entirely free to invent. If we feed into it a function whose image under $(\nabla^2 + k^2)$ is concentrated at a single point, the volume integral on the right ceases to be an integral at all and collapses into the value of U at that point, while the left-hand side retains the form of an integral over the boundary. In one stroke the differential problem becomes the representation formula we were after. Moreover, the non-uniqueness just mentioned is precisely what will allow us to suppress one of the two boundary terms, and with it the over-determination that Poincaré objected to in Kirchhoff's formulation.

The key lies in selecting g as the Green's function $G(\vec{r}, \vec{r}')$, defined by

$$\left(\nabla^2 + k^2\right) G(\vec{r}, \vec{r}') = -4\pi\delta(\vec{r} - \vec{r}'). \tag{5.18}$$

The constant -4π on the right is only a normalisation, chosen so that the factors of the spherical geometry come out tidily and so that the elementary solution turns out to be the outgoing spherical wave with a positive amplitude; what carries the physics is the delta itself, and it is worth explaining why it is there, because the usual answer, that it is mathematically convenient, misses the point entirely.

Recall where a source term belongs in a wave equation. The homogeneous Helmholtz equation asserts that at the point considered no light is being created: whatever amplitude is found there has arrived from somewhere else. A non-vanishing right-hand side asserts the opposite, that light is being poured into the medium at that place, and the number multiplying it measures how much. This is a quantitative statement and not a figure of speech, as one sees by integrating (5.18) over a small ball B_ε centred on $\vec{r}'$ and applying the divergence theorem to the first term,

$$\oint_{\partial B_\varepsilon} \nabla G \cdot d\vec{\sigma} \; + \; k^2 \int_{B_\varepsilon} G \, dV \;=\; -4\pi. \tag{5.19}$$

The volume term will be seen to vanish with ε, so that the flux of ∇G through any sphere surrounding the point equals -4π however small the sphere is made. A quantity whose flux across a closed surface does not depend on how tightly that surface is drawn around the point is exactly what one means by a source of fixed strength, and the delta is its mathematical name. Equation (5.18) therefore reads, in words: *the medium is empty everywhere except at the single point $\vec{r}'$, where a monochromatic emitter of prescribed strength is radiating.*

Such an emitter is a physical idealisation and not a mathematical fiction. Light is emitted by atoms, and an atom is smaller than the wavelength of the light it radiates by some three orders of magnitude; on the scale at which a diffraction pattern is formed, a real elementary emitter is a point to an accuracy far beyond what any experiment can resolve. The delta is the honest description of an object whose size is irrelevant to the phenomenon being described, in the same way that a planet is a point for the purposes of celestial mechanics. And because the medium is linear, so that fields superpose, the point emitter is the atom of the whole theory: a source of any shape is a sum of point emitters weighted by its density, and the field it radiates is the corresponding sum of elementary fields. To solve the problem for a delta is therefore to solve it for every source at once, which is why one solves it first, and it is also why the generalisation to extended sources sketched in Figure 5.3 costs nothing conceptually.

Assumption 9 *The elementary emitter is a point, its linear dimensions being negligible beside the wavelength of the light it radiates, so that its excitation may be represented by a Dirac delta.*

In our context the Green's function admits an immediate optical reading. $G(\vec{r},\vec{r}')$ is the complex amplitude that a point emitter sitting at $\vec{r}'$ produces at the point $\vec{r}$ of the same medium, and the spherical wave (5.21) that we shall obtain below is nothing other than that amplitude. But this is precisely the object that Huygens postulated in 1690, the secondary wavelet emitted by every element of a wavefront (Huygens, 1690). Huygens had to assume the wavelets, and Fresnel had to assume in addition that they interfere and to supply their amplitudes and phases by hand; here the wavelet is not postulated but derived, as the solution of the field equation for an elementary source, and the weight with which each element of the surface contributes is not guessed but delivered by Green's identity. The Helmholtz-Kirchhoff formula (5.22) is, in this sense, the Huygens-Fresnel principle written as a theorem instead of as a hypothesis.

One last remark on the delta will dispose of what is otherwise a genuine perplexity. The source in (5.18) is not placed where the light actually comes from; it is placed at $\vec{r}'$, which is the point at which we wish to know the field. This looks strange until one recalls that the Green's function of the Helmholtz operator is symmetric in its two arguments, $G(\vec{r},\vec{r}') = G(\vec{r}',\vec{r})$, a statement which is nothing but the optical reciprocity theorem: the amplitude that a source at $\vec{r}'$ sends to $\vec{r}$ equals the amplitude that a source at $\vec{r}$ sends to $\vec{r}'$. The delta at the observation point is therefore a probe. We interrogate the region by placing a unit emitter at the point of interest and asking how strongly it illuminates the boundary; reciprocity converts that answer into the statement of how strongly each element of the boundary illuminates the point, and this is the Huygens sum. Without the delta the volume integral in the Helmholtz-Kirchhoff identity would vanish identically and the identity would degenerate into a relation among the boundary data alone, saying nothing whatever about the interior. It is the delta that singles out the observation point and turns an identity into a solution.

Remarkably, we may add an arbitrary number of Green's functions while keeping the solution invariant. What we can do is add an infinity of image charges, but *outside* the integration volume bounded by the surface $\Sigma(V)$, as shown in Figure 5.1.

In this spirit, equation (5.18) becomes

$$(\nabla^2 + k^2)G(\vec{r},\vec{r}') = -4\pi\delta(\vec{r}-\vec{r}') - 4\pi\sum_{i=1}^{\infty}\delta(\vec{r}-\vec{r}_i''). \tag{5.20}$$

The explicit solution for a homogeneous medium of equation (5.18) is

$$G(\vec{r},\vec{r}') = \frac{e^{ik\|\vec{r}-\vec{r}'\|}}{\|\vec{r}-\vec{r}'\|}, \tag{5.21}$$

corresponding to an outgoing spherical wave centred at $\vec{r}'$, as will be proved in (5.25). Substituting G into Green's identity, and using that the volume integral picks up the value $-4\pi U(\vec{r}',\nu)$ on account of the delta, one derives

$$U(\vec{r}',\nu) = -\frac{1}{4\pi}\oint_{\Sigma(V)}\left[U(\vec{r},\nu)\nabla G(\vec{r},\vec{r}') - G(\vec{r},\vec{r}')\nabla U(\vec{r},\nu)\right]\cdot d\vec{\sigma}, \tag{5.22}$$

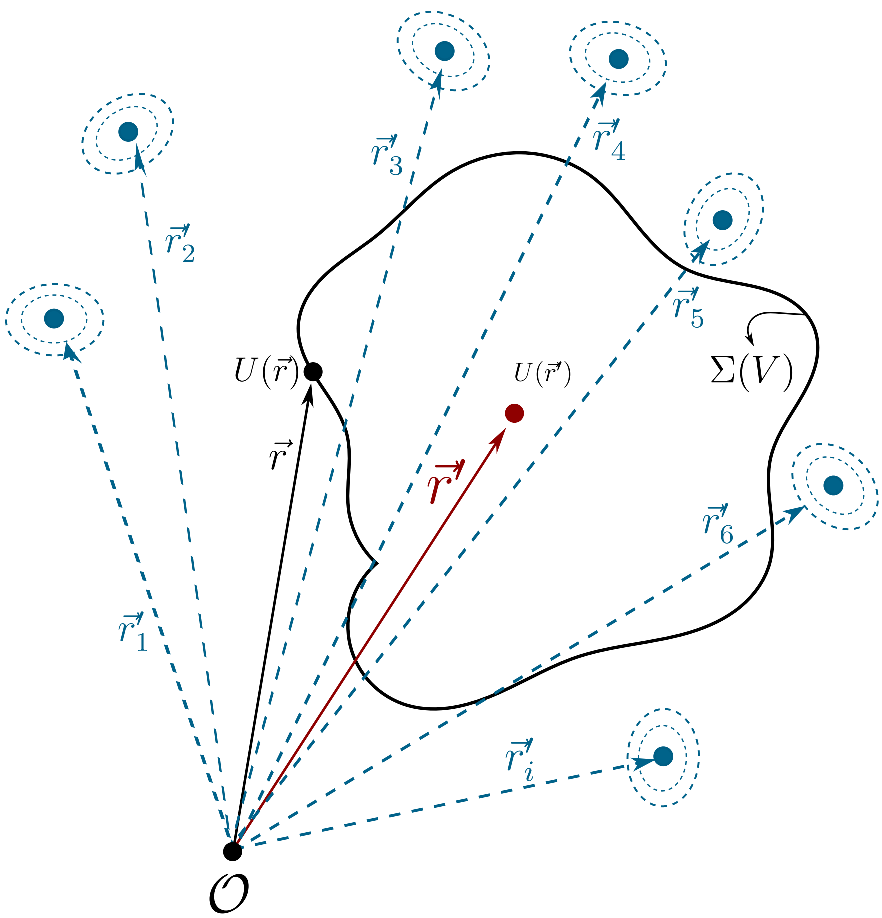


Figure 5.1: On the boundary $\Sigma(V)$ the contribution of the field components $U(\vec{r})$ is shown; inside the region the total field is built from the contribution of all the points on the boundary. One may add infinitely many point Green charges of arbitrary propagation nature; the condition for the invariance of equation (5.18) is that these image charges lie outside $\Sigma(V)$.

known as the Helmholtz-Kirchhoff integral formula. This equation allows $U(\vec{r}', \nu)$ to be computed at any point of the volume V by means of an integral over its boundary $\Sigma(V)$.

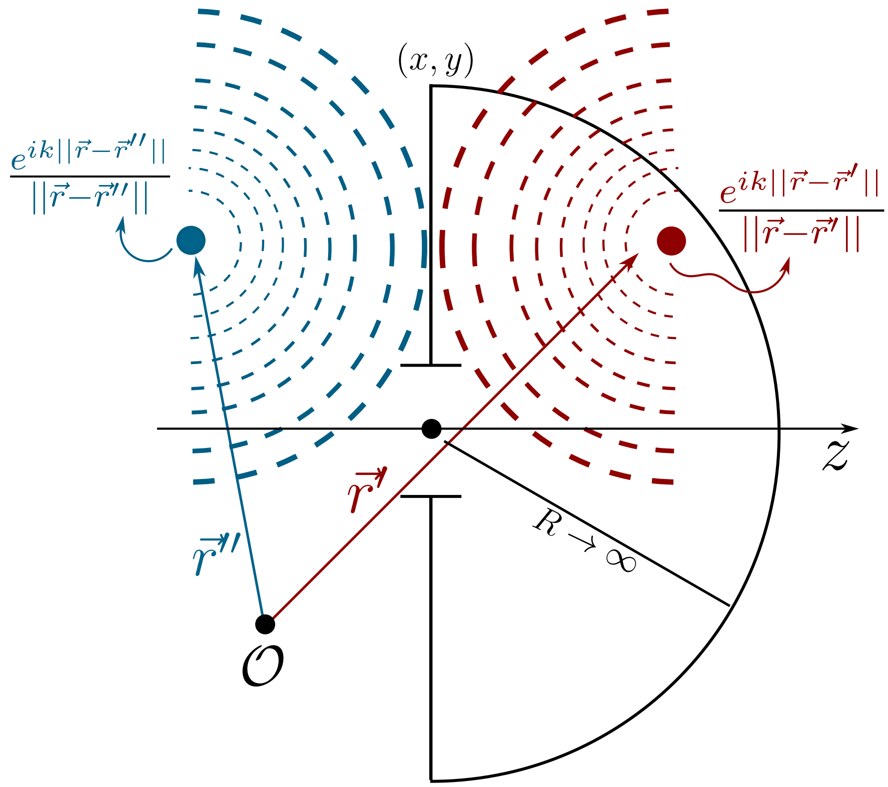


Figure 5.2: Representation of the Green's functions for point sources. Under these boundary conditions the Green's functions behave as point charges, one the mirror image of the other.

Before continuing, it is worth noting that the point-charge nature of the Green's functions $G(\vec{r}, \vec{r}')$ is an assumption taken for granted; there is, however, no restriction on their physical nature. As is well known in electromagnetic theory, radiation sources may have any volume and shape, so our problem (Figure 5.2) can be made more general, as shown in Figure 5.3, and the differential equation for Green's functions with volume would read

$$\left(\nabla^2 + k^2\right) G(\vec{r}, \vec{r}') = -4\pi\rho(\vec{r} - \vec{r}'_+) + 4\pi\rho(\vec{r} - \vec{r}'_-). \tag{5.23}$$

Let us continue with the development for point Green's functions. To simplify the evaluation of the integral, we consider a volume V bounded by an infinite plane Σ_1 ($z = 0$) and a hemisphere Σ_2 of radius R (see Figure 5.2). Closing the surface at infinity is legitimate only if nothing arrives from there, and once again that is a decision about the experiment rather than a consequence of the mathematics: it excludes any wave converging on the aperture from the far side, which is precisely what one arranges in the laboratory by putting every source of light behind the screen and by working in a darkened room.

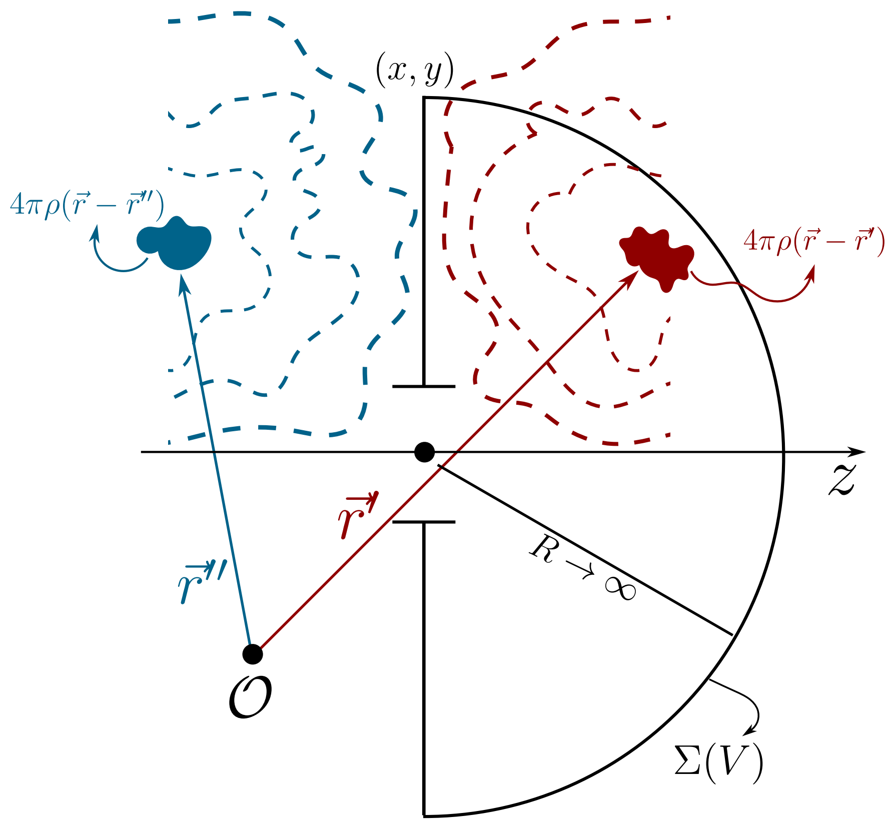


Figure 5.3: Representation of the Green's functions for non-point sources. Under these boundary conditions the Green's functions still behave as mirror images, but now with given volume and geometry; the only constraint on the emission of these sources is that, on the boundary $z = 0$, the total field must vanish, as required by the assumed Dirichlet conditions.

Assumption 10 *No radiation reaches the observation region from infinity, the field on the hemisphere* Σ_2 *being purely outgoing and obeying the Sommerfeld radiation condition* (5.29)*, so that the contribution of* Σ_2 *vanishes as* $R \to \infty$*.*

The Helmholtz-Kirchhoff formula is thereby reduced to

$$U(\vec{r}', \nu) = -\frac{1}{4\pi} \int_{\Sigma_1} \left[U(\vec{r}, \nu) \frac{\partial G}{\partial n} - G(\vec{r}, \vec{r}') \frac{\partial U}{\partial n} \right] d\sigma, \tag{5.24}$$

where $\frac{\partial}{\partial n} = \hat{n} \cdot \nabla$ denotes the normal derivative. However, this formulation requires specifying simultaneously U and $\partial U/\partial n$ on Σ_1, which over-determines the problem. This difficulty motivates the introduction of physically realistic boundary conditions, such as those of Kirchhoff, in order to resolve the ambiguity.

This problem is aggravated by a theorem of complex analysis, attributed to Riemann, which states that if a function and its normal derivative vanish on an open region of a plane, then the function must be identically zero throughout the domain. This restriction motivates the strategic need to eliminate one of the terms in the Helmholtz-Kirchhoff surface integral.

The key is to exploit the degrees of freedom inherent in the choice of the auxiliary Green's function G. Kirchhoff initially proposed the intuitive boundary conditions

- $U = 0$ outside the aperture on Σ (blocked field),
- $\partial U/\partial n = 0$ on opaque regions (vanishing normal derivative).

Although these conditions reproduced the experimental results of diffraction, they introduced mathematical inconsistencies by violating the Cauchy-Riemann uniqueness theorem. Poincaré showed that such assumptions generate contradictions in the mixed derivatives of the field. The resolution came with Sommerfeld, who redefined G by means of a double-layer (mirror) Green's function, ensuring that

- $G = 0$ on Σ (Dirichlet condition),
- $\partial G/\partial n$ remains non-singular,

thus eliminating the redundancy in the boundary conditions and preserving the self-consistency of the formalism.

Sommerfeld's device removes one of the two boundary terms, but it does not remove the need to know the one that survives. The Rayleigh-Sommerfeld formula still requires the value of U at every point of the plane Σ_1, and no theory of diffraction can supply it, because obtaining it would mean solving exactly the electromagnetic problem of the screen that we set out to avoid. What is done instead is to guess it, and the guess is Kirchhoff's. It rests on two separate idealisations, one about the obstacle and one about the field on it, and both are worth stating as what they are.

Assumption 11 *The screen is a plane obstacle of vanishing thickness, perfectly opaque and perfectly absorbing, whose material properties play no part in the problem and whose edges do not perturb the field in their neighbourhood.*

Assumption 12 *Over the aperture the field coincides with the field that would be present had the screen never been introduced, and over the opaque part of the screen it vanishes identically.*

The second of these is the weakest link of the whole construction, and it is instructive to see why. A field that vanishes on part of a plane and equals a smooth incident wave on the rest is discontinuous at the rim of the aperture, so it cannot be an exact solution of the Helmholtz equation; the boundary values we prescribe are, strictly speaking, incompatible with the equation we are solving. The error is confined to a border a few wavelengths wide around the rim, which is why the theory works so well for apertures many wavelengths across and fails for apertures comparable with the wavelength.

5.4.3 The Rayleigh-Sommerfeld formulation

Twice already we have used the explicit form (5.21) of the Green's function without justifying it, and since the whole of the Rayleigh-Sommerfeld construction rests both on the shape of that function and on the freedom to add a second one to it, this is the place to show where it comes from.

Theorem 14 *Let* $R = \|\vec{r} - \vec{r}'\|$. *In a homogeneous, isotropic and unbounded medium the only solution of the Green's equation* (5.18) *that carries energy away from the source is the outgoing spherical wave*

$$G(\vec{r}, \vec{r}') = \frac{e^{ikR}}{R}, \tag{5.25}$$

its amplitude being fixed by the strength assigned to the point emitter and the converging solution being excluded by the radiation condition.

The proof has four steps, and each of them is a physical statement before it is a mathematical one. The first is a matter of symmetry. The medium is homogeneous and isotropic and the source is a single point, so that once the origin has been placed at $\vec{r}'$ nothing in the problem distinguishes one direction of space from another; the solution can depend only on the distance R to the source, and we may write $G = G(R)$.

The second step is to solve the equation away from the source, where it is homogeneous. For a function of R alone the Laplacian reduces to

$$\nabla^2 G = \frac{1}{R^2}\frac{d}{dR}\left(R^2 \frac{dG}{dR}\right) = \frac{1}{R}\frac{d^2}{dR^2}\left(R\,G\right), \tag{5.26}$$

where the second form is the identity (3.25) already established in the third chapter, and which suggests at once the substitution $\chi(R) = R\,G(R)$. With it the homogeneous Helmholtz equation becomes

$$\frac{d^2\chi}{dR^2} + k^2\chi = 0, \tag{5.27}$$

which is the equation of a harmonic oscillator in the variable R and whose general solution is $\chi = A\,e^{ikR} + B\,e^{-ikR}$. Undoing the substitution,

$$G(R) = A\,\frac{e^{ikR}}{R} + B\,\frac{e^{-ikR}}{R}, \tag{5.28}$$

so that the amplitude of a spherically symmetric monochromatic field falls necessarily as the inverse of the distance. This is not an accident of the algebra. The energy crossing a sphere of radius R is proportional to $|G|^2R^2$, and it is only with the $1/R$ dependence that this quantity is independent of R, as the conservation of energy demands in a transparent medium.

The third step selects between the two terms of (5.28). Restoring the temporal factor of the Fourier convention adopted in (5.9), the two solutions read $e^{i(kR-\omega t)}/R$ and $e^{-i(kR+\omega t)}/R$, and their surfaces of constant phase obey respectively $R = ct + \text{const}$ and $R = -ct + \text{const}$. The first expands and the second contracts. Our point is an emitter and not a sink, and the light must leave it and never come back, which is the physical content of the Sommerfeld radiation condition already invoked when the hemisphere Σ_2 was discarded,

$$\lim_{R\to\infty} R\left(\frac{\partial G}{\partial R} - ikG\right) = 0. \tag{5.29}$$

The converging wave violates it, since for it the bracket behaves as $-2ikB\,e^{-ikR}/R$ and the product with R does not tend to zero. Hence $B = 0$.

The fourth step is the only one in which the delta intervenes, and it fixes the amplitude A. Integrate (5.18) over the ball B_ε of radius ε centred on the source, which is what produced (5.19). With $G = A\,e^{ikR}/R$ the volume term is bounded by

$$\left|k^2\int_{B_\varepsilon} G\,dV\right| = \left|4\pi A k^2\int_0^\varepsilon \frac{e^{ikR}}{R}\,R^2\,dR\right| \le 2\pi|A|\,k^2\varepsilon^2 \xrightarrow[\varepsilon\to 0]{} 0,$$

so that the singularity of G is mild enough to be integrable and the entire source strength has to come from the flux term. Since $\partial G/\partial R = A\,e^{ikR}\left(ik/R - 1/R^2\right)$, and the sphere ∂B_ε has area $4\pi\varepsilon^2$ with outward normal $\hat{R}$,

$$\oint_{\partial B_\varepsilon} \nabla G\cdot d\vec{\sigma} = 4\pi\varepsilon^2\,A\,e^{ik\varepsilon}\left(\frac{ik}{\varepsilon} - \frac{1}{\varepsilon^2}\right) = 4\pi A\,e^{ik\varepsilon}\,(ik\varepsilon - 1) \xrightarrow[\varepsilon\to 0]{} -4\pi A,$$

and (5.19) then gives $-4\pi A = -4\pi$, that is $A = 1$, which proves (5.25) and completes the argument.

The sign of the source deserves a word, because it is the reason why (5.18) carries -4π and not $+4\pi$ on the right. The last step of the proof shows that the flux of ∇G out of a small sphere is

$-4\pi A$, so that a positive amplitude A demands a negative constant in the equation; had we written $+4\pi\delta(\vec{r}-\vec{r}')$ we should have been forced to accept $A=-1$, that is $G=-e^{ikR}/R$, and every formula written from (5.21) onwards would have changed sign. The convention adopted here is the one used by Sommerfeld, Born and Wolf and Goodman (Sommerfeld, 1954; Born and Wolf, 1999; Goodman, 2005), and it is the convention for which the diffraction integral comes out with the familiar constant $1/(i\lambda)$. What the computation establishes, beyond the bookkeeping of signs, is that the elementary solution is an outgoing spherical wave centred on the source point whose amplitude is inversely proportional to the distance. This is the Huygens wavelet, now obtained rather than assumed.

Returning to equation (5.22) and choosing the Dirichlet condition for the function G, the second term of the integrand disappears and we are left with

$$U(\vec{r}',\nu) = -\frac{1}{4\pi}\int_{\Sigma_1} \left[U(\vec{r},\nu)\nabla G\cdot\hat{n}\right] d\sigma. \tag{5.30}$$

Here $\hat{n}$ is the normal that points *out* of V, and since V is the half space $z>0$, on the plane Σ_1 that normal points downwards, $\hat{n}=-\hat{z}$. It is worth writing this out explicitly, because it is the sign most easily lost in the whole construction,

$$U(\vec{r}',\nu) = \frac{1}{4\pi}\int_{\Sigma_1} U(\vec{r},\nu)\,\nabla G\cdot\hat{z}\; d\sigma. \tag{5.31}$$

Given the integration surface under consideration, we may express (5.21) as

$$G(\vec{r},\vec{r}') = \frac{e^{ik\|\vec{r}-\vec{r}'\|}}{\|\vec{r}-\vec{r}'\|} - \frac{e^{ik\|\vec{r}-\vec{r}''\|}}{\|\vec{r}-\vec{r}''\|}, \tag{5.32}$$

where

$$\|\vec{r}-\vec{r}''\| = \|\vec{r}-\vec{r}'\|, \qquad x''=x', \quad y''=y', \quad z''=-z', \tag{5.33}$$

since $\vec{r}''$ is the mirror image of $\vec{r}'$ across the plane Σ_1.

That (5.32) is a legitimate Green's function is verified at once with the theorem just proved. The observation point $\vec{r}'$ lies in the half space $z>0$, so its image $\vec{r}''$ lies in $z<0$, outside the volume V; within V the second term is therefore a perfectly regular solution of the homogeneous Helmholtz equation, and adding it alters neither the source at $\vec{r}'$ nor the value of the volume integral, in accordance with the freedom discussed after (5.18). On the plane Σ_1, on the other hand, every point is equidistant from $\vec{r}'$ and from $\vec{r}''$, so that the two spherical waves arrive there with the same amplitude and the same phase and cancel identically, giving $G=0$ over the whole of Σ_1. This is the Dirichlet condition we were looking for, obtained by the same method of images that solved the Michelson interferometer in the previous chapter, and it is what removes the term in $\partial U/\partial n$ from (5.22) and with it the over-determination of Kirchhoff's boundary conditions.

All that (5.31) still requires is the axial derivative of G on the plane. This is the only computation of the section, and since a sign lost here changes the sign of every diffraction formula in the rest of the chapter, we shall carry it out without omitting a single step. To lighten the notation write

$$r_1 = \|\vec{r}-\vec{r}'\|, \qquad r_2 = \|\vec{r}-\vec{r}''\|, \qquad f(\rho) = \frac{e^{ik\rho}}{\rho}, \tag{5.34}$$

so that (5.32) reads simply $G = f(r_1) - f(r_2)$.

First step: the derivative of f. Differentiating the quotient,

$$\frac{df}{d\rho} = \frac{\left(ik\, e^{ik\rho}\right)\rho - e^{ik\rho}}{\rho^2} = \frac{e^{ik\rho}}{\rho}\left(ik - \frac{1}{\rho}\right) = \left(ik - \frac{1}{\rho}\right) f(\rho). \tag{5.35}$$

Second step: the gradient of a distance. With $\vec{r} = (x, y, z)$ the point of integration and $\vec{r}' = (x', y', z')$ held fixed,

$$r_1 = \left[(x-x')^2 + (y-y')^2 + (z-z')^2\right]^{1/2}, \tag{5.36}$$

and differentiating with respect to z, which is the only component we shall need,

$$\frac{\partial r_1}{\partial z} = \frac{1}{2}\,\frac{2\,(z-z')}{[(x-x')^2 + (y-y')^2 + (z-z')^2]^{1/2}} = \frac{z-z'}{r_1}, \tag{5.37}$$

the same computation with $\vec{r}''$ in place of $\vec{r}'$ giving $\partial r_2/\partial z = (z - z'')/r_2$. Carried out for the three components at once these two results read $\nabla r_1 = (\vec{r} - \vec{r}')/r_1$ and $\nabla r_2 = (\vec{r} - \vec{r}'')/r_2$: the gradient of a distance is the unit vector that points away from the centre from which the distance is measured.

Third step: the gradient of G. Applying the chain rule to each term and using (5.35),

$$\nabla G = \left(ik - \frac{1}{r_1}\right)\frac{e^{ikr_1}}{r_1}\,\frac{\vec{r} - \vec{r}'}{r_1} \;-\; \left(ik - \frac{1}{r_2}\right)\frac{e^{ikr_2}}{r_2}\,\frac{\vec{r} - \vec{r}''}{r_2}. \tag{5.38}$$

Fourth step: the projection on the axis. Taking the scalar product of (5.38) with $\hat{z}$, which amounts to keeping the third component of each of the two vectors,

$$\nabla G \cdot \hat{z} = \left(ik - \frac{1}{r_1}\right)\frac{e^{ikr_1}}{r_1}\,\frac{z - z'}{r_1} \;-\; \left(ik - \frac{1}{r_2}\right)\frac{e^{ikr_2}}{r_2}\,\frac{z - z''}{r_2}. \tag{5.39}$$

Fifth step: evaluation on the plane. The integral of (5.31) runs over Σ_1, so (5.39) is needed only at $z = 0$. There the two distances coincide, $r_1 = r_2 \equiv r$, which is the mirror relation already used, and the two axial displacements are

$$\left(z - z'\right)\big|_{z=0} = -z', \qquad \left(z - z''\right)\big|_{z=0} = -z'' = z', \tag{5.40}$$

because $z'' = -z'$. This is the crux of the whole calculation: the source and its image lie on opposite sides of the plane, so their axial direction cosines are equal in magnitude and opposite in sign. Substituting (5.40) into (5.39), the common factor $(ik - 1/r)\,e^{ikr}/r$ may be taken out and what remains is a difference of two opposite numbers,

$$\nabla G \cdot \hat{z}\big|_{\Sigma_1} = \left(ik - \frac{1}{r}\right)\frac{e^{ikr}}{r}\left[\frac{-z'}{r} - \frac{z'}{r}\right] = -\,\frac{2z'}{r}\left(ik - \frac{1}{r}\right)\frac{e^{ikr}}{r}. \tag{5.41}$$

The factor of two is worth a comment. The minus sign that separates the two terms of (5.32) is what makes the two spherical waves cancel on the plane; the very same minus sign, combined with the opposite direction cosines of (5.40), makes their axial derivatives add. The Dirichlet Green's function vanishes on Σ_1 and has there exactly twice the normal derivative of a single wavelet, which is the analytical trace of the fact that the screen behaves as a perfect mirror.

Sixth step: the obliquity factor. It only remains to name the geometrical quantity z'/r. Let θ be the angle between the axis $\hat{z}$ and the vector $\vec{r}' - \vec{r}$ that goes from the point of the aperture to the point of observation, so that

$$\cos\theta = \hat{z} \cdot \frac{\vec{r}' - \vec{r}}{\|\vec{r}' - \vec{r}\|} = \frac{z'}{r}, \tag{5.42}$$

a quantity that is positive because the observation point lies in $z > 0$, and that equals unity for a point of the aperture lying on the axis of observation. With this notation (5.41) takes its final form,

$$\nabla G \cdot \hat{z}|_{\Sigma_1} = -2\left[ik - \frac{1}{r}\right]\frac{e^{ikr}}{r}\cos\theta. \tag{5.43}$$

It is essential to keep in mind which of the two vectors the angle is measured from. Had we written the direction cosine of $\vec{r} - \vec{r}'$, which points from the observation point back to the aperture, we should have obtained the opposite number, and the diffraction integral would have come out with the wrong sign.

Substituting (5.43) into (5.31), the two factors of two combine with the 4π and we obtain

$$U(\vec{r}', \nu) = -\frac{1}{2\pi}\int_\Sigma U(\vec{r}, \nu)\left[ik - \frac{1}{\|\vec{r} - \vec{r}'\|}\right]\frac{e^{ik\|\vec{r}-\vec{r}'\|}}{\|\vec{r} - \vec{r}'\|}\cos\theta\, d\sigma. \tag{5.44}$$

Equation (5.44) is exact within the scalar theory, and it is the last expression in this chapter of which that can be said. To bring it to the form in which it is actually used one discards the term $1/\|\vec{r} - \vec{r}'\|$ against k, which is permissible only when the observation point stands many wavelengths from the aperture. For visible light and laboratory distances the discarded ratio $\lambda/(2\pi\|\vec{r} - \vec{r}'\|)$ is of the order of 10^{-7}, so that the approximation is superb, but it is an approximation and not an identity.

Assumption 13 *The observation point lies many wavelengths away from every point of the aperture, $\|\vec{r} - \vec{r}'\| \gg \lambda$, so that the near field term $1/\|\vec{r} - \vec{r}'\|$ is negligible beside the wavenumber k.*

With it we obtain

$$U(\vec{r}', \nu) = -\frac{ik}{2\pi}\int_\Sigma U(\vec{r}, \nu)\frac{e^{ik\|\vec{r}-\vec{r}'\|}}{\|\vec{r} - \vec{r}'\|}\cos\theta\, d\sigma = \frac{1}{i\lambda}\int_\Sigma U(\vec{r}, \nu)\frac{e^{ik\|\vec{r}-\vec{r}'\|}}{\|\vec{r} - \vec{r}'\|}\cos\theta\, d\sigma, \tag{5.45}$$

where the last equality is nothing but $-ik/2\pi = -i/\lambda = 1/(i\lambda)$. This is a mathematical expression that embodies the Huygens-Fresnel principle: as can be appreciated, there is one spherical wave per point of the integration surface, weighted by the obliquity factor $\cos\theta$ and by the constant $1/(i\lambda)$, which is the constant Fresnel had to postulate and which the derivation now delivers, quarter-wave advance and $1/\lambda$ included.

5.5 The Fresnel approximation

Starting from the field

$$U(\vec{r}') = -\frac{ik}{2\pi}\int_{\Sigma} U(\vec{r})\frac{e^{ik\|\vec{r}-\vec{r}'\|}}{\|\vec{r}-\vec{r}'\|}\cos\theta\,\mathrm{d}\sigma, \tag{5.46}$$

it is possible to carry out the so-called mid-field or Fresnel approximation. The expansion on which it rests is a binomial one, and a binomial series may be truncated only when the quantity expanded is small; here that quantity is the ratio of the transverse separation to the propagation distance, so that truncating amounts to confining the entire problem to a narrow cone about the optical axis.

Assumption 14 *The transverse dimensions of the aperture and of the observation region are small compared with the propagation distance z', and the quartic term omitted from the expansion contributes a phase far below one radian, $k\left[(x-x')^2+(y-y')^2\right]^2/8z'^3 \ll 1$.*

To this end we recall that the integration runs over the plane Σ_1, on which $z=0$, so that the axial separation between the two points is the coordinate z' of the observation point alone,

$$\begin{aligned}
\|\vec{r}-\vec{r}'\| &= \left.\left[(x-x')^2+(y-y')^2+(z-z')^2\right]^{1/2}\right|_{z=0}\\
&= \left[(x-x')^2+(y-y')^2+z'^2\right]^{1/2}\\
&= z'\left(1+\frac{(x-x')^2+(y-y')^2}{z'^2}\right)^{1/2}\\
&\approx z'+\frac{(x-x')^2+(y-y')^2}{2z'}.
\end{aligned} \tag{5.47}$$

It is z', and not z, that measures the propagation distance. The aperture lies at the origin of the axis and it is the observation point that travels, which is the same convention that produced the obliquity factor $\cos\theta = z'/r$ of (5.42) and the evaluation at $z=0$ of (5.40). Setting $z'=0$ and factoring z instead would place the observation point on the screen and make the propagation distance vanish identically. It is clear that, for the factor $\frac{1}{\|\vec{r}-\vec{r}'\|}$, an expansion up to the first term of (5.47) suffices; this is not enough for the factor $e^{ik\|\vec{r}-\vec{r}'\|}$, however, since the latter has oscillatory behaviour over its domain, so both terms are required. The two factors are therefore treated with different degrees of accuracy, and that asymmetry is itself a hypothesis: it presumes that an error of a fraction of a wavelength, harmless in a slowly varying amplitude, is fatal in a phase, which is true precisely because k is enormous on the scale of the aperture.

Assumption 15 *In the amplitude the distance may be replaced by the axial separation, $\|\vec{r}-\vec{r}'\| \approx z'$, while in the phase the quadratic term of (5.47) must be retained.*

We additionally take, for the obliquity factor (5.42), $\cos\theta = \dfrac{z'}{\|\vec{r} - \vec{r}'\|} \approx 1$, which is legitimate in the paraxial regime because the observation point is then close to the axis.

Assumption 16 *The obliquity factor is unity,* $\cos\theta \approx 1$, *every direction of interest making a small angle with the optical axis.*

In this spirit, we obtain an expression for the field of the form

$$U(\vec{r}') = \frac{e^{ikz'}}{i\lambda z'} \int U(\vec{r}) \exp\left(\frac{ik}{2z'}((x - x')^2 + (y - y')^2)\right) d\sigma, \tag{5.48}$$

which describes the propagation or diffraction of the electromagnetic field in the Fresnel regime. The constant $1/(i\lambda z')$, inherited from (5.45), is the one for which the kernel of (5.48) tends to $\delta(\vec{r} - \vec{r}')$ as $z' \to 0$, so that a propagation over no distance at all leaves the field unchanged; this is the check that fixes the sign beyond any doubt. It is worth stressing that equation (5.48) describes a parabolic wave, and although, at least conceptually, such propagation is valid for a mid-field, in practice it has proved more than sufficient for sufficiently short distances.

5.5.1 Example: Fraunhofer diffraction between two confocal spheres

Starting from the Fresnel integral (5.48), *show that between two confocal spheres one obtains Fraunhofer diffraction.*

Before anything else it is convenient to expand the square in the exponent of (5.48). Writing $r^2 = x^2 + y^2$ for the transverse radius on the emitting plane and $r'^2 = x'^2 + y'^2$ for the one on the receiving plane,

$$(x - x')^2 + (y - y')^2 = r^2 + r'^2 - 2\left(xx' + yy'\right), \tag{5.49}$$

so that the Fresnel integral separates into three factors,

$$U(\vec{r}') = \frac{e^{ikz'}}{i\lambda z'} \int U(\vec{r})\, e^{\frac{ik}{2z'}r^2}\, e^{\frac{ik}{2z'}r'^2}\, e^{-\frac{ik}{z'}(xx'+yy')}\, d\sigma. \tag{5.50}$$

The last factor is already the kernel of a Fourier transformation. The obstacle is the pair of quadratic factors, and the whole point of the exercise is that they are not an obstacle at all if one agrees to read the field on a curved surface instead of on a plane.

Consider a spherical cap tangent to a transverse plane at the optical axis, with its centre of curvature on the axis at the signed distance c from the vertex, counted positive in the direction of propagation. The point of the cap at the transverse position $\vec{r}$ is displaced axially from the tangent plane by the sagitta

$$\delta(r) = c - \operatorname{sgn}(c)\sqrt{c^2 - r^2} = c - \operatorname{sgn}(c)\,|c|\sqrt{1 - \frac{r^2}{c^2}} \approx c - c\left(1 - \frac{r^2}{2c^2}\right),$$

where we used $\operatorname{sgn}(c)\,|c| = c$ and expanded the root to first order, which is the same paraxial expansion already performed in (5.47). The two terms in red cancel and there remains

$$\delta(r) = \frac{r^2}{2c}. \tag{5.51}$$

Since a field propagating in the positive direction acquires the factor $e^{ik\delta}$ over an axial displacement δ, the field read on the cap and the field read on the tangent plane are related by

$$u_{\text{cap}}(\vec{r}) = U(\vec{r})\exp\left(\frac{ik\,r^2}{2c}\right). \tag{5.52}$$

Two spheres are said to be **confocal** when each one has its centre of curvature at the vertex of the other. Let therefore $\mathcal{A}$ be the cap tangent to the emitting plane $\mathcal{P}$ at the vertex O and centred at O', and let $\mathcal{F}$ be the cap tangent to the receiving plane $\mathcal{Q}$ at the vertex O' and centred at O, as shown in Figure 5.4. Both radii are then equal to the propagation distance, and with the sign convention of (5.52) the two signed distances are

$$c_{\mathcal{A}} = +z', \qquad c_{\mathcal{F}} = -z', \tag{5.53}$$

the second one being negative because the centre of $\mathcal{F}$ lies upstream of its own vertex.

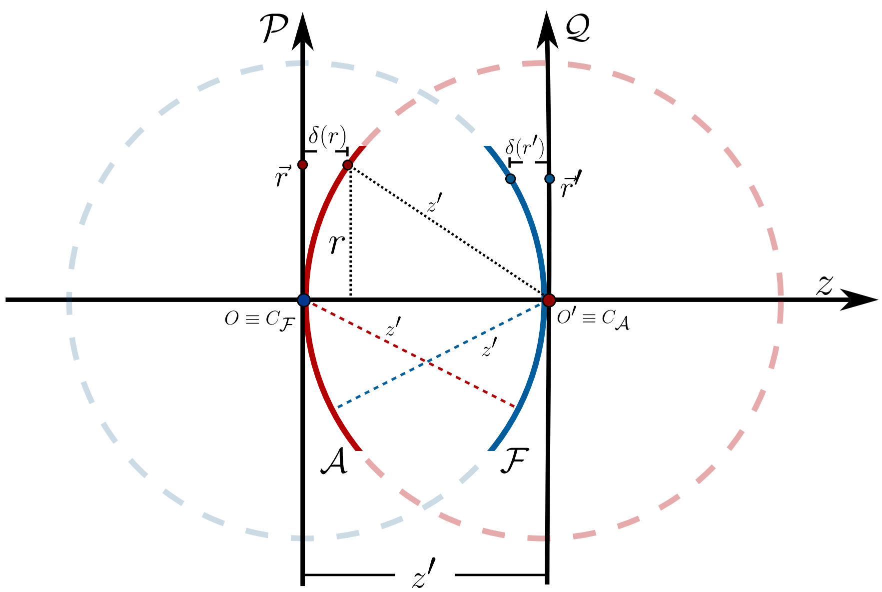


Figure 5.4: Geometry of the two confocal spheres. The cap $\mathcal{A}$ is tangent to the emitting plane $\mathcal{P}$ at the vertex O and has its centre of curvature at O', while the cap $\mathcal{F}$ is tangent to the receiving plane $\mathcal{Q}$ at O' and has its centre of curvature at O, so that both radii equal the propagation distance z'. Read on these two surfaces, the Fresnel propagation reduces to an exact Fourier transformation, which in the language of the fractional Fourier transformation of Section 5.7 is the order $\alpha = \pi/2$.

Applying (5.52) on each side with the radii (5.53),

$$U(\vec{r}) = u_{\mathcal{A}}(\vec{r})\, e^{-\frac{ik}{2z'}r^2}, \qquad u_{\mathcal{F}}(\vec{r}\,') = U(\vec{r}\,')\, e^{-\frac{ik}{2z'}r'^2}, \tag{5.54}$$

and substituting both relations into (5.50),

$$u_{\mathcal{F}}(\vec{r}\,') = \frac{e^{ikz'}}{i\lambda z'}\, e^{-\frac{ik}{2z'}r'^2}\, e^{\frac{ik}{2z'}r'^2} \int u_{\mathcal{A}}(\vec{r})\, e^{-\frac{ik}{2z'}r^2}\, e^{\frac{ik}{2z'}r^2}\, e^{-\frac{ik}{z'}(xx'+yy')}\, d\sigma,$$

where the four exponentials in red cancel in pairs, since in each pair the two exponents are opposite. We are left with

$$\boxed{u_{\mathcal{F}}(\vec{r}\,') = \frac{e^{ikz'}}{i\lambda z'} \int u_{\mathcal{A}}(\vec{r})\, \exp\left[-\frac{ik}{z'}\left(xx'+yy'\right)\right] d\sigma,} \tag{5.55}$$

which is exactly the Fraunhofer expression (5.59) *without the curvature factor*. This proves the assertion.

Three comments are in order, because the result is stronger than it looks.

In the derivation of (5.59) the factor $e^{ikr^2/2z'}$ was *neglected* by appealing to the far field. Here it has not been neglected, it has been *absorbed* by the choice of reference surface, and (5.55) is therefore exact within the Fresnel approximation, for any propagation distance z'. The lesson is that Fraunhofer diffraction is not really a regime of distances but a statement about geometry: it is what Fresnel propagation looks like when the field is read on two confocal spheres. The curvature factor that survives outside the integral in (5.59) is nothing but the price of insisting on reading the field on a plane, and this is precisely the interpretation anticipated in Figure 5.5.

Introducing the spatial frequency associated with the observation point,

$$\vec{\nu} = \frac{\vec{r}\,'}{\lambda z'}, \qquad \frac{k}{z'}\left(xx'+yy'\right) = \frac{2\pi}{\lambda z'}\,\vec{r}\cdot\vec{r}\,' = 2\pi\,\vec{r}\cdot\vec{\nu}, \tag{5.56}$$

equation (5.55) becomes

$$u_{\mathcal{F}}(\lambda z'\,\vec{\nu}) = \frac{e^{ikz'}}{i\lambda z'}\,\widehat{u_{\mathcal{A}}}(\vec{\nu}), \qquad \widehat{u_{\mathcal{A}}}(\vec{\nu}) = \int u_{\mathcal{A}}(\vec{r})\, e^{-2\pi i\,\vec{r}\cdot\vec{\nu}}\, d\sigma, \tag{5.57}$$

so that the observation coordinate on the sphere $\mathcal{F}$ is, up to the scale factor $\lambda z'$, the spatial frequency of the field on the sphere $\mathcal{A}$. This is the precise sense in which a diffracting system performs a Fourier analysis of the object, the statement that Abbe had reached qualitatively in 1873 (Abbe, 1873).

The only approximations used are those already contained in (5.48) together with the first order expansion (5.51) of the sagitta, so the result holds as long as the illuminated region satisfies $r_{\max} \ll z'$, which is the condition for the caps not to depart appreciably from their tangent planes. When this ceases to hold one must abandon the paraxial description, exactly as noted at the end of the following section.

Finally, it is worth recording that this example is the first instance of a much more general fact. Nothing forces the two reference spheres to be confocal, and if they are given a common radius R different from z' the two quadratic factors no longer cancel but combine into a chirp, and the propagation becomes a fractional Fourier transformation of an order fixed by the ratio z'/R. The confocal case treated here is the particular order $\alpha = \pi/2$. This is the subject of Section 5.7.

5.6 The Fraunhofer approximation

Finally, from expression (5.48) a second approximation may be carried out, corresponding to the far field or Fraunhofer regime, in which the observation distances are considerably larger than the area over which one wishes to study the propagated field (whether through a slit or another object). We focus on the factor

$$\exp\left(\frac{ik}{2z'}((x-x')^2+(y-y')^2)\right) = \exp\left(\frac{ik}{2z'}(x^2+y^2)\right)\exp\left(\frac{ik}{2z'}(x'^2+y'^2)\right) \times \exp\left(-\frac{ik}{z'}(xx'+yy')\right), \tag{5.58}$$

where the first of the three exponentials is the quadratic phase accumulated across the aperture itself. Setting it to unity is the defining step of the far field, and the condition for doing so is that the largest phase it can produce over the illuminated region be small.

Assumption 17 *The quadratic phase acquired across the aperture is negligible, $k\,r_{\max}^2/2z' \ll 1$, that is $z' \gg \pi r_{\max}^2/\lambda$, so that the Fresnel number of the aperture is small compared with unity.*

With $\exp\left(\frac{ik}{2z'}(x^2+y^2)\right) \approx 1$ we obtain

$$U(\vec{r}') = \frac{e^{ikz'}}{i\lambda z'}\exp\left(\frac{ik}{2z'}(x'^2+y'^2)\right)\int U(\vec{r})\exp\left(-\frac{ik}{z'}(xx'+yy')\right)d\sigma. \tag{5.59}$$

There we can still appreciate a quadratic phase term outside the integral, usually called the curvature factor, which shows precisely the dephasing induced by the difference of optical path between a plane called the observation screen and a spherical surface of radius z' (the propagation distance of the field along the optical axis) known as the cardinal sphere. The introduction of this spherical surface is rather useful when one wishes to describe Fraunhofer diffraction on a screen as a Fresnel propagation through two confocal cardinal spheres, as shown in Figure 5.5.

This makes more sense once we analyse the effect of each propagation step. It is clear that, without imposing far-field approximations from the outset, propagating the field at $\mathcal{A}$ towards $\mathcal{F}$ with the Fresnel approximation, we would obtain

$$U_{\mathcal{F}}(\vec{r}') = \frac{e^{ikz'}}{i\lambda z'}\exp\left(\frac{ik}{2z'}(x'^2+y'^2)\right)\int u_{\mathcal{A}}\exp\left(\frac{ik}{2z'}(x^2+y^2)\right)\exp\left(-\frac{ik}{z'}(xx'+yy')\right)d\sigma, \tag{5.60}$$

where we may associate $u_{\mathcal{A}}$, i.e. the field on the sphere $\mathcal{A}$, with the field on $\mathcal{P}$ multiplied by its corresponding quadratic factor $u_{\mathcal{P}} = u_{\mathcal{A}}(\vec{r})\exp\left(\dfrac{ik}{2z'}(x^2+y^2)\right)$, while the term $\exp\left(\frac{ik}{2z'}(x'^2+y'^2)\right)$ may be understood as the factor resulting from passing the field from the sphere $\mathcal{F}$ to the screen $\mathcal{Q}$, so that the field on $\mathcal{F}$ would correspond to

$$u_{\mathcal{Q}}(\vec{r}') = \frac{e^{ikz'}}{i\lambda z'}\int u_{\mathcal{P}}\exp\left(-\frac{ik}{z'}(xx'+yy')\right)d\sigma, \tag{5.61}$$

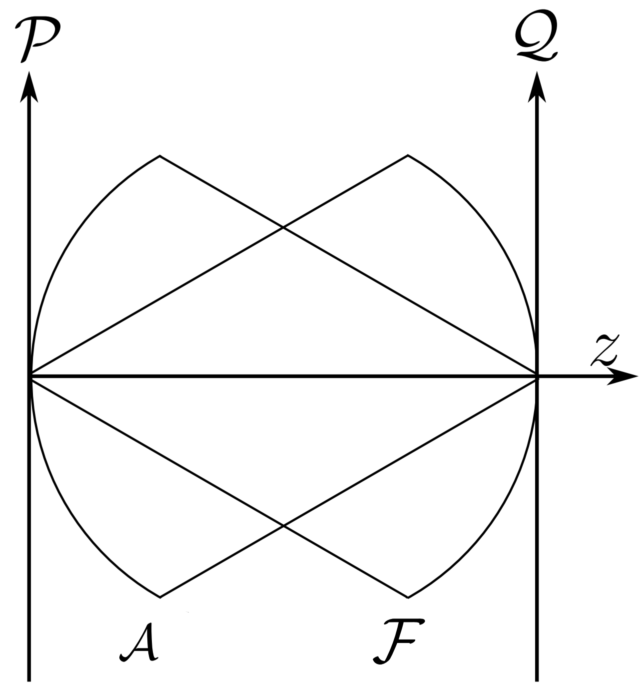


Figure 5.5: Scheme of the confocal cardinal spheres, with radii equal to the propagation distance z' of the field.

thus recovering a Fraunhofer expression for diffraction on $\mathcal{Q}$. It should be noted that one must check whether the study of the propagated field is valid within a paraxial regime, where the field is confined to the neighbourhood of the optical axis and hence the curvature of the cardinal sphere does not differ much from the plane (the screen). In cases where the curvature factor ceases to be negligible, the paraxial regime loses validity and the problem must be addressed from a meta-axial standpoint.

It is worth pausing here to count, because this is the place at which the promise made at the beginning of the chapter falls due. Seventeen assumptions separate Maxwell's equations from the Fraunhofer integral (5.59), and they fall into six families. Five were needed merely to replace the electromagnetic field by a single scalar quantity: the magnetic interaction was discarded, the medium was emptied of charges and currents, the polarisation was frozen first in time and then in space, and the field was declared transverse. A sixth reduced polychromatic light to one spectral component at a time. Four more were spent in building the integral apparatus, by requiring smoothness of the field, by emptying the region of observation of its sources, by shrinking the elementary emitter to a point and by forbidding any radiation to arrive from infinity. Two were spent on the screen, whose material was idealised out of existence and on which the field was then prescribed by hand. One placed the observer many wavelengths from the aperture. Three built the paraxial regime, by truncating the binomial expansion, by treating amplitude and phase with different degrees of accuracy and by straightening the obliquity factor to unity. The seventeenth and last pushed the observer into the far field.

Two lessons follow from the tally. The first is that the elegance of (5.59), a bare Fourier transform of the aperture, is bought rather than given: every one of the seventeen steps has narrowed the class of experiments the formula can describe, and each of them can be violated in a laboratory. The scalar theory fails when the aperture becomes comparable with the wavelength, when the numerical aperture is large, when the screen is a good conductor and the polarisation of the illumination is seen to matter, or when the observation distance is not large enough for the Fresnel number to be small; in every one of those cases the culprit can be traced to a definite item on the list above. The second lesson is the more remarkable, and it is the reason why the theory is taught at all: the formula works anyway, and it works over a range of situations far wider than the assumptions have any right to guarantee. Understanding which of the seventeen is about to break is, in practice, the whole art of applying diffraction theory.

Studying the propagation of the field through Fresnel diffraction is of great importance because, as will be discussed later, it bears a close relationship with what is known as the fractional Fourier transform, a tool that arises naturally when one chains paraxial propagation steps.

5.7 The fractional Fourier transformation

The last section closed with the remark that Fresnel propagation bears a close relationship with the fractional Fourier transformation. That remark is the subject of the present section, and it deserves a careful treatment for two reasons. The first is that the fractional Fourier transformation is not a formal curiosity but the natural language in which paraxial propagation should be written, in the same sense in which the ordinary Fourier transformation is the natural language of the far field. The second is that the object itself, a continuous family of operators interpolating between the identity and the Fourier transformation, has a long and instructive history, one in which the same idea was rediscovered independently in pure analysis, in quantum mechanics, in signal processing and in optics before anybody realised that all four communities were talking about the same one parameter group.

A word on notation before we begin. In (5.48) the propagation distance appeared as z', the axial coordinate of the observation point, the aperture being fixed at the origin of the axis. In the present section the two reference surfaces are treated on the same footing and neither of them is privileged, so we place the emitting surface at the origin and write simply z for the distance to the receiving one. The z of this section is therefore the z' of (5.48), and no axial coordinate of the integration variable appears anywhere in what follows.

5.7.1 A historical context of Fourier analysis and of its fractional powers

The story begins, as so many stories in mathematical physics do, with a vibrating string. In 1747 Jean le Rond d'Alembert wrote down the wave equation of a stretched string and solved it by superposing two travelling waves (d'Alembert, 1747), a solution that Leonhard Euler immediately generalised to arbitrary initial shapes (Euler, 1748). Six years later Daniel Bernoulli proposed something that looked, to his contemporaries, physically obvious and mathematically outrageous at the same time, namely that every possible motion of the string could be written as a sum of the sinusoidal modes of vibration (Bernoulli, 1753). Euler and d'Alembert both rejected the claim. Their objection was not aesthetic but logical, since a sum of sines is a smooth and periodic object while the initial shape of a plucked string has a corner in it, and no eighteenth century mathematician was willing to accept that an infinite sum of analytic functions could reproduce a curve with a kink. The dispute lasted half a century and was never settled by the people who started it.

It was settled, in a sense, by somebody who was not trying to settle it. In 1807 Joseph Fourier submitted to the Institut de France a memoir on the propagation of heat in solids in which he asserted, and used, precisely the representation that Euler had rejected. The referees, Lagrange among them, refused the memoir. Lagrange in particular never accepted the generality of the trigonometric expansion, and Fourier had to wait until 1822 to publish the mature version of his ideas as the *Théorie Analytique de la Chaleur* (Fourier, 1822). What Fourier had understood, and what his referees had not, is that the question of whether the series converges to the function is a separate question from whether the series is useful, and that the coefficients can be computed by an integral regardless of one's philosophical position about convergence. The passage from the series on a finite interval to the integral on the whole line, which is what we now call the Fourier transformation, is already present in that book as a limiting case.

Fourier's boldness left a debt that the nineteenth century spent decades paying. Peter Gustav Lejeune Dirichlet gave in 1829 the first genuine convergence theorem, showing that the series of a piecewise monotone function with finitely many discontinuities converges to the mean of the lateral limits (Dirichlet, 1829). Bernhard Riemann, in his habilitation of 1854 published posthumously in

1867, asked the converse question of which functions admit a trigonometric representation at all, and in doing so was forced to invent a theory of integration (Riemann, 1867). Georg Cantor, studying the uniqueness of trigonometric representations, was led to classify the exceptional sets on which two different series can agree, and out of that classification grew set theory itself (Cantor, 1872). Henri Lebesgue built a new integral partly in order to make the theory of trigonometric series behave (Lebesgue, 1903). It is not an exaggeration to say that a good part of modern analysis exists because Fourier wrote down a formula that nobody could justify.

The theory reached its definitive form in the twentieth century. Michel Plancherel proved in 1910 the identity that bears his name, establishing that the Fourier transformation is an isometry of the space of square integrable functions onto itself (Plancherel, 1910). This is the single most important structural fact about the transformation for our purposes, because an isometry of a Hilbert space onto itself is a unitary operator, and unitary operators are exactly the objects that can be raised to continuous powers. Norbert Wiener (1933) and Edward Titchmarsh (1937) consolidated the analytic theory in the two treatises that defined the subject for a generation, and in 1965 James Cooley and John Tukey rediscovered an algorithm that reduced the numerical cost of the discrete transformation from a quadratic to a logarithmic law and thereby moved Fourier analysis from the blackboard into every laboratory instrument (Cooley and Tukey, 1965).

Optics entered the story from a different door. Ernst Abbe had understood already in 1873, while trying to explain why his microscopes could not resolve arbitrarily fine detail, that the objective of a microscope performs a decomposition of the object into spatial frequencies and that the aperture of the instrument truncates that decomposition (Abbe, 1873). The idea remained essentially qualitative until Pierre-Michel Duffieux published in 1946 the first systematic treatise on the Fourier integral in optics (Duffieux, 1946), which is the direct ancestor of the modern treatment found in Goodman (2005). By the middle of the twentieth century it was a commonplace that the far field of an aperture is the Fourier transform of the aperture, which is the statement we derived in the section on the Fraunhofer approximation.

The idea of a *fractional* power of the Fourier transformation has a separate and much less linear genealogy. Its seed is a purely analytic observation. Charles Hermite had introduced in 1864 the polynomials that carry his name (Hermite, 1864), and Ferdinand Gustav Mehler had found in 1866 the bilinear generating function of those polynomials (Mehler, 1866), a formula that will do all the work for us later in this section. In 1929 Norbert Wiener published a short note in which he observed that the Hermite functions are eigenfunctions of the Fourier transformation with eigenvalues that are the four fourth roots of unity (Wiener, 1929). The observation is elementary once stated, but it is the whole idea, because an operator whose spectrum is known can be raised to any power one likes simply by raising the eigenvalues to that power.

Edward Uhler Condon took that step explicitly in 1937 in a paper whose title says exactly what it does, namely to immerse the Fourier transformation in a continuous group of functional transformations (Condon, 1937). Condon was interested in the group theoretical structure and in the connection with the quantum harmonic oscillator, and he already noticed that the generator of the continuous family is the oscillator Hamiltonian. Two years later Hermann Kober, working from the point of view of integral transformations, constructed roots of the Fourier and Hankel transformations and gave the corresponding kernels (Kober, 1939). Valentine Bargmann arrived at the same family in 1961 from yet another direction, through his Hilbert space of entire functions, where the fractional powers act as simple rotations of the complex variable (Bargmann, 1961). Nicolaas de Bruijn recovered them once more in 1973 in his theory of generalised functions built on the Wigner distribution (de Bruijn, 1973). Each of these authors had the object in his hands and none of them gave it a name that stuck.

The name and the systematic treatment are due to Victor Namias, who in 1980 published the paper that the optics community now takes as the origin of the subject (Namias, 1980). Namias' contribution was not merely terminological. He defined the transformation of fractional order in two equivalent ways, as a differential operator obtained by exponentiating the harmonic oscillator Hamiltonian and as an integral operator with an explicit kernel, showed that the two definitions agree, and then used the resulting tool to solve concrete problems in quantum mechanics that are awkward in the ordinary Fourier language. It is his derivation that we shall reproduce in full in the next subsection.

Namias' treatment was, by his own standards, a working physicist's treatment. The integral he wrote does not converge absolutely for every function one would like to transform, the square root appearing in the normalising constant is written without specifying which branch is meant, and the additivity of the orders is verified on the eigenfunction expansion rather than proved on a well defined domain. These are not pedantic objections, because a careless choice of branch makes the composition law fail by a sign. Adam McBride and Fiona Kerr repaired all of this in 1987 (McBride and Kerr, 1987), giving a definition that converges absolutely on the Schwartz space, fixing the branch in a way that makes the composition law hold for all real orders, and proving that the resulting family is a genuine continuous one parameter group. Kerr extended the construction shortly afterwards to spaces of generalised functions (Kerr, 1988a) and to the space of square integrable functions with applications to differential equations (Kerr, 1988b).

While the analysts were tidying up, the optics community rediscovered the object experimentally. In 1993 David Mendlovic and Haldun Ozaktas published a pair of papers showing that a graded index medium of the appropriate length performs a fractional Fourier transformation on the field that traverses it (Mendlovic and Ozaktas, 1993; Ozaktas and Mendlovic, 1993), and Adolf Lohmann showed in the same year that the fractional order is nothing other than a rotation angle of the Wigner distribution of the field, which gave the whole construction a transparent phase space meaning and an immediate bulk optics implementation with two lenses (Lohmann, 1993). One year later Pierre Pellat-Finet proved the result that concerns us most directly, namely that Fresnel diffraction between two spherical caps is exactly a fractional Fourier transformation, with an order fixed by the geometry (Pellat-Finet, 1994), a result developed at length with Georges Bonnet (Pellat-Finet and Bonnet, 1994) and later incorporated into a full treatise on metaxial and fractional Fourier optics (Pellat-Finet, 2009). In parallel, Luís Almeida introduced the transformation into signal processing as an interpolation between the time and the frequency representations (Almeida, 1994), and the whole body of optical applications was collected by Ozaktas, Zalevsky and Kutay (2001).

It is worth pausing on the fact that the literature does not use a single notion of fractional order, and a reader who moves between papers will meet at least five conventions. The first is the one introduced by Namias, in which the order is an angle α measured in radians, the ordinary Fourier transformation corresponding to $\alpha = \pi/2$ and the identity to $\alpha = 0$. The second, standard in signal processing after Ozaktas, Mendlovic and Lohmann, replaces the angle by a dimensionless order a defined through $\alpha = a\pi/2$, so that the ordinary transformation is the transformation of order one and the parity operator is the transformation of order two. The third is the optical order used by Pellat-Finet, which is again an angle but is tied to a pair of reference spheres rather than to an abstract eigenvalue assignment, and which therefore depends on the geometry chosen to describe the propagation. The fourth is the complex order, introduced by Chung-Chieh Shih (1995b) and realised optically by Luís Bernardo and Olívio Soares (1996), in which the parameter is allowed to leave the real axis and the transformation acquires a Gaussian weighting in addition to its chirp. The fifth is not a convention but a genuine ambiguity, and it deserves emphasis. Since the ordinary Fourier transformation satisfies $\mathcal{F}^4 = \mathcal{I}$, the problem of defining $\mathcal{F}^a$ is the problem of choosing a fourth root of the identity in a continuous way, and that choice is not unique. Namias assigns to the eigenfunction of index n the eigenvalue $e^{-in\alpha}$, which is the choice that makes the generator the harmonic oscillator Hamiltonian, but Shih showed that a different and perfectly consistent assignment, based on weighting the four powers of $\mathcal{F}$, produces an inequivalent family (Shih, 1995a). Gianfranco Cariolaro and collaborators later classified all admissible fractionalisations in a single framework (Cariolaro et al., 1998), and Çağatay Candan, Alper Kutay and Ozaktas solved the corresponding discrete problem (Candan et al., 2000). When a paper speaks of *the* fractional Fourier transformation without further qualification, it almost always means the Namias family, and that is the one we shall construct.

Finally, it should be said that the fractional Fourier transformation is not an isolated object but the simplest nontrivial member of a much larger family. Marcos Moshinsky and Christiane Quesne introduced in 1971 the linear canonical transformations, the unitary representations of the group of linear transformations of phase space that preserve the symplectic form (Moshinsky and Quesne, 1971), and Kurt Bernardo Wolf developed their theory as integral transformations (Wolf, 1979). Stuart Collins had written in 1970 the diffraction integral of an arbitrary paraxial system in terms of the matrix of that system (Collins, 1970), which is the same family seen from the engineering side. In that language the fractional Fourier transformations are exactly the elliptic elements of the group, that is, the rotations of phase space, while free propagation over a plane and the action of a thin lens are parabolic elements and magnifications are hyperbolic elements. The reason the fractional Fourier transformation occupies such a privileged place in optics is that, as we shall prove, once the field is referred to the right pair of spherical surfaces, free propagation becomes a pure rotation.

5.7.2 The fractional Fourier transformation as a differential and as an integral operator

Throughout this subsection we work in one transverse variable, since the two dimensional case follows by taking the tensor product of two identical one dimensional transformations, and we adopt the symmetric convention

$$\mathcal{F}[f](x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(y)\, e^{-ixy}\, \mathrm{d}y, \tag{5.62}$$

which is the convention that makes $\mathcal{F}$ unitary on $L^2(\mathbb{R})$ by the theorem of Plancherel (1910), and which agrees with the sign of the kernel that appeared in the Fraunhofer integral (5.59). Applying (5.62) twice gives the parity operator and four times the identity,

$$\mathcal{F}^2 = \mathcal{P}, \qquad \mathcal{P}f(x) = f(-x), \qquad \mathcal{F}^4 = \mathcal{I}, \tag{5.63}$$

so that the integer powers of $\mathcal{F}$ form a cyclic group of order four. The entire content of this subsection is the construction of a continuous group in which that cyclic group of four elements sits as a subgroup.

Before starting, a point of vocabulary must be settled, because the two words involved are routinely confused and the distinction is essential to everything that follows.

Definition 30 *(Transformation and transform) A* ***transformation*** *is an operator, that is, a map $\mathcal{F}_\alpha : \mathcal{H} \to \mathcal{H}$ from a function space into itself. A* ***transform*** *is the image of one particular element under that map, that is, the function $\mathcal{F}_\alpha f$ obtained by applying the transformation to a given f.*

The distinction is not pedantic. Everything that makes the fractional Fourier construction worth having is a statement about *transformations* and not about transforms: the family $\{\mathcal{F}_\alpha\}_{\alpha\in\mathbb{R}}$ is a one parameter group, its composition law is $\mathcal{F}_\alpha\mathcal{F}_\beta = \mathcal{F}_{\alpha+\beta}$, its generator is the harmonic oscillator Hamiltonian, and its elements are unitary. None of these sentences can even be formulated at the level of a single transform, because a transform is merely a function and carries no group structure whatsoever. In the optical application of the last subsection the point becomes physical rather than linguistic: it is the propagation of the field over a distance that is a transformation, while the field observed on the receiving surface is a transform, and the reason propagation composes so simply is a property of the operators, not of any particular field.

5.7.2.1 The eigenfunctions of the Fourier transformation

The construction rests on a single fact, observed by Wiener in 1929 (Wiener, 1929), namely that the Fourier transformation possesses a complete orthonormal set of eigenfunctions built out of the Hermite polynomials (Hermite, 1864). Let us derive it, since every later step depends on the precise value of the eigenvalues. Recall the generating function of the Hermite polynomials,

$$e^{2xt-t^2} = \sum_{n=0}^{\infty} \frac{H_n(x)}{n!} t^n, \tag{5.64}$$

and define the generating function of the Hermite functions by multiplying (5.64) by the Gaussian $e^{-x^2/2}$,

$$G(x,t) = e^{-x^2/2}\, e^{2xt-t^2} = e^{-x^2/2} \sum_{n=0}^{\infty} \frac{H_n(x)}{n!} t^n. \tag{5.65}$$

We now transform (5.65) with respect to x. Writing the full exponent of the integrand of (5.62),

$$-\frac{x^2}{2}+2xt-t^2-i\xi x=-\frac{x^2}{2}+x\left(2t-i\xi\right)-t^2=$$
$$-\frac{1}{2}\left[x^2-2x\left(2t-i\xi\right)+\left(2t-i\xi\right)^2\right]+\frac{\left(2t-i\xi\right)^2}{2}-t^2,$$

where the two terms in blue have been added and subtracted, which is legitimate because their sum is zero, and which turns the first bracket into a perfect square,

$$=-\frac{1}{2}\left[x-\left(2t-i\xi\right)\right]^2+\frac{\left(2t-i\xi\right)^2}{2}-t^2.$$

The remaining integral is the Gaussian $\int_{-\infty}^{\infty}e^{-\frac{1}{2}(x-a)^2}\mathrm{d}x=\sqrt{2\pi}$, which is valid for complex a by a translation of the integration contour, so that the factor $1/\sqrt{2\pi}$ in (5.62) is cancelled exactly and

$$\mathcal{F}\left[G(\cdot,t)\right](\xi)=\exp\left[\frac{\left(2t-i\xi\right)^2}{2}-t^2\right]. \tag{5.66}$$

It remains to recognise the right hand side. Expanding the square,

$$\frac{4t^2-4it\xi-\xi^2}{2}-t^2=2t^2-2it\xi-\frac{\xi^2}{2}-t^2=-\frac{\xi^2}{2}-2i\xi t+t^2=$$
$$-\frac{\xi^2}{2}+2\xi\left(-it\right)-\left(-it\right)^2,$$

since $-(-it)^2=+t^2$, and comparing with the definition (5.65) we conclude that

$$\mathcal{F}\left[G(\cdot,t)\right](\xi)=G(\xi,-it). \tag{5.67}$$

The generating function is mapped into itself with the parameter rotated by a quarter turn in the complex plane. Expanding both sides of (5.67) in powers of t by means of (5.65),

$$\sum_{n=0}^{\infty}\frac{t^n}{n!}\mathcal{F}\left[H_n\,e^{-x^2/2}\right](\xi)=\sum_{n=0}^{\infty}\frac{\left(-it\right)^n}{n!}H_n(\xi)e^{-\xi^2/2},$$

and identifying the coefficients of t^n, which is legitimate because both series converge for every t in a neighbourhood of the origin, we obtain the result we were after. Introducing the normalised Hermite functions

$$\psi_n(x)=\left(2^n n!\sqrt{\pi}\right)^{-1/2}H_n(x)\,e^{-x^2/2},\qquad \int_{-\infty}^{\infty}\psi_n(x)\psi_m(x)\,\mathrm{d}x=\delta_{nm}, \tag{5.68}$$

which form a complete orthonormal basis of $L^2(\mathbb{R})$, the eigenvalue relation reads

$$\mathcal{F}\left[\psi_n\right]=\left(-i\right)^n\psi_n=e^{-in\pi/2}\,\psi_n. \tag{5.69}$$

Equation (5.69) contains the whole idea. The eigenvalues of the Fourier transformation are $e^{-in\pi/2}$, and the number $\pi/2$ appears there not because of any deep property of the exponential but simply because $\mathcal{F}$ is the fourth root of the identity. Nothing forbids us from replacing $\pi/2$ by an arbitrary angle.

5.7.2.2 Namias' definition and the differential operator

Following Namias (1980), we define the fractional Fourier transformation of order α by prescribing its action on the basis (5.68),

$$\mathcal{F}_\alpha\left[\psi_n\right] = e^{-in\alpha}\,\psi_n, \tag{5.70}$$

and extending it to an arbitrary $f \in L^2(\mathbb{R})$ by linearity and continuity. Writing the expansion of f in the basis,

$$f(x) = \sum_{n=0}^{\infty} c_n\psi_n(x), \qquad c_n = \int_{-\infty}^{\infty} f(y)\psi_n(y)\,\mathrm{d}y, \tag{5.71}$$

the transform of order α is by definition

$$F_\alpha(x) \equiv \mathcal{F}_\alpha\left[f\right](x) = \sum_{n=0}^{\infty} c_n e^{-in\alpha}\psi_n(x). \tag{5.72}$$

The definition is manifestly consistent, since $\alpha = 0$ gives $F_0 = f$ by completeness, $\alpha = \pi/2$ reproduces (5.69), $\alpha = \pi$ gives $\sum c_n(-1)^n\psi_n$, which is the parity operator because ψ_n has the parity of n, and $\alpha = 2\pi$ returns the identity. The family is therefore 2π periodic in α and unitary, since it multiplies the coefficients of an orthonormal expansion by unimodular numbers.

Namias' first observation is that (5.72) can be freed from the basis and written as a differential operator. Differentiating (5.72) with respect to the order,

$$\frac{\partial F_\alpha}{\partial\alpha}(x) = -i\sum_{n=0}^{\infty} n\,c_n e^{-in\alpha}\psi_n(x), \tag{5.73}$$

we are left with the index n standing inside the sum, which is inconvenient because n is not an operator acting on x. The remedy is the differential equation of the Hermite functions. Substituting (5.68) into the Hermite equation $H_n'' - 2xH_n' + 2nH_n = 0$ one obtains, after the elementary computation of the two derivatives of the Gaussian factor,

$$\psi_n''(x) + \left(2n + 1 - x^2\right)\psi_n(x) = 0, \tag{5.74}$$

which is precisely the stationary Schrödinger equation of the quantum harmonic oscillator in natural units. Solving (5.74) for the index,

$$n\,\psi_n(x) = \frac{1}{2}\left[x^2\psi_n(x) - \psi_n''(x) - \psi_n(x)\right], \tag{5.75}$$

and inserting (5.75) into (5.73),

$$\frac{\partial F_\alpha}{\partial\alpha}(x) = -\frac{i}{2}\sum_{n=0}^{\infty} c_n e^{-in\alpha}\left[x^2\psi_n(x) - \psi_n''(x) - \psi_n(x)\right] =$$
$$-\frac{i}{2}\left[x^2\sum_n c_n e^{-in\alpha}\psi_n(x) - \frac{\partial^2}{\partial x^2}\sum_n c_n e^{-in\alpha}\psi_n(x) - \sum_n c_n e^{-in\alpha}\psi_n(x)\right],$$

where the index n has disappeared entirely and each of the three sums is again F_α by (5.72). We therefore arrive at the differential formulation of the fractional Fourier transformation,

$$i\,\frac{\partial F_\alpha}{\partial \alpha} = \frac{1}{2}\left[-\frac{\partial^2}{\partial x^2} + x^2 - 1\right] F_\alpha, \qquad F_\alpha\big|_{\alpha=0} = f. \tag{5.76}$$

Equation (5.76) ***is Namias' definition of the fractional Fourier transformation as a differential operator.*** It is worth reading it twice. It states that the transform of order α of a given function is obtained by letting that function evolve, according to the Schrödinger equation of the harmonic oscillator, for a *time* equal to α, the constant -1 merely removing the zero point energy so that the ground state does not accumulate a phase. In operator language, introducing the oscillator Hamiltonian

$$\hat{H} = \frac{1}{2}\left(-\frac{\partial^2}{\partial x^2} + x^2\right), \qquad \hat{H}\psi_n = \left(n + \frac{1}{2}\right)\psi_n, \tag{5.77}$$

the solution of (5.76) is written at once as the exponential of the generator,

$$\mathcal{F}_\alpha = \exp\left[-i\alpha\left(\hat{H} - \frac{1}{2}\right)\right] = \exp\left[\frac{i\alpha}{2}\left(\frac{\partial^2}{\partial x^2} - x^2 + 1\right)\right], \tag{5.78}$$

which reproduces (5.70) immediately, since $\exp[-i\alpha(\hat{H} - \frac{1}{2})]\psi_n = e^{-i\alpha n}\psi_n$. This identification is the reason the fractional Fourier transformation appears wherever a harmonic oscillator appears, and it explains in retrospect why Condon had already met the same family in 1937 while studying continuous groups of functional transformations (Condon, 1937).

Two structural consequences follow immediately from (5.78). First, since the exponentials of a single generator commute and add their arguments,

$$\mathcal{F}_\alpha \mathcal{F}_\beta = \mathcal{F}_{\alpha+\beta}, \qquad \mathcal{F}_\alpha^{-1} = \mathcal{F}_{-\alpha}, \tag{5.79}$$

so the transformations form a commutative one parameter group. Second, since $\hat{H}$ is self adjoint, $\mathcal{F}_\alpha$ is unitary, and Plancherel's theorem extends to every order,

$$\int_{-\infty}^{\infty} |\mathcal{F}_\alpha[f](x)|^2\,\mathrm{d}x = \int_{-\infty}^{\infty} |f(x)|^2\,\mathrm{d}x, \tag{5.80}$$

which in the optical reading of the next subsection is nothing other than the conservation of the energy carried by the field.

5.7.2.3 The integral form of the transformation

The differential form (5.76) is compact and conceptually transparent, but for computation one wants a kernel. Since the transformation is linear and bounded, it must be representable as

$$\mathcal{F}_\alpha[f](x) = \int_{-\infty}^{\infty} K_\alpha(x, y) f(y)\,\mathrm{d}y, \tag{5.81}$$

and inserting the expansion (5.71) into (5.72) and exchanging the sum with the integral, the kernel is read off as the bilinear spectral sum

$$K_\alpha(x,y) = \sum_{n=0}^{\infty} e^{-in\alpha}\psi_n(x)\psi_n(y). \tag{5.82}$$

Substituting the explicit Hermite functions (5.68),

$$K_\alpha(x,y) = \frac{e^{-\left(x^2+y^2\right)/2}}{\sqrt{\pi}} \sum_{n=0}^{\infty} \frac{H_n(x)H_n(y)}{2^n n!}\, e^{-in\alpha}, \tag{5.83}$$

we are left with a bilinear generating function of Hermite polynomials, which is exactly the object Mehler computed in 1866 (Mehler, 1866),

$$\sum_{n=0}^{\infty} \frac{H_n(x)H_n(y)}{2^n n!}\, t^n = \frac{1}{\sqrt{1-t^2}} \exp\left[\frac{2xyt - \left(x^2+y^2\right)t^2}{1-t^2}\right], \qquad |t|<1. \tag{5.84}$$

We must set $t = e^{-i\alpha}$, which lies on the boundary of the disc of convergence, so (5.84) is to be understood as the limit $t \to e^{-i\alpha}$ taken radially from inside the disc, a limit that exists in the sense of distributions and that is legitimate for every α that is not an integer multiple of π. Let us now carry out the algebra step by step. The denominator factorises as

$$1 - e^{-2i\alpha} = e^{-i\alpha}\left(e^{i\alpha} - e^{-i\alpha}\right) = e^{-i\alpha}\, 2i\sin\alpha, \tag{5.85}$$

so that the crossed term of the exponent of (5.84) becomes

$$\frac{2xy\, e^{-i\alpha}}{e^{-i\alpha}\, 2i\sin\alpha} = \frac{xy}{i\sin\alpha} = -\frac{i\,xy}{\sin\alpha} = -ixy\csc\alpha, \tag{5.86}$$

where the two exponentials in red cancel. For the quadratic term we must not forget the Gaussian prefactor of (5.83), so we collect both contributions,

$$-\frac{x^2+y^2}{2} - \frac{\left(x^2+y^2\right)e^{-2i\alpha}}{e^{-i\alpha}\,2i\sin\alpha} = -\frac{x^2+y^2}{2} - \frac{\left(x^2+y^2\right)e^{-i\alpha}}{2i\sin\alpha} =$$
$$-\frac{x^2+y^2}{2}\left[1 + \frac{e^{-i\alpha}}{i\sin\alpha}\right] = -\frac{x^2+y^2}{2}\cdot\frac{i\sin\alpha + e^{-i\alpha}}{i\sin\alpha},$$

and the numerator of this last fraction collapses beautifully,

$$i\sin\alpha + e^{-i\alpha} = i\sin\alpha + \cos\alpha - i\sin\alpha = \cos\alpha, \tag{5.87}$$

so that, using $-1/i = i$,

$$-\frac{x^2+y^2}{2}\cdot\frac{\cos\alpha}{i\sin\alpha} = \frac{i}{2}\left(x^2+y^2\right)\cot\alpha. \tag{5.88}$$

There remains the normalising constant. Using (5.85) once more,

$$\frac{1}{\sqrt{\pi}}\cdot\frac{1}{\sqrt{1-e^{-2i\alpha}}}=\frac{1}{\sqrt{\pi}}\cdot\frac{e^{i\alpha/2}}{\sqrt{2i\sin\alpha}}=\frac{e^{i\left(\frac{\alpha}{2}-\frac{\pi}{4}\right)}}{\sqrt{2\pi\sin\alpha}}, \tag{5.89}$$

since $1/\sqrt{i}=e^{-i\pi/4}$. This constant is more often written in the compact form that Namias used, and the two agree because

$$1-i\cot\alpha=\frac{\sin\alpha-i\cos\alpha}{\sin\alpha}=\frac{-i\left(\cos\alpha+i\sin\alpha\right)}{\sin\alpha}=\frac{e^{-i\pi/2}e^{i\alpha}}{\sin\alpha}, \tag{5.90}$$

whose square root divided by $\sqrt{2\pi}$ is precisely (5.89). Collecting (5.86), (5.88) and (5.89) we obtain the integral form of the transformation,

$$\boxed{\mathcal{F}_\alpha[f](x)=\sqrt{\frac{1-i\cot\alpha}{2\pi}}\int_{-\infty}^{\infty}\exp\left\{\frac{i}{2}\left(x^2+y^2\right)\cot\alpha-i\,xy\csc\alpha\right\}f(y)\,\mathrm{d}y.} \tag{5.91}$$

Let us verify that (5.91) does what it should. For $\alpha=\pi/2$ we have $\cot\alpha=0$ and $\csc\alpha=1$, so the constant reduces to $1/\sqrt{2\pi}$ and the kernel to e^{-ixy}, recovering the ordinary transformation (5.62) exactly. For $\alpha\to 0$ the coefficient $\cot\alpha$ diverges and the kernel oscillates infinitely fast everywhere except on the diagonal, where the exponent $\frac{i}{2}(x-y)^2\cot\alpha$ vanishes, and a standard stationary phase argument shows that $K_\alpha\to\delta(x-y)$, which is the identity. For $\alpha\to\pi$ the same argument applied to $\frac{i}{2}(x+y)^2\cot\alpha$ gives $K_\pi(x,y)=\delta(x+y)$, the parity operator. The three degenerate orders $\alpha=0,\pi,2\pi$, at which the closed formula (5.91) is meaningless because $\csc\alpha$ blows up, are therefore perfectly well defined in the spectral form (5.82), a discrepancy that we shall have to come back to.

5.7.2.4 The inverse transformation and the index law

The inverse is obtained without any new computation, because (5.82) makes the composition of two transformations transparent. Consider the kernel of the product $\mathcal{F}_\alpha\mathcal{F}_\beta$,

$$\begin{aligned}&\int_{-\infty}^{\infty}K_\alpha(x,s)K_\beta(s,y)\,\mathrm{d}s=\int_{-\infty}^{\infty}\left[\sum_n e^{-in\alpha}\psi_n(x)\psi_n(s)\right]\left[\sum_m e^{-im\beta}\psi_m(s)\psi_m(y)\right]\mathrm{d}s=\\&\sum_n\sum_m e^{-in\alpha}e^{-im\beta}\psi_n(x)\psi_m(y)\underbrace{\int_{-\infty}^{\infty}\psi_n(s)\psi_m(s)\,\mathrm{d}s}_{\delta_{nm}}=\sum_n e^{-in(\alpha+\beta)}\psi_n(x)\psi_n(y),\end{aligned}$$

where the orthonormality (5.68) has collapsed the double sum into a single one, and the last expression is by definition $K_{\alpha+\beta}(x,y)$. We have therefore proved the **index law**

$$\mathcal{F}_\alpha\mathcal{F}_\beta=\mathcal{F}_{\alpha+\beta}, \tag{5.92}$$

which is the property that justifies calling α an order, since orders add under composition exactly as exponents do. Setting $\beta = -\alpha$ and using completeness,

$$K_0(x,y) = \sum_{n=0}^{\infty} \psi_n(x)\psi_n(y) = \delta(x-y), \tag{5.93}$$

we conclude that $\mathcal{F}_\alpha \mathcal{F}_{-\alpha} = \mathcal{I}$, that is,

$$\mathcal{F}_\alpha^{-1} = \mathcal{F}_{-\alpha}, \tag{5.94}$$

so that the inverse transformation is the transformation of the opposite order. Explicitly, replacing α by $-\alpha$ in (5.91) and using $\cot(-\alpha) = -\cot\alpha$ and $\csc(-\alpha) = -\csc\alpha$,

$$f(y) = \sqrt{\frac{1+i\cot\alpha}{2\pi}} \int_{-\infty}^{\infty} \exp\left\{-\frac{i}{2}\left(x^2+y^2\right)\cot\alpha + i\,xy\csc\alpha\right\} F_\alpha(x)\,\mathrm{d}x, \tag{5.95}$$

and comparing (5.95) with (5.91) we see that $K_{-\alpha}(y,x) = \overline{K_\alpha(x,y)}$, which is precisely the statement that $\mathcal{F}_\alpha$ is unitary, in agreement with (5.80).

5.7.2.5 The regularisation of McBride and Kerr

The derivation we have just carried out is Namias' derivation, and as a piece of physics it is complete. As a piece of analysis it has three gaps, all of which were identified and repaired by Adam McBride and Fiona Kerr in 1987 (McBride and Kerr, 1987). Since the modern optical literature uses their version without always saying so, it is worth stating what they changed.

The first gap is the **branch of the square root**. The constant in (5.91) is written as $\sqrt{1-i\cot\alpha}$, and a square root of a complex number is defined only up to a sign. The spectral sum (5.82) has no such ambiguity, so the sign must be fixed by matching the two, and the correct determination is the one already contained in (5.89), which written without ambiguity reads

$$C_\alpha = \frac{\exp\left[i\left(\frac{\alpha}{2} - \frac{\pi}{4}\,\mathrm{sgn}\left(\sin\alpha\right)\right)\right]}{\sqrt{2\pi\left|\sin\alpha\right|}}. \tag{5.96}$$

The discrete factor $\mathrm{sgn}(\sin\alpha)$ is not cosmetic. If one chooses the principal branch naively and composes two transformations whose orders add up to more than π, the index law (5.92) fails by a factor -1. This is the elementary manifestation of a well known phenomenon, namely that the group of transformations is not represented faithfully by the kernels but by a double cover of them, the extra sign being what is called the Maslov index in the theory of canonical transformations (Moshinsky and Quesne, 1971; Wolf, 1979).

The second gap is the **domain and the convergence of the integral**. For a general $f \in L^2(\mathbb{R})$ the integral in (5.91) does not converge absolutely, because the modulus of the kernel is constant in y and the integrand therefore decays no faster than f itself. McBride and Kerr define the transformation first on the Schwartz space $\mathcal{S}$ of rapidly decreasing functions, where absolute convergence is guaranteed, prove that $\mathcal{F}_\alpha$ maps $\mathcal{S}$ onto $\mathcal{S}$ homeomorphically, and only then extend it by duality to the space $\mathcal{S}'$ of tempered distributions. Their technical device for enlarging the domain is to trade one power of decay in the integration variable for one derivative in the observation variable, replacing (5.91) by an expression of the form

$$\mathcal{F}_\alpha[f](x) = C_\alpha\, e^{\frac{i}{2}x^2\cot\alpha}\, \frac{\mathrm{d}}{\mathrm{d}x} \int_{-\infty}^{\infty} \frac{e^{-ixy\csc\alpha}}{-i\, y\csc\alpha}\, e^{\frac{i}{2}y^2\cot\alpha} f(y)\,\mathrm{d}y, \tag{5.97}$$

in which the extra factor $1/y$ makes the integral absolutely convergent for functions that are not themselves integrable, while the derivative taken outside the integral restores the original kernel. Once the resulting operator is shown to be continuous on $\mathcal{S}$, the transposition to $\mathcal{S}'$ is automatic, and the degenerate orders $\alpha = n\pi$ acquire their correct meaning as the distributional kernels $\delta(x \mp y)$ instead of appearing as divergent integrals.

The third gap is that Namias verifies the index law (5.92) on the eigenfunction expansion, which is a formal manipulation of a doubly infinite sum. McBride and Kerr prove it as a theorem for all real α and β on a stated domain, and thereby establish that

$$\{\mathcal{F}_\alpha\}_{\alpha\in\mathbb{R}}, \qquad \mathcal{F}_0 = \mathcal{I}, \qquad \mathcal{F}_{\alpha+2\pi} = \mathcal{F}_\alpha, \tag{5.98}$$

is a genuine continuous one parameter group of homeomorphisms of $\mathcal{S}$, unitary on $L^2(\mathbb{R})$. Kerr afterwards developed the distributional theory in detail (Kerr, 1988a) and the L^2 theory together with applications to differential equations (Kerr, 1988b). With these three repairs, and only with them, is one entitled to manipulate $\mathcal{F}_\alpha$ as an algebraic object, which is what we shall do freely in the next subsection.

5.7.2.6 Why the construction is useful

The two faces of the transformation, the differential and the integral one, are useful in different circumstances, and it is worth separating them.

The **differential form** (5.76) is useful whenever the harmonic oscillator is hiding in the problem, which happens far more often than one would expect. Namias' own motivation was quantum mechanical (Namias, 1980): since $\mathcal{F}_\alpha$ is the evolution operator of the oscillator, the fractional transform of an initial wave packet *is* the wave packet at a later time, so that problems of time evolution become problems of choosing an order, and the well known revival of a wave packet after a full period is simply the 2π periodicity (5.98). More generally, since $\mathcal{F}_\alpha$ interchanges the operators x and $-i\partial_x$ by a rotation of angle α, a differential equation with polynomial coefficients can frequently be rotated into a simpler one, which is the technique Kerr systematised (Kerr, 1988b). The same generator governs the Landau problem, the paraxial wave equation in a quadratic index medium, and the linearised dynamics of any system near a stable equilibrium, so the differential form transfers immediately to all of them.

The **integral form** (5.91) is useful whenever one needs to compute rather than to reason. In signal processing it provides a continuum of representations interpolating between the time domain and the frequency domain, and since a linear chirp is a plane wave in some fractional domain, a chirped signal buried in noise becomes a sharp peak at the right order, which is the basis of chirp detection in radar and sonar and of filtering in fractional domains (Almeida, 1994; Sejdić et al., 2011). Lohmann's identification of the order with a rotation angle of the Wigner distribution (Lohmann, 1993), later placed on a rigorous footing by Mustard (1996), gives the whole construction a geometrical reading in phase space, and the discrete version of Candan, Kutay and Ozaktas (2000) makes it computable with the same logarithmic cost as the ordinary discrete transformation (Cooley and Tukey, 1965). In optics the integral form is what allows a fractional transformation to be built in the laboratory, either with a graded index medium of the appropriate length (Mendlovic and Ozaktas, 1993; Ozaktas and Mendlovic, 1993) or with a pair of lenses (Lohmann, 1993), and it underlies applications to beam propagation, phase retrieval, optical tomography, filtering and image encryption (Ozaktas et al., 2001).

For our purposes, however, the decisive application is none of these. It is the fact, proved in the next subsection, that the Fresnel integral (5.48) is itself a fractional Fourier transformation. The consequence is that the whole of paraxial propagation, which in the plane to plane description of the previous sections looked like a family of integrals labelled by a distance and composing in a complicated way, becomes a one parameter group labelled by an angle and composing by simple addition.

5.7.3 The role of Fourier analysis in the theory of diffraction

We are now in a position to prove the statement announced at the end of the section on the Fraunhofer approximation. The result is due to Pellat-Finet (Pellat-Finet, 1994; Pellat-Finet and Bonnet, 1994) and it may be stated as follows: *Fresnel diffraction between two spherical caps of equal radius is exactly a fractional Fourier transformation, whose order is fixed by the ratio of the propagation distance to the radius of the caps.* Since the transformation acts on two transverse variables we shall use the two dimensional kernel, which is separable and therefore the product of two copies of (5.91),

$$\mathcal{F}_\alpha[g](\vec{v}) = \frac{1 - i\cot\alpha}{2\pi}\int_{\mathbb{R}^2} \exp\left\{\frac{i}{2}\left(u^2+v^2\right)\cot\alpha - i\,\vec{u}\cdot\vec{v}\csc\alpha\right\} g(\vec{u})\,\mathrm{d}^2u, \tag{5.99}$$

where $u^2 = \|\vec{u}\|^2$ and $v^2 = \|\vec{v}\|^2$, the constant being the square of the one dimensional constant because the two dimensional transformation is the tensor product of two identical one dimensional ones.

5.7.3.1 Why a plane cannot be the right reference surface

It is instructive to begin by showing that the naive attempt fails, because the failure tells us exactly what to repair. Expanding the square in the Fresnel integral (5.48),

$$U(\vec{r}\,') = \frac{e^{ikz}}{i\lambda z}\int_{\mathbb{R}^2} U(\vec{r})\exp\left[\frac{ik}{2z}\left(r^2+r'^2\right)\right]\exp\left[-\frac{ik}{z}\,\vec{r}\cdot\vec{r}\,'\right]\mathrm{d}^2r, \tag{5.100}$$

where we used $\|\vec{r}-\vec{r}'\|^2 = r^2 + r'^2 - 2\vec{r}\cdot\vec{r}'$, we see that (5.100) has exactly the algebraic shape of (5.99): two quadratic terms with a common coefficient and one crossed term. Introduce therefore the reduced, dimensionless variables

$$\vec{u} = \frac{\vec{r}}{\omega}, \qquad \vec{v} = \frac{\vec{r}'}{\omega}, \tag{5.101}$$

with ω a scale length to be determined, and identify the coefficients term by term,

$$\frac{k}{2z}\,\omega^2 = \frac{\cot\alpha}{2} \qquad \text{(quadratic terms)}, \tag{5.102}$$

$$\frac{k}{z}\,\omega^2 = \csc\alpha \qquad \text{(crossed term)}. \tag{5.103}$$

Dividing (5.102) by (5.103) the unknown scale ω disappears and we are left with

$$\frac{1}{2} = \frac{\cot\alpha}{2\csc\alpha} = \frac{\cos\alpha}{2} \quad \Longrightarrow \quad \cos\alpha = 1,$$

that is $\alpha = 0$, which is the identity and for which $\csc\alpha$ is infinite, so that (5.103) cannot be satisfied by any finite ω. The conclusion is unambiguous.

Theorem 15 *Fresnel diffraction between two planes is never a fractional Fourier transformation, whatever the order. In the language of the group of linear canonical transformations it is a parabolic element, while the fractional Fourier transformations are the elliptic ones.*

The obstruction is entirely due to the relative weight of the quadratic and the crossed terms, which in (5.100) is fixed at the value one half whereas the kernel (5.99) demands $\cos\alpha/2$. Since the crossed term is the only one that couples the two planes, and is therefore the one we cannot touch, the only freedom we have is to modify the quadratic terms. That is exactly what referring the field to a curved surface does.

5.7.3.2 Reference spheres and the fractional order

Let the emitting plane be at the origin and the receiving plane at the distance z along the axis. Consider a spherical cap tangent to a transverse plane at the axis and whose centre of curvature lies on the axis at the signed distance c from the vertex, counted positive in the direction of propagation. In the paraxial regime the point of the cap at the transverse position $\vec{r}$ is displaced axially from the tangent plane by the sagitta

$$\delta(\vec{r}) = c - \operatorname{sgn}(c)\sqrt{c^2 - r^2} = \frac{r^2}{2c} + \mathcal{O}\left(\frac{r^4}{c^3}\right), \tag{5.104}$$

which is the same expansion we performed in (5.47). Since a field travelling in the positive direction acquires the factor $e^{ik\delta}$ over an axial displacement δ, the field referred to the cap and the field on the tangent plane are related by

$$u_{\text{cap}}(\vec{r}) = U(\vec{r})\exp\left(\frac{ik\,r^2}{2c}\right). \tag{5.105}$$

Let now $\mathcal{A}$ be a cap tangent to the emitting plane with centre of curvature at c_1, and $\mathcal{B}$ a cap tangent to the receiving plane with centre of curvature at the signed distance c_2 from its own vertex. Using (5.105) on both sides,

$$U(\vec{r}) = u_{\mathcal{A}}(\vec{r}) \exp\left(-\frac{ik\,r^2}{2c_1}\right), \qquad u_{\mathcal{B}}(\vec{r}\,') = U(\vec{r}\,') \exp\left(\frac{ik\,r'^2}{2c_2}\right), \tag{5.106}$$

and substituting both relations into (5.100),

$$u_{\mathcal{B}}(\vec{r}\,') = \frac{e^{ikz}}{i\lambda z}\, e^{\frac{ikr'^2}{2c_2}} \int u_{\mathcal{A}}(\vec{r})\, e^{-\frac{ikr^2}{2c_1}} \exp\left[\frac{ik}{2z}\left(r^2 + r'^2\right)\right] e^{-\frac{ik}{z}\vec{r}\cdot\vec{r}\,'}\, \mathrm{d}^2 r =$$
$$\frac{e^{ikz}}{i\lambda z} \exp\left[\frac{ikr'^2}{2}\left(\frac{1}{z} + \frac{1}{c_2}\right)\right] \int u_{\mathcal{A}}(\vec{r}) \exp\left[\frac{ikr^2}{2}\left(\frac{1}{z} - \frac{1}{c_1}\right)\right] e^{-\frac{ik}{z}\vec{r}\cdot\vec{r}\,'}\, \mathrm{d}^2 r,$$

where we have merely collected the coefficients of r^2 and of r'^2. The kernel (5.99) requires the two quadratic coefficients to be equal, which imposes

$$\frac{1}{z} - \frac{1}{c_1} = \frac{1}{z} + \frac{1}{c_2} \quad \Longrightarrow \quad \frac{1}{z} - \frac{1}{c_1} - \frac{1}{z} - \frac{1}{c_2} = 0 \quad \Longrightarrow \quad c_2 = -c_1, \tag{5.107}$$

the two terms in red cancelling identically. We therefore set

$$c_1 = R, \qquad c_2 = -R, \tag{5.108}$$

which describes a perfectly definite geometry: *two spherical caps of the same radius R, the emitting one having its centre of curvature downstream at the abscissa R, and the receiving one, whose vertex sits at the abscissa z, having its centre of curvature upstream at the abscissa $z - R$.* These are the cardinal spheres of radius R associated with the propagation, and their geometry is the one drawn in Figure 5.6.

With (5.108) the two quadratic coefficients collapse into the single quantity

$$Q = \frac{k}{2}\left(\frac{1}{z} - \frac{1}{R}\right), \tag{5.109}$$

and the propagation reads

$$u_{\mathcal{B}}(\vec{r}\,') = \frac{e^{ikz}}{i\lambda z} \int_{\mathbb{R}^2} u_{\mathcal{A}}(\vec{r}) \exp\left[iQ\left(r^2 + r'^2\right)\right] \exp\left[-\frac{ik}{z}\,\vec{r}\cdot\vec{r}\,'\right] \mathrm{d}^2 r, \tag{5.110}$$

which now has exactly the structure of (5.99) with independent quadratic and crossed coefficients. Repeating the identification of the previous subsubsection with the reduced variables (5.101),

$$Q\,\omega^2 = \frac{\cot\alpha}{2} \quad \Longrightarrow \quad \cot\alpha = k\,\omega^2\left(\frac{1}{z} - \frac{1}{R}\right), \tag{5.111}$$
$$\frac{k}{z}\,\omega^2 = \csc\alpha, \tag{5.112}$$

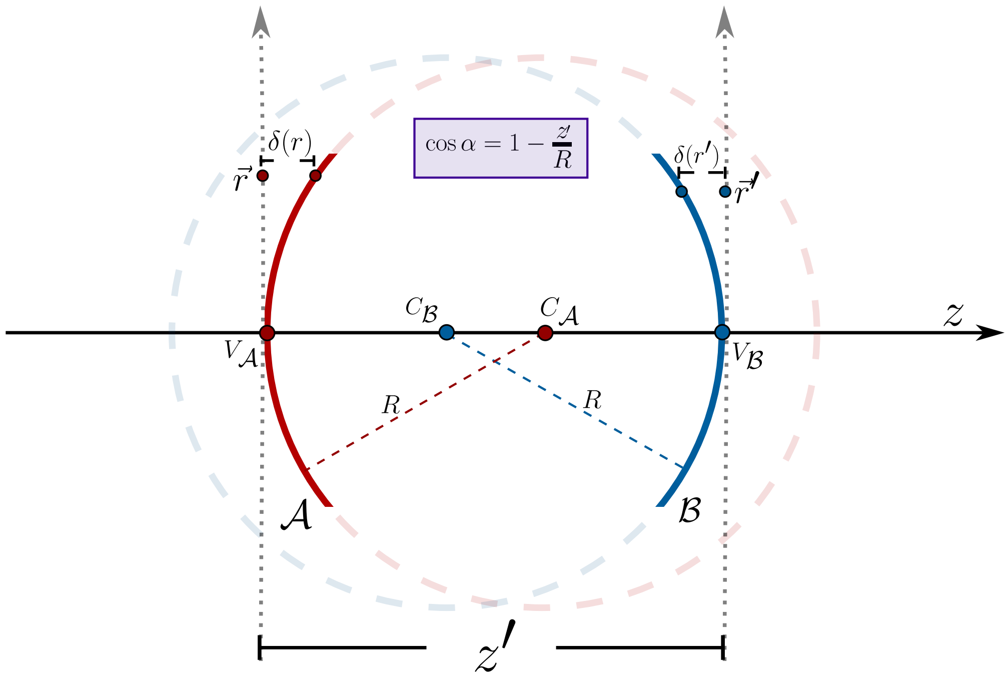


Figure 5.6: The two cardinal spheres of radius R associated with a propagation over the distance z. The emitting cap $\mathcal{A}$ is tangent to its plane at $V_{\mathcal{A}}$ with its centre of curvature downstream at $C_{\mathcal{A}}$, of abscissa R, and the receiving cap $\mathcal{B}$ is tangent to its plane at $V_{\mathcal{B}}$ with its centre upstream at $C_{\mathcal{B}}$, of abscissa $z - R$, so that the two caps bow towards each other. The quantity δ is the sagitta (5.104). The confocal case of Figure 5.4 is recovered when $R = z$, since then $C_{\mathcal{A}}$ falls on $V_{\mathcal{B}}$ and $C_{\mathcal{B}}$ on $V_{\mathcal{A}}$.

and dividing (5.111) by (5.112) the scale ω cancels once again,

$$\cos\alpha = \frac{\cot\alpha}{\csc\alpha} = \frac{k\,\omega^2\left(\dfrac{1}{z} - \dfrac{1}{R}\right)}{k\,\omega^2\,\dfrac{1}{z}} = z\left(\frac{1}{z} - \frac{1}{R}\right),$$

so that we obtain the central result

$$\boxed{\cos\alpha = 1 - \frac{z}{R}.} \tag{5.113}$$

This time there is no contradiction, because the radius R of the reference caps is at our disposal. The scale length follows from (5.112),

$$\omega^2 = \frac{z}{k\sin\alpha} = \frac{\lambda z}{2\pi\sin\alpha}, \tag{5.114}$$

and, eliminating α by means of (5.113) and of $\sin^2\alpha = 1 - \cos^2\alpha$,

$$\sin^2\alpha = 1 - \left(1 - \frac{2z}{R} + \frac{z^2}{R^2}\right) = \frac{z\,(2R - z)}{R^2} \quad\Longrightarrow\quad \sin\alpha = \frac{\sqrt{z\,(2R - z)}}{R},$$

the two units in red cancelling, so that the scale length is expressed entirely in terms of the geometry,

$$\omega^2 = \frac{\lambda R}{2\pi}\sqrt{\frac{z}{2R-z}}. \tag{5.115}$$

5.7.3.3 The normalising constant and the final result

It remains to compare the constants. Changing variables in (5.110) according to (5.101), so that $\mathrm{d}^2 r = \omega^2\,\mathrm{d}^2 u$,

$$u_{\mathcal{B}}(\omega\vec{v}) = \frac{e^{ikz}\,\omega^2}{i\lambda z}\int_{\mathbb{R}^2} u_{\mathcal{A}}(\omega\vec{u})\exp\left[\frac{i}{2}\left(u^2+v^2\right)\cot\alpha - i\,\vec{u}\cdot\vec{v}\csc\alpha\right]\mathrm{d}^2 u, \tag{5.116}$$

and using (5.114) the prefactor simplifies to

$$\frac{\omega^2}{i\lambda z} = \frac{1}{i\,\lambda z}\cdot\frac{\lambda z}{2\pi\sin\alpha} = \frac{1}{2\pi i\sin\alpha} = \frac{-i}{2\pi\sin\alpha}. \tag{5.117}$$

On the other hand the constant of (5.99), rewritten by means of the identity (5.90), is

$$\frac{1-i\cot\alpha}{2\pi} = \frac{1}{2\pi}\cdot\frac{e^{-i\pi/2}e^{i\alpha}}{\sin\alpha} = \frac{-i\,e^{i\alpha}}{2\pi\sin\alpha}. \tag{5.118}$$

Dividing (5.117) by (5.118),

$$\frac{\dfrac{-i}{2\pi\sin\alpha}}{\dfrac{-i\,e^{i\alpha}}{2\pi\sin\alpha}} = \frac{-i}{-i}\,e^{-i\alpha} = e^{-i\alpha},$$

and therefore, collecting everything, we have proved the theorem.

Theorem 16 *(Fresnel diffraction as a fractional Fourier transformation) Let the field be referred to two cardinal spherical caps of radius R separated by the distance z along the axis, in the sense of* (5.108)*, and let the transverse coordinates be reduced by the scale ω of* (5.115)*. Then the Fresnel propagation* (5.48) *is the fractional Fourier transformation*

$$u_{\mathcal{B}}(\omega\vec{v}) = e^{i(kz-\alpha)}\left(\mathcal{F}_\alpha\,u_{\mathcal{A}}\right)(\vec{v}), \tag{5.119}$$

whose order is given by $\cos\alpha = 1 - z/R$*.*

A remark on the residual phase is in order. Apart from the trivial factor e^{ikz}, which merely counts the optical path along the axis, the propagation leaves behind the factor $e^{-i\alpha}$ and nothing else. This is a direct consequence of the normalisation $1/(i\lambda z)$ of the Fresnel integral (5.48), the one for which the kernel tends to $\delta(\vec{r}-\vec{r}\,')$ as $z\to 0$; had the constant been written with the opposite sign, an additive π would have appeared in the exponent of (5.119) with no physical content whatever. With the correct normalisation the order α is exactly the accumulated Gouy phase, a point to which we shall return.

Let us test (5.119) on the three cases we can check independently.

1. $z = 0$. Equation (5.113) gives $\cos\alpha = 1$, hence $\alpha = 0$ and $\mathcal{F}_0 = \mathcal{I}$: the field on the receiving cap is the field on the emitting cap, as it must be.
2. $z = R$. Now $\cos\alpha = 0$, hence $\alpha = \pi/2$ and the transformation is the ordinary Fourier transformation. This is not a coincidence: when $z = R$ the emitting cap has its centre at the abscissa $R = z$, which is the vertex of the receiving cap, and the receiving cap has its centre at $z - R = 0$, which is the vertex of the emitting cap. The two caps are therefore *the confocal cardinal spheres* of Figure 5.5, and we recover exactly the Fraunhofer result (5.59) of the previous section. The far field description is the particular case $\alpha = \pi/2$ of the present one.
3. $z = 2R$. Here $\cos\alpha = -1$, hence $\alpha = \pi$ and $\mathcal{F}_\pi = \mathcal{P}$: the field on the receiving cap is the field on the emitting cap turned upside down. Two consecutive Fourier transformations invert the image, and the geometry says the same thing.

The gain in economy is considerable. In the plane to plane description, chaining two propagations of distances z_1 and z_2 requires composing two Fresnel integrals, and the composite kernel must be recomputed each time. In the present description the composition is the index law (5.92),

$$\mathcal{F}_{\alpha_2}\mathcal{F}_{\alpha_1} = \mathcal{F}_{\alpha_1+\alpha_2}, \tag{5.120}$$

so that propagation is described by the addition of angles. The distances z do not add in any useful way, but the orders do, and this is the precise sense in which α, and not z, is the natural parameter of free paraxial propagation.

5.7.3.4 Physical meaning of a real order

When α is real, (5.113) requires $|1 - z/R| \leq 1$, that is

$$0 \leq z \leq 2R, \tag{5.121}$$

and the order sweeps the interval $[0, \pi]$ monotonically as the observation cap recedes. The reading of α is then completely transparent, and there are three equivalent ways of stating it.

As a degree of diffraction. The order measures how far the field has travelled from the near field towards the far field, on a scale in which $\alpha = 0$ is the aperture itself and $\alpha = \pi/2$ is its Fraunhofer pattern. A field observed at $\alpha = \pi/4$ is halfway between its own distribution and its Fourier transform, in a sense that is not metaphorical but exact, since applying the same transformation twice produces the Fourier transform. The continuum of intermediate regimes that the Fresnel approximation describes only implicitly is here labelled by a single number.

As a rotation of phase space. By the theorem of Lohmann (1993), later established rigorously by Mustard (1996), the transformation $\mathcal{F}_\alpha$ acts on the Wigner distribution of the field simply by rotating the plane of the transverse position and the transverse spatial frequency by the angle α. Free propagation between cardinal spheres is therefore a rigid rotation of phase space, which is why the orders add: rotations compose by adding angles. This also explains geometrically why plane to plane propagation is not a fractional transformation, since in the plane description propagation is a shear of phase space rather than a rotation.

As a Gouy phase. The residual factor $e^{-i\alpha}$ of (5.119) is not an accident. If instead of arbitrary caps one chooses as reference surfaces the wavefronts of a Gaussian beam of Rayleigh range z_R, the order that comes out of the same computation is

$$\alpha = \arctan\left(\frac{z}{z_R}\right), \tag{5.122}$$

which is precisely the accumulated Gouy phase of the beam (Ozaktas and Mendlovic, 1995). The identification is satisfying, since (5.122) tends to $\pi/2$ as $z \to \infty$, recovering once more the statement that the far field is the Fourier transform of the near field, and it shows that the mysterious extra phase of a focused beam is nothing but the order of the fractional transformation the beam has undergone.

One point must be emphasised, because it is a frequent source of confusion. *The order is not a property of the propagation alone.* It is a property of the pair formed by the propagation and the two reference caps, since (5.113) involves both z and R. Given a propagation distance z, any order whatsoever can be obtained by choosing

$$R = \frac{z}{1 - \cos\alpha}, \tag{5.123}$$

and conversely the same pair of caps assigns different orders to different distances. What is intrinsic is not α but the group structure: whatever reference caps are chosen, propagation is represented by an element of the same one parameter group.

5.7.3.5 Physical meaning of a complex order

The condition (5.121) is a real restriction, and it is natural to ask what happens when it is violated. If the radius R of the caps is held fixed while the propagation distance exceeds $2R$, or if the propagation is run backwards so that $z < 0$, then $\cos\alpha$ leaves the interval $[-1, 1]$ and the order becomes complex. Writing $\alpha = \alpha_r + i\alpha_i$ there are two cases.

For $\cos\alpha > 1$, which corresponds to $z < 0$ with $R > 0$, the order is purely imaginary, $\alpha = i\beta$ with

$$\cosh\beta = 1 - \frac{z}{R}. \tag{5.124}$$

For $\cos\alpha < -1$, which corresponds to $z > 2R$, the order is $\alpha = \pi + i\beta$ with $\cosh\beta = z/R - 1$, the additive π carrying the inversion of the image that we already met at $z = 2R$.

The consequence for the kernel is immediate and physically eloquent. Taking the first case and using $\cot(i\beta) = -i\coth\beta$ and $\csc(i\beta) = -i\,\mathrm{csch}\,\beta$, the exponent of (5.99) becomes

$$\begin{aligned}&\frac{i}{2}\left(u^2 + v^2\right)(-i\coth\beta) - i\,\vec{u}\cdot\vec{v}\,(-i\,\mathrm{csch}\,\beta) = \\ &\frac{1}{2}\left(u^2 + v^2\right)\coth\beta - \vec{u}\cdot\vec{v}\,\mathrm{csch}\,\beta,\end{aligned}$$

which is *entirely real*. The kernel has ceased to be a pure chirp and has become a real Gaussian, and since

$$\coth\beta - \operatorname{csch}\beta = \frac{\cosh\beta - 1}{\sinh\beta} > 0 \qquad (\beta > 0),$$

the associated quadratic form is positive definite, so the kernel grows rather than oscillates. This is the analytic signature of an ill posed problem, and it is exactly what one should expect, since $z < 0$ is back propagation and back propagation amplifies the high spatial frequency content of the field without bound. The complex order does not merely describe the difficulty, it quantifies it: the imaginary part β is the exponent of the Gaussian amplification.

More generally, a complex order admits three complementary physical readings.

1. **Group theoretically**, a real order corresponds to an elliptic element of the group of linear canonical transformations, that is, to a rotation of phase space, while a complex order corresponds to an element that also contains a squeeze, that is, a magnification along one direction of phase space and a compression along the other. A complex order therefore announces that the transformation is a rotation composed with a scaling, and the imaginary part measures the scaling.

2. **For Gaussian beams**, the wavefront curvature is not a real number but the complex parameter q defined by $q^{-1} = R^{-1} - i\lambda/(\pi w^2)$, where w is the beam radius. Substituting q for R in (5.113) gives a complex order automatically, whose real part is the phase space rotation and whose imaginary part carries the Gaussian envelope of the beam. Complex order fractional transformations are, for this reason, the natural language of Gaussian beam propagation, and they were realised optically by Shih (1995b) and by Bernardo and Soares (1996).

3. **For apodised systems**, a Gaussian aperture e^{-r^2/a^2} placed in the beam contributes exactly a real Gaussian factor of the kind that a complex order produces. The imaginary part of the order then measures the attenuation of the large $|\vec{u}|$ content of the field, that is, a Gaussian low pass filtering in the fractional domain, which in the image is a loss of resolution. A complex order is, in this reading, the bookkeeping device for soft apertures and for losses.

Finally, it should be stressed that a complex order never signals a physical impossibility. Since (5.123) shows that the order depends on the reference caps, any forward propagation with $z > 0$ can be described with a real order simply by choosing $R \geq z/2$. What a complex order signals is that the reference surfaces chosen are not the ones with respect to which the propagation is a pure rotation. The limiting case of this statement is the plane, for which $R \to \infty$ gives $\cos\alpha \to 1$ and, by (5.114), $\omega \to \infty$: the transformation degenerates, the rotation angle collapses to zero while the scale diverges, and we recover the theorem of the first subsubsection that plane to plane Fresnel propagation lies outside the fractional Fourier family. The whole content of Pellat-Finet's result is that the family is recovered as soon as one agrees to look at the field on a sphere.

Chapter 6

Theory of Coherence

Coherence is the thread that weaves order out of the chaos of random light.

"Time advances in the direction in which correlations increase."
– attributed to Ludwig Boltzmann

6.1 A historical context of coherence theory

At its heart, coherence asks a single, almost childlike question: how much does one thing resemble another, and does that resemblance survive as time goes by? It sounds too simple to matter, and yet the search for a rigorous answer to that question grew, over three centuries, into one of the great unifying ideas of physics: a language that explains why some light sources cast crisp interference fringes while others cast none at all, why a laser beam behaves so differently from a candle flame, how radio astronomers photograph the shadow of a black hole without ever building a telescope the size of the Earth, and why the delicate superpositions inside a quantum computer are so maddeningly hard to keep alive. The story of coherence is really two stories that grew up separately and only later discovered they were talking about the same thing: one begins at a gambling table in seventeenth century France, and the other begins in a darkened optics laboratory. It is worth following both threads from the start.

The first thread is the story of correlation, and it begins, fittingly, with a bet. In 1654 the nobleman and gambler Antoine Gombaud, better known as the Chevalier de Méré, wrote to Blaise Pascal asking how the stakes of an interrupted game of dice should fairly be divided between two players. The resulting correspondence between Pascal and Pierre de Fermat, working out the mathematics of expectation and combinatorics needed to answer such questions, is generally credited as the birth of probability theory. Their ideas matured slowly. In 1713, sixty years later, Jacob Bernoulli's posthumous book *Ars Conjectandi* Bernoulli (1713) proved the law of large numbers, the remarkable fact that if you repeat a random experiment often enough, the observed frequency of an outcome settles down, ever more reliably, around its true theoretical probability. A century later still, in 1812, Pierre Simon Laplace gathered these scattered results, together with his own theory of how new evidence should update our beliefs (what we now call Bayesian inference), into the monumental *Théorie Analytique des Probabilités* Laplace (1812). What was still missing from this whole tradition was a number, a single figure that could say not just how uncertain one quantity is, but how strongly two different quantities move together.

That missing number arrived through an unexpected route: the very practical business of dealing with noisy measurements. In 1809, working out how to fit the imperfect, scatter riddled observations of planetary orbits, Carl Friedrich Gauss introduced the method of least squares in his *Theoria Motus* Gauss (1809), a technique built entirely on the idea that errors cluster statistically around the truth. Decades later, in the 1880s, the polymath Francis Galton, fascinated by how strongly a child's height tends to track its parents', coined the very word "correlation" to describe exactly this kind of joint tendency Galton (1888). It fell to his student Karl Pearson, in 1896, to turn Galton's intuition into a precise formula: the Pearson correlation coefficient Pearson (1896), a single number, always between minus one and plus one, that tells you at a glance whether two quantities march together, drift apart, or ignore each other completely. For the first time, resemblance between two things could be measured rather than merely felt.

The second thread, meanwhile, was unfolding almost entirely without reference to the first, in the study of light itself. In 1801, Thomas Young let sunlight fall through two narrow slits and watched it paint a row of bright and dark bands on a screen behind them Young (1804), the unmistakable signature of two waves sometimes reinforcing and sometimes cancelling one another. For those fringes to appear at all, the light leaving the two slits had to keep a steady, dependable phase relationship with itself, a property nobody yet had a name for, though everybody could see its footprint on the screen. Between 1815 and 1820, Augustin Jean Fresnel built the full mathematical theory of light as a transverse wave, and with it interference and polarisation stopped being curiosities and became consequences of a single physical picture. And yet a puzzle quietly remained. Young's beam, freshly split from a single narrow slit, produced fringes beautifully. But ordinary daylight, or the light of a candle, or of the Sun itself, never produced anything of the sort when simply allowed to interfere with a delayed copy of itself beyond a certain separation. Something about real, everyday light was different from the idealised, perfectly regular wave that Fresnel's mathematics described, and for a hundred years nobody could quite say what.

The answer, when it finally came in the twentieth century, is that ordinary light sources are not single, orderly waves at all. A hot filament, a flame, or the surface of the Sun is an enormous crowd of independent atoms, each one emitting its own short burst of light and then falling silent, entirely out of step with its neighbours, so that the overall wave forms and reforms its phase billions of times each second in a way that is, for all practical purposes, random. Describing such light properly meant marrying the two separate traditions that had grown up in isolation: the statistical machinery of Pascal, Gauss, Galton and Pearson had to be grafted onto the wave optics of Young and Fresnel. That graft is exactly what the word "coherence" now names: not whether a wave has a phase, every wave does, but whether that phase remains predictable enough, for long enough and over a wide enough region, to leave a visible trace when the wave is made to interfere with a copy of itself.

The decisive mathematical step was taken in the 1930s. In 1934 the Dutch physicist Pieter Hendrik van Cittert worked out how the statistical correlations in the light from an incoherent, extended source, a distant star, for instance, change as that light travels away from the source van Cittert (1934), and in 1938 Frits Zernike sharpened and generalised the result into the theorem that today carries both their names Zernike (1938). The van Cittert Zernike theorem makes a beautifully counterintuitive claim: light that starts out utterly incoherent, with every point on the source flickering independently of every other, becomes measurably, predictably more coherent purely by virtue of travelling a long distance. This is not a minor footnote. It is the physical principle that allows astronomers to measure the angular size of stars far too small for any telescope to resolve directly, by comparing the light collected at two separated points and watching how the fringes fade as the separation grows, exactly the trick Albert Michelson and Francis Pease used in 1920 to measure the diameter of the red giant Betelgeuse Michelson and Pease (1921). Decades later, the very same theorem, scaled up from a handful of metres to a network of radio dishes spanning the whole planet, is what let the Event Horizon Telescope collaboration reconstruct, in 2019, the first direct image of a black hole.

Classical coherence theory found its definitive, mature form in the work of Max Born and Emil Wolf, whose treatise *Principles of Optics* Born and Wolf (1999) gathered every strand into a single object, the mutual coherence function, which simply measures how correlated the light wave is between two points in space and two moments in time. Squeezed inside that one function are both of the qualities an experimenter actually cares about: the temporal coherence, essentially how pure a colour the light is and for how long it keeps a steady rhythm with itself, and the spatial coherence, how faithfully the wave stays in step across an extended region rather than just at a single point. Leonard Mandel and Wolf later expanded this picture into the full treatise *Optical Coherence and Quantum Optics* Mandel and Wolf (1995), sharpening the practical notions of coherence time and coherence length that every laboratory now takes for granted. None of this stayed confined to the blackboard. The Michelson interferometer of the 1890s already exploited temporal coherence to measure wavelengths with a precision no ruler could match, and the invention of the laser in 1960, whose light is coherent to a degree no natural source ever achieves, turned coherence from a subject of theoretical fascination into the working principle behind fibre optic communication, precision metrology, holography and modern medicine.

Nor does coherence belong to optics alone. Wherever many independent parts of a system can be made to oscillate in step, something with exactly the same mathematical anatomy appears. In condensed matter physics, superconductivity is, at bottom, a coherence phenomenon: below a critical temperature, an enormous population of paired electrons locks into a single, shared quantum phase across an entire macroscopic sample, which is why a supercurrent flows without any resistance at all. Nuclear magnetic resonance and the magnetic resonance imaging built upon it work by tipping billions of nuclear spins into a common, coherent precession and then watching, with exquisite sensitivity, how quickly that shared rhythm decays. Every atomic clock, and with it the entire Global Positioning System, depends on an oscillator whose phase stays coherent for as long as humanly achievable. Coherence, in short, is not a peculiarity of light; it is what allows countless independent actors, electrons, spins, atoms or photons, to behave, briefly and fragilely, as one.

While all of this classical machinery was being consolidated, an entirely separate and stranger story was unfolding in the quantum world. Its opening scene took place in 1956, when Robert Hanbury Brown and Richard Twiss, trying to measure the angular sizes of stars with radio and then optical detectors, found that photons arriving from a thermal source at two separated detectors were not independent: they tended to arrive in correlated little bunches Hanbury Brown and Twiss (1956). The result seemed to many physicists at the time to be flatly impossible. Photons, everyone assumed, should behave like classical, independent particles once detected, and no such correlation should survive the act of measurement; a real and lively controversy followed before the effect was accepted. Untangling it properly required a fully quantum theory of coherence, and that theory was supplied in 1963, almost simultaneously and independently, by Roy Glauber Glauber (1963) and George Sudarshan Sudarshan (1963). Glauber introduced the coherent states, the quantum states of light that behave as closely as quantum mechanics permits to Fresnel's classical waves, and defined a whole hierarchy of quantum correlation functions capable of explaining Hanbury Brown and Twiss's bunching rigorously, and of predicting its mirror image, photon antibunching, a behaviour so stubbornly non classical that no wave theory of light could ever reproduce it, later confirmed in the laboratory. Sudarshan, working from a different direction, found the precise mathematical boundary, now called the Glauber Sudarshan representation, that separates light whose quantum nature can safely be ignored from light for which it cannot. For this quantum theory of optical coherence, Glauber received the Nobel Prize in 2005, and the field he founded is what we now call quantum optics.

Quantum coherence turned out to matter for reasons that reach far beyond photon counting. It is quantum coherence, the delicate, phase locked superposition of many possible states at once, that makes a quantum bit fundamentally different from an ordinary one, and it is the loss of that coherence through unavoidable contact with the environment, a process physicists call decoherence, that explains why the everyday world around us looks so stubbornly classical even though it is built from quantum parts. Keeping coherence alive for long enough to be useful is, to this day, the single greatest engineering obstacle standing between today's quantum computers and tomorrow's. The same idea reappears in the coldest matter physicists know how to make: in 1995 Eric Cornell, Carl Wieman and Wolfgang Ketterle cooled clouds of atoms to nanokelvin temperatures and watched millions of them collapse into a single, shared quantum state, a Bose Einstein condensate, in effect one enormous coherent matter wave, an achievement recognised with the Nobel Prize in 2001. It even reaches into biology: careful experiments on the light harvesting complexes that plants and certain bacteria use to capture sunlight have found that quantum coherence in the transfer of that energy survives, astonishingly, for measurable fractions of a second at room temperature, hinting that evolution may have stumbled upon and exploited quantum coherence long before physicists gave it a name.

Binding the statistical and the wave pictures of coherence together, classical and quantum alike, is a single piece of mathematics: the Wiener Khinchin theorem, developed independently by Norbert Wiener in 1930 Wiener (1930) and Aleksandr Khinchin in 1934 Khinchin (1934), which states that the correlation of a fluctuating signal with itself and the way that signal's energy is spread across frequency are simply two faces of the same object, translated into one another by a Fourier transform. It is, as later chapters will show, the quiet mathematical hinge on which the whole subject turns.

In the twenty first century coherence has acquired one further identity: it is now understood, and formally quantified, as a genuine physical resource, something that can be created, spent and depleted much like energy or entanglement, with its own rigorous accounting rules inside the broader theory of quantum information Baumgratz et al. (2014). Seen end to end, the history of coherence runs from a seventeenth century argument about dice, through the statistics of inherited height and the interference of sunlight, to the imaging of a black hole and the fragile inner workings of a quantum processor. In its classical guise it explains why stars twinkle the way they do, why a laser can cut steel while a torch cannot, and why an interferometer the size of a planet can photograph something no single telescope ever could. In its quantum guise it explains why the world of atoms feels so unlike the world of tables and chairs, and it now sits at the very centre of the effort to build technologies that think and communicate in ways no classical machine ever could. Coherence, whether counted in correlated dice throws, interfering light waves or entangled qubits, is ultimately always the same thing: order, briefly and remarkably, surviving in a world that constantly conspires to erase it.

6.2 How is statistical similarity quantified?

Suppose we have a completely random phenomenon, such as the electromagnetic radiation emitted by an incandescent bulb that is not in thermodynamic equilibrium, the number of cars in some capital city throughout the day, the motion of an ion in a plasma, or something as simple as throwing a ball into the air under non-ideal conditions.

When we wish to analyse statistically a single object or random variable, the familiar tools appear: the probability density, the mean value, the standard deviation, the kurtosis, and so on. What happens when we want to study at least two random variables and how they relate? When we wish to study the statistics of two or more random variables we must study the statistical similarity between them; but statistical similarity is a rather broad concept, since we may have similarity in the mean values, or in the standard deviations, or in the probability densities. To address this problem we return to linear algebra.

If we have two vectors $\vec{A}, \vec{B} \in \mathbb{R}^n$, the dot product $\vec{A} \cdot \vec{B}$ is interpreted as the magnitude of the projection of $\vec{A}$ onto $\vec{B}$ and vice versa; it can also be read as *how much of* $\vec{A}$ *lies along* $\vec{B}$.

It is worth pausing on what that operation actually does mechanically, because the whole of coherence theory is built on this single gesture. Written out component by component, the dot product is

$$\vec{A} \cdot \vec{B} = \sum_{i=1}^{n} A_i B_i, \tag{6.1}$$

that is, *multiply the two lists entry by entry and add up the results.* Nothing more. And yet the sign of each individual term already carries exactly the information we are after: whenever A_i and B_i lean the same way, their product is positive and pushes the sum upwards; whenever they lean opposite ways, the product is negative and drags it down. If the two vectors tend to agree along most of their components, the positive terms dominate and the total comes out large; if they systematically disagree, the total comes out large but negative; and if they agree just as often as they disagree, the terms cancel one another in the sum and the total collapses to nearly zero. So "multiply the pairs and add" is already, all by itself, a similarity meter, and everything that follows in this chapter is that same gesture applied to progressively richer objects.

This behaviour can be extrapolated to continuous vector spaces, where one usually employs the notion of inner product. The step is smaller than it looks: a function is simply a vector with a continuum of components, since instead of the finite list $A_1, A_2, \ldots, A_n$ we now carry the values $f_A(x)$, one for every x. The index i becomes the label x, the sum over a discrete index becomes an integral over a continuous one, and the dot product becomes the inner product. If f_A, f_B are two elements of the Hilbert space $L^2(\mathbb{R})$ of square integrable functions of one real variable, one can define an inner product as

$$\langle f_A, f_B \rangle = \int_{\mathbb{R}} f_A^*(x) f_B(x) W(x)\, dx, \tag{6.2}$$

where the complex conjugate on the first factor is what keeps $\langle f_A, f_A \rangle$ real and non negative even when the functions are complex, as the optical fields of this chapter will be, and where $W(x)$ is some weight function.[1] The interpretation is still valid: this can be read as how much the function f_A over all the reals coincides with the function f_B over all the reals.

To reach the notion of correlation from linear algebra we use a bit of braket notation, where the ket $|f_A\rangle$ represents a vector of an arbitrary Hilbert space L^2 and the bra $\langle f_A|$ represents an element of the dual of the same Hilbert space,[2] and the inner product of a state with itself, $\langle f_A, f_A\rangle = \langle f_A|f_A\rangle$, accumulates the probability of finding the system anywhere at all; it is the integrand $|f_A(q)|^2$, rather than the completed integral, that plays the role of the probability density of finding f_A in a state $q \pm \triangle q$, where q represents some generalised coordinate. Suppose that $f_A, f_B \in L^2$ have the form

$$|f_A\rangle = \hat{A}\,|\Psi_A\rangle\,, \quad |f_B\rangle = \hat{B}\,|\Psi_B\rangle\,, \tag{6.3}$$

where $\hat{A}, \hat{B}$ are the operators of the physical quantities of $|\Psi_A\rangle\,, |\Psi_B\rangle$, so that we may interpret $|f_A\rangle\,, |f_B\rangle$ as the wave functions steered by their expected value. Using the inner product (6.2) with unit weight function, $W = 1$, the inner product reads

$$\langle f_A|f_B\rangle = \int_{\mathbb{R}^n} \hat{A}\hat{B}\,\langle\Psi_A|\Psi_B\rangle\; d^n q = \int_{\mathbb{R}^n} \hat{A}\hat{B}\,P(A;B)\, d^n q, \tag{6.4}$$

where we see that the similarity between $|f_A\rangle$ and $|f_B\rangle$ can be viewed as the product of $\hat{A}$ with $\hat{B}$ times the probability density of having states A and B.[3] That is, the similarity, or how *correlated* two variables are, can be seen as the integral of the product of the variables times their joint probability density.

With this analogy in mind we can understand correlation as a first measure of the similarity between one or more variables across space and time. If $x_1(\vec{r}, t_1), x_2(\vec{r}', t_2)$ are two random variables and $p_2(x_1, t_1; x_2, t_2)$ is the joint probability density, then the correlation between x_1, x_2 is defined as

$$\Gamma(\vec{r}, \vec{r}'; t_1, t_2) = E\left\{x_1(\vec{r}, t_1) x_2^*(\vec{r}', t_2)\right\} = \iint x_1 x_2^*\; p_2(x_1, t_1; x_2, t_2)\, dx_1 dx_2, \tag{6.5}$$

that is, as the ensemble average of $x_1 \cdot x_2^*$, weighted by how likely each pair of values is to occur together. The conjugation is placed on the second factor here to agree with the convention used for the coherence function in the next section; for real valued variables it does nothing at all, and

[1] $W(x)$ receives this name because, as it says, it gives a weight to each value the integrand may take; there are no uniform contributions, but rather each possible value of the integrand within the domain contributes weighted by the value of W at that point.

[2] The Hilbert space is a particular case of the L^p spaces, the space of p-norm Lebesgue-integrable functionals; for Hilbert $p = 2$, and it has the property that the dual of L^2 is also L^2, which lets quantum mechanics arise naturally in this space.

[3] This last step is deliberately heuristic: we are treating the operators as though they could simply be pulled out of the inner product and replaced by the values they measure, which is only legitimate when acting on their own eigenstates. The aim here is not a derivation but a bridge, to make plausible where the definition that follows comes from; the rigorous treatment belongs to Mandel and Wolf (1995).

for complex ones it merely conjugates Γ as a whole, leaving every magnitude we shall care about untouched.

Structurally this is the same object we have been building all along. The sum over components in (6.1) has simply become an average over the ensemble, and the role of the weight function W in (6.2) is now played by the joint probability density p_2, which is the natural weight to use when the "components" being paired up are the possible outcomes of a random experiment rather than the entries of a list. Correlation, in other words, is the dot product of two random variables, taken in the only sense available when the quantities involved refuse to hold still.

The same sign argument that made the dot product a similarity meter now tells us how to read Γ, and it is worth spelling out the three cases explicitly. If the two variables rise and fall together, so that they are large at the same instants and small at the same instants, then the product $x_1 x_2^*$ is predominantly positive, every realisation of the experiment contributes with the same sign, and the average survives as a large positive number. If they move in strict opposition, one peaking whenever the other troughs, the product is predominantly negative and the average survives as a large negative number, which is just as informative: perfect anticorrelation is a form of perfect similarity, merely with a sign attached. But if the two variables know nothing about each other, then for every realisation in which the product happens to come out positive there is, somewhere in the ensemble, an equally likely realisation in which it comes out negative, the contributions annihilate one another, and the average collapses to zero. **The averaging is therefore not an incidental technicality but the entire mechanism**: it is precisely what destroys the terms that carry no shared information while allowing the terms that do to accumulate.

Two features of the definition (6.5) deserve a comment, because both differ from the Pearson coefficient met in the historical introduction. The first is that we have not subtracted the mean values before multiplying. Subtracting them would produce the covariance, which asks how the *fluctuations* about the average conspire; keeping them, as we do here, asks about the total quantity, and for the optical fields of this chapter the distinction happily evaporates, since a thermal field oscillates symmetrically about zero and its mean vanishes on its own. The second is that we have not divided by anything, so Γ still carries the physical units of the fields it was built from and its size depends on how intense the light happens to be, not only on how similar it is. A correlation of 10^{-6} between two feeble beams may represent far tighter agreement than a correlation of unity between two powerful ones. Repairing this, by dividing Γ by its own value at zero separation, which is precisely the intensity, so that the result is a pure number confined between zero and one that measures similarity and similarity alone, is exactly the step that turns the correlation function into the *complex degree of coherence*, and it is the point the next section works towards.

6.3 Random variables

In the previous section we spoke freely of random variables, of joint probability densities and of ensemble averages, and we did so on purpose, because the aim there was to make the definition of the correlation function plausible rather than to justify it. That debt must now be paid. Coherence theory is, from beginning to end, a theory *about* averages, and an average that is not defined precisely defines nothing precisely. Worse, the objects we shall manipulate are not averages of numbers but averages over an infinite collection of entire field histories, and for such objects naive intuition is not merely imprecise, it is occasionally wrong.

The purpose of this section is therefore to build the notion of a random variable from the ground up, in the measure theoretic language in which it was cast by Kolmogorov (Kolmogorov, 1933), and then to apply it to the vector electromagnetic field, constructing explicitly the probability space on which the whole of the following theory lives. Every ingredient will be given its physical reading as it appears, because each one of them corresponds to something an optician does in a laboratory.

6.3.1 The sample space and the physical meaning of an event

Definition 31 *(Sample space) A* ***sample space*** *is a nonempty set Ω. Its elements $\omega \in \Omega$ are called* ***elementary outcomes*** *or* ***sample points****.*

The temptation is to read ω as a number, and it is essential to resist it. In optics an elementary outcome is not a field value and not a measurement; it is one complete **realisation** of the source, that is, one entire spacetime history that the field could have had. A thermal lamp contains of the order of 10^{20} atoms radiating independently, each with an emission instant and an initial phase that we neither control nor know. One particular assignment of all those microscopic details, together with everything that follows from it by Maxwell's equations, is one ω. The set of all such assignments is Ω.

This is precisely the picture that Figure 6.1 will show in the next section, where several curves ${}^{(i)}E(\vec{r}, t)$ are drawn one above the other: each curve is one ω, and the label i is only a bookkeeping device, since in general Ω is uncountable. What the laboratory delivers on any given afternoon is a single ω, chosen by nature and not revealed to us. The whole of statistical optics consists in extracting statements about Ω from access to one of its points.

Definition 32 *(Event) An* ***event*** *is a subset $A \subseteq \Omega$.*

The physical content of this definition is that an event is a **statement about the realisation that can be declared true or false**. The statement "the field at $\vec{r}_0$ at the instant t_0 exceeded the value E_0" is not a number; it is the set of all realisations for which that happens to be the case. To say that the event occurred is to say that the ω nature selected belongs to that subset. Events and statements are the same thing seen from two sides, and this identification is what allows probability to be built out of set theory.

6.3.2 The sigma algebra, or which questions may be asked

Definition 33 *(Sigma algebra) A* ***sigma algebra*** *on Ω is a collection $\mathcal{F}$ of subsets of Ω such that*

1. *$\Omega \in \mathcal{F}$,*
2. *if $A \in \mathcal{F}$ then $A^c = \Omega \setminus A \in \mathcal{F}$,*
3. *if $A_1, A_2, A_3, \dots \in \mathcal{F}$ then $\bigcup_{n=1}^{\infty} A_n \in \mathcal{F}$.*

*The pair $(\Omega, \mathcal{F})$ is called a **measurable space**.*

A sigma algebra looks like a piece of bookkeeping and is in fact a physical statement. It is the **catalogue of admissible questions**, the list of statements to which the theory is prepared to attach a probability, and the three axioms are exactly the three properties that any honest catalogue of questions must have.

The first says that the question "did anything at all happen?" is admissible. The second says that if we can ask whether an event occurred, we can ask whether it did not, which is the formal expression of the fact that an apparatus capable of detecting something is by that very fact capable of detecting its absence. The third says that if each of a sequence of questions is admissible, so is the question of whether at least one of them holds.

The word *countable* in the third axiom is not a technical convenience but a genuine physical restriction, and it deserves to be pointed out. Any real experimental protocol is a *sequence* of operations: one records a reading, then another, then another. No laboratory has ever performed an uncountable number of operations, and the theory is built so as to promise nothing about what would happen if one could.

One might ask why we do not simply take $\mathcal{F}$ to be the collection of *all* subsets of Ω and be done with it. The answer is that we are not allowed to. Vitali showed in 1905 (Vitali, 1905) that on the real line there exist sets so pathological that no translation invariant notion of size can be assigned to them without contradiction. If we insisted on measuring everything we would be forced into inconsistency, and the sigma algebra is precisely the device that keeps such monsters outside the theory while retaining every set that a physicist would ever have occasion to ask about.

The example we shall need constantly is the following.

Definition 34 *(Borel sigma algebra) The **Borel sigma algebra** $\mathcal{B}(\mathbb{R}^n)$ is the smallest sigma algebra on $\mathbb{R}^n$ containing all the open sets.*

Physically, $\mathcal{B}(\mathbb{R})$ is generated by the questions "does the reading lie between a and b?", together with everything that can be built from countably many such questions by negation and union. This is exactly the class of questions that a real detector, which has a finite resolution and reports an interval rather than a point, is able to answer. The Borel sigma algebra is not an abstraction imposed on physics; it is the mathematical transcription of what a graduated instrument does.

6.3.3 Measure, the Lebesgue measure and the probability measure

Definition 35 *(Measure) A **measure** on a measurable space $(\Omega, \mathcal{F})$ is a function $\mu : \mathcal{F} \to [0, \infty]$ such that $\mu(\emptyset) = 0$ and such that, for every sequence $A_1, A_2, \ldots$ of pairwise disjoint elements of $\mathcal{F}$,*

$$\mu\left(\bigcup_{n=1}^{\infty} A_n\right) = \sum_{n=1}^{\infty} \mu(A_n). \tag{6.6}$$

*The triple $(\Omega, \mathcal{F}, \mu)$ is a **measure space**.*

A measure is a consistent way of assigning a *size*. Condition (6.6) is countable additivity, and it states nothing more exotic than that the size of a whole assembled out of separate pieces is the sum of the sizes of the pieces. That is what the word size means operationally, and a function which violated it would not deserve the name.

Among all measures one is singled out by the geometry of space itself.

Definition 36 *(Lebesgue measure) The **Lebesgue measure** λ is the unique translation invariant measure on $\mathcal{B}(\mathbb{R}^n)$ satisfying $\lambda\left(\prod_{j=1}^{n}[a_j, b_j]\right) = \prod_{j=1}^{n}(b_j - a_j)$.*

The Lebesgue measure (Lebesgue, 1902) is the mathematical form of the words *length*, *area*, *volume* and *duration*. Every integral we have written so far in these notes has been an integral against λ: when we integrated over an aperture in the theory of diffraction we were measuring *how much aperture*, and when we integrate over the response time of a detector we are measuring *how long we looked*. The Lebesgue measure knows about the extent of the domain and it knows nothing whatever about probability.

It is worth insisting on this separation, because in coherence theory both measures appear at once and they do entirely different jobs. **The Lebesgue measure records how long the detector integrated and how large the aperture was; the probability measure records how likely each realisation of the source was.** The time average of a single beam is an integral against λ on the time axis; the ensemble average over realisations is an integral against P on Ω. These are different operations on different spaces, and the fact that they so often agree is not a triviality but a theorem with hypotheses, to which we return at the end of this section.

The Lebesgue measure also supplies the missing justification for the densities of the previous section. When in (6.5) we wrote $p_2(x_1, t_1; x_2, t_2)\, dx_1 dx_2$ we were tacitly assuming that the probability measure on the space of values is absolutely continuous with respect to λ, so that by the theorem of Radon and Nikodym (Nikodym, 1930) it possesses a density. That is a physical hypothesis and not a formality: it asserts that no field value occurs with strictly positive probability, that is, that the source has no infinitely sharp features. A perfectly monochromatic and perfectly stabilised laser would violate it, since its distribution would be concentrated on a circle in the complex plane, a set of zero Lebesgue measure, and no density with respect to the plane would exist.

Definition 37 *(Probability space) A* ***probability measure*** *is a measure P with $P(\Omega) = 1$. The triple $(\Omega, \mathcal{F}, P)$ is called a* ***probability space****.*

The normalisation $P(\Omega) = 1$ is the statement that the source does *something*: some realisation occurs, and the alternatives listed in Ω are exhaustive. Countable additivity, read now in P, is the statement that mutually exclusive alternatives have probabilities that add.

Here lies the crucial physical remark of this subsection. **The probability measure is where the physics of the source resides.** A thermal lamp, a laser and a single photon emitter may perfectly well share the same Ω and the same $\mathcal{F}$, since all three could in principle produce any of the same field histories; what distinguishes them is that they weight those histories differently. To specify a source is to specify P, and every statement that coherence theory will make about light is, in the last analysis, a statement about P.

6.3.4 Random variables and the meaning of measurability

Definition 38 *(Random variable) Let $(\Omega, \mathcal{F}, P)$ be a probability space and $(S, \mathcal{S})$ a measurable space. A map $X : \Omega \to S$ is a* ***random variable*** *if it is* ***measurable****, that is, if*

$$X^{-1}(B) = \{\omega \in \Omega : X(\omega) \in B\} \in \mathcal{F} \qquad \text{for every } B \in \mathcal{S}. \tag{6.7}$$

Two things must be said about this definition, and both are usually said too quickly.

The first is that **a random variable is not a variable that is random**. It is a perfectly deterministic function. Once the realisation ω has been fixed, $X(\omega)$ is a definite element of S, with nothing capricious about it. All the randomness lives in the selection of ω, which is to say in our ignorance of which realisation nature has produced. The field at a point of spacetime is not fluctuating because it is undecided; it is definite in each realisation and we simply do not know which realisation we are in.

The second is that the measurability condition (6.7) is the mathematical statement that **the instrument is a legitimate instrument**. It demands that the question "did the reading fall inside B?" be one of the admissible questions of $\mathcal{F}$. A non measurable map would be an apparatus that reports an answer to a question the theory refuses to assign a probability to, which is a polite way of saying that it reports numbers that mean nothing. Measurability is the licence to exist.

Since the outcome ω is inaccessible, what an experiment actually reconstructs is not P but its image.

Definition 39 *(Law of a random variable) The* ***law*** *or* ***distribution*** *of X is the probability measure P_X on $(S, \mathcal{S})$ defined by the push forward*

$$P_X(B) = P\left(X^{-1}(B)\right), \qquad B \in \mathcal{S}. \tag{6.8}$$

If P_X is absolutely continuous with respect to the Lebesgue measure, its Radon-Nikodym derivative $p = \mathrm{d}P_X/\mathrm{d}\lambda$ is the ***probability density*** *of X.*

The law is what a histogram estimates. Two sources built on completely different sample spaces may possess identical laws for a given observable, and are then indistinguishable by that observable no matter how long one measures. The density p appearing here is exactly the p_2 that was used without apology in (6.5).

Definition 40 *(Expectation) The* ***expectation*** *of an integrable random variable X is the Lebesgue integral of X over the probability space,*

$$E\{X\} = \int_\Omega X(\omega)\,\mathrm{d}P(\omega) = \int_S x\,\mathrm{d}P_X(x) = \int_S x\,p(x)\,\mathrm{d}\lambda(x), \tag{6.9}$$

the second equality being the change of variables induced by (6.8).

Equation (6.9) retroactively legitimises the definition (6.5) of the correlation function. What was written there as an ensemble average and what was written as an integral against a joint density are not two definitions that happen to agree; they are one definition, given on Ω, and its image on the space of values under the push forward. The first form is conceptually primary and the second is the one we can compute with.

There is now a payoff which is worth stating explicitly, because it converts the central analogy of the previous section from a metaphor into a theorem. Consider the set of random variables of finite second moment,

$$L^2(\Omega, \mathcal{F}, P) = \left\{X : \Omega \to \mathbb{C} \;\middle|\; E\left\{|X|^2\right\} < \infty\right\}, \tag{6.10}$$

equipped with the inner product

$$\langle X, Y\rangle = E\left\{XY^*\right\}. \tag{6.11}$$

This is a Hilbert space, and comparing (6.11) with the definition (6.5) of the correlation function we see that **the correlation function is an inner product**, not by analogy but literally, in the space of random variables of finite energy. The dot product with which the previous section began was not an illustrative parallel; it was the same object, waiting for the space to be named. The weight function W of (6.2) has become the measure P, and the sum over components of (6.1) has become the integral over Ω.

From this identification one inequality follows immediately, and it is the one that makes the normalised degree of coherence meaningful. The Cauchy-Schwarz inequality in $L^2(\Omega, \mathcal{F}, P)$ reads

$$\left|E\left\{XY^*\right\}\right|^2 \le E\left\{|X|^2\right\}\, E\left\{|Y|^2\right\}, \tag{6.12}$$

so that the correlation of two random variables can never exceed the geometric mean of their intensities. When we divide by that geometric mean, as anticipated at the end of the previous section, the result is a pure number lying between zero and one. It is, quite literally, the cosine of the angle between two vectors in a Hilbert space, and this is why a degree of coherence larger than unity is not merely unobserved but impossible.

6.3.5 Stochastic processes

A single random variable describes the field at one point of spacetime. To describe it everywhere at once we need a family.

Definition 41 *(Stochastic process) Let T be an index set. A* ***stochastic process*** *is a family $\{X_\tau\}_{\tau\in T}$ of random variables defined on a common probability space $(\Omega, \mathcal{F}, P)$, equivalently a map $X : T \times \Omega \to S$ such that $X(\tau, \cdot)$ is measurable for every $\tau \in T$.*

A stochastic process admits two readings, and the whole intuition of the subject lies in holding both at once (Doob, 1953). If we fix the index τ and let ω vary, we obtain a random variable: a *vertical* slice through Figure 6.1, comparing all realisations at one instant. If instead we fix ω and let τ vary, we obtain a single function of the index, called a **sample path** or **realisation**: a *horizontal* slice, that is, one of the individual curves of the figure. The experimentalist observes horizontal slices and the theory speaks about vertical ones, and reconciling the two is the central practical problem of statistical optics.

6.3.6 The probability space of the electromagnetic field

We can now carry out the construction that this whole section was preparing, and build explicitly the probability space on which coherence theory operates. Let $D \subseteq \mathbb{R}^3 \times \mathbb{R}$ be the region of spacetime in which we intend to describe the field, and recall from the chapter on electromagnetic optics that the field is a complex, three component object, the analytic signal $\vec{\mathcal{E}} = (\mathcal{E}_x, \mathcal{E}_y, \mathcal{E}_z)$.

The sample space. We take

$$\Omega = \left\{\omega : D \to \mathbb{C}^3 \;\middle|\; \omega \text{ satisfies Maxwell's equations in } D\right\}, \tag{6.13}$$

that is, the set of admissible field histories. The restriction is essential and it is physical: nature does not select arbitrary functions of spacetime but only those compatible with the dynamics. **Randomness enters through the sources and never through the propagation**, which remains exactly as deterministic as it was in the previous chapters. A probability measure on an optical field is a measure on the solution space of Maxwell's equations, and this is why, as we shall see in a later subsection, coherence itself obeys a wave equation.

With this choice the field becomes the simplest map imaginable, namely the evaluation of a function at a point,

$$\mathcal{E}_j(\vec{r}, t) : \Omega \to \mathbb{C}, \qquad \omega \mapsto \omega_j(\vec{r}, t), \qquad j \in \{x, y, z\}, \tag{6.14}$$

so that for each spacetime point and each Cartesian component the field is a complex random variable, and the vector $\vec{\mathcal{E}}(\vec{r}, t) : \Omega \to \mathbb{C}^3$ is a $\mathbb{C}^3$ valued random variable. The whole field is the stochastic process indexed by $T = D \times \{x, y, z\}$.

The sigma algebra. We equip Ω with the **cylinder sigma algebra** $\mathcal{F}$, generated by the sets

$$C = \{\omega \in \Omega \mid (\omega(\vec{r}_1, t_1), \dots, \omega(\vec{r}_n, t_n)) \in B\}, \qquad B \in \mathcal{B}(\mathbb{C}^{3n}), \tag{6.15}$$

for every finite collection of spacetime points and every Borel set B.

The physical reading of (6.15) is the most important remark of this section. **A cylinder set is exactly a statement that can be checked by placing a finite number of detectors at a finite number of spacetime points and looking at their readings.** No experiment has ever measured a field everywhere; every experiment samples it at finitely many places and finitely many instants. The sigma algebra generated by the cylinder sets is therefore the collection of all statements decidable by countable sequences of finite measurements, which is to say, the collection of all statements an optical experiment can ever settle. The cylinder construction is not a mathematical convenience, it is the transcription of experimental practice.

The probability measure. It remains to place a measure on this enormous space, and it would be hopeless to attempt it directly. Kolmogorov's extension theorem (Kolmogorov, 1933) shows that it is unnecessary.

Theorem 17 *(Kolmogorov extension theorem) Given a family of finite dimensional distributions $P_{(\vec{r}_1,t_1),\dots,(\vec{r}_n,t_n)}$ on $\mathbb{C}^{3n}$, one for every finite collection of spacetime points, which is consistent under permutation of the points and under marginalisation over any of them, there exists a unique probability measure P on the cylinder sigma algebra $\mathcal{F}$ whose finite dimensional marginals are the given ones.*

The physical content is emancipating. **To specify a source it is enough to specify the joint statistics of any finite set of detector readings**, and this is precisely the data that an experiment can access. We never have to say what P does to the monstrous space of all field histories; we say what the joint distribution of two detectors is, and of three, and so on, taking care only that these prescriptions do not contradict one another, and Kolmogorov guarantees that they knit together into one consistent object. The probability space of an optical field is built out of exactly the information a laboratory can supply.

The finite energy hypothesis. We assume finally that

$$E\left\{|\mathcal{E}_j(\vec{r}, t)|^2\right\} < \infty \qquad \text{for every } j \text{ and every } (\vec{r}, t) \in D, \tag{6.16}$$

which is the statement that the mean energy density is finite, that is, that the source does not radiate infinite power. Physically it is unobjectionable, and mathematically it is exactly the hypothesis that places every field component inside the Hilbert space (6.10). From this point on, every component of the field at every point of spacetime is a vector in one and the same Hilbert space, and the geometry of that space is the geometry of coherence.

The coherence tensor. We may now define, with every symbol having been given a meaning, the object that the rest of this chapter studies. For two spacetime points and two Cartesian components,

$$\Gamma_{jk}(\vec{r},t;\vec{r}\,',t') = E\left\{\mathcal{E}_j(\vec{r},t)\,\mathcal{E}_k^*(\vec{r}\,',t')\right\} = \langle \mathcal{E}_j(\vec{r},t), \mathcal{E}_k(\vec{r}\,',t')\rangle\,, \tag{6.17}$$

a 3×3 matrix attached to every pair of spacetime points, the inner product being the one of (6.11). This is the vector generalisation of the correlation function (6.5), and three of its properties follow at once from the construction.

1. **Hermitian symmetry.** Exchanging the two points and the two indices and conjugating,
$$\Gamma_{kj}^*(\vec{r}\,',t';\vec{r},t) = E\left\{\mathcal{E}_k^*(\vec{r}\,',t')\,\mathcal{E}_j(\vec{r},t)\right\} = \Gamma_{jk}(\vec{r},t;\vec{r}\,',t'), \tag{6.18}$$
since conjugation may be taken inside the expectation, which is an integral of a complex function.

2. **Non negative definiteness.** For any finite set of spacetime points labelled by m and any complex coefficients $a_j^{(m)}$,
$$\sum_{m,n}\sum_{j,k} a_j^{(m)} a_k^{(n)*}\,\Gamma_{jk}(m;n) = E\left\{\left|\sum_m\sum_j a_j^{(m)}\mathcal{E}_j(m)\right|^2\right\} \geq 0, \tag{6.19}$$
because the expectation of a non negative quantity is non negative. Physically this says that no superposition of field components, at whatever points, can carry negative intensity.

3. **Bounded correlation.** By (6.12),
$$\left|\Gamma_{jk}(\vec{r},t;\vec{r}\,',t')\right|^2 \leq \Gamma_{jj}(\vec{r},t;\vec{r},t)\;\Gamma_{kk}(\vec{r}\,',t';\vec{r}\,',t'), \tag{6.20}$$
which is what allows the normalised degree of coherence to be confined to the interval $[0,1]$.

It is worth noticing where polarisation fits into this. Evaluating (6.17) at a single spacetime point, $\vec{r} = \vec{r}\,'$ and $t = t'$, the diagonal entries Γ_{jj} give the mean intensity carried by each Cartesian component while the off diagonal entries measure the correlation between components, and the resulting matrix is exactly the polarisation matrix (3.67) introduced in the chapter on polarisation. **Polarisation is coherence between the components of the field at one point; coherence proper is correlation between the field at different points.** They are the same tensor read in two different regimes, which is the reason a unified treatment of coherence and polarisation is possible at all (Wolf, 2007).

6.3.7 Stationarity, ergodicity and the right to measure

One question remains, and it is the question on which the experimental relevance of everything above depends. The expectation (6.9) is an average over Ω, and Ω is not accessible: nature hands us one realisation. What we can average over is time, along that single realisation. Under what conditions do the two agree?

The first condition concerns the invariance of the source.

Definition 42 *(Stationarity) Let $\theta_\tau : \Omega \to \Omega$ be the time shift defined by $(\theta_\tau \omega)(\vec{r}, t) = \omega(\vec{r}, t + \tau)$. The field is* ***stationary*** *if P is invariant under every such shift,*

$$P\left(\theta_\tau^{-1} A\right) = P(A) \qquad \text{for all } A \in \mathcal{F} \text{ and all } \tau \in \mathbb{R}. \tag{6.21}$$

This is the measure theoretic form of the statement, given in a more elementary way in the next section, that the statistical properties of the light do not depend on when we choose to start the clock. A lamp switched on an hour ago and left alone is stationary; a pulsed source is not. Its immediate consequence is that the coherence tensor (6.17) can depend on the two instants only through their difference, $\Gamma_{jk}(\vec{r}, \vec{r}\,'; t, t') = \Gamma_{jk}(\vec{r}, \vec{r}\,'; \tau)$ with $\tau = t' - t$, which is what makes the Wiener-Khinchin theorem of the following section possible at all.

The second condition is stronger and concerns the indecomposability of the source.

Definition 43 *(Ergodicity) A stationary field is* ***ergodic*** *if every shift invariant event is trivial, that is, if $\theta_\tau^{-1} A = A$ for all τ implies $P(A) = 0$ or $P(A) = 1$.*

Physically, ergodicity says that the source cannot be split into two sub ensembles that never communicate. If it could, a single realisation would remain trapped in one of them forever and could never report on the other. When it holds, Birkhoff's ergodic theorem (Birkhoff, 1931) applies and gives the result we need,

$$\lim_{T\to\infty} \frac{1}{2T} \int_{-T}^{T} \mathcal{E}_j(\vec{r}, t)\, \mathcal{E}_k^*(\vec{r}\,', t + \tau)\, \mathrm{d}t = E\left\{\mathcal{E}_j(\vec{r}, t)\, \mathcal{E}_k^*(\vec{r}\,', t + \tau)\right\} \tag{6.22}$$

for almost every realisation ω.

Equation (6.22) is the licence, and the only licence, to replace the inaccessible ensemble average by the accessible time average of a single beam, and it is the reason the whole theory is experimentally meaningful rather than merely elegant. Notice, finally, what the two sides of (6.22) are: on the left an integral against the Lebesgue measure on the time axis, on the right an integral against the probability measure on Ω. The two measures we were at pains to keep apart at the beginning of this section meet here, in a single identity, and ergodicity is precisely the hypothesis under which they may be exchanged. With that hypothesis in hand we may return to physics and define the coherence function itself.

6.4 The coherence function

The concept of the coherence function is, at the mathematical level, identical to that of the correlation function given in equation (6.5); both represent correlations. However, coherence is a term widely used in the study of electromagnetic radiation; let us see a little of why this is so.

In Figure 6.1 we observe a component of the electric field, represented by $E(\vec{r}, t)$, seen across several observations or realisations. Behaving as a random variable, in each realisation it has a

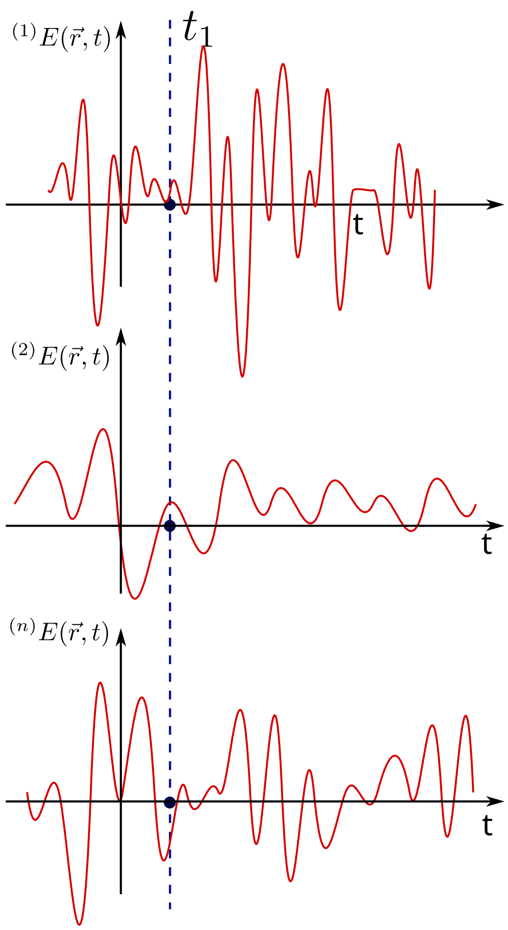


Figure 6.1: Representation of $E(\vec{r}, t)$ as a random variable.

different behaviour in time at a point $\vec{r}$; in general, here we have indexed ${}^{(i)}E(\vec{r},t)$ with integers, but in general there could be an uncountable set of possible states for the electric field. If we look at Figure 6.1 there are two temporal points, t, t', at which we isolate parts of these random oscillations; we could ask how correlated these oscillations are across the realisations over the interval $t'-t$, or in other words, how *coherent* the oscillations are across the realisations.

We speak of coherence when we want to analyse the correlation of electromagnetic radiation between spacetime points. If we have two sources emitting an electric field (for the moment understood as scalar), their coherence function is given by

$$\Gamma(\vec{r},\vec{r}';t,t') = E\left\{\mathcal{E}(\vec{r},t)\mathcal{E}^*(\vec{r}',t')\right\}, \tag{6.23}$$

where we are computing the ensemble average of the components $\mathcal{E}$, which are complex numbers as already discussed in these notes. If we take $\vec{r} = \vec{r}'$ we obtain what is known as the *mutual coherence function*

$$\Gamma(\vec{r}';t,t') = E\left\{\mathcal{E}(\vec{r}',t)\mathcal{E}^*(\vec{r}',t')\right\}. \tag{6.24}$$

We can go even further by including the concept of *statistical stationarity*, but one must be careful since there are several kinds. Formally, a stochastic process $\{X_t\}$ is considered stationary when its statistical properties are invariant under time translations. This condition manifests in different degrees:

1. **Strict (strong) stationarity**: the most demanding form of stationarity requires that all joint probability distributions be time-invariant:

$$F_X(x_{t_1+\tau},\ldots,x_{t_k+\tau}) = F_X(x_{t_1},\ldots,x_{t_k}) \quad \forall \tau, k, t_i, \tag{6.25}$$

   where F_X is the joint distribution function. This implies that:

   - all moments (mean, variance, skewness, etc.) are constant;
   - the marginal distributions are identical at any instant;
   - it is a sufficient but not necessary condition for ARMA models.

2. **Weak (second-order) stationarity**: more practical in applications, it only requires invariance of the first two moments:

$$\begin{cases} \mathbb{E}[X_t] = \mu \quad \text{(constant)} \quad \forall t \\ \mathrm{Var}(X_t) = \sigma^2 \quad \text{(constant)} \quad \forall t \\ \mathrm{Cov}(X_t, X_{t+h}) = \gamma(h) \quad \text{(depends only on } h, \text{ not on } t) \end{cases} \tag{6.26}$$

   This is the definition used in most econometric models (ARIMA, GARCH) and in the WienerKhinchin spectral analysis (which we shall use later).

3. **Trend stationarity (deterministic stationarity)**: a non-stationary process that can be made stationary by removing deterministic components:

$$X_t = \alpha + \beta t + \epsilon_t, \tag{6.27}$$

where ϵ_t is stationary. The non-stationarity here comes exclusively from the temporal trend βt.

We shall use a hypothesis of strong stationarity (although weak stationarity often suffices). In this spirit, if $\tau = t - t'$ we have

$$\Gamma(\vec{r}', t, t') = \Gamma(\vec{r}', t - t') = \Gamma(\vec{r}', \tau). \tag{6.28}$$

As we can see, the mutual coherence at a point $\vec{r}'$ under a stationarity hypothesis gives rise to an explicit dependence on the time difference τ in the ensemble average.

It is worth asking: how necessary is it to analyse the ensemble average; why not take the temporal average? Answering this is **very complicated**, since it touches on the problem of *ergodicity*. Ergodicity, in the sense of ergodic theory, requires that a dynamical system satisfy that the temporal averages coincide with the spatial averages for almost every initial condition. Its rigorous proof is highly non-trivial, to the point that the problems for which ergodicity has been proven with mathematical rigour are few.

From a phenomenological standpoint we shall assume, from now on, an *ergodic hypothesis* in which electromagnetic radiation has a stationary and ergodic character. Under these conditions the mutual coherence function coincides with the temporal average

$$\Gamma(\vec{r}'; t, t') = \int_{\mathbb{R}} \mathcal{E}(\vec{r}', t')\mathcal{E}^*(\vec{r}', t - t')\, dt'. \tag{6.29}$$

The mutual coherence function has several properties, some of which are:

a) $\Gamma(\vec{r}', 0) \geq 0$. This follows from the definition, since $\Gamma(\vec{r}', 0) = \langle |\mathcal{E}(\vec{r}')|^2 \rangle$, i.e. it corresponds to the intensity of the electric field and can only be zero when trivially the field is zero at every instant.

b) $\Gamma(\vec{r}', -\tau) = \Gamma^*(\vec{r}', \tau)$. This is known as the *Hermitian property* of the mutual coherence function, and follows from the fact that $\mathcal{E}(\vec{r}', t)$ is stationary, which allows arbitrary temporal translations from the origin. In this spirit,

$$\Gamma(\vec{r}', \tau) = \langle \mathcal{E}(\vec{r}')\mathcal{E}^*(\vec{r}', t + \tau) \rangle = \langle \mathcal{E}(\vec{r}', t - \tau)\mathcal{E}^*(\vec{r}') \rangle = \Gamma^*(\vec{r}', -\tau). \tag{6.30}$$

c) $|\Gamma(\vec{r}', \tau)| \leq \Gamma(\vec{r}', 0)$. This property follows from the *CauchySchwarz inequality*

$$|\Gamma(\vec{r}', \tau)|^2 = |\, \langle \mathcal{E}(\vec{r}')\mathcal{E}^*(\vec{r}', t+\tau) \rangle \,|^2 \leq \langle \mathcal{E}(\vec{r}', t)\mathcal{E}^*(\vec{r}', t) \rangle \, \langle \mathcal{E}(\vec{r}', t+\tau)\mathcal{E}^*(\vec{r}', t+\tau) \rangle = \Gamma^2(\vec{r}', 0), \tag{6.31}$$

which implies that $|\Gamma(\vec{r}', \tau)|$ can never exceed its initial value $\Gamma(\vec{r}', 0)$. (This property is, let us recall, exclusive to the mutual coherence function; when we consider the coherence between two sources we will see that it is not satisfied - on the contrary, it will increase.)

d) $\Gamma(\vec{r}', \tau)$ is non-negative. We can see this from the definition through the following, rather obvious, inequality

$$\left\langle \left| \int_{T_1}^{T_2} f(t)\mathcal{E}(t)\, dt \right|^2 \right\rangle \geq 0, \tag{6.32}$$

where $f(t)$ is an arbitrary function and T_1, T_2 are arbitrary times. In this way we obtain the inequality

$$\int_{T_1}^{T_2} \int_{T_1}^{T_2} f^*(\vec{r}', t) f(\vec{r}', t')\Gamma(\vec{r}', t'-t)\, dt\, dt' \geq 0. \tag{6.33}$$

e) If the electric field is random, stationary and differentiable, then we can find a velocity $\xi(\vec{r}', t) = \dot{\mathcal{E}}(\vec{r}', t)$, which is stationary, random and of zero mean value, and whose mutual coherence function is

$$\Gamma_\xi(\vec{r}', \tau) = -\ddot{\Gamma}_{\mathcal{E}}(\vec{r}', \tau). \tag{6.34}$$

f) $\Gamma(\vec{r}', \tau)$ is continuous. This holds even when there are jump discontinuities.

Finally, it is worth mentioning that there is a quantity known as the complex degree of coherence, given by

$$\gamma(\vec{r}', \tau) = \frac{\Gamma(\vec{r}', \tau)}{\Gamma(\vec{r}', 0)}, \tag{6.35}$$

and from the CauchySchwarz inequality we have $0 \leq |\gamma(\vec{r}', \tau)| \leq 1$.

6.4.1 Manifestations of coherence: the WienerKhinchin theorem

The mutual coherence function and the power spectral density are two sides of the same coin in the study of light and wave signals. Mutual coherence describes how the fluctuations of a luminous field are correlated in space and time, acting as a map that reveals to what extent the variations at one point of space are synchronised with those at another, even with a certain temporal delay. The power spectral density, in turn, decomposes the energy of the signal into its frequency components, showing which tones or colours - in the case of light - dominate its behaviour. The intimate relation between the two concepts lies in the fact that, through mathematical transformations, it is possible to convert the detailed information on the spacetime correlations captured by mutual coherence into a precise description of how the energy is distributed in the frequency domain, and vice versa. This link allows researchers to jump between complementary perspectives: analysing interference patterns in an interferometer or identifying spectral signatures in a material, all from the same fundamental principles.

We are going to find one of the most important theorems in coherence theory, the *WienerKhinchin theorem*. Note that stationary random signals do not have a Fourier transform; hence we may define the following truncated random signals of the electric field

$$
{}^{\omega}\mathcal{E}_T(\vec{r},t) = \begin{cases} {}^{\omega}\mathcal{E}(\vec{r},t) & -\frac{T}{2} \leq t \leq \frac{T}{2} \\ 0 & \text{otherwise} \end{cases} \tag{6.36}
$$

and we define the spectral power for a realisation ω as the squared magnitude of the Fourier transforms of the electric fields (6.36), that is,

$$
{}^{\omega}S_T(\vec{r},\nu) = \left| \int_{\mathbb{R}}^{\omega} \mathcal{E}_T(\vec{r},t) e^{i2\pi\nu t} dt \right|^2 = \left| \int_{-T/2}^{T/2} {}^{\omega}\mathcal{E}(\vec{r},t) e^{i2\pi\nu t} dt \right|^2 , \tag{6.37}
$$

and the *power spectral density of the event* ω we compute as

$$
\lim_{T\to\infty} \frac{{}^{\omega}S_T(\vec{r},\nu)}{T} = {}^{\omega}S(\vec{r},\nu). \tag{6.38}
$$

Since we already possess a tool to study, in each event or realisation, the power spectral density, then, through its ensemble average, we can find the total density, that is,

$$
S(\vec{r},\nu) = E\left\{ {}^{\omega}S(\vec{r},\nu) \right\}, \tag{6.39}
$$

introducing equations (6.37) and (6.38) we then have

$$
S(\vec{r},\nu) = E\left\{ \lim_{T\to\infty} \left| \frac{1}{T} \int_{-T/2}^{T/2} {}^{\omega}\mathcal{E}(\vec{r},t) e^{i2\pi\nu t} dt \right|^2 \right\}, \tag{6.40}
$$

$$
S(\vec{r},\nu) = \lim_{T\to\infty} \frac{1}{T} \int_{-T/2}^{T/2} \int_{-T/2}^{T/2} E\left\{ \mathcal{E}(\vec{r},t)\mathcal{E}^*(\vec{r}',t') \right\} e^{i2\pi\nu(t-t')} dt\, dt', \tag{6.41}
$$

$$
S(\vec{r},\nu) = \int_{\mathbb{R}} \Gamma(\vec{r},\vec{r}';\tau) e^{i2\pi\nu\tau} d\tau, \tag{6.42}
$$

$$\boxed{S(\vec{r},\nu) = \mathcal{F}\left\{\Gamma(\vec{r},\vec{r}',\tau)\right\}}. \tag{6.43}$$

Result (6.43) reads: the Fourier transform of the coherence function corresponds to the power spectral density, and is known as the WienerKhinchin theorem. Its use is fundamental, since it tells us that by methods such as spectrometry we can find the coherence function of any phenomenon; we can even go further, since this result really stems from something more general - the correlation function - meaning that every stochastic phenomenon has an associated spectrum, and from the measurements or estimates of that spectrum we can compute the correlations in the system.

If we make a simple observation, the optical range of radiation - the visible - is a band of very low frequencies relative to the whole spectrum; the visible runs roughly from 400 nm to 700 nm, and we may say that all of optics is *narrowband*. This obeys the inequality

$$\frac{\triangle\nu}{\nu} \ll 1, \tag{6.44}$$

and, in this narrowband regime, the coherence function takes the form

$$\Gamma(\vec{r};\tau) = \tilde{\Gamma}(\vec{r})e^{i2\pi\tilde{\nu}\tau}, \tag{6.45}$$

which we can represent graphically as shown in Figure 6.2, where one can see the behaviour of the coherence function in the narrowband regime.[4] There one sees an envelope (the observable) enveloping all the fundamental frequencies.

6.4.2 The wave equation of coherence

Let us step a little out of the comfort zone and understand how correlations propagate through space and time. Let $\mathcal{E}^{(r)}(\vec{r},t)$ be some **real** component of the electric field, viewed as a random signal. As is well known from classical electromagnetic theory, it obeys Maxwell's equations and therefore has an associated wave equation, namely

$$\nabla^2\mathcal{E}^{(r)}(\vec{r},t) = \frac{1}{C^2}\frac{\partial^2\mathcal{E}^{(r)}(\vec{r},t)}{\partial t^2}, \tag{6.46}$$

where C is the speed of light in vacuum. We also have, by Fourier analysis,

$$\mathcal{E}(\vec{r},t) = \int_0^\infty \tilde{\mathcal{E}}(\vec{r},\nu)e^{-i2\pi\nu t}d\nu. \tag{6.47}$$

[4]One usually finds in the literature a term similar but not equal to that of narrowband, known as the quasi-monochromaticity condition, written as $\frac{\triangle\lambda}{\lambda} \ll 1$; but this bounds the wavelength so that it is as monochromatic as possible, while the narrowband condition admits several frequencies, provided they are narrow relative to the whole frequency spectrum.

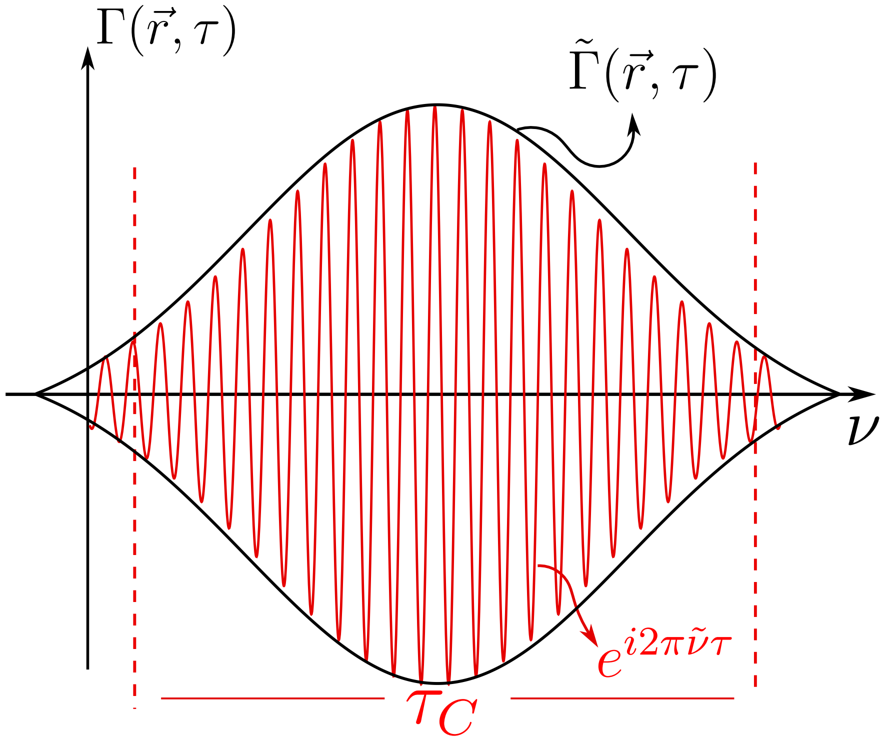


Figure 6.2: Graphical representation of the narrowband coherence function. In black, the envelope $\tilde{\Gamma}(\vec{r},\tau)$ is shown, and in red the oscillations $e^{i2\pi\tilde{\nu}\tau}$. These oscillations are confined within something known as the coherence time τ_C, which we shall see later.

And if we apply the d'Alembertian operator $\Box = \nabla^2 - \frac{1}{C^2}\frac{\partial^2}{\partial t^2}$ to equation (6.47), we find that $\tilde{\mathcal{E}}(\vec{r},\nu)$ has the form of a Helmholtz-type equation, that is,

$$\nabla^2\mathcal{E}(\vec{r},t) - \frac{1}{C^2}\frac{\partial^2\mathcal{E}(\vec{r},t)}{\partial t^2} = \int_0^\infty \left[\nabla^2\tilde{\mathcal{E}}(\vec{r},\nu) + k^2\tilde{\mathcal{E}}(\vec{r},\nu)\right] e^{-i2\pi\nu t} d\nu, \tag{6.48}$$

which leads us to the fact that the electric field, as an analytic and complex signal, also obeys a wave-type equation

$$\nabla^2\mathcal{E}(\vec{r},t) = \frac{1}{C^2}\frac{\partial^2\mathcal{E}(\vec{r},t)}{\partial t^2}. \tag{6.49}$$

Taking the complex conjugate of the previous equation we have

$$\nabla^2\mathcal{E}^*(\vec{r},t) = \frac{1}{C^2}\frac{\partial^2\mathcal{E}^*(\vec{r},t)}{\partial t^2}. \tag{6.50}$$

If we multiply both sides of equation (6.50) by $\mathcal{E}(\vec{r}',t')$, and since we have taken the d'Alembertian with respect to the unprimed variables, using some sign rules we have

$$\nabla^2\left[\mathcal{E}^*(\vec{r},t)\mathcal{E}(\vec{r}',t')\right] = \frac{1}{C^2}\frac{\partial^2}{\partial t^2}\left[\mathcal{E}^*(\vec{r},t)\mathcal{E}(\vec{r}',t')\right]. \tag{6.51}$$

Taking the ensemble average of equation (6.51) over all realisations of the field, this wave equation becomes

$$\nabla^2\Gamma(\vec{r},\vec{r}';t,t') = \frac{1}{C^2}\frac{\partial^2}{\partial t^2}\Gamma(\vec{r},\vec{r}';t,t'), \tag{6.52}$$

and, similarly, using $\Box' = \nabla'^2 - \frac{1}{C^2}\frac{\partial^2}{\partial t'^2}$ we have another wave equation given by

$$\nabla'^2\Gamma(\vec{r},\vec{r}';t,t') = \frac{1}{C^2}\frac{\partial^2}{\partial t'^2}\Gamma(\vec{r},\vec{r}';t,t'). \tag{6.53}$$

Equations (6.52) and (6.53) tell us directly that the coherence functions between two spacetime points evolve in time following a wave-type equation, separately, with respect to each pair of spacetime coordinates - a result that is, without doubt, remarkable. We can take this result even further and study this propagation in the next section.

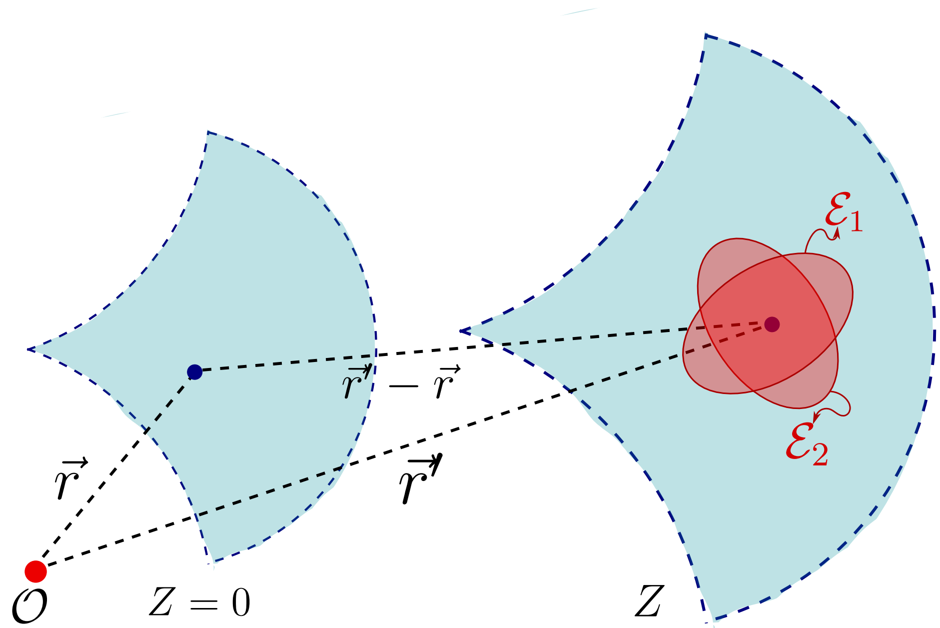


Figure 6.3: Propagation of the electric field from a plane $Z = 0$ to a plane Z, where each point of the wavefront carries its mean polarisation ellipse.

6.4.3 Propagation of coherence and the direction of time: the van CittertZernike theorem

As we know, the electric field obeys a wave-type equation and the orientation of the fields is given by the polarisation of the field; in turn, as we saw in the diffraction chapter (see Section 5.2), at the classical level light fundamentally propagates as spherical waves, as illustrated in Figure 6.3, and at each point of the wavefront the components of the electric field oscillate randomly, but within the mean polarisation ellipse of their representative state. Mathematically we can represent this component as

$$\mathcal{E}_z(\vec{r}_1, \nu) = \frac{i}{\lambda} \int_\Sigma \mathcal{E}_0(\vec{r}, \nu) \frac{e^{ik\|\vec{r}_1 - \vec{r}\|}}{\|\vec{r}_1 - \vec{r}\|} \cos(\hat{z}; \vec{r}_1 - \vec{r})\, d\sigma. \tag{6.54}$$

Using equation (6.47) we have

$$\mathcal{E}_z(\vec{r}_1, t) = \frac{1}{i} \int_{\mathbb{R}^+} \frac{\nu}{V} \int_\Sigma \mathcal{E}_0(\vec{r}, \nu) \frac{\cos(\hat{z}; \vec{r}_1 - \vec{r})}{\|\vec{r}_1 - \vec{r}\|} e^{i2\pi\nu\left(t + \frac{\|\vec{r}_1 - \vec{r}\|}{V}\right)} d\sigma\, d\nu, \tag{6.55}$$

where V is the propagation velocity of the envelope. Rewriting a little more, we have

$$\mathcal{E}_z(\vec{r}_1, t) = \frac{1}{i} \int_\Sigma \frac{\cos(\hat{z}; \vec{r}_1 - \vec{r})}{\|\vec{r}_1 - \vec{r}\|} \int_{\mathbb{R}^+} \frac{\nu}{V} \mathcal{E}_0(\vec{r}, \nu) e^{i2\pi\nu\left(t + \frac{\|\vec{r}_1 - \vec{r}\|}{V}\right)} d\nu\, d\sigma, \tag{6.56}$$

and, taking the narrowband approximation, we have

$$\mathcal{E}_z(\vec{r}_1,t)=\frac{1}{i\tilde{\lambda}}\int_{\Sigma}\mathcal{E}_0\left(\vec{r},t+\frac{\|\vec{r}_1-\vec{r}\|}{V}\right)\frac{\cos\left(\hat{z};\vec{r}_1-\vec{r}\right)}{\|\vec{r}_1-\vec{r}\|}d\sigma. \tag{6.57}$$

We can visualise graphically how we intend to propagate in Figure 6.4, where we observe that we are starting from the coherence function between two points $\vec{r},\vec{r}'$ at instants t,t', respectively, and carrying it to a coherence function at the points $\vec{r}_1,\vec{r}_2$ at two instants t_1,t_2, respectively.

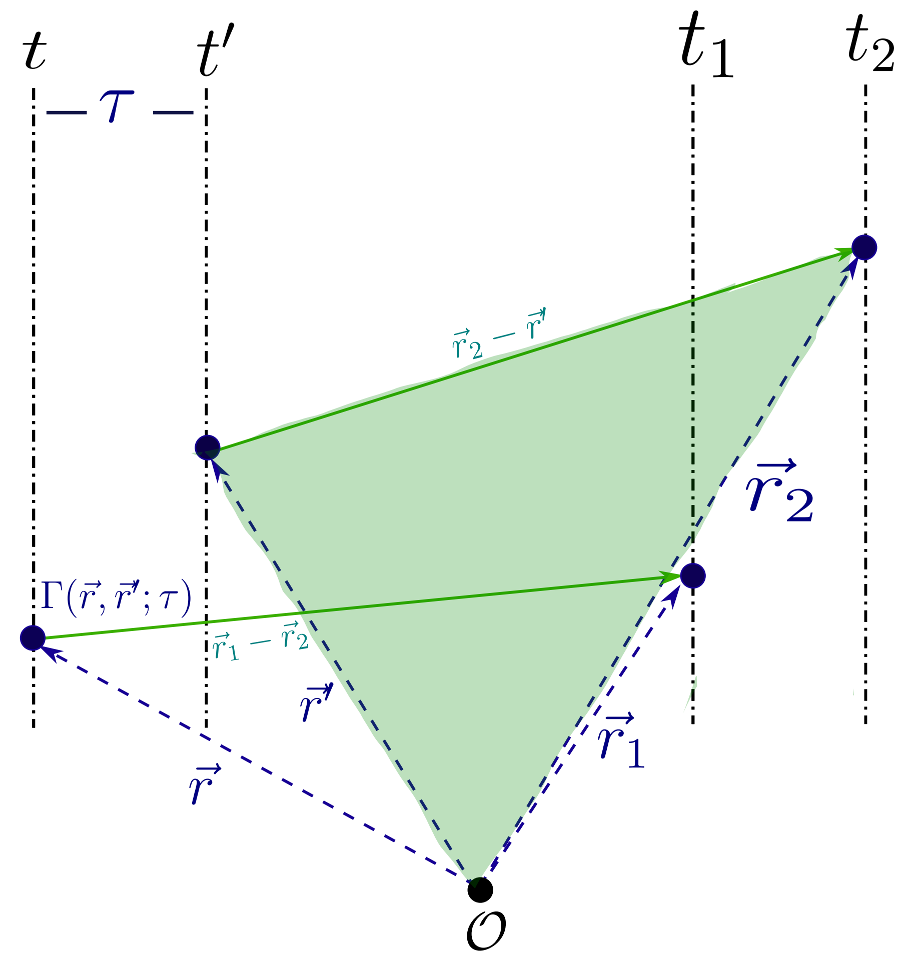


Figure 6.4: Propagation of the coherence function between two points $\vec{r},\vec{r}'$ on the initial plane to the coherence between two points $\vec{r}_1,\vec{r}_2$ on the observation plane.

Starting from equation (6.57) we can compute the coherence function in its propagation as

$$\Gamma_z(\vec{r}_1,\vec{r}_2;\tau)=E\left\{\mathcal{E}_z(\vec{r}_1,t)\mathcal{E}_z^*(\vec{r}_2,t-\tau)\right\}, \tag{6.58}$$

$$\Gamma_z(\vec{r}_1,\vec{r}_2;\tau) = E\left\{\frac{1}{i\tilde{\lambda}}\int_\Sigma \mathcal{E}_0\left(\vec{r}, t+\frac{\|\vec{r}_1-\vec{r}\|}{V}\right)\frac{\cos(\hat{z};\vec{r}_1-\vec{r})}{\|\vec{r}_1-\vec{r}\|}d\sigma\right. \tag{6.59}$$

$$\left.\cdot\frac{1}{-i\tilde{\lambda}}\int_{\Sigma'}\mathcal{E}_0^*\left(\vec{r}', t-\tau+\frac{\|\vec{r}_2-\vec{r}'\|}{V}\right)\frac{\cos(\hat{z};\vec{r}_2-\vec{r}')}{\|\vec{r}_2-\vec{r}'\|}d\sigma'\right\}, \tag{6.60}$$

$$\Gamma_z(\vec{r}_1,\vec{r}_2;\tau) = \frac{1}{\tilde{\lambda}^2}\int_\Sigma\int_{\Sigma'} E\left\{\mathcal{E}_0\left(\vec{r}, t+\frac{\|\vec{r}_1-\vec{r}\|}{V}\right)\mathcal{E}_0^*\left(\vec{r}', t-\tau+\frac{\|\vec{r}_2-\vec{r}'\|}{V}\right)\right\}\frac{\cos(\hat{z};\vec{r}_1-\vec{r})\cos(\hat{z};\vec{r}_2-\vec{r}')}{\|\vec{r}_2-\vec{r}'\|\cdot\|\vec{r}_1-\vec{r}\|}d\sigma\, d\sigma' \tag{6.61}$$

$$\Gamma_z(\vec{r}_1,\vec{r}_2;\tau) = \frac{1}{\tilde{\lambda}^2}\int_\Sigma\int_{\Sigma'}\Gamma_0\left(\vec{r},\vec{r}';\tau+\frac{\|\vec{r}_1-\vec{r}\|+\|\vec{r}_2-\vec{r}'\|}{V}\right)\frac{\cos(\hat{z};\vec{r}_1-\vec{r})\cos(\hat{z};\vec{r}_2-\vec{r}')}{\|\vec{r}_2-\vec{r}'\|\cdot\|\vec{r}_1-\vec{r}\|}d\sigma\, d\sigma' \tag{6.62}$$

The propagation of the random oscillations across n realisations can be visualised in Figure 6.5.

We are going to impose the following condition on the coherence function

$$\Gamma(\vec{r},\vec{r}';\tau) = \Gamma(\vec{r}';\tau)\delta(\vec{r}-\vec{r}'), \tag{6.63}$$

which, in a few words, means that, spatially speaking, the distribution representing the coherence function is point-localised and weighted by the mutual coherence at the point $\vec{r}'$. Imposing this on equation (6.62) we have

$$\Gamma_z(\vec{r}_1,\vec{r}_2;\tau) = \frac{1}{\tilde{\lambda}^2}\int_\Sigma \Gamma_0\left(\vec{r};\tau+\frac{\|\vec{r}-\vec{r}_1\|-\|\vec{r}-\vec{r}_2\|}{V}\right)\frac{\cos(\hat{z};\vec{r}_1-\vec{r})\cos(\hat{z};\vec{r}_2-\vec{r})}{\|\vec{r}_2-\vec{r}\|\cdot\|\vec{r}_1-\vec{r}\|}d\sigma. \tag{6.64}$$

In nature we will not always have narrowband sources; however, we can resort to another argument, namely that most instruments do not have an infinite or broad range but a short one - or, in other words, measurement instruments measure a narrow part of the spectrum. Therefore, including the narrowband condition we have

$$\Gamma_z(\vec{r}_1,\vec{r}_2)e^{i2\pi\tilde{\nu}\tau} = \frac{1}{\tilde{\lambda}^2}\int_\Sigma \Gamma_0(\vec{r})e^{i2\pi\tilde{\nu}\left(\tau+\frac{\|\vec{r}-\vec{r}_1\|-\|\vec{r}-\vec{r}_2\|}{V}\right)}\frac{\cos(\hat{z};\vec{r}_1-\vec{r})\cos(\hat{z};\vec{r}_2-\vec{r})}{\|\vec{r}_2-\vec{r}\|\cdot\|\vec{r}_1-\vec{r}\|}d\sigma. \tag{6.65}$$

Given the coherence time defined by $\tau_C = \int_{\mathbb{R}^+}|\gamma(\vec{r},\tau)|^2 d\tau$, and if we impose a condition of very small times, smaller than a coherence time $\tau \ll \tau_C$, we have

$$\Gamma_z(\vec{r}_1,\vec{r}_2) = \frac{1}{\tilde{\lambda}^2}\int_\Sigma I_0(\vec{r})e^{i2\pi\left(\frac{\|\vec{r}-\vec{r}_1\|-\|\vec{r}-\vec{r}_2\|}{V/\tilde{\nu}}\right)}\frac{\cos(\hat{z};\vec{r}_1-\vec{r})\cos(\hat{z};\vec{r}_2-\vec{r})}{\|\vec{r}_2-\vec{r}\|\cdot\|\vec{r}_1-\vec{r}\|}d\sigma. \tag{6.66}$$

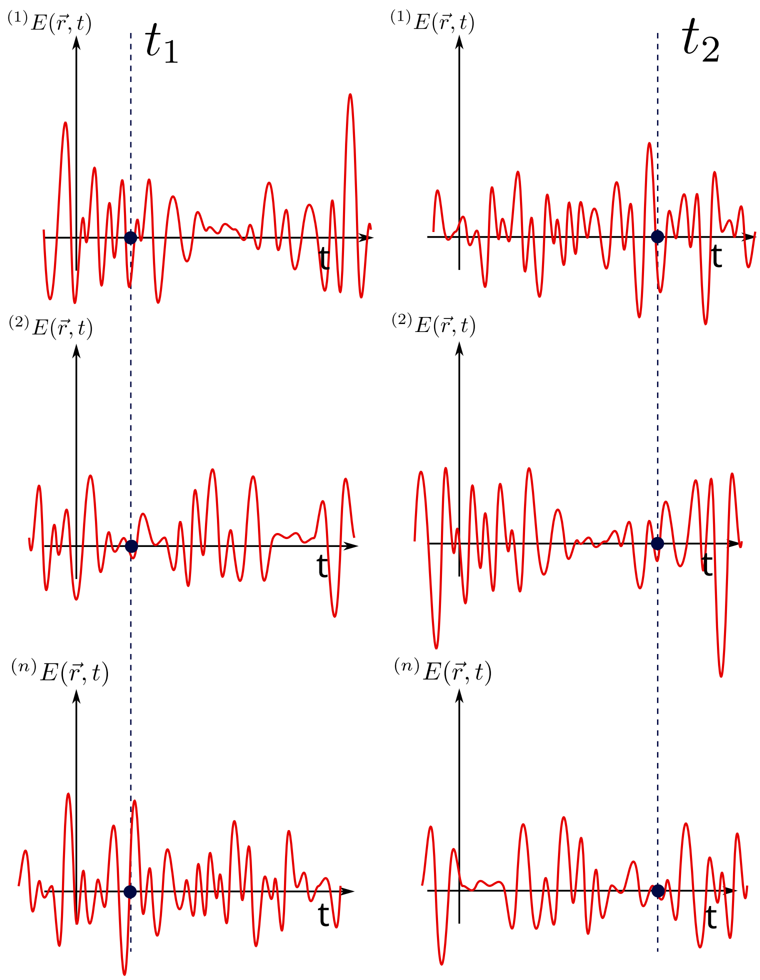


Figure 6.5: Evolution of the realisations upon propagating the electric field.

Taking the far-field approximation we can simplify the problem as follows

$$\Gamma_z(\vec{r}_1,\vec{r}_2)=\frac{1}{\tilde{\lambda}^2 z^2}\int_\Sigma I_0(\vec{r})e^{i2\pi\left(\frac{\|\vec{r}-\vec{r}_1\|-\|\vec{r}-\vec{r}_2\|}{V/\tilde{\nu}}\right)}d\sigma. \tag{6.67}$$

We may now make a Taylor-series expansion, the same one performed in the diffraction chapter. In this spirit the following terms become

$$\|\vec{r}-\vec{r}_1\|=\left(z^2+(x-x_1)^2+(y-y_1)^2\right)^{1/2}=z\left(1+\frac{(x-x_1)^2+(y-y_1)^2}{z^2}\right)^{1/2}\approx z+\frac{(x-x_1)^2+(y-y_1)^2}{2z}, \tag{6.68}$$

$$\|\vec{r}-\vec{r}_2\|\approx z+\frac{(x-x_2)^2+(y-y_2)^2}{2z}. \tag{6.69}$$

Introducing this into integral (6.67) we obtain

$$\Gamma_z(\vec{r}_1,\vec{r}_2)=\frac{1}{\tilde{\lambda}^2 z^2}\int_{\mathbb{R}^2} I_0(x,y)e^{-\frac{i2\pi}{\tilde{\lambda}z}(xx_1+yy_1)}e^{\frac{i2\pi}{\tilde{\lambda}z}(xx_2+yy_2)}e^{-\frac{i\pi}{\tilde{\lambda}z}(x_1^2+y_1^2)}dx\,dy. \tag{6.70}$$

Let us define the vectors $\vec{S}_1=(x_1,y_1)$, $\vec{S}_2=(x_2,y_2)$ and $\vec{S}=(x,y)$. Then

$$\Gamma_z(\vec{S}_1,\vec{S}_2)=\frac{e^{i\frac{\pi}{\tilde{\lambda}z}(\|\vec{S}_1\|^2-\|\vec{S}_2\|^2)}}{\tilde{\lambda}^2 z^2}\int_{\mathbb{R}^2} I_0(\vec{S})e^{-i\frac{2\pi}{\tilde{\lambda}z}\vec{S}\cdot(\vec{S}_1-\vec{S}_2)}dS. \tag{6.71}$$

If we set $\vec{S}_2=\vec{0}$ we arrive at the Fourier integral

$$\boxed{\Gamma_z(\vec{S}_1)=\frac{e^{\frac{i\pi}{\tilde{\lambda}z}\|\vec{S}_1\|^2}}{\tilde{\lambda}^2 z^2}\int_{\mathbb{R}^2}\Gamma_0(\vec{S})e^{-\frac{i2\pi}{\tilde{\lambda}z}\vec{S}\cdot\vec{S}_1}dS}, \tag{6.72}$$

which, basically, means that the correlations, under all the approximations we have made, propagate by means of a Fourier integral; or, what is the same, the van CittertZernike theorem tells us that coherence diffracts - it propagates naturally as a diffraction integral (see Figure 6.6).

How can we tell in which direction time flows? This may sound like a question rather out of context for these notes, but if we could travel back to the golden age of Boltzmann, we would find a heated debate of the time on how statistical physics gives a direction to time, an *arrow*. Paraphrasing Boltzmann, it is said that *time advances in the direction in which correlations increase*. If we look at the van CittertZernike theorem, we can start from two completely uncorrelated sources but, as they propagate, they gradually become correlated; time grows and so does coherence, until a point is reached at which the radiation of those sources is coherent. This is, without doubt, one of the strongest and most beautiful results we can find in statistical physics.

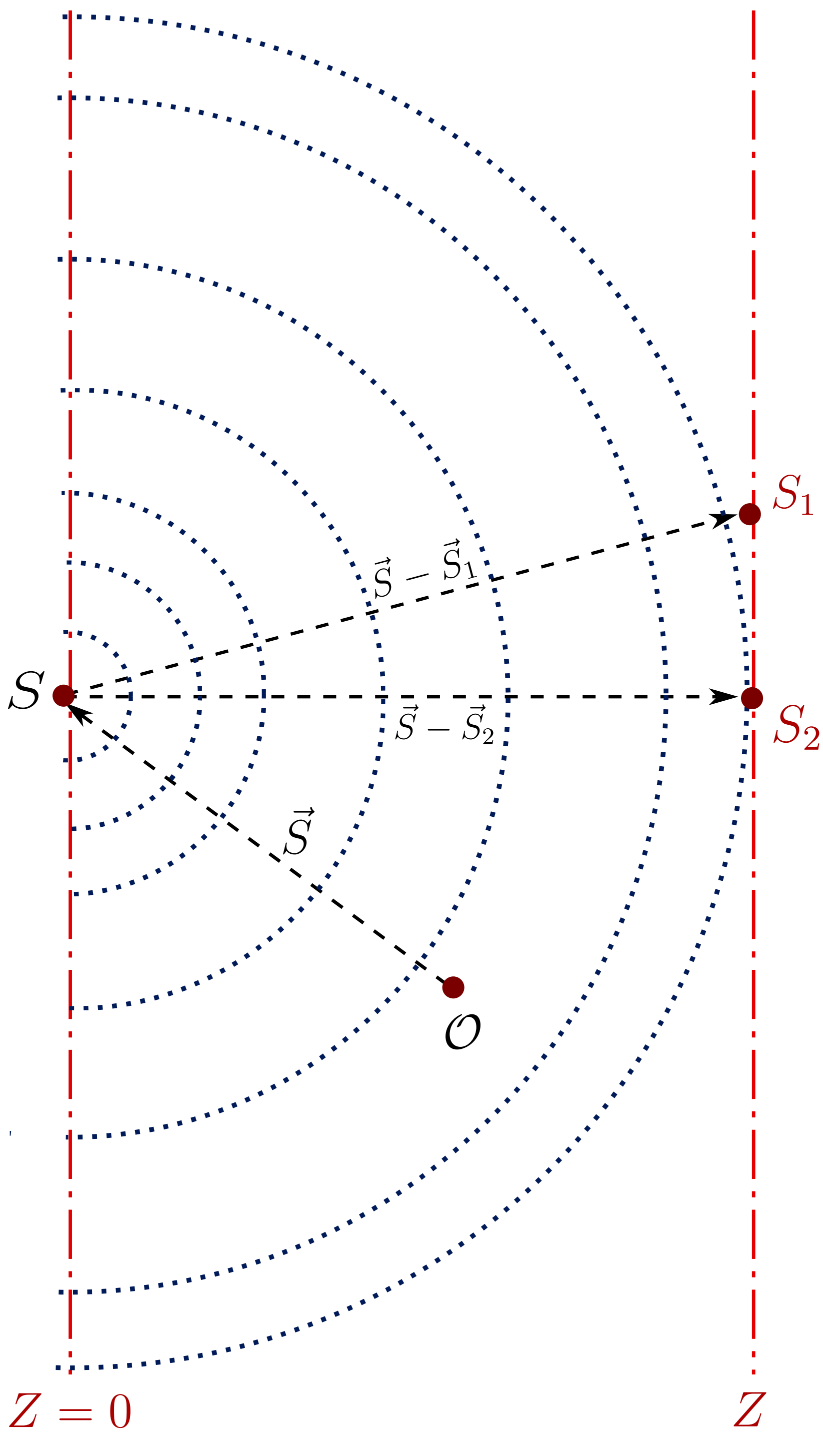


Figure 6.6: The mutual coherence propagated to the far field is the Fourier transform of the intensity distribution $\Gamma_0(\vec{S})$ of the source.

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